

INTRINSIC METRIC DATA AND CANCELLATION LAWS FOR POLYNOMIAL OBSERVATIONS

QIAN QI

ABSTRACT. A polynomial observation germ carries metric information which is not determined by its contact orders. We identify a canonical closed convex set of residual tensors whose positive definite support data are exactly all leading entrance coefficients. For full-rank binary stochastic polynomial observations, its constrained local sections are quadratic quotient forms. We characterize precisely the finite ray probes which determine such a form, prove an adaptive query lower bound over the class of observation metrics, and identify the smaller piecewise-quadratic invariant seen after one-sided endpoint variables are eliminated. For polynomial probability curves we obtain a relative distance factorization, uniform through arbitrary cancellations of a finite residual polynomial. It gives the exact identification wall, every opening along a bounded semialgebraic parameter arc, and the crossover coefficient. The real divisorial order formula computes the exponent; the tensor set retains the metric information which that formula discards. The complete multiplicity theorem, rank-deficient cubic classification, two cubic walls, weighted real comparisons, and their proofs accompany the principal article without deletion.

1. INTRODUCTION

Two different questions arise in a polynomial inverse problem. Near an identifying observation, which joint root displacements are compatible with a small observation error? Near a nonidentifying observation, at what distance does a remote representative enter the observation ball? Orders alone answer neither question sharply. The first also requires a quotient metric on the admissible root directions. The second requires the leading residual vectors, and not only their orders of vanishing.

We use the same construction for both questions. Given a set E of leading residual vectors, form

$$\mathcal{C}(E) = \text{cl}(\text{conv}\{yy^\top : y \in E\} + \text{Sym}_M^+). \quad (1.1)$$

Here Sym_M^+ is the cone of positive semidefinite real symmetric matrices. For every positive definite observation metric H ,

$$\inf_{S \in \mathcal{C}(E)} \text{tr}(HS) = \inf_{y \in E} y^\top Hy.$$

Conversely, these infima, for all $H > 0$, determine $\mathcal{C}(E)$. The substantive issue is to identify the correct leading residual set and to prove that minimization in the original experiment converges to minimization over that set.

Theorem 2.2 solves this issue for an isolated real semialgebraic remote germ. It yields a canonical pair $(\rho, \mathcal{C})$: the exponent is independent of the positive definite norm, and $\mathcal{C}$ is a complete invariant of the leading coefficient as the norm varies. A real monomial presentation computes the pair but is not part of its definition. In particular, no choice of exceptional divisors is declared canonical.

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For the binary stochastic polynomial model, the constrained local leading residuals are affine spaces $La + Dv$. Their metric minima are quadratic forms $v^\top Qv$. Theorem 3.3 characterizes all sufficient finite ray-probe families. The familiar polarization family is optimal for arbitrary positive definite observation metrics, even against deterministic adaptive exact queries. This is an oracle lower bound, not a sampling theorem. For a fixed Hellinger family, the exact condition is a secant-kernel criterion; the larger metric-class lower bound is not silently transferred to that smaller family. One-sided endpoint directions require an additional distinction: the leading spectrum can see only a piecewise-quadratic partial minimum, not the entire matrix. Theorem 3.5 identifies this effective invariant.

Theorem 4.1 is the principal concrete consequence for remote identification. For

$$(x^m, g_a(x)), \quad g_a(x) = \sum_{i=1}^s a_i x^{n_i}, \quad m < n_1 < \cdots < n_s,$$

observed near $(\delta, 0)$, put $z = \delta^{1/m}$. The distance is

$$d_H(a, \delta) = \kappa_H |g_a(z)| (1 + O(z^{n_1-m})), \quad \kappa_H^2 = h_{22} - h_{12}^2/h_{11}.$$

The assertion is a two-sided relative estimate, including the exact zero set, uniformly for bounded coefficients. It therefore remains informative on moving parameter paths where several leading monomials cancel. It classifies the exact wall $g_a(z) = 0$, all nonzero arc openings, and the transition between successive contact orders. A strictly positive polynomial probability realization gives the corresponding Hellinger statement with its Fisher-normalized coefficient.

Relation to the classical order theory. Integral closure, normalized blow-ups, and divisorial order tests already provide the classical mechanism for rational contact exponents; the complex-analytic account and real-analytic appendix in [1] are directly relevant. We do not claim a new valuative criterion for integral closure. Our real sectors encode the feasible inequalities and positive time before orders are taken. Beyond the order, we retain the residual tensor envelope and prove the cancellation-uniform wall formula. The latter is not a consequence of knowing a single exponent. Likewise, the signed real tropical framework of [2] records signs and valuations, whereas cancellation of equal-weight terms depends on their residues. Corollary 4.4 specifies exactly how that distinction controls the wall in this family.

Organization and the accompanying mathematical record. Section 2 constructs the canonical metric invariant. Section 3 develops the binary local theory, observable-law formulation, exact probe criterion, and endpoint elimination. Section 4 proves the uniform polynomial wall classification. Appendix A specifies the real monomial presentation and relates it to the intrinsic invariant. Appendix B records the exact relation to the accompanying results.

The supporting volume is the complete preceding article, not an abridgment. It contains the full root-multiplicity and boundary atlas; exact rank-deficient cubic fibre; inverse through channel-rank loss; weight and endpoint walls; signed second-order and equality-shell computations; real weighted atlas; Puiseux-residue and finite-jet comparisons; separation results; and fixed-format entrance theory. Their original hypotheses and conclusions remain in force. This principal article reorganizes the new argument rather than repeating that entire sequence as its introduction.

2. THE INTRINSIC METRIC CONTACT ENVELOPE

Let N be a compact semialgebraic set, let $R : N \times [0, \delta_0] \rightarrow \mathbb{R}^M$ be continuous and semialgebraic, and suppose that $R(x, 0) = 0$ has just one solution x_* in N . We assume

$$\min_{x \in N} |R(x, \delta)| > 0 \quad (0 < \delta \ll 1), \quad R(x_*, \delta) \longrightarrow 0. \quad (2.1)$$

All residual coordinates and the positive time parameter are fixed. For $H > 0$ set

$$d_H(\delta) = \min_{x \in N} \|R(x, \delta)\|_H, \quad \|y\|_H^2 = y^\top H y.$$

Compact attainment and semialgebraicity imply a positive leading power for d_I . Write

$$d_I(\delta) = a_I \delta^\rho + o(\delta^\rho), \quad a_I > 0, \quad \rho \in \mathbb{Q}_{>0}. \quad (2.2)$$

This elementary one-variable consequence of real curve parametrization is also proved, uniformly on strata, in the accompanying entrance theorem.

Definition 2.1. The leading residual set is

$$\mathcal{E}_\rho = \{y \in \mathbb{R}^M : (0, y) \in \text{cl}\{(\delta, \delta^{-\rho} R(x, \delta)) : x \in N, 0 < \delta < \delta_0\}\}.$$

The metric contact envelope is $\mathcal{C}_\rho = \mathcal{C}(\mathcal{E}_\rho)$ as in (1.1).

The set $\mathcal{E}_\rho$ need not be bounded, convex, or a cone. Taking its convex tensor envelope is a deliberate quotient: the resulting invariant records all positive definite metric minima, not every leading residual vector.

Theorem 2.2 (Complete metric contact data). *Under (2.1), $\mathcal{E}_\rho$ is a nonempty closed semialgebraic set and $\text{dist}(0, \mathcal{E}_\rho) = a_I > 0$. For every $H > 0$,*

$$d_H(\delta) = a_H \delta^\rho + o(\delta^\rho), \quad a_H^2 = \min_{y \in \mathcal{E}_\rho} y^\top H y = \inf_{S \in \mathcal{C}_\rho} \text{tr}(HS) > 0. \quad (2.3)$$

The exponent is the same for all $H > 0$. Moreover,

$$\mathcal{C}_\rho = \bigcap_{H > 0} \{S \in \text{Sym}_M : \text{tr}(HS) \geq a_H^2\}. \quad (2.4)$$

Consequently two such germs have the same leading coefficient for every positive definite metric if and only if their metric contact envelopes agree, after matching their residual coordinates and exponents.

A linear change of residual coordinates $R' = TR$, $T \in \text{GL}_M(\mathbb{R})$, sends $\mathcal{E}_\rho$ to $T\mathcal{E}_\rho$ and $\mathcal{C}_\rho$ to $T\mathcal{C}_\rho T^\top$. A surjective reparametrization of the feasible residual germ leaves these objects unchanged. The function $H \mapsto a_H^2$ is positive homogeneous, concave, monotone in the positive semidefinite order, and locally Lipschitz on Sym_M^{++} .

Proof. The defining closure is semialgebraic and closed, so its zero-time section is closed and semialgebraic. Choose minimizers for d_I . Their normalized residuals are bounded by (2.2); a convergent subsequence yields a point of $\mathcal{E}_\rho$ of norm a_I . Conversely, along every sequence producing a point $y \in \mathcal{E}_\rho$, the inequality $|R(x, \delta)| \geq d_I(\delta)$ gives $|y| \geq a_I$. This proves the first assertions.

Norm equivalence gives constants $0 < c < C$ with $cd_I \leq d_H \leq Cd_I$. The positive semialgebraic function d_H has a leading power; these two inequalities force its exponent to be ρ and its coefficient to be positive. Normalized residuals of its minimizers are bounded, again by norm equivalence. A convergent subsequence has limit $y_H \in \mathcal{E}_\rho$ and $\|y_H\|_H = a_H$. Every other point of $\mathcal{E}_\rho$ is obtained along a sequence, on which $d_H \leq \|R\|_H$ gives $a_H \leq \|y\|_H$. Thus the first minimum in (2.3) is attained. A linear functional bounded below on a set has the same infimum on its convex hull and closure; adding Sym_M^+ cannot lower $\text{tr}(HS)$ when $H > 0$. This proves the second equality.

For completeness, we prove that positive definite functionals suffice in (2.4). The inclusion from left to right is immediate. If $S_0 \notin \mathcal{C}_\rho$, strong separation of a point from a closed convex set gives a nonzero symmetric G with

$$\text{tr}(GS_0) < \inf_{S \in \mathcal{C}_\rho} \text{tr}(GS).$$

Since $\mathcal{C}_\rho + \text{Sym}_M^+ = \mathcal{C}_\rho$, finiteness of the infimum forces $G \geq 0$: otherwise adding a large multiple of $vv^\top$ with $v^\top Gv < 0$ would send it to $-\infty$. Also $\mathcal{C}_\rho \subseteq \text{Sym}_M^+$. For sufficiently small $\varepsilon > 0$ the strict separation persists for $G + \varepsilon I$, because its infimum on $\mathcal{C}_\rho$ is at least the old infimum whereas its value at S_0 tends to the old value. Now $G + \varepsilon I > 0$, proving the reverse inclusion.

Coordinate covariance follows by applying T to the normalized residuals and observing that congruence maps the positive semidefinite cone onto itself. The reparametrization assertion follows from equality of the feasible residual images as germs. The infimum of linear functions of H is concave and positive homogeneous, and each $yy^\top$ is positive semidefinite, giving monotonicity. On a compact metric set $\lambda I \leq H \leq \Lambda I$, fix $y_0 \in \mathcal{E}_\rho$. Every minimizer satisfies $\lambda|y_H|^2 \leq \Lambda|y_0|^2$. Testing the minimum for H' at y_H and conversely gives

$$|a_H^2 - a_{H'}^2| \leq (\Lambda/\lambda)|y_0|^2 \|H - H'\|_{\text{op}}.$$

This proves local Lipschitz continuity. $\square$

Remark 2.3 (What is canonical). The pair $(\rho, \mathcal{C}_\rho)$ is defined in the original real residual space. It depends on the feasible germ and the positive time coordinate. It is not an invariant of an unmarked complex ideal alone. Its first entry records the optimal real contact order. Its second entry is exactly the information needed to vary the leading observation metric. It does not purport to recover nonconvex residual geometry invisible to all such minima.

Proposition 2.4 (Arc form and the optimal exponent). *For R as above,*

$$\rho = \sup_{\gamma} \frac{\min_i \text{ord}(R_i \circ \gamma)}{\text{ord}(\delta \circ \gamma)}, \quad (2.5)$$

where γ ranges over positive-time feasible semialgebraic Puiseux arcs tending to $(x_*, 0)$. The supremum is attained. It is also the least real c for which $\delta^c \leq C|R(x, \delta)|$ throughout the feasible germ, for some $C > 0$.

Proof. The lower estimate $|R| \geq (a_I/2)\delta^\rho$ bounds every arc ratio above by ρ and gives the inequality at $c = \rho$. Semialgebraic choice selects minimizers of d_I as a function of positive time; after shortening the interval they form a Puiseux arc. Their source points tend to x_* , since any other limiting point would be another zero of $R(\cdot, 0)$. Along this arc $|R|$ has order ρ in time. No $c < \rho$ can satisfy the proposed bound there. For a real vector of Puiseux germs, the norm order is the minimum of the coordinate orders, because sums of squares cannot cancel. This proves (2.5). $\square$

3. LOCAL INFORMATION IN BINARY POLYNOMIAL OBSERVATIONS

We now identify the constrained version of the preceding metric construction in an observation model with a spectral target. Fix $D > 0$, $d \geq 1$, and $0 < \pi < 1/2$. Let f_1, f_2 be monic degree- d polynomials with real roots in $[0, D]$. Let $\alpha_1 + \alpha_2 = 1$, $\alpha_a \geq \pi$, and let $U = (u_1, u_2)$ and $V = (v_1, v_2)$ be nonnegative column-stochastic 2×2 matrices. Put

$$K(z) = \sum_{a=1}^2 \alpha_a f_a(z) u_a v_a^\top, \quad q(z) = \mathbf{1}^\top K(z) \mathbf{1}, \quad P_j = K(T_j)/q(T_j), \quad (3.1)$$

where $D < T_1 < \dots < T_\ell$ and $\ell \geq 2d + 1$. The target is the unordered root multiset of $f_1 f_2$, with its bottleneck metric. The observation distance is

$$h_w(P, P')^2 = \sum_j w_j \sum_e (\sqrt{P_{je}} - \sqrt{P'_{je}})^2, \quad w_j > 0, \quad \sum_j w_j = 1.$$

A centre is called full rank here if f_1, f_2 are coprime, U, V are invertible and strictly positive, and both weights are strictly larger than π . Competitors remain in the entire closed model.

Lemma 3.1 (The coefficient inverse). *At a full-rank centre the observation determines the two component polynomials, weights and channels up to a common component permutation. Near its observation the recovered coefficient and stochastic coordinates are Lipschitz, with constants uniform on compact sets of full-rank centres. Their derivative in unconstrained coefficient and stochastic coordinates has a left inverse.*

Proof. First recover the monic normalizer. A homogeneous ambiguity consists of a polynomial e of degree below d and a matrix polynomial H of degree at most d with $H(T_j) = e(T_j)P_j$. The degree- $2d$ polynomial $qH - eK$ vanishes at $2d + 1$ clocks, so it vanishes identically. Applying the invertible second channel to the second marginal gives $q \mid ef_1$ and $q \mid ef_2$. Coprimality implies $q \mid e$, hence $e = 0$ and $H = 0$. Thus the finite linear system for the monic q and K has full column rank. A fixed nonsingular minor supplies an analytic local recovery; subtraction of the equations and the compact coefficient bounds of the closed model give the Lipschitz estimate for all close competitors.

There are two clocks T_i, T_j at which $f_1(T_j)/f_1(T_i) \neq f_2(T_j)/f_2(T_i)$; otherwise a nonzero polynomial of degree at most d would vanish at all the clocks. The matrix $K(T_j)K(T_i)^{-1}$ has these simple eigenvalues and eigenvectors u_1, u_2 . Normalize the eigenvectors by their entry sums. The row sums of $U^{-1}K(z)$ then recover $\alpha_a f_a(z)$; their leading coefficients recover α_a , and their normalized rows recover V . These operations are analytic near the centre. They also apply to every exact competitor and prove uniqueness. A finite cover gives the compact-uniform estimates. Differentiating the analytic recovery gives the asserted left inverse. $\square$

3.1. Observable-law coordinates and their variational costs. Fix a full-rank centre observation p . Mark the distinct clusters by their recovered component, root value, multiplicity, and endpoint status. Within a sufficiently small observation neighbourhood this marking is determined by p , up to the common component permutation. It is not an extra unknown latent label. Multiplicities can be specified by polynomial gcd conditions and endpoint evaluation. They need not be continuous across a change of stratum.

For nearby laws in the model image, isolate the roots near each central root r_j , of multiplicity m_j . Their monic cluster factor is an analytic function of the recovered coefficients. Writing the root displacements as $u_{j,i}$, the quantities

$$s_j = \sum_i u_{j,i}, \quad X_j = \frac{1}{2} \sum_i (u_{j,i} - s_j/m_j)^2$$

are analytic functions of that factor's first two coefficients. Individual repeated roots need not be differentiable. At D reverse the sign of s_j ; at zero keep its sign.

If an interior root is repeated, retain its X_j and retain the signed sums at every endpoint cluster. All interior sums and stochastic variables are free. The retained cone is $\mathcal{K} = \mathbb{R}_+^k$ and the root scale is $t^{1/2}$. If no interior root is repeated, retain all cluster sums; then $\mathcal{K} = \mathbb{R}^{k_0} \times \mathbb{R}_+^{k_1}$, only stochastic variables are free, and the root scale is t . In either case $k \leq 2d$. Let τ_p be the vector of these retained *functions on the observable model image*, centred at p .

For $v \in \mathcal{K}$, define

$$c_p(v) = \lim_{t \downarrow 0} \frac{4}{t^2} \inf \{ h_w(p', p)^2 : p' \text{ is a nearby model law, } \tau_p(p') = tv \}. \quad (3.2)$$

This is a model-dependent variational oracle. It is neither a single empirical statistic nor a claim that latent root constraints can be physically imposed. Its law-space definition removes a choice of nuisance parametrization; it does not remove the model from the definition. Since the centre is identifying, its pattern and the oracle are computable from the exact centre law and the known model. This observation by itself is not a compression theorem.

Proposition 3.2 (The local quotient and the leading root set). *The limits (3.2) exist. If $La + Dv$ is the first observation score in free and retained variables, and $W = \text{diag}(w_j/p_{je})$, then*

$$c_p(v) = v^\top Q v, \quad Q = D^\top W D - D^\top W L (L^\top W L)^{-1} L^\top W D > 0. \quad (3.3)$$

An empty free block contributes zero. These costs are intrinsic to the observable germ with its spectral coordinate and marking. Reparametrizing free directions or adding a free direction to a retained score column leaves them unchanged.

The complete leading root set is the decoding of $\{v \in \mathcal{K} : v^\top Q v \leq 4\}$. In the repeated regime, decode each retained variance v_j into all ordered zero-sum vectors y_j with $\frac{1}{2} \sum_i y_{j,i}^2 = v_j$, and put zero in the other root blocks; endpoint sums are existentially minimized. In the linear regime decode sums into the simple interior increments and all ordered one-sided endpoint increments with those sums. The scaled whole-model root sets converge to these sets in Hausdorff distance, with error $O(\sqrt{t})$ and $O(t)$ respectively, uniformly on compact subsets of a fixed pattern stratum.

Proof. The first coefficient increment in component a is

$$-\sum_{j \in a} s_j \frac{f_a(z)}{z - r_j} - \sum_{\substack{j \in a \text{ repeated} \\ 0 < r_j < D}} X_j \frac{f_a(z)}{(z - r_j)^2}.$$

The simple and double partial fractions at distinct poles are linearly independent. Lemma 3.1 therefore makes the combined score (L, D) injective. Orthogonal projection off $\text{ran } L$ proves (3.3) and positive definiteness.

For the passage from the actual law-space constraint to this minimum, choose at each repeated interior cluster a zero-sum vector y of variance v_j , and use roots $r_j + ta_j/m_j + \sqrt{t} y_i$. Every nonnegative variance has such a vector; one may take $m_j - 1$ equal entries and their negative sum. At endpoints use equal one-sided displacements with the prescribed sum. Give every stochastic variable its free order- t increment. Strict margins at the centre ensure feasibility. These competitors have retained coordinates exactly tv and satisfy

$$F(\xi) - p = t(La + Dv) + O(t^{3/2}), \quad 4h_w(F(\xi), p)^2/t^2 = \|La + Dv\|_W^2 + O(\sqrt{t}).$$

In the linear regime the errors are $O(t^2)$ and $O(t)$. The minimizing free vector is bounded on compact retained sets, giving the upper bound.

Conversely an almost-minimizer has observation error $O(t)$ and therefore coefficient and stochastic errors $O(t)$. Analytic cluster factorization gives cluster coefficient errors of this order. The first coefficient bounds all sums. The real identity for the second centred coefficient, $e_2 = -\frac{1}{2} \sum_i (u_i - s/m)^2$, bounds interior centred displacements by $O(\sqrt{t})$. At endpoints every displacement has the same sign and is bounded by the sum, hence is $O(t)$. Thus the same expansions hold for all almost-minimizers, with bounded free variables, and give the lower bound. Compactness allows minima after restricting to a small closed cluster neighbourhood; the coefficient inverse forces the small-cost minimizers into every smaller such neighbourhood. The limit is independent of that restriction and is the law-space cost in (3.2).

The same bounds apply to all competitors in the observation ball. Divide the root blocks by the stated scale and apply the score expansion. A radial contraction of the retained and free order- t variables by $1 - O(\sqrt{t})$, or $1 - O(t)$ in the linear regime, puts their limiting score in the norm ball of radius two. Repeated root vectors contract by the square root of that factor. This proves the outer Hausdorff bound. For the reverse bound use the feasible root constructions above with the same contraction, choosing its constant to dominate the Taylor remainder. Both estimates are uniform on compact pattern sets by the uniform inverse and strict feasibility margins. Exact identification excludes other limiting components of the whole-model ball. Cluster separation makes the bottleneck metric the maximum of the ordered within-cluster coordinate metrics. This

proves the assertion for the joint set, not merely its diameter. The invariance claims follow from the quotient by $\text{ran } L$ or directly from (3.2). $\square$

The affine residual set $E_v = \{La + Dv : a \in \mathbb{R}^r\}$ has a tensor envelope as in (1.1). Its metric support at W is exactly $c_p(v)$. Thus the local quotient form is a constrained section of the same construction used in Theorem 2.2; no remote order argument is needed to prove the local inverse.

3.2. Which finite probes are sufficient? Let $k \geq 1$, let $\mathcal{K} \subseteq \mathbb{R}^k$ be either cone just described, and let $v_1, \dots, v_m \in \mathcal{K}$. For $H \in \text{Sym}_k$ define the linear sampling map

$$\mathcal{A}_V(H) = (v_1^\top H v_1, \dots, v_m^\top H v_m).$$

For a family $\mathcal{Q} \subseteq \text{Sym}_k^{++}$, its secant set is $\mathcal{Q} - \mathcal{Q} = \{Q - Q' : Q, Q' \in \mathcal{Q}\}$.

Theorem 3.3 (Exact probe criterion and adaptive lower bound). *The ray probes determine $\mathcal{Q}$ on $\mathcal{Q}$ if and only if*

$$\ker \mathcal{A}_V \cap (\mathcal{Q} - \mathcal{Q}) = \{0\}. \quad (3.4)$$

On all positive definite forms this is equivalent to

$$\text{span}\{v_i v_i^\top : 1 \leq i \leq m\} = \text{Sym}_k. \quad (3.5)$$

In particular at least $k(k+1)/2$ ray values are required. The legal probes e_i and $e_i + e_j$, $i < j$, attain this bound. No deterministic adaptive algorithm making fewer exact ray queries in the worst case can recover every positive definite form.

The same lower bound applies to recovery of the complete leading spectral fibre on an endpoint-free pattern, when the class of observation metrics is allowed to vary over all positive definite forms on the observation tangent space. It is not asserted for the fixed Hellinger metric family, for which (3.4) is the exact statement. When endpoint variables are invisible at the leading root scale, the smaller invariant in Theorem 3.5 replaces the full form.

Proof. Two forms produce the same values precisely when their difference is in $\ker \mathcal{A}_V$, proving (3.4). If (3.5) fails, choose a nonzero symmetric H orthogonal to all $v_i v_i^\top$. For any $Q_0 > 0$, both $Q_0 + \varepsilon H$ and $Q_0 - \varepsilon H$ are positive definite for sufficiently small $\varepsilon > 0$, and have the same values. Conversely (3.5) makes the linear map injective. Dimension gives the lower bound. For the specified family,

$$Q_{ii} = c(e_i), \quad Q_{ij} = \frac{1}{2}(c(e_i + e_j) - c(e_i) - c(e_j)),$$

so the bound is attained using vectors in either feasible cone.

For the adaptive statement, run the proposed deterministic algorithm at $Q_0 > 0$ and record its chosen directions. If their number is less than $\dim \text{Sym}_k$, choose $H \neq 0$ orthogonal to every recorded tensor. The two nearby forms $Q_0 \pm \varepsilon H$ give exactly the recorded answers. Induction on the query number shows that the algorithm chooses the same next direction, and makes the same stopping decision, on both inputs. It must therefore output the same answer for two different forms. This argument does not assume continuity of the rule choosing queries.

It remains to justify the observation-model and spectral qualifications. At a fixed binary germ the combined score (L, D) is injective. On its image choose an inner product which makes the free and retained score subspaces orthogonal and gives the retained subspace any prescribed Gram matrix $Q > 0$. Extend it positively to an ambient complement. A constant ambient metric already realizes this tangent metric. Profiling then gives precisely the prescribed Q . This proves realizability over the declared metric class, not over the smaller family obtained by changing only Hellinger design weights.

On an endpoint-free pattern every retained variable is visible in the leading root vector: it is a cluster variance in the repeated regime and an individual simple-root displacement in the linear regime. All v in the retained cone have a decoded root vector. Thus the leading fibre determines

the cone section $\{v \in \mathcal{K} : v^\top Q v \leq 4\}$. Its radial function determines $v^\top Q v$ for all nonzero $v \in \mathcal{K}$. Since $\mathcal{K}$ has nonempty interior, equality there forces equality of the two quadratic polynomials. Two distinct forms in the preceding adversary argument therefore give distinct spectral fibres. This proves the fibre lower bound with its stated restriction. $\square$

Corollary 3.4 (Stability of a sufficient probe family). *Suppose $\mathcal{A}_V$ is injective and let $\sigma_V > 0$ be its least singular value for the Frobenius and Euclidean norms. Least-squares reconstruction from an error vector of norm at most ε gives*

$$\|\hat{Q} - Q\|_{\text{op}} \leq \|\hat{Q} - Q\|_F \leq \varepsilon / \sigma_V.$$

If $Q \geq \lambda I$ and $\eta = \varepsilon / \sigma_V < \lambda$, put $\vartheta = \eta / \lambda$. For a fixed decoder of degree $\gamma \in \{1/2, 1\}$, the reconstructed leading sets satisfy

$$(1 + \vartheta)^{-\gamma/2} \mathcal{L}(Q) \subseteq \mathcal{L}(\hat{Q}) \subseteq (1 - \vartheta)^{-\gamma/2} \mathcal{L}(Q).$$

Their diameters obey the same multiplicative bounds.

Proof. The left inverse of the sampling map has norm $1/\sigma_V$. The form bounds $(1 - \vartheta)Q \leq \hat{Q} \leq (1 + \vartheta)Q$ compare their cone sections by radial factors $(1 \pm \vartheta)^{-1/2}$. The root decoder is homogeneous of degree γ . Inclusion and homogeneity of diameter give the result. $\square$

3.3. The information hidden by one-sided endpoints. In the repeated regime write the retained variables as (x, z) , where $x \in \mathbb{R}_+^r$ are the visible interior variances and $z \in \mathbb{R}_+^e$ are the endpoint sums. The leading root scale erases z . Define

$$G_Q(x) = \min_{z \geq 0} (x, z)^\top Q(x, z). \quad (3.6)$$

Theorem 3.5 (The effective spectral metric). *For $Q > 0$, G_Q is a continuous, positive, homogeneous function of degree two on $\mathbb{R}_+^r \setminus \{0\}$, and is quadratic on each of finitely many closed polyhedral cones covering $\mathbb{R}_+^r$. More explicitly, for a subset A of the endpoint indices put*

$$z_A = -Q_{AA}^{-1} Q_{Ax} x, \quad z_{A^c} = 0.$$

On the cone defined by $x \geq 0$, $z_A \geq 0$, and

$$Q_{bx} x + Q_{bA} z_A \geq 0 \quad (b \notin A),$$

its value is

$$G_Q(x) = x^\top (Q_{xx} - Q_{xA} Q_{AA}^{-1} Q_{Ax}) x. \quad (3.7)$$

The empty subset has the evident convention. The complete leading root fibre with the fixed cluster decoder determines G_Q , and conversely. Hence two matrices yield the same leading spectrum if and only if their effective functions agree on $\mathbb{R}_+^r$; equality of the full matrices is unnecessary.

Proof. Strict convexity and coercivity in z give a unique minimizer. At a minimizer its positive support A satisfies the displayed stationarity and complementary inequalities. Conversely those conditions are the first-order necessary and sufficient conditions for this convex quadratic program. The inverse of the principal positive definite block exists, and substitution gives (3.7). There are finitely many subsets; their conditions are linear homogeneous inequalities in x and cover the cone, including boundaries where several descriptions agree. Continuity follows either from this finite agreeing description or from uniqueness and coercivity of the minimizer. Positive definiteness gives $G_Q(x) \geq \lambda |x|^2$ for some $\lambda > 0$, and scaling the minimizer proves homogeneity.

The spectrum recovers the set $\{x \geq 0 : G_Q(x) \leq 4\}$ by reading the variances of its repeated root blocks. Every variance vector is realizable by ordered zero-sum root blocks. Its radial function recovers G_Q by homogeneity and positivity. Conversely (3.6) and the decoder recover precisely the leading root fibre. This proves both implications. $\square$

Example 3.6. With one visible variance and one endpoint sum, let $Q_b = \begin{pmatrix} 1 & b \\ b & 1 \end{pmatrix}$, $0 \leq b < 1$. Then $G_{Q_b}(x) = x^2$ for $x \geq 0$, although the matrices and their mixed ray costs vary with b . For $-1 < b < 0$, instead $G_{Q_b}(x) = (1 - b^2)x^2$. The finite active-face formula detects this difference. It also explains why a full-matrix lower bound cannot be asserted for every endpoint-decoded fibre.

4. POLYNOMIAL IDENTIFICATION WALLS THROUGH CANCELLATION

Fix integers $1 \leq m < n_1 < \dots < n_s$ and let $a = (a_1, \dots, a_s)$ range over a compact subset of $\mathbb{R}^s$. On a sufficiently small fixed interval $0 \leq x \leq \epsilon$, put

$$g_a(x) = \sum_{i=1}^s a_i x^{n_i}, \quad R_a(x, \delta) = (x^m - \delta, g_a(x)).$$

For $H = (h_{ij}) > 0$, define $d_H(a, \delta) = \min_x \|R_a(x, \delta)\|_H$. Write

$$z = \delta^{1/m}, \quad C(a, z) = g_a(z), \quad \eta = n_1 - m > 0, \quad \kappa_H = (h_{22} - h_{12}^2/h_{11})^{1/2}. \quad (4.1)$$

The scalar C is a polynomial in the ramified time and the model coefficients. It is not a truncated minimum-distance expansion.

Theorem 4.1 (Cancellation-uniform wall factorization). *For every compact coefficient set and compact set of positive definite H , there are $L < \infty$ and $z_0 > 0$ such that for $0 < z < z_0$,*

$$(1 - Lz^\eta)\kappa_H|C(a, z)| \leq d_H(a, z^m) \leq (1 + Lz^\eta)\kappa_H|C(a, z)|. \quad (4.2)$$

In particular $d_H(a, z^m) = 0$ if and only if $C(a, z) = 0$. The estimate is uniform through that exact wall, including coefficient paths on which arbitrarily many leading terms cancel.

For a fixed nonzero coefficient vector, let j be its first nonzero index. Then the entrance exponent and coefficient are n_j/m and $\kappa_H|a_j|$. More generally, along any bounded semialgebraic coefficient path $a(\delta)$, either $C(a(\delta), \delta^{1/m})$ is eventually zero, giving an exact remote representative, or

$$C(a(\delta), \delta^{1/m}) = c\delta^\rho + o(\delta^\rho), \quad c \neq 0, \quad d_H(a(\delta), \delta) = \kappa_H|c|\delta^\rho + o(\delta^\rho). \quad (4.3)$$

Here $\rho \geq n_1/m$. Thus this single finite polynomial classifies every arc opening in the family, not just the generic opening on each fixed-coefficient stratum.

Proof. All constants below are uniform on the compact parameter sets. The trial $x = z$ gives $d_H \leq C_1|C(a, z)|$, and $|C(a, z)| \leq C_2 z^{n_1}$. At any minimizer, norm equivalence therefore gives

$$|x^m - z^m| \leq C_3|C(a, z)| \leq C_4 z^{n_1}.$$

For small z , x/z lies in a fixed neighbourhood of one and $x/z = 1 + O(z^\eta)$. Set $U = x^m - z^m$. The exact divided-difference identity is

$$g_a(x) = C(a, z) + UB_a(x, z), \quad B_a(x, z) = \sum_i a_i \frac{x^{n_i} - z^{n_i}}{x^m - z^m}. \quad (4.4)$$

At $x = z$ use the continuous value $\sum_i a_i(n_i/m)z^{n_i-m}$. For x/z near one, factoring z in each quotient shows $|B_a(x, z)| \leq C_5 z^\eta$. Thus the shear $(U, C) \mapsto (U, C + B_a U)$ changes the H -norm by a relative factor between $1 - C_6 z^\eta$ and $1 + C_6 z^\eta$. This is a uniform operator-norm estimate; it does not divide by C .

Completing the square gives

$$\min_{U \in \mathbb{R}} \|(U, C)\|_H = \kappa_H|C|, \quad U_* = -h_{12}C/h_{11}.$$

Applying the lower shear estimate at a minimizer proves the lower bound in (4.2). For the upper bound take $x_* = (z^m + U_*)^{1/m}$. Since $U_* = O(z^{n_1})$ and $n_1 > m$, this is positive, lies in the fixed interval, and has $x_*/z = 1 + O(z^\eta)$. The same divided-difference and shear estimates apply,

proving the upper bound. If $C = 0$, the trial $x = z$ is exactly a zero of the residual. Conversely a zero residual must have $x = z$ and hence $C = 0$.

For fixed coefficients the first nonzero monomial of C gives the stated order and coefficient. For a bounded semialgebraic path, the substituted scalar is semialgebraic in positive time. It is eventually zero or has a nonzero Puiseux leading term. Boundedness of the coefficients gives $|C| = O(\delta^{n_1/m})$, hence the asserted lower bound on ρ . The uniform relative estimate, rather than an absolute remainder at a generic stratum, proves (4.3) even after cancellation. $\square$

Corollary 4.2 (The entire metric envelope on an opening arc). *On a nonzero arc in (4.3), the leading residual set in the two displayed residual coordinates is*

$$\mathcal{E}_\rho = \{(u, c) : u \in \mathbb{R}\}.$$

Consequently the canonical metric envelope is

$$\text{cl}(\text{conv}\{(u, c)(u, c)^\top : u \in \mathbb{R}\} + \text{Sym}_2^+),$$

and its support data are exactly $a_H^2 = c^2(h_{22} - h_{12}^2/h_{11})$.

Proof. A bounded normalized residual forces $U = O(\delta^\rho)$ and hence $x/z \rightarrow 1$, since $\rho > 1$. Equation (4.4) and $B_a = O(z^\eta)$ make its second normalized coordinate tend to c . Conversely every fixed u is realized by $x = (\delta + u\delta^\rho)^{1/m}$, which is feasible for small positive time. The same identity gives its normalized residual limit (u, c) . The last assertion follows by completing the square, or by Theorem 2.2. $\square$

Corollary 4.3 (Crossover and higher cancellation). *Let $s = 2$, $n_1 = n < n_2 = q$, and fix the second coefficient b . For $a_1 = A\delta^{(q-n)/m}$,*

$$\delta^{-q/m} d_H(a, \delta) \longrightarrow \kappa_H |A + b|$$

uniformly for A in compact sets, including $A = -b$. At $A = -b$ the residual vanishes exactly for all small positive time. If instead

$$a_1 = -b\delta^{(q-n)/m} + c\delta^\lambda, \quad \lambda > (q-n)/m, \quad c \neq 0,$$

then the opening order is $n/m + \lambda$ and its coefficient is $\kappa_H |c|$.

Proof. Substitution in the exact polynomial gives respectively $C = (A + b)\delta^{q/m}$ and $C = c\delta^{n/m+\lambda}$. Apply (4.2). Uniformity includes the zero coefficient because the estimate is relative to $|C|$, with both sides zero on the wall. $\square$

Corollary 4.4 (What valuations and residues determine). *Suppose the nonzero coefficient germs have leading terms $a_i(\delta) = b_i\delta^{\lambda_i} + o(\delta^{\lambda_i})$. Put $r = \min_i(\lambda_i + n_i/m)$. If*

$$\sum_{i: \lambda_i + n_i/m = r} b_i \neq 0,$$

then the opening has order r and coefficient κ_H times the absolute value of this sum. If the sum vanishes, neither the order nor exact membership in the wall is determined by these leading valuations and residues alone; one must evaluate the next nonzero term of the exact scalar C , or determine that it vanishes identically.

Proof. The first assertion is the leading-term rule for the finite sum defining C , followed by Theorem 4.1. For the second, fix $b \neq 0$ in Corollary 4.3. The paths $a_1 = -b\delta^{(q-n)/m}$ and $a_1 = -b\delta^{(q-n)/m} + c\delta^\lambda$ have the same leading valuations and residues, but the first is exact and the second has opening order $n/m + \lambda$. Different $\lambda > (q-n)/m$ give different nonexact orders with the same initial data. $\square$

This does not claim that a fixed finite truncation of an arbitrary parameter path suffices. The finite invariant is the exact model polynomial C ; a fixed-format path has the finite order bounds supplied by the accompanying semialgebraic theory. In the language of real weighted comparison, the residue sum is the obstruction to lifting the putative lowest-order opening, and the cancellation locus is an actual feasible identification wall.

4.1. Strictly positive probability observations. Choose a strictly positive probability vector P_0 with at least three cells and linearly independent zero-sum vectors A, B . On two disjoint sufficiently short closed intervals define

$$P_{\text{base}}(s) = P_0 + As, \quad P_{\text{rem},a}(x) = P_0 + Ax^m + Bg_a(x), \quad s, x \geq 0. \quad (4.5)$$

The interval lengths may be chosen uniformly for bounded a so that all probabilities stay strictly positive. Give the two components distinct constant monic degree-one target polynomials. The exact parameter fibre over P_0 consists of the two origins, even when some or all coefficients vanish. Use the centre path $s = \delta$.

Theorem 4.5 (Hellinger wall with its sharp coefficient). *For the polynomial probability model (4.5), let $d_h(a, \delta)$ be the minimum Hellinger distance from the centre to the remote component. With*

$$\langle Y, Z \rangle_I = \sum_e Y_e Z_e / P_{0,e}, \quad \kappa = \frac{1}{2} \left(\|B\|_I^2 - \frac{\langle A, B \rangle_I^2}{\|A\|_I^2} \right)^{1/2} > 0,$$

there are uniform constants such that

$$(1 - Lz^{\eta_0})\kappa|C(a, z)| \leq d_h(a, z^m) \leq (1 + Lz^{\eta_0})\kappa|C(a, z)|, \quad \eta_0 = \min\{m, n_1 - m\} > 0. \quad (4.6)$$

All exact walls, arc openings and crossover laws of Theorem 4.1 and its corollaries hold with this coefficient. In particular the family realizes each of its successive rational orders and their moving-parameter cancellations while preserving strict positivity and a two-point central exact fibre.

Proof. Let $U = x^m - \delta$ and $V = g_a(x)$. The probability residual is $AU + BV$. Independence and positivity make its Hellinger norm uniformly equivalent to the Euclidean norm of (U, V) on the fixed compact neighbourhood. The trial $x = z$ and the lower norm bound localize every minimizing x exactly as in Theorem 4.1. Thus $x^m = \delta + O(|C|)$, $g_a(x) = O(|C|)$, and both probability vectors are $P_0 + O(\delta)$.

Use the exact identity

$$h(P', P)^2 = \sum_e \frac{(P'_e - P_e)^2}{(\sqrt{P'_e} + \sqrt{P_e})^2}.$$

At the localized points its denominator weights equal $(4P_{0,e})^{-1}(1 + O(\delta))$, uniformly. Hence Hellinger distance is the norm induced on (U, V) by

$$H_0 = \frac{1}{4} \begin{pmatrix} \|A\|_I^2 & \langle A, B \rangle_I \\ \langle A, B \rangle_I & \|B\|_I^2 \end{pmatrix}$$

times a relative factor $1 + O(\delta)$. The localization and divided-difference proof apply to the Hellinger minimizer for the lower bound and to the quadratic minimizing trial for the upper bound. Combining $O(z^{n_1-m})$ and $O(z^m)$ gives (4.6). The Schur complement of H_0 is κ^2 . Finally independence of A, B makes $P_{\text{rem},a}(x) = P_0$ imply $x^m = 0$, and the base equation implies $s = 0$. The exact central fibre and positivity assertions follow. $\square$

Remark 4.6 (Closed observation balls and critical thresholds). The remote component enters a closed observation ball of radius $t(\delta)$ exactly when $d_h(a, \delta) \leq t(\delta)$. The factorization determines this decision when the limiting ratio is different from one, and it identifies the exact wall $C = 0$

without an asymptotic test. When a nonzero threshold and distance have the same leading term, the relative estimate does not assign a side. The accompanying exact semialgebraic sign test for $t - d_h$, or the signed higher-order calculations in the regular cubic cases, remain necessary. Equality is inclusion by compact attainment.

APPENDIX A. REAL MONOMIAL PRESENTATIONS AND THE CLASSICAL ORDER FORMULA

For comparison with a divisorial calculation, suppose the residual is Nash near its feasible set. A finite real monomial presentation in this appendix means a finite disjoint union of compact semialgebraic corner sectors Y_α and a proper surjective semialgebraic map

$$\pi : \coprod_{\alpha} Y_{\alpha} \longrightarrow \{(x, \delta) : x \in N, 0 \leq \delta \leq \delta_1\}$$

onto the germ under consideration, after discarding components with identically zero time and restricting to the closure of the positive-time part. On each sector, in its own dimension, the pullbacks have the form

$$\delta \circ \pi = u(r, z)r^a, \quad R \circ \pi = r^b V(r, z), \quad r_i \geq 0, \quad (\text{A.1})$$

where u is positive, V has no common zero, and these units have positive lower bounds on the compact sectors. A face is used only when it has a relative open part accessible from positive-time feasible points by an arc transverse to that face with the other boundary coordinates positive. All face indices below refer to such sectors and faces. This definition specifies the properness, image, and accessibility used by the calculation; it does not identify every real point of a complex exceptional divisor with a feasible direction.

Simultaneous principalization and real rectilinearization of a *finite* list of ideals and inequalities supply presentations of this kind in the setting treated in the supporting volume. The calculation below is also valid whenever a presentation with these properties is supplied directly. In particular its conclusions do not require an assertion that a single modification monomializes every polynomial in an infinite ring.

Proposition A.1 (Computing the intrinsic pair). *In a presentation (A.1) satisfying the nonexact hypothesis, every accessible index with $a_i = 0$ has $b_i = 0$. The exponent of Theorem 2.2 is*

$$\rho = \max_{\alpha, i: a_i > 0} b_i / a_i.$$

For each $H > 0$, the function

$$\frac{\delta^\rho}{\|R\|_H} \circ \pi = \frac{u^\rho}{\|V\|_H} \prod_i r_i^{\rho a_i - b_i}$$

extends continuously to the zero-time preimage. If M_H is its maximum there, then $0 < M_H < \infty$ and $a_H = M_H^{-1}$. Thus these resolved unit maxima reconstruct the same canonical envelope by (2.4), even when the lists of coordinate divisors in two presentations differ.

Proof. If $a_i = 0 < b_i$, the relative open face with the other coordinates positive maps to a feasible positive-time zero residual, contradicting (2.1). With ρ equal to the displayed maximum, every exponent $\rho a_i - b_i$ is nonnegative. The unit ratio is therefore bounded and continuous. On a transverse arc at a maximizing face it has a positive limit; the image of that face has zero time and zero residual, hence is the unique central solution. It follows that its zero-time maximum is positive and finite.

On each positive time level, surjectivity and compactness give

$$\max_{\delta \circ \pi = \delta} \frac{\delta^\rho}{\|R\|_H} \circ \pi = \frac{\delta^\rho}{d_H(\delta)}.$$

A sequence of maximizers has a convergent subsequence in the compact source, giving the upper limit M_H . Conversely curve selection at a zero-time maximizing point gives an accessible arc on which the unit ratio tends to M_H . Its positive time coordinate is, after restriction, strictly monotone and covers every sufficiently small positive time. This gives the reverse limit and $d_H \sim \delta^\rho/M_H$. It identifies the exponent with the intrinsic one, as does the weighted-average formula for the orders of a lifted arc. Finally Theorem 2.2 reconstructs the envelope from these metric coefficients. $\square$

The exponent in this proposition is the classical relative real contact or Łojasiewicz exponent. The normalized-blow-up and integral-closure theory explains why divisorial orders are the correct algebraic mechanism [1]. Ideals with equal integral closure have equal orders on every arc: an integral equation would otherwise have a unique term of least order. This explains order invariance, but not metric invariance. Multiplying a residual vector by a positive constant leaves its ideal and contact exponent unchanged, while scaling all entrance coefficients and its tensor envelope. Likewise, changing feasible real sectors can remove arcs without changing the ambient complex ideal. We therefore use the intrinsic arc supremum and tensor envelope as the statement of the invariant. No canonical minimal list of real-accessible Rees valuations is asserted for arbitrary inequality-constrained germs.

APPENDIX B. PRESERVATION, SCOPE, AND DEPENDENCE OF THE RESULTS

The present article has three proof dependencies. The canonical metric theorem needs compact semialgebraic minimization and one-variable real parametrization. The local ray-probe theorem needs the binary coefficient inverse and the explicit root-cluster expansion. The polynomial wall factorization needs only the divided-difference identity, localization of minimizers, and a positive definite metric. Thus the last two do not depend on the existence of a general resolution atlas.

The complete preceding article remains the supporting volume. Under the hypotheses and notation of its cubic-fibre theorem, the identifying locus for the rank-deficient cubic family $f = z(z - r)^2$, $g = (z - s)^3$ is

$$\{\det U \neq 0\} \cup \{(3D - r)s - 2rD < 0\} \cup \{(1 - \alpha)(r + 3s)(2r - 3s)^2 - 4\pi r^3 < 0\}.$$

Its direct fibre proof, inverse through rank loss, whole-model local analysis, and both wall calculations are unchanged. The new probability family in Section 4 belongs to the general polynomial observation class used in the accompanying singular-entrance theorem; it is not claimed to be a reparametrization of this cubic binary subfamily. In particular the singular rational openings do not replace the regular cubic order-one calculations.

The original finite constrained-value theorem remains valid with its supplied marking. Section 3 identifies that marking from the exact centre law and gives its costs a precise law-space oracle interpretation. The new lower bound concerns the declared variable-metric class; no sample complexity, operational intervention, or fixed-Hellinger minimality is inferred from it. The endpoint quotient in Theorem 3.5 retains the information actually seen by the spectrum rather than claiming an injective decoder where there is none.

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PROJECTIVE POLYNOMIAL OBSERVATIONS: FINITE SPECTRAL CALIBRATION AND DIVISORIAL ENTRANCES

QIAN QI

ABSTRACT. We study unordered polynomial spectra observed through unknown stochastic channels and a scalar normalization. For the full-rank coprime binary model of fixed degree, we prove that a finite collection of scalar constrained observation costs determines the complete leading spectral fibre, with all channel and mixture parameters free. The costs recover a positive definite quotient form, and the multiplicity pattern decodes its cone section into a joint root set. At degree d , at most $d(2d+1)$ scalar calibrations suffice; perturbations of those values give explicit Hausdorff and diameter bounds. For an isolated Nash remote observation germ, we identify the entrance exponent as the maximal residual-to-time order ratio over accessible real divisors and identify its leading coefficient by a resolved metric-unit maximum. A Puiseux residue theorem relates the real weighted fibre to coordinate real tropicalization, while finite weighted jets imply filtered naturality without an assumed convergence of conjugates. The complete multiplicity law, rank-deficient cubic fibre, inverse through channel-rank loss, weight wall, endpoint wall, and their exact critical-shell calculations are retained. The general semialgebraic atlas and entrance asymptotics remain available beyond the explicit binary normal forms.

1. INTRODUCTION

A normalized polynomial experiment can be locally identifying and yet poorly resolve its roots. It can also admit a remote exact representative which enters a small observation ball while the local inverse remains coercive. These are different phenomena. The first depends on the joint geometry of root splitting and the observation metric; the second depends on contact between a moving observation and a remote image. This article determines both for stochastic polynomial observations.

Our concrete experiment is

$$K(z) = \sum_{a=1}^2 \alpha_a f_a(z) u_a v_a^\top, \quad P_j = K(T_j) / (\mathbf{1}^\top K(T_j) \mathbf{1}),$$

with monic degree- d real-rooted components, roots in $[0, D]$, strictly positive centre weights, and unknown binary stochastic channels. The target is the unordered root multiset of $f_1 f_2$. The observation radius t is weighted Hellinger distance; the spectral modulus is the *diameter of the entire feasible set* in the bottleneck metric, not only its maximal distance from the central spectrum.

Finite information determining a joint root fibre. For coprime components, invertible strictly positive channels and at least $2d+1$ exterior clocks, normalizer recovery and a matrix-pencil inverse control all coefficient and stochastic errors by observation error. Real-root cluster geometry then leaves only two leading regimes: an interior repeated root splits at order $\sqrt{t}$, while simple and one-sided boundary roots move at order t .

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Theorem 4.1 identifies a finite scalar datum which retains the joint metric information lost by scalar orders. On each marked multiplicity pattern there are $k \leq 2d$ retained sum or variance coordinates. For v in their feasible cone, constrain those coordinates to equal tv , minimize over the remaining roots and all stochastic parameters, and set

$$c_\theta(v) = \lim_{t \downarrow 0} \frac{4}{t^2} \min h_w(F_T(\xi), P_\theta)^2.$$

These limits are not formal probes of an arbitrary compact set. They are costs of actual polynomial-root competitors in the full model. We prove that $c_\theta(v) = v^\top Q_\theta v$, with $Q_\theta > 0$, and hence that the $k(k+1)/2$ values

$$c_\theta(e_i), \quad c_\theta(e_i + e_j) \quad (i < j)$$

recover the entire quotient form. The multiplicity pattern decodes $\{v : v^\top Q_\theta v \leq 4\}$ into the complete joint leading root set. Corollary 4.3 gives quantitative metric stability from errors in these finitely many scalars. This finite sufficiency result is the structural starting point of the article.

The underlying multiplicity law is preserved in full. If an interior cluster is repeated, its sharp whole-model modulus is

$$\omega_\theta(t) = C(\theta)\sqrt{t} + O_H(t), \quad C(\theta) = \max_j 2\sqrt{(m_j - 1)/m_j} \kappa_j^{-1/4}.$$

Otherwise the modulus is linear. The complete root sets, endpoint constraints, optimal coefficient orders and compact-uniform errors are in Theorem 3.2. The calibration theorem strengthens their information content without replacing the root-expansion proof by a general definability argument.

Remote geometry and the two cubic walls. The rank-deficient cubic family $f = z(z - r)^2$, $g = (z - s)^3$ has an exact fibre controlled by

$$B = (3D - r)s - 2rD, \quad E = (1 - \alpha)(r + 3s)(2r - 3s)^2 - 4\pi r^3.$$

Its identifying locus is $\{\det U \neq 0\} \cup \{B < 0\} \cup \{E < 0\}$. The complete classification, inverse through rank loss, oblique local Fisher geometry and both identification walls are given by the direct pencil and feasible-chart arguments below. The weight wall and endpoint wall have different nonidentified-side openings; neither is removed by passing to an abstract tangent relation. The exact equality-shell and signed second-order calculations are retained.

For a more general isolated remote germ, let $R(x, \delta) = f(x) - p(\delta)$ and $d(\delta) = \min_x \|R(x, \delta)\|$. In the nonexact Nash case, Theorem 17.1 identifies

$$\rho = \max_{E \text{ real accessible}} \frac{\text{ord}_E(R_1, \dots, R_M)}{\text{ord}_E \delta}, \quad d(\delta) = M^{-1}\delta^\rho + o(\delta^\rho).$$

Here only divisors with positive time order enter the maximum, and M is the maximum of the continuous resolved extension of $\delta^\rho/\|R\|$ on the accessible zero-time preimage. The exponent is thus a relative real contact invariant of a residual ideal and a time ideal, not an invariant of the remote image alone. The coefficient requires their observation-metric units. The formula recovers the singular contact n/m and its sharp coefficient in the existing polynomial probability example, and recovers order one and the cone-projection coefficient in the regular regime. The continuous-semialgebraic family theorem, its rational order lists, its uniform remainders and its exact threshold decisions remain in Theorem 16.1.

Which real weighted object is being measured? The general weighted relation retains real feasibility before imposing a norm and decoding roots. Theorem 13.1 identifies it with the residue of its bounded weighted points over the real algebraic Puiseux field:

$$T_w^+ S = \text{res}\{t^{-w}y : y \in S(\mathbb{R}\langle t \rangle), \text{val } y \geq w\}.$$

Its nonzero coordinate signs and valuations give the corresponding fibre of coordinate real tropicalization [19]; the residue magnitudes, observation-unit section and spectral decoder are retained before that projection. Positive real scale is distinguished from positive coordinate orthant, and the precise sufficient smooth lifting test is stated alongside the comparison with positive tropical lifting [20]. Theorem 13.3 derives the required conjugate limits from finite weighted jets of a Nash isomorphism and its inverse.

The earlier algebraic initial-fibre comparison, scalar distance-function reconstruction, sharp-marginal separation and full binary valuative separation are all retained. The distance-function identity is a general metric fact; it is not the finite model-specific theorem above. Likewise, ordinary valuation data discard positive metric units; the binary separation example makes that mechanism explicit rather than claiming that the mechanism itself is new. The residue comparison positions the object against real tropical geometry without claiming a new tropical fundamental theorem.

Organization and dependencies. The model inverse, multiplicity law, finite calibration, cubic fibre and two walls come first. Their proofs are direct. The relative real atlas, weighted comparisons and singular entrance theory follow, with the complete foundational arguments in the appendix. Definition C.1 specifies the finite complexity datum underlying all fixed-format assertions. Common monomial presentations concern a finite chosen diagram, not every scalar polynomial simultaneously. Appendix C states exactly which concrete proofs do and do not require the general atlas. All proofs and finite calculations from the preceding article are retained; its full text and historical archive also remain in the separate archival volume.

2. THE EXPERIMENT AND THE SPECTRAL METRIC

Fix $D > 0$, a degree $d \geq 1$, and $0 < \pi < 1/2$. A closed binary model point consists of two monic real polynomials f_1, f_2 of degree d , all of whose roots lie in $[0, D]$, weights $\alpha_1 + \alpha_2 = 1$ with $\alpha_a \geq \pi$, and column-stochastic matrices $U = (u_1, u_2)$ and $V = (v_1, v_2)$ with nonnegative entries. Set

$$K(z) = \sum_{a=1}^2 \alpha_a f_a(z) u_a v_a^\top, \quad q(z) = \mathbf{1}^\top K(z) \mathbf{1} = \sum_{a=1}^2 \alpha_a f_a(z), \quad P_j = \frac{K(T_j)}{q(T_j)}. \quad (2.1)$$

The clocks satisfy $D < T_1 < \dots < T_\ell$, and $w_j > 0$, $\sum_j w_j = 1$. Consequently $q(T_j) > 0$, including at every closed-model competitor. The target polynomial is $A = f_1 f_2$. The model is compact in monic coefficients and stochastic entries. Component permutations are identified. The matrices, roots, clocks and design weights are allowed to vary when a family of centres is considered, but every competitor uses its centre's fixed design.

For probability arrays define

$$h_w(P, Q)^2 = \sum_j w_j \sum_e (\sqrt{P_{je}} - \sqrt{Q_{je}})^2.$$

For two multisets $Z = \{z_i\}_{i=1}^n$ and $Z' = \{z'_i\}_{i=1}^n$, put

$$d_\infty(Z, Z') = \min_{\sigma \in \mathfrak{S}_n} \max_i |z_i - z'_{\sigma(i)}|.$$

For real roots this is the maximum discrepancy of their increasing orderings. All diameters in this article are diameters of sets of multisets, not maximal distances from the centre. Write

$$\omega_\theta(t) = \text{diam}\{Z(A_\xi) : h_w(F_T(\xi), P_\theta) \leq t\}. \quad (2.2)$$

A local modulus $\omega_{\theta, \text{loc}}$ restricts the competitors to a specified coefficient neighbourhood of the centre's component pair. This restriction is never implicit.

At a strictly positive datum the Fisher inner product on zero-sum arrays is

$$\langle H, H' \rangle_{I, \theta} = \sum_{j, e} \frac{w_j}{P_{\theta, je}} H_{je} H'_{je}.$$

For component increments p_a of degree below d and stochastic increments $\eta = (\dot{\alpha}_1, \dot{u}_1, \dot{u}_2, \dot{v}_1, \dot{v}_2)$, where channel increments have entry sum zero, define

$$\begin{aligned} \dot{K} &= \sum_a \{ \alpha_a p_a u_a v_a^\top + f_a(\dot{\alpha}_a u_a v_a^\top + \alpha_a \dot{u}_a v_a^\top + \alpha_a u_a \dot{v}_a^\top) \}, \\ \dot{q} &= \sum_a (\alpha_a p_a + \dot{\alpha}_a f_a), \quad S_\theta(p_1, p_2, \eta)_j = \frac{\dot{K}(T_j) - P_{\theta, j} \dot{q}(T_j)}{q(T_j)}. \end{aligned} \quad (2.3)$$

Here $\dot{\alpha}_2 = -\dot{\alpha}_1$. These rational formulas define every Fisher program below in the original experiment coordinates.

Lemma 2.1 (Normalizer recovery). *Suppose f_1, f_2 are coprime, V is invertible, the weights are positive, and $\ell \geq 2d+1$. The observed array determines q and K uniquely among all closed-model competitors. Their coefficients are locally Lipschitz functions of the observations, with uniform constants on compact sets of such centres. No rank assumption on U is required.*

Proof. Let e have degree below d , and let H be a matrix polynomial of degree at most d satisfying $H(T_j) = e(T_j)P_j$. Interpolation of the polynomial $qH - eK$, of degree at most $2d$, gives $qH = eK$. Its second marginal, multiplied by $\text{diag}(\alpha_1, \alpha_2)^{-1}V^{-1}$, gives $q \mid ef_1$ and $q \mid ef_2$. Coprimality implies $q \mid e$, hence $e = 0$ and $H = 0$.

More explicitly, let Π_T project onto the orthogonal complement of the degree- d Vandermonde column space. The map

$$N_\theta : e \longmapsto \Pi_T(e(T_j)P_{\theta, j})_j, \quad \deg e < d,$$

is injective. For a competitor, subtraction of the normalization equations yields $\|N_\theta(q' - q)\| \leq C \max_j \|P'_j - P_j\|$; the competitor normalizer coefficients are bounded by compactness of the root interval. The smallest singular value of N_θ is positive. A bounded left inverse recovers $q' - q$, and Vandermonde interpolation recovers $K' - K$. The singular values and the coefficient bounds have positive uniform margins on compact sets of the stated centres. $\square$

Lemma 2.2 (Full-rank coefficient inverse). *Under Lemma 2.1, suppose also that U is invertible. Then every sufficiently close observation has, up to one component permutation, coefficient and stochastic parameter error bounded by a constant times its observation error. Every exact competitor has the same component polynomials and stochastic parameters. The constants are uniform on compact subsets of these centres.*

Proof. Choose two clocks such that $f_1(T_j)/f_1(T_i) \neq f_2(T_j)/f_2(T_i)$. Otherwise the nonzero polynomial $f_2(T_i)f_1 - f_1(T_i)f_2$, of degree at most d , would vanish at all $2d+1$ clocks. The matrix $K(T_j)K(T_i)^{-1}$ has those two distinct eigenvalues and eigenvectors u_1, u_2 . Normalize the eigenvectors to have entry sum one. Simple-eigenvalue inversion is analytic in a neighbourhood of the centre. Applying the recovered U^{-1} to $K(z)$ and taking row sums recovers $\alpha_a f_a$; leading coefficients and normalized rows recover α_a, V . This construction applies to arbitrary closed-model competitors because their $K'(T_i)$ stays invertible. It also proves exact uniqueness. A finite

neighbourhood cover gives compact-uniform constants. Equivalently, the derivative in monic coefficient and stochastic coordinates has a bounded left inverse. $\square$

3. AN EXPLICIT MULTIPLICITY AND BOUNDARY ATLAS

Fix d, D, π and $\ell \geq 2d + 1$. Let $\mathfrak{M}_d$ be the centres of (2.1) with strict weights, strictly positive invertible U, V , and coprime component polynomials. No restriction is imposed on root multiplicities or on contact with $0, D$. Competitors still range over the entire closed model. Partition $\mathfrak{M}_d$ by the component assignment, multiplicity and endpoint/interior status of each distinct root. At fixed d this is a finite semialgebraic partition. On a pattern stratum write r_j, m_j for its distinct roots and multiplicities, and $a(j) \in \{1, 2\}$ for their component assignments.

For every cluster let s_j denote its first-order sum displacement. It is free at an interior root, nonnegative at 0 , and nonpositive at D . For every repeated interior cluster let $X_j \geq 0$. Define

$$p_a(z) = - \sum_{j:a(j)=a} s_j \frac{f_a(z)}{z - r_j} - \sum_{\substack{j:a(j)=a \\ r_j \in (0, D), m_j \geq 2}} X_j \frac{f_a(z)}{(z - r_j)^2}. \quad (3.1)$$

Insert these polynomials into (2.3). Let L contain the free interior sum and stochastic columns. Let D_0 contain the nonnegative endpoint sum columns (reverse the sign at D) and all X_j columns. With $W = \text{diag}(w_j/P_{je})$, define the projected Gram matrix

$$Q = D_0^\top W D_0 - D_0^\top W L (L^\top W L)^{-1} L^\top W D_0. \quad (3.2)$$

For a repeated interior cluster j , let

$$\kappa_j = \min\{v^\top Q v : v \geq 0, v_{X_j} = 1\}. \quad (3.3)$$

All matrix entries are rational in the root, channel, weight and design coordinates. Empty collections of free or cone columns have their usual zero-dimensional meanings.

Lemma 3.1 (Ordered cluster diameter). *For $m \geq 2$, the set*

$$\{y_1 \leq \dots \leq y_m : \sum_i y_i = 0, \frac{1}{2} \sum_i y_i^2 \leq X\}$$

has diameter $\sqrt{2X(m-1)/m}$ in the maximum coordinate metric.

Proof. Fix a coordinate j . Minimizing the squared norm under its value, ordering and zero sum gives

$$\max y_j = \sqrt{\frac{2X(j-1)}{m(m-j+1)}}, \quad \min y_j = -\sqrt{\frac{2X(m-j)}{mj}}.$$

For the maximum, the last $m-j+1$ coordinates are equal and the first $j-1$ are equal; the minimum is analogous. This also covers the zero endpoint extrema when $j=1$ or $j=m$. Put $a=j-1, b=m-j$. For $a, b \geq 1$, the assertion that the resulting coordinate width is at most $\sqrt{2X(m-1)/m}$ is equivalent to

$$ab(a+b+2)^2 \geq 4(a+1)(b+1).$$

Writing $S = a+b \geq 2$ and $ab \geq 1$, its difference is $ab(S^2 + 4S) - 4S - 4 \geq S^2 - 4 \geq 0$. For $a=0$ or $b=0$ equality holds directly. An extreme vector with one coordinate of absolute value $\sqrt{2X(m-1)/m}$ and the other $m-1$ coordinates equal attains the endpoint width against the zero vector. The diameter of a subset in the maximum metric is its largest coordinate width. This proves the formula, including possible simultaneous constraints on other clusters when it is used below. $\square$

Theorem 3.2 (Multiplicity and boundary laws). *Every centre in $\mathfrak{M}_d$ is exactly identifying. On each root pattern stratum the following whole-model assertions hold uniformly on compact subsets H .*

(a) *If there is a repeated interior cluster, the exponent is $1/2$. The complete leading root set consists of zero clusters at the simple and boundary roots and ordered vectors $y_j \in \mathbb{R}^{m_j}$ at the repeated interior roots satisfying*

$$\sum_i y_{j,i} = 0, \quad X_j = \frac{1}{2} \sum_i y_{j,i}^2, \quad \min_{\text{endpoint sums} \geq 0} v^\top Q v \leq 4. \quad (3.4)$$

The X_j in v are the displayed variances. After grouping and scaling roots by $\sqrt{t}$, the observation ball is within $O_H(\sqrt{t})$ of this set in Hausdorff distance. In particular

$$\omega_\theta(t) = C(\theta)\sqrt{t} + O_H(t), \quad C(\theta) = \max_j 2\sqrt{\frac{m_j - 1}{m_j}} \kappa_j^{-1/4}. \quad (3.5)$$

Every leading point, including diameter-extremizing pairs, is realized by stochastic alternatives.

(b) *If no interior root is repeated, the exponent is 1. For each simple interior cluster take a free real root increment, and for each boundary cluster take an ordered vector of increments with all entries nonnegative at 0 or nonpositive at D . Let s_j be their sums and use (3.1) without X terms. The compact set of these root increments with*

$$\min_\eta \|S_\theta(p_1, p_2, \eta)\|_{L,\theta}^2 \leq 4 \quad (3.6)$$

is the complete leading set $\mathcal{L}_{\text{lin},\theta}$. Its diameter $C_{\text{lin}}(\theta)$ is positive, and

$$\omega_\theta(t) = C_{\text{lin}}(\theta)t + O_H(t^2).$$

The scaled-root Hausdorff error is $O_H(t)$.

(c) *The optimal coefficient orders of a cluster of multiplicity m , with coefficient weight k , are*

$$\beta_k = \begin{cases} 1, & r \in (0, D), \quad k = 1, \\ k/2, & r \in (0, D), \quad 2 \leq k \leq m, \\ k, & r \in \{0, D\}, \quad 1 \leq k \leq m. \end{cases} \quad (3.7)$$

The real order list, boundary and multiplicity pattern, and Fisher active supports admit a common finite semialgebraic refinement. On its square-root strata C^4 is a selected rational function of the root and experiment coordinates; the positive constants on the linear strata are algebraic semialgebraic functions. Both are Nash after refinement. Exact algebraic evaluation uses finite programs.

Proof. Lemma 2.2 gives exact identification and a uniform inverse in coefficients and stochastic parameters. The partial fractions $1/(z - r_j)$ and $1/(z - r_j)^2$ occurring in (3.1) are linearly independent at distinct roots, with their indicated pole orders. The coefficient derivative, and hence its score, is injective on these variables and the free stochastic coordinates. Thus L has full column rank and Q is positive definite whenever it is nonempty. The constrained minima exist, are positive, and have uniformly bounded minimizers on H .

We first obtain coordinates for *all* nearby competitors. Coefficient recovery and analytic coprime cluster factorization give an $O_H(t)$ error in each cluster coefficient and in the stochastic parameters. For an interior cluster, write its exact roots as

$$r_j + ts_j/m_j + \sqrt{t} y_{j,i}, \quad \sum_i y_{j,i} = 0.$$

The first coefficient bounds s_j , and the second bounds $\sum_i y_{j,i}^2$, since for centred real roots their second elementary symmetric function equals $-\frac{1}{2} \sum_i y_{j,i}^2$. At an endpoint all root displacements have the same sign. Their sum is $O_H(t)$, so each is $O_H(t)$. These observations prove the upper bounds in (3.7). For the lower bounds, translate an interior cluster by a nonzero order- t amount for $k = 1$; for $k \geq 2$ split it by $\sqrt{t}$ times a zero-sum real vector with nonzero k th elementary symmetric function. The vector with $m - 1$ equal nonzero coordinates and the last their negative sum has this property for every $k \geq 2$. Its first nonzero coefficient is of order t and the coefficient inverse makes the observation distance comparable to t . At an endpoint displace all roots by the same allowed order- t amount. This proves all lower orders, including the one-sided boundary case. In root-arc parameter ε , the interior split has $\text{ord } G = 4$, $\text{ord } H_k = 2k$, whereas translation or an endpoint arc has orders $2, 2k$. These are admissible witnesses for the real order data, and (3.7) rules out any smaller candidate.

For part (a), expansion of the exact root coordinates gives precisely (3.1) at order t . Terms involving three centred root displacements are $O_H(t^{3/2})$; all higher ones and the bounded mean and endpoint terms satisfy the same bound. Consequently

$$F_T(\xi) = P_\theta + tS_\theta + O_H(t^{3/2}), \quad h_w(F_T(\xi), P_\theta) = \frac{t}{2} \|S_\theta\|_{I,\theta} + O_H(t^{3/2}). \quad (3.8)$$

This follows by Taylor expansion of the square-root probability vector and the reverse triangle inequality; it does not require a nonzero score. Membership in the ball implies $\|S_\theta\| \leq 2 + M\sqrt{t}$. Multiply all sum, variance and stochastic first-order variables by $\rho = 2/(2 + M\sqrt{t})$ and each centred root vector by $\sqrt{\rho}$. This gives a point of (3.4) within $O_H(\sqrt{t})$ of the scaled competitor. Endpoint and simple clusters contribute only $O_H(\sqrt{t})$ in these scaled coordinates.

Conversely take a point of (3.4) and bounded minimizing endpoint and free variables. Use the displayed interior root arcs; at an endpoint realize its sum by equal one-sided displacements, and use the prescribed order- t stochastic increments. Scale first-order variables by $\rho = 1 - M'\sqrt{t}$ and centred roots by $\sqrt{\rho}$. Strict weight and channel margins and cluster separation on H make these genuine closed-model alternatives. For a sufficiently large M' , (3.8) puts them inside the observation ball. This is the reverse Hausdorff inclusion, with the same uniform error.

Homogeneity gives $X_{j,\max} = 2/\sqrt{\kappa_j}$. The marginal projection of the leading set at cluster j includes every ordered zero-sum vector with squared norm at most $2X_{j,\max}$: scale a minimizing feasible vector for $X_j = 1$ and choose the desired shape in that cluster. Lemma 3.1 therefore gives its diameter $2\sqrt{(m_j - 1)/m_j \kappa_j^{-1/4}}$. The diameter in the maximum cluster metric is the largest marginal diameter; no independence of the clusters is used. Cluster separation excludes cross-cluster optimal matchings. Diameters differ by at most twice Hausdorff distance, proving (3.5).

For part (b), every root displacement is $O_H(t)$ by the same sum argument at the endpoints and the simple-root inverse in the interior. Divide the exact displacements by t . They belong to the stated polyhedral sign and ordering cone. Taylor expansion now has remainder $O_H(t^2)$. The first-order score controls their sums and all stochastic variables; on the one-sided endpoint cones the sums control the individual displacements. Hence the root set in (3.6) is compact. The radial argument with $\rho = 1 - Mt$ proves both Hausdorff inclusions with error $O_H(t)$. An allowed nonzero displacement gives positive diameter. This proves (b).

For (c), enumerate the subsets of the cone coordinates other than a fixed X_j . On a support A , put $v_A = -Q_{AA}^{-1}Q_{Aj}$ and require $v_A \geq 0$ and $Q_{bj} + Q_{bA}v_A \geq 0$ off the support. Its value is $Q_{jj} - Q_{jA}Q_{AA}^{-1}Q_{Aj}$. Convexity proves necessity and sufficiency. All tests and selected values are rational, and

$$C^4 = \max_j \frac{16(m_j - 1)^2}{m_j^2 \kappa_j}.$$

There are finitely many patterns and supports. For the linear law, (3.6) is an explicit quadratic inequality on a polyhedral root cone after projecting the free stochastic variables. Its coordinate maxima and minima, and hence its diameter, have finite real algebraic descriptions by (10.5) or by face optimization. Quantifier elimination gives the algebraic value at algebraic input. Stratify the resulting semialgebraic functions and refine with the order lists of Theorem 10.4. Positivity and compact-uniform coercivity ensure continuity within each fixed pattern; Nash refinement gives the final assertion. No continuity of C is asserted across a change of scaling exponent. $\square$

Example 3.3 (A multiplicity crossover). Let $x \in [0, b]$, $b < 1/2$, take $F(x) = (1/2 + x, 1/2 - x)$, and target $A_x(z) = (z - r)^2 - x$. At a fixed centre $a \in (0, b)$, differentiation of the square-root probability vector gives

$$h(F(a + u), F(a)) = \frac{|u|}{\sqrt{1 - 4a^2}} + O_a(u^2).$$

The admissible x interval therefore has endpoints $a \pm t\sqrt{1 - 4a^2} + O_a(t^2)$, and its spectral diameter is

$$\omega_a(t) = \frac{\sqrt{1 - 4a^2}}{\sqrt{a}} t + O_a(t^2).$$

At $a = 0$ it instead has leading law $\sqrt{t}$. More precisely, uniformly for λ in compact subsets of $[0, \infty)$,

$$\frac{\omega_{\lambda t}(t)}{\sqrt{t}} \longrightarrow \sqrt{\lambda + 1} - \sqrt{\max(0, \lambda - 1)}. \quad (3.9)$$

Indeed the feasible interval endpoints are $t \max(0, \lambda - 1) + O(t^2)$ and $t(\lambda + 1) + O(t^2)$, with truncation at zero, and $x \mapsto Z(A_x)$ has ordered roots $r \pm \sqrt{x}$. This example belongs to the general real-graph class of Section 9; it is not a claim about an unobserved restriction of the closed binary model. Here the linear constant really diverges at a multiplicity wall. Section 6 gives a different mechanism.

4. FINITE CALIBRATION OF THE JOINT SPECTRAL FIBRE

The full-rank inverse imposes a restriction on the possible leading fibres which is absent for general compact semialgebraic sets. On each multiplicity stratum a finite list of scalar, constrained observation costs determines the entire joint spectral fibre, not merely its coordinate radii. The constraints specify root sums and variances; mixture weights and both stochastic channels are minimized over.

Throughout this section the centre belongs to $\mathfrak{M}_d$ of Section 3. Distinct clusters are marked by their limiting root and component; their multiplicities and endpoint status form the pattern $\mathcal{P}$. The marking is unambiguous near the centre, up to the already identified component permutation. Fix a small closed cluster neighbourhood B_θ containing the centre in its relative interior. No stochastic coordinate is held fixed in this neighbourhood. Lemma 2.2 implies that every sufficiently small-cost competitor in the whole closed model belongs to this neighbourhood, after the component permutation.

For the roots $r_j + u_{j,i}$ in cluster j put

$$s_j = \sum_{i=1}^{m_j} u_{j,i}, \quad X_j = \frac{1}{2} \sum_{i=1}^{m_j} (u_{j,i} - s_j/m_j)^2.$$

Write $\epsilon_j = 1$ at 0 and $\epsilon_j = -1$ at D ; thus $\epsilon_j s_j \geq 0$ at an endpoint. There are two retained coordinate systems.

If an interior cluster is repeated, retain the X_j at all repeated interior clusters and the $\epsilon_j s_j$ at all endpoint clusters. Denote their number by k and their cone by $\mathcal{K} = \mathbb{R}_+^k$. All interior

sums and all stochastic increments are free variables. If no interior cluster is repeated, retain all cluster sums, with the same sign reversal at D . Then $\mathcal{K} = \mathbb{R}^{k_{\text{int}}} \times \mathbb{R}_+^{k_{\text{end}}}$, and only the stochastic increments are free. In both cases $k \leq 2d$. Let $\gamma = 1/2$ in the first case and $\gamma = 1$ in the second.

For $v \in \mathcal{K}$ and sufficiently small $t > 0$, let $B_\theta(t; v)$ be the competitors in B_θ whose retained coordinates are exactly tv . In particular, prescribing $X_j = tv_j$ does not prescribe the cluster mean or its individual root spacings. Define the scalar calibration cost

$$c_\theta(v) = \lim_{t \downarrow 0} \frac{4}{t^2} \min_{\xi \in B_\theta(t; v)} h_w(F_T(\xi), P_\theta)^2. \quad (4.1)$$

Existence, independence of a sufficiently small B_θ , and the fact that these are actual closed-model alternatives are included in the theorem below. The normalization 4 agrees with (3.8).

Insert the retained and free variables into the score (2.3), using the polynomial increments (3.1) in the repeated regime and its first-sum terms in the linear regime. Write the resulting score as $La + Dv$, with a free. Set

$$Q_\theta = D^\top W D - D^\top W L (L^\top W L)^{-1} L^\top W D, \quad W = \text{diag}(w_j / P_{\theta, j\epsilon}). \quad (4.2)$$

Thus the score is $La + Dv$. An empty free block contributes zero. In the repeated regime this is exactly (3.2), with the same signs and the same free stochastic directions.

Theorem 4.1 (Finite scalar determination). *For every centre in $\mathfrak{M}_d$ the limit (4.1) exists and*

$$c_\theta(v) = \min_a \|La + Dv\|_{L, \theta}^2 = v^\top Q_\theta v, \quad Q_\theta > 0. \quad (4.3)$$

The convergence in (4.1) is uniform for v in compact subsets of $\mathcal{K}$ and for centres in compact subsets of a fixed pattern stratum. Its error is $O(\sqrt{t})$ in the repeated regime and $O(t)$ in the linear regime.

The finite marked scalar datum

$$\mathfrak{C}_\theta = (\mathcal{P}; c_\theta(e_i), c_\theta(e_i + e_j) : 1 \leq i < j \leq k) \quad (4.4)$$

contains $k(k+1)/2$ numbers and determines the complete leading spectral fibre, with its bottleneck metric. It also determines its joint leading cluster-polynomial coefficient image. All mixture weights and channel parameters remain unknown in these calibrations.

More explicitly,

$$(Q_\theta)_{ii} = c_\theta(e_i), \quad (Q_\theta)_{ij} = \frac{c_\theta(e_i + e_j) - c_\theta(e_i) - c_\theta(e_j)}{2}. \quad (4.5)$$

Let $\mathcal{V}_\theta = \{v \in \mathcal{K} : v^\top Q_\theta v \leq 4\}$. In the repeated regime decode v into all ordered, zero-sum vectors y_j with $\frac{1}{2} \sum_i y_{j,i}^2 = v_{X_j}$ at the repeated interior clusters, and zero vectors at every other cluster. Endpoint coordinates of v are existentially quantified. In the linear regime decode interior sums into the corresponding single root increments; at an endpoint use all ordered one-sided increments with signed sum v_j . The union of these decoded vectors is precisely the leading fibre $\mathcal{L}_\theta$ of the whole observation ball, scaled by $t^{-\gamma}$.

Proof. The coefficient inverse gives an injective derivative in monic coefficient and stochastic coordinates. Within either retained/free system the polynomial directions are independent: after division by f_a , their partial fractions have distinct simple poles, and distinct double poles exactly at the repeated interior clusters. Hence the combined score matrix (L, D) is injective. Orthogonal projection off $\text{ran } L$ shows that (4.2) is positive definite and proves the algebraic minimum in (4.3). This also bounds its minimizing free variable a uniformly on the indicated compact sets.

We next prove that this quadratic program is the limit of the actual constrained minima, not a substitute experiment. For a repeated interior cluster and a prescribed $X \geq 0$, choose an ordered

vector y with $\sum_i y_i = 0$ and $\frac{1}{2} \sum_i y_i^2 = X$. Such a vector exists for every X : for example take one entry $-(m-1)b$ and the other $m-1$ entries b , where $b^2 = 2X/(m(m-1))$. Use the exact roots

$$r_j + ta_j/m_j + \sqrt{t} y_{j,i}.$$

At a simple interior root use its order- t sum. At an endpoint realize the prescribed signed sum by equal one-sided displacements. Give the weights and channels their prescribed order- t stochastic increments. Strict interior margins for these parameters and separation of the clusters make all these alternatives feasible for small t . They satisfy the retained constraints exactly. Their observation expansion is

$$F_T(\xi) - P_\theta = t(La + Dv) + O(t^{1+\gamma}), \quad \frac{4}{t^2} h_w(F_T(\xi), P_\theta)^2 = \|La + Dv\|_{L,\theta}^2 + O(t^\gamma), \quad (4.6)$$

where in the linear case all displacements, including the individual endpoint displacements, are $O(t)$. Taking the minimizing a gives the required upper bound.

The constrained sets are closed in the compact cluster neighbourhood and are nonempty by this construction, so their minima are attained. The upper bound makes every minimizing observation error $O(t)$. Lemma 2.2 then makes all coefficient and stochastic errors $O(t)$, uniformly on compact sets of centres and retained vectors. Analytic coprime cluster factorization gives the same bound on cluster coefficients. The first coefficient bounds every free interior sum by $O(t)$. At an endpoint the displacements have one sign, so their prescribed sum controls every displacement. At a repeated interior root the retained centred variance bounds each centred displacement by $O(\sqrt{t})$. Thus every minimizer has bounded free first-order variables and satisfies (4.6), with exactly its prescribed v . Its leading score norm is no smaller than the displayed quadratic minimum. This proves the lower bound, the limit and the uniform errors. The same argument forces every minimizer into every smaller cluster neighbourhood for small t ; the germ and limit do not depend on its initial choice. No channel or weight was fixed during either bound.

Each e_i and $e_i + e_j$ belongs to the relevant cone, including in the linear regime. Polarization therefore recovers the whole matrix from legal scalar calibrations. Theorem 3.2, whose direct root-expansion and recovery argument was used above, identifies the whole-model leading sets by precisely the stated decoders and the constraint $v^\top Q_\theta v \leq 4$. This proves sufficiency for the joint set, not just its diameter. Conversely, the decoder explicitly recovers every leading point and every pair of points. Replacing each root block by $\prod_i (z - y_{j,i})$ gives its joint coefficient image; continuity of the fixed-degree root map gives the inverse on ordered real-rooted blocks. The bottleneck metric is the maximum coordinate metric on these ordered blocks, since separated limiting clusters cannot be matched across one another. This proves the metric assertion. $\square$

Remark 4.2 (The marking and the scope of invariance). The datum is intrinsic to the observation germ with its fixed spectral coordinate and marked multiplicity pattern. Reparametrizing free channel or mean variables leaves $\text{ran } L$ unchanged. Adding a free score column to a retained column leaves its quotient class unchanged. An isometry of the observation tangent metric transports all these classes isometrically. These operations preserve the calibration datum and its decoded metric fibre. A permutation of the marking permutes the datum. An arbitrary change of target coordinate must transport the spectral decoder and its metric; no invariance under an untransported change of units is asserted. The datum need not be minimal: existential elimination of endpoint coordinates may identify two different matrices at the level of the spectral image. We assert sufficiency, not the false converse that equality of spectral images always implies equality of all the calibration numbers.

Corollary 4.3 (Quantitative finite reconstruction). *Use the same pattern and decoder for two positive definite matrices Q, Q' , and suppose $Q \geq \lambda I$ and $\|Q' - Q\|_{\text{op}} \leq \eta < \lambda$. Put $\vartheta = \eta/\lambda$ and*

$\beta = \gamma/2$. Then

$$(1 + \vartheta)^{-\beta} \mathcal{L}(Q) \subseteq \mathcal{L}(Q') \subseteq (1 - \vartheta)^{-\beta} \mathcal{L}(Q). \quad (4.7)$$

If $R = \max_{y \in \mathcal{L}(Q)} \|y\|_\infty$, their Hausdorff distance is at most

$$R \max\{1 - (1 + \vartheta)^{-\beta}, (1 - \vartheta)^{-\beta} - 1\}.$$

Their diameters satisfy the same two multiplicative bounds as (4.7). If each number in (4.4) is known with error at most e , the matrix reconstructed by (4.5) has operator-norm error at most $3ke/2$.

Proof. The hypotheses give $(1 - \vartheta)Q \leq Q' \leq (1 + \vartheta)Q$. The corresponding cone sections therefore lie between the radial rescalings by $(1 + \vartheta)^{-1/2}$ and $(1 - \vartheta)^{-1/2}$. Their root decoders are homogeneous of degree γ : variances scale as squared roots in the repeated regime, and sums scale as roots in the linear regime. This proves (4.7). Compare each point with its radial rescaling to obtain the Hausdorff estimate; diameter is monotone under set inclusion and homogeneous under dilation. Finally a diagonal entry has error at most e and an off-diagonal entry at most $3e/2$. The maximal absolute row sum bounds the operator norm by $3ke/2$. $\square$

The preceding theorem is not the distance-function reconstruction of an arbitrary compact set. It first derives a quadratic quotient from the full binary coefficient inverse, then realizes a finite collection of its probes by feasible polynomial-root experiments, and finally decodes that quotient into a constrained joint spectrum. The metric reconstruction Theorem 12.3 remains useful for leading sets outside this class; no quadratic shape is assumed in the general real atlas.

5. A RANK-DEFICIENT CUBIC STRATUM

Let $\mathfrak{F}$ denote the degree-three centres with

$$f = z(z - r)^2, \quad g = (z - s)^3, \quad 0 < s < r < D,$$

strict weights $\pi < \alpha < 1 - \pi$, strictly positive U, V , $\det V \neq 0$, and at least seven clocks. The first channel may have either rank. Put $\gamma = 1 - \alpha$ and

$$A = (r + 3s)(2r - 3s)^2, \quad B = (3D - r)s - 2rD, \quad E = \gamma A - 4\pi r^3. \quad (5.1)$$

Here A is a scalar, not the target polynomial.

Theorem 5.1 (The entire cubic fibre). *The exact spectral fibre is a singleton precisely on*

$$\mathfrak{I} = \{\det U \neq 0\} \cup \{B < 0\} \cup \{E < 0\}. \quad (5.2)$$

On its complement, besides (f, g) all component-polynomial pairs are, up to permutation,

$$(f, h_c), \quad h_c = (1 - c)f + cg, \quad c_* \leq c \leq \min(c_D, \gamma/\pi), \quad (5.3)$$

where

$$c_* = \frac{4r^3}{A}, \quad b = \frac{rs}{3s - 2r}, \quad R(z) = \frac{f(z)}{g(z)}, \quad c_D = \frac{R(D)}{R(D) - 1}.$$

The interval is nonempty even at equality in either constraint, and every one of its pairs has a positive stochastic realization. Conditional on the component pair, that realization is unique.

Proof. Lemma 2.1 fixes q, K for every competitor. If U is invertible, Lemma 2.2 applies; in particular $\det K / \det K_3 = fg$ gives spectral identification.

If $\det U = 0$, stochasticity gives $u_1 = u_2 = u$. The second marginal $V \operatorname{diag}(\alpha, \gamma)(f, g)^\top$ has coefficient rank two. Every competitor therefore has invertible second channel and its two monic polynomials lie in the affine pencil h_c . Expansion gives

$$\operatorname{disc}(h_c) = c(c - 1)^2(r - s)^3(Ac - 4r^3), \quad h_c(0) = -cs^3, \quad (5.4)$$

and

$$\frac{R'(z)}{R(z)} = \frac{(2r-3s)z+rs}{z(z-r)(z-s)}, \quad A = 4r^3 - 27s^2(r-s) < 4r^3.$$

Nonnegative roots exclude $c < 0$, and the discriminant excludes $0 < c < 1$. For $c > 1$ there is a root in (s, r) and no root in $[0, s]$. The other two admissible roots must be above r , solving $R(z) = c/(c-1) > 1$. They exist only for $s > 2r/3$. In that case R increases to its maximum at b and then decreases to one. The double-root value is $c_* > 1$; the larger root increases from b to D as c increases from c_* to c_D . Thus the whole real-rooted pencil in $[0, D]$ is $\{0, 1\}$, with $[c_*, c_D]$ added exactly when $b \leq D$.

For a competing pair with pencil coordinates c_1, c_2 , the weights obey $\alpha'_1 c_1 + \alpha'_2 c_2 = \gamma \in (0, 1)$. One coordinate must be zero and the other is one or a value $c \geq c_*$. The latter weight is γ/c , so the floor imposes $c \leq \gamma/\pi$; the other weight is automatically at least $\alpha > \pi$. Conversely put

$$\beta_f = \alpha v_1 + \gamma(1 - 1/c)v_2, \quad \beta_h = \gamma v_2/c.$$

Both vectors are positive and $\beta_f f + \beta_h h_c = \alpha v_1 f + \gamma v_2 g$. Their entry sums are the stated weights; normalizing them and keeping first channel (u, u) gives the realization. Coefficient functionals dual to the independent polynomials recover the two rank-one coefficient matrices uniquely, hence their weights and normalized marginals. Finally $b \leq D$ is equivalent to $B \geq 0$ and, on this set, $c_* \leq \gamma/\pi$ is equivalent to $E \geq 0$. $\square$

5.1. Quantitative inversion across the rank locus. All coefficient norms below are Euclidean, including the leading coefficient when a monic polynomial is represented as a vector. Constants are specified from a fixed compact $K_0 \subset \mathfrak{F}$. Norm comparisons between these finite-dimensional spaces are included in these constants. Put

$$\delta = \max_j \|P'_j - P_{\theta,j}\|_F.$$

Use coefficient order $(1, z, z^2, z^3)$. The two components form $F_{\text{coef}} \in \mathbb{R}^{2 \times 4}$, with one polynomial per row, and the second-marginal coefficient matrix is $B_\theta = V \text{diag}(\alpha, \gamma) F_{\text{coef}} \in \mathbb{R}^{2 \times 4}$. Its two rows are the scalar coordinate polynomials of that marginal. Here σ_2 is its smaller singular value and matrix norms are Euclidean operator norms (or the corresponding fixed norm comparisons). Set

$$\beta_0 = \min_{K_0} \sigma_2(B_\theta) > 0, \quad d_{fg} = \min_{K_0} \|g - f\|_{\text{coef}} > 0.$$

Normalizer recovery supplies C_B with $\|B' - B_\theta\| \leq C_B \delta$. Let K_d bound $\|\text{diag}(\alpha') F'_{\text{coef}}\|$ on all closed-model competitors for the designs in K_0 , and set

$$W_0 = \frac{2K_d}{\pi\beta_0}, \quad C_f = \max_{K_0} \|f\|_{\text{coef}}, \quad K_{\text{aff}} = W_0 C_B (1 + C_f). \quad (5.5)$$

Proposition 5.2 (From the second marginal to the pencil). *If $\delta \leq \beta_0/(2C_B)$, every closed competitor has invertible V' and its component polynomials satisfy*

$$f'_b = h_{c_b} + e_b, \quad \deg e_b < 3, \quad \|e_b\|_{\text{coef}} \leq K_{\text{aff}} \delta. \quad (5.6)$$

Moreover $\|\text{diag}(\alpha')^{-1} V'^{-1}\| \leq W_0$. If the ordered component pair is within ρ of (f, g) , then

$$|c_1|, |c_2 - 1| \leq (\rho + K_{\text{aff}} \delta)/d_{fg}. \quad (5.7)$$

No lower bound on $|\det U|$ is used.

Proof. Since $B' = V' \text{diag}(\alpha') F'_{\text{coef}}$,

$$\beta_0/2 \leq \sigma_2(B') \leq \sigma_{\min}(V') K_d.$$

This proves the inverse bound, using $\alpha'_a \geq \pi$. Write C_θ for the matrix with coefficient rows f, g and put

$$T = \text{diag}(\alpha')^{-1} V'^{-1} V \text{diag}(\alpha, \gamma), \quad E_0 = \text{diag}(\alpha')^{-1} V'^{-1} (B' - B_\theta).$$

Then $F'_{\text{coef}} = TC_\theta + E_0$ and $\|E_0\| \leq W_0 C_B \delta$. If a row of T is (l, c) , monicity says $l + c + (E_0)_{\text{lead}} = 1$. Subtracting $(E_0)_{\text{lead}} f$ from that error row gives the exact decomposition $f'_b = (1 - c)f + cg + e_b$, with the bound in (5.5) and zero leading coefficient in e_b . Finally $h_c - f = c(g - f)$ and $h_c - g = (c - 1)(g - f)$ imply (5.7) by the triangle inequality. $\square$

Lemma 5.3 (Two robust isolation estimates). *There are explicit positive $r_f, r_g, \varepsilon_0, C_0, C_1$, depending only on K_0 , with the following property. If $h_c + e$ is monic with all roots in $[0, D]$, $\deg e < 3$ and $\|e\|_{\text{coef}} \leq \varepsilon$, then*

$$|c| \leq r_f, \quad \varepsilon \leq \varepsilon_0 \implies |c| \leq C_0 \varepsilon, \quad |1 - c| \leq r_g, \quad \varepsilon \leq \varepsilon_0 \implies |1 - c| \leq C_1 \varepsilon.$$

The constants require only root-separation and coefficient bounds on K_0 , not a sign or a nonzero margin for B, E .

Proof. Let $s_0 = \min_{K_0} s > 0$ and $d_0 = \min_{K_0} 4r^3(r - s)^3 > 0$. Choose L_D as a Lipschitz bound for the discriminant on a coefficient neighbourhood of $\{h_c : \theta \in K_0, |c| \leq 1\}$, and R_D such that

$$|\text{disc}(h_c) + 4r^3(r - s)^3 c| \leq R_D c^2 \quad (|c| \leq 1).$$

These constants are finite because the discriminant is polynomial. Take $r_f \leq \min(1, d_0/(2R_D))$, omitting the second bound if $R_D = 0$. The endpoint condition gives $-cs^3 + e(0) \leq 0$, hence $c \geq -\varepsilon/s_0^3$. For $0 \leq c \leq r_f$, nonnegative discriminant gives $0 \leq -d_0 c + R_D c^2 + L_D \varepsilon$, hence $c \leq 2L_D \varepsilon/d_0$. One may take $C_0 = \max(s_0^{-3}, 2L_D/d_0)$.

Near $c = 1$, put $v = 1 - c$ and translate $z = s + x$. There are exact identities

$$h_c(s + x) = x^3 + vA_2x^2 + vA_1x + vA_0, \quad A_2 = 3s - 2r, \quad A_1 = f'(s), \quad A_0 = f(s) = s(r - s)^2.$$

Let L_s bound coefficient translation of e on K_0 , set $E_s = L_s \varepsilon$, and write the perturbed coefficients as $u_i = vA_i + e_i$, $|e_i| \leq E_s$. Choose $M \geq \max_{K_0} (|A_1|, |A_2|)$ and $a_* = \min_{K_0} A_0 > 0$. Depressing the perturbed cubic by its own quadratic coefficient gives

$$p = u_1 - u_2^2/3, \quad q = u_0 - u_1u_2/3 + 2u_2^3/27.$$

For $u = |v| + E_s \leq 1$,

$$|p| \leq C_p u, \quad |q - A_0 v| \leq E_s + C_q u^2, \quad C_p = M + 1 + (M + 1)^2/3, \quad C_q = (M + 1)^2/3 + 2(M + 1)^3/27.$$

Real-rootedness is exactly $p \leq 0$ and $|q| \leq c_{\text{cub}}(-p)^{3/2}$ with $c_{\text{cub}} = 2/(3\sqrt{3})$. Thus

$$a_* |v| \leq E_s + C_q u^2 + c_{\text{cub}} C_p^{3/2} u^{3/2}. \quad (5.8)$$

Choose

$$\eta = \min\{1, a_*/(4C_q), (a_*/(4c_{\text{cub}} C_p^{3/2}))^2\}, \quad r_g = \eta/2, \quad \varepsilon_0 \leq \eta/(2L_s).$$

Then $u \leq \eta$, the last two terms in (5.8) are at most $a_* u/2$, and $|v| \leq (2/a_* + 1)E_s$. Take $C_1 = (2/a_* + 1)L_s$. Decrease ε_0 also to stay in the discriminant neighbourhood used in the first argument. Every threshold is determined by the stated positive lower margins and finite polynomial upper bounds. $\square$

Let d_{par} be Euclidean distance in monic coefficients and stochastic entries, minimized over the component permutation.

Lemma 5.4 (A local inverse through the walls). *For every compact $K_0 \subset \mathfrak{F}$, there exist $\rho, M, t_0 > 0$ such that a closed-model competitor whose component pair is within ρ of (f, g) , up to permutation, satisfies*

$$d_{\text{par}}(\xi, \theta) \leq M \max_j \|F_T(\xi)_j - P_{\theta,j}\|_F \quad (5.9)$$

whenever the observation error is below t_0 . These constants require no lower bound on $|\det U|$, $-B$, or $-E$. For a compact $H \subset \mathfrak{J}$ and small fixed ρ , the remote separation

$$\eta_H(\rho) = \min_{\theta \in H, \xi \text{ in the closed model } d_{\text{par}}(\xi, \theta) \geq \rho} h_w(F_T(\xi), P_\theta) \quad (5.10)$$

is positive if the minimizing set is nonempty. The whole-model inverse therefore holds uniformly on H .

Proof. Let $r_* = \min(r_f, r_g)$ from Lemma 5.3 and choose

$$\rho \leq d_{fg} r_*/4, \quad t_0 \leq \min\{\beta_0/(2C_B), \rho/K_{\text{aff}}, \varepsilon_0/K_{\text{aff}}\}.$$

Proposition 5.2 gives an error at most $\varepsilon = K_{\text{aff}}\delta$ in the affine pencil and pencil coordinates in the required neighbourhoods. The two isolation estimates then give $|c_1| + |c_2 - 1| \leq (C_0 + C_1)K_{\text{aff}}\delta$. Together with (5.6), this controls both polynomial coefficient vectors by a common multiple of δ .

Fix the component permutation placing the close ordered pair near (f, g) ; separation of f, g makes this choice unambiguous for a smaller ρ . To recover the stochastic parameters without dividing by $\det U$, choose coefficient functionals ℓ_1, ℓ_2 dual to f, g . Their norms are bounded by the inverse Gram matrix of this pair, uniformly on K_0 . The matrix $T'_{ij} = \ell_i(f'_j)$ is $I + O_{K_0}(\delta)$. Decrease t_0 so that $\|T' - I\| \leq 1/2$; then $\|(T')^{-1}\| \leq 2$. Apply the two functionals entrywise to K' . Inverting T' recovers $M'_a = \alpha'_a u'_a v'_a{}^\top$ with $O_{K_0}(\delta)$ error from M_a . Entry sums give the weights, and row and column sums divided by those weights give the two channels. These divisions have lower bound π . This proves (5.9) with constants constructed from normalizer, second-marginal, Gram, root and weight margins only. In particular no full-rank inverse has been continued through its singular determinant.

For the last assertion the feasible set in (5.10) is compact, allowing the centre's design to vary in H . A zero minimum would give an exact competitor separated from that centre. The entire fibre and conditional uniqueness in Theorem 5.1 exclude this on H . Small whole-model observation balls thus lie in the local neighbourhood. It is this exclusion radius, rather than a local coefficient constant, that may collapse at a remote wall. $\square$

5.2. The complete local Fisher fibre. Use $a, X, Y \geq 0$, free b, c , and five free stochastic coordinates η . The component jets are

$$p_f = -a(z - r)^2 - bz(z - r) - Xz, \quad p_g = -c(z - s)^2 - Y(z - s). \quad (5.11)$$

Let L be the seven free score columns and D_0 the three columns indexed by $0, x, y$ for (a, X, Y) . Define Q by (3.2) and

$$J(X, Y) = \min_{a \geq 0} (a, X, Y) Q(a, X, Y)^\top, \quad \kappa_x = \min_{Y \geq 0} J(1, Y), \quad \kappa_y = \min_{X \geq 0} J(X, 1). \quad (5.12)$$

Proposition 5.5 (Coercivity and finite supports). *Throughout $\mathfrak{F}$, L has rank seven, $Q_{00} > 0$, and κ_x, κ_y are positive and locally Lipschitz. For every compact $K_0 \subset \mathfrak{F}$,*

$$\|S_\theta(a, b, c, \eta, X, Y)\|_{I, \theta} \geq c_{K_0} \|(a, b, c, \eta, X, Y)\| \quad (a, X, Y \geq 0).$$

Each κ_i is given by four finite support tests. For $A \subseteq \{0, x, y\} \setminus \{i\}$ with Q_{AA} positive definite, put

$$v_A = -Q_{AA}^{-1} Q_{Ai}, \quad k_{i,A} = Q_{ii} - Q_{iA} Q_{AA}^{-1} Q_{Ai}.$$

The test is $v_A \geq 0$ and $Q_{ji} + Q_{jA}v_A \geq 0$ for inactive j . The empty support has value Q_{ii} . At least one test succeeds and all successful tests give κ_i .

Proof. Every feasible jet is realized by roots

$$ta, \quad r + tb/2 \pm \sqrt{tX}, \quad s + tc/3 + \sqrt{t}y_i, \quad \sum_i y_i = 0, \quad \frac{1}{2} \sum_i y_i^2 = Y,$$

and order- t stochastic increments. Their observation increment is $tS + O(t^{3/2})$. A zero score, combined with Lemma 5.4, forces the component jets and η to vanish. Coefficient comparison in (5.11) then forces every variable to vanish. Compactness of the feasible unit sphere gives coercivity, uniformly on K_0 . This also proves independence of the free columns, $Q_{00} > 0$, and existence, positivity and boundedness of the constrained minimizers.

Choose a minimizer with minimal positive support in the projected score columns. Those supported columns are independent: a nonzero dependence would allow a signed change of positive coefficients, preserving the represented vector, until one coefficient vanished. On an independent support, stationarity and the inactive inequalities are exactly the stated tests. Conversely their KKT conditions prove optimality by convexity. This handles a singular full Q without inverting it. Uniformly bounded minimizers and smooth score dependence give local Lipschitz dependence by evaluating each problem at a minimizer of the other problem. $\square$

Theorem 5.6 (Local normal form and whole-model restriction). *At every centre of $\mathfrak{F}$ the local leading root set is*

$$(\{0\}, \{-\sqrt{X}, \sqrt{X}\}, Z(z^3 - Yz - Z)), \quad X, Y \geq 0, \quad J(X, Y) \leq 4, \quad 27Z^2 \leq 4Y^3. \quad (5.13)$$

On compact subsets of $\mathfrak{F}$, scaled local observation balls converge to this set with Hausdorff error $O(\sqrt{t})$. Hence

$$\omega_{\theta, \text{loc}}(t) = C_{\text{loc}}(\theta)\sqrt{t} + O(t), \quad C_{\text{loc}} = \max\{\sqrt{2}\kappa_x^{-1/4}, 2\sqrt{2/3}\kappa_y^{-1/4}\}. \quad (5.14)$$

The error is compact-uniform, and C_{loc} is positive and locally Lipschitz on all of $\mathfrak{F}$. On compact subsets of $\mathfrak{J}$ these statements hold for the whole-model modulus. The support tests, (5.2), and $9\kappa_y - 16\kappa_x$ give a finite semialgebraic partition. On its constant choices,

$$C_{\text{loc}}^4 = \max\{4/\kappa_x, 64/(9\kappa_y)\}$$

is one rational function, and C_{loc} is Nash after refinement.

Proof. The local inverse bounds coefficients and stochastic increments by $O(t)$ for every local competitor. Its first component has exact roots $ta, r + tb/2 \pm \sqrt{tX}$ with bounded a, b, X and $a, X \geq 0$. Its second component has exact roots $s + tc/3 + \sqrt{t}y_i$ with $\sum_i y_i = 0$ and bounded squared sum. Put $Y = \frac{1}{2} \sum_i y_i^2$ and $Z = \prod_i y_i$. The real-cubic condition is exactly $Y \geq 0, 27Z^2 \leq 4Y^3$. Expansion gives $F = P + tS + O(t^{3/2})$ and $h_w = t\|S\|_I/2 + O(t^{3/2})$ uniformly on these bounded coordinates. Radial contraction of all first-order variables, together with $Z \mapsto \rho^{3/2}Z$, proves the outer Hausdorff inclusion. Conversely choose bounded minimizing free and endpoint variables and realize the displayed root arcs with $\rho = 1 - M\sqrt{t}$. They lie in the observation ball and give the inner inclusion. The inequality boundary and all joint cubic coefficient relations are therefore realized.

Homogeneity gives $X_{\max} = 2/\sqrt{\kappa_x}$ and $Y_{\max} = 2/\sqrt{\kappa_y}$. The double-cluster diameter is $\sqrt{X_{\max}}$. Lemma 3.1 with $m = 3$ gives the triple diameter $2\sqrt{Y_{\max}/3}$, attained also by $(-2h, h, h)$ and $(-h, -h, 2h)$ with $h = \sqrt{Y_{\max}/3}$. The maximum of marginal diameters gives (5.14). Proposition 5.5, cluster separation and the positive remote exclusion in Lemma 5.4 give the remaining claims. The local formula remains valid at $B = 0$ or $E = 0$, but its identification with the whole-model modulus does not. $\square$

Example 5.7 (The earlier rational phase values). For $D = 1$, $\pi = 1/10$, $r = 3/5$, $s = 1/5$, $\alpha = 2/5$, $u_1 = u_2 = (1/3, 2/3)^\top$, $v_1 = (1/4, 3/4)^\top$, $v_2 = (3/4, 1/4)^\top$, clocks $(6/5, 7/5, 8/5, 9/5, 2, 5/2, 3)$ and equal weights, $B = -18/25$. The empty supports are admissible, the double cluster dominates, and exact rational evaluation gives

$$C^4 = \frac{78550423766145732273576408862601811503774391446}{385279214348552940073664077156875}, \quad 3778.707 < C < 3778.708.$$

Changing only u_2 to $(2/5, 3/5)^\top$ gives the triple-dominant phase, $59.970 < C < 59.971$. The exact arithmetic records for both values are retained in the archival companion. These large values are not evidence for an unspecified wall blow-up.

Corollary 5.8 (Centre-known local risk). *Let $H \subset \mathfrak{J}$ be compact, fix $0 < c \leq 1/(8\sqrt{2})$, and observe independent categorical samples at the design clocks with $n_j/N \rightarrow w_j$ uniformly on H (integer rounding suffices). For squared spectral loss on the specified neighbourhood $h_w(F_T(\xi), P_\theta) \leq cN^{-1/2}$, with the centre known, the minimax risk $\mathcal{R}_N(\theta, c)$ satisfies uniformly on H*

$$\frac{3c}{32}C_{\text{loc}}(\theta)^2 - O_H(N^{-1/4}) \leq \sqrt{N}\mathcal{R}_N(\theta, c) \leq cC_{\text{loc}}(\theta)^2 + O_H(N^{-1/4}).$$

Proof. A constant decision in the specified spectral ball has loss at most its squared diameter. A diameter pair in that ball has weighted Hellinger distance at most $2cN^{-1/2}$. Tensorization bounds its product Hellinger distance by $2c\sqrt{\max_j n_j/(Nw_j)}$. Uniform convergence and the positive minimum of the design weights on H make the maximum ratio at most 2 for all sufficiently large N , uniformly on H . Thus the product distance is at most $2c\sqrt{2} \leq 1/4$. Total variation is bounded by this Hellinger distance in the present convention. The two-point testing bound for squared metric loss is $d^2(1 - \text{TV})/8$, hence at least $3d^2/32$. Substitute (5.14). The centre and its neighbourhood are known; unknown-centre adaptation and honest confidence sets across a wall are not asserted. $\square$

6. A REMOTE BRANCH AT THE STOCHASTIC WEIGHT WALL

Fix r, s, D with $B > 0$, positive u, v_1, v_2 with $v_1 \neq v_2$, and $U = (u, u)$, $V = (v_1, v_2)$. Assume

$$\gamma_0 = \pi c_*, \quad \alpha_0 = 1 - \pi c_*, \quad \pi < \alpha_0 < 1 - \pi. \quad (6.1)$$

The design is fixed. Let θ_δ be the original (f, g) centre with weight $\alpha = \alpha_0 + \delta$. Then $E = -A\delta$, so $\delta > 0$ is the identified side. Put

$$a = \frac{4rs}{r+3s}, \quad b = \frac{rs}{3s-2r}, \quad h_* = (1 - c_*)f + c_*g = (z - a)(z - b)^2. \quad (6.2)$$

Substitution gives $s < a < r < b < D$. At $\delta = 0$ the two exact component pairs are (f, g) and (f, h_*) . The remote pair has weight π on h_* and second-channel columns

$$v_f^* = \frac{\alpha_0 v_1 + (\gamma_0 - \pi)v_2}{1 - \pi}, \quad v_h^* = v_2. \quad (6.3)$$

Let ξ_* be this point; its observation is $P_0 = P_{\theta_0}$. Its second channel is invertible because $v_f^* - v_2 = \alpha_0(v_1 - v_2)/(1 - \pi) \neq 0$. The sorted base and remote spectra are respectively

$$(0, s, s, s, r, r), \quad (0, a, r, r, b, b),$$

whose distance is

$$D_* = \max(r - s, b - r) > 0. \quad (6.4)$$

Thus a zero-length additional pencil interval already has a positive exact-fibre spectral diameter.

6.1. A complete real corner chart at the remote point. Use the eleven coordinates $v = (a_0, b_0, X, c_0, d_0, Y, k, \dot{u}_f, \dot{u}_h, \dot{v}_f, \dot{v}_h)$, where a binary channel increment is represented by its one free zero-sum coordinate. Set

$$\begin{aligned} p_f &= -a_0(z - r)^2 - b_0z(z - r) - Xz, \\ p_h &= -c_0(z - b)^2 - d_0(z - a)(z - b) - Y(z - a), \\ \mathcal{T}_* &= \{v : a_0, X, Y, k \geq 0\}. \end{aligned} \tag{6.5}$$

The remaining seven coordinates are free. Here k is the increment of the weight on h_* ; the weight on f has increment $-k$. Let J_*v be the full coefficient and stochastic vector with polynomial increments p_f, p_h and these stochastic increments.

Proposition 6.1 (Remote tangent cone and converse realization). *In a neighbourhood of ξ_* , modulo the one component permutation, the closed model has a Nash coordinate chart from a neighbourhood of zero in $\mathcal{T}_*$. Its derivative is the invertible linear map J_* . Consequently its coefficient–stochastic tangent cone is exactly $J_*(\mathcal{T}_*)$. Every normalized secant has a subsequential limit in this cone, and every vector of the cone is realized by a feasible arc. For bounded $v \in \mathcal{T}_*$ the arcs can be chosen with*

$$\xi(\tau, v) = \xi_* + \tau J_*v + O(\tau^2), \quad F_T(\xi(\tau, v)) = P_0 + \tau S_*v + O(\tau^2), \tag{6.6}$$

uniformly on bounded coordinate sets. Here S_ is the score (2.3) at the remote point. All assertions are uniform on compact subsets of the wall family with its stated strict margins.*

Proof. Factor each nearby polynomial into its separated simple and quadratic factors. Coprime factorization is analytic in coefficients. Write the simple root of the first component as u , the mean of its quadratic roots as $r + v_f/2$, and their squared half-separation as x . Write the corresponding coordinates of the second component as $a + w$, $b + v_h/2$ and y . Thus the exact polynomials are

$$(z - u)((z - r - v_f/2)^2 - x), \quad (z - a - w)((z - b - v_h/2)^2 - y).$$

These six coordinates are Nash functions of the coefficients near the remote pair. Conversely the displayed formulas are polynomial in the six coordinates. Their derivative has the three columns in (6.5) for each component. They are independent: for the first polynomial, evaluation at 0 and r first determines a_0 and X , and then the remaining coefficient determines b_0 ; for the second, evaluation at a and b determines c_0 and Y , and then d_0 . The inverse function theorem gives the asserted local coefficient chart.

The first simple root is feasible exactly when $u \geq 0$, locally. Its upper endpoint is inactive. All other cluster centres are strictly inside $(0, D)$, so their interval constraints are inactive in this neighbourhood. Real-rootedness of the two quadratic factors is exactly $x, y \geq 0$. The weight on h_* is at its floor, giving $k \geq 0$; the other floor has strict margin $1 - 2\pi > 0$. Channel entries are strictly positive, so only their linear column-sum equations remain active. This proves that the complete feasible chart is the indicated orthant times seven free coordinates, not merely an outer cone. Separated factors are unique. The alternative component labelling is a second, identical chart under the permutation quotient and introduces no extra directions.

For the secant assertion, let τ be the norm of a coefficient and stochastic displacement. Analytic factorization bounds the simple and mean shifts by $O(\tau)$. The quadratic coefficients bound x, y by $O(\tau)$ as well, since variance is mean squared minus the constant coefficient. Dividing the exact chart coordinates by τ gives a bounded vector in $\mathcal{T}_*$. Its subsequential limits give all normalized tangent secants. In the norm-normalized case J_*v has norm one in the limit, so no nonzero secant disappears in the chart.

Conversely set $u = \tau a_0$, $v_f = \tau b_0$, $x = \tau X$, $w = \tau c_0$, $v_h = \tau d_0$, $y = \tau Y$, choose weight $\pi + \tau k$ on h , and take the specified order- τ channel increments. The roots are

$$\tau a_0, \quad r + \tau b_0/2 \pm \sqrt{\tau X}, \quad a + \tau c_0, \quad b + \tau d_0/2 \pm \sqrt{\tau Y}.$$

They remain feasible for small τ , uniformly for bounded coordinates. They are analytic in $\sqrt{\tau}$; their polynomial coefficients are analytic in τ . In fact the exact identities are

$$f_\tau - f - \tau p_f = \tau^2 \{a_0 b_0(z - r) + a_0 X + b_0^2 z/4\} - \tau^3 a_0 b_0^2/4, \quad (6.7)$$

$$h_\tau - h_* - \tau p_h = \tau^2 \{c_0 d_0(z - b) + c_0 Y + d_0^2(z - a)/4\} - \tau^3 c_0 d_0^2/4. \quad (6.8)$$

There is no order- $\tau^{3/2}$ coefficient term at these simple/double clusters. Taylor expansion of the rational observation map, whose normalizers stay positive, proves (6.6). Compact strict root, channel, weight and clock margins make all inverse chart and derivative bounds uniform as stated. $\square$

6.2. The constrained entrance program. Put $B_0 = \frac{d}{d\delta} P_{\theta_\delta} \Big|_{\delta=0}$ and use the Fisher norm at P_0 . Define

$$\mu_E = \frac{1}{2} \min_{v \in \mathcal{T}_*} \|S_* v - B_0\|_{I,0}. \quad (6.9)$$

This is an explicit finite-dimensional program on the cone just proved to be the full feasible tangent cone.

Lemma 6.2 (Positive entrance separation). *The score has no nonzero zero on $\mathcal{T}_*$, and B_0 is not in its image. Thus $\mu_E > 0$ and its minimum is attained. Eliminating seven free variables gives sixteen independent-support KKT tests on four nonnegative variables. For algebraic input μ_E is real algebraic.*

Proof. Suppose first that $S_* v = B_0$. The differentiated normalizer equations, using Lemma 2.1 at the coprime remote pair with invertible second channel, imply $\dot{q} = f - g$ and $\dot{K} = \frac{d}{d\delta} K_{\theta_\delta} \Big|_0$. The derivative of the second marginal belongs to $\text{span}(f, g)$. Its stochastic derivative terms already belong to that span; the remote second-channel weight matrix is invertible. Hence $p_f, p_h \in \text{span}(f, g)$. Since their degrees are below three, write $p_f = t_f(g - f)$ and $p_h = t_h(g - f)$.

Evaluation at 0 gives $t_f = a_0 r^2/s^3 \geq 0$, while evaluation at r gives $t_f = -Xr/(r - s)^3 \leq 0$. Therefore $t_f = 0$ and $p_f = 0$. At b the second polynomial gives

$$t_h = \frac{Y(b - a)}{f(b) - g(b)} \geq 0,$$

because $R(b) > 1$. Differentiation of the monic normalizer now gives

$$\pi t_h + c_* k = -1, \quad (6.10)$$

an impossibility. For a zero score, the same argument has right-hand side zero and implies $t_h = k = 0$. All polynomial jets vanish. Coefficient functionals dual to f, h_* then imply that each coefficient matrix has zero derivative; taking its row and column sums forces all four channel derivatives to vanish. The independent chart columns in Proposition 6.1 force $v = 0$.

The intersection of $\mathcal{T}_*$ with the unit sphere is compact, so $\|S_* v\|_{I,0} \geq c\|v\|$ on the cone. Thus the objective in (6.9) tends to infinity with $\|v\|$; its minimum is attained and positive, since equality $S_* v = B_0$ was excluded.

Let $\mathbf{L}_*$ be the seven free columns, $\mathbf{D}_*$ the four nonnegative columns, $W = \text{diag}(w_j/P_{0,j_e})$, and $\mathcal{P}$ the Fisher-orthogonal projection off $\text{im } \mathbf{L}_*$. Put

$$Q_* = \mathbf{D}_*^\top W \mathcal{P} \mathbf{D}_*, \quad g_* = \mathbf{D}_*^\top W \mathcal{P} B_0, \quad b_* = B_0^\top W \mathcal{P} B_0.$$

Coercivity implies full rank of the free columns, and

$$4\mu_E^2 = \min_{z \geq 0} \{z^\top Q_* z - 2g_*^\top z + b_*\}. \quad (6.11)$$

For each of the sixteen supports A with $Q_{*,AA}$ positive definite, set $z_A = Q_{*,AA}^{-1}g_{*,A}$, require $z_A \geq 0$ and $(Q_* z - g_*)_j \geq 0$ off A , and take value $b_* - g_{*,A}^\top Q_{*,AA}^{-1}g_{*,A}$. Include the empty support. The minimal-positive-support argument of Proposition 5.5 proves completeness, including singular Q_* . All entries are rational in the wall coordinates; take the positive square root after selecting the exact minimum. $\square$

Lemma 6.3 (Local coefficient control at the remote point). *In a coefficient and stochastic neighbourhood of ξ_* in the closed model,*

$$d_{\text{par}}(\xi, \xi_*) \leq C \|F_T(\xi) - P_0\|.$$

Every competitor at parameter distance $O(\delta)$ has bounded exact chart coordinates at scale δ and observation expansion $P_0 + \delta S_ v + O(\delta^2)$. Conversely each bounded cone vector has such a feasible realization.*

Proof. If the inverse failed, normalize a sequence of displacements by their norm. Proposition 6.1 gives a nonzero limit $J_* v$ with $v \in \mathcal{T}_*$ and zero score, contradicting Lemma 6.2. Equivalently in its chart, Taylor's theorem and cone coercivity give $\|F_T(\Phi(w)) - P_0\| \geq c\|w\| - C\|w\|^2 \geq c\|w\|/2$ for small feasible w . The chart is locally bi-Lipschitz in coefficients and stochastic entries, which proves the inverse and the stated bounded coordinate assertion. The expansions and converse are (6.6). This proof does not assume the inverse in order to identify the tangent cone. $\square$

6.3. The two-scale law with a quadratic entrance remainder. Choose disjoint small closed coefficient and stochastic neighbourhoods N_0, N_* of the two exact representatives at the wall, modulo component permutation, with ξ_* in the relative interior of N_* . Put

$$d_{\text{rem}}(\delta) = \min_{\xi \in N_*} h_w(F_T(\xi), P_{\theta_\delta}). \quad (6.12)$$

Their boundaries have a positive observation separation from P_0 .

Theorem 6.4 (Weight-wall transition). *Under (6.1) and $B > 0$,*

$$d_{\text{rem}}(\delta) = \mu_E \delta + O(\delta^2) \quad (\delta \downarrow 0). \quad (6.13)$$

The minimum is independent of the sufficiently small neighbourhood N_ for all sufficiently small δ . The base local constant $C_{\text{loc}}(\theta_\delta)$ has the finite positive limit $C_{\text{loc}}(\theta_0)$; neither κ_x nor κ_y vanishes at this wall. For fixed $\lambda > 0$ the whole-model modulus satisfies*

$$0 < \lambda < \mu_E : \quad \omega_{\theta_\delta}(\lambda\delta) = C_{\text{loc}}(\theta_0)\sqrt{\lambda\delta} + O_\lambda(\delta), \quad (6.14)$$

$$\lambda > \mu_E : \quad \omega_{\theta_\delta}(\lambda\delta) = D_* + O_\lambda(\sqrt{\delta}). \quad (6.15)$$

The exact entrance curve for the closed ball is $t = d_{\text{rem}}(\delta)$, including equality. The layer not decided by the first-order threshold alone has width $O(\delta^2)$ in t .

On the nonidentified side, for small $\varepsilon > 0$,

$$\omega_{\theta_{-\varepsilon}}(0) = \max \left\{ r - s, \ b - r + \sqrt{\frac{\chi\varepsilon}{\pi}} + O(\varepsilon) \right\}, \quad \chi = \frac{f(b) - g(b)}{b - a} > 0. \quad (6.16)$$

If $b - r < r - s$, the exact diameter stays equal to $r - s$ initially; if $b - r \geq r - s$, it opens above D_ at the displayed square-root rate.*

Proof. Theorem 5.1 gives exactly the two representatives at $\delta = 0$. Compactness gives a positive observation gap outside $N_0 \cup N_*$. Since ξ_* itself is at distance $O(\delta)$ from P_{θ_δ} , a minimizer in (6.12) has observation $O(\delta)$ from P_0 and lies away from the boundary of N_* . Lemma 6.3 bounds its coefficient and stochastic displacement by $O(\delta)$. Its exact chart coordinates are therefore δv for bounded $v \in \mathcal{T}_*$. Expansion at the common positive probability array and the reverse triangle inequality give

$$h_w(F_T(\xi), P_{\theta_\delta}) = \frac{\delta}{2} \|S_* v - B_0\|_{I,0} + O(\delta^2), \quad (6.17)$$

uniformly on those bounded coordinates. The lower bound in (6.13) follows by minimizing the leading norm. Conversely a minimizer of (6.9), realized by the exact chart, gives the upper bound. The error is quadratic because of (6.7)–(6.8), not merely an appeal to a generic root-continuity bound. This also implies the weaker $O(\delta^{3/2})$ entrance estimate. The same localization puts all minimizers inside any fixed smaller remote neighbourhood for small δ , proving independence of its choice.

Proposition 5.5 and Theorem 5.6 apply on a compact base path segment including θ_0 ; this pair stays locally coercive there. Thus C_{loc} is positive and locally Lipschitz through the wall. If $\lambda < \mu_E$, the remote neighbourhood is excluded from the radius- $\lambda\delta$ ball for small δ . The exterior gap excludes every other competitor, so the whole ball is the base local ball. Formula (5.14) gives (6.14). Replacing $C_{\text{loc}}(\theta_\delta)$ by $C_{\text{loc}}(\theta_0)$ costs only $O(\delta^{3/2})$.

If $\lambda > \mu_E$, a remote minimizer and the base centre belong to the ball. Every competitor belongs to one of the two neighbourhoods. The two coefficient inverses put its roots within $O_\lambda(\sqrt{\delta})$ of the corresponding exact spectrum at the wall. The bottleneck triangle inequality gives both bounds in (6.15). Compact attainment makes remote membership exactly $t \geq d_{\text{rem}}(\delta)$, including equality. The width of the unresolved first-order layer follows from (6.13); it does not decide the sign of the difference on the critical ray.

For $\theta_{-\varepsilon}$ the extra exact pencil interval is $[c_*, c_* + \varepsilon/\pi]$, since $c_D > c_*$ is fixed. At $c = c_* + \sigma$, expansion at $z = b + u$ gives

$$h_c(b + u) = (b - a)u^2 + u^3 + \sigma(g(b) - f(b)) + O(\sigma|u|).$$

Since $R(b) > 1$, $\chi > 0$ and the two roots near b are $b \pm \sqrt{\chi\sigma} + O(\sigma)$. The third root remains in (s, r) . The larger root increases with c by the monotonic branch of R above its maximum. Comparing sorted lists, the distance to the base spectrum is exactly $\max(r - s, b_+(c) - r)$. Two remote spectra are only $O(\sqrt{\varepsilon})$ apart, less than D_* for small ε . The exact-fibre diameter is therefore attained against the base at the upper pencil endpoint. This proves (6.16) and its two cases. $\square$

Remark 6.5 (The critical ray). At $t = \mu_E\delta$, remote membership is decided by the sign of $d_{\text{rem}}(\delta) - \mu_E\delta$. For algebraic input, (6.12) has a compact first-order minimum formula. Elimination and real algebraic branch isolation determine its Puiseux expansion. The first nonzero higher coefficient decides the critical ray; if the difference is identically zero, the closed ball includes the remote branch. No strict regime is assigned to that ray by convention.

Remark 6.6 (The geometric wall). At $B = 0$ with $E > 0$, the remote double root is $b = D$, and the additional pencil interval again consists of one point at positive spectral distance. On the nonidentified side put $\ell_0 = D - b$. Expansion at the strict maximum of R gives

$$c_D - c_* = \frac{-R''(b)}{2(R(b) - 1)^2} \ell_0^2 + O(\ell_0^3), \quad \ell_0 = \frac{B}{3s - 2r}.$$

The coefficient is positive and the weight floor is inactive. This interval is born with quadratic length in geometric clearance, rather than linear length in the weight parameter. The base local constants remain finite on compact subsets of $\mathfrak{F}$. A two-scale entrance law at that endpoint

requires a different cone with one-sided root coordinates. The complete entrance theorem here is for the weight wall.

Example 6.7 (An exact wall datum). Take $D = 1$, $r = 3/5$, $s = 11/20$, $\pi = 1/10$, the rank-one columns and clocks in Section 5, and equal design weights. Then

$$\alpha_0 = \frac{547}{675}, \quad c_* = \frac{256}{135}, \quad a = \frac{44}{75}, \quad b = \frac{11}{15}, \quad B = \frac{3}{25}, \quad D_* = \frac{2}{15}, \quad \chi = \frac{3}{64},$$

and $v_f^* = (1457, 3403)^\top / 4860$. Exact rational evaluation gives

$$5975.429249985072 < C_{\text{loc}}(\theta_0) < 5975.429249985073, \\ 2.131024914 \cdot 10^{-6} < \mu_E < 2.131024915 \cdot 10^{-6}.$$

The empty positive support is admissible in each base program and in the remote program. The ancillary exact records retain the rational Gram matrices, squared entrance constant, fourth power of the local constant and isolating intervals. The decimals follow from rational power comparisons, not optimization tolerances. They illustrate two different, poorly conditioned scales; they do not test the universal quantifiers of the family theorems.

Three stability questions. Multiplicity and one-sided boundary constraints determine the weights in Theorem 3.2. Exact identification in the cubic family is instead determined by (5.2). Conditioning of a fixed local law is measured by the positive Fisher programs. At the weight wall, identification fails while those programs stay coercive and the local square-root exponent persists. In Example 3.3, by contrast, the exponent changes and the linear constant diverges. The joint atlas retains these distinct limiting operations rather than identifying their strata.

7. FINITE REMOTE ENTRANCES AND THE CRITICAL WINDOW

The preceding weight wall is an instance of a more general phenomenon. A regular feasible corner, rather than invertibility of the ambient score, is the relevant local hypothesis. We formulate the result in the square-root probability embedding, so that the distance in its statement is exactly the Hellinger distance and not a quadratic approximation to it.

Let X be compact and semialgebraic, let F be a continuous probability map, and let T be a continuous monic-polynomial root-multiset target of fixed degree. Quotient a finite representation symmetry when necessary. Suppose

$$F^{-1}(P_0) = \{x_0, x_1, \dots, x_m\}. \quad (7.1)$$

The points are representatives in this quotient. Let $x_\delta \rightarrow x_0$ be a feasible centre path, $P_\delta = F(x_\delta)$, and suppose P_0 is strictly positive. The design weights are fixed. Define

$$E(P) = (\sqrt{w_j} \sqrt{P_{je}})_{j,e}, \quad p_0 = E(P_0), \quad p_\delta = E(P_\delta) = p_0 + \delta b + \delta^2 c + o(\delta^2).$$

The Euclidean distance between two embedded arrays is h_w .

For each $j \geq 1$, assume that a full relative neighbourhood of x_j in X has coordinates

$$\psi_j : K_j \cap B_{r_j}(0) \longrightarrow X, \quad \psi_j(0) = x_j,$$

where K_j is a closed convex polyhedral cone and ψ_j is a Nash parametrization, a homeomorphism onto that relative neighbourhood. Its inverse need not be differentiable at the vertex. Assume that $E \circ F \circ \psi_j$ has a C^2 extension to an ambient neighbourhood of zero. Write

$$E(F(\psi_j(z))) = p_0 + J_j z + Q_j(z, z) + o(\|z\|^2), \quad \|J_j v\| \geq a_j \|v\| \quad (v \in K_j), \quad a_j > 0. \quad (7.2)$$

Here $Q_j(z, z)$ denotes one half of the second derivative. The score may have a kernel in the ambient vector space; the last inequality is required only on the feasible cone. Choose small disjoint closed neighbourhoods $N_0, \dots, N_m$, with boundaries avoiding the exact fibre, and put

$$d_j(\delta) = \min_{x \in N_j} h_w(F(x), P_\delta), \quad \mu_j = \min_{v \in K_j} \|J_j v - b\|.$$

Theorem 7.1 (Remote entrance for regular corners). *Assume (7.1)–(7.2) and $\mu_j > 0$ for $1 \leq j \leq m$. Then*

$$d_j(\delta) = \mu_j \delta + O(\delta^2) \quad (\delta \downarrow 0). \quad (7.3)$$

For small δ the minimum is unchanged on replacing N_j by any fixed smaller relative neighbourhood of x_j . If $\lambda > 0$ differs from every μ_j , the target image of the radius- $\lambda\delta$ ball converges in Hausdorff distance to

$$\mathcal{D}_\lambda = \{T(x_0)\} \cup \{T(x_j) : \mu_j < \lambda\}. \quad (7.4)$$

Its diameter consequently tends to $\text{diam } \mathcal{D}_\lambda$. For $\lambda < \min_j \mu_j$, the whole ball is the base local ball for small δ . Any compact-uniform local asymptotic law at the base therefore remains valid in this regime.

Proof. Fix a remote index and omit it from the notation. Taylor's theorem and cone coercivity give, after reducing r ,

$$\|E(F(\psi(z))) - p_0\| \geq a\|z\|/2 \quad (z \in K, \|z\| < r).$$

The competitor $z = 0$ gives an upper bound $O(\delta)$ for $d(\delta)$. Compactness and (7.1) exclude the boundary of N from minimizing sequences. A minimizer thus has $z = O(\delta)$: its distance to p_0 is at most $d(\delta) + \|p_\delta - p_0\|$. Write $z = \delta v$. The vectors $v \in K$ are uniformly bounded, and

$$E(F(\psi(\delta v))) - p_\delta = \delta(Jv - b) + \delta^2\{Q(v, v) - c\} + o(\delta^2)$$

uniformly on bounded sets. Minimizing the leading norm gives the lower bound in (7.3). The function $v \mapsto \|Jv - b\|$ is coercive on K , so its minimum is attained; inserting a minimizer in the exact chart gives the upper bound. The same localization shows independence of the fixed neighbourhood.

Outside the union of the N_j the continuous function $h_w(F(x), P_0)$ has a positive minimum. This remains an exclusion gap for balls of radius $\lambda\delta$ about P_δ . In a remote neighbourhood, membership is possible exactly when $t \geq d_j(\delta)$, by compact attainment. Thus a remote neighbourhood is excluded if $\lambda < \mu_j$ and contains a minimizer if $\lambda > \mu_j$. Every sequence of remaining competitors converges, after subselection, to an exact representative; its target converges to that representative's target. Conversely x_δ supplies $T(x_0)$ in the limit and each included remote minimizer supplies $T(x_j)$. Finiteness gives Hausdorff convergence to (7.4). The diameter conclusion follows from the elementary bound by twice the Hausdorff distance. When no remote neighbourhood is included, all competitors are in N_0 , which proves the last assertion. $\square$

The coefficient of the next term can be determined without assuming a unique minimizer or an invertible ambient score. This is useful at a change of active face of a constrained Fisher program.

Theorem 7.2 (The second-order entrance coefficient). *Under the hypotheses of Theorem 7.1, set*

$$M_j = \operatorname{argmin}_{v \in K_j} \|J_j v - b\|.$$

This is a nonempty compact set. The residual $r_j = J_j v - b$ is the same for all $v \in M_j$, and has norm μ_j . Define

$$k_j = \frac{1}{\mu_j} \min_{v \in M_j} \langle r_j, Q_j(v, v) - c \rangle. \quad (7.5)$$

Then

$$d_j(\delta) = \mu_j \delta + k_j \delta^2 + o(\delta^2). \quad (7.6)$$

At radius $t = \mu_j \delta + s \delta^2$, the j th remote neighbourhood is excluded for $s < k_j$ and included for $s > k_j$, for all sufficiently small positive δ . Equality requires the next nonzero term or the exact minimum inequality; it is not assigned to a regime by convention. For algebraic Nash input, μ_j and k_j are real algebraic numbers computable by real quantifier elimination and real-root isolation.

Proof. Coercivity bounds all minimizers. The image $J_j K_j$ is a closed convex polyhedral cone. The squared distance to b has a unique minimizing image point: if two different image points minimized it, their midpoint would have strictly smaller squared distance. This proves the assertion about r_j without requiring uniqueness of its preimage.

Again suppress j . Localization in the preceding proof bounds all scaled minimizers in one compact subset of K , also containing M . On that set the squared normalized distance has the uniform expansion

$$f_\delta(v) = \|Jv - b\|^2 + 2\delta \langle Jv - b, Q(v, v) - c \rangle + o(\delta).$$

Every limit of minimizers of f_δ belongs to M , by comparison with any point of M . The first term is at least μ^2 everywhere. The lower bound for the first-order increment of $\min f_\delta$ is therefore the minimum of the displayed second term over M . For the upper bound evaluate f_δ at a point of M attaining that minimum. Compactness and continuity justify both extrema. Consequently

$$\min f_\delta = \mu^2 + 2\delta \min_{v \in M} \langle r, Q(v, v) - c \rangle + o(\delta).$$

Taking the positive square root and multiplying by δ proves (7.6). Comparing with the proposed radius gives the strict cases. The minimum defining M and the minimum in (7.5) have first-order polynomial formulas with compact attainment; the latter may equivalently be restricted to a sufficiently large algebraic ball. Elimination and isolation give algebraic outputs. The formula is intrinsic to the distance and the fixed centre path, since its value equals the coefficient in (7.6), independently of the chosen chart. $\square$

Corollary 7.3 (Application to the cubic weight wall). *The remote point of Section 6 satisfies the preceding hypotheses with $m = 1$, $K_1 = \mathcal{T}_*$ and $\mu_1 = \mu_E$. In particular its entrance has the expansion*

$$d_{\text{rem}}(\delta) = \mu_E \delta + k_E \delta^2 + o(\delta^2),$$

where k_E is given by (7.5) in the exact remote chart of Proposition 6.1. This formula does not assert a common sign for k_E on the weight-wall family.

Proof. Theorem 5.1 gives the two exact representatives. The remote chart and its converse feasible realization are Proposition 6.1; its smooth coefficient formulas and positive normalizers give the required C^2 observation embedding. In these coordinates

$$J_1 v = \left(\frac{\sqrt{w_j}}{2\sqrt{P_{0,j_e}}} (S_* v)_{j_e} \right)_{j,e}, \quad b = \left(\frac{\sqrt{w_j}}{2\sqrt{P_{0,j_e}}} (B_0)_{j_e} \right)_{j,e}.$$

Lemma 6.2 is exactly cone coercivity and positive separation in this embedding. Hence the first minimization is (6.9). The two second-order derivatives in (7.5) are taken from the rational coefficient chart and the square-root probability map, not from root coordinates with an unproved differentiability at a multiple root. Theorems 7.1 and 7.2 apply. $\square$

Example 7.4 (Both signs of the critical correction). Let $e_0 = (1, 1)/\sqrt{2}$, $e_1 = (-1, 1)/\sqrt{2}$ and

$$q(s) = \frac{(1 - s^2)e_0 + 2se_1}{1 + s^2}, \quad P(s) = (q_1(s)^2, q_2(s)^2).$$

For sufficiently small $|s|$ both coordinates of $q(s)$ are positive. Take a base interval with observation $P(s)$ and a disjoint remote interval $0 \leq u \leq e$ with observation $P(-u/2)$. Give the two intervals distinct simple-root targets at zero, for example $z - (1/4 + s)$ and $z - (3/4 + u)$, and choose e small enough to keep all roots in $[0, 1]$. The exact fibre at $P(0)$ has two points. Choose the centre $s_\delta = (\delta + a\delta^2)/2$, where a is fixed. At the remote point $K = \mathbb{R}_+$, $Jv = -ve_1$, $b = e_1$, $c = ae_1 - e_0/2$, $M = \{0\}$ and $\mu = 1$. Formula (7.5) gives $k = a$. Indeed the nearest remote point is $u = 0$ for small $\delta > 0$, since on this short circle arc the chord distance increases with the separation of the stereographic parameters. Directly,

$$d_{\text{rem}}(\delta) = \frac{2s_\delta}{\sqrt{1 + s_\delta^2}} = \delta + a\delta^2 + O(\delta^3).$$

Thus either sign of the second-order correction occurs in an algebraic probability experiment satisfying all hypotheses. For $a = 0$ the quadratic coefficient vanishes and the next term is $-\delta^3/8$.

These results apply in any coefficient dimension and to any fixed number of latent components once their stated finite-fibre and feasible-cone hypotheses have been verified. They are not a classification of all rank-deficient singularities. The endpoint double root at the geometric wall requires a different parametrization and a different separation argument, supplied in Section 8.

8. THE GEOMETRIC ENDPOINT WALL

We now complete a second wall mechanism in the same rank-deficient cubic experiment. Fix $D > r > 0$, strict weights $\alpha, \gamma = 1 - \alpha$, positive $U = (u, u)$ and invertible positive V , and the exterior design. Put

$$s_0 = \frac{2rD}{3D - r}, \quad s_\delta = s_0 - \delta, \quad f = z(z - r)^2, \quad g_\delta = (z - s_\delta)^3. \quad (8.1)$$

Then $2r/3 < s_0 < r$. Assume the strict weight margin

$$E_0 = \gamma A(s_0) - 4\pi r^3 > 0, \quad A(s) = (r + 3s)(2r - 3s)^2. \quad (8.2)$$

Let θ_δ be this base centre and P_δ its observation. Here $B(\theta_\delta) = -(3D - r)\delta$. Thus positive δ is the identified side, while the weight constraint stays inactive locally. Write $c_* = 4r^3/A(s_0)$ and

$$a = \frac{4rs_0}{r + 3s_0}, \quad h_* = (1 - c_*)f + c_*g_0 = (z - a)(z - D)^2, \quad \beta = \gamma/c_*.$$

The remote weight on h_* is $\beta > \pi$, and its other weight is $1 - \beta > \pi$. Its first channel is (u, u) and its second channel is

$$v_f^* = \frac{\alpha v_1 + (\gamma - \beta)v_2}{1 - \beta}, \quad v_h^* = v_2. \quad (8.3)$$

Both columns are positive and distinct. Denote this remote point by ξ_B . Theorem 5.1 gives exactly the base and this remote representative at $\delta = 0$.

Proposition 8.1 (The endpoint-root parametrization). *A full relative neighbourhood of ξ_B , modulo the component permutation, has the following exact feasible parametrization:*

$$\begin{aligned} f_v(z) &= (z - a_0)((z - r - b_0/2)^2 - X), \\ h_v(z) &= (z - a - c_0)(z - D - d_1)(z - D - d_2), \\ K_B &= \{a_0, X \geq 0, \quad d_1 \leq d_2 \leq 0\}. \end{aligned} \quad (8.4)$$

The weight on h is $\beta + k$, and the four binary channel coordinates have their unrestricted small increments. The other coordinates b_0, c_0, k are free. This is a polynomial parametrization with

continuous semialgebraic inverse; no differentiability of its inverse at $d_1 = d_2$ is asserted. Its first polynomial jets are

$$p_f = -a_0(z - r)^2 - b_0z(z - r) - Xz, \quad p_h = -c_0(z - D)^2 - (d_1 + d_2)(z - a)(z - D). \quad (8.5)$$

Let S_B be the score at ξ_B applied to these jets and the stochastic increments. Then S_B has no nonzero zero on K_B . If $B_B = \frac{d}{d\delta}P_\delta|_0$, then $B_B \notin S_B K_B$. In particular

$$\mu_B = \frac{1}{2} \min_{v \in K_B} \|S_B v - B_B\|_{I,0} > 0. \quad (8.6)$$

The minimum is attained. After seven free coordinates are eliminated, it is a quadratic program with three nonnegative coordinates and eight independent-support tests.

Proof. Separated simple and double factors give the first component's simple root, quadratic mean and squared half-separation as analytic functions of its coefficients. At the second component the separated simple root is analytic; the two roots near D have a unique increasing ordering and depend continuously and semialgebraically on the coefficients. Their feasibility is exactly $d_1 \leq d_2 \leq 0$. All other root bounds, weights and channel inequalities have strict margins, except $a_0 \geq 0$ and $X \geq 0$. This proves exact coverage and injectivity of the parametrization, not merely containment of tangent directions. Its polynomial formulas give a uniform $O(\tau^2)$ coefficient remainder on substituting τv for bounded $v \in K_B$. Positive normalizers and probabilities give the same remainder for the observation embedding.

We prove the two score assertions directly. Put $\dot{g} = 3(z - s_0)^2$, the derivative of g_δ at zero. Equality of remote and base scores implies equality of their derivatives of q, K by the differentiated normalizer argument in Lemma 2.1. The coefficient second marginal at the remote point is $(1 - \beta)v_f^* f + \beta v_2 h_*$. Its two coefficient columns are independent. Modulo $\text{span}(f, g_0)$, equality of its derivative with $\gamma v_2 \dot{g}$ gives

$$p_f = t_f(g_0 - f), \quad p_h = c_* \dot{g} + t_h(g_0 - f).$$

The polynomials have the stated degrees, so no additional monic direction occurs. Evaluation of the first identity at 0 and r gives $t_f = a_0 r^2 / s_0^3 \geq 0$ and $t_f = -Xr / (r - s_0)^3 \leq 0$. Thus $t_f = 0$ and $p_f = 0$.

To interpret the second identity, consider the algebraic double-root pencil curve, without imposing its endpoint constraint,

$$\bar{h}_\delta = (1 - c_*(s_\delta))f + c_*(s_\delta)g_\delta = (z - a(s_\delta))(z - b(s_\delta))^2, \quad b(s) = \frac{rs}{3s - 2r}.$$

At zero, $b(s_0) = D$ and

$$\dot{b} = \frac{d}{d\delta}b(s_0 - \delta)\Big|_0 = \frac{2r^2}{(3s_0 - 2r)^2} > 0. \quad (8.7)$$

Both $p_h(D)$ and $\dot{h}(D)$ vanish. Since $g_0(D) - f(D) \neq 0$, this determines t_h uniquely and implies $p_h = \dot{h}$. Differentiation in z at D now gives

$$-(d_1 + d_2)(D - a) = p'_h(D) = -2(D - a)\dot{b}.$$

This would require $d_1 + d_2 = 2\dot{b} > 0$, contrary to feasibility. Therefore $S_B v = B_B$ is impossible.

For a zero score the same second-marginal argument instead gives $p_f = t_f(g_0 - f)$ and $p_h = t_h(g_0 - f)$. The first polynomial again vanishes. Evaluating the second at D forces $t_h = 0$. Its coefficients then give $c_0 = 0$ and $d_1 + d_2 = 0$, whence $d_1 = d_2 = 0$ on K_B . The first polynomial's independent columns give $a_0 = b_0 = X = 0$. Coefficient functionals dual to f, h_* recover zero derivatives of both coefficient matrices, and their entry, row and column sums give zero weight and channel increments. Thus the entire cone vector is zero. Compactness of the feasible unit

sphere yields $\|S_B v\|_{I,0} \geq c\|v\|$ on K_B . The objective in (8.6) is consequently coercive and has a positive attained minimum.

Finally the score depends on d_1, d_2 only through $w = -(d_1 + d_2) \geq 0$. Every $w \geq 0$ is realized, for example by $d_1 = d_2 = -w/2$. The remaining nonnegative coordinates are a_0, X ; there are seven free coordinates. Their columns are independent by the zero-score argument. Fisher-orthogonal projection off those columns reduces the problem to three nonnegative variables. The minimal-positive-support argument of Proposition 5.5 gives the eight support tests, including the empty support and only inverting positive definite supported Gram matrices. This remains valid when the full projected Gram matrix is singular. $\square$

Theorem 8.2 (The geometric-wall two-scale law). *Under (8.1)–(8.2), the remote entrance satisfies*

$$d_{\text{rem},B}(\delta) = \mu_B \delta + k_B \delta^2 + o(\delta^2), \quad \mu_B > 0, \quad (8.8)$$

where k_B is given by Theorem 7.2 in the parametrization (8.4). In particular, for fixed $\lambda > 0$,

$$\lambda < \mu_B : \quad \omega_{\theta_\delta}(\lambda\delta) = C_{\text{loc}}(\theta_0)\sqrt{\lambda\delta} + O_\lambda(\delta), \quad (8.9)$$

$$\lambda > \mu_B : \quad \omega_{\theta_\delta}(\lambda\delta) = D - r + O_\lambda(\sqrt{\delta}). \quad (8.10)$$

The base local Fisher constant stays positive and finite at the wall. The strict cases in the critical window $t = \mu_B \delta + s\delta^2$ are decided by $s - k_B$.

On the nonidentified side $\theta_{-\varepsilon}$, for all sufficiently small $\varepsilon > 0$, the additional exact pencil interval has length

$$c_D(s_0 + \varepsilon) - c_*(s_0 + \varepsilon) = \Lambda_B \varepsilon^2 + O(\varepsilon^3), \quad \Lambda_B = \frac{-R''_{s_0}(D)}{2(R_{s_0}(D) - 1)^2} b^2 > 0, \quad (8.11)$$

where $R_s(z) = f(z)/(z - s)^3$. Nevertheless its exact whole-model spectral diameter is constant:

$$\omega_{\theta_{-\varepsilon}}(0) = D - r. \quad (8.12)$$

Thus the two wall mechanisms differ both in the size of the newborn pencil interval and in the opening of its exact spectral diameter.

Proof. The endpoint parametrization and its score separation satisfy the hypotheses of Theorems 7.1 and 7.2. The derivative of the square-root observation embedding is the score multiplied entrywise by $\sqrt{w_j}/(2\sqrt{P_{0,j\varepsilon}})$, so its entrance minimum is precisely (8.6). This proves (8.8) and the critical-window assertion, including possible nonuniqueness of the minimizing endpoint-root split.

The sorted spectra at the wall are $(0, s_0, s_0, s_0, r, r)$ and $(0, a, r, r, D, D)$. Their distance is $\max(r - s_0, D - r) = D - r$, since $r - s_0 = r(D - r)/(3D - r) < D - r$. The local inverse and local Fisher law of Section 5 are uniform on a compact segment of the base path through zero. Below the entrance threshold all competitors are in that base neighbourhood, proving (8.9). Above it, the localization proof for the remote cone bounds its exact coordinates by $O(\delta)$. The remote endpoint roots move by $O(\delta)$ and its interior double cluster by $O(\sqrt{\delta})$; the base inverse gives the same $O(\sqrt{\delta})$ root bound. A remote minimizer and the centre are both feasible, and the exterior observation gap excludes every other region. The bottleneck triangle inequality proves (8.10).

For the last assertions, $b(s_0 + \varepsilon) = D - b\varepsilon + O(\varepsilon^2)$. The function R_s has a strict nondegenerate maximum at $b(s)$ near the wall. Taylor expansion of $R_s(D)$ there, followed by $x \mapsto x/(x - 1)$, gives

$$c_D(s) - c_*(s) = \frac{-R''_s(b(s))}{2(R_s(b(s)) - 1)^2} (D - b(s))^2 + O((D - b(s))^3).$$

This proves (8.11). The strict weight margin keeps $\gamma/\pi > c_D(s)$ for small ε , so the entire interval $[c_*(s), c_D(s)]$ is feasible. Its upper endpoint has a root exactly at D . A remote spectrum has

increasing ordering $(0, a_c, r, r, b_-(c), b_+(c))$, with $s < a_c < r < b_-(c) \leq b_+(c) \leq D$. Its distance to the base is exactly $\max(r - s, b_+(c) - r) \leq D - r$, and equality is attained at $c_D(s)$. The remote spectra converge uniformly to one spectrum as $\varepsilon \downarrow 0$; their mutual diameter is therefore less than $D - r$ for small ε . Theorem 5.1 excludes any other exact component. This proves (8.12). $\square$

8.1. Exact critical-ray examples. We record two algebraic specializations of the second-order formula. Use the clocks $(6/5, 7/5, 8/5, 9/5, 2, 5/2, 3)$ with equal weights, $u_1 = u_2 = (1/3, 2/3)^\top$, and $v_1 = (1/4, 3/4)^\top$, $v_2 = (3/4, 1/4)^\top$.

Example 8.3 (An endpoint wall with a positive critical correction). Take $D = 1$, $r = 3/5$, $s_0 = 1/2$, $\alpha = 2/5$, $\pi = 1/10$. The remote coordinates and the interval-opening coefficient are

$$a = \frac{4}{7}, \quad c_* = \frac{32}{7}, \quad \beta = \frac{21}{160}, \quad v_f^* = \frac{1}{556}(289, 267)^\top, \quad \dot{b} = 8, \quad \Lambda_B = \frac{38400}{49}.$$

The projected three-variable program has the empty support as its only successful independent-support test. Its linear vector is strictly negative entrywise, so the minimizer has all three nonnegative coordinates zero. The free score has rank seven; hence the first-order minimizer is unique, including the endpoint split. Exact rational arithmetic and positive-root isolation give

$$0.000063237134157102 < \mu_B < 0.000063237134157103, \\ 0.071857358263070 < k_B < 0.071857358263071.$$

Consequently the remote branch is excluded on $t = \mu_B \delta$ for all sufficiently small positive δ . On the opposite side the exact spectral diameter is $2/5$, despite the nonzero quadratic pencil-interval opening.

Example 8.4 (The weight-wall critical ray). At the weight-wall datum of Example 6.7, the four entries of the projected linear vector g_* are strictly negative. The empty cone support is therefore the unique first-order minimizer, with its seven free coordinates recovered by the inverse free Gram matrix. The second-order formula gives

$$0.011285405417542 < k_E < 0.011285405417543.$$

Thus the critical ray $t = \mu_E \delta$ also excludes the remote branch for this datum when $\delta > 0$ is sufficiently small. This sign is a property of this specialization, not a convention at an unresolved ray and not a sign assertion for the whole wall family.

For reproducibility, the finite arithmetic behind both examples can be specified without numerical differentiation. Write the exact remote and base probability expansions at the selected first-order minimizer as

$$P_{\text{rem}}(\delta) = P + \delta A + \delta^2 C + O(\delta^3), \quad P_{\text{base}}(\delta) = P + \delta B + \delta^2 D + O(\delta^3).$$

The rational coefficients are obtained by multiplying the exact polynomial, weight and channel jets and dividing by the normalizer. The square-root expansion then gives

$$\mu^2 = \frac{1}{4} \sum_{j,e} \frac{w_j (A_{je} - B_{je})^2}{P_{je}}, \\ \mu k = \sum_{j,e} w_j \left\{ \frac{(A_{je} - B_{je})(C_{je} - D_{je})}{4P_{je}} - \frac{(A_{je} - B_{je})(A_{je}^2 - B_{je}^2)}{16P_{je}^2} \right\}. \quad (8.13)$$

Strict empty-support KKT inequalities and positive semidefiniteness of the projected Gram matrix prove uniqueness of the cone optimum even when that matrix is singular: the difference of objectives is $z^\top Q z - 2g^\top z > 0$ for every nonzero $z \geq 0$. The rational values of μ^2 , μk , the Gram data and isolating intervals are retained with the exact regression. The interval tests square

rational endpoints and compare integers after clearing positive denominators. They certify these finite specializations only.

9. REAL MONOMIAL DATA IN PARAMETER FAMILIES

We work first with a compact semialgebraic model X , a continuous semialgebraic probability map F , and a continuous semialgebraic monic target polynomial A_x of fixed degree n . Complex coefficients are represented by real and imaginary parts. Suppose the target is constant on $F^{-1}(P)$, with value $A_0(z) = \prod_r (z - r)^{m_r}$. Disjoint isolating discs and compactness give, on a sufficiently small closed observation neighbourhood, unique monic cluster factors

$$A_{r,x}(r + z) = z^{m_r} + \sum_{a < m_r} c_{r,a}(x) z^a, \quad k_{r,a} = m_r - a. \quad (9.1)$$

Multiplication of coprime monic factors has invertible differential. The cluster factors are consequently analytic functions of the total coefficients. Their graphs with the isolating conditions are semialgebraic.

Definition 9.1 (Input and families). An algebraic input is a finite first-order polynomial presentation of X, F, A, P , design weights, isolating data, denominator exclusions and branch choices. A real algebraic constant is encoded by an integer polynomial and a rational isolating interval; a complex constant is encoded by two real algebraic constants. A fixed-format family has a semialgebraic parameter set Θ , a semialgebraic $X \subset \Theta \times K$ with K compact and compact fibres, continuous semialgebraic maps F, A on X , and a semialgebraic choice $P_\theta \in F_\theta(X_\theta)$. Degrees and numbers of coordinates are fixed. Effectivity refers to algebraic presentations and numerical outputs to algebraic parameter values. No complexity bound is asserted.

Adjoin nonnegative square-root coordinates to the observation graph and write

$$R_{je} = \sqrt{w_j}(\sqrt{F(x)_{je}} - \sqrt{P_{je}}), \quad G = \sum_{j,e} R_{je}^2, \quad H_{r,a} = (\Re c_{r,a})^2 + (\Im c_{r,a})^2.$$

They are polynomial functions on the resulting compact real graph Γ . Zero probability cells are permitted. No passage to a Fisher quadratic form is used in this construction.

Definition 9.2 (Real monomial presentation). A real monomial presentation is a finite collection of smooth real algebraic sources and proper maps, collectively covering a neighbourhood of $G = 0$ in Γ , with finite algebraic etale coordinate presentations. On each source component either $G \equiv 0$ and all $H \equiv 0$, or

$$G = u \prod_i |y_i|^{a_i}, \quad H_{r,a} \equiv 0 \quad \text{or} \quad H_{r,a} = v_{r,a} \prod_i |y_i|^{b_{r,a,i}}. \quad (9.2)$$

The units have positive upper and lower bounds on smaller closed real coordinate boxes above $G = 0$. A contributing divisor has $a_i > 0$ and a real point away from the other divisors, with an admissible transversal on which $G > 0$. An identically zero H has order $+\infty$; a component $G \equiv 0$ is recorded separately and contributes no ratio. Properness belongs to the source maps, not to an open coordinate box. The finite encoding of an adapted coordinate presentation is specified in Appendix A.

Lemma 9.3 (Spreading a finite resolution certificate). *Let $S = \text{Spec } A$ be an integral affine variety over a characteristic-zero field, let $K = \text{Frac}(A)$, and let $Y \rightarrow S$ be of finite type. Suppose that over K there is a finite reduced closed chain*

$$Y_{0,K} = Y_{K,\text{red}} \supset Y_{1,K} \supset \cdots \supset Y_{q,K} = \emptyset,$$

projective maps $Z_{j,K} \rightarrow Y_{j,K}$ with smooth sources, specified inverse isomorphisms over $Y_{j,K} \setminus Y_{j+1,K}$, and finite smooth-centre principalizations of specified functions on those sources. There is a

nonempty open S° on which the following hold on every fibre. The closed chain is flat with geometrically reduced fibres and ends in the empty set; the inverse maps remain inverse over its actual successive complements. Sources and centres are relatively smooth, principalization blow-ups commute with fibre formation, and the final sources have relative normal-crossing divisors with fixed integer monomial orders on a finite adapted etale atlas. Zero identities and inverse units persist. A nonzero monomial cannot become identically zero on a component of a retained fibre. For algebraic input, such finite data and an exceptional set are effectively constructible.

Proof. Lemma A.2 descends the complete diagram, including the actual open complements and both inverse identities, and imposes flatness and geometric reducedness. Lemma A.1 proves the needed base change for all powers of every centre ideal after one localization per finite collection of charts. The explicit bad-locus construction in Appendix A treats relative smoothness, normal-crossing intersections, chart coverage, units and zero flags separately, and proves that each discarded algebraic image has proper closure. The nonzero-specialization assertion follows from relative codimension one and the inverse units, not from generic nonvanishing alone. The constructive certificate specification in Appendix B proves the last assertion. $\square$

Theorem 9.4 (Relative real principalization). *On the exact-target identifying locus of a fixed-format family there is a finite semialgebraic stratification with fixed cluster multiplicities, a fixed finite collection of algebraic source chart presentations, fixed integer orders and fixed real-accessibility flags as in Definition 9.2. The source maps are fibrewise proper real covers. Charts, units and box sizes may vary with the parameter. On compact subsets of a stratum the cover is usable over a common sufficiently small observation neighbourhood. The construction is effective for the algebraic presentations of Definition 9.1.*

Proof. Lemma A.3 algebraizes the proper enlarged graph by square-slack lifts without discarding strict exclusions or real branches. Resolve the reduced generic fibre and its successive reduced singular loci, then principalize the nonzero functions on the smooth sources. Lemma 9.3 spreads this finite certificate. The explicit exceptional algebraic loci in Appendix A are reconstructed on reduced bases of strictly smaller dimension, giving finite termination. Lemma A.4 proves both inclusions in the exact real coverage assertion, including lower-dimensional real loci, and descends the selected label branches. Lemma A.5 supplies the real transversals and the finite real flag partition. That partition is not confused with the algebraic dimension-drop recursion. The final proof in Appendix A supplies the uniform neighbourhood and unit bounds by properness over compact parameter subsets. These arguments prove the asserted cover on the original graph, rather than on an algebraic relaxation of its inequalities. $\square$

Proposition 9.5 (Fixed-centre real cover). *Every graph Γ above has a real monomial presentation. Exact pieces and lower-dimensional real loci are included. Every recorded divisor has a real admissible attaining arc.*

Proof. Apply Theorem 9.4 to a one-point base and use Lemma A.5. $\square$

Corollary 9.6 (Intrinsic scalar orders). *For each nonrigid cluster coefficient put*

$$\beta_{r,a} = \min_{\text{recorded real divisors}} \frac{b_{r,a,i}}{a_i}, \quad \alpha = \min_{r,a} \frac{\beta_{r,a}}{k_{r,a}}. \quad (9.3)$$

Either the target is constant on an observation neighbourhood, or $\alpha > 0$ is rational and $\omega_P(t) \asymp t^\alpha$. Every finite coefficient minimum is attained by an admissible arc. The numbers are independent of the monomial presentation.

Proof. The complete real divisor list and the monomial inequalities in Lemma A.5 give $H \leq CG^\beta$, hence $|c_{r,a}| \leq Ch_w^\beta$. A minimizing transversal satisfies $G \asymp u^{a_i}$ and $H \asymp u^{b_i}$ and attains

the exponent. The inclusion $\{G = 0\} \subseteq \{H = 0\}$ forces positive contributing orders. If all coefficients vanish on an observation neighbourhood, the target is locally rigid. Otherwise, for a monic polynomial of degree m , its greatest root modulus is comparable to $\max_{a < m} |c_a|^{1/(m-a)}$. Elementary symmetric identities give one bound; dominance of z^m outside a fixed multiple of this maximum gives the other. The centre lies in every ball, and the diameter is between its greatest centre displacement and twice that displacement. This proves the order of ω_P . The optimal coefficient inequality is intrinsic, so its attained exponent is independent of the presentation. The scalar divisor-ratio calculation is classical; compare [6]. $\square$

10. JOINT SPECIALIZATION AND UNIFORM SPECTRAL ASYMPTOTICS

On a nonrigid identified stratum write $\alpha = p/q$ in lowest terms. Let $\Psi(\theta, \varepsilon, x, U, V)$ denote graph membership and

$$R_{je}(\theta, x) = \varepsilon^q U_{je}, \quad c_{r,a}(\theta, x) = \varepsilon^{pk_{r,a}} V_{r,a}, \quad \sum_{j,e} U_{je}^2 \leq 1. \quad (10.1)$$

Complex equalities denote two real equalities. Define the joint family by the fixed-centre formula

$$(\theta, U, V) \in \mathcal{J} \iff \theta \in S \text{ and } \forall \rho > 0 \exists \varepsilon, x, U', V' : \quad (10.2)$$

$$0 < \varepsilon < \rho, \quad \Psi(\theta, \varepsilon, x, U', V'), \quad \|(U', V') - (U, V)\|^2 < \rho^2.$$

Put $\mathcal{J}_0(\theta) = \mathcal{J}_\theta$ and $\mathcal{C}_0(\theta) = \text{pr}_V \mathcal{J}_0(\theta)$. There is no quantified θ' in this formula. Closure with a drifting centre is a different operation, and taking independent closures in the coefficient coordinates is also a different operation.

Theorem 10.1 (The definable joint initial family). *The sets $\mathcal{J}_0(\theta)$ and $\mathcal{C}_0(\theta)$ form semialgebraic families of nonempty compact sets. The coefficient images of the radius- t observation ball, scaled by $c_{r,a}/t^{\alpha k_{r,a}}$, converge in Hausdorff distance to $\mathcal{C}_0(\theta)$ for every fixed θ . If*

$$Z_r(V) = Z \left(z^{m_r} + \sum_{a < m_r} V_{r,a} z^a \right), \quad \mathcal{L}_0(\theta) = \{(Z_r(V))_r : V \in \mathcal{C}_0(\theta)\},$$

then the scaled cluster-root images converge to $\mathcal{L}_0(\theta)$, and

$$\omega_\theta(t) = C(\theta)t^\alpha + o(t^\alpha), \quad C(\theta) = \max_{V, V' \in \mathcal{C}_0(\theta)} \max_r d_\infty(Z_r(V), Z_r(V')) \in (0, \infty). \quad (10.3)$$

This is the diameter of a joint feasible image, not that of a Cartesian product of scalar envelopes.

Proof. Corollary 9.6 bounds every scaled coefficient for small t ; U is already bounded. The fixed-centre closure is nonempty and compact. Quantifier elimination proves semialgebraicity with θ as a free parameter. Compactness of the graph fibre lets one pass to convergent subsequences of the unscaled graph variables when projecting an initial point, so projection loses no limiting coefficient point.

Fix θ and a point of the outer coefficient limit. Its distance to the positive-radius coefficient section is a bounded nonnegative semialgebraic function of t , with a subsequence tending to zero. The one-variable monotonicity theorem makes its limit zero. A finite net of the compact outer limit gives the inner Hausdorff inclusion. Failure of the outer inclusion would give a bounded convergent subsequence of scaled coefficients outside the defining limit, a contradiction. This proves convergence of the entire ball. Continuity of monic root multisets on bounded coefficient sets gives the corresponding root-set convergence.

Let Δ be the least separation of distinct base roots. For small t , each cluster lies in its $\Delta/4$ disc. A matching within clusters has error tending to zero; a cross-cluster matching has an edge of length at least $\Delta/2$. An optimal matching therefore stays within clusters. Translation and scaling there are exact. Taking the diameter gives (10.3). An attaining coefficient transversal

prevents the limiting diameter from being zero. This also explains why all inequality faces and joint relations in the closure remain relevant. $\square$

Proposition 10.2 (Semialgebraic decay on strata). *Let $e(\theta, t) \geq 0$ be semialgebraic for $0 < t < \tau(\theta)$, with $\tau > 0$ semialgebraic, and suppose $e(\theta, t) \rightarrow 0$ for each fixed θ . After a finite Nash stratification, on every stratum a rational $\nu > 0$ and a positive continuous semialgebraic $\tau_0(\theta)$ satisfy*

$$0 < t < \tau_0(\theta) \implies e(\theta, t) \leq t^\nu.$$

The estimate is uniform on compact subsets; ν need not be sharp.

Proof. Lemma B.1 gives a finite positive rational candidate list from the polynomial supports of a quantifier-free graph formula, including specialized zero coefficients and the eventual-zero case. Choose ν below its positive candidates. The complete argument in Appendix B selects a genuinely positive semialgebraic threshold, makes it continuous, and takes its minimum only on a compact subset of a stratum. No uniform rate is inferred from pointwise convergence alone. $\square$

Theorem 10.3 (Uniform spectral power law). *The identified nonrigid locus has a finite Nash stratification on which α is fixed, $\mathcal{L}_0$ has constant semialgebraic topological type, and $C(\theta)$ is positive and Nash. For every compact subset H of a stratum, for some $\nu > 0$,*

$$d_H(\mathcal{L}_t(\theta), \mathcal{L}_0(\theta)) = O_H(t^\nu), \quad \omega_\theta(t) = C(\theta)t^\alpha + O_H(t^{\alpha+\nu}). \quad (10.4)$$

Here $\mathcal{L}_t$ is the exact clustered root image scaled by $t^{-\alpha}$ and d_H uses the maximum cluster bottleneck metric.

Proof. Represent each root multiset by all its labelled arrays through elementary symmetric identities. The formula

$$M_d(z, z') = \bigvee_{\sigma \in \mathfrak{S}_m} \bigwedge_{i=1}^m |z_i - z'_{\sigma(i)}|^2 \leq d^2 \quad (10.5)$$

expresses bottleneck distance at most d . Use the conjunction over clusters, and real and imaginary parts for complex roots. The two inclusions

$$(\forall z \in \mathcal{L}_t)(\exists z' \in \mathcal{L}_0)M_d(z, z'), \quad (\forall z' \in \mathcal{L}_0)(\exists z \in \mathcal{L}_t)M_d(z, z')$$

and minimality over $d \geq 0$ define the Hausdorff error. Compactness gives attainment. Rational powers of t are encoded by positive algebraic roots, so this error is semialgebraic. Theorem 10.1 proves its pointwise limit zero; Proposition 10.2 gives the compact-uniform power bound.

The diameter has a first-order formula saying that all pairs have distance at most C and one pair attains it. It is therefore semialgebraic and positive. Lemma B.2 gives its Nash refinement. Refine also by positive cluster separation and observation-radius choices, so that compact minima exclude cross-cluster matching uniformly. Hardt triviality supplies the topological types separately. Finally diameters differ by at most twice Hausdorff distance, which gives the second estimate. Topological triviality has supplied neither the error rate nor the Nash dependence. $\square$

Theorem 10.4 (A finite real spectral atlas). *A fixed-format family admits a finite semialgebraic stratification retaining the following data: the semialgebraic type and dimension of the exact target coefficient fibre and its root image; on the identified locus, rigidity or a positive rational exponent, cluster multiplicities, and a fixed real monomial order list with accessibility and minimizing indices; on its nonrigid part, the joint initial coefficient and root-set types, a Nash leading diameter and the compact-uniform power remainder; and finite algebraic descriptions of these objects. On a subfamily with a proved Fisher-cone representation, the active supports and choices of leading constants can be retained in the same refinement. On a nonidentified stratum the modulus tends to the positive diameter of the exact target root image. No general leading set is asserted to be quadratic, and no local leading constant excludes a remote exact component.*

Proof. The exact coefficient fibre is $\{A_x : F_\theta(x) = P_\theta\}$, not the parameter fibre itself. Apply Hardt triviality to its compact image and root graph. Singleton images are selected by a first-order formula. At fixed θ , compactness shows that decreasing observation balls converge in target image to the exact image, so their diameters converge to its diameter. On the singleton locus apply Theorem 9.4, Corollary 9.6, and Theorems 10.1 and 10.3. These are separate constructions entering one finite refinement. A proved Fisher subfamily contributes only finitely many additional support and comparison conditions. Effective descriptions are provided by the following theorem and Appendix B. $\square$

11. EFFECTIVE OUTPUT AND THE ROLE OF COMPUTATION

Theorem 11.1 (Effective algebraic specialization). *For an algebraic input on the identified locus, finite algorithms compute local rigidity, the nonrigid exponent α , quantifier-free presentations of $\mathcal{S}_0, \mathcal{C}_0, \mathcal{L}_0$, and an integer polynomial with a rational isolating interval for C . Exact identification itself is decidable. In algebraically presented parameter families the corresponding strata and formulas are effectively describable. These assertions are independent of any finite example regression.*

Proof. Appendix B gives the complete input–output procedure. It constructs each maximum $m_H(s) = \max\{H : G \leq s\}$ by elimination, extracts a finite support-pair list, and tests the comparability formula (B.1) for every candidate. The unique true candidate selects the actual positive maximum-function branch, while a separate formula detects eventual zero. No unspecified branch-isolation oracle is used. Substitute the resulting p/q in (10.2), eliminate, adjoin roots and use (10.5) to compute the leading diameter. Real-root isolation and field-norm conversion provide its integer polynomial and isolating interval. With free parameters the same finite tests compute strata, followed by Nash refinement. A second, explicit finite-certificate search for the geometric presentations is given in that appendix; spectral outputs do not depend on executing it. $\square$

The symbolic calculations accompanying the explicit polynomial examples test identities and specified finite optimization problems. They do not implement the universal algorithms above and are not proof certificates for relative principalization, uniform asymptotics, or arbitrary degree.

12. THE CONSTRAINED WEIGHTED DEFORMATION

We identify the object of Section 10 before discussing its metric consequences. The datum, the weight filtration and the observation norm are part of the object. A change of parametrization is not a change of any of these three data.

Fix an identified nonrigid datum. Let S be the germ at the origin of the *image* of the actual observation–coefficient relation

$$x \mapsto (R(x), c(x)).$$

All real and imaginary parts are included. The defining inequalities and selected real branches are retained in this image. Set $w = (q, \dots, q, (pk_{r,a})_{r,a})$ when $\alpha = p/q$, and write $D_\varepsilon y = (\varepsilon^{w_i} y_i)_i$. Define

$$\mathfrak{D}_w^+(S) = \overline{\{(\varepsilon, D_\varepsilon^{-1}y) : \varepsilon > 0, y \in S\}}, \quad T_w^+ S = \{v : (0, v) \in \mathfrak{D}_w^+(S)\}. \quad (12.1)$$

Closure is ordinary real closure, locally in the displayed coordinates. Replacing a representative of the germ by a smaller representative does not change its zero fibre. The superscript emphasizes positive real specialization, not a new algebraic deformation operation.

Theorem 12.1 (Comparison and real liftability). *The fixed-centre joint initial fibre satisfies*

$$\mathcal{S}_0 = T_w^+ S \cap \{(U, V) : \|U\| \leq 1\}, \quad \mathcal{C}_0 = \text{pr}_V \mathcal{S}_0. \quad (12.2)$$

Thus it is the observation-unit section of the positive-real weighted tangent relation. It depends on the relation germ, not on a resolution, a square-slack presentation or a proper surjective presentation of that relation.

Suppose an algebraic presentation of the relation has ideal I over $\mathbb{R}$, with its actual real branch conditions specified separately. Form the algebraic weighted deformation by substitution $y_i = \varepsilon^{w_i} v_i$ and saturation by ε . Its zero-fibre ideal is the weight-initial ideal of I , with the minimum-weight convention. The set $T_w^+ S$ consists exactly of those real points of this algebraic zero fibre which have a positive-real semialgebraic lift satisfying the original branch conditions. Merely taking all real points of the algebraic initial fibre can give a strictly larger set, even when I is real prime.

Proof. The forward inclusion in (12.2) follows from the defining substitutions (10.1). For the reverse inclusion take a sequence $(\varepsilon_n, U_n, V_n)$ in the positive deformation with limit $(0, U, V)$ and $\|U\| \leq 1$. If $\|U\| < 1$, it eventually satisfies the unit constraint. At a boundary point put

$$a_n = \max(1, \|U_n\|^{1/q}), \quad \varepsilon'_n = a_n \varepsilon_n.$$

Then $a_n \rightarrow 1$, and the same unscaled relation point has coordinates $U'_n = a_n^{-q} U_n$, $V'_{r,a,n} = a_n^{-pk_{r,a}} V_{r,a,n}$ at ε'_n . These satisfy the unit constraint and have the same limit. This proves the boundary inclusion, which need not follow by interchanging an arbitrary intersection and closure. The coefficient bounds in Corollary 9.6 make this section compact. Image germs are unchanged by the stated surjective presentations, proving the presentation assertion.

Algebraically, contract the ideal generated by $f(D_\varepsilon v)$, $f \in I$, from $\mathbb{R}[\varepsilon, \varepsilon^{-1}, v]$ to $\mathbb{R}[\varepsilon, v]$. This is precisely saturation by ε . For each f , divide $f(D_\varepsilon v)$ by its lowest power of ε ; its value at zero is the minimum-weight part of f . Taking all $f \in I$, or a weight-adapted Groebner basis, shows that the special-fibre ideal is $\text{in}_w I$. This is the usual associated-graded construction, not an additional resolution theorem.

Every real limiting point satisfies the saturated equations. Conversely, a real point is in the constrained zero fibre precisely when it is a limit from the specified positive real part. Semialgebraic curve selection supplies an arc in that part; its positive first coordinate can be used as parameter after a finite Puiseux reparametrization. This proves the liftability characterization. For strictness, take $S = \{(x, y) : y^2 = x^3\}$ at the origin with weights $(1, 1)$. The saturated equation is $v_y^2 = \varepsilon v_x^3$. Its algebraic zero fibre has real points $v_y = 0$, with either sign of v_x . Positive real liftability instead requires $v_x \geq 0$, and every such point is realized. The ideal $(y^2 - x^3)$ is real prime: its coordinate ring embeds in $\mathbb{R}[t]$ by $(x, y) = (t^2, t^3)$. Thus even real primality does not recover the missing accessibility condition from the initial ideal alone. $\square$

For weight one the algebraic construction is the tangent-cone instance of the normal-cone construction; see [12, 18]. Weighted initial ideals are treated in [15]. These algebraic objects do not, by themselves, specify the positive real liftable part or the observation-unit section. Inner-metric tangent limits, as in [2], retain different data: here one rescales a constrained *outer* relation and subsequently applies the specified root map. No identification with an inner-metric tangent cone is asserted.

Theorem 12.2 (Filtered naturality). *Let ϕ be a semialgebraic homeomorphism between two relation germs, fixing the origin. Suppose, locally uniformly on bounded sets,*

$$\phi_\varepsilon = D_\varepsilon^{-1} \phi D_\varepsilon \longrightarrow \phi_0, \quad (\phi^{-1})_\varepsilon \longrightarrow \phi_0^{-1},$$

where ϕ_0 is a homeomorphism. Then $T_w^+ \phi(S) = \phi_0(T_w^+ S)$. If the observation block of ϕ_0 preserves its norm, the corresponding unit sections are also carried onto each other. The same statement holds with the norm transported by ϕ_0 . With the target decoding transported at the same time, the leading spectral image and its diameter are unchanged.

Proof. For a bounded convergent sequence of rescaled relation points v_n , uniform convergence on a compact neighbourhood gives $\phi_{\varepsilon_n}(v_n) \rightarrow \phi_0(\lim v_n)$. This proves one inclusion of tangent relations. Apply the inverse hypothesis for the other. Theorem 12.1 then identifies their unit sections; the last assertion is equality of the decoded images. For example, weighted analytic coordinate changes whose invertible leading graded map is ϕ_0 and whose remaining terms have strictly larger weight satisfy the stated convergence, as do their filtered inverses. $\square$

The transport of the metric is essential. Equivalent observation norms preserve optimal orders, but need not preserve a leading diameter. Section 15 gives this distinction inside the full binary experiment, without adding a discrete experimental label.

12.1. Exactly which scalar probes recover the joint set. Write C_t for the coefficient section scaled as in Theorem 10.1. For $a \in \mathbb{R}^N$ let

$$m_t(a) = \min_{v \in C_t} \|v - a\|^2. \quad (12.3)$$

The dimension N includes both parts of complex coefficients.

Theorem 12.3 (Quadratic reconstruction and finite witnesses). *The limits $m_0(a) = \lim_{t \downarrow 0} m_t(a)$ exist locally uniformly in a and satisfy*

$$m_0(a) = \text{dist}(a, \mathcal{C}_0)^2, \quad \mathcal{C}_0 = \bigcap_{a \in \mathbb{Q}^N} \{v : \|v - a\|^2 \geq m_0(a)\}. \quad (12.4)$$

In particular these scalar probes, unlike their vanishing orders alone, determine the joint coefficient set and hence its decoded spectral set. For compact nonempty sets C, D in a fixed compact ball B , and a finite η -net $A \subset B$ (with an arbitrarily slightly enlarged ball if rational points are required),

$$\max_{a \in A} |\text{dist}(a, C) - \text{dist}(a, D)| \leq d_H(C, D) \leq \max_{a \in A} |\text{dist}(a, C) - \text{dist}(a, D)| + 2\eta. \quad (12.5)$$

For algebraic input one quantifier-free formula with a free describes $m_0(a)$. Distinct initial fibres therefore admit an effectively searchable rational probe with unequal limiting values.

Proof. For compact sets the distance functions differ uniformly by at most their Hausdorff distance. Squaring is uniformly Lipschitz on bounded distance intervals. Theorem 10.1 therefore proves the first assertion. A point of $\mathcal{C}_0$ satisfies every inequality in (12.4). If $v \notin \mathcal{C}_0$, put $d = \text{dist}(v, \mathcal{C}_0) > 0$ and choose rational a with $\|a - v\| < d/3$. Then $\text{dist}(a, \mathcal{C}_0) > 2d/3 > \|a - v\|$, excluding v .

The left inequality in (12.5) is the distance-function bound. The Hausdorff distance is attained as $\text{dist}(x, D)$ for some $x \in C$, or with C, D exchanged. At such an x it equals $|\text{dist}(x, C) - \text{dist}(x, D)|$. Each distance function is one-Lipschitz; replacing x by a net point proves the other inequality.

Finally, membership in C_t , minimization in (12.3), and the limit are first-order formulas over a real closed field, with a free. Theorem 11.1 and quantifier elimination yield one finite formula for their limit graph. If $C \neq D$, the strict separating inequality above persists on an open set of probe centres. Enumeration of rational points, with exact comparison of the resulting algebraic values, must find a witness. The enumeration concerns a witness only, not an approximation to the whole set by scalar coordinate envelopes. $\square$

All affine support values determine only a closed convex hull. Quadratic distance probes determine the actual compact set, including its nonconvex real faces. The reconstruction theorem thus gives a precise sufficiency statement to accompany, rather than overstate, the insufficiency statements of Sections 14 and 15.

13. REAL RESIDUES AND FILTERED CHANGES OF OBSERVATION

We compare the weighted relation with signed valuation geometry before imposing its observation metric. Throughout, positive-real refers to positive values of the degeneration parameter. It does not restrict the coordinates of a real leading coefficient to the positive orthant.

Let $\mathbb{K} = \mathbb{R}\langle t \rangle$ be the real closed field of real algebraic Puiseux series, ordered by positive leading coefficient, with t positive infinitesimal. Write val for its rational valuation, $\mathcal{O}$ for the valuation ring, and $\text{res} : \mathcal{O} \rightarrow \mathbb{R}$ for its residue map. We use the convention $\text{val}(0) = +\infty$. A semialgebraic germ and its defining inequalities are interpreted over $\mathbb{K}$ by the same real-closed-field formulas.

Let $S \subset \mathbb{R}^n$ be the actual semialgebraic observation-coefficient relation germ at zero and let $w \in \mathbb{Q}_{>0}^n$. Put

$$\mathcal{B}_w(S) = \{y \in S(\mathbb{K}) : \text{val}(y_i) \geq w_i \text{ for all } i\}.$$

Only the germ matters, since every point of this set is infinitesimal. The coefficientwise residue below is taken after multiplication by t^{-w_i} .

Theorem 13.1 (Weighted real residue comparison). *The positive-real weighted tangent relation satisfies*

$$T_w^+ S = \text{res}(t^{-w} \mathcal{B}_w(S)). \quad (13.1)$$

For every sign vector $\sigma \in \{-, +\}^n$, its part with all coordinates nonzero has the exact comparison

$$\begin{aligned} \exists v \in T_w^+ S \cap (\mathbb{R}^*)^n, \quad \text{sgn } v = \sigma \\ \iff \exists y \in S(\mathbb{K}), \quad \text{val } y = w, \quad \text{sgn } y = \sigma. \end{aligned} \quad (13.2)$$

Thus forgetting the nonzero residue magnitudes gives the fixed-weight fibre of the coordinate signed valuation map. Imposing $S \subset \mathbb{R}_{>0}^n$ gives its positive-orthant specialization. Zero coordinates on the left of (13.1) also retain lifts of higher valuation; they must not be discarded by requiring equality of every coordinate valuation.

For the observation and coefficient blocks of Theorem 12.1, the manuscript's joint object is consequently the following operation:

$$\boxed{\mathcal{B}_w(S) \xrightarrow{\text{res}(t^{-w} \cdot)} T_w^+ S \xrightarrow{\|U\| \leq 1} \mathcal{I}_0 \xrightarrow{\text{target projection and roots}} \mathcal{L}_\theta.} \quad (13.3)$$

The observation-unit section and spectral decoder in this formula are additional metric data, not part of the coordinate signed valuation.

Proof. A point on the right of (13.1) is represented by algebraic Puiseux series. After a common positive ramification these series converge on a small positive interval. Transfer of the finite semialgebraic defining formula, or equivalently the eventual signs of its convergent algebraic functions, makes the resulting real points belong to S . Their rescaled limit is their displayed residue, proving one inclusion.

Conversely let $(0, v)$ lie in the closure of $\{(\epsilon, \epsilon^{-w} y) : \epsilon > 0, y \in S\}$. This is a semialgebraic set after clearing the weight denominators. Real curve selection gives a semialgebraic arc with positive first coordinate approaching $(0, v)$. Its first coordinate is nonconstant and eventually strictly monotone. Reparametrize by that coordinate. Each component is an algebraic Puiseux germ in ϵ , so it gives a point of $\mathbb{K}$ after renaming ϵ as t . Boundedness of the rescaled arc says $\text{val } y_i \geq w_i$, and its limit is $\text{res}(t^{-w} y)$. This proves equality. A rescaled coordinate has nonzero residue precisely when its valuation is exactly its prescribed weight; its sign is then the residue sign. This proves (13.2). The observation-unit identity in (13.3) is Theorem 12.1, including its boundary rescaling argument at $\|U\| = 1$; it is not an unproved interchange of closure and intersection with a closed ball. $\square$

Jell–Scheiderer–Yu [19] develop real tropicalization for semialgebraic sets over an ordered nonarchimedean real closed field and identify real analytification with an inverse limit of such real tropicalizations. Under the normalization $|t| = e^{-1}$, the right side of (13.2) is the fibre of their *coordinate* signed valuation map at $(\sigma_i e^{-w_i})_i$. Formula (13.1) retains the full residues before that projection. We make no claim that the full inverse-limit real analytification loses the same information as one coordinate tropicalization. The fixed observation norm and root decoder are specified here independently of that inverse-limit construction.

Positive tropical geometry asks for positive lifts, rather than all real points of a complex initial variety. The initial-ideal and lifting criteria studied by Rose–Telek [20] are directly relevant in that setting. In the present setting positivity applies to the scale; coordinate orthants and all feasibility inequalities must be kept separately. The following familiar nonsingular criterion records exactly when an initial-point calculation alone suffices locally.

Proposition 13.2 (A sufficient real lifting test). *After clearing weight denominators, suppose a local presentation of the weighted deformation is*

$$F(t, z) = 0, \quad g_1(t, z) > 0, \dots, g_s(t, z) > 0,$$

with Nash functions, and suppose these are the complete local equations and inequalities of the actual deformed relation. If $F(0, v) = 0$, $D_z F(0, v)$ has full row rank, and $g_i(0, v) > 0$ for every i , then $v \in T_w^+ S$. This is a sufficient test, not an equivalence at a singular initial point or at an active inequality.

Proof. Choose as many z coordinates as there are equations so that the corresponding Jacobian minor is invertible, and keep the remaining coordinates fixed at their values in v . The Nash implicit-function theorem gives $z(t)$ with $z(0) = v$ and $F(t, z(t)) = 0$. The strict inequalities persist for small positive t . Undoing the weighted substitution gives a real lift in the actual relation. $\square$

13.1. Naturality from finite weighted jets. We next derive the convergence needed for naturality from a condition on the presentation, instead of assuming that convergence. For $w \in \mathbb{Q}_{>0}^n$ and $w' \in \mathbb{Q}_{>0}^n$, call a Nash isomorphism germ $h : (\mathbb{R}^n, 0) \rightarrow (\mathbb{R}^n, 0)$ *bi-filtered* if every nonzero Taylor monomial y^α of h_j has $w \cdot \alpha \geq w'_j$, and the inverse satisfies the corresponding condition with w, w' reversed. Only finitely many monomials have weight below any w'_j ; the vanishing conditions are therefore conditions on finite jets. Let $h_{[w, w']}$ consist of the terms with equality of weights.

Theorem 13.3 (Finite-jet filtered naturality). *A bi-filtered Nash isomorphism satisfies, locally uniformly on bounded sets,*

$$D_{t, w'}^{-1} h D_{t, w} = h_{[w, w']} + O(t^\eta), \quad \eta > 0, \quad (13.4)$$

where $D_{t, w} = \text{diag}(t^{w_i})$. The analogous statement holds for h^{-1} , and its weighted initial map is the inverse polynomial map of $h_{[w, w']}$. Consequently

$$T_{w'}^+(h(S)) = h_{[w, w]}(T_w^+ S).$$

Weighted initial maps preserve composition. They commute with the residue comparison (13.1). A positive ramification $t = s^q$ replaces w by qw and leaves the residue set unchanged. When the observation-unit section and spectral decoder are transported, so is the resulting metric fibre. Equality of untransported bottleneck diameters requires that the transported target map be an isometry.

Proof. Choose a common denominator for all the weights. In a finite Taylor polynomial, every term of weight strictly larger than its output weight has a gap bounded below by a positive

reciprocal integer. Such terms are $O(t^\eta)$ on bounded rescaled sets. Choose the Taylor order N so large that $(N+1) \min_i w_i > \max_j w'_j + \eta$. The analytic Taylor remainder is $O(\|D_{t,w} z\|^{N+1})$ uniformly for bounded z , and hence has the same bound after division by the output weights. This proves (13.4). The same argument applies to the inverse. Substitution in the identities $h^{-1}h = hh^{-1} = \text{id}$ shows that their limiting polynomial maps are mutually inverse; boundedness needed for substitution follows from the uniform estimates just proved. The same argument proves preservation of composition.

Apply these limits to any rescaled arc in S and then apply the inverse to obtain both inclusions of the tangent-set identity. Alternatively, valuation compatibility of the Taylor series and reduction modulo the positive-valuation ideal give

$$\text{res}(t^{-w'} h(y)) = h_{[w,w']}(\text{res}(t^{-w} y)),$$

which proves commutation with (13.1). Ramification follows by the substitution $t = s^q$. Finally a transported section and decoder have the same feasible decoded points by their definitions; the metric is preserved exactly when its transport is respected. $\square$

For example, at weights $(1, 2)$ the map $h(y_1, y_2) = (y_1, y_2 + y_1^2)$ is bi-filtered and its initial map is nonlinear. The theorem thus includes weighted changes beyond ordinary linear similarities, while stating precisely which metric data must be transported. Neither the residue comparison nor the implicit-function lifting test is presented as a new tropical fundamental theorem. The observation-specific finite determination is Theorem 4.1; the comparison specifies the real object to which that theorem applies.

14. SCALAR ENVELOPES DO NOT DETERMINE THE SPECTRAL INVARIANT

The distinction between the scalar orders and the joint initial fibre can be made exact. The following examples have not only the same optimal orders but the same individual coefficient envelopes at every sufficiently small radius. Their leading spectral diameters are nevertheless different. The assertion concerns the scalar data of Corollary 9.6 and separate coefficient envelopes; it does not concern the collection of value functions of all possible combinations of the coefficients.

Theorem 14.1 (Separation from marginal scalar data). *There are two compact semialgebraic probability experiments with a unique exact target, real quadratic targets with roots in $[0, 1]$, and identical coefficient orders and individual coefficient envelopes, for which*

$$\omega_A(t) = (1 + \sqrt{5})t, \quad \omega_B(t) = \frac{1 + \sqrt{17}}{2}t \quad (0 \leq t \leq 1/20).$$

In both experiments the coefficient orders are $(\beta_1, \beta_2) = (1, 2)$ and the spectral exponent is one. On each source arc the orders of (G, H_1, H_2) are $(2, 2, 4)$. Thus the order lists and the separate optimal coefficient envelopes do not determine the joint initial root set or its diameter.

Proof. Put $r = 1/2$ and $e = 1/20$. In each experiment the source is the disjoint union of two closed intervals $0 \leq \rho \leq e$. Its probability map and datum are

$$F(\rho) = ((1 - \rho^2/2)^2, \rho^2(1 - \rho^2/4)), \quad P_0 = (1, 0).$$

The entries are nonnegative and sum to one. Since $\sqrt{F_1} = 1 - \rho^2/2$ on this interval,

$$h(F(\rho), P_0)^2 = 2 - 2\sqrt{F_1} = \rho^2.$$

Associate with an interval labelled (u, v) the target

$$A_{u,v,\rho}(z) = (z - r)^2 + u\rho(z - r) + v\rho^2.$$

Use the label sets

$$\mathcal{E}_A = \{(1, -1), (2, -4)\}, \quad \mathcal{E}_B = \{(1, -4), (2, -1)\}.$$

These are compact semialgebraic experiments in the sense of Definition 9.1; the discrete label is simply an additional source coordinate. All exact targets are $(z - r)^2$. The two root increments on an interval are

$$\rho \ell(u, v), \quad \rho q(u, v), \quad \ell(u, v) = \frac{-u - \sqrt{u^2 - 4v}}{2} < 0 < q(u, v) = \frac{-u + \sqrt{u^2 - 4v}}{2}. \quad (14.1)$$

They lie in $[0, 1]$ after translation by r , since the greatest absolute increment is at most $(1 + \sqrt{5})/20 < 1/2$.

Write $c_1 = u\rho$ and $c_0 = v\rho^2$. In either experiment the exact marginal images over the radius- t ball are

$$\{c_1\} = [0, 2t], \quad \{c_0\} = [-4t^2, 0].$$

In particular $\max |c_1| = 2t$ and $\max |c_0| = 4t^2$, not just to leading order. Moreover $G = \rho^2$, $H_1 = u^2\rho^2$, and $H_2 = v^2\rho^4$. The order assertions follow. Even the unordered lists of individual squared units agree: they are $\{1, 4\}$ and $\{1, 16\}$. Their association across the two coordinates is different.

For $t > 0$, put $\lambda = \rho/t$. The scaled coefficient image is already independent of t :

$$\mathcal{C}_{0,E} = \bigcup_{(u,v) \in \mathcal{E}_E} \{(u\lambda, v\lambda^2) : 0 \leq \lambda \leq 1\}, \quad E \in \{A, B\}. \quad (14.2)$$

This is its joint initial coefficient set. The ordered root image consists of $(\lambda \ell(u, v), \lambda q(u, v))$. Its first coordinate is nonpositive, its second nonnegative, and the zero vector belongs to it. The diameter in the maximum metric is the larger of its two coordinate widths. Since $u > 0$, $-\ell(u, v) > q(u, v)$ on each interval. Therefore

$$C_E = \max_{(u,v) \in \mathcal{E}_E} \frac{u + \sqrt{u^2 - 4v}}{2}.$$

For A the maximum is $1 + \sqrt{5}$. For B it is $(1 + \sqrt{17})/2$, which exceeds $1 + \sqrt{2}$. The inequalities are strict and these two constants differ. Translation and multiplication by t in (14.1) prove the asserted exact diameter identities. No product approximation of (14.2) has been used. $\square$

The example isolates the information carried by the joint specialization: relations between weighted coefficients survive both real closure and the root map. A finite list of scalar exponents cannot recover these relations, even when augmented by the separate sharp envelopes. Conversely the full joint set determines the diameter by Theorem 10.1; no new scalar resolution theorem is required for this distinction.

15. IDENTICAL SCALAR ORDERS IN THE BINARY EXPERIMENT

The example in Section 14 keeps sharp marginal ranges fixed. We now keep a different, much larger valiative object fixed inside the full experiment of Section 2: all real analytic arc orders of the observation discrepancy and every polynomial scalar function of the target coefficients. Leading units are not included in this datum. The distinction is necessary, since Theorem 12.3 shows how richer scalar values recover the joint set.

Take degree one, $D = 1$, $\pi = 1/10$, clocks $(2, 3, 4)$ and weights $1/3$. Consider the two strictly interior full-rank centres θ_e , with $e \in \{1/4, 1/8\}$, given by

$$f = z - \frac{1}{4}, \quad g = z - \frac{3}{4}, \quad \alpha = \frac{1}{2}, \quad U_e = \begin{pmatrix} \frac{1}{2} + e & \frac{1}{2} - e \\ \frac{1}{2} - e & \frac{1}{2} + e \end{pmatrix}, \quad V = \begin{pmatrix} \frac{1}{4} & \frac{3}{4} \\ \frac{3}{4} & \frac{1}{4} \end{pmatrix}. \quad (15.1)$$

All competitors still range over the original closed binary model; no parameter is held known. In a common small open coordinate ball write

$$z = (\dot{r}, \dot{s}, \dot{\alpha}, \dot{u}_1, \dot{u}_2, \dot{v}_1, \dot{v}_2).$$

The first entries of the binary channel columns are the independent channel coordinates. Let $G_e(z)$ be squared Hellinger discrepancy from θ_e , and let $\Delta c(z)$ be the two target coefficient increments. The latter polynomial map is the same at both centres.

Theorem 15.1 (Valuative equivalence without metric equivalence). *For every nonconstant real analytic arc $z(\tau)$ through zero,*

$$\text{ord}_\tau G_{1/4}(z(\tau)) = \text{ord}_\tau G_{1/8}(z(\tau)) = 2 \min_i \text{ord}_\tau z_i(\tau). \quad (15.2)$$

For every polynomial H in the target coefficient increments, including all mixed polynomial combinations, the orders of $H(\Delta c(z(\tau)))$ agree at the two centres, with the same convention for identically zero functions. The corresponding accessible real divisorial order data can be computed on one common real monomial presentation and agree as well.

Nevertheless the two whole-model leading spectral sets are distinct. Their exponents are one, and their squared leading diameters are

$$C_{1/4}^2 = \frac{129609775953}{25280}, \quad C_{1/8}^2 = \frac{10058351786055}{1396384}. \quad (15.3)$$

Thus the complete scalar arc-order information just specified does not determine the joint metric initial fibre, even within the principal binary observation model.

Proof. Both channels are invertible and the component polynomials are coprime. Lemma 2.2 implies that the observation derivative in these seven coordinates is injective. Since all probability cells are positive, the square-root embedding is analytic and

$$G_e(z) = \frac{1}{4} z^\top H_e z + O(\|z\|^3), \quad H_e > 0.$$

It follows either by this expansion or by two-sided comparison with $\|z\|^2$ that (15.2) holds. The coefficient functions are literally identical in the common coordinates, proving their order assertion for every H .

For the divisorial assertion, first blow up the maximal ideal at zero. On every real affine chart the pullback of G_e is the square of an exceptional coordinate times a strictly positive analytic unit near the real exceptional locus. Indeed the leading factor is the positive quadratic form H_e evaluated on a nonzero real projective direction. These units remain nonzero under every further real modification. Principalize any finite specified collection of the common coefficient functions on this same space. The exceptional orders and their real accessibility are the same for both experiments. Alternatively every accessible divisor has a real transversal, so its observation order already follows from (15.2). The assertion concerns real accessible data; no equality of complex zero divisors of the two positive quadratic forms is used. Every additional finite collection of polynomial combinations is handled in the same way.

We compute the metric sets. Let S_e be the 12-by-7 score matrix from (2.3), at the three clocks, in the coordinate order above. Set

$$H_e = S_e^\top \text{diag}((3P_{e,je})^{-1}) S_e, \quad M_e = (H_e^{-1})_{\{1,2\},\{1,2\}}.$$

Profiling the five free nuisance coordinates in the positive definite quadratic form shows that the leading pair of root increments is exactly

$$L_e = \{x \in \mathbb{R}^2 : x^\top M_e^{-1} x \leq 4\}. \quad (15.4)$$

For completeness, the lower bound on G_e bounds all competitor coordinates by $O(t)$. Division by t and Taylor expansion give the outer inclusion in (15.4). Every vector in its interior is realized

by its linear parameter arc, which is feasible because the centre is interior. Boundary vectors are limits of their radial contractions. This proves Hausdorff convergence. Exact uniqueness and compactness, from Lemma 2.2, exclude competitors outside this coordinate neighbourhood at sufficiently small radii. The limit is consequently a whole-model limit.

Here the entries of the matrix are obtained by differentiating the explicit rational functions

$$P_{j,ab} = \frac{\alpha(T_j - r)u_{1a}v_{1b} + (1 - \alpha)(T_j - s)u_{2a}v_{2b}}{\alpha(T_j - r) + (1 - \alpha)(T_j - s)}.$$

Rational elimination gives

$$M_{1/4} = \frac{1}{404480} \begin{pmatrix} 129609775953 & 79361335413 \\ 79361335413 & 48691215153 \end{pmatrix}, \quad (15.5)$$

$$M_{1/8} = \frac{1}{22342144} \begin{pmatrix} 10058351786055 & 6157532670525 \\ 6157532670525 & 3781629625095 \end{pmatrix}. \quad (15.6)$$

These equalities can be checked solely by forming H_e from the preceding rational display and multiplying its inverse; no fitted numerical covariance is involved. For separated real roots the bottleneck metric is the maximum norm of their ordered increments. A centrally symmetric ellipsoid (15.4) has maximum-norm diameter $4 \max_i \sqrt{(M_e)_{ii}}$. The first diagonal entry is the larger in each displayed matrix. This proves (15.3). The values are unequal, so neither the leading root sets nor their coefficient images can be equal. $\square$

The example fixes all orders of the target scalar functions, not their sharp leading values. Those values contain the positive units erased by valuation. It therefore complements the sharp-marginal example without claiming that arbitrary scalar information is insufficient. The relative real atlas remains useful for finding the common weights; its units and its joint relation are necessary for recovering the metric section at those weights.

16. SINGULAR ENTRANCES IN SEMIALGEBRAIC FAMILIES

The regular-corner calculation of Section 7 does not exhaust remote identification. We now describe entrance orders without a corner parametrization or an injective observation derivative. The required finiteness is finiteness of the exact parameter fibre, modulo any already specified finite symmetry, not smoothness of its representatives.

Let $X \subset \Theta \times K$ be semialgebraic with compact fibres in a fixed compact set K . Let $f_\theta : X_\theta \rightarrow \mathbb{R}^M$ and $A_\theta : X_\theta \rightarrow \mathbb{C}[z]$ be continuous semialgebraic maps, the latter monic of fixed degree. For the probability experiment, f is its weighted square-root embedding. Let

$$p_\theta(\delta) = f_\theta(x_\theta(\delta)), \quad x_\theta(\delta) \longrightarrow x_{0,\theta}, \quad \delta \downarrow 0,$$

be a semialgebraic centre path with p_θ extending C^1 to zero. All data, including this path, have fixed format. Assume $f_\theta^{-1}(p_\theta(0))$ is finite. After a finite semialgebraic partition its elements can be listed by continuous semialgebraic sections $x_{0,\theta}, \dots, x_{N,\theta}$; subsequent refinements will be understood. Choose pairwise disjoint small closed relative neighbourhoods $N_{j,\theta}$ of these points, and put

$$d_j(\theta, \delta) = \min_{x \in N_{j,\theta}} \|f_\theta(x) - p_\theta(\delta)\|. \quad (16.1)$$

The neighbourhoods can be chosen semialgebraically, for example using closed balls of radius smaller than a quarter of the least separation of the listed fibre points.

Theorem 16.1 (Finite entrance orders and exact membership). *There is a finite Nash stratification of the parameter set with the following properties. For each j either d_j is identically zero for*

all sufficiently small positive δ , or there are a rational $\rho_j \geq 1$ and a positive Nash function a_j such that, for every compact subset H of the stratum,

$$d_j(\theta, \delta) = a_j(\theta)\delta^{\rho_j} + O_H(\delta^{\rho_j + \nu_j}), \quad \nu_j > 0. \quad (16.2)$$

The nonzero exponents belong to a finite rational list determined by the fixed-format presentation. The germ of d_j is independent of the sufficiently small isolating neighbourhood.

For any specified positive semialgebraic threshold $t_\theta(\delta) \rightarrow 0$, a further finite partition decides, for each j , whether $t_\theta - d_j$ is eventually positive, negative, or identically zero. The remote component is included precisely in the first and third cases. At every fixed parameter the feasible target root sets converge in Hausdorff distance to

$$\bigcup_{j: d_j \leq t_\theta \text{ eventually}} \{Z(A_\theta(x_{j,\theta}))\}. \quad (16.3)$$

All the strata, exponents, leading-coefficient graphs and membership decisions are effectively describable for algebraic input. No regular-corner assumption is made in these assertions.

Proof. First fix a parameter. The boundary of an isolating neighbourhood contains no point of the exact fibre. Its observation distance from $p(0)$ has a positive minimum. The same is true of the complement of smaller neighbourhoods of all the exact points. Minima over these compact sets are semialgebraic functions of the parameter. Refinement makes them continuous and positive, so the resulting separation bounds are uniform on compact subsets of a stratum. Radii and the finitely many fibre sections admit the same refinement. Finite semialgebraic choice of the fibre points follows by cell decomposition of their finite-fibre graph.

The competitor x_j gives $d_j \leq \|p(\delta) - p(0)\| = O(\delta)$. If a sequence of minimizers did not tend to x_j , compactness would give a different point of the exact fibre in N_j . This is impossible. The same argument for the annulus between two smaller isolating neighbourhoods shows that their minima are equal for all sufficiently small δ . Their distance germs, not just their leading powers, are therefore independent of this choice.

The graph of (16.1) is a first-order formula: there is a point attaining the displayed squared distance, and every point of N_j has at least that squared distance. Quantifier elimination makes the graph semialgebraic. An eventually nonzero nonnegative one-variable semialgebraic function has an expansion $a\delta^\rho + o(\delta^\rho)$ with $a > 0$ and rational ρ . For clarity about finiteness in families, apply Lemma B.1 to a quantifier-free graph formula for d_j . Among the finitely many nonzero specialized polynomial supports, two lowest-order monomials must balance on a positive graph branch. Their exponent differences give a finite rational candidate list. Eventual vanishing is tested separately. The comparability test used in (B.1) selects the actual exponent by a first-order formula. The inequality $d_j = O(\delta)$ forces $\rho \geq 1$.

The limit $a = \lim d_j/\delta^\rho$ also has a first-order graph. Refine until it is positive and Nash. On such a stratum the error $|d_j/(a\delta^\rho) - 1|$ tends to zero on each fibre. Proposition 10.2 gives a positive rational power bound and a positive continuous radius. Taking their bounds and the bounds on a on compact subsets proves (16.2). Thus neither a metric rate nor uniformity has been deduced from topological triviality.

For the threshold assertion, a one-variable semialgebraic function has constant sign or is zero on some terminal interval $(0, \eta)$. Each of these alternatives has a first-order formula with the parameter free, so a finite partition decides them. The sign may be determined by a higher Puiseux coefficient when leading terms coincide; no finite truncation is declared sufficient before the sign test has been made. If the difference is identically zero, compact attainment in (16.1) puts a point of N_j in the closed ball. Thus equality is inclusion, not an assigned side of a critical ray.

The exterior separation already proved forces every small-radius competitor into one of the N_j . Within an included neighbourhood, all feasible points tend to its unique exact point, while a minimizer provides at least one feasible point for every sufficiently small δ . Continuity of the target and of finite-degree roots proves both Hausdorff inclusions in (16.3). The centre path ensures the union is nonempty. Finally every selection, limiting graph and sign test used above is a finite real-algebraic formula. The finite-support algorithm and real-root isolation in Theorem 11.1 prove the stated effectivity. $\square$

This theorem describes singular *entrance orders*, not a classification of singularities of the source. Its proof uses the relative semialgebraic machinery for a conclusion not supplied by a regular tangent-cone projection: entrances tangent to the centre path may occur at strictly higher rational orders.

Theorem 16.2 (Realization of every rational contact order). *Every rational $\rho \geq 1$ occurs as a positive remote entrance exponent in a compact semialgebraic model with strictly positive probability observations, a two-point exact fibre and distinct polynomial targets on its two representatives. For $\rho = n/m > 1$ the probability maps can be chosen polynomial in their respective branch parameters, of degree at most n . Hence there is no universal finite list independent of format, although Theorem 16.1 gives a finite list for every fixed-format family.*

Proof. Choose a strictly positive probability vector P_0 with at least three cells and linearly independent zero-sum vectors A, B such that $\langle A, B \rangle_I = 0$, where $\langle H, H' \rangle_I = \sum_e H_e H'_e / P_{0,e}$. Let the model be the disjoint union of two sufficiently short closed intervals, with maps

$$P_{\text{base}}(s) = P_0 + As, \quad P_{\text{rem}}(u) = P_0 + Au^m + Bu^n, \quad s, u \geq 0.$$

All probabilities are positive after shortening the intervals. Independence of A, B makes the exact fibre at P_0 precisely the two origins. Give the two intervals different constant monic degree-one targets. Take the centre path $s = \delta$ and use Hellinger distance.

Put $x = u^m$ and $\rho = n/m > 1$. The trial $x = \delta$ gives an entrance upper bound $O(\delta^\rho)$. Hellinger distance is locally comparable to Euclidean probability distance, and independence gives

$$|x - \delta| + x^\rho \leq C \|A(x - \delta) + Bx^\rho\|.$$

Every minimizer thus has $x = \delta + O(\delta^\rho)$. Write $x = \delta + v\delta^\rho$; the relevant v are bounded. The square-root embedding, expanded uniformly at P_0 , gives

$$\delta^{-\rho} h(P_{\text{rem}}(u), P_{\text{base}}(\delta)) = \frac{1}{2} \|vA + B\|_I + o(1).$$

Every bounded v is feasible for small δ since $\rho > 1$. The minimum of the right side is $\|B\|_I/2 > 0$ by the chosen orthogonality. This proves the desired exponent and leading coefficient. For $\rho = 1$, use instead $P_{\text{rem}}(u) = P_0 + Bu$, with the same orthogonal A, B . Projection onto the nonnegative B ray gives the positive coefficient $\|A\|_I/2$. These constructions concern the stated general class of probability models; they do not assert that every exponent occurs in the cubic binary family. $\square$

16.1. An algebraic test for the regular regime. The following criterion derives the corner hypotheses from a local presentation. It also separates two ways for that regime to fail: singular feasible geometry and tangency of the centre path to a regular observation image.

Proposition 16.3 (A checkable corner criterion). *At a remote representative x_* suppose the model is locally*

$$X = \{x : h(x) = 0, g_i(x) \geq 0\},$$

with Nash functions. Assume $Dh(x_)$ has full row rank and the differentials of the active g_i , restricted to $\ker Dh(x_*)$, are linearly independent. Put*

$$K_* = \{v : Dh(x_*)v = 0, Dg_i(x_*)v \geq 0 \text{ for active } i\}.$$

Suppose f has a C^2 Nash extension near x_* , the centre admits $p(\delta) = p(0) + \delta b + \delta^2 c + o(\delta^2)$, and

$$\ker Df(x_*) \cap K_* = \{0\}, \quad p'(0) \notin Df(x_*)K_*. \quad (16.4)$$

Then a full Nash corner chart exists, its observation derivative is cone-coercive, and the entrance has exponent one. The first and second coefficients are exactly those of Theorems 7.1 and 7.2. For algebraic input all these hypotheses are finite algebraic or first-order tests.

Proof. The Nash implicit-function theorem first supplies coordinates on $h = 0$. The independent active functions can then be used as the first coordinates on this manifold, completed by free coordinates. In these coordinates the feasible germ is exactly an orthant times a Euclidean space; inactive inequalities have a strict margin. The derivative of the inverse chart identifies this orthant cone with K_* . The first condition in (16.4), compactness of its unit section and continuity give a positive coercivity constant. Its linear image is a closed polyhedral cone, so the second condition gives a strictly positive attained distance from $p'(0)$ to that image. The existing regular-corner theorems now apply, including nonunique preimages of the first projected point. Matrix ranks are tested by minors; both conditions in (16.4) are first-order statements in linear equations and inequalities. Nash branch specifications and their derivatives at algebraic points are encoded by their polynomial equations and isolating conditions. $\square$

Metric projection, constrained value functions and second-order sensitivity have a substantial existing theory [16, 8, 9]. The regular-corner expansion is used here in that role, with its proof and normalization retained. The additional statements are the fixed-format classification and exact membership rule without its regularity hypotheses, the realization theorem, and the algebraic verification of the regular regime. The two cubic walls of Sections 6 and 8 are explicit realizations of the linear regime; they are not offered as an exhaustive list of singular entrance geometries.

17. DIVISORIAL GEOMETRY OF A REMOTE ENTRANCE

We now identify the singularity datum behind a nonzero entrance order. The datum is a pair: the ideal of the observation residual and the ideal of the centre-path parameter, together with the accessible real sectors. It is not the singularity of the remote image alone. The norm on the residual supplies additional unit data for the leading coefficient.

Fix a remote representative and a sufficiently small compact isolating neighbourhood N . Write

$$R(x, \delta) = f(x) - p(\delta), \quad \mathfrak{a} = (R_1, \dots, R_M), \quad \mathfrak{t} = (\delta), \quad d(\delta) = \min_{x \in N} \|R(x, \delta)\|.$$

In this section N is semialgebraic, f is Nash on a neighbourhood of N , and p is Nash near zero. Assume that $R(x, 0) = 0$ has just one solution in N , that it lies in the relative interior of N , and that $d(\delta) > 0$ for all sufficiently small positive δ . We also assume $p(0) = f(x_*)$ at that solution. This is the nonexact alternative of Theorem 16.1 in the Nash category; its broader continuous-semialgebraic statement is retained unchanged. A Nash path after a positive ramification $\delta = s^q$ is treated by applying the following construction to s and dividing its order by q .

We use simultaneous real principalization of the *finite* pair $(\mathfrak{a}, \mathfrak{t})$, with rectilinearization of the inequalities of $N \times [0, \delta_0]$. The standard resolution and rectilinearization results [3, 4] give, after shrinking and covering a compact preimage, finitely many closed real sectors with coordinates (r, z) , $r_i \geq 0$, on which

$$\delta = u(r, z) \prod_i r_i^{a_i}, \quad \|R\| = v(r, z) \prod_i r_i^{b_i}, \quad a_i, b_i \in \mathbb{N}, \quad u, v > 0. \quad (17.1)$$

The units and their inverses are bounded on these sectors. To obtain v , principalize the residual ideal, write its pulled-back generators as $r^b U_1, \dots, r^b U_M$ with no common zero, and take $v = (\sum U_i^2)^{1/2}$. The charts parametrize the actual closed feasible set, not the complex zero locus

of its equations. Only sectors in the closure of their positive-time interiors are retained. We call their coordinate faces *accessible real divisors*. If a chart has a lower-dimensional source component, coordinates and faces are taken in that component; identically zero-time components are discarded. These operations are on the finite diagram just specified.

Theorem 17.1 (Divisorial contact and the metric unit). *In any such finite real monomial presentation, an accessible index with $a_i = 0$ has $b_i = 0$. Define*

$$\rho = \max_{\text{accessible } i, a_i > 0} \frac{b_i}{a_i}. \quad (17.2)$$

Then ρ is positive and independent of the presentation. It has the intrinsic real-arc characterization

$$\rho = \sup_{\gamma} \frac{\text{ord}(\mathbf{a} \circ \gamma)}{\text{ord}(\delta \circ \gamma)}, \quad (17.3)$$

where γ ranges over feasible positive-time semialgebraic Puiseux arcs tending to $(x_, 0)$, and the order of an ideal is the minimum order of its displayed generators. The supremum is attained by an arc transverse to a suitable accessible divisor.*

On each resolved sector the function $\delta^\rho / \|R\|$ extends continuously to time zero as

$$\Lambda(r, z) = \frac{u(r, z)^\rho}{v(r, z)} \prod_i r_i^{\rho a_i - b_i}. \quad (17.4)$$

Let M be its maximum on the compact accessible zero-time preimage, allowing all the sectors. Then $0 < M < \infty$ and

$$d(\delta) = M^{-1} \delta^\rho + o(\delta^\rho). \quad (17.5)$$

Equivalently, ρ is the least real number c for which $\delta^c \leq C \|R(x, \delta)\|$ throughout the feasible germ, for some $C > 0$. Thus the entrance exponent is a relative real Łojasiewicz contact exponent of $(\mathbf{a}, \mathbf{t})$, and the entrance coefficient is the reciprocal of a resolved unit maximum.

Proof. All assertions concern the finite real sectors of (17.1). If $a_i = 0 < b_i$, set $r_i = 0$ while keeping the other boundary coordinates small and positive. The time is positive and the residual vanishes. The image is feasible, which contradicts $d(\delta) > 0$. Thus $b_i = 0$ whenever $a_i = 0$. There are positive-time points with both time and residual tending to zero, for instance the fixed competitor x_* . Consequently at least one accessible ratio b_i/a_i is positive.

For ρ in (17.2), every exponent $\rho a_i - b_i$ is nonnegative. The bounded positive units therefore make (17.4) continuous and bounded on each compact sector. Choose an index attaining the maximal ratio and approach a generic point of its face, with all other boundary coordinates positive. The ratio has a strictly positive value there. Its image at time zero has zero residual and therefore is $(x_*, 0)$. It follows that its maximum M over the zero-time preimage is positive and finite.

For an arbitrary feasible Puiseux arc, properness and semialgebraic selection give a Puiseux lift after further ramification. On a sector containing a final subarc, write $\beta_i = \text{ord } r_i \geq 0$; coordinates tending to a nonzero value have order zero. Coordinates with positive a_i cannot vanish identically along a positive-time arc. Coordinates with positive b_i cannot vanish identically either, since the residual is nonzero there. The units have order zero. Hence

$$\frac{\text{ord}(\mathbf{a} \circ \gamma)}{\text{ord}(\delta \circ \gamma)} = \frac{\sum_i b_i \beta_i}{\sum_i a_i \beta_i} \leq \max_{a_i > 0} \frac{b_i}{a_i}.$$

A transverse arc at a generic maximizing face gives equality. This proves (17.3). It proves independence of the presentation without identifying a real sector with all real points of a complex exceptional divisor.

For $\delta > 0$, surjectivity onto the feasible set and compact attainment give the exact identity

$$\max_{\text{resolved time}=\delta} \Lambda = \frac{\delta^\rho}{d(\delta)}. \quad (17.6)$$

Every sequence of maximizers on its left has a subsequence converging to the compact zero-time preimage, so its upper limit is at most M . For the reverse bound, take a zero-time point attaining M . It belongs to the closure of the positive-time part. Curve selection supplies a positive-time semialgebraic arc tending to it. Its time coordinate is nonconstant and, after shortening the arc, strictly monotone, with image containing every sufficiently small positive time. Evaluation along this arc gives a lower limit at least M in (17.6). This proves (17.5). Finally the bounded extension of Λ gives $\delta^\rho \leq C\|R\|$. On a transverse maximizing arc the ratio $\delta^c/\|R\|$ is unbounded for $c < \rho$. This proves the last characterization. $\square$

Corollary 17.2 (Integral closure and the role of the norm). *Replacing the residual ideal by an ideal with the same integral closure leaves (17.3) unchanged, with the same feasible real sectors and time parameter. The leading coefficient need not be preserved by this replacement. If the centre path extends C^1 to zero, then $\rho \geq 1$.*

Proof. If h is integral over an ideal I , an integral equation has the form $h^n + a_1 h^{n-1} + \dots + a_n = 0$, with $a_j \in I^j$. Along an arc, $\text{ord } h < \text{ord } I$ would make h^n the unique lowest-order term, an impossibility. Ideals with the same integral closure thus have the same minimum order on every arc, and (17.3) applies. This is the valuative mechanism behind the classical connection with Łojasiewicz exponents [21]. Multiplication of every residual generator by a positive constant preserves the ideal but multiplies its norm and the entrance coefficient by that constant. The latter requires the unit in (17.4), not just the unnormed ideal. Finally x_* is an admissible competitor and $\|R(x_*, \delta)\| = \|p(0) - p(\delta)\| = O(\delta)$. Together with (17.5), this forces $\rho \geq 1$. $\square$

17.1. How the contact pair recovers the existing examples. The singular realization in Theorem 16.2 has, in probability coordinates, the residual

$$A(u^m - \delta) + Bu^n, \quad n > m.$$

The sector following the tangent contact can be parametrized by

$$\delta = u^m - u^n z = u^m(1 - u^{n-m}z).$$

There its two independent residual coordinates have common order n , whereas time has order m . Thus $(a, b) = (m, n)$ and the accessible contact ratio is n/m . No larger ratio is possible: if $u^m < \delta/2$ then $|u^m - \delta| \geq \delta/2$, whereas if $u^m \geq \delta/2$ then $u^n \geq (\delta/2)^{n/m}$. Independence of A, B and local equivalence of the probability and Hellinger metrics give a lower bound by a positive multiple of $\delta^{n/m}$ in both cases. In the tangent sector the resolved Hellinger unit at $u = 0$ is

$$\frac{1}{2}\|zA + B\|_I.$$

Its reciprocal maximum is $2/\|B\|_I$ when $\langle A, B \rangle_I = 0$, recovering the exact coefficient $\|B\|_I/2$. This computation identifies the divisor and its metric unit; it does not merely read a Puiseux exponent from an already minimized scalar function.

In the regular regime of Proposition 16.3, cone coercivity localizes minimizing displacements to order δ . The ordinary blow-up $x - x_* = \delta v$ has residual $\delta(Jv - b) + O(\delta^2)$ on the feasible cone. Separation makes its leading norm positive; the relevant contact is $(a, b) = (1, 1)$, and the unit optimization gives $\min_v \|Jv - b\| = \mu$. This recovers the first coefficient of Theorem 7.1. The signed second coefficient and the exact equality-shell test still require the second-order and exact feasible-chart arguments there. In particular, (17.5) does not assign a side when a threshold has the same leading term as $d(\delta)$.

For the two cubic identification walls, the direct feasible charts prove precisely this regular localization, even where the channel rank drops. Their entrance exponents are therefore one, while their metric units and nonidentified-side openings differ. The divisorial formula explains what the exponent measures; it does not replace the exact cubic fibre classification, divide by a vanishing determinant, or suppress either wall mechanism.

APPENDIX A. FINITE ALGEBRAIC CERTIFICATES AND REAL DESCENT

This appendix supplies the full family argument used in Lemma 9.3 and Theorem 9.4. A variety in the algebraic construction is a reduced separated scheme of finite type over a characteristic-zero field. Smoothness is geometric smoothness. For an effective input the ground constants belong to a specified finite real algebraic extension of $\mathbb{Q}$; generic fibres are presented over its finitely generated function fields. No algorithmic assertion about an arbitrary computable field is intended. Real realizations are subsequently taken in $\mathbb{R}$, with the original sign and branch conditions retained.

An algebraic coordinate presentation consists of finitely many affine coordinate rings, localized by specified nonzero elements, transition morphisms and their inverse identities, and a coordinate map whose relative Jacobian is invertible. An adapted presentation also lists each divisor equation, its expression as a coordinate times a unit, and the monomial identities for the functions. Projective maps are recorded by homogeneous equations or by their finite blow-up sequences. These are finite polynomial data, not a numerical collection of coordinate boxes.

A.1. The Rees algebra and a finite diagram.

Lemma A.1 (One localization for all powers of an ideal). *Let A be a noetherian domain, B an algebra of finite type over A , and $I \subset B$ a finitely generated ideal. There is $a \in A \setminus \{0\}$ such that, for every A_a -algebra A' , with $B' = B_a \otimes_{A_a} A'$,*

$$I_a^n \otimes_{A_a} A' \xrightarrow{\sim} (I_a B')^n \quad (n \geq 0), \quad \left(\bigoplus_{n \geq 0} I_a^n \right) \otimes_{A_a} A' \cong \bigoplus_{n \geq 0} (I_a B')^n.$$

The first isomorphism identifies both sides as submodules of B' . Consequently the blow-up in I_a commutes with this base change. One localization suffices for any specified finite sequence of centre ideals on finitely many affine charts of successive blow-ups.

Proof. The associated graded algebra $\text{gr}_I B = \bigoplus_{n \geq 0} I^n / I^{n+1}$ is of finite type over A : it is generated by B/I in degree zero and the images of a finite set of generators of I in degree one. Since A is noetherian, these algebras are finitely presented. Apply generic freeness [17, Tag 051S] to B as a module over itself and to $\text{gr}_I B$ as a module over itself. After inverting the product of the two resulting elements, both are flat over A_a . Each graded piece is a direct summand as an A_a -module, hence flat. In particular B_a / I_a is flat. Induction using

$$0 \longrightarrow I_a^{n-1} / I_a^n \longrightarrow B_a / I_a^n \longrightarrow B_a / I_a^{n-1} \longrightarrow 0$$

shows that B_a / I_a^n is flat for every $n \geq 1$. Tensoring $0 \rightarrow I_a^n \rightarrow B_a \rightarrow B_a / I_a^n \rightarrow 0$ with A' is therefore exact on the left. Its image in B' is the ideal generated by products of n images of elements of I_a , namely $(I_a B')^n$. The isomorphisms respect multiplication and grading, proving the Rees algebra assertion. Relative Proj commutes with base change; this proves the blow-up assertion. For a finite sequence, apply the argument to the actual centre ideal on every affine chart at each stage and multiply the finitely many localizing elements. Previously proved identities survive further localization. No countable intersection of open sets is used. $\square$

Lemma A.2 (Spreading the entire stratifying diagram). *Let A be a finitely generated integral algebra over a characteristic-zero field k , and let $K = \text{Frac}(A)$. Suppose $Y \rightarrow \text{Spec } A$ is of finite type and a generic certificate consists of a reduced closed chain $Y_{0,K} = Y_{K,\text{red}} \supset \cdots \supset Y_{q,K} = \emptyset$,*

projective $Z_{j,K} \rightarrow Y_{j,K}$, and specified inverse isomorphisms above $E_{j,K} = Y_{j,K} \setminus Y_{j+1,K}$. Then, after replacing A by A_a for one nonzero a , the chain can be chosen with $Y_0 = Y_{\text{red}}$, all its members flat with geometrically reduced fibres, and the stated isomorphisms hold above the actual complements $E_j = Y_j \setminus Y_{j+1}$. In particular they hold above $(E_j)_s$ on every fibre. The union of these complements is the whole reduced fibre; no constancy of the number of irreducible components is required.

Proof. First replace Y by its reduction. Take reduced closures of the generic $Y_{j,K}$ inside Y and use $Y_q = \emptyset$. Clear denominators in the projective equations and affine presentations of the generic maps. All schemes involved are of finite presentation over the noetherian base. The finite-presentation limit lemma [17, Tag 01ZM, Lemma 32.10.1(1)–(3)] descends objects, morphisms, and equalities of morphisms. Apply it to the inverse maps on the generic complements, their inclusions, and both inverse identities. The complements used on the model are $Y_j \setminus Y_{j+1}$; their generic fibres are the prescribed complements. The descended inverse maps and equalities therefore give isomorphisms on these actual open subschemes after a further localization, not merely on unspecified subsets with the same generic points. Finite affine covers and their intersection diagrams are included in this application of the limit lemma. Thus chart coverage, as well as the formula for each map, descends.

Each reduced generic member is geometrically reduced because K has characteristic zero. Generic freeness makes each member flat on a dense open set. The generic-fibre reducedness theorem [17, Section 37.26, Lemma 37.26.4] then permits a further common shrink so that all fibres of these finitely many members are geometrically reduced. This is an imposed property, not an optional replacement of scheme structure after specializing. Finally, on every fibre the sets $Y_{j,s} \setminus Y_{j+1,s}$ form a disjoint partition of $Y_{0,s}$ because the chain ends in the empty set. For any point there is a unique index at which it leaves this finite descending chain. The corresponding isomorphism covers that point, including points on a special fibre component. This argument neither identifies fibre components with generic components nor assumes their number to be constant. $\square$

A.2. Relative smoothness, normal crossings, and exceptional sets. We now prove Lemma 9.3 with a precise list of the further conditions. Separate the finitely many open-and-closed components of each smooth generic source according to the identically zero functions. These components and their idempotent identities are part of the finite diagram. On a component on which a function is declared zero, descend that zero identity. On every other component principalize the product of the remaining functions, recording all factors and units.

The absolute input is characteristic-zero resolution and principalization [4, 5]. More precisely, the variety resolution of [5, Theorem 1.1] supplies the smooth sources; the marked-ideal resolution of [5, Theorem 1.3], applied with mark one, supplies principalization together with the normal-crossing exceptional boundary. A product which is a monomial times a unit makes each of its factors a monomial times a unit on the regular local rings. This use is over the field K . Compatibility with arbitrary special fibres is supplied by Lemma A.1, not by a claim that functoriality for smooth maps implies arbitrary base-change compatibility.

Spread the finite sequence of centres and its adapted charts. The following finite conditions are imposed before retaining the open base. The source and centre smooth loci are the loci on which their finite presentations satisfy the relative smoothness criterion. On a smooth source the intersections of the designated divisor equations must be smooth of their prescribed relative codimensions; in adapted coordinates this says that the appropriate relative differentials are independent. This condition is imposed for every subset of the finite divisor list. Each coordinate chart must be etale, each declared inverse unit must multiply its unit to one, and the finite charts must cover their source. All scheme-theoretic identities in these statements are spread

by the finite-diagram lemma. Their remaining rank conditions are open algebraic conditions; complements of chart unions are constructible algebraic sets.

For clarity, the discarded algebraic parameter set is the union of the closures of images of the following bad sets, together with the localizing hypersurfaces already used:

- (1) the original irreducible components not dominating the current reduced integral base;
- (2) nonsmooth points of a source or centre, and failures of the specified relative dimension or centre codimension;
- (3) failures of smoothness or codimension of any designated divisor intersection, or failure of an adapted etale Jacobian;
- (4) the complement of the chosen finite source charts, and any zero of a function declared invertible;
- (5) failures of a descended zero identity, a chart transition, or an inverse isomorphism on an actual complement in the closed chain.

For items (4)–(5) the bad set is empty once the relevant identities and coverage have descended; it is harmless to include it explicitly in the certificate. Every bad set in this list has empty generic fibre: this is respectively dominance, generic smoothness in the given certificate, generic normal crossings, generic coverage and invertibility, or the specified generic identities. Images are constructible by Chevalley’s theorem [17, Tag 054K]. A constructible subset of an integral noetherian space which is dense contains a nonempty open subset and hence the generic point. Therefore the closure of each of these images is proper. This justification applies also when $Y \rightarrow S$ is not proper.

After their removal, apply Lemma A.1 on every spread blow-up chart. The resulting smooth sources and centres, relative normal crossings, unit identities and finite chart covers hold on every retained fibre. A nonzero monomial cannot specialize to the zero function on a fibre component: its unit is invertible and its finitely many divisors each have relative codimension one. Their union cannot contain a component of the smooth fibre. An identically zero generic function remains zero by its descended identity. The open-and-closed source decomposition prevents a change of these flags through an unlabelled component intersection. Flatness of the closed chain and smoothness of the sources exclude additional vertical components on the retained open set. Coverage of any special fibre components is, more directly, the partition argument in Lemma A.2. This proves all fibrewise assertions of Lemma 9.3.

Each discarded algebraic subset has strictly smaller dimension than its current integral base. Replace it by its reduced irreducible components and reconstruct the entire generic certificate on each component. There are finitely many components at each step and depth at most $\dim S + 1$. This gives finite termination. The exceptional list is not required to be minimal. Importantly, a change of a *real* accessibility flag need not lie in a proper algebraic subset: two full-dimensional real chambers can have different flags. Such changes are treated by finite real quantifier-elimination partitions below, not by this dimension argument.

A.3. Exact real algebraization.

Lemma A.3 (The square-slack cover with all branch conditions). *Let S be a semialgebraic base and $\Gamma \rightarrow S$ a semialgebraic proper map realized as a graph in $S \times \mathbb{R}^N$. The graph includes all auxiliary functions, sign choices and denominator exclusions. After a finite base refinement, it admits finitely many algebraic square-slack lifts whose real points over S project exactly onto Γ . Projection is proper over compact subsets of S . Every later map factoring through these lifts projects only to points of Γ .*

Proof. Let C be the ordinary semialgebraic closure of Γ in the ambient real algebraic space. Properness implies that Γ is relatively closed in $S \times \mathbb{R}^N$: a convergent sequence over a convergent

base sequence lies over a compact subset of S and has a graph limit there. It follows that $C \cap (S \times \mathbb{R}^N) = \Gamma$. A closed semialgebraic set is a finite union of basic closed semialgebraic sets [7, Theorem 2.7.2]. For a piece

$$C_l = \{x : f_{l,a}(x) = 0, p_{l,b}(x) \geq 0\}$$

use the algebraic set

$$Y_l = \{(x, s) : f_{l,a}(x) = 0, p_{l,b}(x) = s_b^2\}.$$

Every real point of Y_l projects into C_l , and each point of C_l has a real lift. Over S , the union therefore projects exactly to Γ . Over a compact base subset the graph is compact. The slack coordinates are bounded by the square roots of maxima of the finitely many polynomials there, and the lift is closed; hence the lifted real set is compact. This proves properness and the last assertion.

Strict exclusions are encoded in the complete graph formula before this construction. For a denominator excluded everywhere on that compact graph, $d \neq 0$ may also be retained by adjoining w with $dw = 1$, together with any required sign of d . Nonnegative square-root branches are specified in the graph, not selected after taking an algebraic closure. If adjoining such coordinates destroys properness on the initial base, first apply Hardt triviality to the enlarged graph, whose individual fibres are compact, and refine the base. A trivialization with compact fibre makes the restricted graph map proper. Closure is taken only *after* this step. Consequently no point with a vanishing excluded denominator is reintroduced over the retained stratum. General Boolean branch conditions are instead kept in the complete graph formula: one does not split a compact fibre into nonclosed pieces, adjoin unbounded reciprocals on those pieces, and then assume their fibres are compact. The equality $C \cap (S \times \mathbb{R}^N) = \Gamma$ already preserves all such conditions without that operation. $\square$

Lemma A.4 (Real branch selection and simultaneous descent). *The root labels and Nash graph functions used in the cluster construction can be carried on finitely many selected real Nash branches of algebraic base covers. On a finite semialgebraic refinement, the proper resolved square-slack sources cover exactly the original graph in every fibre. Descent by root permutations preserves the entire joint coefficient and root relation, not merely its scalar order list.*

Proof. Refine a Nash stratum by the regular locus of each integral component of its algebraic closure. Its lower-dimensional singular parts are treated recursively. A Nash function is algebraic over the coordinate function field. Adjoin the finitely many graph coordinates needed here to that field. The extension is finite and separable in characteristic zero. Choose equations and invert their leading coefficients and discriminants; a further localization gives a finite etale algebraic cover. Include the original graph formula to select the intended real branches. Finite semialgebraic cell decomposition makes each selected graph a Nash section and separates its alternatives. Complex sheets and real sheets failing the original formula are not retained. Thus monodromy is handled by base refinement, rather than by globally labelling a nontrivial cover. Exceptional leading coefficients and discriminants are handled by the reduced lower-dimensional base recursion. A finite union of selected sections has compact inverse image over compact subsets of a stratum.

Apply Lemma A.3 on each selected branch. For a real point y of a lifted fibre, choose the index with $y \in Y_{j,s} \setminus Y_{j+1,s}$ in the spread singular-locus chain. The inverse isomorphism of Lemma A.2 supplies a real point of its smooth source above y . A blow-up of a smooth real variety in a smooth centre of positive codimension c has fibre $\mathbb{RP}^{c-1}$ at a real centre point and is an isomorphism away from the centre. In local smooth coordinates the centre is a coordinate subspace, which proves both assertions directly. Such a blow-up therefore does not lose real points. Composing the finitely many principalizations and taking the finite union over j proves real surjectivity. A real singular point with no top-dimensional real branch is covered by a later member of the chain.

Projectivity of these maps and properness of the slack projection prove properness of the real cover.

Conversely every real resolved point factors through a real slack point on the selected branch, and hence projects to the original graph by Lemma A.3. This proves equality, not just inclusion, of the covered real fibres. A root-label permutation simultaneously permutes cluster indices, their coefficient coordinates and root arrays. The equations defining Ψ and the fixed-centre formula (10.2) are equivariant for this simultaneous action. A finite union commutes with taking a fibrewise closure; projection to the finite quotient therefore preserves the joint relation and its root image. Descent never pairs one coefficient from one sheet with another coefficient from a different sheet. The bottleneck metric is invariant under these permutations. $\square$

A.4. Real accessibility and completeness of the order list.

Lemma A.5 (Real-accessible divisors). *On a resolved square-slack source with the above relative adapted coordinates, accessibility of a divisor is a first-order real condition on the parameter. Every recorded divisor has an admissible real transversal attaining its monomial orders. Every real degeneration relevant to a finite coefficient order is controlled by one of the recorded divisors. Thus the minimum in Corollary 9.6 is both an upper-inequality exponent and an attained exponent.*

Proof. In a chart with source equations $E(\theta, y) = 0$, chart exclusions $D(\theta, y) \neq 0$, and adapted divisor coordinates $y_1, \dots, y_l$, the flag for i with $a_i > 0$ is the formula

$$\exists y : E(\theta, y) = 0, \quad D(\theta, y) \neq 0, \quad y_i = 0, \quad \prod_{h \neq i, h \leq l} y_h \neq 0, \quad \det J_{\text{et}}(\theta, y) \neq 0. \quad (\text{A.1})$$

Inverse units and the selected real base formula are included in E, D . Charts are the ones meeting the inverse image of $G = 0$. Since the full real slack fibre already represents the admissible graph, there is no additional unrecorded sign restriction on source directions. At a point satisfying (A.1), the real inverse function theorem gives a transversal with $y_i = t$ and other adapted coordinates fixed. For small nonzero t , $G > 0$, and every point projects into the original graph. The unit bounds give $G \asymp |t|^{a_i}$ and $H \asymp |t|^{b_i}$ for each nonzero H on that source component.

A real degeneration with $G \rightarrow 0$ has a lifted accumulation point above $G = 0$, by properness. Some adapted chart contains that point. If it lies on several divisors, relative normal crossings allows a small real perturbation *on any one of them* avoiding all the others. Hence each divisor of positive G order meeting this real chart has a point of the form (A.1). In a smaller box, with β the minimum of its recorded ratios, $b_i \geq \beta a_i$ for $a_i > 0$ and $b_i \geq 0$ otherwise. The monomial identities and positive unit bounds imply $H \leq CG^\beta$ throughout the box, including its intersections. Equivalently, along a lifted arc the order ratio is a weighted average of the b_i/a_i with nonnegative additional terms from indices with $a_i = 0$; it cannot be smaller than their minimum. A finite cover near the compact exact fibre gives this inequality for all competitors. The transversal for a minimizing divisor attains it. If G is identically zero on a source with real points, the exact-target condition gives $H = 0$ there. A smooth fibre component with a real point has a Zariski-dense real open subset. Thus a nonzero monomial H cannot occur on such an exact real component; real-empty components contribute no order. This handles zero flags as well as the nonzero ratios.

Finally (A.1) is a finite polynomial formula. Quantifier elimination partitions the base by all its truth values. These semialgebraic pieces may have full dimension. Nash refinement makes the choices compatible with the selected base branches. This partition is finite without an algebraic dimension-drop assertion. $\square$

Completion of Theorem 9.4. Stratify the enlarged graph and the exact cluster data as in Lemma A.3, and carry labels as in Lemma A.4. Over an integral algebraic base resolve the

reduced generic fibre, then its reduced singular locus, and continue until the empty set. Dimension decreases in this chain. Principalize on its smooth sources the product of the nonzero functions. Lemmas A.1 and A.2, followed by the explicit bad-locus construction, spread this certificate over a nonempty open base with all fibrewise properties. Repeat on the finitely many reduced exceptional components. Lemma A.4 proves exact real coverage after descent, and Lemma A.5 proves the accessibility assertions and the completeness of the orders.

For a fixed fibre, cover the compact inverse image of $G = 0$ by finitely many smaller real coordinate boxes with positive lower and finite upper unit bounds. Properness implies that a sufficiently small observation neighbourhood is covered: otherwise a sequence with $G \rightarrow 0$ outside the chosen source neighbourhood would have a lifted accumulation point above $G = 0$ outside it. The same argument works above a compact base subset, since the selected label graph, the slack lift and the projective resolved maps are proper there. Alternatively their positive radii and unit bounds can be selected semialgebraically and made continuous by finite refinement. Their extrema on a compact parameter subset give common neighbourhoods and bounds. Fibrewise proper source maps must not be confused with the smaller, nonproper coordinate boxes. This proves all existence and compact-uniform assertions. The effective construction for algebraic input is detailed at the end of Appendix B. $\square$

APPENDIX B. FINITE SUPPORT ORDERS, UNIFORM DECAY, AND EFFECTIVE OUTPUT

The metric and effective parts can be proved without making a parameter-dependent choice of a Puiseux branch. Only the ordinary one-variable algebraic-branch theorem is used to prove that a finite candidate list is exhaustive. Selection from that list is then a first-order decision problem.

Lemma B.1 (A finite support test for the order). *Let $e(\theta, t) \geq 0$ be a semialgebraic function, defined for $0 < t < \tau(\theta)$, where $\tau > 0$ is semialgebraic, and suppose $e(\theta, t) \rightarrow 0$ for fixed θ . A quantifier-free graph formula supplies a finite computable set $\mathcal{B} \subset \mathbb{Q}_{>0}$ such that for every θ either $e(\theta, t)$ is eventually zero or exactly one $b \in \mathcal{B}$ satisfies*

$$\exists c, C, a > 0 \forall t : 0 < t < a \implies ct^b \leq e(\theta, t) \leq Ct^b, \quad a < \tau(\theta). \quad (\text{B.1})$$

Each of these alternatives is an effective semialgebraic condition for algebraic input. No real exponent is quantified over.

Proof. Eliminate quantifiers in the graph formula and let $P_1, \dots, P_N$ be its finite list of polynomials in (θ, t, y) . Fix θ and discard those specializing to the zero polynomial in (t, y) . At a graph point some remaining polynomial must vanish. Otherwise all nonzero polynomial signs are unchanged on a small two-dimensional neighbourhood; the zero specialized polynomials impose constant truth values there. The graph formula would then contain that neighbourhood, contradicting single-valuedness. For each remaining P_i , the set $\{t : P_i(\theta, t, e(\theta, t)) = 0\}$ is semialgebraic in one variable. These finitely many sets cover a punctured interval. After their finitely many endpoints are removed, at least one vanishing equation holds on an entire interval $(0, a)$. A polynomial independent of y cannot vanish there unless it was one of the discarded zero polynomials.

By one-variable semialgebraic decomposition, on a smaller interval the function is either identically zero or positive, continuous, and a single real algebraic branch of such an equation. The ordinary algebraic Puiseux theorem gives, in the positive case, $e(\theta, t) = a_\theta t^{b_\theta} + o(t^{b_\theta})$ with $a_\theta > 0$ and rational $b_\theta > 0$; see [7, 1]. Write its vanishing equation as $\sum p_{ij}(\theta) t^i y^j = 0$. Inserting the first term shows that $\min(i + b_\theta j)$ is achieved at least twice among nonzero specialized coefficients. Otherwise its unique lowest term could not cancel. Therefore

$$b_\theta = (i - i')/(j' - j) > 0$$

for two distinct support points of one of the original P_i . Take $\mathcal{B}$ to be all such positive ratios. Specialization may remove support points but cannot add a pair to this list. This proves finiteness uniformly in θ , without following algebraic branches through parameter specializations.

Eventual zero is the formula $\exists a > 0 \forall t (0 < t < a \Rightarrow e(\theta, t) = 0)$, with $a < \tau(\theta)$. For each candidate, (B.1) is a first-order polynomial formula: replace $t^{p/q}$ by its unique positive root, or put $t = s^q$ throughout. Quantifier elimination decides it. The actual branch expansion supplies one true candidate; two distinct powers cannot be bounded above and below by positive multiples of each other as $t \downarrow 0$. Exactly one candidate is therefore true in the nonzero case. This also tracks the branch selected by the original Boolean graph formula, including all specialized coefficients and inequality faces, rather than selecting a root of a polynomial without checking graph membership. $\square$

Full proof of Proposition 10.2. Use Lemma B.1. If the candidate set is nonempty, choose a positive rational $\nu < \min \mathcal{B}$; if it is empty use $\nu = 1$. On every parameter fibre $e(\theta, t) \leq t^\nu$ eventually, including the eventual-zero case. The set of pairs (θ, a) with $0 < a \leq \min(1, \tau(\theta))$ and this inequality for all $0 < t < a$ is semialgebraic and has a nonempty positive fibre at every parameter. Semialgebraic choice selects a positive $a = \tau_0(\theta)$, not a possibly vanishing limit value at the boundary of a stratum. A finite Nash stratification makes this selected function continuous. Its minimum on a compact subset of a stratum is strictly positive; the estimate is uniform there. The exponent may be smaller than the sharp order, which is exactly what is needed for the uniform remainder. $\square$

Lemma B.2 (Nash refinement of graph functions). *Finitely many semialgebraic functions on a semialgebraic set admit a common finite Nash stratification on which each function is Nash. For algebraic presentations this refinement is effective.*

Proof. Take their combined graph. Semialgebraic Nash stratification, refined by the ranks of the coordinate projection and by a compatible stratification of its image, is finite; these are the usual graph and rank-stratification constructions in [7, 1]. On a graph stratum the projection is injective. If its constant rank were less than the dimension of that stratum, the constant-rank theorem would give a positive-dimensional local fibre, contradicting injectivity. The projection thus has full rank onto its image stratum. Its local inverse is analytic by the inverse function theorem and is semialgebraic because its graph is semialgebraic; hence it is Nash. Finitely refine images to separate overlapping strata and repeat on their lower-dimensional boundaries. This proves the claimed finite partition and gives the precise reason that graph stratification, rather than topological triviality alone, yields Nash functions. Effective semialgebraic decomposition and rank tests give the algebraic version of the construction. $\square$

The application to Theorem 10.3 is now direct. The polynomial matching formula (10.5), with two quantified inclusions and minimality over the distance, defines the Hausdorff error as a semialgebraic function of (θ, t) . Theorem 10.1 proves its pointwise convergence to zero. Proposition 10.2 gives a compact-uniform power estimate. Lemma B.2 applied to the diameter and the positive separation and radius functions gives the asserted Nash refinement. Hardt triviality is used separately for the topological type of the leading fibres. The inequality $|\text{diam } A - \text{diam } B| \leq 2d_H(A, B)$ proves the diameter remainder.

B.1. An explicit terminating spectral algorithm. Here is a proof of Theorem 11.1 with its data representations and branch-selection step made explicit. The input is the polynomial formula of Definition 9.1; each algebraic constant has its integer polynomial and rational isolating interval. Arithmetic is in a primitive-element representation of their finite real algebraic field. All quantified formulas below are subjected to real quantifier elimination as in [1]. Finite root permutations are literal finite disjunctions, and complex coordinates are pairs of real coordinates.

First decide exact identification by the formula that all x, x' with $F(x) = F(x') = P$ have equal target coefficients. On this locus construct the cluster graph using its polynomial factorization equations and the specified isolating conditions. For each $H = |c|^2$, form the graph of

$$m_H(s) = \max\{H(x) : x \in \Gamma, G(x) \leq s\}$$

by the conjunction of the universal upper-bound formula and an existential attainment formula. Compactness gives the maximum; compactness and $\{G = 0\} \subset \{H = 0\}$ give $m_H(s) \rightarrow 0$. Apply quantifier elimination to this graph. Construct its finite support-pair list as in Lemma B.1, test eventual zero, and test (B.1) for each rational candidate. The unique true candidate is β_H . This is a finite algorithm for selecting the *actual maximum-function branch*. It needs no unspecified command to isolate an algebraic branch at the boundary. If all maxima are eventually zero the target is locally rigid; otherwise set $\alpha = \min_H \beta_H / k_H$ over the nonzero maxima.

For a family, keep θ free in exactly these tests. Their finitely many truth sets give the parameter pieces and include all zero specializations. Nash-refine them using Lemma B.2. The supports are finite before specialization, so this procedure still terminates without a bound on a real-valued exponent.

Next substitute $\alpha = p/q$ in the fixed-centre formula (10.2). Elimination computes a quantifier-free presentation of $\mathcal{J}_0$ and its projection $\mathcal{C}_0$. Adjoin the labelled roots by elementary symmetric identities to obtain $\mathcal{L}_0$. Use (10.5) to express that all pairs of elements have distance at most C and that some pair attains C . Compactness makes this a formula for one positive real number. Univariate real-root isolation applied to its eliminated formula selects that number over the input field. Taking a field norm gives a nonzero integer polynomial vanishing at C ; renewed root isolation and the same formula select a rational isolating interval for the intended root. The same graph formulas with free parameters compute the family of constants, and the preceding Nash and decay constructions compute a compatible refinement and an admissible rational remainder exponent. No complexity bound is claimed.

B.2. Effective geometric certificates without an implicit oracle. The geometric output can also be given a terminating, though generally inefficient, specification. Over the field just described, enumerate finite algebraic diagrams and finite semialgebraic base partitions in increasing finite encoding length, refining a fixed semialgebraic triviality partition of the enlarged graph with compact fibres. Properness on those base pieces follows from that trivialization and is preserved under further base restriction. A candidate consists of square-slack presentations, reduced closed chains with inverse-map identities, smooth sources with specified projective maps, finite centre ideals and adapted etale charts, integer monomial orders, inverse units and selected real branch formulas. Verify polynomial identities and centre blow-up charts by ideal arithmetic; verify relative smoothness, chart ranks and divisor intersections by their Jacobian criteria. Properness is certified by the specified projective construction and the square-slack argument, not by numerical bounds on open charts. Verify partition coverage, fibrewise real surjectivity, zero flags and accessibility by real quantifier elimination. These are finite formulas: for example coverage is $\forall \theta \forall x \in \Gamma_\theta \exists y$ in one of the finitely many source charts with the stated projection. Each positive neighbourhood and unit bound can likewise be represented by a semialgebraic choice graph and its universal verification formula.

The existence proof in Appendix A, performed over finite algebraic and rational-function extensions of the input field, produces a certificate with such a finite encoding. An exhaustive search therefore eventually finds a verified certificate. Invalid candidates are rejected by the finite tests. This is an explicit input–output existence algorithm; it is not a claim that the accompanying regression script implements a practical resolution package. The direct maximum-function algorithm above computes the spectral outputs without this enumeration. Thus neither

effectivity route depends on an unrepresented branch oracle, and neither is inferred from a finite list of example calculations.

APPENDIX C. FORMAT, FINITE DIAGRAMS, AND THEOREM DEPENDENCIES

We make explicit two conventions used in the general semialgebraic statements. Neither convention alters the whole-model quantifiers in the concrete binary theorems.

Definition C.1 (Geometric and arithmetic format). A fixed-format family is specified by a fixed finite list of formulas for the feasible-set graph, observation graph, monic target-coefficient graph, centre-path graph, and any isolating neighbourhoods or threshold functions used in the assertion. Its geometric format is

$$\mathcal{F} = (n_\Theta, n_X, n_{\text{aux}}, M, d_A, s, D, L).$$

Here the first three numbers bound the parameter, source and auxiliary variables in the entire list; M is the observation dimension; d_A is the target degree; s bounds the number of polynomial atoms; D bounds their total degrees; and L bounds the length of their Boolean and quantifier syntax trees, with the quantifier blocks included. Rational functions are encoded by graph equations and denominator sign conditions. A Nash branch is encoded by its polynomial equations and real branch inequalities. Square roots, root constraints, path ramifications and neighbourhood choices are included as auxiliary graph data, not treated as zero-cost primitive operations.

Coefficients may range over real parameters for geometric finiteness. For effective algebraic input, each fixed coefficient is specified by a square-free integer polynomial and a rational isolating interval, with a consistent common real algebraic embedding for joint arithmetic. Bounds on their degrees and binary encoding lengths may be appended as the arithmetic format. No bit-complexity estimate is claimed here. A new threshold or a selected scalar polynomial must be included in the finite list before a uniform bound is asserted for it.

Proposition C.2 (An explicit meaning of the finite order list). *For the format in Definition C.1, there is a computable integer $B(\mathcal{F})$ such that every nonzero exponent in Theorem 16.1 belongs to*

$$\{p/q : 1 \leq p, q \leq B(\mathcal{F})\}.$$

When the centre path is C^1 , only the members at least one occur. The integer can be obtained from a real quantifier-elimination degree bound applied to the actual minimum graph formula; it is not an unquantified appeal to a notion of bounded format.

Proof. The graph formula for the attained minimum uses one copy of the source variables for an attaining point, another for comparison with all competitors, the already specified graph formulas, and an equation for the squared norm. Its numbers of variables, atoms and degrees are bounded by computable functions of $\mathcal{F}$. Apply real quantifier elimination [1] and let B bound the degrees of all output polynomials, increasing it if necessary to a positive integer. Fix a parameter. On a final positive interval the graph of a nonzero minimum lies on the zero set of a nonzero specialized output polynomial $P(\delta, z)$: otherwise a sign cell containing the graph would have interior in the (δ, z) plane, contrary to its being a graph. There are only finitely many cells and polynomials, so one final branch and a nonzero polynomial can be selected. Identically zero specialized polynomials are discarded first.

Substitute $z = a\delta^\rho + o(\delta^\rho)$, $a > 0$, using the one-variable semialgebraic Puiseux theorem. At least two distinct monomials $\delta^i z^j$ have the lowest order; a unique lowest term could not cancel. They must have different j , since otherwise equality of orders would force equal i . Thus $\rho = (i - i')/(j' - j)$, with nonzero integer denominator and numerator of absolute value at most B . Positivity of ρ gives the asserted finite list after reduction. Vanishing coefficients only remove monomials and do not enlarge this bound. The C^1 lower bound follows from the fixed exact competitor as in Theorem 16.1. $\square$

C.1. Finite scalar families, not one modification for all polynomials. A common monomial presentation is asserted for every *finite selected* family of functions, ideals and inequalities, after a common refinement depending on that family. It is not asserted that one fixed modification simultaneously monomializes the infinite ring of all polynomial scalar functions of the target coefficients. This convention applies in particular to the divisorial-order interpretation of Theorem 15.1. The theorem's comparison along every analytic arc and for every scalar polynomial remains valid pointwise for each polynomial; it does not require a simultaneous principalization for all of them.

There is a simple reason for the distinction. In a plane, the infinitely many germs $x - cy$, $c \in \mathbb{R}$, have distinct strict transforms. No fixed finite normal-crossings divisor can contain the zero divisors of all of them. A finite chosen list can be included in a resolution diagram; its size and degrees then enter $\mathcal{F}$. The contact formula in Theorem 17.1 uses only the finite residual/time pair and feasibility inequalities. The finite calibration theorem uses exactly the probes listed in (4.4).

C.2. Dependency of the concrete results. The normalizer recovery and full-rank coefficient inverse are direct interpolation and matrix-pencil arguments. The multiplicity and boundary laws use these inverses, real-root cluster expansions and finite Fisher projection. The finite calibration theorem uses the same inverse to prove the convergence of *actual* constrained minima and their whole-model recovery; its stability bound is finite-dimensional. None of these proofs needs the relative resolution atlas. The semialgebraic refinement of their algebraic output functions is the separate organizing use of real algebraic geometry.

The exact cubic fibre and the inverse through rank loss are direct affine-pencil and second-marginal arguments. The weight and endpoint walls require their different exact feasible charts and the regular conic expansion. Those proofs, including equality-shell membership and the signed second correction, do not depend on a general singular resolution theorem. They have therefore been placed before the general weighted framework in the principal article.

The general atlas is essential to the uniform weighted output theorem for arbitrary specified semialgebraic observation relations, beyond the explicit binary normal forms. Real curve selection and algebraic Puiseux germs give the residue comparison. Simultaneous principalization of the residual/time pair supplies the divisorial entrance formula; the scalar semialgebraic entrance theorem supplies the broader continuous-semialgebraic statement and uniform family refinements. These are distinct inputs and conclusions. We do not claim that the existence of a general atlas is necessary for a computation already proved by the concrete pencil or Fisher arguments.

APPENDIX D. EXACT OUTPUTS AND FINITE ARITHMETIC

Three different assertions should not be conflated. The existence of the relative real atlas is a geometric theorem. The first-order procedures produce spectral or entrance outputs from algebraic presentations. The rational calculations of specific Fisher programs are finite specializations, not implementations or proofs of the entire atlas.

For the direct procedures the input is a finite list of polynomials with rational or real algebraic coefficients, a Boolean formula of polynomial sign conditions and the quantifiers specifying the model and its maps. A real algebraic coefficient is represented by its integer polynomial and a rational isolating interval. Rational powers use a nonnegative auxiliary variable and its integer-power equation. Compact minimum formulas have the form

$$\exists x [\Phi(x) \wedge Q(x) = z] \wedge \forall y [\Phi(y) \Rightarrow Q(y) \geq z].$$

The sign or limit formulas retain the centre as a free fixed parameter. Elimination produces a finite output formula; at algebraic parameter values, root isolation gives the numerical algebraic

outputs. In particular, the quadratic-probe graph and the entrance minimum graph fit this encoding without any projective blow-up representation.

The separate certificate search in Appendix B concerns geometric presentations. Its existence-by-enumeration role is supplementary: the direct outputs above do not wait for that search, and no run of the finite example regression is described as having constructed such a certificate. This separation leaves the geometric proofs and the direct algorithmic conclusions intact.

For the two cubic critical examples, use the coordinate orders

$$(a_0, b_0, X, c_0, w, k, u_f, u_h, v_f, v_h) \quad \text{and} \quad (a_0, b_0, X, c_0, d_0, Y, k, u_f, u_h, v_f, v_h),$$

respectively. At the endpoint, $w = -(d_1 + d_2) \geq 0$; for a first-order representative choose $d_1 = d_2 = -w/2$. The nonnegative coordinates are (a_0, X, w) and (a_0, X, Y, k) . Both displayed specializations have the empty cone support, so $w = 0$ at their endpoint optimum and the endpoint split is uniquely zero. Thus the representative choice introduces no second-order ambiguity into those exact examples.

Let L and D be respectively the seven free score columns and the cone columns, $W = \text{diag}(w_j/P_{je})$, and let B be the base probability velocity. All entries are rational at the stated examples. Set

$$G_L = L^\top W L, \quad b_f = G_L^{-1} L^\top W B, \quad \tilde{D} = D - L G_L^{-1} L^\top W D, \\ Q = \tilde{D}^\top W \tilde{D}, \quad g = \tilde{D}^\top W (B - L b_f), \quad b_* = (B - L b_f)^\top W (B - L b_f).$$

The exact checks give $\text{rank } L = 7$, $\text{rank } Q = 2$, $g < 0$ entrywise and nonnegative principal minors of Q . Every nonempty independent-support KKT test fails, whereas the empty support succeeds. Moreover $z^\top Q z - 2g^\top z > 0$ for every nonzero $z \geq 0$. This proves uniqueness of the cone optimum despite the singular projected Gram matrix. The free optimum is b_f and $\mu^2 = b_*/4$. Multiplication and division of the exact probability jets then give μk by (8.13). For positive rational interval endpoints l, u , the tests for μ and k are respectively

$$l^2 < \mu^2 < u^2, \quad l^2 \mu^2 < (\mu k)^2 < u^2 \mu^2, \quad \mu k > 0.$$

After clearing positive denominators these are integer comparisons. The same exact arithmetic forms and inverts the two Fisher matrices in (15.5)–(15.6). The ancillary record gives the full rational Gram matrices, linear vectors, optimizers and interval tests; reproducibility of these finite operations is separate from the mathematical arguments above.

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PROJECTIVE POLYNOMIAL OBSERVATIONS: REAL WEIGHTED DEGENERATIONS AND IDENTIFICATION WALLS

QIAN QI

ABSTRACT. We study unordered polynomial spectra observed through stochastic channels and an unknown scalar normalization. The joint leading coefficient fibre is identified with the observation-unit section of a positive-real weighted tangent relation. We prove its presentation invariance, filtered naturality and reconstruction from quadratic scalar probes; algebraic initial ideals alone can fail to retain its real liftable part. Within the full binary observation model, two experiments have identical observation orders on every real analytic arc and identical orders for every polynomial scalar function of the target coefficients, but different leading spectral diameters. A relative real atlas supplies finite rational weights, Nash leading diameters and compact-uniform power remainders. For finite exact fibres, remote entrance orders also form finite rational lists in fixed-format semialgebraic families, without a regular-corner hypothesis; every rational order at least one is realizable as format varies. Algebraic tests recover the regular linear regime and its second-order coefficient. We retain the complete multiplicity law for the fixed-degree, full-rank, coprime binary model, the exact rank-deficient cubic fibre, and the distinct weight-wall and endpoint-wall mechanisms. These computations separate local root instability from the entrance of a remote exact target.

1. INTRODUCTION

A singular inverse problem involves more than the orders at which its coordinates vanish. The real relations among their leading coefficients, the norm imposed by observation and the possibility of remote exact representatives all affect the feasible target set. We study these data for a normalized polynomial experiment. Its target is an unordered spectrum, so the object to be determined is a set of root multisets and its bottleneck diameter, rather than independent coefficient bounds.

There are two distinct degenerations. Near an identified datum, multiple roots require different weights on different polynomial coefficients. Near an identification wall, a distant parameter representative can enter the observation ball while the local inverse remains coercive. The article develops a constrained weighted relation for the first phenomenon and a finite-fibre entrance theory for the second. Their connection is semialgebraic: both must retain real feasibility before taking their metric output. Neither is determined by an unconstrained complex algebraic specialization.

The joint relation and its scalar information. At a datum with a unique exact target, write $c_{r,a}$ for the coefficient of degree a in the factor of the target belonging to a root of multiplicity m_r . Put $k_{r,a} = m_r - a$ and

$$R_{je} = \sqrt{w_j}(\sqrt{F(x)_{je}} - \sqrt{P_{je}}), \quad G = \|R\|^2, \quad H_{r,a} = |c_{r,a}|^2.$$

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Real monomial orders give $\alpha = \min_{r,a,i} b_{r,a,i}/(a_i k_{r,a}) = p/q$, where only accessible real divisors contribute. Hold the datum fixed and make the simultaneous substitutions

$$R = \varepsilon^q U, \quad c_{r,a} = \varepsilon^{p k_{r,a}} V_{r,a}.$$

Theorem 6.1 identifies the resulting initial fibre as

$$\mathcal{J}_0 = T_w^+ S \cap \{\|U\| \leq 1\}, \quad \mathcal{C}_0 = \text{pr}_V \mathcal{J}_0,$$

where S is the actual real relation germ and $T_w^+ S$ its positive-real weighted tangent relation. The algebraic weighted deformation is the usual saturated initial construction. What must additionally be retained is its positive real liftable part and its specified observation-unit section. The cusp $y^2 = x^3$ already shows why taking all real points of an algebraic initial fibre is insufficient.

Theorem 6.2 proves invariance under filtered changes of presentation, with the observation norm and target decoding transported. Theorem 6.3 supplies an exact converse to indiscriminate scalar-insufficiency claims: the limiting minima of $\|V - a\|^2$, for rational a , reconstruct the compact joint fibre. Finite nets give quantitative witnesses for its Hausdorff distance from another candidate. Thus specified scalar values can suffice even though orders, or separate coordinate envelopes, do not.

The latter distinction holds inside the principal experiment, not only in a disjoint-label example. Theorem 8.1 gives two interior full-rank binary experiments with all parameters unknown. Their observation orders coincide along *every* real analytic arc in common parameter coordinates. The orders of every polynomial scalar function of the target coefficients also coincide, as do their accessible real divisorial order data. Their squared leading spectral diameters are nevertheless

$$\frac{129609775953}{25280} \quad \text{and} \quad \frac{10058351786055}{1396384}.$$

The proof gives the two exact profiled Fisher ellipsoids. It isolates precisely the information lost on discarding positive leading units. The sharp-marginal separation theorem is retained separately; neither assertion denies reconstruction from the quadratic probes above.

Finite atlases and singular entrances. Theorem 3.4 supplies a finite relative real monomial presentation of the actual constrained graph. Its proof retains the Rees base-change, finite-diagram spreading, real-cover descent and accessibility arguments in Appendix A. The fixed-centre Hausdorff theorem then gives

$$\omega_\theta(t) = C(\theta)t^\alpha + O_H(t^{\alpha+\nu}), \quad C(\theta) = \text{diam}\{(Z_r(V))_r : V \in \mathcal{C}_0(\theta)\},$$

uniformly on compact subsets H of finite Nash strata. The exponent is constant on each stratum and the positive leading diameter is Nash. No leading set is assumed to be quadratic. Hardt triviality supplies topological information; the finite-support and threshold arguments, not triviality, supply the metric remainder.

For a remote representative x_j of a finite exact fibre, define $d_j(\theta, \delta)$ as its least observation distance from a specified moving centre. Theorem 13.1 proves, with no corner or derivative-rank hypothesis, the alternative of eventual exact membership or

$$d_j(\theta, \delta) = a_j(\theta)\delta^{\rho_j} + O_H(\delta^{\rho_j+\nu_j}), \quad \rho_j \in \mathbb{Q}, \quad \rho_j \geq 1,$$

for a C^1 centre path. The exponents form a finite list for fixed format. It also decides all semialgebraic critical thresholds by the exact sign of $t - d_j$, including equality, and determines the limiting union of feasible exact target spectra. This applies the general atlas machinery beyond the direct regular-corner calculation. Theorem 13.2 realizes every rational $\rho \geq 1$ in polynomial probability models as their format varies. Proposition 13.3 derives the regular regime from algebraic rank and cone-separation tests.

The complete binary and cubic laws. The experiment has two degree- d component polynomials with roots in $[0, D]$, positive mixture weights and binary stochastic channels. With coprime components, strictly positive invertible channels and at least $2d + 1$ exterior clocks, Theorem 9.2 determines the whole-model leading set for every multiplicity and boundary pattern. A repeated interior cluster has exponent $1/2$ and leading constant

$$C = \max_j 2\sqrt{(m_j - 1)/m_j} \kappa_j^{-1/4},$$

where κ_j is a projected Fisher-cone minimum. Otherwise the exponent is one. These are fixed-degree, full-rank, coprime binary assertions, with competitors in the entire closed model.

For the rank-deficient cubic family $f = z(z - r)^2$, $g = (z - s)^3$, the exact fibre is governed by

$$B = (3D - r)s - 2rD, \quad E = (1 - \alpha)(r + 3s)(2r - 3s)^2 - 4\pi r^3.$$

The identifying locus is $\{\det U \neq 0\} \cup \{B < 0\} \cup \{E < 0\}$. Theorem 10.1 classifies the entire fibre. The inverse through rank loss uses second-marginal recovery and robust pencil isolation rather than division by $\det U$, B or E . Its local Fisher law remains coercive at both walls.

Theorems 11.4 and 14.2 treat respectively the weight wall and the endpoint wall. Their exact feasible charts are different, but both have linear remote entrance and retain a finite positive local square-root constant. The regular value expansion is

$$d_j(\delta) = \mu_j \delta + k_j \delta^2 + o(\delta^2), \quad k_j = \frac{1}{\mu_j} \min_{v \in M_j} \langle J_j v - b, Q_j(v, v) - c \rangle.$$

The residual is common on the set M_j of first-order minimizers; uniqueness of its preimage is unnecessary. On the nonidentified side, the weight-wall pencil interval opens linearly and may produce a square-root spectral opening, whereas the endpoint-wall interval opens quadratically and its exact spectral diameter remains $D - r$. The algebraic critical examples and both signs of the general second correction remain part of the article.

Relation to existing theories. Resolution, real algebraic geometry and scalar valiative exponent calculations are established inputs [3, 4, 5, 6, 1, 7, 11, 13, 14]. We do not claim a new resolution algorithm. Normal cones and weighted initial degenerations supply the algebraic comparison [12, 18, 15]; real liftability and the observation section are identified explicitly in Theorem 6.1. Subanalytic inner-metric tangent limits [2] address a different metric object from this weighted outer relation with a prescribed target map. Variational geometry and optimization sensitivity [16, 8, 9] provide the setting for the regular-corner calculation. Our singular entrance theorem removes its corner hypothesis rather than claiming that metric projection is itself new. Nonstandard statistical rates in finite mixtures [10] concern estimation of a mixing distribution; the present deterministic whole-model modulus also records a remote exact target at positive spectral distance. Such an entrance must be established from its exact fibre and is not implied by a local statistical rate.

The effective claims used for spectral and entrance outputs are the direct first-order algorithms of Theorem 5.1 and Theorem 13.1. The supplementary geometric certificate search is retained in Appendix B; it is not needed to execute these output algorithms and no complexity or practical resolution implementation is claimed. Appendix C makes this distinction explicit. Exact arithmetic checks certify the displayed finite examples, not the universal quantifiers of the theorems.

2. THE EXPERIMENT AND THE SPECTRAL METRIC

Fix $D > 0$, a degree $d \geq 1$, and $0 < \pi < 1/2$. A closed binary model point consists of two monic real polynomials f_1, f_2 of degree d , all of whose roots lie in $[0, D]$, weights $\alpha_1 + \alpha_2 = 1$ with

$\alpha_a \geq \pi$, and column-stochastic matrices $U = (u_1, u_2)$ and $V = (v_1, v_2)$ with nonnegative entries. Set

$$K(z) = \sum_{a=1}^2 \alpha_a f_a(z) u_a v_a^\top, \quad q(z) = \mathbf{1}^\top K(z) \mathbf{1} = \sum_{a=1}^2 \alpha_a f_a(z), \quad P_j = \frac{K(T_j)}{q(T_j)}. \quad (2.1)$$

The clocks satisfy $D < T_1 < \dots < T_\ell$, and $w_j > 0$, $\sum_j w_j = 1$. Consequently $q(T_j) > 0$, including at every closed-model competitor. The target polynomial is $A = f_1 f_2$. The model is compact in monic coefficients and stochastic entries. Component permutations are identified. The matrices, roots, clocks and design weights are allowed to vary when a family of centres is considered, but every competitor uses its centre's fixed design.

For probability arrays define

$$h_w(P, Q)^2 = \sum_j w_j \sum_e (\sqrt{P_{je}} - \sqrt{Q_{je}})^2.$$

For two multisets $Z = \{z_i\}_{i=1}^n$ and $Z' = \{z'_i\}_{i=1}^n$, put

$$d_\infty(Z, Z') = \min_{\sigma \in \mathfrak{S}_n} \max_i |z_i - z'_{\sigma(i)}|.$$

For real roots this is the maximum discrepancy of their increasing orderings. All diameters in this article are diameters of sets of multisets, not maximal distances from the centre. Write

$$\omega_\theta(t) = \text{diam}\{Z(A_\xi) : h_w(F_T(\xi), P_\theta) \leq t\}. \quad (2.2)$$

A local modulus $\omega_{\theta, \text{loc}}$ restricts the competitors to a specified coefficient neighbourhood of the centre's component pair. This restriction is never implicit.

At a strictly positive datum the Fisher inner product on zero-sum arrays is

$$\langle H, H' \rangle_{I, \theta} = \sum_{j,e} \frac{w_j}{P_{\theta, je}} H_{je} H'_{je}.$$

For component increments p_a of degree below d and stochastic increments $\eta = (\dot{\alpha}_1, \dot{u}_1, \dot{u}_2, \dot{v}_1, \dot{v}_2)$, where channel increments have entry sum zero, define

$$\begin{aligned} \dot{K} &= \sum_a \{ \alpha_a p_a u_a v_a^\top + f_a(\dot{\alpha}_a u_a v_a^\top + \alpha_a \dot{u}_a v_a^\top + \alpha_a u_a \dot{v}_a^\top) \}, \\ \dot{q} &= \sum_a (\alpha_a p_a + \dot{\alpha}_a f_a), \quad S_\theta(p_1, p_2, \eta)_j = \frac{\dot{K}(T_j) - P_{\theta, j} \dot{q}(T_j)}{q(T_j)}. \end{aligned} \quad (2.3)$$

Here $\dot{\alpha}_2 = -\dot{\alpha}_1$. These rational formulas define every Fisher program below in the original experiment coordinates.

Lemma 2.1 (Normalizer recovery). *Suppose f_1, f_2 are coprime, V is invertible, the weights are positive, and $\ell \geq 2d + 1$. The observed array determines q and K uniquely among all closed-model competitors. Their coefficients are locally Lipschitz functions of the observations, with uniform constants on compact sets of such centres. No rank assumption on U is required.*

Proof. Let e have degree below d , and let H be a matrix polynomial of degree at most d satisfying $H(T_j) = e(T_j) P_j$. Interpolation of the polynomial $qH - eK$, of degree at most $2d$, gives $qH = eK$. Its second marginal, multiplied by $\text{diag}(\alpha_1, \alpha_2)^{-1} V^{-1}$, gives $q \mid e f_1$ and $q \mid e f_2$. Coprimality implies $q \mid e$, hence $e = 0$ and $H = 0$.

More explicitly, let Π_T project onto the orthogonal complement of the degree- d Vandermonde column space. The map

$$N_\theta : e \longmapsto \Pi_T(e(T_j) P_{\theta, j})_j, \quad \deg e < d,$$

is injective. For a competitor, subtraction of the normalization equations yields $\|N_\theta(q' - q)\| \leq C \max_j \|P'_j - P_j\|$; the competitor normalizer coefficients are bounded by compactness of the root

interval. The smallest singular value of N_θ is positive. A bounded left inverse recovers $q' - q$, and Vandermonde interpolation recovers $K' - K$. The singular values and the coefficient bounds have positive uniform margins on compact sets of the stated centres. $\square$

Lemma 2.2 (Full-rank coefficient inverse). *Under Lemma 2.1, suppose also that U is invertible. Then every sufficiently close observation has, up to one component permutation, coefficient and stochastic parameter error bounded by a constant times its observation error. Every exact competitor has the same component polynomials and stochastic parameters. The constants are uniform on compact subsets of these centres.*

Proof. Choose two clocks such that $f_1(T_j)/f_1(T_i) \neq f_2(T_j)/f_2(T_i)$. Otherwise the nonzero polynomial $f_2(T_i)f_1 - f_1(T_i)f_2$, of degree at most d , would vanish at all $2d + 1$ clocks. The matrix $K(T_j)K(T_i)^{-1}$ has those two distinct eigenvalues and eigenvectors u_1, u_2 . Normalize the eigenvectors to have entry sum one. Simple-eigenvalue inversion is analytic in a neighbourhood of the centre. Applying the recovered U^{-1} to $K(z)$ and taking row sums recovers $\alpha_a f_a$; leading coefficients and normalized rows recover α_a, V . This construction applies to arbitrary closed-model competitors because their $K'(T_i)$ stays invertible. It also proves exact uniqueness. A finite neighbourhood cover gives compact-uniform constants. Equivalently, the derivative in monic coefficient and stochastic coordinates has a bounded left inverse. $\square$

3. REAL MONOMIAL DATA IN PARAMETER FAMILIES

We work first with a compact semialgebraic model X , a continuous semialgebraic probability map F , and a continuous semialgebraic monic target polynomial A_x of fixed degree n . Complex coefficients are represented by real and imaginary parts. Suppose the target is constant on $F^{-1}(P)$, with value $A_0(z) = \prod_r (z - r)^{m_r}$. Disjoint isolating discs and compactness give, on a sufficiently small closed observation neighbourhood, unique monic cluster factors

$$A_{r,x}(r + z) = z^{m_r} + \sum_{a < m_r} c_{r,a}(x)z^a, \quad k_{r,a} = m_r - a. \quad (3.1)$$

Multiplication of coprime monic factors has invertible differential. The cluster factors are consequently analytic functions of the total coefficients. Their graphs with the isolating conditions are semialgebraic.

Definition 3.1 (Input and families). An algebraic input is a finite first-order polynomial presentation of X, F, A, P , design weights, isolating data, denominator exclusions and branch choices. A real algebraic constant is encoded by an integer polynomial and a rational isolating interval; a complex constant is encoded by two real algebraic constants. A fixed-format family has a semialgebraic parameter set Θ , a semialgebraic $X \subset \Theta \times K$ with K compact and compact fibres, continuous semialgebraic maps F, A on X , and a semialgebraic choice $P_\theta \in F_\theta(X_\theta)$. Degrees and numbers of coordinates are fixed. Effectivity refers to algebraic presentations and numerical outputs to algebraic parameter values. No complexity bound is asserted.

Adjoin nonnegative square-root coordinates to the observation graph and write

$$R_{je} = \sqrt{w_j}(\sqrt{F(x)_{je}} - \sqrt{P_{je}}), \quad G = \sum_{j,e} R_{je}^2, \quad H_{r,a} = (\Re c_{r,a})^2 + (\Im c_{r,a})^2.$$

They are polynomial functions on the resulting compact real graph Γ . Zero probability cells are permitted. No passage to a Fisher quadratic form is used in this construction.

Definition 3.2 (Real monomial presentation). A real monomial presentation is a finite collection of smooth real algebraic sources and proper maps, collectively covering a neighbourhood of $G = 0$

in Γ , with finite algebraic étale coordinate presentations. On each source component either $G \equiv 0$ and all $H \equiv 0$, or

$$G = u \prod_i |y_i|^{a_i}, \quad H_{r,a} \equiv 0 \quad \text{or} \quad H_{r,a} = v_{r,a} \prod_i |y_i|^{b_{r,a,i}}. \quad (3.2)$$

The units have positive upper and lower bounds on smaller closed real coordinate boxes above $G = 0$. A contributing divisor has $a_i > 0$ and a real point away from the other divisors, with an admissible transversal on which $G > 0$. An identically zero H has order $+\infty$; a component $G \equiv 0$ is recorded separately and contributes no ratio. Properness belongs to the source maps, not to an open coordinate box. The finite encoding of an adapted coordinate presentation is specified in Appendix A.

Lemma 3.3 (Spreading a finite resolution certificate). *Let $S = \text{Spec } A$ be an integral affine variety over a characteristic-zero field, let $K = \text{Frac}(A)$, and let $Y \rightarrow S$ be of finite type. Suppose that over K there is a finite reduced closed chain*

$$Y_{0,K} = Y_{K,\text{red}} \supset Y_{1,K} \supset \cdots \supset Y_{q,K} = \emptyset,$$

projective maps $Z_{j,K} \rightarrow Y_{j,K}$ with smooth sources, specified inverse isomorphisms over $Y_{j,K} \setminus Y_{j+1,K}$, and finite smooth-centre principalizations of specified functions on those sources. There is a nonempty open S° on which the following hold on every fibre. The closed chain is flat with geometrically reduced fibres and ends in the empty set; the inverse maps remain inverse over its actual successive complements. Sources and centres are relatively smooth, principalization blow-ups commute with fibre formation, and the final sources have relative normal-crossing divisors with fixed integer monomial orders on a finite adapted étale atlas. Zero identities and inverse units persist. A nonzero monomial cannot become identically zero on a component of a retained fibre. For algebraic input, such finite data and an exceptional set are effectively constructible.

Proof. Lemma A.2 descends the complete diagram, including the actual open complements and both inverse identities, and imposes flatness and geometric reducedness. Lemma A.1 proves the needed base change for all powers of every centre ideal after one localization per finite collection of charts. The explicit bad-locus construction in Appendix A treats relative smoothness, normal-crossing intersections, chart coverage, units and zero flags separately, and proves that each discarded algebraic image has proper closure. The nonzero-specialization assertion follows from relative codimension one and the inverse units, not from generic nonvanishing alone. The constructive certificate specification in Appendix B proves the last assertion. $\square$

Theorem 3.4 (Relative real principalization). *On the exact-target identifying locus of a fixed-format family there is a finite semialgebraic stratification with fixed cluster multiplicities, a fixed finite collection of algebraic source chart presentations, fixed integer orders and fixed real-accessibility flags as in Definition 3.2. The source maps are fibrewise proper real covers. Charts, units and box sizes may vary with the parameter. On compact subsets of a stratum the cover is usable over a common sufficiently small observation neighbourhood. The construction is effective for the algebraic presentations of Definition 3.1.*

Proof. Lemma A.3 algebraizes the proper enlarged graph by square-slack lifts without discarding strict exclusions or real branches. Resolve the reduced generic fibre and its successive reduced singular loci, then principalize the nonzero functions on the smooth sources. Lemma 3.3 spreads this finite certificate. The explicit exceptional algebraic loci in Appendix A are reconstructed on reduced bases of strictly smaller dimension, giving finite termination. Lemma A.4 proves both inclusions in the exact real coverage assertion, including lower-dimensional real loci, and descends the selected label branches. Lemma A.5 supplies the real transversals and the finite real flag partition. That partition is not confused with the algebraic dimension-drop recursion. The final

proof in Appendix A supplies the uniform neighbourhood and unit bounds by properness over compact parameter subsets. These arguments prove the asserted cover on the original graph, rather than on an algebraic relaxation of its inequalities. $\square$

Proposition 3.5 (Fixed-centre real cover). *Every graph Γ above has a real monomial presentation. Exact pieces and lower-dimensional real loci are included. Every recorded divisor has a real admissible attaining arc.*

Proof. Apply Theorem 3.4 to a one-point base and use Lemma A.5. $\square$

Corollary 3.6 (Intrinsic scalar orders). *For each nonrigid cluster coefficient put*

$$\beta_{r,a} = \min_{\text{recorded real divisors}} \frac{b_{r,a,i}}{a_i}, \quad \alpha = \min_{r,a} \frac{\beta_{r,a}}{k_{r,a}}. \quad (3.3)$$

Either the target is constant on an observation neighbourhood, or $\alpha > 0$ is rational and $\omega_P(t) \asymp t^\alpha$. Every finite coefficient minimum is attained by an admissible arc. The numbers are independent of the monomial presentation.

Proof. The complete real divisor list and the monomial inequalities in Lemma A.5 give $H \leq CG^\beta$, hence $|c_{r,a}| \leq Ch_w^\beta$. A minimizing transversal satisfies $G \asymp u^{a_i}$ and $H \asymp u^{b_i}$ and attains the exponent. The inclusion $\{G = 0\} \subseteq \{H = 0\}$ forces positive contributing orders. If all coefficients vanish on an observation neighbourhood, the target is locally rigid. Otherwise, for a monic polynomial of degree m , its greatest root modulus is comparable to $\max_{a < m} |c_a|^{1/(m-a)}$. Elementary symmetric identities give one bound; dominance of z^m outside a fixed multiple of this maximum gives the other. The centre lies in every ball, and the diameter is between its greatest centre displacement and twice that displacement. This proves the order of ω_P . The optimal coefficient inequality is intrinsic, so its attained exponent is independent of the presentation. The scalar divisor-ratio calculation is classical; compare [6]. $\square$

4. JOINT SPECIALIZATION AND UNIFORM SPECTRAL ASYMPTOTICS

On a nonrigid identified stratum write $\alpha = p/q$ in lowest terms. Let $\Psi(\theta, \varepsilon, x, U, V)$ denote graph membership and

$$R_{je}(\theta, x) = \varepsilon^q U_{je}, \quad c_{r,a}(\theta, x) = \varepsilon^{pk_{r,a}} V_{r,a}, \quad \sum_{j,e} U_{je}^2 \leq 1. \quad (4.1)$$

Complex equalities denote two real equalities. Define the joint family by the fixed-centre formula

$$(\theta, U, V) \in \mathcal{J} \iff \theta \in S \text{ and } \forall \rho > 0 \exists \varepsilon, x, U', V' : \quad (4.2)$$

$$0 < \varepsilon < \rho, \quad \Psi(\theta, \varepsilon, x, U', V'), \quad \|(U', V') - (U, V)\|^2 < \rho^2.$$

Put $\mathcal{J}_0(\theta) = \mathcal{J}_\theta$ and $\mathcal{C}_0(\theta) = \text{pr}_V \mathcal{J}_0(\theta)$. There is no quantified θ' in this formula. Closure with a drifting centre is a different operation, and taking independent closures in the coefficient coordinates is also a different operation.

Theorem 4.1 (The definable joint initial family). *The sets $\mathcal{J}_0(\theta)$ and $\mathcal{C}_0(\theta)$ form semialgebraic families of nonempty compact sets. The coefficient images of the radius- t observation ball, scaled by $c_{r,a}/t^{\alpha k_{r,a}}$, converge in Hausdorff distance to $\mathcal{C}_0(\theta)$ for every fixed θ . If*

$$Z_r(V) = Z \left(z^{m_r} + \sum_{a < m_r} V_{r,a} z^a \right), \quad \mathcal{L}_0(\theta) = \{(Z_r(V))_r : V \in \mathcal{C}_0(\theta)\},$$

then the scaled cluster-root images converge to $\mathcal{L}_0(\theta)$, and

$$\omega_\theta(t) = C(\theta)t^\alpha + o(t^\alpha), \quad C(\theta) = \max_{V, V' \in \mathcal{C}_0(\theta)} \min_r d_\infty(Z_r(V), Z_r(V')) \in (0, \infty). \quad (4.3)$$

This is the diameter of a joint feasible image, not that of a Cartesian product of scalar envelopes.

Proof. Corollary 3.6 bounds every scaled coefficient for small t ; U is already bounded. The fixed-centre closure is nonempty and compact. Quantifier elimination proves semialgebraicity with θ as a free parameter. Compactness of the graph fibre lets one pass to convergent subsequences of the unscaled graph variables when projecting an initial point, so projection loses no limiting coefficient point.

Fix θ and a point of the outer coefficient limit. Its distance to the positive-radius coefficient section is a bounded nonnegative semialgebraic function of t , with a subsequence tending to zero. The one-variable monotonicity theorem makes its limit zero. A finite net of the compact outer limit gives the inner Hausdorff inclusion. Failure of the outer inclusion would give a bounded convergent subsequence of scaled coefficients outside the defining limit, a contradiction. This proves convergence of the entire ball. Continuity of monic root multisets on bounded coefficient sets gives the corresponding root-set convergence.

Let Δ be the least separation of distinct base roots. For small t , each cluster lies in its $\Delta/4$ disc. A matching within clusters has error tending to zero; a cross-cluster matching has an edge of length at least $\Delta/2$. An optimal matching therefore stays within clusters. Translation and scaling there are exact. Taking the diameter gives (4.3). An attaining coefficient transversal prevents the limiting diameter from being zero. This also explains why all inequality faces and joint relations in the closure remain relevant. $\square$

Proposition 4.2 (Semialgebraic decay on strata). *Let $e(\theta, t) \geq 0$ be semialgebraic for $0 < t < \tau(\theta)$, with $\tau > 0$ semialgebraic, and suppose $e(\theta, t) \rightarrow 0$ for each fixed θ . After a finite Nash stratification, on every stratum a rational $\nu > 0$ and a positive continuous semialgebraic $\tau_0(\theta)$ satisfy*

$$0 < t < \tau_0(\theta) \implies e(\theta, t) \leq t^\nu.$$

The estimate is uniform on compact subsets; ν need not be sharp.

Proof. Lemma B.1 gives a finite positive rational candidate list from the polynomial supports of a quantifier-free graph formula, including specialized zero coefficients and the eventual-zero case. Choose ν below its positive candidates. The complete argument in Appendix B selects a genuinely positive semialgebraic threshold, makes it continuous, and takes its minimum only on a compact subset of a stratum. No uniform rate is inferred from pointwise convergence alone. $\square$

Theorem 4.3 (Uniform spectral power law). *The identified nonrigid locus has a finite Nash stratification on which α is fixed, $\mathcal{L}_0$ has constant semialgebraic topological type, and $C(\theta)$ is positive and Nash. For every compact subset H of a stratum, for some $\nu > 0$,*

$$d_H(\mathcal{L}_t(\theta), \mathcal{L}_0(\theta)) = O_H(t^\nu), \quad \omega_\theta(t) = C(\theta)t^\alpha + O_H(t^{\alpha+\nu}). \quad (4.4)$$

Here $\mathcal{L}_t$ is the exact clustered root image scaled by $t^{-\alpha}$ and d_H uses the maximum cluster bottleneck metric.

Proof. Represent each root multiset by all its labelled arrays through elementary symmetric identities. The formula

$$M_d(z, z') = \bigvee_{\sigma \in \mathfrak{S}_m} \bigwedge_{i=1}^m |z_i - z'_{\sigma(i)}|^2 \leq d^2 \quad (4.5)$$

expresses bottleneck distance at most d . Use the conjunction over clusters, and real and imaginary parts for complex roots. The two inclusions

$$(\forall z \in \mathcal{L}_t)(\exists z' \in \mathcal{L}_0)M_d(z, z'), \quad (\forall z' \in \mathcal{L}_0)(\exists z \in \mathcal{L}_t)M_d(z, z')$$

and minimality over $d \geq 0$ define the Hausdorff error. Compactness gives attainment. Rational powers of t are encoded by positive algebraic roots, so this error is semialgebraic. Theorem 4.1 proves its pointwise limit zero; Proposition 4.2 gives the compact-uniform power bound.

The diameter has a first-order formula saying that all pairs have distance at most C and one pair attains it. It is therefore semialgebraic and positive. Lemma B.2 gives its Nash refinement. Refine also by positive cluster separation and observation-radius choices, so that compact minima exclude cross-cluster matching uniformly. Hardt triviality supplies the topological types separately. Finally diameters differ by at most twice Hausdorff distance, which gives the second estimate. Topological triviality has supplied neither the error rate nor the Nash dependence. $\square$

Theorem 4.4 (A finite real spectral atlas). *A fixed-format family admits a finite semialgebraic stratification retaining the following data: the semialgebraic type and dimension of the exact target coefficient fibre and its root image; on the identified locus, rigidity or a positive rational exponent, cluster multiplicities, and a fixed real monomial order list with accessibility and minimizing indices; on its nonrigid part, the joint initial coefficient and root-set types, a Nash leading diameter and the compact-uniform power remainder; and finite algebraic descriptions of these objects. On a subfamily with a proved Fisher-cone representation, the active supports and choices of leading constants can be retained in the same refinement. On a nonidentified stratum the modulus tends to the positive diameter of the exact target root image. No general leading set is asserted to be quadratic, and no local leading constant excludes a remote exact component.*

Proof. The exact coefficient fibre is $\{A_x : F_\theta(x) = P_\theta\}$, not the parameter fibre itself. Apply Hardt triviality to its compact image and root graph. Singleton images are selected by a first-order formula. At fixed θ , compactness shows that decreasing observation balls converge in target image to the exact image, so their diameters converge to its diameter. On the singleton locus apply Theorem 3.4, Corollary 3.6, and Theorems 4.1 and 4.3. These are separate constructions entering one finite refinement. A proved Fisher subfamily contributes only finitely many additional support and comparison conditions. Effective descriptions are provided by the following theorem and Appendix B. $\square$

5. EFFECTIVE OUTPUT AND THE ROLE OF COMPUTATION

Theorem 5.1 (Effective algebraic specialization). *For an algebraic input on the identified locus, finite algorithms compute local rigidity, the nonrigid exponent α , quantifier-free presentations of $\mathcal{I}_0, \mathcal{C}_0, \mathcal{L}_0$, and an integer polynomial with a rational isolating interval for C . Exact identification itself is decidable. In algebraically presented parameter families the corresponding strata and formulas are effectively describable. These assertions are independent of any finite example regression.*

Proof. Appendix B gives the complete input–output procedure. It constructs each maximum $m_H(s) = \max\{H : G \leq s\}$ by elimination, extracts a finite support-pair list, and tests the comparability formula (B.1) for every candidate. The unique true candidate selects the actual positive maximum-function branch, while a separate formula detects eventual zero. No unspecified branch-isolation oracle is used. Substitute the resulting p/q in (4.2), eliminate, adjoin roots and use (4.5) to compute the leading diameter. Real-root isolation and field-norm conversion provide its integer polynomial and isolating interval. With free parameters the same finite tests compute strata, followed by Nash refinement. A second, explicit finite-certificate search for the geometric presentations is given in that appendix; spectral outputs do not depend on executing it. $\square$

The symbolic calculations accompanying the explicit polynomial examples test identities and specified finite optimization problems. They do not implement the universal algorithms above and are not proof certificates for relative principalization, uniform asymptotics, or arbitrary degree.

6. THE CONSTRAINED WEIGHTED DEFORMATION

We identify the object of Section 4 before discussing its metric consequences. The datum, the weight filtration and the observation norm are part of the object. A change of parametrization is not a change of any of these three data.

Fix an identified nonrigid datum. Let S be the germ at the origin of the *image* of the actual observation-coefficient relation

$$x \mapsto (R(x), c(x)).$$

All real and imaginary parts are included. The defining inequalities and selected real branches are retained in this image. Set $w = (q, \dots, q, (pk_{r,a})_{r,a})$ when $\alpha = p/q$, and write $D_\varepsilon y = (\varepsilon^{w_i} y_i)_i$. Define

$$\mathfrak{D}_w^+(S) = \overline{\{(\varepsilon, D_\varepsilon^{-1}y) : \varepsilon > 0, y \in S\}}, \quad T_w^+ S = \{v : (0, v) \in \mathfrak{D}_w^+(S)\}. \quad (6.1)$$

Closure is ordinary real closure, locally in the displayed coordinates. Replacing a representative of the germ by a smaller representative does not change its zero fibre. The superscript emphasizes positive real specialization, not a new algebraic deformation operation.

Theorem 6.1 (Comparison and real liftability). *The fixed-centre joint initial fibre satisfies*

$$\mathcal{J}_0 = T_w^+ S \cap \{(U, V) : \|U\| \leq 1\}, \quad \mathcal{C}_0 = \text{pr}_V \mathcal{J}_0. \quad (6.2)$$

Thus it is the observation-unit section of the positive-real weighted tangent relation. It depends on the relation germ, not on a resolution, a square-slack presentation or a proper surjective presentation of that relation.

Suppose an algebraic presentation of the relation has ideal I over $\mathbb{R}$, with its actual real branch conditions specified separately. Form the algebraic weighted deformation by substitution $y_i = \varepsilon^{w_i} v_i$ and saturation by ε . Its zero-fibre ideal is the weight-initial ideal of I , with the minimum-weight convention. The set $T_w^+ S$ consists exactly of those real points of this algebraic zero fibre which have a positive-real semialgebraic lift satisfying the original branch conditions. Merely taking all real points of the algebraic initial fibre can give a strictly larger set, even when I is real prime.

Proof. The forward inclusion in (6.2) follows from the defining substitutions (4.1). For the reverse inclusion take a sequence $(\varepsilon_n, U_n, V_n)$ in the positive deformation with limit $(0, U, V)$ and $\|U\| \leq 1$. If $\|U\| < 1$, it eventually satisfies the unit constraint. At a boundary point put

$$a_n = \max(1, \|U_n\|^{1/q}), \quad \varepsilon'_n = a_n \varepsilon_n.$$

Then $a_n \rightarrow 1$, and the same unscaled relation point has coordinates $U'_n = a_n^{-q} U_n$, $V'_{r,a,n} = a_n^{-pk_{r,a}} V_{r,a,n}$ at ε'_n . These satisfy the unit constraint and have the same limit. This proves the boundary inclusion, which need not follow by interchanging an arbitrary intersection and closure. The coefficient bounds in Corollary 3.6 make this section compact. Image germs are unchanged by the stated surjective presentations, proving the presentation assertion.

Algebraically, contract the ideal generated by $f(D_\varepsilon v)$, $f \in I$, from $\mathbb{R}[\varepsilon, \varepsilon^{-1}, v]$ to $\mathbb{R}[\varepsilon, v]$. This is precisely saturation by ε . For each f , divide $f(D_\varepsilon v)$ by its lowest power of ε ; its value at zero is the minimum-weight part of f . Taking all $f \in I$, or a weight-adapted Groebner basis, shows that the special-fibre ideal is $\text{in}_w I$. This is the usual associated-graded construction, not an additional resolution theorem.

Every real limiting point satisfies the saturated equations. Conversely, a real point is in the constrained zero fibre precisely when it is a limit from the specified positive real part. Semialgebraic curve selection supplies an arc in that part; its positive first coordinate can be used as parameter after a finite Puiseux reparametrization. This proves the liftability characterization. For strictness, take $S = \{(x, y) : y^2 = x^3\}$ at the origin with weights $(1, 1)$. The saturated equation is $v_y^2 = \varepsilon v_x^3$. Its algebraic zero fibre has real points $v_y = 0$, with either sign of v_x . Positive real liftability

instead requires $v_x \geq 0$, and every such point is realized. The ideal $(y^2 - x^3)$ is real prime: its coordinate ring embeds in $\mathbb{R}[t]$ by $(x, y) = (t^2, t^3)$. Thus even real primality does not recover the missing accessibility condition from the initial ideal alone. $\square$

For weight one the algebraic construction is the tangent-cone instance of the normal-cone construction; see [12, 18]. Weighted initial ideals are treated in [15]. These algebraic objects do not, by themselves, specify the positive real liftable part or the observation-unit section. Inner-metric tangent limits, as in [2], retain different data: here one rescales a constrained *outer* relation and subsequently applies the specified root map. No identification with an inner-metric tangent cone is asserted.

Theorem 6.2 (Filtered naturality). *Let ϕ be a semialgebraic homeomorphism between two relation germs, fixing the origin. Suppose, locally uniformly on bounded sets,*

$$\phi_\varepsilon = D_\varepsilon^{-1} \phi D_\varepsilon \longrightarrow \phi_0, \quad (\phi^{-1})_\varepsilon \longrightarrow \phi_0^{-1},$$

where ϕ_0 is a homeomorphism. Then $T_w^+ \phi(S) = \phi_0(T_w^+ S)$. If the observation block of ϕ_0 preserves its norm, the corresponding unit sections are also carried onto each other. The same statement holds with the norm transported by ϕ_0 . With the target decoding transported at the same time, the leading spectral image and its diameter are unchanged.

Proof. For a bounded convergent sequence of rescaled relation points v_n , uniform convergence on a compact neighbourhood gives $\phi_{\varepsilon_n}(v_n) \rightarrow \phi_0(\lim v_n)$. This proves one inclusion of tangent relations. Apply the inverse hypothesis for the other. Theorem 6.1 then identifies their unit sections; the last assertion is equality of the decoded images. For example, weighted analytic coordinate changes whose invertible leading graded map is ϕ_0 and whose remaining terms have strictly larger weight satisfy the stated convergence, as do their filtered inverses. $\square$

The transport of the metric is essential. Equivalent observation norms preserve optimal orders, but need not preserve a leading diameter. Section 8 gives this distinction inside the full binary experiment, without adding a discrete experimental label.

6.1. Exactly which scalar probes recover the joint set. Write C_t for the coefficient section scaled as in Theorem 4.1. For $a \in \mathbb{R}^N$ let

$$m_t(a) = \min_{v \in C_t} \|v - a\|^2. \quad (6.3)$$

The dimension N includes both parts of complex coefficients.

Theorem 6.3 (Quadratic reconstruction and finite witnesses). *The limits $m_0(a) = \lim_{t \downarrow 0} m_t(a)$ exist locally uniformly in a and satisfy*

$$m_0(a) = \text{dist}(a, \mathcal{C}_0)^2, \quad \mathcal{C}_0 = \bigcap_{a \in \mathbb{Q}^N} \{v : \|v - a\|^2 \geq m_0(a)\}. \quad (6.4)$$

In particular these scalar probes, unlike their vanishing orders alone, determine the joint coefficient set and hence its decoded spectral set. For compact nonempty sets C, D in a fixed compact ball B , and a finite η -net $A \subset B$ (with an arbitrarily slightly enlarged ball if rational points are required),

$$\max_{a \in A} |\text{dist}(a, C) - \text{dist}(a, D)| \leq d_H(C, D) \leq \max_{a \in A} |\text{dist}(a, C) - \text{dist}(a, D)| + 2\eta. \quad (6.5)$$

For algebraic input one quantifier-free formula with a free describes $m_0(a)$. Distinct initial fibres therefore admit an effectively searchable rational probe with unequal limiting values.

Proof. For compact sets the distance functions differ uniformly by at most their Hausdorff distance. Squaring is uniformly Lipschitz on bounded distance intervals. Theorem 4.1 therefore proves the first assertion. A point of $\mathcal{C}_0$ satisfies every inequality in (6.4). If $v \notin \mathcal{C}_0$, put $d = \text{dist}(v, \mathcal{C}_0) > 0$ and choose rational a with $\|a - v\| < d/3$. Then $\text{dist}(a, \mathcal{C}_0) > 2d/3 > \|a - v\|$, excluding v .

The left inequality in (6.5) is the distance-function bound. The Hausdorff distance is attained as $\text{dist}(x, D)$ for some $x \in C$, or with C, D exchanged. At such an x it equals $|\text{dist}(x, C) - \text{dist}(x, D)|$. Each distance function is one-Lipschitz; replacing x by a net point proves the other inequality.

Finally, membership in C_t , minimization in (6.3), and the limit are first-order formulas over a real closed field, with a free. Theorem 5.1 and quantifier elimination yield one finite formula for their limit graph. If $C \neq D$, the strict separating inequality above persists on an open set of probe centres. Enumeration of rational points, with exact comparison of the resulting algebraic values, must find a witness. The enumeration concerns a witness only, not an approximation to the whole set by scalar coordinate envelopes. $\square$

All affine support values determine only a closed convex hull. Quadratic distance probes determine the actual compact set, including its nonconvex real faces. The reconstruction theorem thus gives a precise sufficiency statement to accompany, rather than overstate, the insufficiency statements of Sections 7 and 8.

7. SCALAR ENVELOPES DO NOT DETERMINE THE SPECTRAL INVARIANT

The distinction between the scalar orders and the joint initial fibre can be made exact. The following examples have not only the same optimal orders but the same individual coefficient envelopes at every sufficiently small radius. Their leading spectral diameters are nevertheless different. The assertion concerns the scalar data of Corollary 3.6 and separate coefficient envelopes; it does not concern the collection of value functions of all possible combinations of the coefficients.

Theorem 7.1 (Separation from marginal scalar data). *There are two compact semialgebraic probability experiments with a unique exact target, real quadratic targets with roots in $[0, 1]$, and identical coefficient orders and individual coefficient envelopes, for which*

$$\omega_A(t) = (1 + \sqrt{5})t, \quad \omega_B(t) = \frac{1 + \sqrt{17}}{2}t \quad (0 \leq t \leq 1/20).$$

In both experiments the coefficient orders are $(\beta_1, \beta_2) = (1, 2)$ and the spectral exponent is one. On each source arc the orders of (G, H_1, H_2) are $(2, 2, 4)$. Thus the order lists and the separate optimal coefficient envelopes do not determine the joint initial root set or its diameter.

Proof. Put $r = 1/2$ and $e = 1/20$. In each experiment the source is the disjoint union of two closed intervals $0 \leq \rho \leq e$. Its probability map and datum are

$$F(\rho) = ((1 - \rho^2/2)^2, \rho^2(1 - \rho^2/4)), \quad P_0 = (1, 0).$$

The entries are nonnegative and sum to one. Since $\sqrt{F_1} = 1 - \rho^2/2$ on this interval,

$$h(F(\rho), P_0)^2 = 2 - 2\sqrt{F_1} = \rho^2.$$

Associate with an interval labelled (u, v) the target

$$A_{u,v,\rho}(z) = (z - r)^2 + u\rho(z - r) + v\rho^2.$$

Use the label sets

$$\mathcal{E}_A = \{(1, -1), (2, -4)\}, \quad \mathcal{E}_B = \{(1, -4), (2, -1)\}.$$

These are compact semialgebraic experiments in the sense of Definition 3.1; the discrete label is simply an additional source coordinate. All exact targets are $(z - r)^2$. The two root increments on an interval are

$$\rho \ell(u, v), \quad \rho q(u, v), \quad \ell(u, v) = \frac{-u - \sqrt{u^2 - 4v}}{2} < 0 < q(u, v) = \frac{-u + \sqrt{u^2 - 4v}}{2}. \quad (7.1)$$

They lie in $[0, 1]$ after translation by r , since the greatest absolute increment is at most $(1 + \sqrt{5})/20 < 1/2$.

Write $c_1 = u\rho$ and $c_0 = v\rho^2$. In either experiment the exact marginal images over the radius- t ball are

$$\{c_1\} = [0, 2t], \quad \{c_0\} = [-4t^2, 0].$$

In particular $\max |c_1| = 2t$ and $\max |c_0| = 4t^2$, not just to leading order. Moreover $G = \rho^2$, $H_1 = u^2\rho^2$, and $H_2 = v^2\rho^4$. The order assertions follow. Even the unordered lists of individual squared units agree: they are $\{1, 4\}$ and $\{1, 16\}$. Their association across the two coordinates is different.

For $t > 0$, put $\lambda = \rho/t$. The scaled coefficient image is already independent of t :

$$\mathcal{C}_{0,E} = \bigcup_{(u,v) \in \mathcal{E}_E} \{(u\lambda, v\lambda^2) : 0 \leq \lambda \leq 1\}, \quad E \in \{A, B\}. \quad (7.2)$$

This is its joint initial coefficient set. The ordered root image consists of $(\lambda\ell(u, v), \lambda q(u, v))$. Its first coordinate is nonpositive, its second nonnegative, and the zero vector belongs to it. The diameter in the maximum metric is the larger of its two coordinate widths. Since $u > 0$, $-\ell(u, v) > q(u, v)$ on each interval. Therefore

$$C_E = \max_{(u,v) \in \mathcal{E}_E} \frac{u + \sqrt{u^2 - 4v}}{2}.$$

For A the maximum is $1 + \sqrt{5}$. For B it is $(1 + \sqrt{17})/2$, which exceeds $1 + \sqrt{2}$. The inequalities are strict and these two constants differ. Translation and multiplication by t in (7.1) prove the asserted exact diameter identities. No product approximation of (7.2) has been used. $\square$

The example isolates the information carried by the joint specialization: relations between weighted coefficients survive both real closure and the root map. A finite list of scalar exponents cannot recover these relations, even when augmented by the separate sharp envelopes. Conversely the full joint set determines the diameter by Theorem 4.1; no new scalar resolution theorem is required for this distinction.

8. IDENTICAL SCALAR ORDERS IN THE BINARY EXPERIMENT

The example in Section 7 keeps sharp marginal ranges fixed. We now keep a different, much larger valiative object fixed inside the full experiment of Section 2: all real analytic arc orders of the observation discrepancy and every polynomial scalar function of the target coefficients. Leading units are not included in this datum. The distinction is necessary, since Theorem 6.3 shows how richer scalar values recover the joint set.

Take degree one, $D = 1$, $\pi = 1/10$, clocks $(2, 3, 4)$ and weights $1/3$. Consider the two strictly interior full-rank centres θ_e , with $e \in \{1/4, 1/8\}$, given by

$$f = z - \frac{1}{4}, \quad g = z - \frac{3}{4}, \quad \alpha = \frac{1}{2}, \quad U_e = \begin{pmatrix} \frac{1}{2} + e & \frac{1}{2} - e \\ \frac{1}{2} - e & \frac{1}{2} + e \end{pmatrix}, \quad V = \begin{pmatrix} \frac{1}{4} & \frac{3}{4} \\ \frac{3}{4} & \frac{1}{4} \end{pmatrix}. \quad (8.1)$$

All competitors still range over the original closed binary model; no parameter is held known. In a common small open coordinate ball write

$$z = (\dot{r}, \dot{s}, \dot{\alpha}, \dot{u}_1, \dot{u}_2, \dot{v}_1, \dot{v}_2).$$

The first entries of the binary channel columns are the independent channel coordinates. Let $G_e(z)$ be squared Hellinger discrepancy from θ_e , and let $\Delta c(z)$ be the two target coefficient increments. The latter polynomial map is the same at both centres.

Theorem 8.1 (Valuative equivalence without metric equivalence). *For every nonconstant real analytic arc $z(\tau)$ through zero,*

$$\text{ord}_\tau G_{1/4}(z(\tau)) = \text{ord}_\tau G_{1/8}(z(\tau)) = 2 \min_i \text{ord}_\tau z_i(\tau). \quad (8.2)$$

For every polynomial H in the target coefficient increments, including all mixed polynomial combinations, the orders of $H(\Delta c(z(\tau)))$ agree at the two centres, with the same convention for identically zero functions. The corresponding accessible real divisorial order data can be computed on one common real monomial presentation and agree as well.

Nevertheless the two whole-model leading spectral sets are distinct. Their exponents are one, and their squared leading diameters are

$$C_{1/4}^2 = \frac{129609775953}{25280}, \quad C_{1/8}^2 = \frac{10058351786055}{1396384}. \quad (8.3)$$

Thus the complete scalar arc-order information just specified does not determine the joint metric initial fibre, even within the principal binary observation model.

Proof. Both channels are invertible and the component polynomials are coprime. Lemma 2.2 implies that the observation derivative in these seven coordinates is injective. Since all probability cells are positive, the square-root embedding is analytic and

$$G_e(z) = \frac{1}{4} z^\top H_e z + O(\|z\|^3), \quad H_e > 0.$$

It follows either by this expansion or by two-sided comparison with $\|z\|^2$ that (8.2) holds. The coefficient functions are literally identical in the common coordinates, proving their order assertion for every H .

For the divisorial assertion, first blow up the maximal ideal at zero. On every real affine chart the pullback of G_e is the square of an exceptional coordinate times a strictly positive analytic unit near the real exceptional locus. Indeed the leading factor is the positive quadratic form H_e evaluated on a nonzero real projective direction. These units remain nonzero under every further real modification. Principalize any finite specified collection of the common coefficient functions on this same space. The exceptional orders and their real accessibility are the same for both experiments. Alternatively every accessible divisor has a real transversal, so its observation order already follows from (8.2). The assertion concerns real accessible data; no equality of complex zero divisors of the two positive quadratic forms is used. Every additional finite collection of polynomial combinations is handled in the same way.

We compute the metric sets. Let S_e be the 12-by-7 score matrix from (2.3), at the three clocks, in the coordinate order above. Set

$$H_e = S_e^\top \text{diag}((3P_{e,je})^{-1}) S_e, \quad M_e = (H_e^{-1})_{\{1,2\},\{1,2\}}.$$

Profiling the five free nuisance coordinates in the positive definite quadratic form shows that the leading pair of root increments is exactly

$$L_e = \{x \in \mathbb{R}^2 : x^\top M_e^{-1} x \leq 4\}. \quad (8.4)$$

For completeness, the lower bound on G_e bounds all competitor coordinates by $O(t)$. Division by t and Taylor expansion give the outer inclusion in (8.4). Every vector in its interior is realized by its linear parameter arc, which is feasible because the centre is interior. Boundary vectors are limits of their radial contractions. This proves Hausdorff convergence. Exact uniqueness and compactness, from Lemma 2.2, exclude competitors outside this coordinate neighbourhood at sufficiently small radii. The limit is consequently a whole-model limit.

Here the entries of the matrix are obtained by differentiating the explicit rational functions

$$P_{j,ab} = \frac{\alpha(T_j - r)u_{1a}v_{1b} + (1 - \alpha)(T_j - s)u_{2a}v_{2b}}{\alpha(T_j - r) + (1 - \alpha)(T_j - s)}.$$

Rational elimination gives

$$M_{1/4} = \frac{1}{404480} \begin{pmatrix} 129609775953 & 79361335413 \\ 79361335413 & 48691215153 \end{pmatrix}, \quad (8.5)$$

$$M_{1/8} = \frac{1}{22342144} \begin{pmatrix} 10058351786055 & 6157532670525 \\ 6157532670525 & 3781629625095 \end{pmatrix}. \quad (8.6)$$

These equalities can be checked solely by forming H_e from the preceding rational display and multiplying its inverse; no fitted numerical covariance is involved. For separated real roots the bottleneck metric is the maximum norm of their ordered increments. A centrally symmetric ellipsoid (8.4) has maximum-norm diameter $4 \max_i \sqrt{(M_e)_{ii}}$. The first diagonal entry is the larger in each displayed matrix. This proves (8.3). The values are unequal, so neither the leading root sets nor their coefficient images can be equal. $\square$

The example fixes all orders of the target scalar functions, not their sharp leading values. Those values contain the positive units erased by valuation. It therefore complements the sharp-marginal example without claiming that arbitrary scalar information is insufficient. The relative real atlas remains useful for finding the common weights; its units and its joint relation are necessary for recovering the metric section at those weights.

9. AN EXPLICIT MULTIPLICITY AND BOUNDARY ATLAS

Fix d, D, π and $\ell \geq 2d + 1$. Let $\mathfrak{M}_d$ be the centres of (2.1) with strict weights, strictly positive invertible U, V , and coprime component polynomials. No restriction is imposed on root multiplicities or on contact with $0, D$. Competitors still range over the entire closed model. Partition $\mathfrak{M}_d$ by the component assignment, multiplicity and endpoint/interior status of each distinct root. At fixed d this is a finite semialgebraic partition. On a pattern stratum write r_j, m_j for its distinct roots and multiplicities, and $a(j) \in \{1, 2\}$ for their component assignments.

For every cluster let s_j denote its first-order sum displacement. It is free at an interior root, nonnegative at 0 , and nonpositive at D . For every repeated interior cluster let $X_j \geq 0$. Define

$$p_a(z) = - \sum_{j:a(j)=a} s_j \frac{f_a(z)}{z - r_j} - \sum_{\substack{j:a(j)=a \\ r_j \in (0,D), m_j \geq 2}} X_j \frac{f_a(z)}{(z - r_j)^2}. \quad (9.1)$$

Insert these polynomials into (2.3). Let L contain the free interior sum and stochastic columns. Let D_0 contain the nonnegative endpoint sum columns (reverse the sign at D) and all X_j columns. With $W = \text{diag}(w_j/P_{je})$, define the projected Gram matrix

$$Q = D_0^T W D_0 - D_0^T W L (L^T W L)^{-1} L^T W D_0. \quad (9.2)$$

For a repeated interior cluster j , let

$$\kappa_j = \min\{v^T Q v : v \geq 0, v_{X_j} = 1\}. \quad (9.3)$$

All matrix entries are rational in the root, channel, weight and design coordinates. Empty collections of free or cone columns have their usual zero-dimensional meanings.

Lemma 9.1 (Ordered cluster diameter). *For $m \geq 2$, the set*

$$\{y_1 \leq \cdots \leq y_m : \sum_i y_i = 0, \frac{1}{2} \sum_i y_i^2 \leq X\}$$

has diameter $\sqrt{2X(m-1)/m}$ in the maximum coordinate metric.

Proof. Fix a coordinate j . Minimizing the squared norm under its value, ordering and zero sum gives

$$\max y_j = \sqrt{\frac{2X(j-1)}{m(m-j+1)}}, \quad \min y_j = -\sqrt{\frac{2X(m-j)}{mj}}.$$

For the maximum, the last $m-j+1$ coordinates are equal and the first $j-1$ are equal; the minimum is analogous. This also covers the zero endpoint extrema when $j=1$ or $j=m$. Put $a=j-1, b=m-j$. For $a, b \geq 1$, the assertion that the resulting coordinate width is at most $\sqrt{2X(m-1)/m}$ is equivalent to

$$ab(a+b+2)^2 \geq 4(a+1)(b+1).$$

Writing $S = a+b \geq 2$ and $ab \geq 1$, its difference is $ab(S^2 + 4S) - 4S - 4 \geq S^2 - 4 \geq 0$. For $a=0$ or $b=0$ equality holds directly. An extreme vector with one coordinate of absolute value $\sqrt{2X(m-1)/m}$ and the other $m-1$ coordinates equal attains the endpoint width against the zero vector. The diameter of a subset in the maximum metric is its largest coordinate width. This proves the formula, including possible simultaneous constraints on other clusters when it is used below. $\square$

Theorem 9.2 (Multiplicity and boundary laws). *Every centre in $\mathfrak{M}_d$ is exactly identifying. On each root pattern stratum the following whole-model assertions hold uniformly on compact subsets H .*

(a) *If there is a repeated interior cluster, the exponent is $1/2$. The complete leading root set consists of zero clusters at the simple and boundary roots and ordered vectors $y_j \in \mathbb{R}^{m_j}$ at the repeated interior roots satisfying*

$$\sum_i y_{j,i} = 0, \quad X_j = \frac{1}{2} \sum_i y_{j,i}^2, \quad \min_{\text{endpoint sums} \geq 0} v^\top Q v \leq 4. \quad (9.4)$$

The X_j in v are the displayed variances. After grouping and scaling roots by $\sqrt{t}$, the observation ball is within $O_H(\sqrt{t})$ of this set in Hausdorff distance. In particular

$$\omega_\theta(t) = C(\theta)\sqrt{t} + O_H(t), \quad C(\theta) = \max_j 2\sqrt{\frac{m_j-1}{m_j}} \kappa_j^{-1/4}. \quad (9.5)$$

Every leading point, including diameter-extremizing pairs, is realized by stochastic alternatives.

(b) *If no interior root is repeated, the exponent is 1. For each simple interior cluster take a free real root increment, and for each boundary cluster take an ordered vector of increments with all entries nonnegative at 0 or nonpositive at D . Let s_j be their sums and use (9.1) without X terms. The compact set of these root increments with*

$$\min_\eta \|S_\theta(p_1, p_2, \eta)\|_{I, \theta}^2 \leq 4 \quad (9.6)$$

is the complete leading set $\mathcal{L}_{\text{lin}, \theta}$. Its diameter $C_{\text{lin}}(\theta)$ is positive, and

$$\omega_\theta(t) = C_{\text{lin}}(\theta)t + O_H(t^2).$$

The scaled-root Hausdorff error is $O_H(t)$.

(c) *The optimal coefficient orders of a cluster of multiplicity m , with coefficient weight k , are*

$$\beta_k = \begin{cases} 1, & r \in (0, D), \quad k = 1, \\ k/2, & r \in (0, D), \quad 2 \leq k \leq m, \\ k, & r \in \{0, D\}, \quad 1 \leq k \leq m. \end{cases} \quad (9.7)$$

The real order list, boundary and multiplicity pattern, and Fisher active supports admit a common finite semialgebraic refinement. On its square-root strata C^4 is a selected rational function of the root and experiment coordinates; the positive constants on the linear strata are algebraic semialgebraic functions. Both are Nash after refinement. Exact algebraic evaluation uses finite programs.

Proof. Lemma 2.2 gives exact identification and a uniform inverse in coefficients and stochastic parameters. The partial fractions $1/(z - r_j)$ and $1/(z - r_j)^2$ occurring in (9.1) are linearly independent at distinct roots, with their indicated pole orders. The coefficient derivative, and hence its score, is injective on these variables and the free stochastic coordinates. Thus L has full column rank and Q is positive definite whenever it is nonempty. The constrained minima exist, are positive, and have uniformly bounded minimizers on H .

We first obtain coordinates for *all* nearby competitors. Coefficient recovery and analytic coprime cluster factorization give an $O_H(t)$ error in each cluster coefficient and in the stochastic parameters. For an interior cluster, write its exact roots as

$$r_j + ts_j/m_j + \sqrt{t} y_{j,i}, \quad \sum_i y_{j,i} = 0.$$

The first coefficient bounds s_j , and the second bounds $\sum_i y_{j,i}^2$, since for centred real roots their second elementary symmetric function equals $-\frac{1}{2} \sum_i y_{j,i}^2$. At an endpoint all root displacements have the same sign. Their sum is $O_H(t)$, so each is $O_H(t)$. These observations prove the upper bounds in (9.7). For the lower bounds, translate an interior cluster by a nonzero order- t amount for $k = 1$; for $k \geq 2$ split it by $\sqrt{t}$ times a zero-sum real vector with nonzero k th elementary symmetric function. The vector with $m - 1$ equal nonzero coordinates and the last their negative sum has this property for every $k \geq 2$. Its first nonzero coefficient is of order t and the coefficient inverse makes the observation distance comparable to t . At an endpoint displace all roots by the same allowed order- t amount. This proves all lower orders, including the one-sided boundary case. In root-arc parameter ε , the interior split has $\text{ord } G = 4$, $\text{ord } H_k = 2k$, whereas translation or an endpoint arc has orders $2, 2k$. These are admissible witnesses for the real order data, and (9.7) rules out any smaller candidate.

For part (a), expansion of the exact root coordinates gives precisely (9.1) at order t . Terms involving three centred root displacements are $O_H(t^{3/2})$; all higher ones and the bounded mean and endpoint terms satisfy the same bound. Consequently

$$F_T(\xi) = P_\theta + tS_\theta + O_H(t^{3/2}), \quad h_w(F_T(\xi), P_\theta) = \frac{t}{2} \|S_\theta\|_{I,\theta} + O_H(t^{3/2}). \quad (9.8)$$

This follows by Taylor expansion of the square-root probability vector and the reverse triangle inequality; it does not require a nonzero score. Membership in the ball implies $\|S_\theta\| \leq 2 + M\sqrt{t}$. Multiply all sum, variance and stochastic first-order variables by $\rho = 2/(2 + M\sqrt{t})$ and each centred root vector by $\sqrt{\rho}$. This gives a point of (9.4) within $O_H(\sqrt{t})$ of the scaled competitor. Endpoint and simple clusters contribute only $O_H(\sqrt{t})$ in these scaled coordinates.

Conversely take a point of (9.4) and bounded minimizing endpoint and free variables. Use the displayed interior root arcs; at an endpoint realize its sum by equal one-sided displacements, and use the prescribed order- t stochastic increments. Scale first-order variables by $\rho = 1 - M'\sqrt{t}$ and centred roots by $\sqrt{\rho}$. Strict weight and channel margins and cluster separation on H make these genuine closed-model alternatives. For a sufficiently large M' , (9.8) puts them inside the observation ball. This is the reverse Hausdorff inclusion, with the same uniform error.

Homogeneity gives $X_{j,\max} = 2/\sqrt{\kappa_j}$. The marginal projection of the leading set at cluster j includes every ordered zero-sum vector with squared norm at most $2X_{j,\max}$: scale a minimizing feasible vector for $X_j = 1$ and choose the desired shape in that cluster. Lemma 9.1 therefore

gives its diameter $2\sqrt{(m_j - 1)/m_j \kappa_j^{-1/4}}$. The diameter in the maximum cluster metric is the largest marginal diameter; no independence of the clusters is used. Cluster separation excludes cross-cluster optimal matchings. Diameters differ by at most twice Hausdorff distance, proving (9.5).

For part (b), every root displacement is $O_H(t)$ by the same sum argument at the endpoints and the simple-root inverse in the interior. Divide the exact displacements by t . They belong to the stated polyhedral sign and ordering cone. Taylor expansion now has remainder $O_H(t^2)$. The first-order score controls their sums and all stochastic variables; on the one-sided endpoint cones the sums control the individual displacements. Hence the root set in (9.6) is compact. The radial argument with $\rho = 1 - Mt$ proves both Hausdorff inclusions with error $O_H(t)$. An allowed nonzero displacement gives positive diameter. This proves (b).

For (c), enumerate the subsets of the cone coordinates other than a fixed X_j . On a support A , put $v_A = -Q_{AA}^{-1}Q_{Aj}$ and require $v_A \geq 0$ and $Q_{bj} + Q_{bA}v_A \geq 0$ off the support. Its value is $Q_{jj} - Q_{jA}Q_{AA}^{-1}Q_{Aj}$. Convexity proves necessity and sufficiency. All tests and selected values are rational, and

$$C^4 = \max_j \frac{16(m_j - 1)^2}{m_j^2 \kappa_j}.$$

There are finitely many patterns and supports. For the linear law, (9.6) is an explicit quadratic inequality on a polyhedral root cone after projecting the free stochastic variables. Its coordinate maxima and minima, and hence its diameter, have finite real algebraic descriptions by (4.5) or by face optimization. Quantifier elimination gives the algebraic value at algebraic input. Stratify the resulting semialgebraic functions and refine with the order lists of Theorem 4.4. Positivity and compact-uniform coercivity ensure continuity within each fixed pattern; Nash refinement gives the final assertion. No continuity of C is asserted across a change of scaling exponent. $\square$

Example 9.3 (A multiplicity crossover). Let $x \in [0, b]$, $b < 1/2$, take $F(x) = (1/2 + x, 1/2 - x)$, and target $A_x(z) = (z - r)^2 - x$. At a fixed centre $a \in (0, b)$, differentiation of the square-root probability vector gives

$$h(F(a + u), F(a)) = \frac{|u|}{\sqrt{1 - 4a^2}} + O_a(u^2).$$

The admissible x interval therefore has endpoints $a \pm t\sqrt{1 - 4a^2} + O_a(t^2)$, and its spectral diameter is

$$\omega_a(t) = \frac{\sqrt{1 - 4a^2}}{\sqrt{a}} t + O_a(t^2).$$

At $a = 0$ it instead has leading law $\sqrt{t}$. More precisely, uniformly for λ in compact subsets of $[0, \infty)$,

$$\frac{\omega_{\lambda t}(t)}{\sqrt{t}} \rightarrow \sqrt{\lambda + 1} - \sqrt{\max(0, \lambda - 1)}. \quad (9.9)$$

Indeed the feasible interval endpoints are $t \max(0, \lambda - 1) + O(t^2)$ and $t(\lambda + 1) + O(t^2)$, with truncation at zero, and $x \mapsto Z(A_x)$ has ordered roots $r \pm \sqrt{x}$. This example belongs to the general real-graph class of Section 3; it is not a claim about an unobserved restriction of the closed binary model. Here the linear constant really diverges at a multiplicity wall. Section 11 gives a different mechanism.

10. A RANK-DEFICIENT CUBIC STRATUM

Let $\mathfrak{F}$ denote the degree-three centres with

$$f = z(z - r)^2, \quad g = (z - s)^3, \quad 0 < s < r < D,$$

strict weights $\pi < \alpha < 1 - \pi$, strictly positive U, V , $\det V \neq 0$, and at least seven clocks. The first channel may have either rank. Put $\gamma = 1 - \alpha$ and

$$A = (r + 3s)(2r - 3s)^2, \quad B = (3D - r)s - 2rD, \quad E = \gamma A - 4\pi r^3. \quad (10.1)$$

Here A is a scalar, not the target polynomial.

Theorem 10.1 (The entire cubic fibre). *The exact spectral fibre is a singleton precisely on*

$$\mathfrak{I} = \{\det U \neq 0\} \cup \{B < 0\} \cup \{E < 0\}. \quad (10.2)$$

On its complement, besides (f, g) all component-polynomial pairs are, up to permutation,

$$(f, h_c), \quad h_c = (1 - c)f + cg, \quad c_* \leq c \leq \min(c_D, \gamma/\pi), \quad (10.3)$$

where

$$c_* = \frac{4r^3}{A}, \quad b = \frac{rs}{3s - 2r}, \quad R(z) = \frac{f(z)}{g(z)}, \quad c_D = \frac{R(D)}{R(D) - 1}.$$

The interval is nonempty even at equality in either constraint, and every one of its pairs has a positive stochastic realization. Conditional on the component pair, that realization is unique.

Proof. Lemma 2.1 fixes q, K for every competitor. If U is invertible, Lemma 2.2 applies; in particular $\det K / \det K_3 = fg$ gives spectral identification.

If $\det U = 0$, stochasticity gives $u_1 = u_2 = u$. The second marginal $V \operatorname{diag}(\alpha, \gamma)(f, g)^\top$ has coefficient rank two. Every competitor therefore has invertible second channel and its two monic polynomials lie in the affine pencil h_c . Expansion gives

$$\operatorname{disc}(h_c) = c(c - 1)^2(r - s)^3(4r^3 - Ac), \quad h_c(0) = -cs^3, \quad (10.4)$$

and

$$\frac{R'(z)}{R(z)} = \frac{(2r - 3s)z + rs}{z(z - r)(z - s)}, \quad A = 4r^3 - 27s^2(r - s) < 4r^3.$$

Nonnegative roots exclude $c < 0$, and the discriminant excludes $0 < c < 1$. For $c > 1$ there is a root in (s, r) and no root in $[0, s]$. The other two admissible roots must be above r , solving $R(z) = c/(c - 1) > 1$. They exist only for $s > 2r/3$. In that case R increases to its maximum at b and then decreases to one. The double-root value is $c_* > 1$; the larger root increases from b to D as c increases from c_* to c_D . Thus the whole real-rooted pencil in $[0, D]$ is $\{0, 1\}$, with $[c_*, c_D]$ added exactly when $b \leq D$.

For a competing pair with pencil coordinates c_1, c_2 , the weights obey $\alpha'_1 c_1 + \alpha'_2 c_2 = \gamma \in (0, 1)$. One coordinate must be zero and the other is one or a value $c \geq c_*$. The latter weight is γ/c , so the floor imposes $c \leq \gamma/\pi$; the other weight is automatically at least $\alpha > \pi$. Conversely put

$$\beta_f = \alpha v_1 + \gamma(1 - 1/c)v_2, \quad \beta_h = \gamma v_2/c.$$

Both vectors are positive and $\beta_f f + \beta_h h_c = \alpha v_1 f + \gamma v_2 g$. Their entry sums are the stated weights; normalizing them and keeping first channel (u, u) gives the realization. Coefficient functionals dual to the independent polynomials recover the two rank-one coefficient matrices uniquely, hence their weights and normalized marginals. Finally $b \leq D$ is equivalent to $B \geq 0$ and, on this set, $c_* \leq \gamma/\pi$ is equivalent to $E \geq 0$. $\square$

10.1. Quantitative inversion across the rank locus. All coefficient norms below are Euclidean, including the leading coefficient when a monic polynomial is represented as a vector. Constants are specified from a fixed compact $K_0 \subset \mathfrak{F}$. Norm comparisons between these finite-dimensional spaces are included in these constants. Put

$$\delta = \max_j \|P'_j - P_{\theta, j}\|_F.$$

Use coefficient order $(1, z, z^2, z^3)$. The two components form $F_{\text{coef}} \in \mathbb{R}^{2 \times 4}$, with one polynomial per row, and the second-marginal coefficient matrix is $\mathbf{B}_\theta = V \text{diag}(\alpha, \gamma) F_{\text{coef}} \in \mathbb{R}^{2 \times 4}$. Its two rows are the scalar coordinate polynomials of that marginal. Here σ_2 is its smaller singular value and matrix norms are Euclidean operator norms (or the corresponding fixed norm comparisons). Set

$$\beta_0 = \min_{K_0} \sigma_2(\mathbf{B}_\theta) > 0, \quad d_{fg} = \min_{K_0} \|g - f\|_{\text{coef}} > 0.$$

Normalizer recovery supplies C_B with $\|\mathbf{B}' - \mathbf{B}_\theta\| \leq C_B \delta$. Let K_d bound $\|\text{diag}(\alpha') F'_{\text{coef}}\|$ on all closed-model competitors for the designs in K_0 , and set

$$W_0 = \frac{2K_d}{\pi\beta_0}, \quad C_f = \max_{K_0} \|f\|_{\text{coef}}, \quad K_{\text{aff}} = W_0 C_B (1 + C_f). \quad (10.5)$$

Proposition 10.2 (From the second marginal to the pencil). *If $\delta \leq \beta_0/(2C_B)$, every closed competitor has invertible V' and its component polynomials satisfy*

$$f'_b = h_{c_b} + e_b, \quad \deg e_b < 3, \quad \|e_b\|_{\text{coef}} \leq K_{\text{aff}} \delta. \quad (10.6)$$

Moreover $\|\text{diag}(\alpha')^{-1} V'^{-1}\| \leq W_0$. If the ordered component pair is within ρ of (f, g) , then

$$|c_1|, |c_2 - 1| \leq (\rho + K_{\text{aff}} \delta)/d_{fg}. \quad (10.7)$$

No lower bound on $|\det U|$ is used.

Proof. Since $\mathbf{B}' = V' \text{diag}(\alpha') F'_{\text{coef}}$,

$$\beta_0/2 \leq \sigma_2(\mathbf{B}') \leq \sigma_{\min}(V') K_d.$$

This proves the inverse bound, using $\alpha'_a \geq \pi$. Write C_θ for the matrix with coefficient rows f, g and put

$$T = \text{diag}(\alpha')^{-1} V'^{-1} V \text{diag}(\alpha, \gamma), \quad E_0 = \text{diag}(\alpha')^{-1} V'^{-1} (\mathbf{B}' - \mathbf{B}_\theta).$$

Then $F'_{\text{coef}} = T C_\theta + E_0$ and $\|E_0\| \leq W_0 C_B \delta$. If a row of T is (l, c) , monicity says $l + c + (E_0)_{\text{lead}} = 1$. Subtracting $(E_0)_{\text{lead}} f$ from that error row gives the exact decomposition $f'_b = (1 - c)f + cg + e_b$, with the bound in (10.5) and zero leading coefficient in e_b . Finally $h_c - f = c(g - f)$ and $h_c - g = (c - 1)(g - f)$ imply (10.7) by the triangle inequality. $\square$

Lemma 10.3 (Two robust isolation estimates). *There are explicit positive $r_f, r_g, \varepsilon_0, C_0, C_1$, depending only on K_0 , with the following property. If $h_c + e$ is monic with all roots in $[0, D]$, $\deg e < 3$ and $\|e\|_{\text{coef}} \leq \varepsilon$, then*

$$|c| \leq r_f, \quad \varepsilon \leq \varepsilon_0 \implies |c| \leq C_0 \varepsilon, \quad |1 - c| \leq r_g, \quad \varepsilon \leq \varepsilon_0 \implies |1 - c| \leq C_1 \varepsilon.$$

The constants require only root-separation and coefficient bounds on K_0 , not a sign or a nonzero margin for B, E .

Proof. Let $s_0 = \min_{K_0} s > 0$ and $d_0 = \min_{K_0} 4r^3(r - s)^3 > 0$. Choose L_D as a Lipschitz bound for the discriminant on a coefficient neighbourhood of $\{h_c : \theta \in K_0, |c| \leq 1\}$, and R_D such that

$$|\text{disc}(h_c) + 4r^3(r - s)^3 c| \leq R_D c^2 \quad (|c| \leq 1).$$

These constants are finite because the discriminant is polynomial. Take $r_f \leq \min(1, d_0/(2R_D))$, omitting the second bound if $R_D = 0$. The endpoint condition gives $-cs^3 + e(0) \leq 0$, hence $c \geq -\varepsilon/s_0^3$. For $0 \leq c \leq r_f$, nonnegative discriminant gives $0 \leq -d_0 c + R_D c^2 + L_D \varepsilon$, hence $c \leq 2L_D \varepsilon/d_0$. One may take $C_0 = \max(s_0^{-3}, 2L_D/d_0)$.

Near $c = 1$, put $v = 1 - c$ and translate $z = s + x$. There are exact identities

$$h_c(s + x) = x^3 + vA_2 x^2 + vA_1 x + vA_0, \quad A_2 = 3s - 2r, \quad A_1 = f'(s), \quad A_0 = f(s) = s(r - s)^2.$$

Let L_s bound coefficient translation of e on K_0 , set $E_s = L_s \varepsilon$, and write the perturbed coefficients as $u_i = vA_i + e_i$, $|e_i| \leq E_s$. Choose $M \geq \max_{K_0}(|A_1|, |A_2|)$ and $a_* = \min_{K_0} A_0 > 0$. Depressing the perturbed cubic by its own quadratic coefficient gives

$$p = u_1 - u_2^2/3, \quad q = u_0 - u_1 u_2/3 + 2u_2^3/27.$$

For $u = |v| + E_s \leq 1$,

$$|p| \leq C_p u, \quad |q - A_0 v| \leq E_s + C_q u^2, \quad C_p = M+1+(M+1)^2/3, \quad C_q = (M+1)^2/3 + 2(M+1)^3/27.$$

Real-rootedness is exactly $p \leq 0$ and $|q| \leq c_{\text{cub}}(-p)^{3/2}$ with $c_{\text{cub}} = 2/(3\sqrt{3})$. Thus

$$a_* |v| \leq E_s + C_q u^2 + c_{\text{cub}} C_p^{3/2} u^{3/2}. \quad (10.8)$$

Choose

$$\eta = \min\{1, a_*/(4C_q), (a_*/(4c_{\text{cub}} C_p^{3/2}))^2\}, \quad r_g = \eta/2, \quad \varepsilon_0 \leq \eta/(2L_s).$$

Then $u \leq \eta$, the last two terms in (10.8) are at most $a_* u/2$, and $|v| \leq (2/a_* + 1)E_s$. Take $C_1 = (2/a_* + 1)L_s$. Decrease ε_0 also to stay in the discriminant neighbourhood used in the first argument. Every threshold is determined by the stated positive lower margins and finite polynomial upper bounds. $\square$

Let d_{par} be Euclidean distance in monic coefficients and stochastic entries, minimized over the component permutation.

Lemma 10.4 (A local inverse through the walls). *For every compact $K_0 \subset \mathfrak{F}$, there exist $\rho, M, t_0 > 0$ such that a closed-model competitor whose component pair is within ρ of (f, g) , up to permutation, satisfies*

$$d_{\text{par}}(\xi, \theta) \leq M \max_j \|F_T(\xi)_j - P_{\theta,j}\|_F \quad (10.9)$$

whenever the observation error is below t_0 . These constants require no lower bound on $|\det U|$, $-B$, or $-E$. For a compact $H \subset \mathfrak{J}$ and small fixed ρ , the remote separation

$$\eta_H(\rho) = \min_{\theta \in H, \xi \text{ in the closed model}, d_{\text{par}}(\xi, \theta) \geq \rho} h_w(F_T(\xi), P_\theta) \quad (10.10)$$

is positive if the minimizing set is nonempty. The whole-model inverse therefore holds uniformly on H .

Proof. Let $r_* = \min(r_f, r_g)$ from Lemma 10.3 and choose

$$\rho \leq d_{fg} r_*/4, \quad t_0 \leq \min\{\beta_0/(2C_B), \rho/K_{\text{aff}}, \varepsilon_0/K_{\text{aff}}\}.$$

Proposition 10.2 gives an error at most $\varepsilon = K_{\text{aff}} \delta$ in the affine pencil and pencil coordinates in the required neighbourhoods. The two isolation estimates then give $|c_1| + |c_2 - 1| \leq (C_0 + C_1)K_{\text{aff}} \delta$. Together with (10.6), this controls both polynomial coefficient vectors by a common multiple of δ .

Fix the component permutation placing the close ordered pair near (f, g) ; separation of f, g makes this choice unambiguous for a smaller ρ . To recover the stochastic parameters without dividing by $\det U$, choose coefficient functionals ℓ_1, ℓ_2 dual to f, g . Their norms are bounded by the inverse Gram matrix of this pair, uniformly on K_0 . The matrix $T'_{ij} = \ell_i(f'_j)$ is $I + O_{K_0}(\delta)$. Decrease t_0 so that $\|T' - I\| \leq 1/2$; then $\|(T')^{-1}\| \leq 2$. Apply the two functionals entrywise to K' . Inverting T' recovers $M'_a = \alpha'_a u'_a v'_a{}^\top$ with $O_{K_0}(\delta)$ error from M_a . Entry sums give the weights, and row and column sums divided by those weights give the two channels. These divisions have lower bound π . This proves (10.9) with constants constructed from normalizer, second-marginal, Gram, root and weight margins only. In particular no full-rank inverse has been continued through its singular determinant.

For the last assertion the feasible set in (10.10) is compact, allowing the centre's design to vary in H . A zero minimum would give an exact competitor separated from that centre. The

entire fibre and conditional uniqueness in Theorem 10.1 exclude this on H . Small whole-model observation balls thus lie in the local neighbourhood. It is this exclusion radius, rather than a local coefficient constant, that may collapse at a remote wall. $\square$

10.2. The complete local Fisher fibre. Use $a, X, Y \geq 0$, free b, c , and five free stochastic coordinates η . The component jets are

$$p_f = -a(z-r)^2 - bz(z-r) - Xz, \quad p_g = -c(z-s)^2 - Y(z-s). \quad (10.11)$$

Let L be the seven free score columns and D_0 the three columns indexed by $0, x, y$ for (a, X, Y) . Define Q by (9.2) and

$$J(X, Y) = \min_{a \geq 0} (a, X, Y)Q(a, X, Y)^\top, \quad \kappa_x = \min_{Y \geq 0} J(1, Y), \quad \kappa_y = \min_{X \geq 0} J(X, 1). \quad (10.12)$$

Proposition 10.5 (Coercivity and finite supports). *Throughout $\mathfrak{F}$, L has rank seven, $Q_{00} > 0$, and κ_x, κ_y are positive and locally Lipschitz. For every compact $K_0 \subset \mathfrak{F}$,*

$$\|S_\theta(a, b, c, \eta, X, Y)\|_{I, \theta} \geq c_{K_0} \|(a, b, c, \eta, X, Y)\| \quad (a, X, Y \geq 0).$$

Each κ_i is given by four finite support tests. For $A \subseteq \{0, x, y\} \setminus \{i\}$ with Q_{AA} positive definite, put

$$v_A = -Q_{AA}^{-1}Q_{Ai}, \quad k_{i,A} = Q_{ii} - Q_{iA}Q_{AA}^{-1}Q_{Ai}.$$

The test is $v_A \geq 0$ and $Q_{ji} + Q_{jA}v_A \geq 0$ for inactive j . The empty support has value Q_{ii} . At least one test succeeds and all successful tests give κ_i .

Proof. Every feasible jet is realized by roots

$$ta, \quad r + tb/2 \pm \sqrt{tX}, \quad s + tc/3 + \sqrt{t}y_i, \quad \sum_i y_i = 0, \quad \frac{1}{2} \sum_i y_i^2 = Y,$$

and order- t stochastic increments. Their observation increment is $tS + O(t^{3/2})$. A zero score, combined with Lemma 10.4, forces the component jets and η to vanish. Coefficient comparison in (10.11) then forces every variable to vanish. Compactness of the feasible unit sphere gives coercivity, uniformly on K_0 . This also proves independence of the free columns, $Q_{00} > 0$, and existence, positivity and boundedness of the constrained minimizers.

Choose a minimizer with minimal positive support in the projected score columns. Those supported columns are independent: a nonzero dependence would allow a signed change of positive coefficients, preserving the represented vector, until one coefficient vanished. On an independent support, stationarity and the inactive inequalities are exactly the stated tests. Conversely their KKT conditions prove optimality by convexity. This handles a singular full Q without inverting it. Uniformly bounded minimizers and smooth score dependence give local Lipschitz dependence by evaluating each problem at a minimizer of the other problem. $\square$

Theorem 10.6 (Local normal form and whole-model restriction). *At every centre of $\mathfrak{F}$ the local leading root set is*

$$(\{0\}, \{-\sqrt{X}, \sqrt{X}\}, Z(z^3 - Yz - Z)), \quad X, Y \geq 0, \quad J(X, Y) \leq 4, \quad 27Z^2 \leq 4Y^3. \quad (10.13)$$

On compact subsets of $\mathfrak{F}$, scaled local observation balls converge to this set with Hausdorff error $O(\sqrt{t})$. Hence

$$\omega_{\theta, \text{loc}}(t) = C_{\text{loc}}(\theta)\sqrt{t} + O(t), \quad C_{\text{loc}} = \max\{\sqrt{2}\kappa_x^{-1/4}, 2\sqrt{2/3}\kappa_y^{-1/4}\}. \quad (10.14)$$

The error is compact-uniform, and C_{loc} is positive and locally Lipschitz on all of $\mathfrak{F}$. On compact subsets of $\mathfrak{J}$ these statements hold for the whole-model modulus. The support tests, (10.2), and $9\kappa_y - 16\kappa_x$ give a finite semialgebraic partition. On its constant choices,

$$C_{\text{loc}}^4 = \max\{4/\kappa_x, 64/(9\kappa_y)\}$$

is one rational function, and C_{loc} is Nash after refinement.

Proof. The local inverse bounds coefficients and stochastic increments by $O(t)$ for every local competitor. Its first component has exact roots $ta, r + tb/2 \pm \sqrt{tX}$ with bounded a, b, X and $a, X \geq 0$. Its second component has exact roots $s + tc/3 + \sqrt{t}y_i$ with $\sum_i y_i = 0$ and bounded squared sum. Put $Y = \frac{1}{2} \sum_i y_i^2$ and $Z = \prod_i y_i$. The real-cubic condition is exactly $Y \geq 0, 27Z^2 \leq 4Y^3$. Expansion gives $F = P + tS + O(t^{3/2})$ and $h_w = t\|S\|_I/2 + O(t^{3/2})$ uniformly on these bounded coordinates. Radial contraction of all first-order variables, together with $Z \mapsto \rho^{3/2}Z$, proves the outer Hausdorff inclusion. Conversely choose bounded minimizing free and endpoint variables and realize the displayed root arcs with $\rho = 1 - M\sqrt{t}$. They lie in the observation ball and give the inner inclusion. The inequality boundary and all joint cubic coefficient relations are therefore realized.

Homogeneity gives $X_{\max} = 2/\sqrt{\kappa_x}$ and $Y_{\max} = 2/\sqrt{\kappa_y}$. The double-cluster diameter is $\sqrt{X_{\max}}$. Lemma 9.1 with $m = 3$ gives the triple diameter $2\sqrt{Y_{\max}/3}$, attained also by $(-2h, h, h)$ and $(-h, -h, 2h)$ with $h = \sqrt{Y_{\max}/3}$. The maximum of marginal diameters gives (10.14). Proposition 10.5, cluster separation and the positive remote exclusion in Lemma 10.4 give the remaining claims. The local formula remains valid at $B = 0$ or $E = 0$, but its identification with the whole-model modulus does not. $\square$

Example 10.7 (The earlier rational phase values). For $D = 1$, $\pi = 1/10$, $r = 3/5$, $s = 1/5$, $\alpha = 2/5$, $u_1 = u_2 = (1/3, 2/3)^\top$, $v_1 = (1/4, 3/4)^\top$, $v_2 = (3/4, 1/4)^\top$, clocks $(6/5, 7/5, 8/5, 9/5, 2, 5/2, 3)$ and equal weights, $B = -18/25$. The empty supports are admissible, the double cluster dominates, and exact rational evaluation gives

$$C^4 = \frac{78550423766145732273576408862601811503774391446}{385279214348552940073664077156875}, \quad 3778.707 < C < 3778.708.$$

Changing only u_2 to $(2/5, 3/5)^\top$ gives the triple-dominant phase, $59.970 < C < 59.971$. The exact arithmetic records for both values are retained in the archival companion. These large values are not evidence for an unspecified wall blow-up.

Corollary 10.8 (Centre-known local risk). *Let $H \subset \mathfrak{J}$ be compact, fix $0 < c \leq 1/(8\sqrt{2})$, and observe independent categorical samples at the design clocks with $n_j/N \rightarrow w_j$ uniformly on H (integer rounding suffices). For squared spectral loss on the specified neighbourhood $h_w(F_T(\xi), P_\theta) \leq cN^{-1/2}$, with the centre known, the minimax risk $\mathcal{R}_N(\theta, c)$ satisfies uniformly on H*

$$\frac{3c}{32}C_{\text{loc}}(\theta)^2 - O_H(N^{-1/4}) \leq \sqrt{N}\mathcal{R}_N(\theta, c) \leq cC_{\text{loc}}(\theta)^2 + O_H(N^{-1/4}).$$

Proof. A constant decision in the specified spectral ball has loss at most its squared diameter. A diameter pair in that ball has weighted Hellinger distance at most $2cN^{-1/2}$. Tensorization bounds its product Hellinger distance by $2c\sqrt{\max_j n_j/(Nw_j)}$. Uniform convergence and the positive minimum of the design weights on H make the maximum ratio at most 2 for all sufficiently large N , uniformly on H . Thus the product distance is at most $2c\sqrt{2} \leq 1/4$. Total variation is bounded by this Hellinger distance in the present convention. The two-point testing bound for squared metric loss is $d^2(1 - \text{TV})/8$, hence at least $3d^2/32$. Substitute (10.14). The centre and its neighbourhood are known; unknown-centre adaptation and honest confidence sets across a wall are not asserted. $\square$

11. A REMOTE BRANCH AT THE STOCHASTIC WEIGHT WALL

Fix r, s, D with $B > 0$, positive u, v_1, v_2 with $v_1 \neq v_2$, and $U = (u, u)$, $V = (v_1, v_2)$. Assume

$$\gamma_0 = \pi c_*, \quad \alpha_0 = 1 - \pi c_*, \quad \pi < \alpha_0 < 1 - \pi. \quad (11.1)$$

The design is fixed. Let θ_δ be the original (f, g) centre with weight $\alpha = \alpha_0 + \delta$. Then $E = -A\delta$, so $\delta > 0$ is the identified side. Put

$$a = \frac{4rs}{r+3s}, \quad b = \frac{rs}{3s-2r}, \quad h_* = (1-c_*)f + c_*g = (z-a)(z-b)^2. \quad (11.2)$$

Substitution gives $s < a < r < b < D$. At $\delta = 0$ the two exact component pairs are (f, g) and (f, h_*) . The remote pair has weight π on h_* and second-channel columns

$$v_f^* = \frac{\alpha_0 v_1 + (\gamma_0 - \pi)v_2}{1 - \pi}, \quad v_h^* = v_2. \quad (11.3)$$

Let ξ_* be this point; its observation is $P_0 = P_{\theta_0}$. Its second channel is invertible because $v_f^* - v_2 = \alpha_0(v_1 - v_2)/(1 - \pi) \neq 0$. The sorted base and remote spectra are respectively

$$(0, s, s, s, r, r), \quad (0, a, r, r, b, b),$$

whose distance is

$$D_* = \max(r - s, b - r) > 0. \quad (11.4)$$

Thus a zero-length additional pencil interval already has a positive exact-fibre spectral diameter.

11.1. A complete real corner chart at the remote point. Use the eleven coordinates $v = (a_0, b_0, X, c_0, d_0, Y, k, \dot{u}_f, \dot{u}_h, \dot{v}_f, \dot{v}_h)$, where a binary channel increment is represented by its one free zero-sum coordinate. Set

$$\begin{aligned} p_f &= -a_0(z-r)^2 - b_0z(z-r) - Xz, \\ p_h &= -c_0(z-b)^2 - d_0(z-a)(z-b) - Y(z-a), \\ \mathcal{T}_* &= \{v : a_0, X, Y, k \geq 0\}. \end{aligned} \quad (11.5)$$

The remaining seven coordinates are free. Here k is the increment of the weight on h_* ; the weight on f has increment $-k$. Let J_*v be the full coefficient and stochastic vector with polynomial increments p_f, p_h and these stochastic increments.

Proposition 11.1 (Remote tangent cone and converse realization). *In a neighbourhood of ξ_* , modulo the one component permutation, the closed model has a Nash coordinate chart from a neighbourhood of zero in $\mathcal{T}_*$. Its derivative is the invertible linear map J_* . Consequently its coefficient-stochastic tangent cone is exactly $J_*(\mathcal{T}_*)$. Every normalized secant has a subsequential limit in this cone, and every vector of the cone is realized by a feasible arc. For bounded $v \in \mathcal{T}_*$ the arcs can be chosen with*

$$\xi(\tau, v) = \xi_* + \tau J_*v + O(\tau^2), \quad F_T(\xi(\tau, v)) = P_0 + \tau S_*v + O(\tau^2), \quad (11.6)$$

uniformly on bounded coordinate sets. Here S_ is the score (2.3) at the remote point. All assertions are uniform on compact subsets of the wall family with its stated strict margins.*

Proof. Factor each nearby polynomial into its separated simple and quadratic factors. Coprime factorization is analytic in coefficients. Write the simple root of the first component as u , the mean of its quadratic roots as $r + v_f/2$, and their squared half-separation as x . Write the corresponding coordinates of the second component as $a + w$, $b + v_h/2$ and y . Thus the exact polynomials are

$$(z-u)((z-r-v_f/2)^2-x), \quad (z-a-w)((z-b-v_h/2)^2-y).$$

These six coordinates are Nash functions of the coefficients near the remote pair. Conversely the displayed formulas are polynomial in the six coordinates. Their derivative has the three columns in (11.5) for each component. They are independent: for the first polynomial, evaluation at 0 and r first determines a_0 and X , and then the remaining coefficient determines b_0 ; for the second, evaluation at a and b determines c_0 and Y , and then d_0 . The inverse function theorem gives the asserted local coefficient chart.

The first simple root is feasible exactly when $u \geq 0$, locally. Its upper endpoint is inactive. All other cluster centres are strictly inside $(0, D)$, so their interval constraints are inactive in this neighbourhood. Real-rootedness of the two quadratic factors is exactly $x, y \geq 0$. The weight on h_* is at its floor, giving $k \geq 0$; the other floor has strict margin $1 - 2\pi > 0$. Channel entries are strictly positive, so only their linear column-sum equations remain active. This proves that the complete feasible chart is the indicated orthant times seven free coordinates, not merely an outer cone. Separated factors are unique. The alternative component labelling is a second, identical chart under the permutation quotient and introduces no extra directions.

For the secant assertion, let τ be the norm of a coefficient and stochastic displacement. Analytic factorization bounds the simple and mean shifts by $O(\tau)$. The quadratic coefficients bound x, y by $O(\tau)$ as well, since variance is mean squared minus the constant coefficient. Dividing the exact chart coordinates by τ gives a bounded vector in $\mathcal{T}_*$. Its subsequential limits give all normalized tangent secants. In the norm-normalized case J_*v has norm one in the limit, so no nonzero secant disappears in the chart.

Conversely set $u = \tau a_0$, $v_f = \tau b_0$, $x = \tau X$, $w = \tau c_0$, $v_h = \tau d_0$, $y = \tau Y$, choose weight $\pi + \tau k$ on h , and take the specified order- τ channel increments. The roots are

$$\tau a_0, \quad r + \tau b_0/2 \pm \sqrt{\tau X}, \quad a + \tau c_0, \quad b + \tau d_0/2 \pm \sqrt{\tau Y}.$$

They remain feasible for small τ , uniformly for bounded coordinates. They are analytic in $\sqrt{\tau}$; their polynomial coefficients are analytic in τ . In fact the exact identities are

$$f_\tau - f - \tau p_f = \tau^2 \{a_0 b_0(z - r) + a_0 X + b_0^2 z/4\} - \tau^3 a_0 b_0^2/4, \quad (11.7)$$

$$h_\tau - h_* - \tau p_h = \tau^2 \{c_0 d_0(z - b) + c_0 Y + d_0^2(z - a)/4\} - \tau^3 c_0 d_0^2/4. \quad (11.8)$$

There is no order- $\tau^{3/2}$ coefficient term at these simple/double clusters. Taylor expansion of the rational observation map, whose normalizers stay positive, proves (11.6). Compact strict root, channel, weight and clock margins make all inverse chart and derivative bounds uniform as stated. $\square$

11.2. The constrained entrance program. Put $B_0 = \frac{d}{d\delta} P_{\theta_\delta} \Big|_{\delta=0}$ and use the Fisher norm at P_0 . Define

$$\mu_E = \frac{1}{2} \min_{v \in \mathcal{T}_*} \|S_* v - B_0\|_{I,0}. \quad (11.9)$$

This is an explicit finite-dimensional program on the cone just proved to be the full feasible tangent cone.

Lemma 11.2 (Positive entrance separation). *The score has no nonzero zero on $\mathcal{T}_*$, and B_0 is not in its image. Thus $\mu_E > 0$ and its minimum is attained. Eliminating seven free variables gives sixteen independent-support KKT tests on four nonnegative variables. For algebraic input μ_E is real algebraic.*

Proof. Suppose first that $S_* v = B_0$. The differentiated normalizer equations, using Lemma 2.1 at the coprime remote pair with invertible second channel, imply $\dot{q} = f - g$ and $\dot{K} = \frac{d}{d\delta} K_{\theta_\delta} \Big|_0$. The derivative of the second marginal belongs to $\text{span}(f, g)$. Its stochastic derivative terms already belong to that span; the remote second-channel weight matrix is invertible. Hence $p_f, p_h \in \text{span}(f, g)$. Since their degrees are below three, write $p_f = t_f(g - f)$ and $p_h = t_h(g - f)$.

Evaluation at 0 gives $t_f = a_0 r^2/s^3 \geq 0$, while evaluation at r gives $t_f = -Xr/(r - s)^3 \leq 0$. Therefore $t_f = 0$ and $p_f = 0$. At b the second polynomial gives

$$t_h = \frac{Y(b - a)}{f(b) - g(b)} \geq 0,$$

because $R(b) > 1$. Differentiation of the monic normalizer now gives

$$\pi t_h + c_* k = -1, \quad (11.10)$$

an impossibility. For a zero score, the same argument has right-hand side zero and implies $t_h = k = 0$. All polynomial jets vanish. Coefficient functionals dual to f, h_* then imply that each coefficient matrix has zero derivative; taking its row and column sums forces all four channel derivatives to vanish. The independent chart columns in Proposition 11.1 force $v = 0$.

The intersection of $\mathcal{T}_*$ with the unit sphere is compact, so $\|S_* v\|_{L,0} \geq c\|v\|$ on the cone. Thus the objective in (11.9) tends to infinity with $\|v\|$; its minimum is attained and positive, since equality $S_* v = B_0$ was excluded.

Let L_* be the seven free columns, D_* the four nonnegative columns, $W = \text{diag}(w_j/P_{0,j}e)$, and $\mathcal{P}$ the Fisher-orthogonal projection off $\text{im } L_*$. Put

$$Q_* = D_*^\top W \mathcal{P} D_*, \quad g_* = D_*^\top W \mathcal{P} B_0, \quad b_* = B_0^\top W \mathcal{P} B_0.$$

Coercivity implies full rank of the free columns, and

$$4\mu_E^2 = \min_{z \geq 0} \{z^\top Q_* z - 2g_*^\top z + b_*\}. \quad (11.11)$$

For each of the sixteen supports A with $Q_{*,AA}$ positive definite, set $z_A = Q_{*,AA}^{-1} g_{*,A}$, require $z_A \geq 0$ and $(Q_* z - g_*)_j \geq 0$ off A , and take value $b_* - g_{*,A}^\top Q_{*,AA}^{-1} g_{*,A}$. Include the empty support. The minimal-positive-support argument of Proposition 10.5 proves completeness, including singular Q_* . All entries are rational in the wall coordinates; take the positive square root after selecting the exact minimum. $\square$

Lemma 11.3 (Local coefficient control at the remote point). *In a coefficient and stochastic neighbourhood of ξ_* in the closed model,*

$$d_{\text{par}}(\xi, \xi_*) \leq C \|F_T(\xi) - P_0\|.$$

Every competitor at parameter distance $O(\delta)$ has bounded exact chart coordinates at scale δ and observation expansion $P_0 + \delta S_ v + O(\delta^2)$. Conversely each bounded cone vector has such a feasible realization.*

Proof. If the inverse failed, normalize a sequence of displacements by their norm. Proposition 11.1 gives a nonzero limit $J_* v$ with $v \in \mathcal{T}_*$ and zero score, contradicting Lemma 11.2. Equivalently in its chart, Taylor's theorem and cone coercivity give $\|F_T(\Phi(w)) - P_0\| \geq c\|w\| - C\|w\|^2 \geq c\|w\|/2$ for small feasible w . The chart is locally bi-Lipschitz in coefficients and stochastic entries, which proves the inverse and the stated bounded coordinate assertion. The expansions and converse are (11.6). This proof does not assume the inverse in order to identify the tangent cone. $\square$

11.3. The two-scale law with a quadratic entrance remainder. Choose disjoint small closed coefficient and stochastic neighbourhoods N_0, N_* of the two exact representatives at the wall, modulo component permutation, with ξ_* in the relative interior of N_* . Put

$$d_{\text{rem}}(\delta) = \min_{\xi \in N_*} h_w(F_T(\xi), P_{\theta_\delta}). \quad (11.12)$$

Their boundaries have a positive observation separation from P_0 .

Theorem 11.4 (Weight-wall transition). *Under (11.1) and $B > 0$,*

$$d_{\text{rem}}(\delta) = \mu_E \delta + O(\delta^2) \quad (\delta \downarrow 0). \quad (11.13)$$

The minimum is independent of the sufficiently small neighbourhood N_ for all sufficiently small δ . The base local constant $C_{\text{loc}}(\theta_\delta)$ has the finite positive limit $C_{\text{loc}}(\theta_0)$; neither κ_x nor κ_y vanishes*

at this wall. For fixed $\lambda > 0$ the whole-model modulus satisfies

$$0 < \lambda < \mu_E : \quad \omega_{\theta_\delta}(\lambda\delta) = C_{\text{loc}}(\theta_0)\sqrt{\lambda\delta} + O_\lambda(\delta), \quad (11.14)$$

$$\lambda > \mu_E : \quad \omega_{\theta_\delta}(\lambda\delta) = D_* + O_\lambda(\sqrt{\delta}). \quad (11.15)$$

The exact entrance curve for the closed ball is $t = d_{\text{rem}}(\delta)$, including equality. The layer not decided by the first-order threshold alone has width $O(\delta^2)$ in t .

On the nonidentified side, for small $\varepsilon > 0$,

$$\omega_{\theta_{-\varepsilon}}(0) = \max \left\{ r - s, b - r + \sqrt{\frac{\chi\varepsilon}{\pi}} + O(\varepsilon) \right\}, \quad \chi = \frac{f(b) - g(b)}{b - a} > 0. \quad (11.16)$$

If $b - r < r - s$, the exact diameter stays equal to $r - s$ initially; if $b - r \geq r - s$, it opens above D_* at the displayed square-root rate.

Proof. Theorem 10.1 gives exactly the two representatives at $\delta = 0$. Compactness gives a positive observation gap outside $N_0 \cup N_*$. Since ξ_* itself is at distance $O(\delta)$ from P_{θ_δ} , a minimizer in (11.12) has observation $O(\delta)$ from P_0 and lies away from the boundary of N_* . Lemma 11.3 bounds its coefficient and stochastic displacement by $O(\delta)$. Its exact chart coordinates are therefore δv for bounded $v \in \mathcal{T}_*$. Expansion at the common positive probability array and the reverse triangle inequality give

$$h_w(F_T(\xi), P_{\theta_\delta}) = \frac{\delta}{2} \|S_* v - B_0\|_{I,0} + O(\delta^2), \quad (11.17)$$

uniformly on those bounded coordinates. The lower bound in (11.13) follows by minimizing the leading norm. Conversely a minimizer of (11.9), realized by the exact chart, gives the upper bound. The error is quadratic because of (11.7)–(11.8), not merely an appeal to a generic root-continuity bound. This also implies the weaker $O(\delta^{3/2})$ entrance estimate. The same localization puts all minimizers inside any fixed smaller remote neighbourhood for small δ , proving independence of its choice.

Proposition 10.5 and Theorem 10.6 apply on a compact base path segment including θ_0 ; this pair stays locally coercive there. Thus C_{loc} is positive and locally Lipschitz through the wall. If $\lambda < \mu_E$, the remote neighbourhood is excluded from the radius- $\lambda\delta$ ball for small δ . The exterior gap excludes every other competitor, so the whole ball is the base local ball. Formula (10.14) gives (11.14). Replacing $C_{\text{loc}}(\theta_\delta)$ by $C_{\text{loc}}(\theta_0)$ costs only $O(\delta^{3/2})$.

If $\lambda > \mu_E$, a remote minimizer and the base centre belong to the ball. Every competitor belongs to one of the two neighbourhoods. The two coefficient inverses put its roots within $O_\lambda(\sqrt{\delta})$ of the corresponding exact spectrum at the wall. The bottleneck triangle inequality gives both bounds in (11.15). Compact attainment makes remote membership exactly $t \geq d_{\text{rem}}(\delta)$, including equality. The width of the unresolved first-order layer follows from (11.13); it does not decide the sign of the difference on the critical ray.

For $\theta_{-\varepsilon}$ the extra exact pencil interval is $[c_*, c_* + \varepsilon/\pi]$, since $c_D > c_*$ is fixed. At $c = c_* + \sigma$, expansion at $z = b + u$ gives

$$h_c(b + u) = (b - a)u^2 + u^3 + \sigma(g(b) - f(b)) + O(\sigma|u|).$$

Since $R(b) > 1$, $\chi > 0$ and the two roots near b are $b \pm \sqrt{\chi\sigma} + O(\sigma)$. The third root remains in (s, r) . The larger root increases with c by the monotonic branch of R above its maximum. Comparing sorted lists, the distance to the base spectrum is exactly $\max(r - s, b_+(c) - r)$. Two remote spectra are only $O(\sqrt{\varepsilon})$ apart, less than D_* for small ε . The exact-fibre diameter is therefore attained against the base at the upper pencil endpoint. This proves (11.16) and its two cases. $\square$

Remark 11.5 (The critical ray). At $t = \mu_E \delta$, remote membership is decided by the sign of $d_{\text{rem}}(\delta) - \mu_E \delta$. For algebraic input, (11.12) has a compact first-order minimum formula. Elimination and real algebraic branch isolation determine its Puiseux expansion. The first nonzero higher coefficient decides the critical ray; if the difference is identically zero, the closed ball includes the remote branch. No strict regime is assigned to that ray by convention.

Remark 11.6 (The geometric wall). At $B = 0$ with $E > 0$, the remote double root is $b = D$, and the additional pencil interval again consists of one point at positive spectral distance. On the nonidentified side put $\ell_0 = D - b$. Expansion at the strict maximum of R gives

$$c_D - c_* = \frac{-R''(b)}{2(R(b) - 1)^2} \ell_0^2 + O(\ell_0^3), \quad \ell_0 = \frac{B}{3s - 2r}.$$

The coefficient is positive and the weight floor is inactive. This interval is born with quadratic length in geometric clearance, rather than linear length in the weight parameter. The base local constants remain finite on compact subsets of $\mathfrak{F}$. A two-scale entrance law at that endpoint requires a different cone with one-sided root coordinates. The complete entrance theorem here is for the weight wall.

Example 11.7 (An exact wall datum). Take $D = 1$, $r = 3/5$, $s = 11/20$, $\pi = 1/10$, the rank-one columns and clocks in Section 10, and equal design weights. Then

$$\alpha_0 = \frac{547}{675}, \quad c_* = \frac{256}{135}, \quad a = \frac{44}{75}, \quad b = \frac{11}{15}, \quad B = \frac{3}{25}, \quad D_* = \frac{2}{15}, \quad \chi = \frac{3}{64},$$

and $v_f^* = (1457, 3403)^T / 4860$. Exact rational evaluation gives

$$5975.429249985072 < C_{\text{loc}}(\theta_0) < 5975.429249985073, \\ 2.131024914 \cdot 10^{-6} < \mu_E < 2.131024915 \cdot 10^{-6}.$$

The empty positive support is admissible in each base program and in the remote program. The ancillary exact records retain the rational Gram matrices, squared entrance constant, fourth power of the local constant and isolating intervals. The decimals follow from rational power comparisons, not optimization tolerances. They illustrate two different, poorly conditioned scales; they do not test the universal quantifiers of the family theorems.

Three stability questions. Multiplicity and one-sided boundary constraints determine the weights in Theorem 9.2. Exact identification in the cubic family is instead determined by (10.2). Conditioning of a fixed local law is measured by the positive Fisher programs. At the weight wall, identification fails while those programs stay coercive and the local square-root exponent persists. In Example 9.3, by contrast, the exponent changes and the linear constant diverges. The joint atlas retains these distinct limiting operations rather than identifying their strata.

12. FINITE REMOTE ENTRANCES AND THE CRITICAL WINDOW

The preceding weight wall is an instance of a more general phenomenon. A regular feasible corner, rather than invertibility of the ambient score, is the relevant local hypothesis. We formulate the result in the square-root probability embedding, so that the distance in its statement is exactly the Hellinger distance and not a quadratic approximation to it.

Let X be compact and semialgebraic, let F be a continuous probability map, and let T be a continuous monic-polynomial root-multiset target of fixed degree. Quotient a finite representation symmetry when necessary. Suppose

$$F^{-1}(P_0) = \{x_0, x_1, \dots, x_m\}. \quad (12.1)$$

The points are representatives in this quotient. Let $x_\delta \rightarrow x_0$ be a feasible centre path, $P_\delta = F(x_\delta)$, and suppose P_0 is strictly positive. The design weights are fixed. Define

$$E(P) = (\sqrt{w_j} \sqrt{P_{je}})_{j,e}, \quad p_0 = E(P_0), \quad p_\delta = E(P_\delta) = p_0 + \delta b + \delta^2 c + o(\delta^2).$$

The Euclidean distance between two embedded arrays is h_w .

For each $j \geq 1$, assume that a full relative neighbourhood of x_j in X has coordinates

$$\psi_j : K_j \cap B_{r_j}(0) \rightarrow X, \quad \psi_j(0) = x_j,$$

where K_j is a closed convex polyhedral cone and ψ_j is a Nash parametrization, a homeomorphism onto that relative neighbourhood. Its inverse need not be differentiable at the vertex. Assume that $E \circ F \circ \psi_j$ has a C^2 extension to an ambient neighbourhood of zero. Write

$$E(F(\psi_j(z))) = p_0 + J_j z + Q_j(z, z) + o(\|z\|^2), \quad \|J_j v\| \geq a_j \|v\| \quad (v \in K_j), \quad a_j > 0. \quad (12.2)$$

Here $Q_j(z, z)$ denotes one half of the second derivative. The score may have a kernel in the ambient vector space; the last inequality is required only on the feasible cone. Choose small disjoint closed neighbourhoods $N_0, \dots, N_m$, with boundaries avoiding the exact fibre, and put

$$d_j(\delta) = \min_{x \in N_j} h_w(F(x), P_\delta), \quad \mu_j = \min_{v \in K_j} \|J_j v - b\|.$$

Theorem 12.1 (Remote entrance for regular corners). *Assume (12.1)–(12.2) and $\mu_j > 0$ for $1 \leq j \leq m$. Then*

$$d_j(\delta) = \mu_j \delta + O(\delta^2) \quad (\delta \downarrow 0). \quad (12.3)$$

For small δ the minimum is unchanged on replacing N_j by any fixed smaller relative neighbourhood of x_j . If $\lambda > 0$ differs from every μ_j , the target image of the radius- $\lambda\delta$ ball converges in Hausdorff distance to

$$\mathcal{D}_\lambda = \{T(x_0)\} \cup \{T(x_j) : \mu_j < \lambda\}. \quad (12.4)$$

Its diameter consequently tends to $\text{diam } \mathcal{D}_\lambda$. For $\lambda < \min_j \mu_j$, the whole ball is the base local ball for small δ . Any compact-uniform local asymptotic law at the base therefore remains valid in this regime.

Proof. Fix a remote index and omit it from the notation. Taylor's theorem and cone coercivity give, after reducing r ,

$$\|E(F(\psi(z))) - p_0\| \geq a\|z\|/2 \quad (z \in K, \|z\| < r).$$

The competitor $z = 0$ gives an upper bound $O(\delta)$ for $d(\delta)$. Compactness and (12.1) exclude the boundary of N from minimizing sequences. A minimizer thus has $z = O(\delta)$: its distance to p_0 is at most $d(\delta) + \|p_\delta - p_0\|$. Write $z = \delta v$. The vectors $v \in K$ are uniformly bounded, and

$$E(F(\psi(\delta v))) - p_\delta = \delta(Jv - b) + \delta^2\{Q(v, v) - c\} + o(\delta^2)$$

uniformly on bounded sets. Minimizing the leading norm gives the lower bound in (12.3). The function $v \mapsto \|Jv - b\|$ is coercive on K , so its minimum is attained; inserting a minimizer in the exact chart gives the upper bound. The same localization shows independence of the fixed neighbourhood.

Outside the union of the N_j the continuous function $h_w(F(x), P_0)$ has a positive minimum. This remains an exclusion gap for balls of radius $\lambda\delta$ about P_δ . In a remote neighbourhood, membership is possible exactly when $t \geq d_j(\delta)$, by compact attainment. Thus a remote neighbourhood is excluded if $\lambda < \mu_j$ and contains a minimizer if $\lambda > \mu_j$. Every sequence of remaining competitors converges, after subselection, to an exact representative; its target converges to that representative's target. Conversely x_δ supplies $T(x_0)$ in the limit and each included remote minimizer supplies $T(x_j)$. Finiteness gives Hausdorff convergence to (12.4). The diameter conclusion follows from the

elementary bound by twice the Hausdorff distance. When no remote neighbourhood is included, all competitors are in N_0 , which proves the last assertion. $\square$

The coefficient of the next term can be determined without assuming a unique minimizer or an invertible ambient score. This is useful at a change of active face of a constrained Fisher program.

Theorem 12.2 (The second-order entrance coefficient). *Under the hypotheses of Theorem 12.1, set*

$$M_j = \operatorname{argmin}_{v \in K_j} \|J_j v - b\|.$$

This is a nonempty compact set. The residual $r_j = J_j v - b$ is the same for all $v \in M_j$, and has norm μ_j . Define

$$k_j = \frac{1}{\mu_j} \min_{v \in M_j} \langle r_j, Q_j(v, v) - c \rangle. \quad (12.5)$$

Then

$$d_j(\delta) = \mu_j \delta + k_j \delta^2 + o(\delta^2). \quad (12.6)$$

At radius $t = \mu_j \delta + s \delta^2$, the j th remote neighbourhood is excluded for $s < k_j$ and included for $s > k_j$, for all sufficiently small positive δ . Equality requires the next nonzero term or the exact minimum inequality; it is not assigned to a regime by convention. For algebraic Nash input, μ_j and k_j are real algebraic numbers computable by real quantifier elimination and real-root isolation.

Proof. Coercivity bounds all minimizers. The image $J_j K_j$ is a closed convex polyhedral cone. The squared distance to b has a unique minimizing image point: if two different image points minimized it, their midpoint would have strictly smaller squared distance. This proves the assertion about r_j without requiring uniqueness of its preimage.

Again suppress j . Localization in the preceding proof bounds all scaled minimizers in one compact subset of K , also containing M . On that set the squared normalized distance has the uniform expansion

$$f_\delta(v) = \|Jv - b\|^2 + 2\delta \langle Jv - b, Q(v, v) - c \rangle + o(\delta).$$

Every limit of minimizers of f_δ belongs to M , by comparison with any point of M . The first term is at least μ^2 everywhere. The lower bound for the first-order increment of $\min f_\delta$ is therefore the minimum of the displayed second term over M . For the upper bound evaluate f_δ at a point of M attaining that minimum. Compactness and continuity justify both extrema. Consequently

$$\min f_\delta = \mu^2 + 2\delta \min_{v \in M} \langle r, Q(v, v) - c \rangle + o(\delta).$$

Taking the positive square root and multiplying by δ proves (12.6). Comparing with the proposed radius gives the strict cases. The minimum defining M and the minimum in (12.5) have first-order polynomial formulas with compact attainment; the latter may equivalently be restricted to a sufficiently large algebraic ball. Elimination and isolation give algebraic outputs. The formula is intrinsic to the distance and the fixed centre path, since its value equals the coefficient in (12.6), independently of the chosen chart. $\square$

Corollary 12.3 (Application to the cubic weight wall). *The remote point of Section 11 satisfies the preceding hypotheses with $m = 1$, $K_1 = \mathcal{T}_*$ and $\mu_1 = \mu_E$. In particular its entrance has the expansion*

$$d_{\text{rem}}(\delta) = \mu_E \delta + k_E \delta^2 + o(\delta^2),$$

where k_E is given by (12.5) in the exact remote chart of Proposition 11.1. This formula does not assert a common sign for k_E on the weight-wall family.

Proof. Theorem 10.1 gives the two exact representatives. The remote chart and its converse feasible realization are Proposition 11.1; its smooth coefficient formulas and positive normalizers give the required C^2 observation embedding. In these coordinates

$$J_1 v = \left(\frac{\sqrt{w_j}}{2\sqrt{P_{0,je}}} (S_* v)_{je} \right)_{j,e}, \quad b = \left(\frac{\sqrt{w_j}}{2\sqrt{P_{0,je}}} (B_0)_{je} \right)_{j,e}.$$

Lemma 11.2 is exactly cone coercivity and positive separation in this embedding. Hence the first minimization is (11.9). The two second-order derivatives in (12.5) are taken from the rational coefficient chart and the square-root probability map, not from root coordinates with an unproved differentiability at a multiple root. Theorems 12.1 and 12.2 apply. $\square$

Example 12.4 (Both signs of the critical correction). Let $e_0 = (1, 1)/\sqrt{2}$, $e_1 = (-1, 1)/\sqrt{2}$ and

$$q(s) = \frac{(1 - s^2)e_0 + 2se_1}{1 + s^2}, \quad P(s) = (q_1(s)^2, q_2(s)^2).$$

For sufficiently small $|s|$ both coordinates of $q(s)$ are positive. Take a base interval with observation $P(s)$ and a disjoint remote interval $0 \leq u \leq e$ with observation $P(-u/2)$. Give the two intervals distinct simple-root targets at zero, for example $z - (1/4 + s)$ and $z - (3/4 + u)$, and choose e small enough to keep all roots in $[0, 1]$. The exact fibre at $P(0)$ has two points. Choose the centre $s_\delta = (\delta + a\delta^2)/2$, where a is fixed. At the remote point $K = \mathbb{R}_+$, $Jv = -ve_1$, $b = e_1$, $c = ae_1 - e_0/2$, $M = \{0\}$ and $\mu = 1$. Formula (12.5) gives $k = a$. Indeed the nearest remote point is $u = 0$ for small $\delta > 0$, since on this short circle arc the chord distance increases with the separation of the stereographic parameters. Directly,

$$d_{\text{rem}}(\delta) = \frac{2s_\delta}{\sqrt{1 + s_\delta^2}} = \delta + a\delta^2 + O(\delta^3).$$

Thus either sign of the second-order correction occurs in an algebraic probability experiment satisfying all hypotheses. For $a = 0$ the quadratic coefficient vanishes and the next term is $-\delta^3/8$.

These results apply in any coefficient dimension and to any fixed number of latent components once their stated finite-fibre and feasible-cone hypotheses have been verified. They are not a classification of all rank-deficient singularities. The endpoint double root at the geometric wall requires a different parametrization and a different separation argument, supplied in Section 14.

13. SINGULAR ENTRANCES IN SEMIALGEBRAIC FAMILIES

The regular-corner calculation of Section 12 does not exhaust remote identification. We now describe entrance orders without a corner parametrization or an injective observation derivative. The required finiteness is finiteness of the exact parameter fibre, modulo any already specified finite symmetry, not smoothness of its representatives.

Let $X \subset \Theta \times K$ be semialgebraic with compact fibres in a fixed compact set K . Let $f_\theta : X_\theta \rightarrow \mathbb{R}^M$ and $A_\theta : X_\theta \rightarrow \mathbb{C}[z]$ be continuous semialgebraic maps, the latter monic of fixed degree. For the probability experiment, f is its weighted square-root embedding. Let

$$p_\theta(\delta) = f_\theta(x_\theta(\delta)), \quad x_\theta(\delta) \longrightarrow x_{0,\theta}, \quad \delta \downarrow 0,$$

be a semialgebraic centre path with p_θ extending C^1 to zero. All data, including this path, have fixed format. Assume $f_\theta^{-1}(p_\theta(0))$ is finite. After a finite semialgebraic partition its elements can be listed by continuous semialgebraic sections $x_{0,\theta}, \dots, x_{N,\theta}$; subsequent refinements will be understood. Choose pairwise disjoint small closed relative neighbourhoods $N_{j,\theta}$ of these points, and put

$$d_j(\theta, \delta) = \min_{x \in N_{j,\theta}} \|f_\theta(x) - p_\theta(\delta)\|. \quad (13.1)$$

The neighbourhoods can be chosen semialgebraically, for example using closed balls of radius smaller than a quarter of the least separation of the listed fibre points.

Theorem 13.1 (Finite entrance orders and exact membership). *There is a finite Nash stratification of the parameter set with the following properties. For each j either d_j is identically zero for all sufficiently small positive δ , or there are a rational $\rho_j \geq 1$ and a positive Nash function a_j such that, for every compact subset H of the stratum,*

$$d_j(\theta, \delta) = a_j(\theta)\delta^{\rho_j} + O_H(\delta^{\rho_j + \nu_j}), \quad \nu_j > 0. \quad (13.2)$$

The nonzero exponents belong to a finite rational list determined by the fixed-format presentation. The germ of d_j is independent of the sufficiently small isolating neighbourhood.

For any specified positive semialgebraic threshold $t_\theta(\delta) \rightarrow 0$, a further finite partition decides, for each j , whether $t_\theta - d_j$ is eventually positive, negative, or identically zero. The remote component is included precisely in the first and third cases. At every fixed parameter the feasible target root sets converge in Hausdorff distance to

$$\bigcup_{j: d_j \leq t_\theta \text{ eventually}} \{Z(A_\theta(x_{j,\theta}))\}. \quad (13.3)$$

All the strata, exponents, leading-coefficient graphs and membership decisions are effectively describable for algebraic input. No regular-corner assumption is made in these assertions.

Proof. First fix a parameter. The boundary of an isolating neighbourhood contains no point of the exact fibre. Its observation distance from $p(0)$ has a positive minimum. The same is true of the complement of smaller neighbourhoods of all the exact points. Minima over these compact sets are semialgebraic functions of the parameter. Refinement makes them continuous and positive, so the resulting separation bounds are uniform on compact subsets of a stratum. Radii and the finitely many fibre sections admit the same refinement. Finite semialgebraic choice of the fibre points follows by cell decomposition of their finite-fibre graph.

The competitor x_j gives $d_j \leq \|p(\delta) - p(0)\| = O(\delta)$. If a sequence of minimizers did not tend to x_j , compactness would give a different point of the exact fibre in N_j . This is impossible. The same argument for the annulus between two smaller isolating neighbourhoods shows that their minima are equal for all sufficiently small δ . Their distance germs, not just their leading powers, are therefore independent of this choice.

The graph of (13.1) is a first-order formula: there is a point attaining the displayed squared distance, and every point of N_j has at least that squared distance. Quantifier elimination makes the graph semialgebraic. An eventually nonzero nonnegative one-variable semialgebraic function has an expansion $a\delta^\rho + o(\delta^\rho)$ with $a > 0$ and rational ρ . For clarity about finiteness in families, apply Lemma B.1 to a quantifier-free graph formula for d_j . Among the finitely many nonzero specialized polynomial supports, two lowest-order monomials must balance on a positive graph branch. Their exponent differences give a finite rational candidate list. Eventual vanishing is tested separately. The comparability test used in (B.1) selects the actual exponent by a first-order formula. The inequality $d_j = O(\delta)$ forces $\rho \geq 1$.

The limit $a = \lim d_j/\delta^\rho$ also has a first-order graph. Refine until it is positive and Nash. On such a stratum the error $|d_j/(a\delta^\rho) - 1|$ tends to zero on each fibre. Proposition 4.2 gives a positive rational power bound and a positive continuous radius. Taking their bounds and the bounds on a on compact subsets proves (13.2). Thus neither a metric rate nor uniformity has been deduced from topological triviality.

For the threshold assertion, a one-variable semialgebraic function has constant sign or is zero on some terminal interval $(0, \eta)$. Each of these alternatives has a first-order formula with the parameter free, so a finite partition decides them. The sign may be determined by a higher Puiseux coefficient when leading terms coincide; no finite truncation is declared sufficient before

the sign test has been made. If the difference is identically zero, compact attainment in (13.1) puts a point of N_j in the closed ball. Thus equality is inclusion, not an assigned side of a critical ray.

The exterior separation already proved forces every small-radius competitor into one of the N_j . Within an included neighbourhood, all feasible points tend to its unique exact point, while a minimizer provides at least one feasible point for every sufficiently small δ . Continuity of the target and of finite-degree roots proves both Hausdorff inclusions in (13.3). The centre path ensures the union is nonempty. Finally every selection, limiting graph and sign test used above is a finite real-algebraic formula. The finite-support algorithm and real-root isolation in Theorem 5.1 prove the stated effectivity. $\square$

This theorem describes singular *entrance orders*, not a classification of singularities of the source. Its proof uses the relative semialgebraic machinery for a conclusion not supplied by a regular tangent-cone projection: entrances tangent to the centre path may occur at strictly higher rational orders.

Theorem 13.2 (Realization of every rational contact order). *Every rational $\rho \geq 1$ occurs as a positive remote entrance exponent in a compact semialgebraic model with strictly positive probability observations, a two-point exact fibre and distinct polynomial targets on its two representatives. For $\rho = n/m > 1$ the probability maps can be chosen polynomial in their respective branch parameters, of degree at most n . Hence there is no universal finite list independent of format, although Theorem 13.1 gives a finite list for every fixed-format family.*

Proof. Choose a strictly positive probability vector P_0 with at least three cells and linearly independent zero-sum vectors A, B such that $\langle A, B \rangle_I = 0$, where $\langle H, H' \rangle_I = \sum_e H_e H'_e / P_{0,e}$. Let the model be the disjoint union of two sufficiently short closed intervals, with maps

$$P_{\text{base}}(s) = P_0 + As, \quad P_{\text{rem}}(u) = P_0 + Au^m + Bu^n, \quad s, u \geq 0.$$

All probabilities are positive after shortening the intervals. Independence of A, B makes the exact fibre at P_0 precisely the two origins. Give the two intervals different constant monic degree-one targets. Take the centre path $s = \delta$ and use Hellinger distance.

Put $x = u^m$ and $\rho = n/m > 1$. The trial $x = \delta$ gives an entrance upper bound $O(\delta^\rho)$. Hellinger distance is locally comparable to Euclidean probability distance, and independence gives

$$|x - \delta| + x^\rho \leq C \|A(x - \delta) + Bx^\rho\|.$$

Every minimizer thus has $x = \delta + O(\delta^\rho)$. Write $x = \delta + v\delta^\rho$; the relevant v are bounded. The square-root embedding, expanded uniformly at P_0 , gives

$$\delta^{-\rho} h(P_{\text{rem}}(u), P_{\text{base}}(\delta)) = \frac{1}{2} \|vA + B\|_I + o(1).$$

Every bounded v is feasible for small δ since $\rho > 1$. The minimum of the right side is $\|B\|_I/2 > 0$ by the chosen orthogonality. This proves the desired exponent and leading coefficient. For $\rho = 1$, use instead $P_{\text{rem}}(u) = P_0 + Bu$, with the same orthogonal A, B . Projection onto the nonnegative B ray gives the positive coefficient $\|A\|_I/2$. These constructions concern the stated general class of probability models; they do not assert that every exponent occurs in the cubic binary family. $\square$

13.1. An algebraic test for the regular regime. The following criterion derives the corner hypotheses from a local presentation. It also separates two ways for that regime to fail: singular feasible geometry and tangency of the centre path to a regular observation image.

Proposition 13.3 (A checkable corner criterion). *At a remote representative x_* suppose the model is locally*

$$X = \{x : h(x) = 0, g_i(x) \geq 0\},$$

with Nash functions. Assume $Dh(x_*)$ has full row rank and the differentials of the active g_i , restricted to $\ker Dh(x_*)$, are linearly independent. Put

$$K_* = \{v : Dh(x_*)v = 0, \quad Dg_i(x_*)v \geq 0 \text{ for active } i\}.$$

Suppose f has a C^2 Nash extension near x_* , the centre admits $p(\delta) = p(0) + \delta b + \delta^2 c + o(\delta^2)$, and

$$\ker Df(x_*) \cap K_* = \{0\}, \quad p'(0) \notin Df(x_*)K_*. \quad (13.4)$$

Then a full Nash corner chart exists, its observation derivative is cone-coercive, and the entrance has exponent one. The first and second coefficients are exactly those of Theorems 12.1 and 12.2. For algebraic input all these hypotheses are finite algebraic or first-order tests.

Proof. The Nash implicit-function theorem first supplies coordinates on $h = 0$. The independent active functions can then be used as the first coordinates on this manifold, completed by free coordinates. In these coordinates the feasible germ is exactly an orthant times a Euclidean space; inactive inequalities have a strict margin. The derivative of the inverse chart identifies this orthant cone with K_* . The first condition in (13.4), compactness of its unit section and continuity give a positive coercivity constant. Its linear image is a closed polyhedral cone, so the second condition gives a strictly positive attained distance from $p'(0)$ to that image. The existing regular-corner theorems now apply, including nonunique preimages of the first projected point. Matrix ranks are tested by minors; both conditions in (13.4) are first-order statements in linear equations and inequalities. Nash branch specifications and their derivatives at algebraic points are encoded by their polynomial equations and isolating conditions. $\square$

Metric projection, constrained value functions and second-order sensitivity have a substantial existing theory [16, 8, 9]. The regular-corner expansion is used here in that role, with its proof and normalization retained. The additional statements are the fixed-format classification and exact membership rule without its regularity hypotheses, the realization theorem, and the algebraic verification of the regular regime. The two cubic walls of Sections 11 and 14 are explicit realizations of the linear regime; they are not offered as an exhaustive list of singular entrance geometries.

14. THE GEOMETRIC ENDPOINT WALL

We now complete a second wall mechanism in the same rank-deficient cubic experiment. Fix $D > r > 0$, strict weights $\alpha, \gamma = 1 - \alpha$, positive $U = (u, u)$ and invertible positive V , and the exterior design. Put

$$s_0 = \frac{2rD}{3D - r}, \quad s_\delta = s_0 - \delta, \quad f = z(z - r)^2, \quad g_\delta = (z - s_\delta)^3. \quad (14.1)$$

Then $2r/3 < s_0 < r$. Assume the strict weight margin

$$E_0 = \gamma A(s_0) - 4\pi r^3 > 0, \quad A(s) = (r + 3s)(2r - 3s)^2. \quad (14.2)$$

Let θ_δ be this base centre and P_δ its observation. Here $B(\theta_\delta) = -(3D - r)\delta$. Thus positive δ is the identified side, while the weight constraint stays inactive locally. Write $c_* = 4r^3/A(s_0)$ and

$$a = \frac{4rs_0}{r + 3s_0}, \quad h_* = (1 - c_*)f + c_*g_0 = (z - a)(z - D)^2, \quad \beta = \gamma/c_*.$$

The remote weight on h_* is $\beta > \pi$, and its other weight is $1 - \beta > \pi$. Its first channel is (u, u) and its second channel is

$$v_f^* = \frac{\alpha v_1 + (\gamma - \beta)v_2}{1 - \beta}, \quad v_h^* = v_2. \quad (14.3)$$

Both columns are positive and distinct. Denote this remote point by ξ_B . Theorem 10.1 gives exactly the base and this remote representative at $\delta = 0$.

Proposition 14.1 (The endpoint-root parametrization). *A full relative neighbourhood of ξ_B , modulo the component permutation, has the following exact feasible parametrization:*

$$\begin{aligned} f_v(z) &= (z - a_0)((z - r - b_0/2)^2 - X), \\ h_v(z) &= (z - a - c_0)(z - D - d_1)(z - D - d_2), \\ K_B &= \{a_0, X \geq 0, \quad d_1 \leq d_2 \leq 0\}. \end{aligned} \quad (14.4)$$

The weight on h is $\beta + k$, and the four binary channel coordinates have their unrestricted small increments. The other coordinates b_0, c_0, k are free. This is a polynomial parametrization with continuous semialgebraic inverse; no differentiability of its inverse at $d_1 = d_2$ is asserted. Its first polynomial jets are

$$p_f = -a_0(z - r)^2 - b_0z(z - r) - Xz, \quad p_h = -c_0(z - D)^2 - (d_1 + d_2)(z - a)(z - D). \quad (14.5)$$

Let S_B be the score at ξ_B applied to these jets and the stochastic increments. Then S_B has no nonzero zero on K_B . If $B_B = \frac{d}{d\delta}P_\delta|_0$, then $B_B \notin S_B K_B$. In particular

$$\mu_B = \frac{1}{2} \min_{v \in K_B} \|S_B v - B_B\|_{I,0} > 0. \quad (14.6)$$

The minimum is attained. After seven free coordinates are eliminated, it is a quadratic program with three nonnegative coordinates and eight independent-support tests.

Proof. Separated simple and double factors give the first component's simple root, quadratic mean and squared half-separation as analytic functions of its coefficients. At the second component the separated simple root is analytic; the two roots near D have a unique increasing ordering and depend continuously and semialgebraically on the coefficients. Their feasibility is exactly $d_1 \leq d_2 \leq 0$. All other root bounds, weights and channel inequalities have strict margins, except $a_0 \geq 0$ and $X \geq 0$. This proves exact coverage and injectivity of the parametrization, not merely containment of tangent directions. Its polynomial formulas give a uniform $O(\tau^2)$ coefficient remainder on substituting τv for bounded $v \in K_B$. Positive normalizers and probabilities give the same remainder for the observation embedding.

We prove the two score assertions directly. Put $\dot{g} = 3(z - s_0)^2$, the derivative of g_δ at zero. Equality of remote and base scores implies equality of their derivatives of q, K by the differentiated normalizer argument in Lemma 2.1. The coefficient second marginal at the remote point is $(1 - \beta)v_f^* f + \beta v_2 h_*$. Its two coefficient columns are independent. Modulo $\text{span}(f, g_0)$, equality of its derivative with $\gamma v_2 \dot{g}$ gives

$$p_f = t_f(g_0 - f), \quad p_h = c_* \dot{g} + t_h(g_0 - f).$$

The polynomials have the stated degrees, so no additional monic direction occurs. Evaluation of the first identity at 0 and r gives $t_f = a_0 r^2 / s_0^3 \geq 0$ and $t_f = -Xr / (r - s_0)^3 \leq 0$. Thus $t_f = 0$ and $p_f = 0$.

To interpret the second identity, consider the algebraic double-root pencil curve, without imposing its endpoint constraint,

$$\bar{h}_\delta = (1 - c_*(s_\delta))f + c_*(s_\delta)g_\delta = (z - a(s_\delta))(z - b(s_\delta))^2, \quad b(s) = \frac{rs}{3s - 2r}.$$

At zero, $b(s_0) = D$ and

$$\dot{b} = \frac{d}{d\delta}b(s_0 - \delta)\Big|_0 = \frac{2r^2}{(3s_0 - 2r)^2} > 0. \quad (14.7)$$

Both $p_h(D)$ and $\dot{h}(D)$ vanish. Since $g_0(D) - f(D) \neq 0$, this determines t_h uniquely and implies $p_h = \dot{h}$. Differentiation in z at D now gives

$$-(d_1 + d_2)(D - a) = p'_h(D) = -2(D - a)\dot{b}.$$

This would require $d_1 + d_2 = 2\dot{b} > 0$, contrary to feasibility. Therefore $S_B v = B_B$ is impossible.

For a zero score the same second-marginal argument instead gives $p_f = t_f(g_0 - f)$ and $p_h = t_h(g_0 - f)$. The first polynomial again vanishes. Evaluating the second at D forces $t_h = 0$. Its coefficients then give $c_0 = 0$ and $d_1 + d_2 = 0$, whence $d_1 = d_2 = 0$ on K_B . The first polynomial's independent columns give $a_0 = b_0 = X = 0$. Coefficient functionals dual to f, h_* recover zero derivatives of both coefficient matrices, and their entry, row and column sums give zero weight and channel increments. Thus the entire cone vector is zero. Compactness of the feasible unit sphere yields $\|S_B v\|_{I,0} \geq c\|v\|$ on K_B . The objective in (14.6) is consequently coercive and has a positive attained minimum.

Finally the score depends on d_1, d_2 only through $w = -(d_1 + d_2) \geq 0$. Every $w \geq 0$ is realized, for example by $d_1 = d_2 = -w/2$. The remaining nonnegative coordinates are a_0, X ; there are seven free coordinates. Their columns are independent by the zero-score argument. Fisher-orthogonal projection off those columns reduces the problem to three nonnegative variables. The minimal-positive-support argument of Proposition 10.5 gives the eight support tests, including the empty support and only inverting positive definite supported Gram matrices. This remains valid when the full projected Gram matrix is singular. $\square$

Theorem 14.2 (The geometric-wall two-scale law). *Under (14.1)–(14.2), the remote entrance satisfies*

$$d_{\text{rem},B}(\delta) = \mu_B \delta + k_B \delta^2 + o(\delta^2), \quad \mu_B > 0, \quad (14.8)$$

where k_B is given by Theorem 12.2 in the parametrization (14.4). In particular, for fixed $\lambda > 0$,

$$\lambda < \mu_B : \quad \omega_{\theta_\delta}(\lambda\delta) = C_{\text{loc}}(\theta_0)\sqrt{\lambda\delta} + O_\lambda(\delta), \quad (14.9)$$

$$\lambda > \mu_B : \quad \omega_{\theta_\delta}(\lambda\delta) = D - r + O_\lambda(\sqrt{\delta}). \quad (14.10)$$

The base local Fisher constant stays positive and finite at the wall. The strict cases in the critical window $t = \mu_B \delta + s\delta^2$ are decided by $s - k_B$.

On the nonidentified side $\theta_{-\varepsilon}$, for all sufficiently small $\varepsilon > 0$, the additional exact pencil interval has length

$$c_D(s_0 + \varepsilon) - c_*(s_0 + \varepsilon) = \Lambda_B \varepsilon^2 + O(\varepsilon^3), \quad \Lambda_B = \frac{-R''_{s_0}(D)}{2(R_{s_0}(D) - 1)^2} \dot{b}^2 > 0, \quad (14.11)$$

where $R_s(z) = f(z)/(z - s)^3$. Nevertheless its exact whole-model spectral diameter is constant:

$$\omega_{\theta_{-\varepsilon}}(0) = D - r. \quad (14.12)$$

Thus the two wall mechanisms differ both in the size of the newborn pencil interval and in the opening of its exact spectral diameter.

Proof. The endpoint parametrization and its score separation satisfy the hypotheses of Theorems 12.1 and 12.2. The derivative of the square-root observation embedding is the score multiplied entrywise by $\sqrt{w_j}/(2\sqrt{P_{0,je}})$, so its entrance minimum is precisely (14.6). This proves (14.8) and the critical-window assertion, including possible nonuniqueness of the minimizing endpoint-root split.

The sorted spectra at the wall are $(0, s_0, s_0, s_0, r, r)$ and $(0, a, r, r, D, D)$. Their distance is $\max(r - s_0, D - r) = D - r$, since $r - s_0 = r(D - r)/(3D - r) < D - r$. The local inverse and local Fisher law of Section 10 are uniform on a compact segment of the base path through zero. Below the entrance threshold all competitors are in that base neighbourhood, proving (14.9). Above it, the localization proof for the remote cone bounds its exact coordinates by $O(\delta)$. The remote endpoint roots move by $O(\delta)$ and its interior double cluster by $O(\sqrt{\delta})$; the base inverse gives the same $O(\sqrt{\delta})$ root bound. A remote minimizer and the centre are both feasible, and the exterior observation gap excludes every other region. The bottleneck triangle inequality proves (14.10).

For the last assertions, $b(s_0 + \varepsilon) = D - \dot{b}\varepsilon + O(\varepsilon^2)$. The function R_s has a strict nondegenerate maximum at $b(s)$ near the wall. Taylor expansion of $R_s(D)$ there, followed by $x \mapsto x/(x-1)$, gives

$$c_D(s) - c_*(s) = \frac{-R_s''(b(s))}{2(R_s(b(s)) - 1)^2} (D - b(s))^2 + O((D - b(s))^3).$$

This proves (14.11). The strict weight margin keeps $\gamma/\pi > c_D(s)$ for small ε , so the entire interval $[c_*(s), c_D(s)]$ is feasible. Its upper endpoint has a root exactly at D . A remote spectrum has increasing ordering $(0, a_c, r, r, b_-(c), b_+(c))$, with $s < a_c < r < b_-(c) \leq b_+(c) \leq D$. Its distance to the base is exactly $\max(r - s, b_+(c) - r) \leq D - r$, and equality is attained at $c_D(s)$. The remote spectra converge uniformly to one spectrum as $\varepsilon \downarrow 0$; their mutual diameter is therefore less than $D - r$ for small ε . Theorem 10.1 excludes any other exact component. This proves (14.12). $\square$

14.1. Exact critical-ray examples. We record two algebraic specializations of the second-order formula. Use the clocks $(6/5, 7/5, 8/5, 9/5, 2, 5/2, 3)$ with equal weights, $u_1 = u_2 = (1/3, 2/3)^\top$, and $v_1 = (1/4, 3/4)^\top$, $v_2 = (3/4, 1/4)^\top$.

Example 14.3 (An endpoint wall with a positive critical correction). Take $D = 1$, $r = 3/5$, $s_0 = 1/2$, $\alpha = 2/5$, $\pi = 1/10$. The remote coordinates and the interval-opening coefficient are

$$a = \frac{4}{7}, \quad c_* = \frac{32}{7}, \quad \beta = \frac{21}{160}, \quad v_f^* = \frac{1}{556} (289, 267)^\top, \quad \dot{b} = 8, \quad \Lambda_B = \frac{38400}{49}.$$

The projected three-variable program has the empty support as its only successful independent-support test. Its linear vector is strictly negative entrywise, so the minimizer has all three nonnegative coordinates zero. The free score has rank seven; hence the first-order minimizer is unique, including the endpoint split. Exact rational arithmetic and positive-root isolation give

$$\begin{aligned} 0.000063237134157102 &< \mu_B < 0.000063237134157103, \\ 0.071857358263070 &< k_B < 0.071857358263071. \end{aligned}$$

Consequently the remote branch is excluded on $t = \mu_B \delta$ for all sufficiently small positive δ . On the opposite side the exact spectral diameter is $2/5$, despite the nonzero quadratic pencil-interval opening.

Example 14.4 (The weight-wall critical ray). At the weight-wall datum of Example 11.7, the four entries of the projected linear vector g_* are strictly negative. The empty cone support is therefore the unique first-order minimizer, with its seven free coordinates recovered by the inverse free Gram matrix. The second-order formula gives

$$0.011285405417542 < k_E < 0.011285405417543.$$

Thus the critical ray $t = \mu_E \delta$ also excludes the remote branch for this datum when $\delta > 0$ is sufficiently small. This sign is a property of this specialization, not a convention at an unresolved ray and not a sign assertion for the whole wall family.

For reproducibility, the finite arithmetic behind both examples can be specified without numerical differentiation. Write the exact remote and base probability expansions at the selected first-order minimizer as

$$P_{\text{rem}}(\delta) = P + \delta A + \delta^2 C + O(\delta^3), \quad P_{\text{base}}(\delta) = P + \delta B + \delta^2 D + O(\delta^3).$$

The rational coefficients are obtained by multiplying the exact polynomial, weight and channel jets and dividing by the normalizer. The square-root expansion then gives

$$\begin{aligned}\mu^2 &= \frac{1}{4} \sum_{j,e} \frac{w_j (A_{je} - B_{je})^2}{P_{je}}, \\ \mu k &= \sum_{j,e} w_j \left\{ \frac{(A_{je} - B_{je})(C_{je} - D_{je})}{4P_{je}} - \frac{(A_{je} - B_{je})(A_{je}^2 - B_{je}^2)}{16P_{je}^2} \right\}.\end{aligned}\quad (14.13)$$

Strict empty-support KKT inequalities and positive semidefiniteness of the projected Gram matrix prove uniqueness of the cone optimum even when that matrix is singular: the difference of objectives is $z^\top Qz - 2g^\top z > 0$ for every nonzero $z \geq 0$. The rational values of μ^2 , μk , the Gram data and isolating intervals are retained with the exact regression. The interval tests square rational endpoints and compare integers after clearing positive denominators. They certify these finite specializations only.

APPENDIX A. FINITE ALGEBRAIC CERTIFICATES AND REAL DESCENT

This appendix supplies the full family argument used in Lemma 3.3 and Theorem 3.4. A variety in the algebraic construction is a reduced separated scheme of finite type over a characteristic-zero field. Smoothness is geometric smoothness. For an effective input the ground constants belong to a specified finite real algebraic extension of $\mathbb{Q}$; generic fibres are presented over its finitely generated function fields. No algorithmic assertion about an arbitrary computable field is intended. Real realizations are subsequently taken in $\mathbb{R}$, with the original sign and branch conditions retained.

An algebraic coordinate presentation consists of finitely many affine coordinate rings, localized by specified nonzero elements, transition morphisms and their inverse identities, and a coordinate map whose relative Jacobian is invertible. An adapted presentation also lists each divisor equation, its expression as a coordinate times a unit, and the monomial identities for the functions. Projective maps are recorded by homogeneous equations or by their finite blow-up sequences. These are finite polynomial data, not a numerical collection of coordinate boxes.

A.1. The Rees algebra and a finite diagram.

Lemma A.1 (One localization for all powers of an ideal). *Let A be a noetherian domain, B an algebra of finite type over A , and $I \subset B$ a finitely generated ideal. There is $a \in A \setminus \{0\}$ such that, for every A_a -algebra A' , with $B' = B_a \otimes_{A_a} A'$,*

$$I_a^n \otimes_{A_a} A' \xrightarrow{\sim} (I_a B')^n \quad (n \geq 0), \quad \left(\bigoplus_{n \geq 0} I_a^n \right) \otimes_{A_a} A' \cong \bigoplus_{n \geq 0} (I_a B')^n.$$

The first isomorphism identifies both sides as submodules of B' . Consequently the blow-up in I_a commutes with this base change. One localization suffices for any specified finite sequence of centre ideals on finitely many affine charts of successive blow-ups.

Proof. The associated graded algebra $\text{gr}_I B = \bigoplus_{n \geq 0} I^n / I^{n+1}$ is of finite type over A : it is generated by B/I in degree zero and the images of a finite set of generators of I in degree one. Since A is noetherian, these algebras are finitely presented. Apply generic freeness [17, Tag 051S] to B as a module over itself and to $\text{gr}_I B$ as a module over itself. After inverting the product of the two resulting elements, both are flat over A_a . Each graded piece is a direct summand as an A_a -module, hence flat. In particular B_a / I_a is flat. Induction using

$$0 \longrightarrow I_a^{n-1} / I_a^n \longrightarrow B_a / I_a^n \longrightarrow B_a / I_a^{n-1} \longrightarrow 0$$

shows that B_a/I_a^n is flat for every $n \geq 1$. Tensoring $0 \rightarrow I_a^n \rightarrow B_a \rightarrow B_a/I_a^n \rightarrow 0$ with A' is therefore exact on the left. Its image in B' is the ideal generated by products of n images of elements of I_a , namely $(I_a B')^n$. The isomorphisms respect multiplication and grading, proving the Rees algebra assertion. Relative Proj commutes with base change; this proves the blow-up assertion. For a finite sequence, apply the argument to the actual centre ideal on every affine chart at each stage and multiply the finitely many localizing elements. Previously proved identities survive further localization. No countable intersection of open sets is used. $\square$

Lemma A.2 (Spreading the entire stratifying diagram). *Let A be a finitely generated integral algebra over a characteristic-zero field k , and let $K = \text{Frac}(A)$. Suppose $Y \rightarrow \text{Spec } A$ is of finite type and a generic certificate consists of a reduced closed chain $Y_{0,K} = Y_{K,\text{red}} \supset \cdots \supset Y_{q,K} = \emptyset$, projective $Z_{j,K} \rightarrow Y_{j,K}$, and specified inverse isomorphisms above $E_{j,K} = Y_{j,K} \setminus Y_{j+1,K}$. Then, after replacing A by A_a for one nonzero a , the chain can be chosen with $Y_0 = Y_{\text{red}}$, all its members flat with geometrically reduced fibres, and the stated isomorphisms hold above the actual complements $E_j = Y_j \setminus Y_{j+1}$. In particular they hold above $(E_j)_s$ on every fibre. The union of these complements is the whole reduced fibre; no constancy of the number of irreducible components is required.*

Proof. First replace Y by its reduction. Take reduced closures of the generic $Y_{j,K}$ inside Y and use $Y_q = \emptyset$. Clear denominators in the projective equations and affine presentations of the generic maps. All schemes involved are of finite presentation over the noetherian base. The finite-presentation limit lemma [17, Tag 01ZM, Lemma 32.10.1(1)–(3)] descends objects, morphisms, and equalities of morphisms. Apply it to the inverse maps on the generic complements, their inclusions, and both inverse identities. The complements used on the model are $Y_j \setminus Y_{j+1}$; their generic fibres are the prescribed complements. The descended inverse maps and equalities therefore give isomorphisms on these actual open subschemes after a further localization, not merely on unspecified subsets with the same generic points. Finite affine covers and their intersection diagrams are included in this application of the limit lemma. Thus chart coverage, as well as the formula for each map, descends.

Each reduced generic member is geometrically reduced because K has characteristic zero. Generic freeness makes each member flat on a dense open set. The generic-fibre reducedness theorem [17, Section 37.26, Lemma 37.26.4] then permits a further common shrink so that all fibres of these finitely many members are geometrically reduced. This is an imposed property, not an optional replacement of scheme structure after specializing. Finally, on every fibre the sets $Y_{j,s} \setminus Y_{j+1,s}$ form a disjoint partition of $Y_{0,s}$ because the chain ends in the empty set. For any point there is a unique index at which it leaves this finite descending chain. The corresponding isomorphism covers that point, including points on a special fibre component. This argument neither identifies fibre components with generic components nor assumes their number to be constant. $\square$

A.2. Relative smoothness, normal crossings, and exceptional sets. We now prove Lemma 3.3 with a precise list of the further conditions. Separate the finitely many open-and-closed components of each smooth generic source according to the identically zero functions. These components and their idempotent identities are part of the finite diagram. On a component on which a function is declared zero, descend that zero identity. On every other component principalize the product of the remaining functions, recording all factors and units.

The absolute input is characteristic-zero resolution and principalization [4, 5]. More precisely, the variety resolution of [5, Theorem 1.1] supplies the smooth sources; the marked-ideal resolution of [5, Theorem 1.3], applied with mark one, supplies principalization together with the normal-crossing exceptional boundary. A product which is a monomial times a unit makes each of its factors a monomial times a unit on the regular local rings. This use is over the field K .

Compatibility with arbitrary special fibres is supplied by Lemma A.1, not by a claim that functoriality for smooth maps implies arbitrary base-change compatibility.

Spread the finite sequence of centres and its adapted charts. The following finite conditions are imposed before retaining the open base. The source and centre smooth loci are the loci on which their finite presentations satisfy the relative smoothness criterion. On a smooth source the intersections of the designated divisor equations must be smooth of their prescribed relative codimensions; in adapted coordinates this says that the appropriate relative differentials are independent. This condition is imposed for every subset of the finite divisor list. Each coordinate chart must be etale, each declared inverse unit must multiply its unit to one, and the finite charts must cover their source. All scheme-theoretic identities in these statements are spread by the finite-diagram lemma. Their remaining rank conditions are open algebraic conditions; complements of chart unions are constructible algebraic sets.

For clarity, the discarded algebraic parameter set is the union of the closures of images of the following bad sets, together with the localizing hypersurfaces already used:

- (1) the original irreducible components not dominating the current reduced integral base;
- (2) nonsmooth points of a source or centre, and failures of the specified relative dimension or centre codimension;
- (3) failures of smoothness or codimension of any designated divisor intersection, or failure of an adapted etale Jacobian;
- (4) the complement of the chosen finite source charts, and any zero of a function declared invertible;
- (5) failures of a descended zero identity, a chart transition, or an inverse isomorphism on an actual complement in the closed chain.

For items (4)–(5) the bad set is empty once the relevant identities and coverage have descended; it is harmless to include it explicitly in the certificate. Every bad set in this list has empty generic fibre: this is respectively dominance, generic smoothness in the given certificate, generic normal crossings, generic coverage and invertibility, or the specified generic identities. Images are constructible by Chevalley’s theorem [17, Tag 054K]. A constructible subset of an integral noetherian space which is dense contains a nonempty open subset and hence the generic point. Therefore the closure of each of these images is proper. This justification applies also when $Y \rightarrow S$ is not proper.

After their removal, apply Lemma A.1 on every spread blow-up chart. The resulting smooth sources and centres, relative normal crossings, unit identities and finite chart covers hold on every retained fibre. A nonzero monomial cannot specialize to the zero function on a fibre component: its unit is invertible and its finitely many divisors each have relative codimension one. Their union cannot contain a component of the smooth fibre. An identically zero generic function remains zero by its descended identity. The open-and-closed source decomposition prevents a change of these flags through an unlabelled component intersection. Flatness of the closed chain and smoothness of the sources exclude additional vertical components on the retained open set. Coverage of any special fibre components is, more directly, the partition argument in Lemma A.2. This proves all fibrewise assertions of Lemma 3.3.

Each discarded algebraic subset has strictly smaller dimension than its current integral base. Replace it by its reduced irreducible components and reconstruct the entire generic certificate on each component. There are finitely many components at each step and depth at most $\dim S + 1$. This gives finite termination. The exceptional list is not required to be minimal. Importantly, a change of a *real* accessibility flag need not lie in a proper algebraic subset: two full-dimensional real chambers can have different flags. Such changes are treated by finite real quantifier-elimination partitions below, not by this dimension argument.

A.3. Exact real algebraization.

Lemma A.3 (The square-slack cover with all branch conditions). *Let S be a semialgebraic base and $\Gamma \rightarrow S$ a semialgebraic proper map realized as a graph in $S \times \mathbb{R}^N$. The graph includes all auxiliary functions, sign choices and denominator exclusions. After a finite base refinement, it admits finitely many algebraic square-slack lifts whose real points over S project exactly onto Γ . Projection is proper over compact subsets of S . Every later map factoring through these lifts projects only to points of Γ .*

Proof. Let C be the ordinary semialgebraic closure of Γ in the ambient real algebraic space. Properness implies that Γ is relatively closed in $S \times \mathbb{R}^N$: a convergent sequence over a convergent base sequence lies over a compact subset of S and has a graph limit there. It follows that $C \cap (S \times \mathbb{R}^N) = \Gamma$. A closed semialgebraic set is a finite union of basic closed semialgebraic sets [7, Theorem 2.7.2]. For a piece

$$C_l = \{x : f_{l,a}(x) = 0, p_{l,b}(x) \geq 0\}$$

use the algebraic set

$$Y_l = \{(x, s) : f_{l,a}(x) = 0, p_{l,b}(x) = s_b^2\}.$$

Every real point of Y_l projects into C_l , and each point of C_l has a real lift. Over S , the union therefore projects exactly to Γ . Over a compact base subset the graph is compact. The slack coordinates are bounded by the square roots of maxima of the finitely many polynomials there, and the lift is closed; hence the lifted real set is compact. This proves properness and the last assertion.

Strict exclusions are encoded in the complete graph formula before this construction. For a denominator excluded everywhere on that compact graph, $d \neq 0$ may also be retained by adjoining w with $dw = 1$, together with any required sign of d . Nonnegative square-root branches are specified in the graph, not selected after taking an algebraic closure. If adjoining such coordinates destroys properness on the initial base, first apply Hardt triviality to the enlarged graph, whose individual fibres are compact, and refine the base. A trivialization with compact fibre makes the restricted graph map proper. Closure is taken only *after* this step. Consequently no point with a vanishing excluded denominator is reintroduced over the retained stratum. General Boolean branch conditions are instead kept in the complete graph formula: one does not split a compact fibre into nonclosed pieces, adjoin unbounded reciprocals on those pieces, and then assume their fibres are compact. The equality $C \cap (S \times \mathbb{R}^N) = \Gamma$ already preserves all such conditions without that operation. $\square$

Lemma A.4 (Real branch selection and simultaneous descent). *The root labels and Nash graph functions used in the cluster construction can be carried on finitely many selected real Nash branches of algebraic base covers. On a finite semialgebraic refinement, the proper resolved square-slack sources cover exactly the original graph in every fibre. Descent by root permutations preserves the entire joint coefficient and root relation, not merely its scalar order list.*

Proof. Refine a Nash stratum by the regular locus of each integral component of its algebraic closure. Its lower-dimensional singular parts are treated recursively. A Nash function is algebraic over the coordinate function field. Adjoin the finitely many graph coordinates needed here to that field. The extension is finite and separable in characteristic zero. Choose equations and invert their leading coefficients and discriminants; a further localization gives a finite etale algebraic cover. Include the original graph formula to select the intended real branches. Finite semialgebraic cell decomposition makes each selected graph a Nash section and separates its alternatives. Complex sheets and real sheets failing the original formula are not retained. Thus monodromy is handled by base refinement, rather than by globally labelling a nontrivial cover. Exceptional leading

coefficients and discriminants are handled by the reduced lower-dimensional base recursion. A finite union of selected sections has compact inverse image over compact subsets of a stratum.

Apply Lemma A.3 on each selected branch. For a real point y of a lifted fibre, choose the index with $y \in Y_{j,s} \setminus Y_{j+1,s}$ in the spread singular-locus chain. The inverse isomorphism of Lemma A.2 supplies a real point of its smooth source above y . A blow-up of a smooth real variety in a smooth centre of positive codimension c has fibre $\mathbb{RP}^{c-1}$ at a real centre point and is an isomorphism away from the centre. In local smooth coordinates the centre is a coordinate subspace, which proves both assertions directly. Such a blow-up therefore does not lose real points. Composing the finitely many principalizations and taking the finite union over j proves real surjectivity. A real singular point with no top-dimensional real branch is covered by a later member of the chain. Projectivity of these maps and properness of the slack projection prove properness of the real cover.

Conversely every real resolved point factors through a real slack point on the selected branch, and hence projects to the original graph by Lemma A.3. This proves equality, not just inclusion, of the covered real fibres. A root-label permutation simultaneously permutes cluster indices, their coefficient coordinates and root arrays. The equations defining Ψ and the fixed-centre formula (4.2) are equivariant for this simultaneous action. A finite union commutes with taking a fibrewise closure; projection to the finite quotient therefore preserves the joint relation and its root image. Descent never pairs one coefficient from one sheet with another coefficient from a different sheet. The bottleneck metric is invariant under these permutations. $\square$

A.4. Real accessibility and completeness of the order list.

Lemma A.5 (Real-accessible divisors). *On a resolved square-slack source with the above relative adapted coordinates, accessibility of a divisor is a first-order real condition on the parameter. Every recorded divisor has an admissible real transversal attaining its monomial orders. Every real degeneration relevant to a finite coefficient order is controlled by one of the recorded divisors. Thus the minimum in Corollary 3.6 is both an upper-inequality exponent and an attained exponent.*

Proof. In a chart with source equations $E(\theta, y) = 0$, chart exclusions $D(\theta, y) \neq 0$, and adapted divisor coordinates $y_1, \dots, y_l$, the flag for i with $a_i > 0$ is the formula

$$\exists y : E(\theta, y) = 0, \quad D(\theta, y) \neq 0, \quad y_i = 0, \quad \prod_{h \neq i, h \leq l} y_h \neq 0, \quad \det J_{\text{et}}(\theta, y) \neq 0. \quad (\text{A.1})$$

Inverse units and the selected real base formula are included in E, D . Charts are the ones meeting the inverse image of $G = 0$. Since the full real slack fibre already represents the admissible graph, there is no additional unrecorded sign restriction on source directions. At a point satisfying (A.1), the real inverse function theorem gives a transversal with $y_i = t$ and other adapted coordinates fixed. For small nonzero t , $G > 0$, and every point projects into the original graph. The unit bounds give $G \asymp |t|^{a_i}$ and $H \asymp |t|^{b_i}$ for each nonzero H on that source component.

A real degeneration with $G \rightarrow 0$ has a lifted accumulation point above $G = 0$, by properness. Some adapted chart contains that point. If it lies on several divisors, relative normal crossings allows a small real perturbation *on any one of them* avoiding all the others. Hence each divisor of positive G order meeting this real chart has a point of the form (A.1). In a smaller box, with β the minimum of its recorded ratios, $b_i \geq \beta a_i$ for $a_i > 0$ and $b_i \geq 0$ otherwise. The monomial identities and positive unit bounds imply $H \leq CG^\beta$ throughout the box, including its intersections. Equivalently, along a lifted arc the order ratio is a weighted average of the b_i/a_i with nonnegative additional terms from indices with $a_i = 0$; it cannot be smaller than their minimum. A finite cover near the compact exact fibre gives this inequality for all competitors. The transversal for a minimizing divisor attains it. If G is identically zero on a source with real points, the exact-target condition gives $H = 0$ there. A smooth fibre component with a real point

has a Zariski-dense real open subset. Thus a nonzero monomial H cannot occur on such an exact real component; real-empty components contribute no order. This handles zero flags as well as the nonzero ratios.

Finally (A.1) is a finite polynomial formula. Quantifier elimination partitions the base by all its truth values. These semialgebraic pieces may have full dimension. Nash refinement makes the choices compatible with the selected base branches. This partition is finite without an algebraic dimension-drop assertion. $\square$

Completion of Theorem 3.4. Stratify the enlarged graph and the exact cluster data as in Lemma A.3, and carry labels as in Lemma A.4. Over an integral algebraic base resolve the reduced generic fibre, then its reduced singular locus, and continue until the empty set. Dimension decreases in this chain. Principalize on its smooth sources the product of the nonzero functions. Lemmas A.1 and A.2, followed by the explicit bad-locus construction, spread this certificate over a nonempty open base with all fibrewise properties. Repeat on the finitely many reduced exceptional components. Lemma A.4 proves exact real coverage after descent, and Lemma A.5 proves the accessibility assertions and the completeness of the orders.

For a fixed fibre, cover the compact inverse image of $G = 0$ by finitely many smaller real coordinate boxes with positive lower and finite upper unit bounds. Properness implies that a sufficiently small observation neighbourhood is covered: otherwise a sequence with $G \rightarrow 0$ outside the chosen source neighbourhood would have a lifted accumulation point above $G = 0$ outside it. The same argument works above a compact base subset, since the selected label graph, the slack lift and the projective resolved maps are proper there. Alternatively their positive radii and unit bounds can be selected semialgebraically and made continuous by finite refinement. Their extrema on a compact parameter subset give common neighbourhoods and bounds. Fibrewise proper source maps must not be confused with the smaller, nonproper coordinate boxes. This proves all existence and compact-uniform assertions. The effective construction for algebraic input is detailed at the end of Appendix B. $\square$

APPENDIX B. FINITE SUPPORT ORDERS, UNIFORM DECAY, AND EFFECTIVE OUTPUT

The metric and effective parts can be proved without making a parameter-dependent choice of a Puiseux branch. Only the ordinary one-variable algebraic-branch theorem is used to prove that a finite candidate list is exhaustive. Selection from that list is then a first-order decision problem.

Lemma B.1 (A finite support test for the order). *Let $e(\theta, t) \geq 0$ be a semialgebraic function, defined for $0 < t < \tau(\theta)$, where $\tau > 0$ is semialgebraic, and suppose $e(\theta, t) \rightarrow 0$ for fixed θ . A quantifier-free graph formula supplies a finite computable set $\mathcal{B} \subset \mathbb{Q}_{>0}$ such that for every θ either $e(\theta, t)$ is eventually zero or exactly one $b \in \mathcal{B}$ satisfies*

$$\exists c, C, a > 0 \forall t : \quad 0 < t < a \implies ct^b \leq e(\theta, t) \leq Ct^b, \quad a < \tau(\theta). \quad (\text{B.1})$$

Each of these alternatives is an effective semialgebraic condition for algebraic input. No real exponent is quantified over.

Proof. Eliminate quantifiers in the graph formula and let $P_1, \dots, P_N$ be its finite list of polynomials in (θ, t, y) . Fix θ and discard those specializing to the zero polynomial in (t, y) . At a graph point some remaining polynomial must vanish. Otherwise all nonzero polynomial signs are unchanged on a small two-dimensional neighbourhood; the zero specialized polynomials impose constant truth values there. The graph formula would then contain that neighbourhood, contradicting single-valuedness. For each remaining P_i , the set $\{t : P_i(\theta, t, e(\theta, t)) = 0\}$ is semialgebraic in one variable. These finitely many sets cover a punctured interval. After their finitely many endpoints are removed, at least one vanishing equation holds on an entire interval $(0, a)$. A polynomial independent of y cannot vanish there unless it was one of the discarded zero polynomials.

By one-variable semialgebraic decomposition, on a smaller interval the function is either identically zero or positive, continuous, and a single real algebraic branch of such an equation. The ordinary algebraic Puiseux theorem gives, in the positive case, $e(\theta, t) = a_\theta t^{b_\theta} + o(t^{b_\theta})$ with $a_\theta > 0$ and rational $b_\theta > 0$; see [7, 1]. Write its vanishing equation as $\sum p_{ij}(\theta) t^i y^j = 0$. Inserting the first term shows that $\min(i + b_\theta j)$ is achieved at least twice among nonzero specialized coefficients. Otherwise its unique lowest term could not cancel. Therefore

$$b_\theta = (i - i')/(j' - j) > 0$$

for two distinct support points of one of the original P_i . Take $\mathcal{B}$ to be all such positive ratios. Specialization may remove support points but cannot add a pair to this list. This proves finiteness uniformly in θ , without following algebraic branches through parameter specializations.

Eventual zero is the formula $\exists a > 0 \forall t (0 < t < a \Rightarrow e(\theta, t) = 0)$, with $a < \tau(\theta)$. For each candidate, (B.1) is a first-order polynomial formula: replace $t^{p/q}$ by its unique positive root, or put $t = s^q$ throughout. Quantifier elimination decides it. The actual branch expansion supplies one true candidate; two distinct powers cannot be bounded above and below by positive multiples of each other as $t \downarrow 0$. Exactly one candidate is therefore true in the nonzero case. This also tracks the branch selected by the original Boolean graph formula, including all specialized coefficients and inequality faces, rather than selecting a root of a polynomial without checking graph membership. $\square$

Full proof of Proposition 4.2. Use Lemma B.1. If the candidate set is nonempty, choose a positive rational $\nu < \min \mathcal{B}$; if it is empty use $\nu = 1$. On every parameter fibre $e(\theta, t) \leq t^\nu$ eventually, including the eventual-zero case. The set of pairs (θ, a) with $0 < a \leq \min(1, \tau(\theta))$ and this inequality for all $0 < t < a$ is semialgebraic and has a nonempty positive fibre at every parameter. Semialgebraic choice selects a positive $a = \tau_0(\theta)$, not a possibly vanishing limit value at the boundary of a stratum. A finite Nash stratification makes this selected function continuous. Its minimum on a compact subset of a stratum is strictly positive; the estimate is uniform there. The exponent may be smaller than the sharp order, which is exactly what is needed for the uniform remainder. $\square$

Lemma B.2 (Nash refinement of graph functions). *Finitely many semialgebraic functions on a semialgebraic set admit a common finite Nash stratification on which each function is Nash. For algebraic presentations this refinement is effective.*

Proof. Take their combined graph. Semialgebraic Nash stratification, refined by the ranks of the coordinate projection and by a compatible stratification of its image, is finite; these are the usual graph and rank-stratification constructions in [7, 1]. On a graph stratum the projection is injective. If its constant rank were less than the dimension of that stratum, the constant-rank theorem would give a positive-dimensional local fibre, contradicting injectivity. The projection thus has full rank onto its image stratum. Its local inverse is analytic by the inverse function theorem and is semialgebraic because its graph is semialgebraic; hence it is Nash. Finitely refine images to separate overlapping strata and repeat on their lower-dimensional boundaries. This proves the claimed finite partition and gives the precise reason that graph stratification, rather than topological triviality alone, yields Nash functions. Effective semialgebraic decomposition and rank tests give the algebraic version of the construction. $\square$

The application to Theorem 4.3 is now direct. The polynomial matching formula (4.5), with two quantified inclusions and minimality over the distance, defines the Hausdorff error as a semialgebraic function of (θ, t) . Theorem 4.1 proves its pointwise convergence to zero. Proposition 4.2 gives a compact-uniform power estimate. Lemma B.2 applied to the diameter and the positive separation and radius functions gives the asserted Nash refinement. Hardt triviality is

used separately for the topological type of the leading fibres. The inequality $|\text{diam } A - \text{diam } B| \leq 2d_H(A, B)$ proves the diameter remainder.

B.1. An explicit terminating spectral algorithm. Here is a proof of Theorem 5.1 with its data representations and branch-selection step made explicit. The input is the polynomial formula of Definition 3.1; each algebraic constant has its integer polynomial and rational isolating interval. Arithmetic is in a primitive-element representation of their finite real algebraic field. All quantified formulas below are subjected to real quantifier elimination as in [1]. Finite root permutations are literal finite disjunctions, and complex coordinates are pairs of real coordinates.

First decide exact identification by the formula that all x, x' with $F(x) = F(x') = P$ have equal target coefficients. On this locus construct the cluster graph using its polynomial factorization equations and the specified isolating conditions. For each $H = |c|^2$, form the graph of

$$m_H(s) = \max\{H(x) : x \in \Gamma, G(x) \leq s\}$$

by the conjunction of the universal upper-bound formula and an existential attainment formula. Compactness gives the maximum; compactness and $\{G = 0\} \subset \{H = 0\}$ give $m_H(s) \rightarrow 0$. Apply quantifier elimination to this graph. Construct its finite support-pair list as in Lemma B.1, test eventual zero, and test (B.1) for each rational candidate. The unique true candidate is β_H . This is a finite algorithm for selecting the *actual maximum-function branch*. It needs no unspecified command to isolate an algebraic branch at the boundary. If all maxima are eventually zero the target is locally rigid; otherwise set $\alpha = \min_H \beta_H/k_H$ over the nonzero maxima.

For a family, keep θ free in exactly these tests. Their finitely many truth sets give the parameter pieces and include all zero specializations. Nash-refine them using Lemma B.2. The supports are finite before specialization, so this procedure still terminates without a bound on a real-valued exponent.

Next substitute $\alpha = p/q$ in the fixed-centre formula (4.2). Elimination computes a quantifier-free presentation of $\mathcal{J}_0$ and its projection $\mathcal{C}_0$. Adjoin the labelled roots by elementary symmetric identities to obtain $\mathcal{L}_0$. Use (4.5) to express that all pairs of elements have distance at most C and that some pair attains C . Compactness makes this a formula for one positive real number. Univariate real-root isolation applied to its eliminated formula selects that number over the input field. Taking a field norm gives a nonzero integer polynomial vanishing at C ; renewed root isolation and the same formula select a rational isolating interval for the intended root. The same graph formulas with free parameters compute the family of constants, and the preceding Nash and decay constructions compute a compatible refinement and an admissible rational remainder exponent. No complexity bound is claimed.

B.2. Effective geometric certificates without an implicit oracle. The geometric output can also be given a terminating, though generally inefficient, specification. Over the field just described, enumerate finite algebraic diagrams and finite semialgebraic base partitions in increasing finite encoding length, refining a fixed semialgebraic triviality partition of the enlarged graph with compact fibres. Properness on those base pieces follows from that trivialization and is preserved under further base restriction. A candidate consists of square-slack presentations, reduced closed chains with inverse-map identities, smooth sources with specified projective maps, finite centre ideals and adapted etale charts, integer monomial orders, inverse units and selected real branch formulas. Verify polynomial identities and centre blow-up charts by ideal arithmetic; verify relative smoothness, chart ranks and divisor intersections by their Jacobian criteria. Properness is certified by the specified projective construction and the square-slack argument, not by numerical bounds on open charts. Verify partition coverage, fibrewise real surjectivity, zero flags and accessibility by real quantifier elimination. These are finite formulas: for example coverage is $\forall \theta \forall x \in \Gamma_\theta \exists y$ in one of the finitely many source charts with the stated projection. Each positive

neighbourhood and unit bound can likewise be represented by a semialgebraic choice graph and its universal verification formula.

The existence proof in Appendix A, performed over finite algebraic and rational-function extensions of the input field, produces a certificate with such a finite encoding. An exhaustive search therefore eventually finds a verified certificate. Invalid candidates are rejected by the finite tests. This is an explicit input–output existence algorithm; it is not a claim that the accompanying regression script implements a practical resolution package. The direct maximum-function algorithm above computes the spectral outputs without this enumeration. Thus neither effectivity route depends on an unrepresented branch oracle, and neither is inferred from a finite list of example calculations.

APPENDIX C. EXACT OUTPUTS AND FINITE ARITHMETIC

Three different assertions should not be conflated. The existence of the relative real atlas is a geometric theorem. The first-order procedures produce spectral or entrance outputs from algebraic presentations. The rational calculations of specific Fisher programs are finite specializations, not implementations or proofs of the entire atlas.

For the direct procedures the input is a finite list of polynomials with rational or real algebraic coefficients, a Boolean formula of polynomial sign conditions and the quantifiers specifying the model and its maps. A real algebraic coefficient is represented by its integer polynomial and a rational isolating interval. Rational powers use a nonnegative auxiliary variable and its integer-power equation. Compact minimum formulas have the form

$$\exists x [\Phi(x) \wedge Q(x) = z] \wedge \forall y [\Phi(y) \Rightarrow Q(y) \geq z].$$

The sign or limit formulas retain the centre as a free fixed parameter. Elimination produces a finite output formula; at algebraic parameter values, root isolation gives the numerical algebraic outputs. In particular, the quadratic-probe graph and the entrance minimum graph fit this encoding without any projective blow-up representation.

The separate certificate search in Appendix B concerns geometric presentations. Its existence-by-enumeration role is supplementary: the direct outputs above do not wait for that search, and no run of the finite example regression is described as having constructed such a certificate. This separation leaves the geometric proofs and the direct algorithmic conclusions intact.

For the two cubic critical examples, use the coordinate orders

$$(a_0, b_0, X, c_0, w, k, u_f, u_h, v_f, v_h) \quad \text{and} \quad (a_0, b_0, X, c_0, d_0, Y, k, u_f, u_h, v_f, v_h),$$

respectively. At the endpoint, $w = -(d_1 + d_2) \geq 0$; for a first-order representative choose $d_1 = d_2 = -w/2$. The nonnegative coordinates are (a_0, X, w) and (a_0, X, Y, k) . Both displayed specializations have the empty cone support, so $w = 0$ at their endpoint optimum and the endpoint split is uniquely zero. Thus the representative choice introduces no second-order ambiguity into those exact examples.

Let L and D be respectively the seven free score columns and the cone columns, $W = \text{diag}(w_j/P_{je})$, and let B be the base probability velocity. All entries are rational at the stated examples. Set

$$G_L = L^\top W L, \quad b_f = G_L^{-1} L^\top W B, \quad \tilde{D} = D - L G_L^{-1} L^\top W D, \\ Q = \tilde{D}^\top W \tilde{D}, \quad g = \tilde{D}^\top W (B - L b_f), \quad b_* = (B - L b_f)^\top W (B - L b_f).$$

The exact checks give $\text{rank } L = 7$, $\text{rank } Q = 2$, $g < 0$ entrywise and nonnegative principal minors of Q . Every nonempty independent-support KKT test fails, whereas the empty support succeeds. Moreover $z^\top Q z - 2g^\top z > 0$ for every nonzero $z \geq 0$. This proves uniqueness of the cone optimum despite the singular projected Gram matrix. The free optimum is b_f and $\mu^2 = b_*/4$. Multiplication

and division of the exact probability jets then give μk by (14.13). For positive rational interval endpoints l, u , the tests for μ and k are respectively

$$l^2 < \mu^2 < u^2, \quad l^2 \mu^2 < (\mu k)^2 < u^2 \mu^2, \quad \mu k > 0.$$

After clearing positive denominators these are integer comparisons. The same exact arithmetic forms and inverts the two Fisher matrices in (8.5)–(8.6). The ancillary record gives the full rational Gram matrices, linear vectors, optimizers and interval tests; reproducibility of these finite operations is separate from the mathematical arguments above.

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PROJECTIVE POLYNOMIAL OBSERVATIONS: JOINT SPECTRAL INVARIANTS AND IDENTIFICATION WALLS

QIAN QI

ABSTRACT. We study unordered polynomial spectra observed through stochastic channels and an unknown scalar normalization. For compact semialgebraic models, a finite relative real monomial presentation determines a joint weighted initial coefficient fibre, with the datum held fixed. Its root-multiset image gives the exact leading bottleneck diameter. A finite Nash stratification in fixed-format families supports constant rational exponents, Nash leading diameters and compact-uniform power remainders; algebraic presentations admit finite exact algorithms. Two real polynomial experiments with identical individual coefficient envelopes have different leading spectral diameters, separating the joint invariant from marginal scalar data. We compute every multiplicity and boundary pattern in the fixed-degree, full-rank, coprime binary model and classify the whole exact fibre of a rank-deficient cubic family. For finite exact fibres with cone-coercive remote parametrizations, we prove an entrance law and a second-order formula, allowing nonunique first-order minimizers. Both the stochastic weight wall and the geometric endpoint wall satisfy these hypotheses. Their remote entrance scales are linear while the local Fisher constants stay finite; on the opposite side their exact spectral diameters open differently. Explicit rational examples decide the critical rays.

1. INTRODUCTION

An unordered spectrum is not recovered by bounding its polynomial coefficients independently. Near a multiple root, relations among the coefficients survive rescaling and constrain the limiting root multiset. In a normalized stochastic experiment the admissible relations also include inequalities, rank constraints and an unobserved scalar normalization. A second phenomenon is global: a component of the exact fibre can enter from a positive distance in coefficient space while the local inverse at the original component remains uniformly conditioned. This article separates these two problems and connects them by a real spectral atlas.

The joint invariant. Let X be a compact semialgebraic model, F its continuous semialgebraic probability map, and A_x a continuous monic target polynomial. At a datum P with a unique exact target, write $c_{r,a}$ for the coefficient of degree a in the monic factor belonging to a root r of multiplicity m_r . Set $G = h_w(F(x), P)^2$ and $H_{r,a} = |c_{r,a}|^2$ on the actual real observation graph. A real monomial presentation gives

$$G = u \prod_i |y_i|^{a_i}, \quad H_{r,a} = v_{r,a} \prod_i |y_i|^{b_{r,a,i}}, \quad \alpha = \min_{r,a,i} \frac{b_{r,a,i}}{a_i(m_r - a)}.$$

Only divisors with an admissible real transversal enter the minimum. The scalar divisor-ratio calculation is classical. The additional object is obtained by the simultaneous substitutions, for $\alpha = p/q$,

$$R = \varepsilon^q U, \quad c_{r,a} = \varepsilon^{p(m_r - a)} V_{r,a}, \quad \|U\| \leq 1,$$

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followed by real closure with P fixed. Its coefficient projection $\mathcal{C}_0$ retains all joint relations and inequality faces. Theorem 4.1 identifies the exact leading diameter as

$$C(P) = \max_{V, V' \in \mathcal{C}_0} \max_r d_\infty \left(Z \left(z^{m_r} + \sum_{a < m_r} V_{r,a} z^a \right), Z \left(z^{m_r} + \sum_{a < m_r} V'_{r,a} z^a \right) \right).$$

In particular, the construction is neither a product of marginal coefficient bounds nor a closure in which the centre is allowed to drift.

The distinction is quantitative. Theorem 6.1 gives two compact real-rooted quadratic experiments having identical observation functions, identical scalar order triples $(2, 2, 4)$ on each component, and identical exact individual coefficient ranges at every small radius. Nevertheless their spectral diameters are respectively

$$(1 + \sqrt{5})t \quad \text{and} \quad \frac{1 + \sqrt{17}}{2} t.$$

The difference is the pairing of the leading coefficients. Thus even sharp individual scalar envelopes do not determine the spectral constant. This assertion concerns marginal data; it does not identify the joint invariant with information irrecoverable from every possible scalar function of all the coefficients.

Relative real geometry and effective output. Theorem 3.4 constructs a finite fibrewise proper real monomial cover in fixed-format families. Its proof keeps four interfaces separate. A complete finite diagram, including inverse maps on the actual successive complements of a closed chain, is spread over an algebraic base. A single generic localization for each finite collection of charts makes the Rees algebras commute with every fibre base change. Square-slack lifts and selected real algebraic branches preserve the original graph, including strict exclusions. Finally, real accessibility is a finite first-order condition with an attaining transversal. Special algebraic fibres are reconstructed by dimension descent; changes between real chambers are handled by a separate finite semialgebraic partition. Appendix A proves these interfaces explicitly.

The joint initial family is defined by a formula in which the parameter is free but fixed inside the closure quantifiers. The semialgebraic error between its root image and the scaled positive-radius image tends to zero on each fibre. A finite support argument and positive continuous threshold selection then give, on every compact subset H of a finite Nash stratum,

$$\omega_\theta(t) = C(\theta)t^\alpha + O_H(t^{\alpha+\nu}), \quad \alpha, \nu > 0, \quad \alpha \in \mathbb{Q}, \quad C \text{ positive and Nash.}$$

Hardt triviality is used for topological types and proper restrictions, not for a metric rate. Theorem 5.1 gives finite algorithms for algebraic inputs. In particular, its order algorithm tests finitely many rational candidates by quantifier elimination; it does not invoke an unspecified procedure for choosing a Puiseux branch. The detailed representation and termination arguments are in Appendix B.

Explicit binary polynomial laws. The experiment in Section 2 has two monic degree- d component polynomials with roots in $[0, D]$, positive mixture weights, and binary column-stochastic channels. With coprime components, strictly positive invertible channels and at least $2d + 1$ exterior clocks, Theorem 7.2 gives the whole-model leading set for every multiplicity and endpoint/interior pattern. A repeated interior cluster has exponent $1/2$ and leading constant

$$C = \max_j 2\sqrt{(m_j - 1)/m_j} \kappa_j^{-1/4},$$

where each κ_j is an explicit projected Fisher-cone minimum. With no repeated interior cluster the exponent is one. Competitors range over the entire closed model, not just the root-pattern

stratum. These are fixed-degree, full-rank, coprime binary assertions; no classification of arbitrary multicomponent singularities is implicit.

Rank deficiency requires a separate inverse. For $f = z(z - r)^2$, $g = (z - s)^3$, $0 < s < r < D$, positive channels and invertible second channel, put

$$B = (3D - r)s - 2rD, \quad E = (1 - \alpha)(r + 3s)(2r - 3s)^2 - 4\pi r^3.$$

The entire exact fibre is classified in Theorem 8.1. The identifying locus is $\{\det U \neq 0\} \cup \{B < 0\} \cup \{E < 0\}$. A quantitative inverse through rank changes uses the second marginal and two robust pencil-isolation inequalities, never division by $\det U$, B or E . The resulting local Fisher law remains coercive through both identification walls. Global identification can fail there because the exclusion radius of a remote component collapses, rather than because the local constant diverges.

Remote entrances and two different walls. For a compact model with a finite exact fibre, Theorem 10.1 describes the entrance of each remote representative admitting an exact polyhedral-cone parametrization and a cone-coercive first observation derivative. Ambient score injectivity is not required. In the square-root probability embedding, write the remote map as $p_0 + Jz + Q(z, z) + o(\|z\|^2)$ and the centre path as $p_0 + \delta b + \delta^2 c + o(\delta^2)$. The entrance coefficient is $\mu = \min_{v \in K} \|Jv - b\|$. If $\mu > 0$, then

$$d_{\text{rem}}(\delta) = \mu\delta + k\delta^2 + o(\delta^2), \quad k = \frac{1}{\mu} \min_{v \in M} \langle Jv - b, Q(v, v) - c \rangle, \quad M = \arg \min_{v \in K} \|Jv - b\|.$$

The residual $Jv - b$ is common to all $v \in M$, although the minimizer itself need not be unique. Theorem 10.2 proves this formula and the strict regimes in the second-order critical window. Several remote representatives give a finite sequence of target-set entrances. Both signs of k occur in an elementary probability example.

At the cubic weight wall $E = 0$, $B > 0$, the remote pair is $(f, (z - a)(z - b)^2)$ with $s < a < r < b < D$. Theorem 9.4 gives its first entrance law; Corollary 10.3 gives the second-order coefficient. At the geometric wall $B = 0$, $E > 0$, the remote double root is instead at D . Ordered endpoint-root coordinates, not the interior variance cone, give its complete feasible parametrization. Theorem 11.2 proves the two-scale entrance for a transverse fixed-design path through this wall. In both cases the entrance is linear and the base local square-root law stays uniformly conditioned. On the opposite side the weight-wall pencil interval opens linearly and can produce a square-root increase in exact spectral diameter. The geometric-wall interval opens quadratically, yet its exact diameter remains $D - r$. These are two completed, distinct mechanisms; their intersection and arbitrary higher-rank singularities are not being identified with either local cone.

Classical inputs and scope of the contribution. Resolution and principalization, scalar divisor ratios, quantifier elimination, semialgebraic triviality and algebraic one-variable asymptotics are established inputs [2, 3, 4, 5, 1, 6, 9]. Algorithmic equiresolution in families is studied in [7]. The finite-diagram and real-cover arguments here specify how these inputs apply to the actual constrained graph. They are not claims to a new resolution algorithm. Hà's current version [8] already gives finite scalar testing in Theorem 5.8, a non-toric finite-candidate chamber principle in Theorem 9.9, and a toric family wall-chamber statement under the hypotheses of Theorem 12.3. A finite chamber vocabulary is therefore not the novelty claimed here. The joint real leading set, its separation from individual scalar envelopes, and the remote-entrance laws with explicit weight and endpoint realizations are the additional metric statements. No general leading set is asserted to be quadratic.

Sections 3–6 establish the general joint invariant, after Section 2 fixes the experiment. Sections 7–9 develop the explicit multiplicity, rank-deficient cubic and weight-wall laws. Sections 10 and

11 give the structural entrance theorem and the geometric wall. The proof appendices supply the complete relative and algorithmic interfaces. Finite exact diagnostics accompany the displayed algebraic examples; they are neither a substitute for these proofs nor a machine verification of their universal quantifiers.

2. THE EXPERIMENT AND THE SPECTRAL METRIC

Fix $D > 0$, a degree $d \geq 1$, and $0 < \pi < 1/2$. A closed binary model point consists of two monic real polynomials f_1, f_2 of degree d , all of whose roots lie in $[0, D]$, weights $\alpha_1 + \alpha_2 = 1$ with $\alpha_a \geq \pi$, and column-stochastic matrices $U = (u_1, u_2)$ and $V = (v_1, v_2)$ with nonnegative entries. Set

$$K(z) = \sum_{a=1}^2 \alpha_a f_a(z) u_a v_a^\top, \quad q(z) = \mathbf{1}^\top K(z) \mathbf{1} = \sum_{a=1}^2 \alpha_a f_a(z), \quad P_j = \frac{K(T_j)}{q(T_j)}. \quad (2.1)$$

The clocks satisfy $D < T_1 < \dots < T_\ell$, and $w_j > 0$, $\sum_j w_j = 1$. Consequently $q(T_j) > 0$, including at every closed-model competitor. The target polynomial is $A = f_1 f_2$. The model is compact in monic coefficients and stochastic entries. Component permutations are identified. The matrices, roots, clocks and design weights are allowed to vary when a family of centres is considered, but every competitor uses its centre's fixed design.

For probability arrays define

$$h_w(P, Q)^2 = \sum_j w_j \sum_e (\sqrt{P_{je}} - \sqrt{Q_{je}})^2.$$

For two multisets $Z = \{z_i\}_{i=1}^n$ and $Z' = \{z'_i\}_{i=1}^n$, put

$$d_\infty(Z, Z') = \min_{\sigma \in \mathfrak{S}_n} \max_i |z_i - z'_{\sigma(i)}|.$$

For real roots this is the maximum discrepancy of their increasing orderings. All diameters in this article are diameters of sets of multisets, not maximal distances from the centre. Write

$$\omega_\theta(t) = \text{diam}\{Z(A_\xi) : h_w(F_T(\xi), P_\theta) \leq t\}. \quad (2.2)$$

A local modulus $\omega_{\theta, \text{loc}}$ restricts the competitors to a specified coefficient neighbourhood of the centre's component pair. This restriction is never implicit.

At a strictly positive datum the Fisher inner product on zero-sum arrays is

$$\langle H, H' \rangle_{I, \theta} = \sum_{j, e} \frac{w_j}{P_{\theta, je}} H_{je} H'_{je}.$$

For component increments p_a of degree below d and stochastic increments $\eta = (\dot{\alpha}_1, \dot{u}_1, \dot{u}_2, \dot{v}_1, \dot{v}_2)$, where channel increments have entry sum zero, define

$$\begin{aligned} \dot{K} &= \sum_a \{ \alpha_a p_a u_a v_a^\top + f_a(\dot{\alpha}_a u_a v_a^\top + \alpha_a \dot{u}_a v_a^\top + \alpha_a u_a \dot{v}_a^\top) \}, \\ \dot{q} &= \sum_a (\alpha_a p_a + \dot{\alpha}_a f_a), \quad S_\theta(p_1, p_2, \eta)_j = \frac{\dot{K}(T_j) - P_{\theta, j} \dot{q}(T_j)}{q(T_j)}. \end{aligned} \quad (2.3)$$

Here $\dot{\alpha}_2 = -\dot{\alpha}_1$. These rational formulas define every Fisher program below in the original experiment coordinates.

Lemma 2.1 (Normalizer recovery). *Suppose f_1, f_2 are coprime, V is invertible, the weights are positive, and $\ell \geq 2d + 1$. The observed array determines q and K uniquely among all closed-model competitors. Their coefficients are locally Lipschitz functions of the observations, with uniform constants on compact sets of such centres. No rank assumption on U is required.*

Proof. Let e have degree below d , and let H be a matrix polynomial of degree at most d satisfying $H(T_j) = e(T_j)P_j$. Interpolation of the polynomial $qH - eK$, of degree at most $2d$, gives $qH = eK$. Its second marginal, multiplied by $\text{diag}(\alpha_1, \alpha_2)^{-1}V^{-1}$, gives $q \mid ef_1$ and $q \mid ef_2$. Coprimality implies $q \mid e$, hence $e = 0$ and $H = 0$.

More explicitly, let Π_T project onto the orthogonal complement of the degree- d Vandermonde column space. The map

$$N_\theta : e \mapsto \Pi_T(e(T_j)P_{\theta,j})_j, \quad \deg e < d,$$

is injective. For a competitor, subtraction of the normalization equations yields $\|N_\theta(q' - q)\| \leq C \max_j \|P'_j - P_j\|$; the competitor normalizer coefficients are bounded by compactness of the root interval. The smallest singular value of N_θ is positive. A bounded left inverse recovers $q' - q$, and Vandermonde interpolation recovers $K' - K$. The singular values and the coefficient bounds have positive uniform margins on compact sets of the stated centres. $\square$

Lemma 2.2 (Full-rank coefficient inverse). *Under Lemma 2.1, suppose also that U is invertible. Then every sufficiently close observation has, up to one component permutation, coefficient and stochastic parameter error bounded by a constant times its observation error. Every exact competitor has the same component polynomials and stochastic parameters. The constants are uniform on compact subsets of these centres.*

Proof. Choose two clocks such that $f_1(T_j)/f_1(T_i) \neq f_2(T_j)/f_2(T_i)$. Otherwise the nonzero polynomial $f_2(T_i)f_1 - f_1(T_i)f_2$, of degree at most d , would vanish at all $2d + 1$ clocks. The matrix $K(T_j)K(T_i)^{-1}$ has those two distinct eigenvalues and eigenvectors u_1, u_2 . Normalize the eigenvectors to have entry sum one. Simple-eigenvalue inversion is analytic in a neighbourhood of the centre. Applying the recovered U^{-1} to $K(z)$ and taking row sums recovers $\alpha_a f_a$; leading coefficients and normalized rows recover α_a, V . This construction applies to arbitrary closed-model competitors because their $K'(T_i)$ stays invertible. It also proves exact uniqueness. A finite neighbourhood cover gives compact-uniform constants. Equivalently, the derivative in monic coefficient and stochastic coordinates has a bounded left inverse. $\square$

3. REAL MONOMIAL DATA IN PARAMETER FAMILIES

We work first with a compact semialgebraic model X , a continuous semialgebraic probability map F , and a continuous semialgebraic monic target polynomial A_x of fixed degree n . Complex coefficients are represented by real and imaginary parts. Suppose the target is constant on $F^{-1}(P)$, with value $A_0(z) = \prod_r (z - r)^{m_r}$. Disjoint isolating discs and compactness give, on a sufficiently small closed observation neighbourhood, unique monic cluster factors

$$A_{r,x}(r + z) = z^{m_r} + \sum_{a < m_r} c_{r,a}(x)z^a, \quad k_{r,a} = m_r - a. \quad (3.1)$$

Multiplication of coprime monic factors has invertible differential. The cluster factors are consequently analytic functions of the total coefficients. Their graphs with the isolating conditions are semialgebraic.

Definition 3.1 (Input and families). An algebraic input is a finite first-order polynomial presentation of X, F, A, P , design weights, isolating data, denominator exclusions and branch choices. A real algebraic constant is encoded by an integer polynomial and a rational isolating interval; a complex constant is encoded by two real algebraic constants. A fixed-format family has a semialgebraic parameter set Θ , a semialgebraic $X \subset \Theta \times K$ with K compact and compact fibres, continuous semialgebraic maps F, A on X , and a semialgebraic choice $P_\theta \in F_\theta(X_\theta)$. Degrees and numbers of coordinates are fixed. Effectivity refers to algebraic presentations and numerical outputs to algebraic parameter values. No complexity bound is asserted.

Adjoin nonnegative square-root coordinates to the observation graph and write

$$R_{je} = \sqrt{w_j}(\sqrt{F(x)_{je}} - \sqrt{P_{je}}), \quad G = \sum_{j,e} R_{je}^2, \quad H_{r,a} = (\Re c_{r,a})^2 + (\Im c_{r,a})^2.$$

They are polynomial functions on the resulting compact real graph Γ . Zero probability cells are permitted. No passage to a Fisher quadratic form is used in this construction.

Definition 3.2 (Real monomial presentation). A real monomial presentation is a finite collection of smooth real algebraic sources and proper maps, collectively covering a neighbourhood of $G = 0$ in Γ , with finite algebraic etale coordinate presentations. On each source component either $G \equiv 0$ and all $H \equiv 0$, or

$$G = u \prod_i |y_i|^{a_i}, \quad H_{r,a} \equiv 0 \quad \text{or} \quad H_{r,a} = v_{r,a} \prod_i |y_i|^{b_{r,a,i}}. \quad (3.2)$$

The units have positive upper and lower bounds on smaller closed real coordinate boxes above $G = 0$. A contributing divisor has $a_i > 0$ and a real point away from the other divisors, with an admissible transversal on which $G > 0$. An identically zero H has order $+\infty$; a component $G \equiv 0$ is recorded separately and contributes no ratio. Properness belongs to the source maps, not to an open coordinate box. The finite encoding of an adapted coordinate presentation is specified in Appendix A.

Lemma 3.3 (Spreading a finite resolution certificate). *Let $S = \text{Spec } A$ be an integral affine variety over a characteristic-zero field, let $K = \text{Frac}(A)$, and let $Y \rightarrow S$ be of finite type. Suppose that over K there is a finite reduced closed chain*

$$Y_{0,K} = Y_{K,\text{red}} \supset Y_{1,K} \supset \cdots \supset Y_{q,K} = \emptyset,$$

projective maps $Z_{j,K} \rightarrow Y_{j,K}$ with smooth sources, specified inverse isomorphisms over $Y_{j,K} \setminus Y_{j+1,K}$, and finite smooth-centre principalizations of specified functions on those sources. There is a nonempty open S° on which the following hold on every fibre. The closed chain is flat with geometrically reduced fibres and ends in the empty set; the inverse maps remain inverse over its actual successive complements. Sources and centres are relatively smooth, principalization blow-ups commute with fibre formation, and the final sources have relative normal-crossing divisors with fixed integer monomial orders on a finite adapted etale atlas. Zero identities and inverse units persist. A nonzero monomial cannot become identically zero on a component of a retained fibre. For algebraic input, such finite data and an exceptional set are effectively constructible.

Proof. Lemma A.2 descends the complete diagram, including the actual open complements and both inverse identities, and imposes flatness and geometric reducedness. Lemma A.1 proves the needed base change for all powers of every centre ideal after one localization per finite collection of charts. The explicit bad-locus construction in Appendix A treats relative smoothness, normal-crossing intersections, chart coverage, units and zero flags separately, and proves that each discarded algebraic image has proper closure. The nonzero-specialization assertion follows from relative codimension one and the inverse units, not from generic nonvanishing alone. The constructive certificate specification in Appendix B proves the last assertion. $\square$

Theorem 3.4 (Relative real principalization). *On the exact-target identifying locus of a fixed-format family there is a finite semialgebraic stratification with fixed cluster multiplicities, a fixed finite collection of algebraic source chart presentations, fixed integer orders and fixed real-accessibility flags as in Definition 3.2. The source maps are fibrewise proper real covers. Charts, units and box sizes may vary with the parameter. On compact subsets of a stratum the cover is usable over a common sufficiently small observation neighbourhood. The construction is effective for the algebraic presentations of Definition 3.1.*

Proof. Lemma A.3 algebraizes the proper enlarged graph by square-slack lifts without discarding strict exclusions or real branches. Resolve the reduced generic fibre and its successive reduced singular loci, then principalize the nonzero functions on the smooth sources. Lemma 3.3 spreads this finite certificate. The explicit exceptional algebraic loci in Appendix A are reconstructed on reduced bases of strictly smaller dimension, giving finite termination. Lemma A.4 proves both inclusions in the exact real coverage assertion, including lower-dimensional real loci, and descends the selected label branches. Lemma A.5 supplies the real transversals and the finite real flag partition. That partition is not confused with the algebraic dimension-drop recursion. The final proof in Appendix A supplies the uniform neighbourhood and unit bounds by properness over compact parameter subsets. These arguments prove the asserted cover on the original graph, rather than on an algebraic relaxation of its inequalities. $\square$

Proposition 3.5 (Fixed-centre real cover). *Every graph Γ above has a real monomial presentation. Exact pieces and lower-dimensional real loci are included. Every recorded divisor has a real admissible attaining arc.*

Proof. Apply Theorem 3.4 to a one-point base and use Lemma A.5. $\square$

Corollary 3.6 (Intrinsic scalar orders). *For each nonrigid cluster coefficient put*

$$\beta_{r,a} = \min_{\text{recorded real divisors}} \frac{b_{r,a,i}}{a_i}, \quad \alpha = \min_{r,a} \frac{\beta_{r,a}}{k_{r,a}}. \quad (3.3)$$

Either the target is constant on an observation neighbourhood, or $\alpha > 0$ is rational and $\omega_P(t) \asymp t^\alpha$. Every finite coefficient minimum is attained by an admissible arc. The numbers are independent of the monomial presentation.

Proof. The complete real divisor list and the monomial inequalities in Lemma A.5 give $H \leq CG^\beta$, hence $|c_{r,a}| \leq Ch_w^\beta$. A minimizing transversal satisfies $G \asymp u^{a_i}$ and $H \asymp u^{b_i}$ and attains the exponent. The inclusion $\{G = 0\} \subseteq \{H = 0\}$ forces positive contributing orders. If all coefficients vanish on an observation neighbourhood, the target is locally rigid. Otherwise, for a monic polynomial of degree m , its greatest root modulus is comparable to $\max_{a < m} |c_a|^{1/(m-a)}$. Elementary symmetric identities give one bound; dominance of z^m outside a fixed multiple of this maximum gives the other. The centre lies in every ball, and the diameter is between its greatest centre displacement and twice that displacement. This proves the order of ω_P . The optimal coefficient inequality is intrinsic, so its attained exponent is independent of the presentation. The scalar divisor-ratio calculation is classical; compare [5]. $\square$

4. JOINT SPECIALIZATION AND UNIFORM SPECTRAL ASYMPTOTICS

On a nonrigid identified stratum write $\alpha = p/q$ in lowest terms. Let $\Psi(\theta, \varepsilon, x, U, V)$ denote graph membership and

$$R_{je}(\theta, x) = \varepsilon^q U_{je}, \quad c_{r,a}(\theta, x) = \varepsilon^{pk_{r,a}} V_{r,a}, \quad \sum_{j,e} U_{je}^2 \leq 1. \quad (4.1)$$

Complex equalities denote two real equalities. Define the joint family by the fixed-centre formula

$$(\theta, U, V) \in \mathcal{J} \iff \theta \in S \text{ and } \forall \rho > 0 \exists \varepsilon, x, U', V' : \quad (4.2)$$

$$0 < \varepsilon < \rho, \quad \Psi(\theta, \varepsilon, x, U', V'), \quad \|(U', V') - (U, V)\|^2 < \rho^2.$$

Put $\mathcal{J}_0(\theta) = \mathcal{J}_\theta$ and $\mathcal{C}_0(\theta) = \text{pr}_V \mathcal{J}_0(\theta)$. There is no quantified θ' in this formula. Closure with a drifting centre is a different operation, and taking independent closures in the coefficient coordinates is also a different operation.

Theorem 4.1 (The definable joint initial family). *The sets $\mathcal{I}_0(\theta)$ and $\mathcal{C}_0(\theta)$ form semialgebraic families of nonempty compact sets. The coefficient images of the radius- t observation ball, scaled by $c_{r,a}/t^{\alpha_{k_r,a}}$, converge in Hausdorff distance to $\mathcal{C}_0(\theta)$ for every fixed θ . If*

$$Z_r(V) = Z\left(z^{m_r} + \sum_{a < m_r} V_{r,a} z^a\right), \quad \mathcal{L}_0(\theta) = \{(Z_r(V))_r : V \in \mathcal{C}_0(\theta)\},$$

then the scaled cluster-root images converge to $\mathcal{L}_0(\theta)$, and

$$\omega_\theta(t) = C(\theta)t^\alpha + o(t^\alpha), \quad C(\theta) = \max_{V, V' \in \mathcal{C}_0(\theta)} \max_r d_\infty(Z_r(V), Z_r(V')) \in (0, \infty). \quad (4.3)$$

This is the diameter of a joint feasible image, not that of a Cartesian product of scalar envelopes.

Proof. Corollary 3.6 bounds every scaled coefficient for small t ; U is already bounded. The fixed-centre closure is nonempty and compact. Quantifier elimination proves semialgebraicity with θ as a free parameter. Compactness of the graph fibre lets one pass to convergent subsequences of the unscaled graph variables when projecting an initial point, so projection loses no limiting coefficient point.

Fix θ and a point of the outer coefficient limit. Its distance to the positive-radius coefficient section is a bounded nonnegative semialgebraic function of t , with a subsequence tending to zero. The one-variable monotonicity theorem makes its limit zero. A finite net of the compact outer limit gives the inner Hausdorff inclusion. Failure of the outer inclusion would give a bounded convergent subsequence of scaled coefficients outside the defining limit, a contradiction. This proves convergence of the entire ball. Continuity of monic root multisets on bounded coefficient sets gives the corresponding root-set convergence.

Let Δ be the least separation of distinct base roots. For small t , each cluster lies in its $\Delta/4$ disc. A matching within clusters has error tending to zero; a cross-cluster matching has an edge of length at least $\Delta/2$. An optimal matching therefore stays within clusters. Translation and scaling there are exact. Taking the diameter gives (4.3). An attaining coefficient transversal prevents the limiting diameter from being zero. This also explains why all inequality faces and joint relations in the closure remain relevant. $\square$

Proposition 4.2 (Semialgebraic decay on strata). *Let $e(\theta, t) \geq 0$ be semialgebraic for $0 < t < \tau(\theta)$, with $\tau > 0$ semialgebraic, and suppose $e(\theta, t) \rightarrow 0$ for each fixed θ . After a finite Nash stratification, on every stratum a rational $\nu > 0$ and a positive continuous semialgebraic $\tau_0(\theta)$ satisfy*

$$0 < t < \tau_0(\theta) \implies e(\theta, t) \leq t^\nu.$$

The estimate is uniform on compact subsets; ν need not be sharp.

Proof. Lemma B.1 gives a finite positive rational candidate list from the polynomial supports of a quantifier-free graph formula, including specialized zero coefficients and the eventual-zero case. Choose ν below its positive candidates. The complete argument in Appendix B selects a genuinely positive semialgebraic threshold, makes it continuous, and takes its minimum only on a compact subset of a stratum. No uniform rate is inferred from pointwise convergence alone. $\square$

Theorem 4.3 (Uniform spectral power law). *The identified nonrigid locus has a finite Nash stratification on which α is fixed, $\mathcal{L}_0$ has constant semialgebraic topological type, and $C(\theta)$ is positive and Nash. For every compact subset H of a stratum, for some $\nu > 0$,*

$$d_H(\mathcal{L}_t(\theta), \mathcal{L}_0(\theta)) = O_H(t^\nu), \quad \omega_\theta(t) = C(\theta)t^\alpha + O_H(t^{\alpha+\nu}). \quad (4.4)$$

Here $\mathcal{L}_t$ is the exact clustered root image scaled by $t^{-\alpha}$ and d_H uses the maximum cluster bottleneck metric.

Proof. Represent each root multiset by all its labelled arrays through elementary symmetric identities. The formula

$$M_d(z, z') = \bigvee_{\sigma \in \mathfrak{S}_m} \bigwedge_{i=1}^m |z_i - z'_{\sigma(i)}|^2 \leq d^2 \quad (4.5)$$

expresses bottleneck distance at most d . Use the conjunction over clusters, and real and imaginary parts for complex roots. The two inclusions

$$(\forall z \in \mathcal{L}_t)(\exists z' \in \mathcal{L}_0)M_d(z, z'), \quad (\forall z' \in \mathcal{L}_0)(\exists z \in \mathcal{L}_t)M_d(z, z')$$

and minimality over $d \geq 0$ define the Hausdorff error. Compactness gives attainment. Rational powers of t are encoded by positive algebraic roots, so this error is semialgebraic. Theorem 4.1 proves its pointwise limit zero; Proposition 4.2 gives the compact-uniform power bound.

The diameter has a first-order formula saying that all pairs have distance at most C and one pair attains it. It is therefore semialgebraic and positive. Lemma B.2 gives its Nash refinement. Refine also by positive cluster separation and observation-radius choices, so that compact minima exclude cross-cluster matching uniformly. Hardt triviality supplies the topological types separately. Finally diameters differ by at most twice Hausdorff distance, which gives the second estimate. Topological triviality has supplied neither the error rate nor the Nash dependence. $\square$

Theorem 4.4 (A finite real spectral atlas). *A fixed-format family admits a finite semialgebraic stratification retaining the following data: the semialgebraic type and dimension of the exact target coefficient fibre and its root image; on the identified locus, rigidity or a positive rational exponent, cluster multiplicities, and a fixed real monomial order list with accessibility and minimizing indices; on its nonrigid part, the joint initial coefficient and root-set types, a Nash leading diameter and the compact-uniform power remainder; and finite algebraic descriptions of these objects. On a subfamily with a proved Fisher-cone representation, the active supports and choices of leading constants can be retained in the same refinement. On a nonidentified stratum the modulus tends to the positive diameter of the exact target root image. No general leading set is asserted to be quadratic, and no local leading constant excludes a remote exact component.*

Proof. The exact coefficient fibre is $\{A_x : F_\theta(x) = P_\theta\}$, not the parameter fibre itself. Apply Hardt triviality to its compact image and root graph. Singleton images are selected by a first-order formula. At fixed θ , compactness shows that decreasing observation balls converge in target image to the exact image, so their diameters converge to its diameter. On the singleton locus apply Theorem 3.4, Corollary 3.6, and Theorems 4.1 and 4.3. These are separate constructions entering one finite refinement. A proved Fisher subfamily contributes only finitely many additional support and comparison conditions. Effective descriptions are provided by the following theorem and Appendix B. $\square$

5. EFFECTIVE OUTPUT AND THE ROLE OF COMPUTATION

Theorem 5.1 (Effective algebraic specialization). *For an algebraic input on the identified locus, finite algorithms compute local rigidity, the nonrigid exponent α , quantifier-free presentations of $\mathcal{S}_0, \mathcal{C}_0, \mathcal{L}_0$, and an integer polynomial with a rational isolating interval for C . Exact identification itself is decidable. In algebraically presented parameter families the corresponding strata and formulas are effectively describable. These assertions are independent of any finite example regression.*

Proof. Appendix B gives the complete input–output procedure. It constructs each maximum $m_H(s) = \max\{H : G \leq s\}$ by elimination, extracts a finite support-pair list, and tests the comparability formula (B.1) for every candidate. The unique true candidate selects the actual positive maximum-function branch, while a separate formula detects eventual zero. No unspecified

branch-isolation oracle is used. Substitute the resulting p/q in (4.2), eliminate, adjoin roots and use (4.5) to compute the leading diameter. Real-root isolation and field-norm conversion provide its integer polynomial and isolating interval. With free parameters the same finite tests compute strata, followed by Nash refinement. A second, explicit finite-certificate search for the geometric presentations is given in that appendix; spectral outputs do not depend on executing it. $\square$

The symbolic calculations accompanying the explicit polynomial examples test identities and specified finite optimization problems. They do not implement the universal algorithms above and are not proof certificates for relative principalization, uniform asymptotics, or arbitrary degree.

6. SCALAR ENVELOPES DO NOT DETERMINE THE SPECTRAL INVARIANT

The distinction between the scalar orders and the joint initial fibre can be made exact. The following examples have not only the same optimal orders but the same individual coefficient envelopes at every sufficiently small radius. Their leading spectral diameters are nevertheless different. The assertion concerns the scalar data of Corollary 3.6 and separate coefficient envelopes; it does not concern the collection of value functions of all possible combinations of the coefficients.

Theorem 6.1 (Separation from marginal scalar data). *There are two compact semialgebraic probability experiments with a unique exact target, real quadratic targets with roots in $[0, 1]$, and identical coefficient orders and individual coefficient envelopes, for which*

$$\omega_A(t) = (1 + \sqrt{5})t, \quad \omega_B(t) = \frac{1 + \sqrt{17}}{2}t \quad (0 \leq t \leq 1/20).$$

In both experiments the coefficient orders are $(\beta_1, \beta_2) = (1, 2)$ and the spectral exponent is one. On each source arc the orders of (G, H_1, H_2) are $(2, 2, 4)$. Thus the order lists and the separate optimal coefficient envelopes do not determine the joint initial root set or its diameter.

Proof. Put $r = 1/2$ and $e = 1/20$. In each experiment the source is the disjoint union of two closed intervals $0 \leq \rho \leq e$. Its probability map and datum are

$$F(\rho) = ((1 - \rho^2/2)^2, \rho^2(1 - \rho^2/4)), \quad P_0 = (1, 0).$$

The entries are nonnegative and sum to one. Since $\sqrt{F_1} = 1 - \rho^2/2$ on this interval,

$$h(F(\rho), P_0)^2 = 2 - 2\sqrt{F_1} = \rho^2.$$

Associate with an interval labelled (u, v) the target

$$A_{u,v,\rho}(z) = (z - r)^2 + u\rho(z - r) + v\rho^2.$$

Use the label sets

$$\mathcal{E}_A = \{(1, -1), (2, -4)\}, \quad \mathcal{E}_B = \{(1, -4), (2, -1)\}.$$

These are compact semialgebraic experiments in the sense of Definition 3.1; the discrete label is simply an additional source coordinate. All exact targets are $(z - r)^2$. The two root increments on an interval are

$$\rho \ell(u, v), \quad \rho q(u, v), \quad \ell(u, v) = \frac{-u - \sqrt{u^2 - 4v}}{2} < 0 < q(u, v) = \frac{-u + \sqrt{u^2 - 4v}}{2}. \quad (6.1)$$

They lie in $[0, 1]$ after translation by r , since the greatest absolute increment is at most $(1 + \sqrt{5})/20 < 1/2$.

Write $c_1 = u\rho$ and $c_0 = v\rho^2$. In either experiment the exact marginal images over the radius- t ball are

$$\{c_1\} = [0, 2t], \quad \{c_0\} = [-4t^2, 0].$$

In particular $\max |c_1| = 2t$ and $\max |c_0| = 4t^2$, not just to leading order. Moreover $G = \rho^2$, $H_1 = u^2 \rho^2$, and $H_2 = v^2 \rho^4$. The order assertions follow. Even the unordered lists of individual squared units agree: they are $\{1, 4\}$ and $\{1, 16\}$. Their association across the two coordinates is different.

For $t > 0$, put $\lambda = \rho/t$. The scaled coefficient image is already independent of t :

$$\mathcal{C}_{0,E} = \bigcup_{(u,v) \in \mathcal{E}_E} \{(u\lambda, v\lambda^2) : 0 \leq \lambda \leq 1\}, \quad E \in \{A, B\}. \quad (6.2)$$

This is its joint initial coefficient set. The ordered root image consists of $(\lambda \ell(u, v), \lambda q(u, v))$. Its first coordinate is nonpositive, its second nonnegative, and the zero vector belongs to it. The diameter in the maximum metric is the larger of its two coordinate widths. Since $u > 0$, $-\ell(u, v) > q(u, v)$ on each interval. Therefore

$$C_E = \max_{(u,v) \in \mathcal{E}_E} \frac{u + \sqrt{u^2 - 4v}}{2}.$$

For A the maximum is $1 + \sqrt{5}$. For B it is $(1 + \sqrt{17})/2$, which exceeds $1 + \sqrt{5}$. The inequalities are strict and these two constants differ. Translation and multiplication by t in (6.1) prove the asserted exact diameter identities. No product approximation of (6.2) has been used. $\square$

The example isolates the information carried by the joint specialization: relations between weighted coefficients survive both real closure and the root map. A finite list of scalar exponents cannot recover these relations, even when augmented by the separate sharp envelopes. Conversely the full joint set determines the diameter by Theorem 4.1; no new scalar resolution theorem is required for this distinction.

7. AN EXPLICIT MULTIPLICITY AND BOUNDARY ATLAS

Fix d, D, π and $\ell \geq 2d + 1$. Let $\mathfrak{M}_d$ be the centres of (2.1) with strict weights, strictly positive invertible U, V , and coprime component polynomials. No restriction is imposed on root multiplicities or on contact with $0, D$. Competitors still range over the entire closed model. Partition $\mathfrak{M}_d$ by the component assignment, multiplicity and endpoint/interior status of each distinct root. At fixed d this is a finite semialgebraic partition. On a pattern stratum write r_j, m_j for its distinct roots and multiplicities, and $a(j) \in \{1, 2\}$ for their component assignments.

For every cluster let s_j denote its first-order sum displacement. It is free at an interior root, nonnegative at 0 , and nonpositive at D . For every repeated interior cluster let $X_j \geq 0$. Define

$$p_a(z) = - \sum_{j:a(j)=a} s_j \frac{f_a(z)}{z - r_j} - \sum_{\substack{j:a(j)=a \\ r_j \in (0,D), m_j \geq 2}} X_j \frac{f_a(z)}{(z - r_j)^2}. \quad (7.1)$$

Insert these polynomials into (2.3). Let L contain the free interior sum and stochastic columns. Let D_0 contain the nonnegative endpoint sum columns (reverse the sign at D) and all X_j columns. With $W = \text{diag}(w_j/P_{je})$, define the projected Gram matrix

$$Q = D_0^\top W D_0 - D_0^\top W L (L^\top W L)^{-1} L^\top W D_0. \quad (7.2)$$

For a repeated interior cluster j , let

$$\kappa_j = \min\{v^\top Q v : v \geq 0, v_{X_j} = 1\}. \quad (7.3)$$

All matrix entries are rational in the root, channel, weight and design coordinates. Empty collections of free or cone columns have their usual zero-dimensional meanings.

Lemma 7.1 (Ordered cluster diameter). *For $m \geq 2$, the set*

$$\{y_1 \leq \dots \leq y_m : \sum_i y_i = 0, \frac{1}{2} \sum_i y_i^2 \leq X\}$$

has diameter $\sqrt{2X(m-1)/m}$ in the maximum coordinate metric.

Proof. Fix a coordinate j . Minimizing the squared norm under its value, ordering and zero sum gives

$$\max y_j = \sqrt{\frac{2X(j-1)}{m(m-j+1)}}, \quad \min y_j = -\sqrt{\frac{2X(m-j)}{mj}}.$$

For the maximum, the last $m-j+1$ coordinates are equal and the first $j-1$ are equal; the minimum is analogous. This also covers the zero endpoint extrema when $j=1$ or $j=m$. Put $a=j-1, b=m-j$. For $a, b \geq 1$, the assertion that the resulting coordinate width is at most $\sqrt{2X(m-1)/m}$ is equivalent to

$$ab(a+b+2)^2 \geq 4(a+1)(b+1).$$

Writing $S = a+b \geq 2$ and $ab \geq 1$, its difference is $ab(S^2 + 4S) - 4S - 4 \geq S^2 - 4 \geq 0$. For $a=0$ or $b=0$ equality holds directly. An extreme vector with one coordinate of absolute value $\sqrt{2X(m-1)/m}$ and the other $m-1$ coordinates equal attains the endpoint width against the zero vector. The diameter of a subset in the maximum metric is its largest coordinate width. This proves the formula, including possible simultaneous constraints on other clusters when it is used below. $\square$

Theorem 7.2 (Multiplicity and boundary laws). *Every centre in $\mathfrak{M}_d$ is exactly identifying. On each root pattern stratum the following whole-model assertions hold uniformly on compact subsets H .*

(a) *If there is a repeated interior cluster, the exponent is $1/2$. The complete leading root set consists of zero clusters at the simple and boundary roots and ordered vectors $y_j \in \mathbb{R}^{m_j}$ at the repeated interior roots satisfying*

$$\sum_i y_{j,i} = 0, \quad X_j = \frac{1}{2} \sum_i y_{j,i}^2, \quad \min_{\text{endpoint sums} \geq 0} v^\top Q v \leq 4. \quad (7.4)$$

The X_j in v are the displayed variances. After grouping and scaling roots by $\sqrt{t}$, the observation ball is within $O_H(\sqrt{t})$ of this set in Hausdorff distance. In particular

$$\omega_\theta(t) = C(\theta)\sqrt{t} + O_H(t), \quad C(\theta) = \max_j 2\sqrt{\frac{m_j-1}{m_j}} \kappa_j^{-1/4}. \quad (7.5)$$

Every leading point, including diameter-extremizing pairs, is realized by stochastic alternatives.

(b) *If no interior root is repeated, the exponent is 1. For each simple interior cluster take a free real root increment, and for each boundary cluster take an ordered vector of increments with all entries nonnegative at 0 or nonpositive at D . Let s_j be their sums and use (7.1) without X terms. The compact set of these root increments with*

$$\min_\eta \|S_\theta(p_1, p_2, \eta)\|_{I, \theta}^2 \leq 4 \quad (7.6)$$

is the complete leading set $\mathcal{L}_{\text{lin}, \theta}$. Its diameter $C_{\text{lin}}(\theta)$ is positive, and

$$\omega_\theta(t) = C_{\text{lin}}(\theta)t + O_H(t^2).$$

The scaled-root Hausdorff error is $O_H(t)$.

(c) The optimal coefficient orders of a cluster of multiplicity m , with coefficient weight k , are

$$\beta_k = \begin{cases} 1, & r \in (0, D), \quad k = 1, \\ k/2, & r \in (0, D), \quad 2 \leq k \leq m, \\ k, & r \in \{0, D\}, \quad 1 \leq k \leq m. \end{cases} \quad (7.7)$$

The real order list, boundary and multiplicity pattern, and Fisher active supports admit a common finite semialgebraic refinement. On its square-root strata C^4 is a selected rational function of the root and experiment coordinates; the positive constants on the linear strata are algebraic semialgebraic functions. Both are Nash after refinement. Exact algebraic evaluation uses finite programs.

Proof. Lemma 2.2 gives exact identification and a uniform inverse in coefficients and stochastic parameters. The partial fractions $1/(z - r_j)$ and $1/(z - r_j)^2$ occurring in (7.1) are linearly independent at distinct roots, with their indicated pole orders. The coefficient derivative, and hence its score, is injective on these variables and the free stochastic coordinates. Thus L has full column rank and Q is positive definite whenever it is nonempty. The constrained minima exist, are positive, and have uniformly bounded minimizers on H .

We first obtain coordinates for *all* nearby competitors. Coefficient recovery and analytic coprime cluster factorization give an $O_H(t)$ error in each cluster coefficient and in the stochastic parameters. For an interior cluster, write its exact roots as

$$r_j + ts_j/m_j + \sqrt{t}y_{j,i}, \quad \sum_i y_{j,i} = 0.$$

The first coefficient bounds s_j , and the second bounds $\sum_i y_{j,i}^2$, since for centred real roots their second elementary symmetric function equals $-\frac{1}{2} \sum_i y_{j,i}^2$. At an endpoint all root displacements have the same sign. Their sum is $O_H(t)$, so each is $O_H(t)$. These observations prove the upper bounds in (7.7). For the lower bounds, translate an interior cluster by a nonzero order- t amount for $k = 1$; for $k \geq 2$ split it by $\sqrt{t}$ times a zero-sum real vector with nonzero k th elementary symmetric function. The vector with $m - 1$ equal nonzero coordinates and the last their negative sum has this property for every $k \geq 2$. Its first nonzero coefficient is of order t and the coefficient inverse makes the observation distance comparable to t . At an endpoint displace all roots by the same allowed order- t amount. This proves all lower orders, including the one-sided boundary case. In root-arc parameter ε , the interior split has $\text{ord } G = 4$, $\text{ord } H_k = 2k$, whereas translation or an endpoint arc has orders $2, 2k$. These are admissible witnesses for the real order data, and (7.7) rules out any smaller candidate.

For part (a), expansion of the exact root coordinates gives precisely (7.1) at order t . Terms involving three centred root displacements are $O_H(t^{3/2})$; all higher ones and the bounded mean and endpoint terms satisfy the same bound. Consequently

$$F_T(\xi) = P_\theta + tS_\theta + O_H(t^{3/2}), \quad h_w(F_T(\xi), P_\theta) = \frac{t}{2} \|S_\theta\|_{I,\theta} + O_H(t^{3/2}). \quad (7.8)$$

This follows by Taylor expansion of the square-root probability vector and the reverse triangle inequality; it does not require a nonzero score. Membership in the ball implies $\|S_\theta\| \leq 2 + M\sqrt{t}$. Multiply all sum, variance and stochastic first-order variables by $\rho = 2/(2 + M\sqrt{t})$ and each centred root vector by $\sqrt{\rho}$. This gives a point of (7.4) within $O_H(\sqrt{t})$ of the scaled competitor. Endpoint and simple clusters contribute only $O_H(\sqrt{t})$ in these scaled coordinates.

Conversely take a point of (7.4) and bounded minimizing endpoint and free variables. Use the displayed interior root arcs; at an endpoint realize its sum by equal one-sided displacements, and use the prescribed order- t stochastic increments. Scale first-order variables by $\rho = 1 - M'\sqrt{t}$ and centred roots by $\sqrt{\rho}$. Strict weight and channel margins and cluster separation on H make

these genuine closed-model alternatives. For a sufficiently large M' , (7.8) puts them inside the observation ball. This is the reverse Hausdorff inclusion, with the same uniform error.

Homogeneity gives $X_{j,\max} = 2/\sqrt{\kappa_j}$. The marginal projection of the leading set at cluster j includes every ordered zero-sum vector with squared norm at most $2X_{j,\max}$: scale a minimizing feasible vector for $X_j = 1$ and choose the desired shape in that cluster. Lemma 7.1 therefore gives its diameter $2\sqrt{(m_j - 1)/m_j \kappa_j^{-1/4}}$. The diameter in the maximum cluster metric is the largest marginal diameter; no independence of the clusters is used. Cluster separation excludes cross-cluster optimal matchings. Diameters differ by at most twice Hausdorff distance, proving (7.5).

For part (b), every root displacement is $O_H(t)$ by the same sum argument at the endpoints and the simple-root inverse in the interior. Divide the exact displacements by t . They belong to the stated polyhedral sign and ordering cone. Taylor expansion now has remainder $O_H(t^2)$. The first-order score controls their sums and all stochastic variables; on the one-sided endpoint cones the sums control the individual displacements. Hence the root set in (7.6) is compact. The radial argument with $\rho = 1 - Mt$ proves both Hausdorff inclusions with error $O_H(t)$. An allowed nonzero displacement gives positive diameter. This proves (b).

For (c), enumerate the subsets of the cone coordinates other than a fixed X_j . On a support A , put $v_A = -Q_{AA}^{-1}Q_{Aj}$ and require $v_A \geq 0$ and $Q_{bj} + Q_{bA}v_A \geq 0$ off the support. Its value is $Q_{jj} - Q_{jA}Q_{AA}^{-1}Q_{Aj}$. Convexity proves necessity and sufficiency. All tests and selected values are rational, and

$$C^4 = \max_j \frac{16(m_j - 1)^2}{m_j^2 \kappa_j}.$$

There are finitely many patterns and supports. For the linear law, (7.6) is an explicit quadratic inequality on a polyhedral root cone after projecting the free stochastic variables. Its coordinate maxima and minima, and hence its diameter, have finite real algebraic descriptions by (4.5) or by face optimization. Quantifier elimination gives the algebraic value at algebraic input. Stratify the resulting semialgebraic functions and refine with the order lists of Theorem 4.4. Positivity and compact-uniform coercivity ensure continuity within each fixed pattern; Nash refinement gives the final assertion. No continuity of C is asserted across a change of scaling exponent. $\square$

Example 7.3 (A multiplicity crossover). Let $x \in [0, b]$, $b < 1/2$, take $F(x) = (1/2 + x, 1/2 - x)$, and target $A_x(z) = (z - r)^2 - x$. At a fixed centre $a \in (0, b)$, differentiation of the square-root probability vector gives

$$h(F(a + u), F(a)) = \frac{|u|}{\sqrt{1 - 4a^2}} + O_a(u^2).$$

The admissible x interval therefore has endpoints $a \pm t\sqrt{1 - 4a^2} + O_a(t^2)$, and its spectral diameter is

$$\omega_a(t) = \frac{\sqrt{1 - 4a^2}}{\sqrt{a}} t + O_a(t^2).$$

At $a = 0$ it instead has leading law $\sqrt{t}$. More precisely, uniformly for λ in compact subsets of $[0, \infty)$,

$$\frac{\omega_{\lambda t}(t)}{\sqrt{t}} \rightarrow \sqrt{\lambda + 1} - \sqrt{\max(0, \lambda - 1)}. \quad (7.9)$$

Indeed the feasible interval endpoints are $t \max(0, \lambda - 1) + O(t^2)$ and $t(\lambda + 1) + O(t^2)$, with truncation at zero, and $x \mapsto Z(A_x)$ has ordered roots $r \pm \sqrt{x}$. This example belongs to the general real-graph class of Section 3; it is not a claim about an unobserved restriction of the closed binary

model. Here the linear constant really diverges at a multiplicity wall. Section 9 gives a different mechanism.

8. A RANK-DEFICIENT CUBIC STRATUM

Let $\mathfrak{F}$ denote the degree-three centres with

$$f = z(z - r)^2, \quad g = (z - s)^3, \quad 0 < s < r < D,$$

strict weights $\pi < \alpha < 1 - \pi$, strictly positive U, V , $\det V \neq 0$, and at least seven clocks. The first channel may have either rank. Put $\gamma = 1 - \alpha$ and

$$A = (r + 3s)(2r - 3s)^2, \quad B = (3D - r)s - 2rD, \quad E = \gamma A - 4\pi r^3. \quad (8.1)$$

Here A is a scalar, not the target polynomial.

Theorem 8.1 (The entire cubic fibre). *The exact spectral fibre is a singleton precisely on*

$$\mathfrak{I} = \{\det U \neq 0\} \cup \{B < 0\} \cup \{E < 0\}. \quad (8.2)$$

On its complement, besides (f, g) all component-polynomial pairs are, up to permutation,

$$(f, h_c), \quad h_c = (1 - c)f + cg, \quad c_* \leq c \leq \min(c_D, \gamma/\pi), \quad (8.3)$$

where

$$c_* = \frac{4r^3}{A}, \quad b = \frac{rs}{3s - 2r}, \quad R(z) = \frac{f(z)}{g(z)}, \quad c_D = \frac{R(D)}{R(D) - 1}.$$

The interval is nonempty even at equality in either constraint, and every one of its pairs has a positive stochastic realization. Conditional on the component pair, that realization is unique.

Proof. Lemma 2.1 fixes q, K for every competitor. If U is invertible, Lemma 2.2 applies; in particular $\det K / \det K_3 = fg$ gives spectral identification.

If $\det U = 0$, stochasticity gives $u_1 = u_2 = u$. The second marginal $V \operatorname{diag}(\alpha, \gamma)(f, g)^\top$ has coefficient rank two. Every competitor therefore has invertible second channel and its two monic polynomials lie in the affine pencil h_c . Expansion gives

$$\operatorname{disc}(h_c) = c(c - 1)^2(r - s)^3(4r^3 - 3cs^2), \quad h_c(0) = -cs^3, \quad (8.4)$$

and

$$\frac{R'(z)}{R(z)} = \frac{(2r - 3s)z + rs}{z(z - r)(z - s)}, \quad A = 4r^3 - 27s^2(r - s) < 4r^3.$$

Nonnegative roots exclude $c < 0$, and the discriminant excludes $0 < c < 1$. For $c > 1$ there is a root in (s, r) and no root in $[0, s]$. The other two admissible roots must be above r , solving $R(z) = c/(c - 1) > 1$. They exist only for $s > 2r/3$. In that case R increases to its maximum at b and then decreases to one. The double-root value is $c_* > 1$; the larger root increases from b to D as c increases from c_* to c_D . Thus the whole real-rooted pencil in $[0, D]$ is $\{0, 1\}$, with $[c_*, c_D]$ added exactly when $b \leq D$.

For a competing pair with pencil coordinates c_1, c_2 , the weights obey $\alpha'_1 c_1 + \alpha'_2 c_2 = \gamma \in (0, 1)$. One coordinate must be zero and the other is one or a value $c \geq c_*$. The latter weight is γ/c , so the floor imposes $c \leq \gamma/\pi$; the other weight is automatically at least $\alpha > \pi$. Conversely put

$$\beta_f = \alpha v_1 + \gamma(1 - 1/c)v_2, \quad \beta_h = \gamma v_2/c.$$

Both vectors are positive and $\beta_f f + \beta_h h_c = \alpha v_1 f + \gamma v_2 g$. Their entry sums are the stated weights; normalizing them and keeping first channel (u, u) gives the realization. Coefficient functionals dual to the independent polynomials recover the two rank-one coefficient matrices uniquely, hence their weights and normalized marginals. Finally $b \leq D$ is equivalent to $B \geq 0$ and, on this set, $c_* \leq \gamma/\pi$ is equivalent to $E \geq 0$. $\square$

8.1. Quantitative inversion across the rank locus. All coefficient norms below are Euclidean, including the leading coefficient when a monic polynomial is represented as a vector. Constants are specified from a fixed compact $K_0 \subset \mathfrak{F}$. Norm comparisons between these finite-dimensional spaces are included in these constants. Put

$$\delta = \max_j \|P'_j - P_{\theta,j}\|_F.$$

Use coefficient order $(1, z, z^2, z^3)$. The two components form $F_{\text{coef}} \in \mathbb{R}^{2 \times 4}$, with one polynomial per row, and the second-marginal coefficient matrix is $\mathbf{B}_\theta = V \text{diag}(\alpha, \gamma) F_{\text{coef}} \in \mathbb{R}^{2 \times 4}$. Its two rows are the scalar coordinate polynomials of that marginal. Here σ_2 is its smaller singular value and matrix norms are Euclidean operator norms (or the corresponding fixed norm comparisons). Set

$$\beta_0 = \min_{K_0} \sigma_2(\mathbf{B}_\theta) > 0, \quad d_{fg} = \min_{K_0} \|g - f\|_{\text{coef}} > 0.$$

Normalizer recovery supplies C_B with $\|\mathbf{B}' - \mathbf{B}_\theta\| \leq C_B \delta$. Let K_d bound $\|\text{diag}(\alpha') F'_{\text{coef}}\|$ on all closed-model competitors for the designs in K_0 , and set

$$W_0 = \frac{2K_d}{\pi\beta_0}, \quad C_f = \max_{K_0} \|f\|_{\text{coef}}, \quad K_{\text{aff}} = W_0 C_B (1 + C_f). \quad (8.5)$$

Proposition 8.2 (From the second marginal to the pencil). *If $\delta \leq \beta_0/(2C_B)$, every closed competitor has invertible V' and its component polynomials satisfy*

$$f'_b = h_{c_b} + e_b, \quad \deg e_b < 3, \quad \|e_b\|_{\text{coef}} \leq K_{\text{aff}} \delta. \quad (8.6)$$

Moreover $\|\text{diag}(\alpha')^{-1} V'^{-1}\| \leq W_0$. If the ordered component pair is within ρ of (f, g) , then

$$|c_1|, |c_2 - 1| \leq (\rho + K_{\text{aff}} \delta)/d_{fg}. \quad (8.7)$$

No lower bound on $|\det U|$ is used.

Proof. Since $\mathbf{B}' = V' \text{diag}(\alpha') F'_{\text{coef}}$,

$$\beta_0/2 \leq \sigma_2(\mathbf{B}') \leq \sigma_{\min}(V') K_d.$$

This proves the inverse bound, using $\alpha'_a \geq \pi$. Write C_θ for the matrix with coefficient rows f, g and put

$$T = \text{diag}(\alpha')^{-1} V'^{-1} V \text{diag}(\alpha, \gamma), \quad E_0 = \text{diag}(\alpha')^{-1} V'^{-1} (\mathbf{B}' - \mathbf{B}_\theta).$$

Then $F'_{\text{coef}} = T C_\theta + E_0$ and $\|E_0\| \leq W_0 C_B \delta$. If a row of T is (l, c) , monicity says $l + c + (E_0)_{\text{lead}} = 1$. Subtracting $(E_0)_{\text{lead}} f$ from that error row gives the exact decomposition $f'_b = (1 - c)f + cg + e_b$, with the bound in (8.5) and zero leading coefficient in e_b . Finally $h_c - f = c(g - f)$ and $h_c - g = (c - 1)(g - f)$ imply (8.7) by the triangle inequality. $\square$

Lemma 8.3 (Two robust isolation estimates). *There are explicit positive $r_f, r_g, \varepsilon_0, C_0, C_1$, depending only on K_0 , with the following property. If $h_c + e$ is monic with all roots in $[0, D]$, $\deg e < 3$ and $\|e\|_{\text{coef}} \leq \varepsilon$, then*

$$|c| \leq r_f, \quad \varepsilon \leq \varepsilon_0 \implies |c| \leq C_0 \varepsilon, \quad |1 - c| \leq r_g, \quad \varepsilon \leq \varepsilon_0 \implies |1 - c| \leq C_1 \varepsilon.$$

The constants require only root-separation and coefficient bounds on K_0 , not a sign or a nonzero margin for B, E .

Proof. Let $s_0 = \min_{K_0} s > 0$ and $d_0 = \min_{K_0} 4r^3(r - s)^3 > 0$. Choose L_D as a Lipschitz bound for the discriminant on a coefficient neighbourhood of $\{h_c : \theta \in K_0, |c| \leq 1\}$, and R_D such that

$$|\text{disc}(h_c) + 4r^3(r - s)^3 c| \leq R_D c^2 \quad (|c| \leq 1).$$

These constants are finite because the discriminant is polynomial. Take $r_f \leq \min(1, d_0/(2R_D))$, omitting the second bound if $R_D = 0$. The endpoint condition gives $-cs^3 + e(0) \leq 0$, hence

$c \geq -\varepsilon/s_0^3$. For $0 \leq c \leq r_f$, nonnegative discriminant gives $0 \leq -d_0c + R_Dc^2 + L_D\varepsilon$, hence $c \leq 2L_D\varepsilon/d_0$. One may take $C_0 = \max(s_0^{-3}, 2L_D/d_0)$.

Near $c = 1$, put $v = 1 - c$ and translate $z = s + x$. There are exact identities

$$h_c(s+x) = x^3 + vA_2x^2 + vA_1x + vA_0, \quad A_2 = 3s - 2r, \quad A_1 = f'(s), \quad A_0 = f(s) = s(r-s)^2.$$

Let L_s bound coefficient translation of e on K_0 , set $E_s = L_s\varepsilon$, and write the perturbed coefficients as $u_i = vA_i + e_i$, $|e_i| \leq E_s$. Choose $M \geq \max_{K_0}(|A_1|, |A_2|)$ and $a_* = \min_{K_0} A_0 > 0$. Depressing the perturbed cubic by its own quadratic coefficient gives

$$p = u_1 - u_2^2/3, \quad q = u_0 - u_1u_2/3 + 2u_2^3/27.$$

For $u = |v| + E_s \leq 1$,

$$|p| \leq C_p u, \quad |q - A_0v| \leq E_s + C_q u^2, \quad C_p = M+1+(M+1)^2/3, \quad C_q = (M+1)^2/3 + 2(M+1)^3/27.$$

Real-rootedness is exactly $p \leq 0$ and $|q| \leq c_{\text{cub}}(-p)^{3/2}$ with $c_{\text{cub}} = 2/(3\sqrt{3})$. Thus

$$a_*|v| \leq E_s + C_q u^2 + c_{\text{cub}} C_p^{3/2} u^{3/2}. \quad (8.8)$$

Choose

$$\eta = \min\{1, a_*/(4C_q), (a_*/(4c_{\text{cub}}C_p^{3/2}))^2\}, \quad r_g = \eta/2, \quad \varepsilon_0 \leq \eta/(2L_s).$$

Then $u \leq \eta$, the last two terms in (8.8) are at most $a_*u/2$, and $|v| \leq (2/a_* + 1)E_s$. Take $C_1 = (2/a_* + 1)L_s$. Decrease ε_0 also to stay in the discriminant neighbourhood used in the first argument. Every threshold is determined by the stated positive lower margins and finite polynomial upper bounds. $\square$

Let d_{par} be Euclidean distance in monic coefficients and stochastic entries, minimized over the component permutation.

Lemma 8.4 (A local inverse through the walls). *For every compact $K_0 \subset \mathfrak{F}$, there exist $\rho, M, t_0 > 0$ such that a closed-model competitor whose component pair is within ρ of (f, g) , up to permutation, satisfies*

$$d_{\text{par}}(\xi, \theta) \leq M \max_j \|F_T(\xi)_j - P_{\theta,j}\|_F \quad (8.9)$$

whenever the observation error is below t_0 . These constants require no lower bound on $|\det U|$, $-B$, or $-E$. For a compact $H \subset \mathfrak{J}$ and small fixed ρ , the remote separation

$$\eta_H(\rho) = \min_{\theta \in H, \xi \text{ in the closed model } d_{\text{par}}(\xi, \theta) \geq \rho} h_w(F_T(\xi), P_\theta) \quad (8.10)$$

is positive if the minimizing set is nonempty. The whole-model inverse therefore holds uniformly on H .

Proof. Let $r_* = \min(r_f, r_g)$ from Lemma 8.3 and choose

$$\rho \leq d_{fg}r_*/4, \quad t_0 \leq \min\{\beta_0/(2C_B), \rho/K_{\text{aff}}, \varepsilon_0/K_{\text{aff}}\}.$$

Proposition 8.2 gives an error at most $\varepsilon = K_{\text{aff}}\delta$ in the affine pencil and pencil coordinates in the required neighbourhoods. The two isolation estimates then give $|c_1| + |c_2 - 1| \leq (C_0 + C_1)K_{\text{aff}}\delta$. Together with (8.6), this controls both polynomial coefficient vectors by a common multiple of δ .

Fix the component permutation placing the close ordered pair near (f, g) ; separation of f, g makes this choice unambiguous for a smaller ρ . To recover the stochastic parameters without dividing by $\det U$, choose coefficient functionals ℓ_1, ℓ_2 dual to f, g . Their norms are bounded by the inverse Gram matrix of this pair, uniformly on K_0 . The matrix $T'_{ij} = \ell_i(f'_j)$ is $I + O_{K_0}(\delta)$. Decrease t_0 so that $\|T' - I\| \leq 1/2$; then $\|(T')^{-1}\| \leq 2$. Apply the two functionals entrywise to K' . Inverting T' recovers $M'_a = \alpha'_a u'_a v'_a{}^\top$ with $O_{K_0}(\delta)$ error from M_a . Entry sums give the weights, and row and column sums divided by those weights give the two channels. These divisions have

lower bound π . This proves (8.9) with constants constructed from normalizer, second-marginal, Gram, root and weight margins only. In particular no full-rank inverse has been continued through its singular determinant.

For the last assertion the feasible set in (8.10) is compact, allowing the centre's design to vary in H . A zero minimum would give an exact competitor separated from that centre. The entire fibre and conditional uniqueness in Theorem 8.1 exclude this on H . Small whole-model observation balls thus lie in the local neighbourhood. It is this exclusion radius, rather than a local coefficient constant, that may collapse at a remote wall. $\square$

8.2. The complete local Fisher fibre. Use $a, X, Y \geq 0$, free b, c , and five free stochastic coordinates η . The component jets are

$$p_f = -a(z-r)^2 - bz(z-r) - Xz, \quad p_g = -c(z-s)^2 - Y(z-s). \quad (8.11)$$

Let L be the seven free score columns and D_0 the three columns indexed by $0, x, y$ for (a, X, Y) . Define Q by (7.2) and

$$J(X, Y) = \min_{a \geq 0} (a, X, Y)Q(a, X, Y)^\top, \quad \kappa_x = \min_{Y \geq 0} J(1, Y), \quad \kappa_y = \min_{X \geq 0} J(X, 1). \quad (8.12)$$

Proposition 8.5 (Coercivity and finite supports). *Throughout $\mathfrak{F}$, L has rank seven, $Q_{00} > 0$, and κ_x, κ_y are positive and locally Lipschitz. For every compact $K_0 \subset \mathfrak{F}$,*

$$\|S_\theta(a, b, c, \eta, X, Y)\|_{L, \theta} \geq c_{K_0} \|(a, b, c, \eta, X, Y)\| \quad (a, X, Y \geq 0).$$

Each κ_i is given by four finite support tests. For $A \subseteq \{0, x, y\} \setminus \{i\}$ with Q_{AA} positive definite, put

$$v_A = -Q_{AA}^{-1}Q_{Ai}, \quad k_{i,A} = Q_{ii} - Q_{iA}Q_{AA}^{-1}Q_{Ai}.$$

The test is $v_A \geq 0$ and $Q_{ji} + Q_{jA}v_A \geq 0$ for inactive j . The empty support has value Q_{ii} . At least one test succeeds and all successful tests give κ_i .

Proof. Every feasible jet is realized by roots

$$ta, \quad r + tb/2 \pm \sqrt{tX}, \quad s + tc/3 + \sqrt{t}y_i, \quad \sum_i y_i = 0, \quad \frac{1}{2} \sum_i y_i^2 = Y,$$

and order- t stochastic increments. Their observation increment is $tS + O(t^{3/2})$. A zero score, combined with Lemma 8.4, forces the component jets and η to vanish. Coefficient comparison in (8.11) then forces every variable to vanish. Compactness of the feasible unit sphere gives coercivity, uniformly on K_0 . This also proves independence of the free columns, $Q_{00} > 0$, and existence, positivity and boundedness of the constrained minimizers.

Choose a minimizer with minimal positive support in the projected score columns. Those supported columns are independent: a nonzero dependence would allow a signed change of positive coefficients, preserving the represented vector, until one coefficient vanished. On an independent support, stationarity and the inactive inequalities are exactly the stated tests. Conversely their KKT conditions prove optimality by convexity. This handles a singular full Q without inverting it. Uniformly bounded minimizers and smooth score dependence give local Lipschitz dependence by evaluating each problem at a minimizer of the other problem. $\square$

Theorem 8.6 (Local normal form and whole-model restriction). *At every centre of $\mathfrak{F}$ the local leading root set is*

$$(\{0\}, \{-\sqrt{X}, \sqrt{X}\}, Z(z^3 - Yz - Z)), \quad X, Y \geq 0, \quad J(X, Y) \leq 4, \quad 27Z^2 \leq 4Y^3. \quad (8.13)$$

On compact subsets of $\mathfrak{F}$, scaled local observation balls converge to this set with Hausdorff error $O(\sqrt{t})$. Hence

$$\omega_{\theta, \text{loc}}(t) = C_{\text{loc}}(\theta)\sqrt{t} + O(t), \quad C_{\text{loc}} = \max\{\sqrt{2}\kappa_x^{-1/4}, 2\sqrt{2/3}\kappa_y^{-1/4}\}. \quad (8.14)$$

The error is compact-uniform, and C_{loc} is positive and locally Lipschitz on all of $\mathfrak{F}$. On compact subsets of $\mathfrak{I}$ these statements hold for the whole-model modulus. The support tests, (8.2), and $9\kappa_y - 16\kappa_x$ give a finite semialgebraic partition. On its constant choices,

$$C_{\text{loc}}^4 = \max\{4/\kappa_x, 64/(9\kappa_y)\}$$

is one rational function, and C_{loc} is Nash after refinement.

Proof. The local inverse bounds coefficients and stochastic increments by $O(t)$ for every local competitor. Its first component has exact roots $ta, r + tb/2 \pm \sqrt{tX}$ with bounded a, b, X and $a, X \geq 0$. Its second component has exact roots $s + tc/3 + \sqrt{t}y_i$ with $\sum_i y_i = 0$ and bounded squared sum. Put $Y = \frac{1}{2}\sum_i y_i^2$ and $Z = \prod_i y_i$. The real-cubic condition is exactly $Y \geq 0, 27Z^2 \leq 4Y^3$. Expansion gives $F = P + tS + O(t^{3/2})$ and $h_w = t\|S\|_I/2 + O(t^{3/2})$ uniformly on these bounded coordinates. Radial contraction of all first-order variables, together with $Z \mapsto \rho^{3/2}Z$, proves the outer Hausdorff inclusion. Conversely choose bounded minimizing free and endpoint variables and realize the displayed root arcs with $\rho = 1 - M\sqrt{t}$. They lie in the observation ball and give the inner inclusion. The inequality boundary and all joint cubic coefficient relations are therefore realized.

Homogeneity gives $X_{\text{max}} = 2/\sqrt{\kappa_x}$ and $Y_{\text{max}} = 2/\sqrt{\kappa_y}$. The double-cluster diameter is $\sqrt{X_{\text{max}}}$. Lemma 7.1 with $m = 3$ gives the triple diameter $2\sqrt{Y_{\text{max}}/3}$, attained also by $(-2h, h, h)$ and $(-h, -h, 2h)$ with $h = \sqrt{Y_{\text{max}}/3}$. The maximum of marginal diameters gives (8.14). Proposition 8.5, cluster separation and the positive remote exclusion in Lemma 8.4 give the remaining claims. The local formula remains valid at $B = 0$ or $E = 0$, but its identification with the whole-model modulus does not. $\square$

Example 8.7 (The earlier rational phase values). For $D = 1, \pi = 1/10, r = 3/5, s = 1/5, \alpha = 2/5, u_1 = u_2 = (1/3, 2/3)^\top, v_1 = (1/4, 3/4)^\top, v_2 = (3/4, 1/4)^\top$, clocks $(6/5, 7/5, 8/5, 9/5, 2, 5/2, 3)$ and equal weights, $B = -18/25$. The empty supports are admissible, the double cluster dominates, and exact rational evaluation gives

$$C^4 = \frac{78550423766145732273576408862601811503774391446}{385279214348552940073664077156875}, \quad 3778.707 < C < 3778.708.$$

Changing only u_2 to $(2/5, 3/5)^\top$ gives the triple-dominant phase, $59.970 < C < 59.971$. The exact arithmetic records for both values are retained in the archival companion. These large values are not evidence for an unspecified wall blow-up.

Corollary 8.8 (Centre-known local risk). *Let $H \subset \mathfrak{I}$ be compact, fix $0 < c \leq 1/(8\sqrt{2})$, and observe independent categorical samples at the design clocks with $n_j/N \rightarrow w_j$ uniformly on H (integer rounding suffices). For squared spectral loss on the specified neighbourhood $h_w(F_T(\xi), P_\theta) \leq cN^{-1/2}$, with the centre known, the minimax risk $\mathcal{R}_N(\theta, c)$ satisfies uniformly on H*

$$\frac{3c}{32}C_{\text{loc}}(\theta)^2 - O_H(N^{-1/4}) \leq \sqrt{N}\mathcal{R}_N(\theta, c) \leq cC_{\text{loc}}(\theta)^2 + O_H(N^{-1/4}).$$

Proof. A constant decision in the specified spectral ball has loss at most its squared diameter. A diameter pair in that ball has weighted Hellinger distance at most $2cN^{-1/2}$. Tensorization bounds its product Hellinger distance by $2c\sqrt{\max_j n_j/(Nw_j)}$. Uniform convergence and the positive minimum of the design weights on H make the maximum ratio at most 2 for all sufficiently large N , uniformly on H . Thus the product distance is at most $2c\sqrt{2} \leq 1/4$. Total variation is bounded by this Hellinger distance in the present convention. The two-point testing bound for squared metric loss is $d^2(1 - \text{TV})/8$, hence at least $3d^2/32$. Substitute (8.14). The centre and its neighbourhood are known; unknown-centre adaptation and honest confidence sets across a wall are not asserted. $\square$

9. A REMOTE BRANCH AT THE STOCHASTIC WEIGHT WALL

Fix r, s, D with $B > 0$, positive u, v_1, v_2 with $v_1 \neq v_2$, and $U = (u, u)$, $V = (v_1, v_2)$. Assume

$$\gamma_0 = \pi c_*, \quad \alpha_0 = 1 - \pi c_*, \quad \pi < \alpha_0 < 1 - \pi. \quad (9.1)$$

The design is fixed. Let θ_δ be the original (f, g) centre with weight $\alpha = \alpha_0 + \delta$. Then $E = -A\delta$, so $\delta > 0$ is the identified side. Put

$$a = \frac{4rs}{r+3s}, \quad b = \frac{rs}{3s-2r}, \quad h_* = (1-c_*)f + c_*g = (z-a)(z-b)^2. \quad (9.2)$$

Substitution gives $s < a < r < b < D$. At $\delta = 0$ the two exact component pairs are (f, g) and (f, h_*) . The remote pair has weight π on h_* and second-channel columns

$$v_f^* = \frac{\alpha_0 v_1 + (\gamma_0 - \pi)v_2}{1 - \pi}, \quad v_h^* = v_2. \quad (9.3)$$

Let ξ_* be this point; its observation is $P_0 = P_{\theta_0}$. Its second channel is invertible because $v_f^* - v_2 = \alpha_0(v_1 - v_2)/(1 - \pi) \neq 0$. The sorted base and remote spectra are respectively

$$(0, s, s, s, r, r), \quad (0, a, r, r, b, b),$$

whose distance is

$$D_* = \max(r - s, b - r) > 0. \quad (9.4)$$

Thus a zero-length additional pencil interval already has a positive exact-fibre spectral diameter.

9.1. A complete real corner chart at the remote point. Use the eleven coordinates $v = (a_0, b_0, X, c_0, d_0, Y, k, \dot{u}_f, \dot{u}_h, \dot{v}_f, \dot{v}_h)$, where a binary channel increment is represented by its one free zero-sum coordinate. Set

$$\begin{aligned} p_f &= -a_0(z-r)^2 - b_0z(z-r) - Xz, \\ p_h &= -c_0(z-b)^2 - d_0(z-a)(z-b) - Y(z-a), \\ \mathcal{T}_* &= \{v : a_0, X, Y, k \geq 0\}. \end{aligned} \quad (9.5)$$

The remaining seven coordinates are free. Here k is the increment of the weight on h_* ; the weight on f has increment $-k$. Let J_*v be the full coefficient and stochastic vector with polynomial increments p_f, p_h and these stochastic increments.

Proposition 9.1 (Remote tangent cone and converse realization). *In a neighbourhood of ξ_* , modulo the one component permutation, the closed model has a Nash coordinate chart from a neighbourhood of zero in $\mathcal{T}_*$. Its derivative is the invertible linear map J_* . Consequently its coefficient-stochastic tangent cone is exactly $J_*(\mathcal{T}_*)$. Every normalized secant has a subsequential limit in this cone, and every vector of the cone is realized by a feasible arc. For bounded $v \in \mathcal{T}_*$ the arcs can be chosen with*

$$\xi(\tau, v) = \xi_* + \tau J_*v + O(\tau^2), \quad F_T(\xi(\tau, v)) = P_0 + \tau S_*v + O(\tau^2), \quad (9.6)$$

uniformly on bounded coordinate sets. Here S_ is the score (2.3) at the remote point. All assertions are uniform on compact subsets of the wall family with its stated strict margins.*

Proof. Factor each nearby polynomial into its separated simple and quadratic factors. Coprime factorization is analytic in coefficients. Write the simple root of the first component as u , the mean of its quadratic roots as $r + v_f/2$, and their squared half-separation as x . Write the corresponding coordinates of the second component as $a + w$, $b + v_h/2$ and y . Thus the exact polynomials are

$$(z-u)((z-r-v_f/2)^2-x), \quad (z-a-w)((z-b-v_h/2)^2-y).$$

These six coordinates are Nash functions of the coefficients near the remote pair. Conversely the displayed formulas are polynomial in the six coordinates. Their derivative has the three columns

in (9.5) for each component. They are independent: for the first polynomial, evaluation at 0 and r first determines a_0 and X , and then the remaining coefficient determines b_0 ; for the second, evaluation at a and b determines c_0 and Y , and then d_0 . The inverse function theorem gives the asserted local coefficient chart.

The first simple root is feasible exactly when $u \geq 0$, locally. Its upper endpoint is inactive. All other cluster centres are strictly inside $(0, D)$, so their interval constraints are inactive in this neighbourhood. Real-rootedness of the two quadratic factors is exactly $x, y \geq 0$. The weight on h_* is at its floor, giving $k \geq 0$; the other floor has strict margin $1 - 2\pi > 0$. Channel entries are strictly positive, so only their linear column-sum equations remain active. This proves that the complete feasible chart is the indicated orthant times seven free coordinates, not merely an outer cone. Separated factors are unique. The alternative component labelling is a second, identical chart under the permutation quotient and introduces no extra directions.

For the secant assertion, let τ be the norm of a coefficient and stochastic displacement. Analytic factorization bounds the simple and mean shifts by $O(\tau)$. The quadratic coefficients bound x, y by $O(\tau)$ as well, since variance is mean squared minus the constant coefficient. Dividing the exact chart coordinates by τ gives a bounded vector in $\mathcal{T}_*$. Its subsequential limits give all normalized tangent secants. In the norm-normalized case J_*v has norm one in the limit, so no nonzero secant disappears in the chart.

Conversely set $u = \tau a_0$, $v_f = \tau b_0$, $x = \tau X$, $w = \tau c_0$, $v_h = \tau d_0$, $y = \tau Y$, choose weight $\pi + \tau k$ on h , and take the specified order- τ channel increments. The roots are

$$\tau a_0, \quad r + \tau b_0/2 \pm \sqrt{\tau X}, \quad a + \tau c_0, \quad b + \tau d_0/2 \pm \sqrt{\tau Y}.$$

They remain feasible for small τ , uniformly for bounded coordinates. They are analytic in $\sqrt{\tau}$; their polynomial coefficients are analytic in τ . In fact the exact identities are

$$f_\tau - f - \tau p_f = \tau^2 \{a_0 b_0(z - r) + a_0 X + b_0^2 z/4\} - \tau^3 a_0 b_0^2/4, \quad (9.7)$$

$$h_\tau - h_* - \tau p_h = \tau^2 \{c_0 d_0(z - b) + c_0 Y + d_0^2(z - a)/4\} - \tau^3 c_0 d_0^2/4. \quad (9.8)$$

There is no order- $\tau^{3/2}$ coefficient term at these simple/double clusters. Taylor expansion of the rational observation map, whose normalizers stay positive, proves (9.6). Compact strict root, channel, weight and clock margins make all inverse chart and derivative bounds uniform as stated. $\square$

9.2. The constrained entrance program. Put $B_0 = \frac{d}{d\delta} P_{\theta_\delta} \Big|_{\delta=0}$ and use the Fisher norm at P_0 . Define

$$\mu_E = \frac{1}{2} \min_{v \in \mathcal{T}_*} \|S_* v - B_0\|_{I,0}. \quad (9.9)$$

This is an explicit finite-dimensional program on the cone just proved to be the full feasible tangent cone.

Lemma 9.2 (Positive entrance separation). *The score has no nonzero zero on $\mathcal{T}_*$, and B_0 is not in its image. Thus $\mu_E > 0$ and its minimum is attained. Eliminating seven free variables gives sixteen independent-support KKT tests on four nonnegative variables. For algebraic input μ_E is real algebraic.*

Proof. Suppose first that $S_* v = B_0$. The differentiated normalizer equations, using Lemma 2.1 at the coprime remote pair with invertible second channel, imply $\dot{q} = f - g$ and $\dot{K} = \frac{d}{d\delta} K_{\theta_\delta} \Big|_0$. The derivative of the second marginal belongs to $\text{span}(f, g)$. Its stochastic derivative terms already belong to that span; the remote second-channel weight matrix is invertible. Hence $p_f, p_h \in \text{span}(f, g)$. Since their degrees are below three, write $p_f = t_f(g - f)$ and $p_h = t_h(g - f)$.

Evaluation at 0 gives $t_f = a_0 r^2 / s^3 \geq 0$, while evaluation at r gives $t_f = -Xr / (r - s)^3 \leq 0$. Therefore $t_f = 0$ and $p_f = 0$. At b the second polynomial gives

$$t_h = \frac{Y(b - a)}{f(b) - g(b)} \geq 0,$$

because $R(b) > 1$. Differentiation of the monic normalizer now gives

$$\pi t_h + c_* k = -1, \quad (9.10)$$

an impossibility. For a zero score, the same argument has right-hand side zero and implies $t_h = k = 0$. All polynomial jets vanish. Coefficient functionals dual to f, h_* then imply that each coefficient matrix has zero derivative; taking its row and column sums forces all four channel derivatives to vanish. The independent chart columns in Proposition 9.1 force $v = 0$.

The intersection of $\mathcal{T}_*$ with the unit sphere is compact, so $\|S_* v\|_{I,0} \geq c\|v\|$ on the cone. Thus the objective in (9.9) tends to infinity with $\|v\|$; its minimum is attained and positive, since equality $S_* v = B_0$ was excluded.

Let L_* be the seven free columns, D_* the four nonnegative columns, $W = \text{diag}(w_j / P_{0,je})$, and $\mathcal{P}$ the Fisher-orthogonal projection off $\text{im } L_*$. Put

$$Q_* = D_*^\top W \mathcal{P} D_*, \quad g_* = D_*^\top W \mathcal{P} B_0, \quad b_* = B_0^\top W \mathcal{P} B_0.$$

Coercivity implies full rank of the free columns, and

$$4\mu_E^2 = \min_{z \geq 0} \{z^\top Q_* z - 2g_*^\top z + b_*\}. \quad (9.11)$$

For each of the sixteen supports A with $Q_{*,AA}$ positive definite, set $z_A = Q_{*,AA}^{-1} g_{*,A}$, require $z_A \geq 0$ and $(Q_* z - g_*)_j \geq 0$ off A , and take value $b_* - g_{*,A}^\top Q_{*,AA}^{-1} g_{*,A}$. Include the empty support. The minimal-positive-support argument of Proposition 8.5 proves completeness, including singular Q_* . All entries are rational in the wall coordinates; take the positive square root after selecting the exact minimum. $\square$

Lemma 9.3 (Local coefficient control at the remote point). *In a coefficient and stochastic neighbourhood of ξ_* in the closed model,*

$$d_{\text{par}}(\xi, \xi_*) \leq C \|F_T(\xi) - P_0\|.$$

Every competitor at parameter distance $O(\delta)$ has bounded exact chart coordinates at scale δ and observation expansion $P_0 + \delta S_ v + O(\delta^2)$. Conversely each bounded cone vector has such a feasible realization.*

Proof. If the inverse failed, normalize a sequence of displacements by their norm. Proposition 9.1 gives a nonzero limit $J_* v$ with $v \in \mathcal{T}_*$ and zero score, contradicting Lemma 9.2. Equivalently in its chart, Taylor's theorem and cone coercivity give $\|F_T(\Phi(w)) - P_0\| \geq c\|w\| - C\|w\|^2 \geq c\|w\|/2$ for small feasible w . The chart is locally bi-Lipschitz in coefficients and stochastic entries, which proves the inverse and the stated bounded coordinate assertion. The expansions and converse are (9.6). This proof does not assume the inverse in order to identify the tangent cone. $\square$

9.3. The two-scale law with a quadratic entrance remainder. Choose disjoint small closed coefficient and stochastic neighbourhoods N_0, N_* of the two exact representatives at the wall, modulo component permutation, with ξ_* in the relative interior of N_* . Put

$$d_{\text{rem}}(\delta) = \min_{\xi \in N_*} h_w(F_T(\xi), P_{\theta_\delta}). \quad (9.12)$$

Their boundaries have a positive observation separation from P_0 .

Theorem 9.4 (Weight-wall transition). *Under (9.1) and $B > 0$,*

$$d_{\text{rem}}(\delta) = \mu_E \delta + O(\delta^2) \quad (\delta \downarrow 0). \quad (9.13)$$

The minimum is independent of the sufficiently small neighbourhood N_ for all sufficiently small δ . The base local constant $C_{\text{loc}}(\theta_\delta)$ has the finite positive limit $C_{\text{loc}}(\theta_0)$; neither κ_x nor κ_y vanishes at this wall. For fixed $\lambda > 0$ the whole-model modulus satisfies*

$$0 < \lambda < \mu_E : \quad \omega_{\theta_\delta}(\lambda\delta) = C_{\text{loc}}(\theta_0)\sqrt{\lambda\delta} + O_\lambda(\delta), \quad (9.14)$$

$$\lambda > \mu_E : \quad \omega_{\theta_\delta}(\lambda\delta) = D_* + O_\lambda(\sqrt{\delta}). \quad (9.15)$$

The exact entrance curve for the closed ball is $t = d_{\text{rem}}(\delta)$, including equality. The layer not decided by the first-order threshold alone has width $O(\delta^2)$ in t .

On the nonidentified side, for small $\varepsilon > 0$,

$$\omega_{\theta_{-\varepsilon}}(0) = \max \left\{ r - s, \ b - r + \sqrt{\frac{\chi\varepsilon}{\pi}} + O(\varepsilon) \right\}, \quad \chi = \frac{f(b) - g(b)}{b - a} > 0. \quad (9.16)$$

If $b - r < r - s$, the exact diameter stays equal to $r - s$ initially; if $b - r \geq r - s$, it opens above D_ at the displayed square-root rate.*

Proof. Theorem 8.1 gives exactly the two representatives at $\delta = 0$. Compactness gives a positive observation gap outside $N_0 \cup N_*$. Since ξ_* itself is at distance $O(\delta)$ from P_{θ_δ} , a minimizer in (9.12) has observation $O(\delta)$ from P_0 and lies away from the boundary of N_* . Lemma 9.3 bounds its coefficient and stochastic displacement by $O(\delta)$. Its exact chart coordinates are therefore δv for bounded $v \in \mathcal{T}_*$. Expansion at the common positive probability array and the reverse triangle inequality give

$$h_w(F_T(\xi), P_{\theta_\delta}) = \frac{\delta}{2} \|S_* v - B_0\|_{L,0} + O(\delta^2), \quad (9.17)$$

uniformly on those bounded coordinates. The lower bound in (9.13) follows by minimizing the leading norm. Conversely a minimizer of (9.9), realized by the exact chart, gives the upper bound. The error is quadratic because of (9.7)–(9.8), not merely an appeal to a generic root-continuity bound. This also implies the weaker $O(\delta^{3/2})$ entrance estimate. The same localization puts all minimizers inside any fixed smaller remote neighbourhood for small δ , proving independence of its choice.

Proposition 8.5 and Theorem 8.6 apply on a compact base path segment including θ_0 ; this pair stays locally coercive there. Thus C_{loc} is positive and locally Lipschitz through the wall. If $\lambda < \mu_E$, the remote neighbourhood is excluded from the radius- $\lambda\delta$ ball for small δ . The exterior gap excludes every other competitor, so the whole ball is the base local ball. Formula (8.14) gives (9.14). Replacing $C_{\text{loc}}(\theta_\delta)$ by $C_{\text{loc}}(\theta_0)$ costs only $O(\delta^{3/2})$.

If $\lambda > \mu_E$, a remote minimizer and the base centre belong to the ball. Every competitor belongs to one of the two neighbourhoods. The two coefficient inverses put its roots within $O_\lambda(\sqrt{\delta})$ of the corresponding exact spectrum at the wall. The bottleneck triangle inequality gives both bounds in (9.15). Compact attainment makes remote membership exactly $t \geq d_{\text{rem}}(\delta)$, including equality. The width of the unresolved first-order layer follows from (9.13); it does not decide the sign of the difference on the critical ray.

For $\theta_{-\varepsilon}$ the extra exact pencil interval is $[c_*, c_* + \varepsilon/\pi]$, since $c_D > c_*$ is fixed. At $c = c_* + \sigma$, expansion at $z = b + u$ gives

$$h_c(b + u) = (b - a)u^2 + u^3 + \sigma(g(b) - f(b)) + O(\sigma|u|).$$

Since $R(b) > 1$, $\chi > 0$ and the two roots near b are $b \pm \sqrt{\chi\sigma} + O(\sigma)$. The third root remains in (s, r) . The larger root increases with c by the monotonic branch of R above its maximum. Comparing sorted lists, the distance to the base spectrum is exactly $\max(r - s, b_+(c) - r)$. Two

remote spectra are only $O(\sqrt{\varepsilon})$ apart, less than D_* for small ε . The exact-fibre diameter is therefore attained against the base at the upper pencil endpoint. This proves (9.16) and its two cases. $\square$

Remark 9.5 (The critical ray). At $t = \mu_E \delta$, remote membership is decided by the sign of $d_{\text{rem}}(\delta) - \mu_E \delta$. For algebraic input, (9.12) has a compact first-order minimum formula. Elimination and real algebraic branch isolation determine its Puiseux expansion. The first nonzero higher coefficient decides the critical ray; if the difference is identically zero, the closed ball includes the remote branch. No strict regime is assigned to that ray by convention.

Remark 9.6 (The geometric wall). At $B = 0$ with $E > 0$, the remote double root is $b = D$, and the additional pencil interval again consists of one point at positive spectral distance. On the nonidentified side put $\ell_0 = D - b$. Expansion at the strict maximum of R gives

$$c_D - c_* = \frac{-R''(b)}{2(R(b) - 1)^2} \ell_0^2 + O(\ell_0^3), \quad \ell_0 = \frac{B}{3s - 2r}.$$

The coefficient is positive and the weight floor is inactive. This interval is born with quadratic length in geometric clearance, rather than linear length in the weight parameter. The base local constants remain finite on compact subsets of $\mathfrak{F}$. A two-scale entrance law at that endpoint requires a different cone with one-sided root coordinates. The complete entrance theorem here is for the weight wall.

Example 9.7 (An exact wall datum). Take $D = 1$, $r = 3/5$, $s = 11/20$, $\pi = 1/10$, the rank-one columns and clocks in Section 8, and equal design weights. Then

$$\alpha_0 = \frac{547}{675}, \quad c_* = \frac{256}{135}, \quad a = \frac{44}{75}, \quad b = \frac{11}{15}, \quad B = \frac{3}{25}, \quad D_* = \frac{2}{15}, \quad \chi = \frac{3}{64},$$

and $v_f^* = (1457, 3403)^\top / 4860$. Exact rational evaluation gives

$$5975.429249985072 < C_{\text{loc}}(\theta_0) < 5975.429249985073, \\ 2.131024914 \cdot 10^{-6} < \mu_E < 2.131024915 \cdot 10^{-6}.$$

The empty positive support is admissible in each base program and in the remote program. The ancillary exact records retain the rational Gram matrices, squared entrance constant, fourth power of the local constant and isolating intervals. The decimals follow from rational power comparisons, not optimization tolerances. They illustrate two different, poorly conditioned scales; they do not test the universal quantifiers of the family theorems.

Three stability questions. Multiplicity and one-sided boundary constraints determine the weights in Theorem 7.2. Exact identification in the cubic family is instead determined by (8.2). Conditioning of a fixed local law is measured by the positive Fisher programs. At the weight wall, identification fails while those programs stay coercive and the local square-root exponent persists. In Example 7.3, by contrast, the exponent changes and the linear constant diverges. The joint atlas retains these distinct limiting operations rather than identifying their strata.

10. FINITE REMOTE ENTRANCES AND THE CRITICAL WINDOW

The preceding weight wall is an instance of a more general phenomenon. A regular feasible corner, rather than invertibility of the ambient score, is the relevant local hypothesis. We formulate the result in the square-root probability embedding, so that the distance in its statement is exactly the Hellinger distance and not a quadratic approximation to it.

Let X be compact and semialgebraic, let F be a continuous probability map, and let T be a continuous monic-polynomial root-multiset target of fixed degree. Quotient a finite representation symmetry when necessary. Suppose

$$F^{-1}(P_0) = \{x_0, x_1, \dots, x_m\}. \quad (10.1)$$

The points are representatives in this quotient. Let $x_\delta \rightarrow x_0$ be a feasible centre path, $P_\delta = F(x_\delta)$, and suppose P_0 is strictly positive. The design weights are fixed. Define

$$E(P) = (\sqrt{w_j} \sqrt{P_{je}})_{j,e}, \quad p_0 = E(P_0), \quad p_\delta = E(P_\delta) = p_0 + \delta b + \delta^2 c + o(\delta^2).$$

The Euclidean distance between two embedded arrays is h_w .

For each $j \geq 1$, assume that a full relative neighbourhood of x_j in X has coordinates

$$\psi_j : K_j \cap B_{r_j}(0) \rightarrow X, \quad \psi_j(0) = x_j,$$

where K_j is a closed convex polyhedral cone and ψ_j is a Nash parametrization, a homeomorphism onto that relative neighbourhood. Its inverse need not be differentiable at the vertex. Assume that $E \circ F \circ \psi_j$ has a C^2 extension to an ambient neighbourhood of zero. Write

$$E(F(\psi_j(z))) = p_0 + J_j z + Q_j(z, z) + o(\|z\|^2), \quad \|J_j v\| \geq a_j \|v\| \quad (v \in K_j), \quad a_j > 0. \quad (10.2)$$

Here $Q_j(z, z)$ denotes one half of the second derivative. The score may have a kernel in the ambient vector space; the last inequality is required only on the feasible cone. Choose small disjoint closed neighbourhoods $N_0, \dots, N_m$, with boundaries avoiding the exact fibre, and put

$$d_j(\delta) = \min_{x \in N_j} h_w(F(x), P_\delta), \quad \mu_j = \min_{v \in K_j} \|J_j v - b\|.$$

Theorem 10.1 (Remote entrance for regular corners). *Assume (10.1)–(10.2) and $\mu_j > 0$ for $1 \leq j \leq m$. Then*

$$d_j(\delta) = \mu_j \delta + O(\delta^2) \quad (\delta \downarrow 0). \quad (10.3)$$

For small δ the minimum is unchanged on replacing N_j by any fixed smaller relative neighbourhood of x_j . If $\lambda > 0$ differs from every μ_j , the target image of the radius- $\lambda\delta$ ball converges in Hausdorff distance to

$$\mathcal{D}_\lambda = \{T(x_0)\} \cup \{T(x_j) : \mu_j < \lambda\}. \quad (10.4)$$

Its diameter consequently tends to $\text{diam } \mathcal{D}_\lambda$. For $\lambda < \min_j \mu_j$, the whole ball is the base local ball for small δ . Any compact-uniform local asymptotic law at the base therefore remains valid in this regime.

Proof. Fix a remote index and omit it from the notation. Taylor's theorem and cone coercivity give, after reducing r ,

$$\|E(F(\psi(z))) - p_0\| \geq a\|z\|/2 \quad (z \in K, \|z\| < r).$$

The competitor $z = 0$ gives an upper bound $O(\delta)$ for $d(\delta)$. Compactness and (10.1) exclude the boundary of N from minimizing sequences. A minimizer thus has $z = O(\delta)$: its distance to p_0 is at most $d(\delta) + \|p_\delta - p_0\|$. Write $z = \delta v$. The vectors $v \in K$ are uniformly bounded, and

$$E(F(\psi(\delta v))) - p_\delta = \delta(Jv - b) + \delta^2\{Q(v, v) - c\} + o(\delta^2)$$

uniformly on bounded sets. Minimizing the leading norm gives the lower bound in (10.3). The function $v \mapsto \|Jv - b\|$ is coercive on K , so its minimum is attained; inserting a minimizer in the exact chart gives the upper bound. The same localization shows independence of the fixed neighbourhood.

Outside the union of the N_j the continuous function $h_w(F(x), P_0)$ has a positive minimum. This remains an exclusion gap for balls of radius $\lambda\delta$ about P_δ . In a remote neighbourhood, membership is possible exactly when $t \geq d_j(\delta)$, by compact attainment. Thus a remote neighbourhood is excluded

if $\lambda < \mu_j$ and contains a minimizer if $\lambda > \mu_j$. Every sequence of remaining competitors converges, after subselection, to an exact representative; its target converges to that representative's target. Conversely x_δ supplies $T(x_0)$ in the limit and each included remote minimizer supplies $T(x_j)$. Finiteness gives Hausdorff convergence to (10.4). The diameter conclusion follows from the elementary bound by twice the Hausdorff distance. When no remote neighbourhood is included, all competitors are in N_0 , which proves the last assertion. $\square$

The coefficient of the next term can be determined without assuming a unique minimizer or an invertible ambient score. This is useful at a change of active face of a constrained Fisher program.

Theorem 10.2 (The second-order entrance coefficient). *Under the hypotheses of Theorem 10.1, set*

$$M_j = \operatorname{argmin}_{v \in K_j} \|J_j v - b\|.$$

This is a nonempty compact set. The residual $r_j = J_j v - b$ is the same for all $v \in M_j$, and has norm μ_j . Define

$$k_j = \frac{1}{\mu_j} \min_{v \in M_j} \langle r_j, Q_j(v, v) - c \rangle. \quad (10.5)$$

Then

$$d_j(\delta) = \mu_j \delta + k_j \delta^2 + o(\delta^2). \quad (10.6)$$

At radius $t = \mu_j \delta + s \delta^2$, the j th remote neighbourhood is excluded for $s < k_j$ and included for $s > k_j$, for all sufficiently small positive δ . Equality requires the next nonzero term or the exact minimum inequality; it is not assigned to a regime by convention. For algebraic Nash input, μ_j and k_j are real algebraic numbers computable by real quantifier elimination and real-root isolation.

Proof. Coercivity bounds all minimizers. The image $J_j K_j$ is a closed convex polyhedral cone. The squared distance to b has a unique minimizing image point: if two different image points minimized it, their midpoint would have strictly smaller squared distance. This proves the assertion about r_j without requiring uniqueness of its preimage.

Again suppress j . Localization in the preceding proof bounds all scaled minimizers in one compact subset of K , also containing M . On that set the squared normalized distance has the uniform expansion

$$f_\delta(v) = \|Jv - b\|^2 + 2\delta \langle Jv - b, Q(v, v) - c \rangle + o(\delta).$$

Every limit of minimizers of f_δ belongs to M , by comparison with any point of M . The first term is at least μ^2 everywhere. The lower bound for the first-order increment of $\min f_\delta$ is therefore the minimum of the displayed second term over M . For the upper bound evaluate f_δ at a point of M attaining that minimum. Compactness and continuity justify both extrema. Consequently

$$\min f_\delta = \mu^2 + 2\delta \min_{v \in M} \langle r, Q(v, v) - c \rangle + o(\delta).$$

Taking the positive square root and multiplying by δ proves (10.6). Comparing with the proposed radius gives the strict cases. The minimum defining M and the minimum in (10.5) have first-order polynomial formulas with compact attainment; the latter may equivalently be restricted to a sufficiently large algebraic ball. Elimination and isolation give algebraic outputs. The formula is intrinsic to the distance and the fixed centre path, since its value equals the coefficient in (10.6), independently of the chosen chart. $\square$

Corollary 10.3 (Application to the cubic weight wall). *The remote point of Section 9 satisfies the preceding hypotheses with $m = 1$, $K_1 = \mathcal{T}_*$ and $\mu_1 = \mu_E$. In particular its entrance has the expansion*

$$d_{\text{rem}}(\delta) = \mu_E \delta + k_E \delta^2 + o(\delta^2),$$

where k_E is given by (10.5) in the exact remote chart of Proposition 9.1. This formula does not assert a common sign for k_E on the weight-wall family.

Proof. Theorem 8.1 gives the two exact representatives. The remote chart and its converse feasible realization are Proposition 9.1; its smooth coefficient formulas and positive normalizers give the required C^2 observation embedding. In these coordinates

$$J_1 v = \left(\frac{\sqrt{w_j}}{2\sqrt{P_{0,je}}} (S_* v)_{je} \right)_{j,e}, \quad b = \left(\frac{\sqrt{w_j}}{2\sqrt{P_{0,je}}} (B_0)_{je} \right)_{j,e}.$$

Lemma 9.2 is exactly cone coercivity and positive separation in this embedding. Hence the first minimization is (9.9). The two second-order derivatives in (10.5) are taken from the rational coefficient chart and the square-root probability map, not from root coordinates with an unproved differentiability at a multiple root. Theorems 10.1 and 10.2 apply. $\square$

Example 10.4 (Both signs of the critical correction). Let $e_0 = (1, 1)/\sqrt{2}$, $e_1 = (-1, 1)/\sqrt{2}$ and

$$q(s) = \frac{(1 - s^2)e_0 + 2se_1}{1 + s^2}, \quad P(s) = (q_1(s)^2, q_2(s)^2).$$

For sufficiently small $|s|$ both coordinates of $q(s)$ are positive. Take a base interval with observation $P(s)$ and a disjoint remote interval $0 \leq u \leq e$ with observation $P(-u/2)$. Give the two intervals distinct simple-root targets at zero, for example $z - (1/4 + s)$ and $z - (3/4 + u)$, and choose e small enough to keep all roots in $[0, 1]$. The exact fibre at $P(0)$ has two points. Choose the centre $s_\delta = (\delta + a\delta^2)/2$, where a is fixed. At the remote point $K = \mathbb{R}_+$, $Jv = -ve_1$, $b = e_1$, $c = ae_1 - e_0/2$, $M = \{0\}$ and $\mu = 1$. Formula (10.5) gives $k = a$. Indeed the nearest remote point is $u = 0$ for small $\delta > 0$, since on this short circle arc the chord distance increases with the separation of the stereographic parameters. Directly,

$$d_{\text{rem}}(\delta) = \frac{2s_\delta}{\sqrt{1 + s_\delta^2}} = \delta + a\delta^2 + O(\delta^3).$$

Thus either sign of the second-order correction occurs in an algebraic probability experiment satisfying all hypotheses. For $a = 0$ the quadratic coefficient vanishes and the next term is $-\delta^3/8$.

These results apply in any coefficient dimension and to any fixed number of latent components once their stated finite-fibre and feasible-cone hypotheses have been verified. They are not a classification of all rank-deficient singularities. The endpoint double root at the geometric wall requires a different parametrization and a different separation argument, supplied in Section 11.

11. THE GEOMETRIC ENDPOINT WALL

We now complete a second wall mechanism in the same rank-deficient cubic experiment. Fix $D > r > 0$, strict weights $\alpha, \gamma = 1 - \alpha$, positive $U = (u, u)$ and invertible positive V , and the exterior design. Put

$$s_0 = \frac{2rD}{3D - r}, \quad s_\delta = s_0 - \delta, \quad f = z(z - r)^2, \quad g_\delta = (z - s_\delta)^3. \quad (11.1)$$

Then $2r/3 < s_0 < r$. Assume the strict weight margin

$$E_0 = \gamma A(s_0) - 4\pi r^3 > 0, \quad A(s) = (r + 3s)(2r - 3s)^2. \quad (11.2)$$

Let θ_δ be this base centre and P_δ its observation. Here $B(\theta_\delta) = -(3D - r)\delta$. Thus positive δ is the identified side, while the weight constraint stays inactive locally. Write $c_* = 4r^3/A(s_0)$ and

$$a = \frac{4rs_0}{r + 3s_0}, \quad h_* = (1 - c_*)f + c_*g_0 = (z - a)(z - D)^2, \quad \beta = \gamma/c_*.$$

The remote weight on h_* is $\beta > \pi$, and its other weight is $1 - \beta > \pi$. Its first channel is (u, u) and its second channel is

$$v_f^* = \frac{\alpha v_1 + (\gamma - \beta)v_2}{1 - \beta}, \quad v_h^* = v_2. \quad (11.3)$$

Both columns are positive and distinct. Denote this remote point by ξ_B . Theorem 8.1 gives exactly the base and this remote representative at $\delta = 0$.

Proposition 11.1 (The endpoint-root parametrization). *A full relative neighbourhood of ξ_B , modulo the component permutation, has the following exact feasible parametrization:*

$$\begin{aligned} f_v(z) &= (z - a_0)((z - r - b_0/2)^2 - X), \\ h_v(z) &= (z - a - c_0)(z - D - d_1)(z - D - d_2), \\ K_B &= \{a_0, X \geq 0, \quad d_1 \leq d_2 \leq 0\}. \end{aligned} \quad (11.4)$$

The weight on h is $\beta + k$, and the four binary channel coordinates have their unrestricted small increments. The other coordinates b_0, c_0, k are free. This is a polynomial parametrization with continuous semialgebraic inverse; no differentiability of its inverse at $d_1 = d_2$ is asserted. Its first polynomial jets are

$$p_f = -a_0(z - r)^2 - b_0z(z - r) - Xz, \quad p_h = -c_0(z - D)^2 - (d_1 + d_2)(z - a)(z - D). \quad (11.5)$$

Let S_B be the score at ξ_B applied to these jets and the stochastic increments. Then S_B has no nonzero zero on K_B . If $B_B = \frac{d}{d\delta}P_\delta|_0$, then $B_B \notin S_B K_B$. In particular

$$\mu_B = \frac{1}{2} \min_{v \in K_B} \|S_B v - B_B\|_{I,0} > 0. \quad (11.6)$$

The minimum is attained. After seven free coordinates are eliminated, it is a quadratic program with three nonnegative coordinates and eight independent-support tests.

Proof. Separated simple and double factors give the first component's simple root, quadratic mean and squared half-separation as analytic functions of its coefficients. At the second component the separated simple root is analytic; the two roots near D have a unique increasing ordering and depend continuously and semialgebraically on the coefficients. Their feasibility is exactly $d_1 \leq d_2 \leq 0$. All other root bounds, weights and channel inequalities have strict margins, except $a_0 \geq 0$ and $X \geq 0$. This proves exact coverage and injectivity of the parametrization, not merely containment of tangent directions. Its polynomial formulas give a uniform $O(\tau^2)$ coefficient remainder on substituting τv for bounded $v \in K_B$. Positive normalizers and probabilities give the same remainder for the observation embedding.

We prove the two score assertions directly. Put $\dot{g} = 3(z - s_0)^2$, the derivative of g_δ at zero. Equality of remote and base scores implies equality of their derivatives of q, K by the differentiated normalizer argument in Lemma 2.1. The coefficient second marginal at the remote point is $(1 - \beta)v_f^*f + \beta v_2 h_*$. Its two coefficient columns are independent. Modulo $\text{span}(f, g_0)$, equality of its derivative with $\gamma v_2 \dot{g}$ gives

$$p_f = t_f(g_0 - f), \quad p_h = c_* \dot{g} + t_h(g_0 - f).$$

The polynomials have the stated degrees, so no additional monic direction occurs. Evaluation of the first identity at 0 and r gives $t_f = a_0 r^2 / s_0^3 \geq 0$ and $t_f = -Xr / (r - s_0)^3 \leq 0$. Thus $t_f = 0$ and $p_f = 0$.

To interpret the second identity, consider the algebraic double-root pencil curve, without imposing its endpoint constraint,

$$\bar{h}_\delta = (1 - c_*(s_\delta))f + c_*(s_\delta)g_\delta = (z - a(s_\delta))(z - b(s_\delta))^2, \quad b(s) = \frac{rs}{3s - 2r}.$$

At zero, $b(s_0) = D$ and

$$\dot{b} = \left. \frac{d}{d\delta} b(s_0 - \delta) \right|_0 = \frac{2r^2}{(3s_0 - 2r)^2} > 0. \quad (11.7)$$

Both $p_h(D)$ and $\dot{h}(D)$ vanish. Since $g_0(D) - f(D) \neq 0$, this determines t_h uniquely and implies $p_h = \dot{h}$. Differentiation in z at D now gives

$$-(d_1 + d_2)(D - a) = p'_h(D) = -2(D - a)\dot{b}.$$

This would require $d_1 + d_2 = 2\dot{b} > 0$, contrary to feasibility. Therefore $S_B v = B_B$ is impossible.

For a zero score the same second-marginal argument instead gives $p_f = t_f(g_0 - f)$ and $p_h = t_h(g_0 - f)$. The first polynomial again vanishes. Evaluating the second at D forces $t_h = 0$. Its coefficients then give $c_0 = 0$ and $d_1 + d_2 = 0$, whence $d_1 = d_2 = 0$ on K_B . The first polynomial's independent columns give $a_0 = b_0 = X = 0$. Coefficient functionals dual to f, h_* recover zero derivatives of both coefficient matrices, and their entry, row and column sums give zero weight and channel increments. Thus the entire cone vector is zero. Compactness of the feasible unit sphere yields $\|S_B v\|_{L,0} \geq c\|v\|$ on K_B . The objective in (11.6) is consequently coercive and has a positive attained minimum.

Finally the score depends on d_1, d_2 only through $w = -(d_1 + d_2) \geq 0$. Every $w \geq 0$ is realized, for example by $d_1 = d_2 = -w/2$. The remaining nonnegative coordinates are a_0, X ; there are seven free coordinates. Their columns are independent by the zero-score argument. Fisher-orthogonal projection off those columns reduces the problem to three nonnegative variables. The minimal-positive-support argument of Proposition 8.5 gives the eight support tests, including the empty support and only inverting positive definite supported Gram matrices. This remains valid when the full projected Gram matrix is singular. $\square$

Theorem 11.2 (The geometric-wall two-scale law). *Under (11.1)–(11.2), the remote entrance satisfies*

$$d_{\text{rem},B}(\delta) = \mu_B \delta + k_B \delta^2 + o(\delta^2), \quad \mu_B > 0, \quad (11.8)$$

where k_B is given by Theorem 10.2 in the parametrization (11.4). In particular, for fixed $\lambda > 0$,

$$\lambda < \mu_B: \quad \omega_{\theta_\delta}(\lambda\delta) = C_{\text{loc}}(\theta_0)\sqrt{\lambda\delta} + O_\lambda(\delta), \quad (11.9)$$

$$\lambda > \mu_B: \quad \omega_{\theta_\delta}(\lambda\delta) = D - r + O_\lambda(\sqrt{\delta}). \quad (11.10)$$

The base local Fisher constant stays positive and finite at the wall. The strict cases in the critical window $t = \mu_B \delta + s\delta^2$ are decided by $s - k_B$.

On the nonidentified side $\theta_{-\varepsilon}$, for all sufficiently small $\varepsilon > 0$, the additional exact pencil interval has length

$$c_D(s_0 + \varepsilon) - c_*(s_0 + \varepsilon) = \Lambda_B \varepsilon^2 + O(\varepsilon^3), \quad \Lambda_B = \frac{-R''_{s_0}(D)}{2(R_{s_0}(D) - 1)^2} \dot{b}^2 > 0, \quad (11.11)$$

where $R_s(z) = f(z)/(z - s)^3$. Nevertheless its exact whole-model spectral diameter is constant:

$$\omega_{\theta_{-\varepsilon}}(0) = D - r. \quad (11.12)$$

Thus the two wall mechanisms differ both in the size of the newborn pencil interval and in the opening of its exact spectral diameter.

Proof. The endpoint parametrization and its score separation satisfy the hypotheses of Theorems 10.1 and 10.2. The derivative of the square-root observation embedding is the score multiplied entrywise by $\sqrt{w_j}/(2\sqrt{P_{0,je}})$, so its entrance minimum is precisely (11.6). This proves (11.8) and the critical-window assertion, including possible nonuniqueness of the minimizing endpoint-root split.

The sorted spectra at the wall are $(0, s_0, s_0, s_0, r, r)$ and $(0, a, r, r, D, D)$. Their distance is $\max(r - s_0, D - r) = D - r$, since $r - s_0 = r(D - r)/(3D - r) < D - r$. The local inverse and local Fisher law of Section 8 are uniform on a compact segment of the base path through zero. Below the entrance threshold all competitors are in that base neighbourhood, proving (11.9). Above it, the localization proof for the remote cone bounds its exact coordinates by $O(\delta)$. The remote endpoint roots move by $O(\delta)$ and its interior double cluster by $O(\sqrt{\delta})$; the base inverse gives the same $O(\sqrt{\delta})$ root bound. A remote minimizer and the centre are both feasible, and the exterior observation gap excludes every other region. The bottleneck triangle inequality proves (11.10).

For the last assertions, $b(s_0 + \varepsilon) = D - \dot{b}\varepsilon + O(\varepsilon^2)$. The function R_s has a strict nondegenerate maximum at $b(s)$ near the wall. Taylor expansion of $R_s(D)$ there, followed by $x \mapsto x/(x - 1)$, gives

$$c_D(s) - c_*(s) = \frac{-R_s''(b(s))}{2(R_s(b(s)) - 1)^2}(D - b(s))^2 + O((D - b(s))^3).$$

This proves (11.11). The strict weight margin keeps $\gamma/\pi > c_D(s)$ for small ε , so the entire interval $[c_*(s), c_D(s)]$ is feasible. Its upper endpoint has a root exactly at D . A remote spectrum has increasing ordering $(0, a_c, r, r, b_-(c), b_+(c))$, with $s < a_c < r < b_-(c) \leq b_+(c) \leq D$. Its distance to the base is exactly $\max(r - s, b_+(c) - r) \leq D - r$, and equality is attained at $c_D(s)$. The remote spectra converge uniformly to one spectrum as $\varepsilon \downarrow 0$; their mutual diameter is therefore less than $D - r$ for small ε . Theorem 8.1 excludes any other exact component. This proves (11.12). $\square$

11.1. Exact critical-ray examples. We record two algebraic specializations of the second-order formula. Use the clocks $(6/5, 7/5, 8/5, 9/5, 2, 5/2, 3)$ with equal weights, $u_1 = u_2 = (1/3, 2/3)^\top$, and $v_1 = (1/4, 3/4)^\top$, $v_2 = (3/4, 1/4)^\top$.

Example 11.3 (An endpoint wall with a positive critical correction). Take $D = 1$, $r = 3/5$, $s_0 = 1/2$, $\alpha = 2/5$, $\pi = 1/10$. The remote coordinates and the interval-opening coefficient are

$$a = \frac{4}{7}, \quad c_* = \frac{32}{7}, \quad \beta = \frac{21}{160}, \quad v_f^* = \frac{1}{556}(289, 267)^\top, \quad \dot{b} = 8, \quad \Lambda_B = \frac{38400}{49}.$$

The projected three-variable program has the empty support as its only successful independent-support test. Its linear vector is strictly negative entrywise, so the minimizer has all three nonnegative coordinates zero. The free score has rank seven; hence the first-order minimizer is unique, including the endpoint split. Exact rational arithmetic and positive-root isolation give

$$0.000063237134157102 < \mu_B < 0.000063237134157103, \\ 0.071857358263070 < k_B < 0.071857358263071.$$

Consequently the remote branch is excluded on $t = \mu_B \delta$ for all sufficiently small positive δ . On the opposite side the exact spectral diameter is $2/5$, despite the nonzero quadratic pencil-interval opening.

Example 11.4 (The weight-wall critical ray). At the weight-wall datum of Example 9.7, the four entries of the projected linear vector g_* are strictly negative. The empty cone support is therefore the unique first-order minimizer, with its seven free coordinates recovered by the inverse free Gram matrix. The second-order formula gives

$$0.011285405417542 < k_E < 0.011285405417543.$$

Thus the critical ray $t = \mu_E \delta$ also excludes the remote branch for this datum when $\delta > 0$ is sufficiently small. This sign is a property of this specialization, not a convention at an unresolved ray and not a sign assertion for the whole wall family.

For reproducibility, the finite arithmetic behind both examples can be specified without numerical differentiation. Write the exact remote and base probability expansions at the selected first-order minimizer as

$$P_{\text{rem}}(\delta) = P + \delta A + \delta^2 C + O(\delta^3), \quad P_{\text{base}}(\delta) = P + \delta B + \delta^2 D + O(\delta^3).$$

The rational coefficients are obtained by multiplying the exact polynomial, weight and channel jets and dividing by the normalizer. The square-root expansion then gives

$$\begin{aligned} \mu^2 &= \frac{1}{4} \sum_{j,e} \frac{w_j (A_{je} - B_{je})^2}{P_{je}}, \\ \mu k &= \sum_{j,e} w_j \left\{ \frac{(A_{je} - B_{je})(C_{je} - D_{je})}{4P_{je}} - \frac{(A_{je} - B_{je})(A_{je}^2 - B_{je}^2)}{16P_{je}^2} \right\}. \end{aligned} \quad (11.13)$$

Strict empty-support KKT inequalities and positive semidefiniteness of the projected Gram matrix prove uniqueness of the cone optimum even when that matrix is singular: the difference of objectives is $z^\top Q z - 2g^\top z > 0$ for every nonzero $z \geq 0$. The rational values of μ^2 , μk , the Gram data and isolating intervals are retained with the exact regression. The interval tests square rational endpoints and compare integers after clearing positive denominators. They certify these finite specializations only.

APPENDIX A. FINITE ALGEBRAIC CERTIFICATES AND REAL DESCENT

This appendix supplies the full family argument used in Lemma 3.3 and Theorem 3.4. A variety in the algebraic construction is a reduced separated scheme of finite type over a characteristic-zero field. Smoothness is geometric smoothness. For an effective input the ground constants belong to a specified finite real algebraic extension of $\mathbb{Q}$; generic fibres are presented over its finitely generated function fields. No algorithmic assertion about an arbitrary computable field is intended. Real realizations are subsequently taken in $\mathbb{R}$, with the original sign and branch conditions retained.

An algebraic coordinate presentation consists of finitely many affine coordinate rings, localized by specified nonzero elements, transition morphisms and their inverse identities, and a coordinate map whose relative Jacobian is invertible. An adapted presentation also lists each divisor equation, its expression as a coordinate times a unit, and the monomial identities for the functions. Projective maps are recorded by homogeneous equations or by their finite blow-up sequences. These are finite polynomial data, not a numerical collection of coordinate boxes.

A.1. The Rees algebra and a finite diagram.

Lemma A.1 (One localization for all powers of an ideal). *Let A be a noetherian domain, B an algebra of finite type over A , and $I \subset B$ a finitely generated ideal. There is $a \in A \setminus \{0\}$ such that, for every A_a -algebra A' , with $B' = B_a \otimes_{A_a} A'$,*

$$I_a^n \otimes_{A_a} A' \xrightarrow{\sim} (I_a B')^n \quad (n \geq 0), \quad \left(\bigoplus_{n \geq 0} I_a^n \right) \otimes_{A_a} A' \cong \bigoplus_{n \geq 0} (I_a B')^n.$$

The first isomorphism identifies both sides as submodules of B' . Consequently the blow-up in I_a commutes with this base change. One localization suffices for any specified finite sequence of centre ideals on finitely many affine charts of successive blow-ups.

Proof. The associated graded algebra $\text{gr}_I B = \bigoplus_{n \geq 0} I^n / I^{n+1}$ is of finite type over A : it is generated by B/I in degree zero and the images of a finite set of generators of I in degree one. Since A is noetherian, these algebras are finitely presented. Apply generic freeness [10, Tag 051S] to B as a module over itself and to $\text{gr}_I B$ as a module over itself. After inverting the product of

the two resulting elements, both are flat over A_a . Each graded piece is a direct summand as an A_a -module, hence flat. In particular B_a/I_a is flat. Induction using

$$0 \longrightarrow I_a^{n-1}/I_a^n \longrightarrow B_a/I_a^n \longrightarrow B_a/I_a^{n-1} \longrightarrow 0$$

shows that B_a/I_a^n is flat for every $n \geq 1$. Tensoring $0 \rightarrow I_a^n \rightarrow B_a \rightarrow B_a/I_a^n \rightarrow 0$ with A' is therefore exact on the left. Its image in B' is the ideal generated by products of n images of elements of I_a , namely $(I_a B')^n$. The isomorphisms respect multiplication and grading, proving the Rees algebra assertion. Relative Proj commutes with base change; this proves the blow-up assertion. For a finite sequence, apply the argument to the actual centre ideal on every affine chart at each stage and multiply the finitely many localizing elements. Previously proved identities survive further localization. No countable intersection of open sets is used. $\square$

Lemma A.2 (Spreading the entire stratifying diagram). *Let A be a finitely generated integral algebra over a characteristic-zero field k , and let $K = \text{Frac}(A)$. Suppose $Y \rightarrow \text{Spec } A$ is of finite type and a generic certificate consists of a reduced closed chain $Y_{0,K} = Y_{K,\text{red}} \supset \cdots \supset Y_{q,K} = \emptyset$, projective $Z_{j,K} \rightarrow Y_{j,K}$, and specified inverse isomorphisms above $E_{j,K} = Y_{j,K} \setminus Y_{j+1,K}$. Then, after replacing A by A_a for one nonzero a , the chain can be chosen with $Y_0 = Y_{\text{red}}$, all its members flat with geometrically reduced fibres, and the stated isomorphisms hold above the actual complements $E_j = Y_j \setminus Y_{j+1}$. In particular they hold above $(E_j)_s$ on every fibre. The union of these complements is the whole reduced fibre; no constancy of the number of irreducible components is required.*

Proof. First replace Y by its reduction. Take reduced closures of the generic $Y_{j,K}$ inside Y and use $Y_q = \emptyset$. Clear denominators in the projective equations and affine presentations of the generic maps. All schemes involved are of finite presentation over the noetherian base. The finite-presentation limit lemma [10, Tag 01ZM, Lemma 32.10.1(1)–(3)] descends objects, morphisms, and equalities of morphisms. Apply it to the inverse maps on the generic complements, their inclusions, and both inverse identities. The complements used on the model are $Y_j \setminus Y_{j+1}$; their generic fibres are the prescribed complements. The descended inverse maps and equalities therefore give isomorphisms on these actual open subschemes after a further localization, not merely on unspecified subsets with the same generic points. Finite affine covers and their intersection diagrams are included in this application of the limit lemma. Thus chart coverage, as well as the formula for each map, descends.

Each reduced generic member is geometrically reduced because K has characteristic zero. Generic freeness makes each member flat on a dense open set. The generic-fibre reducedness theorem [10, Section 37.26, Lemma 37.26.4] then permits a further common shrink so that all fibres of these finitely many members are geometrically reduced. This is an imposed property, not an optional replacement of scheme structure after specializing. Finally, on every fibre the sets $Y_{j,s} \setminus Y_{j+1,s}$ form a disjoint partition of $Y_{0,s}$ because the chain ends in the empty set. For any point there is a unique index at which it leaves this finite descending chain. The corresponding isomorphism covers that point, including points on a special fibre component. This argument neither identifies fibre components with generic components nor assumes their number to be constant. $\square$

A.2. Relative smoothness, normal crossings, and exceptional sets. We now prove Lemma 3.3 with a precise list of the further conditions. Separate the finitely many open-and-closed components of each smooth generic source according to the identically zero functions. These components and their idempotent identities are part of the finite diagram. On a component on which a function is declared zero, descend that zero identity. On every other component principalize the product of the remaining functions, recording all factors and units.

The absolute input is characteristic-zero resolution and principalization [3, 4]. More precisely, the variety resolution of [4, Theorem 1.1] supplies the smooth sources; the marked-ideal resolution of [4, Theorem 1.3], applied with mark one, supplies principalization together with the normal-crossing exceptional boundary. A product which is a monomial times a unit makes each of its factors a monomial times a unit on the regular local rings. This use is over the field K . Compatibility with arbitrary special fibres is supplied by Lemma A.1, not by a claim that functoriality for smooth maps implies arbitrary base-change compatibility.

Spread the finite sequence of centres and its adapted charts. The following finite conditions are imposed before retaining the open base. The source and centre smooth loci are the loci on which their finite presentations satisfy the relative smoothness criterion. On a smooth source the intersections of the designated divisor equations must be smooth of their prescribed relative codimensions; in adapted coordinates this says that the appropriate relative differentials are independent. This condition is imposed for every subset of the finite divisor list. Each coordinate chart must be etale, each declared inverse unit must multiply its unit to one, and the finite charts must cover their source. All scheme-theoretic identities in these statements are spread by the finite-diagram lemma. Their remaining rank conditions are open algebraic conditions; complements of chart unions are constructible algebraic sets.

For clarity, the discarded algebraic parameter set is the union of the closures of images of the following bad sets, together with the localizing hypersurfaces already used:

- (1) the original irreducible components not dominating the current reduced integral base;
- (2) nonsmooth points of a source or centre, and failures of the specified relative dimension or centre codimension;
- (3) failures of smoothness or codimension of any designated divisor intersection, or failure of an adapted etale Jacobian;
- (4) the complement of the chosen finite source charts, and any zero of a function declared invertible;
- (5) failures of a descended zero identity, a chart transition, or an inverse isomorphism on an actual complement in the closed chain.

For items (4)–(5) the bad set is empty once the relevant identities and coverage have descended; it is harmless to include it explicitly in the certificate. Every bad set in this list has empty generic fibre: this is respectively dominance, generic smoothness in the given certificate, generic normal crossings, generic coverage and invertibility, or the specified generic identities. Images are constructible by Chevalley’s theorem [10, Tag 054K]. A constructible subset of an integral noetherian space which is dense contains a nonempty open subset and hence the generic point. Therefore the closure of each of these images is proper. This justification applies also when $Y \rightarrow S$ is not proper.

After their removal, apply Lemma A.1 on every spread blow-up chart. The resulting smooth sources and centres, relative normal crossings, unit identities and finite chart covers hold on every retained fibre. A nonzero monomial cannot specialize to the zero function on a fibre component: its unit is invertible and its finitely many divisors each have relative codimension one. Their union cannot contain a component of the smooth fibre. An identically zero generic function remains zero by its descended identity. The open-and-closed source decomposition prevents a change of these flags through an unlabelled component intersection. Flatness of the closed chain and smoothness of the sources exclude additional vertical components on the retained open set. Coverage of any special fibre components is, more directly, the partition argument in Lemma A.2. This proves all fibrewise assertions of Lemma 3.3.

Each discarded algebraic subset has strictly smaller dimension than its current integral base. Replace it by its reduced irreducible components and reconstruct the entire generic certificate on each component. There are finitely many components at each step and depth at most $\dim S + 1$.

This gives finite termination. The exceptional list is not required to be minimal. Importantly, a change of a *real* accessibility flag need not lie in a proper algebraic subset: two full-dimensional real chambers can have different flags. Such changes are treated by finite real quantifier-elimination partitions below, not by this dimension argument.

A.3. Exact real algebraization.

Lemma A.3 (The square-slack cover with all branch conditions). *Let S be a semialgebraic base and $\Gamma \rightarrow S$ a semialgebraic proper map realized as a graph in $S \times \mathbb{R}^N$. The graph includes all auxiliary functions, sign choices and denominator exclusions. After a finite base refinement, it admits finitely many algebraic square-slack lifts whose real points over S project exactly onto Γ . Projection is proper over compact subsets of S . Every later map factoring through these lifts projects only to points of Γ .*

Proof. Let C be the ordinary semialgebraic closure of Γ in the ambient real algebraic space. Properness implies that Γ is relatively closed in $S \times \mathbb{R}^N$: a convergent sequence over a convergent base sequence lies over a compact subset of S and has a graph limit there. It follows that $C \cap (S \times \mathbb{R}^N) = \Gamma$. A closed semialgebraic set is a finite union of basic closed semialgebraic sets [6, Theorem 2.7.2]. For a piece

$$C_l = \{x : f_{l,a}(x) = 0, p_{l,b}(x) \geq 0\}$$

use the algebraic set

$$Y_l = \{(x, s) : f_{l,a}(x) = 0, p_{l,b}(x) = s_b^2\}.$$

Every real point of Y_l projects into C_l , and each point of C_l has a real lift. Over S , the union therefore projects exactly to Γ . Over a compact base subset the graph is compact. The slack coordinates are bounded by the square roots of maxima of the finitely many polynomials there, and the lift is closed; hence the lifted real set is compact. This proves properness and the last assertion.

Strict exclusions are encoded in the complete graph formula before this construction. For a denominator excluded everywhere on that compact graph, $d \neq 0$ may also be retained by adjoining w with $dw = 1$, together with any required sign of d . Nonnegative square-root branches are specified in the graph, not selected after taking an algebraic closure. If adjoining such coordinates destroys properness on the initial base, first apply Hardt triviality to the enlarged graph, whose individual fibres are compact, and refine the base. A trivialization with compact fibre makes the restricted graph map proper. Closure is taken only *after* this step. Consequently no point with a vanishing excluded denominator is reintroduced over the retained stratum. General Boolean branch conditions are instead kept in the complete graph formula: one does not split a compact fibre into nonclosed pieces, adjoin unbounded reciprocals on those pieces, and then assume their fibres are compact. The equality $C \cap (S \times \mathbb{R}^N) = \Gamma$ already preserves all such conditions without that operation. $\square$

Lemma A.4 (Real branch selection and simultaneous descent). *The root labels and Nash graph functions used in the cluster construction can be carried on finitely many selected real Nash branches of algebraic base covers. On a finite semialgebraic refinement, the proper resolved square-slack sources cover exactly the original graph in every fibre. Descent by root permutations preserves the entire joint coefficient and root relation, not merely its scalar order list.*

Proof. Refine a Nash stratum by the regular locus of each integral component of its algebraic closure. Its lower-dimensional singular parts are treated recursively. A Nash function is algebraic over the coordinate function field. Adjoin the finitely many graph coordinates needed here to that field. The extension is finite and separable in characteristic zero. Choose equations and invert their leading coefficients and discriminants; a further localization gives a finite etale algebraic cover.

Include the original graph formula to select the intended real branches. Finite semialgebraic cell decomposition makes each selected graph a Nash section and separates its alternatives. Complex sheets and real sheets failing the original formula are not retained. Thus monodromy is handled by base refinement, rather than by globally labelling a nontrivial cover. Exceptional leading coefficients and discriminants are handled by the reduced lower-dimensional base recursion. A finite union of selected sections has compact inverse image over compact subsets of a stratum.

Apply Lemma A.3 on each selected branch. For a real point y of a lifted fibre, choose the index with $y \in Y_{j,s} \setminus Y_{j+1,s}$ in the spread singular-locus chain. The inverse isomorphism of Lemma A.2 supplies a real point of its smooth source above y . A blow-up of a smooth real variety in a smooth centre of positive codimension c has fibre $\mathbb{RP}^{c-1}$ at a real centre point and is an isomorphism away from the centre. In local smooth coordinates the centre is a coordinate subspace, which proves both assertions directly. Such a blow-up therefore does not lose real points. Composing the finitely many principalizations and taking the finite union over j proves real surjectivity. A real singular point with no top-dimensional real branch is covered by a later member of the chain. Projectivity of these maps and properness of the slack projection prove properness of the real cover.

Conversely every real resolved point factors through a real slack point on the selected branch, and hence projects to the original graph by Lemma A.3. This proves equality, not just inclusion, of the covered real fibres. A root-label permutation simultaneously permutes cluster indices, their coefficient coordinates and root arrays. The equations defining Ψ and the fixed-centre formula (4.2) are equivariant for this simultaneous action. A finite union commutes with taking a fibrewise closure; projection to the finite quotient therefore preserves the joint relation and its root image. Descent never pairs one coefficient from one sheet with another coefficient from a different sheet. The bottleneck metric is invariant under these permutations. $\square$

A.4. Real accessibility and completeness of the order list.

Lemma A.5 (Real-accessible divisors). *On a resolved square-slack source with the above relative adapted coordinates, accessibility of a divisor is a first-order real condition on the parameter. Every recorded divisor has an admissible real transversal attaining its monomial orders. Every real degeneration relevant to a finite coefficient order is controlled by one of the recorded divisors. Thus the minimum in Corollary 3.6 is both an upper-inequality exponent and an attained exponent.*

Proof. In a chart with source equations $E(\theta, y) = 0$, chart exclusions $D(\theta, y) \neq 0$, and adapted divisor coordinates $y_1, \dots, y_l$, the flag for i with $a_i > 0$ is the formula

$$\exists y : E(\theta, y) = 0, \quad D(\theta, y) \neq 0, \quad y_i = 0, \quad \prod_{h \neq i, h \leq l} y_h \neq 0, \quad \det J_{\text{et}}(\theta, y) \neq 0. \quad (\text{A.1})$$

Inverse units and the selected real base formula are included in E, D . Charts are the ones meeting the inverse image of $G = 0$. Since the full real slack fibre already represents the admissible graph, there is no additional unrecorded sign restriction on source directions. At a point satisfying (A.1), the real inverse function theorem gives a transversal with $y_i = t$ and other adapted coordinates fixed. For small nonzero t , $G > 0$, and every point projects into the original graph. The unit bounds give $G \asymp |t|^{a_i}$ and $H \asymp |t|^{b_i}$ for each nonzero H on that source component.

A real degeneration with $G \rightarrow 0$ has a lifted accumulation point above $G = 0$, by properness. Some adapted chart contains that point. If it lies on several divisors, relative normal crossings allows a small real perturbation *on any one of them* avoiding all the others. Hence each divisor of positive G order meeting this real chart has a point of the form (A.1). In a smaller box, with β the minimum of its recorded ratios, $b_i \geq \beta a_i$ for $a_i > 0$ and $b_i \geq 0$ otherwise. The monomial identities and positive unit bounds imply $H \leq CG^\beta$ throughout the box, including its intersections. Equivalently, along a lifted arc the order ratio is a weighted average of the

b_i/a_i with nonnegative additional terms from indices with $a_i = 0$; it cannot be smaller than their minimum. A finite cover near the compact exact fibre gives this inequality for all competitors. The transversal for a minimizing divisor attains it. If G is identically zero on a source with real points, the exact-target condition gives $H = 0$ there. A smooth fibre component with a real point has a Zariski-dense real open subset. Thus a nonzero monomial H cannot occur on such an exact real component; real-empty components contribute no order. This handles zero flags as well as the nonzero ratios.

Finally (A.1) is a finite polynomial formula. Quantifier elimination partitions the base by all its truth values. These semialgebraic pieces may have full dimension. Nash refinement makes the choices compatible with the selected base branches. This partition is finite without an algebraic dimension-drop assertion. $\square$

Completion of Theorem 3.4. Stratify the enlarged graph and the exact cluster data as in Lemma A.3, and carry labels as in Lemma A.4. Over an integral algebraic base resolve the reduced generic fibre, then its reduced singular locus, and continue until the empty set. Dimension decreases in this chain. Principalize on its smooth sources the product of the nonzero functions. Lemmas A.1 and A.2, followed by the explicit bad-locus construction, spread this certificate over a nonempty open base with all fibrewise properties. Repeat on the finitely many reduced exceptional components. Lemma A.4 proves exact real coverage after descent, and Lemma A.5 proves the accessibility assertions and the completeness of the orders.

For a fixed fibre, cover the compact inverse image of $G = 0$ by finitely many smaller real coordinate boxes with positive lower and finite upper unit bounds. Properness implies that a sufficiently small observation neighbourhood is covered: otherwise a sequence with $G \rightarrow 0$ outside the chosen source neighbourhood would have a lifted accumulation point above $G = 0$ outside it. The same argument works above a compact base subset, since the selected label graph, the slack lift and the projective resolved maps are proper there. Alternatively their positive radii and unit bounds can be selected semialgebraically and made continuous by finite refinement. Their extrema on a compact parameter subset give common neighbourhoods and bounds. Fibrewise proper source maps must not be confused with the smaller, nonproper coordinate boxes. This proves all existence and compact-uniform assertions. The effective construction for algebraic input is detailed at the end of Appendix B. $\square$

APPENDIX B. FINITE SUPPORT ORDERS, UNIFORM DECAY, AND EFFECTIVE OUTPUT

The metric and effective parts can be proved without making a parameter-dependent choice of a Puiseux branch. Only the ordinary one-variable algebraic-branch theorem is used to prove that a finite candidate list is exhaustive. Selection from that list is then a first-order decision problem.

Lemma B.1 (A finite support test for the order). *Let $e(\theta, t) \geq 0$ be a semialgebraic function, defined for $0 < t < \tau(\theta)$, where $\tau > 0$ is semialgebraic, and suppose $e(\theta, t) \rightarrow 0$ for fixed θ . A quantifier-free graph formula supplies a finite computable set $\mathcal{B} \subset \mathbb{Q}_{>0}$ such that for every θ either $e(\theta, t)$ is eventually zero or exactly one $b \in \mathcal{B}$ satisfies*

$$\exists c, C, a > 0 \forall t : \quad 0 < t < a \implies ct^b \leq e(\theta, t) \leq Ct^b, \quad a < \tau(\theta). \quad (\text{B.1})$$

Each of these alternatives is an effective semialgebraic condition for algebraic input. No real exponent is quantified over.

Proof. Eliminate quantifiers in the graph formula and let $P_1, \dots, P_N$ be its finite list of polynomials in (θ, t, y) . Fix θ and discard those specializing to the zero polynomial in (t, y) . At a graph point some remaining polynomial must vanish. Otherwise all nonzero polynomial signs are unchanged on a small two-dimensional neighbourhood; the zero specialized polynomials impose constant truth values there. The graph formula would then contain that neighbourhood, contradicting

single-valuedness. For each remaining P_i , the set $\{t : P_i(\theta, t, e(\theta, t)) = 0\}$ is semialgebraic in one variable. These finitely many sets cover a punctured interval. After their finitely many endpoints are removed, at least one vanishing equation holds on an entire interval $(0, a)$. A polynomial independent of y cannot vanish there unless it was one of the discarded zero polynomials.

By one-variable semialgebraic decomposition, on a smaller interval the function is either identically zero or positive, continuous, and a single real algebraic branch of such an equation. The ordinary algebraic Puiseux theorem gives, in the positive case, $e(\theta, t) = a_\theta t^{b_\theta} + o(t^{b_\theta})$ with $a_\theta > 0$ and rational $b_\theta > 0$; see [6, 1]. Write its vanishing equation as $\sum p_{ij}(\theta) t^i y^j = 0$. Inserting the first term shows that $\min(i + b_\theta j)$ is achieved at least twice among nonzero specialized coefficients. Otherwise its unique lowest term could not cancel. Therefore

$$b_\theta = (i - i')/(j' - j) > 0$$

for two distinct support points of one of the original P_i . Take $\mathcal{B}$ to be all such positive ratios. Specialization may remove support points but cannot add a pair to this list. This proves finiteness uniformly in θ , without following algebraic branches through parameter specializations.

Eventual zero is the formula $\exists a > 0 \forall t (0 < t < a \Rightarrow e(\theta, t) = 0)$, with $a < \tau(\theta)$. For each candidate, (B.1) is a first-order polynomial formula: replace $t^{p/q}$ by its unique positive root, or put $t = s^q$ throughout. Quantifier elimination decides it. The actual branch expansion supplies one true candidate; two distinct powers cannot be bounded above and below by positive multiples of each other as $t \downarrow 0$. Exactly one candidate is therefore true in the nonzero case. This also tracks the branch selected by the original Boolean graph formula, including all specialized coefficients and inequality faces, rather than selecting a root of a polynomial without checking graph membership. $\square$

Full proof of Proposition 4.2. Use Lemma B.1. If the candidate set is nonempty, choose a positive rational $\nu < \min \mathcal{B}$; if it is empty use $\nu = 1$. On every parameter fibre $e(\theta, t) \leq t^\nu$ eventually, including the eventual-zero case. The set of pairs (θ, a) with $0 < a \leq \min(1, \tau(\theta))$ and this inequality for all $0 < t < a$ is semialgebraic and has a nonempty positive fibre at every parameter. Semialgebraic choice selects a positive $a = \tau_0(\theta)$, not a possibly vanishing limit value at the boundary of a stratum. A finite Nash stratification makes this selected function continuous. Its minimum on a compact subset of a stratum is strictly positive; the estimate is uniform there. The exponent may be smaller than the sharp order, which is exactly what is needed for the uniform remainder. $\square$

Lemma B.2 (Nash refinement of graph functions). *Finitely many semialgebraic functions on a semialgebraic set admit a common finite Nash stratification on which each function is Nash. For algebraic presentations this refinement is effective.*

Proof. Take their combined graph. Semialgebraic Nash stratification, refined by the ranks of the coordinate projection and by a compatible stratification of its image, is finite; these are the usual graph and rank-stratification constructions in [6, 1]. On a graph stratum the projection is injective. If its constant rank were less than the dimension of that stratum, the constant-rank theorem would give a positive-dimensional local fibre, contradicting injectivity. The projection thus has full rank onto its image stratum. Its local inverse is analytic by the inverse function theorem and is semialgebraic because its graph is semialgebraic; hence it is Nash. Finitely refine images to separate overlapping strata and repeat on their lower-dimensional boundaries. This proves the claimed finite partition and gives the precise reason that graph stratification, rather than topological triviality alone, yields Nash functions. Effective semialgebraic decomposition and rank tests give the algebraic version of the construction. $\square$

The application to Theorem 4.3 is now direct. The polynomial matching formula (4.5), with two quantified inclusions and minimality over the distance, defines the Hausdorff error

as a semialgebraic function of (θ, t) . Theorem 4.1 proves its pointwise convergence to zero. Proposition 4.2 gives a compact-uniform power estimate. Lemma B.2 applied to the diameter and the positive separation and radius functions gives the asserted Nash refinement. Hardt triviality is used separately for the topological type of the leading fibres. The inequality $|\text{diam } A - \text{diam } B| \leq 2d_H(A, B)$ proves the diameter remainder.

B.1. An explicit terminating spectral algorithm. Here is a proof of Theorem 5.1 with its data representations and branch-selection step made explicit. The input is the polynomial formula of Definition 3.1; each algebraic constant has its integer polynomial and rational isolating interval. Arithmetic is in a primitive-element representation of their finite real algebraic field. All quantified formulas below are subjected to real quantifier elimination as in [1]. Finite root permutations are literal finite disjunctions, and complex coordinates are pairs of real coordinates.

First decide exact identification by the formula that all x, x' with $F(x) = F(x') = P$ have equal target coefficients. On this locus construct the cluster graph using its polynomial factorization equations and the specified isolating conditions. For each $H = |c|^2$, form the graph of

$$m_H(s) = \max\{H(x) : x \in \Gamma, G(x) \leq s\}$$

by the conjunction of the universal upper-bound formula and an existential attainment formula. Compactness gives the maximum; compactness and $\{G = 0\} \subset \{H = 0\}$ give $m_H(s) \rightarrow 0$. Apply quantifier elimination to this graph. Construct its finite support-pair list as in Lemma B.1, test eventual zero, and test (B.1) for each rational candidate. The unique true candidate is β_H . This is a finite algorithm for selecting the *actual maximum-function branch*. It needs no unspecified command to isolate an algebraic branch at the boundary. If all maxima are eventually zero the target is locally rigid; otherwise set $\alpha = \min_H \beta_H/k_H$ over the nonzero maxima.

For a family, keep θ free in exactly these tests. Their finitely many truth sets give the parameter pieces and include all zero specializations. Nash-refine them using Lemma B.2. The supports are finite before specialization, so this procedure still terminates without a bound on a real-valued exponent.

Next substitute $\alpha = p/q$ in the fixed-centre formula (4.2). Elimination computes a quantifier-free presentation of $\mathcal{J}_0$ and its projection $\mathcal{C}_0$. Adjoin the labelled roots by elementary symmetric identities to obtain $\mathcal{L}_0$. Use (4.5) to express that all pairs of elements have distance at most C and that some pair attains C . Compactness makes this a formula for one positive real number. Univariate real-root isolation applied to its eliminated formula selects that number over the input field. Taking a field norm gives a nonzero integer polynomial vanishing at C ; renewed root isolation and the same formula select a rational isolating interval for the intended root. The same graph formulas with free parameters compute the family of constants, and the preceding Nash and decay constructions compute a compatible refinement and an admissible rational remainder exponent. No complexity bound is claimed.

B.2. Effective geometric certificates without an implicit oracle. The geometric output can also be given a terminating, though generally inefficient, specification. Over the field just described, enumerate finite algebraic diagrams and finite semialgebraic base partitions in increasing finite encoding length, refining a fixed semialgebraic triviality partition of the enlarged graph with compact fibres. Properness on those base pieces follows from that trivialization and is preserved under further base restriction. A candidate consists of square-slack presentations, reduced closed chains with inverse-map identities, smooth sources with specified projective maps, finite centre ideals and adapted etale charts, integer monomial orders, inverse units and selected real branch formulas. Verify polynomial identities and centre blow-up charts by ideal arithmetic; verify relative smoothness, chart ranks and divisor intersections by their Jacobian criteria. Properness is certified by the specified projective construction and the square-slack argument, not by numerical

bounds on open charts. Verify partition coverage, fibrewise real surjectivity, zero flags and accessibility by real quantifier elimination. These are finite formulas: for example coverage is $\forall \theta \forall x \in \Gamma_\theta \exists y$ in one of the finitely many source charts with the stated projection. Each positive neighbourhood and unit bound can likewise be represented by a semialgebraic choice graph and its universal verification formula.

The existence proof in Appendix A, performed over finite algebraic and rational-function extensions of the input field, produces a certificate with such a finite encoding. An exhaustive search therefore eventually finds a verified certificate. Invalid candidates are rejected by the finite tests. This is an explicit input–output existence algorithm; it is not a claim that the accompanying regression script implements a practical resolution package. The direct maximum-function algorithm above computes the spectral outputs without this enumeration. Thus neither effectivity route depends on an unrepresented branch oracle, and neither is inferred from a finite list of example calculations.

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PROJECTIVE POLYNOMIAL OBSERVATIONS: JOINT SPECTRAL ATLASES AND A WEIGHT-WALL TRANSITION

QIAN QI

ABSTRACT. We study unordered polynomial spectra observed through stochastic channels and an unknown scalar normalization. For compact semialgebraic models, we construct a finite relative real monomial presentation and the joint weighted initial coefficient fibre, with the datum held fixed in the specialization. Its root-multiset image gives the exact leading bottleneck diameter. In fixed-format families, a finite Nash stratification supports constant rational exponents and compact-uniform power remainders. Algebraic inputs admit finite exact descriptions of the leading sets and constants. In the full-rank coprime binary model at fixed degree, these objects have explicit formulas for every multiplicity and boundary pattern. For a rank-deficient cubic family, we classify the entire exact fibre and prove a coefficient inverse uniform through channel-rank changes and identification walls. At a weight wall, a remote feasible corner enters at distance $\mu_E \delta + O(\delta^2)$, while the base local Fisher constant stays positive and finite. The resulting two-scale spectral transition separates loss of identification from deterioration of local conditioning.

1. INTRODUCTION

A polynomial spectrum is an unordered collection, and its singular perturbations are governed by relations among coefficients rather than by independent coefficient bounds. When the observation is additionally normalized and constrained by stochastic inequalities, an invisible ambient direction need not correspond to any admissible alternative. Two questions must therefore be kept separate: the geometry of all nearby real observation fibres and the entrance of an exact component which is remote in coefficient space. We give a joint real specialization for the first question and an explicit stochastic transition for the second.

Let X be a compact semialgebraic model, F its continuous semialgebraic probability map, and A_x a continuous monic target polynomial. For a datum P with a unique exact target, let $c_{r,a}$ be the coefficient of degree a in the monic cluster at a root r of multiplicity m_r . On the real observation graph set $G = h_w(F(x), P)^2$ and $H_{r,a} = |c_{r,a}|^2$. Real monomial orders give the intrinsic nonrigid exponent

$$\alpha_P = \min_{r,a,i} \frac{b_{r,a,i}}{a_i(m_r - a)}.$$

The scalar order calculation is classical. The leading spectral invariant is not that finite list alone. It is the compact set obtained by the *simultaneous* substitutions

$$R = \varepsilon^q U, \quad c_{r,a} = \varepsilon^{p(m_r - a)} V_{r,a}, \quad \|U\| \leq 1, \quad \alpha_P = p/q,$$

and real closure with P fixed. Theorem 4.1 takes its cluster-root image and identifies its exact bottleneck diameter. In particular the construction retains inequality faces and relations among coefficients that separate optimal scalar envelopes cannot recover.

For parameter families, three different ingredients are required. Theorem 3.4 constructs relative real monomial data, including special zero functions, vertical components, and lower-dimensional

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real loci. Lemma 3.3 supplies the finite algebraic spreading certificate on which that construction relies. Theorem 4.1 defines the leading family by an explicit fibrewise quantified formula. Finally Proposition 4.2 derives a uniform positive decay power from finitely many polynomial supports and a continuous threshold selection. Together they yield, on each compact subset H of a finite Nash stratum,

$$\omega_\theta(t) = C(\theta)t^\alpha + O_H(t^{\alpha+\nu}), \quad \alpha, \nu > 0, \quad \alpha \in \mathbb{Q}, \quad C \text{ positive and Nash.}$$

Topological triviality is used for the topology of the fibres; it is not used as a substitute for any of these metric or algebraic arguments. Theorem 5.1 separately describes finite elimination procedures for algebraic inputs.

The normalized experiment in Section 2 has two monic component polynomials with roots in $[0, D]$, positive mixture weights, and binary column-stochastic channels. For fixed degree d , coprime components and strictly positive invertible channels, at least $2d + 1$ exterior clocks give an explicit whole-model atlas for every root multiplicity and endpoint/interior pattern (Theorem 6.2). An interior repeated cluster gives exponent $1/2$ and constant

$$C = \max_j 2\sqrt{(m_j - 1)/m_j} \kappa_j^{-1/4},$$

where κ_j is a specified finite Fisher-cone minimum. With no repeated interior cluster the exponent is one and the leading set is polyhedral with a quadratic observation constraint. These assertions concern all closed-model competitors, not only perturbations in the chosen root-pattern stratum.

Rank deficiency requires a different inverse argument. For $f = z(z - r)^2$, $g = (z - s)^3$, $0 < s < r < D$, positive channels and invertible second channel, define

$$B = (3D - r)s - 2rD, \quad E = (1 - \alpha)(r + 3s)(2r - 3s)^2 - 4\pi r^3.$$

The exact identifying locus is $\{\det U \neq 0\} \cup \{B > 0\} \cup \{E < 0\}$ (Theorem 7.1). Proposition 7.2 and Lemma 7.3 give a quantitative coefficient inverse through the rank locus, without dividing by $\det U$, B or E . The complete local square-root law is then computed by four support tests for each of two constants. It extends through the identification walls as a *local* law.

At the weight wall $E = 0$ with $B > 0$, the additional exact pair is $(f, (z - a)(z - b)^2)$ with $s < a < r < b < D$. Its spectrum is a positive distance D_* from the base spectrum. Proposition 8.1 constructs an exact real corner chart at this remote point. Theorem 8.4 gives its observation entrance distance $d_{\text{rem}}(\delta) = \mu_E \delta + O(\delta^2)$ with $\mu_E > 0$. For $t = \lambda \delta$ the whole-model diameter tends to zero at the base square-root scale when $\lambda < \mu_E$ and tends to D_* when $\lambda > \mu_E$. On the other side the exact remote interval opens with an explicit square-root spectral term. The local Fisher constant remains finite throughout. This is distinct from a multiplicity crossover, where even the local scaling exponent changes.

Classical inputs and the additional invariant. Table 1 separates scalar and topological inputs from the joint metric output. Resolution, principalization and divisor-ratio calculations are not claimed as new constructions. Related algorithmic stratification in families is developed in [7]; our use requires a proper cover of the actual real inequality graph and the finite specialization certificate of Lemma 3.3.

The comparison with Hà uses version 2 of March 14, 2026. In particular, Theorem 9.9 already supplies a non-toric finite-candidate chamber principle; such a principle is not an absent antecedent. Even granting these scalar results, they do not specify the simultaneous real coefficient relations, their root-multiset image with its exact diameter, the normalization and stochastic tangent constraints, or the distance at which a remote exact component enters an observation ball. These are the outputs retained in the present joint atlas and transition calculation.

TABLE 1. Inputs retained in the comparison.

Result	Hypothesis or construction	Output used here
Bierstone–Milman [2, 3, 4]	Real uniformization; characteristic-zero resolution and principalization	Proper smooth sources and monomial coordinate data.
Bivià–Ausina–Encinas [5]	Resolution of the relevant ideal data	Scalar orders from divisorial ratios.
Hardt [9]	A semialgebraic map	Finite topological triviality, not an error rate.
Hà [8, Theorem 5.8]	Finite testing for integral closures and positive asymptotic orders	A finite-max formula for a filtration exponent.
Hà [8, Theorem 9.9]	A common finite candidate set with continuous orders and positive denominators	Finite locally closed strata with constant maximizing set.
Hà [8, Theorem 12.3]	A common toric model, constant Newton data and uniform nondegeneracy	A scalar family wall–chamber description.

Proof organization. Section 2 fixes the experiment and proves normalizer recovery. Sections 3 and 4 prove the relative and joint statements, followed by their effective algebraic interface in Section 5. Section 6 gives the explicit full-rank multiplicity atlas. Sections 7 and 8 treat the rank-crossing inverse and the remote weight-wall transition. The finite exact arithmetic for the displayed examples is ancillary evidence for those examples, not a verification of the universal family theorems.

2. THE EXPERIMENT AND THE SPECTRAL METRIC

Fix $D > 0$, a degree $d \geq 1$, and $0 < \pi < 1/2$. A closed binary model point consists of two monic real polynomials f_1, f_2 of degree d , all of whose roots lie in $[0, D]$, weights $\alpha_1 + \alpha_2 = 1$ with $\alpha_a \geq \pi$, and column-stochastic matrices $U = (u_1, u_2)$ and $V = (v_1, v_2)$ with nonnegative entries. Set

$$K(z) = \sum_{a=1}^2 \alpha_a f_a(z) u_a v_a^T, \quad q(z) = \mathbf{1}^T K(z) \mathbf{1} = \sum_{a=1}^2 \alpha_a f_a(z), \quad P_j = \frac{K(T_j)}{q(T_j)}. \quad (2.1)$$

The clocks satisfy $D < T_1 < \dots < T_\ell$, and $w_j > 0$, $\sum_j w_j = 1$. Consequently $q(T_j) > 0$, including at every closed-model competitor. The target polynomial is $A = f_1 f_2$. The model is compact in monic coefficients and stochastic entries. Component permutations are identified. The matrices, roots, clocks and design weights are allowed to vary when a family of centres is considered, but every competitor uses its centre’s fixed design.

For probability arrays define

$$h_w(P, Q)^2 = \sum_j w_j \sum_e (\sqrt{P_{je}} - \sqrt{Q_{je}})^2.$$

For two multisets $Z = \{z_i\}_{i=1}^n$ and $Z' = \{z'_i\}_{i=1}^n$, put

$$d_\infty(Z, Z') = \min_{\sigma \in \mathfrak{S}_n} \max_i |z_i - z'_{\sigma(i)}|.$$

For real roots this is the maximum discrepancy of their increasing orderings. All diameters in this article are diameters of sets of multisets, not maximal distances from the centre. Write

$$\omega_\theta(t) = \text{diam}\{Z(A_\xi) : h_w(F_T(\xi), P_\theta) \leq t\}. \quad (2.2)$$

A local modulus $\omega_{\theta, \text{loc}}$ restricts the competitors to a specified coefficient neighbourhood of the centre’s component pair. This restriction is never implicit.

At a strictly positive datum the Fisher inner product on zero-sum arrays is

$$\langle H, H' \rangle_{I, \theta} = \sum_{j, e} \frac{w_j}{P_{\theta, je}} H_{je} H'_{je}.$$

For component increments p_a of degree below d and stochastic increments $\eta = (\dot{\alpha}_1, \dot{u}_1, \dot{u}_2, \dot{v}_1, \dot{v}_2)$, where channel increments have entry sum zero, define

$$\begin{aligned} \dot{K} &= \sum_a \{ \alpha_a p_a u_a v_a^\top + f_a (\dot{\alpha}_a u_a v_a^\top + \alpha_a \dot{u}_a v_a^\top + \alpha_a u_a \dot{v}_a^\top) \}, \\ \dot{q} &= \sum_a (\alpha_a p_a + \dot{\alpha}_a f_a), \quad S_\theta(p_1, p_2, \eta)_j = \frac{\dot{K}(T_j) - P_{\theta, j} \dot{q}(T_j)}{q(T_j)}. \end{aligned} \quad (2.3)$$

Here $\dot{\alpha}_2 = -\dot{\alpha}_1$. These rational formulas define every Fisher program below in the original experiment coordinates.

Lemma 2.1 (Normalizer recovery). *Suppose f_1, f_2 are coprime, V is invertible, the weights are positive, and $\ell \geq 2d + 1$. The observed array determines q and K uniquely among all closed-model competitors. Their coefficients are locally Lipschitz functions of the observations, with uniform constants on compact sets of such centres. No rank assumption on U is required.*

Proof. Let e have degree below d , and let H be a matrix polynomial of degree at most d satisfying $H(T_j) = e(T_j)P_j$. Interpolation of the polynomial $qH - eK$, of degree at most $2d$, gives $qH = eK$. Its second marginal, multiplied by $\text{diag}(\alpha_1, \alpha_2)^{-1}V^{-1}$, gives $q \mid ef_1$ and $q \mid ef_2$. Coprimality implies $q \mid e$, hence $e = 0$ and $H = 0$.

More explicitly, let Π_T project onto the orthogonal complement of the degree- d Vandermonde column space. The map

$$N_\theta : e \mapsto \Pi_T(e(T_j)P_{\theta, j})_j, \quad \deg e < d,$$

is injective. For a competitor, subtraction of the normalization equations yields $\|N_\theta(q' - q)\| \leq C \max_j \|P'_j - P_j\|$; the competitor normalizer coefficients are bounded by compactness of the root interval. The smallest singular value of N_θ is positive. A bounded left inverse recovers $q' - q$, and Vandermonde interpolation recovers $K' - K$. The singular values and the coefficient bounds have positive uniform margins on compact sets of the stated centres. $\square$

Lemma 2.2 (Full-rank coefficient inverse). *Under Lemma 2.1, suppose also that U is invertible. Then every sufficiently close observation has, up to one component permutation, coefficient and stochastic parameter error bounded by a constant times its observation error. Every exact competitor has the same component polynomials and stochastic parameters. The constants are uniform on compact subsets of these centres.*

Proof. Choose two clocks such that $f_1(T_j)/f_1(T_i) \neq f_2(T_j)/f_2(T_i)$. Otherwise the nonzero polynomial $f_2(T_i)f_1 - f_1(T_i)f_2$, of degree at most d , would vanish at all $2d + 1$ clocks. The matrix $K(T_j)K(T_i)^{-1}$ has those two distinct eigenvalues and eigenvectors u_1, u_2 . Normalize the eigenvectors to have entry sum one. Simple-eigenvalue inversion is analytic in a neighbourhood of the centre. Applying the recovered U^{-1} to $K(z)$ and taking row sums recovers $\alpha_a f_a$; leading coefficients and normalized rows recover α_a, V . This construction applies to arbitrary closed-model competitors because their $K'(T_i)$ stays invertible. It also proves exact uniqueness. A finite neighbourhood cover gives compact-uniform constants. Equivalently, the derivative in monic coefficient and stochastic coordinates has a bounded left inverse. $\square$

3. REAL MONOMIAL DATA IN PARAMETER FAMILIES

We first work with a compact semialgebraic model X , a continuous semialgebraic probability map F , and a continuous semialgebraic monic polynomial A_x of fixed degree n . Complex

coefficients are represented by their real and imaginary parts. Suppose $A_x = A_0$ on $F^{-1}(P)$, and write $A_0(z) = \prod_r (z - r)^{m_r}$. Disjoint isolating discs and compactness give, on a sufficiently small closed observation neighbourhood, unique monic cluster factors

$$A_{r,x}(r + z) = z^{m_r} + \sum_{a < m_r} c_{r,a}(x) z^a, \quad k_{r,a} = m_r - a. \quad (3.1)$$

Multiplication of coprime monic factors has invertible differential. Consequently these factors are analytic in the total coefficients; their graphs, including the isolating conditions, are semialgebraic.

Definition 3.1 (Input and families). An algebraic input is a finite first-order polynomial presentation of X, F, A, P , the design weights and the isolating data. A real constant is given by an integer polynomial and a rational isolating interval; a complex constant is a pair of such real constants. All inequalities, denominator exclusions and choices of algebraic branches are included. A fixed-format family has a semialgebraic parameter space Θ , a semialgebraic $X \subset \Theta \times K$ with K compact and compact fibres X_θ , continuous semialgebraic maps F, A on X , and a semialgebraic choice $P_\theta \in F_\theta(X_\theta)$. The degrees and numbers of coordinates are fixed. Numerical algebraic output refers to algebraic parameter values. No complexity bound is asserted.

Adjoin nonnegative square-root coordinates and set

$$R_{je} = \sqrt{w_j}(\sqrt{F(x)_{je}} - \sqrt{P_{je}}), \quad G = \sum_{j,e} R_{je}^2, \quad H_{r,a} = (\Re c_{r,a})^2 + (\Im c_{r,a})^2.$$

They are polynomial functions on the resulting compact real graph Γ . This presentation includes zero probability cells.

Definition 3.2 (Real monomial presentation). Such a presentation is a finite collection of smooth real algebraic sources and proper maps, collectively covering a neighbourhood of $G = 0$ in Γ , with finite algebraic etale coordinate presentations. Smaller real coordinate boxes are taken about the inverse image of $G = 0$. On each algebraic source component either $G \equiv 0$, in which case all $H \equiv 0$, or

$$G = u \prod_i |y_i|^{a_i}, \quad H_{r,a} \equiv 0 \quad \text{or} \quad H_{r,a} = v_{r,a} \prod_i |y_i|^{b_{r,a,i}}. \quad (3.2)$$

The units are positive and bounded above and below on the smaller closed boxes. A contributing divisor has $a_i > 0$ and a real point away from the other divisors, with an admissible real transversal on which $G > 0$. An identically zero H has order $+\infty$; an exact component $G \equiv 0$ is recorded separately and never enters an order ratio. Properness is asserted of the source maps, not of an open coordinate box by itself.

3.1. The algebraic spreading interface. We use characteristic-zero resolution and principalization in the forms of [4, Theorems 1.1 and 1.3]. The first resolves a variety over any characteristic-zero field by finitely many smooth-centre blow-ups, isomorphically off its singular locus. The second principalizes a nonzero ideal on a smooth variety, compatibly with a normal-crossing boundary. Applied to a product of finitely many nonzero functions, it makes every factor a monomial times a unit in common coordinates. These are absolute theorems over a field. The following lemma specifies the additional family argument; arbitrary specialization is not inferred from functoriality for smooth maps.

Lemma 3.3 (Spreading a finite resolution certificate). *Let S be an integral affine variety over a characteristic-zero field k , let $K = k(S)$, and let $Y \rightarrow S$ be of finite type. Suppose that over K one has a finite descending chain of reduced closed subvarieties*

$$Y_{0,K} = Y_{K,\text{red}} \supset Y_{1,K} \supset \cdots \supset Y_{q,K} = \emptyset,$$

projective maps $Z_{j,K} \rightarrow Y_{j,K}$ with smooth sources, isomorphisms over $Y_{j,K} \setminus Y_{j+1,K}$, and, on each $Z_{j,K}$, a finite sequence of smooth-centre principalizations of specified functions. There is a nonempty algebraic open subset $S^\circ \subset S$ on which these data extend with the following properties on every fibre:

- (1) the closed chain ends in the empty set and the indicated regular-locus maps remain isomorphisms;
- (2) each $Z_j \rightarrow S^\circ$ and each centre in its subsequent principalizations is smooth, and the principalization blow-ups commute with formation of the fibre;
- (3) the final sources are smooth over S° , all indicated divisor intersections are smooth of their prescribed relative codimensions, and the same monomial identities, integer orders, and invertible units hold on a finite relative etale atlas;
- (4) functions identically zero on a generic source component remain identically zero on that component; a function represented by a unit times a relative normal-crossing monomial cannot become identically zero on a component of a retained fibre.

The exceptional base and all displayed data can be constructed from finite algebraic presentations over an effective field k .

Proof. Replace S by a dense smooth open subset. Every scheme, morphism, closed immersion, inverse morphism and unit in the generic certificate uses finitely many polynomial equations with coefficients in K . Clear their denominators, take closures of the Y_j , and spread the projective maps by their projective equations. Spread the inverse maps on $Y_j \setminus Y_{j+1}$ as well. Their two composition identities, and the equality $Y_q = \emptyset$, hold after one further localization: on each affine chart the relevant equality is an equality of finitely presented ideals or morphisms that holds after tensoring with K . The finitely many denominators witnessing those equalities can be inverted simultaneously. The reduced structure is immaterial to real points; fibrewise reducedness may also be imposed on the generic geometrically reduced pieces after shrinking.

For clarity about blow-up base change, use the actual ideal of each spread centre. On its finitely many affine charts, generic freeness [10, Tag 051S] is applied to the ambient algebra and to the associated graded algebra $\bigoplus_{m \geq 0} I^m / I^{m+1}$, a finite-type algebra. After localization both are flat over the base. The successive exact sequences of the I -adic filtration show that every quotient by I^m is flat. Thus formation of every I^m commutes with passage to a fibre. Formation of the Rees algebra and its projective spectrum therefore commutes with that passage. Repeat this argument for the finitely many blow-ups. This explicitly rules out an unsupported base-change assertion for blowing up a singular family.

The nonsmooth locus of each source or centre, and the loci where a divisor intersection has the wrong rank or codimension, are determined by finitely many Jacobian and nonvanishing conditions on affine charts. For the final smooth sources, choose generic etale coordinates adapted to the normal-crossing divisors. Spread their equations, their etale Jacobians, the identities $f = u \prod y_i^{n_i}$, and the inverse units. The complement of these finitely many charts is also included in the bad locus. Every such bad locus has empty generic fibre. Its image in S is constructible by Chevalley's theorem [10, Tag 054K] and omits the generic point. A constructible subset of an integral noetherian space which is dense contains an open subset, and hence the generic point. The closure of each bad image is therefore proper. Remove the union of these finitely many closures. This argument uses constructibility, not properness of the original map $Y \rightarrow S$.

All the asserted ranks, identities and coverings now hold fibrewise. A generically zero function is retained by its zero identity. A nonzero monomial identity has an invertible unit and divisors of relative codimension one; their union contains no fibre component. Such a function cannot become identically zero on that component. Any parameter where that happens belongs to the removed locus.

Ideal arithmetic, localization, the singular-locus tests, elimination for constructible images, and generic flatness can be performed by finite algebraic calculations. For example, generic Groebner bases with their nonzero denominators give a valid open set; it need not be the largest one. The lemma claims a constructible certificate and an open set, not a canonical minimal exceptional set. $\square$

Theorem 3.4 (Relative real principalization). *On the locus of a fixed-format family where the exact target polynomial is unique, there is a finite semialgebraic stratification with fixed cluster multiplicities, a fixed finite collection of algebraic source chart presentations, and fixed integer orders and real-accessibility flags as in Definition 3.2. The source maps are fibrewise proper real covers. Charts, units and the sizes of coordinate boxes may vary with the parameter. On every compact subset of a stratum the cover is usable over a common sufficiently small observation neighbourhood. The construction is effective for the presentations of Definition 3.1.*

Proof. *The real graph and its algebraic base.* Apply semialgebraic triviality to the graph family and refine the base into Nash strata. Since its fibres are compact, a product trivialization makes the restricted graph map proper: the inverse image of a compact set is homeomorphic to that set times a compact fibre. Partition by cluster multiplicities and select disjoint isolating discs. Their radii and a smaller observation radius can be chosen positive semialgebraically. Refine until these choices and the labelled distinct roots are Nash. The chosen cluster graph over a stratum is again proper after this smaller closed observation restriction.

A Nash stratum is an open semialgebraic subset of the smooth real locus of its Zariski closure after a finite refinement. Use the corresponding integral algebraic component S as base and retain the original real sign conditions throughout. When a root label is algebraic over the parameter functions, adjoin its coordinates and defining equations. After removal of the discriminant locus this is a finite algebraic base cover. Real branch selection on Nash cells selects precisely the intended labelled roots; parameters on its exceptional locus are treated in the induction below. No choice of a complex branch is counted as a real parameter lift.

The graph restricted to this real base is relatively closed. Its ordinary semialgebraic closure in the ambient algebraic space has the same fibres over the stratum. Express that closed set as a finite union of basic closed sets. On each piece replace every inequality $p \geq 0$ by $p = s_p^2$. Projection of real points of this algebraic lift is exactly the basic closed set, not a superset obtained by discarding signs. Over a compact subset of the stratum the original graph is compact; the slack coordinates are bounded by the square roots of maxima of polynomials on that compact graph. The real lift is consequently proper there. All subsequent maps factor through this lift. This is the compatibility interface for the stochastic inequalities and square roots.

Construction over the generic fibre. For each lifted algebraic family over S , remove components not dominating S temporarily and pass to $K = k(S)$. Work with its reduced generic fibre. Resolve that fibre, isomorphically over its regular locus. Resolve its reduced singular locus separately, then its singular locus, and continue. Dimension strictly decreases, so this gives a finite closed chain ending in the empty set, and maps of precisely the type in Lemma 3.3. On each smooth source separate the functions that are identically zero and principalize the product of all the remaining G, H . A product with normal crossings gives simultaneous monomial identities for every factor. The absolute constructions over K are finite, so Lemma 3.3 applies.

Coverage of every retained real fibre. Let y be a real point of a lifted fibre over S° . Choose the unique first member of the closed chain for which $y \in Y_j \setminus Y_{j+1}$. The spread isomorphism gives a real point of the smooth fibre Z_j above y . Each subsequent principalization blows up a smooth real centre in a smooth real source. Above a real centre point its fibre is a real projective space, which is nonempty; elsewhere it is an isomorphism. These blow-ups do not lose real points. Their composition, and the finite disjoint union over j , is proper onto the real lifted fibre. A real

singular point with no real generic branch is covered by a later member of the chain. It is not silently lost in a resolution of the top-dimensional component.

Special and vertical components. The closures of the images of nondominating original components are proper algebraic subsets of S . Add them to the exceptional set of Lemma 3.3. On the retained open set the spread chain and its isomorphisms cover every fibre point, including any component of a fibre: this is stronger than merely resolving a generic component. Parameters at which a function becomes identically zero, a chart ceases to cover, or an additional vertical component occurs are either already covered by those fibrewise certificates or belong to the exceptional set. Recompute the lift and the entire generic construction on each irreducible component of that exceptional set, with its reduced base. Its dimension is smaller. There are finitely many components at each stage and at most $\dim S + 1$ stages along a branch. This proves finite termination and a finite total collection of source charts and orders. One does not reuse generic nonzero flags on a special fibre.

Descent, accessibility and neighbourhood sizes. Restrict the finite algebraic label covers to the selected real Nash branches. On each such branch the fibre is the original labelled graph; projecting back changes neither an integer order nor a real transversal. If several labels are retained, take their finite union and the finite permutation quotient. Real-empty source components are discarded. On a smooth component with real points, the exact-fibre condition forces every H to vanish identically whenever G does, since its real regular locus is Zariski dense. The existence of a generic real point of a proposed divisor, away from all other divisors and in the chosen coordinate chart, is a first-order condition in the parameter and source coordinates. Partition the base by these finitely many conditions. At such a point an inverse étale chart gives a real transversal $y_i = t$ with other coordinates fixed. All its points still project through the square-slack lift, so it is admissible. A complex divisor with no such real point is not recorded.

For a fixed parameter, compactness of the inverse image of $G = 0$ gives finitely many smaller coordinate boxes on which the units have positive bounds. Properness then gives $G \leq \varepsilon_0$ inside their images: a contrary sequence with $G \rightarrow 0$ would have a lifted accumulation point above $G = 0$ outside the chosen neighbourhood. Over a compact parameter set the same argument applies to the proper family of source maps. Equivalently, positive radii and unit bounds may be selected semialgebraically and made continuous by refinement. Their minima and maxima on that compact set provide the required uniform bounds.

Absolute constructive resolution, the finite spreading certificate, induction on base dimension, and real quantifier elimination provide all these representations. Coordinate-box selection can alternatively be implemented by enumeration of algebraic neighbourhoods and verification of covering and unit bounds by real elimination; existence and compactness prove termination. This completes the construction rather than assuming a simultaneous resolution on an unstratified base. $\square$

Proposition 3.5 (Fixed-centre real cover). *Every graph Γ above has a real monomial presentation. Exact pieces and lower-dimensional real loci are included. Every recorded divisor has a real admissible attaining arc.*

Proof. Apply Theorem 3.4 to a one-point base. The real-coverage and transversal parts of its proof give the stated conclusions. $\square$

Corollary 3.6 (Intrinsic scalar orders). *For a nonrigid cluster coefficient put*

$$\beta_{r,a} = \min_{\text{recorded real divisors}} \frac{b_{r,a,i}}{a_i}, \quad \alpha = \min_{r,a} \frac{\beta_{r,a}}{k_{r,a}}. \quad (3.3)$$

Either the target is constant on an observation neighbourhood, or $\alpha > 0$ is rational and $\omega_P(t) \asymp t^\alpha$. Every finite coefficient minimum is attained by an admissible arc. The numbers are independent of the monomial presentation.

Proof. On each nonexact box, $b_i \geq \beta a_i$ for $a_i > 0$ and $b_i \geq 0$ otherwise. Unit bounds give $H \leq CG^\beta$, hence $|c_{r,a}| \leq Ch_w^\beta$. A minimizing real transversal has $G \asymp u^{a_i}$ and $H \asymp u^{b_i}$ and attains this order. The inclusion $\{G = 0\} \subseteq \{H = 0\}$ forces positive relevant orders. If no nonzero H occurs near $G = 0$, the target is locally rigid. For a monic polynomial of degree m , its largest root modulus is comparable to $\max_{a < m} |c_a|^{1/(m-a)}$: elementary symmetric identities give one inequality and domination of the lower terms by z^m outside a fixed multiple of this maximum gives the other. The centre lies in every ball, so diameter is between maximal centre displacement and twice that displacement. Finally the optimal exponent in the real coefficient inequality is intrinsic, proving independence. The divisor-ratio calculation itself is classical; compare [5]. $\square$

4. JOINT SPECIALIZATION AND UNIFORM SPECTRAL ASYMPTOTICS

4.1. Closure with the parameter fixed. On a nonrigid identified stratum write $\alpha = p/q$ in lowest terms. Let $\Psi(\theta, \varepsilon, x, U, V)$ denote graph membership and

$$R_{je}(\theta, x) = \varepsilon^q U_{je}, \quad c_{r,a}(\theta, x) = \varepsilon^{pk_{r,a}} V_{r,a}, \quad \sum_{j,e} U_{je}^2 \leq 1. \quad (4.1)$$

Complex equalities stand for two real equalities. Define the family by

$$(\theta, U, V) \in \mathcal{J} \iff \theta \in S \text{ and } \forall \varepsilon, x, U', V' : \quad 0 < \varepsilon < \rho, \quad \Psi(\theta, \varepsilon, x, U', V'), \quad \|(U', V') - (U, V)\|^2 < \rho^2. \quad (4.2)$$

Write $\mathcal{J}_0(\theta) = \mathcal{J}_\theta$ and $\mathcal{C}_0(\theta) = \text{pr}_V \mathcal{J}_0(\theta)$. There is no quantified θ' in this formula. Taking closure of the union while allowing the centre to drift would be a different operation.

Theorem 4.1 (The definable joint initial family). *The sets $\mathcal{J}_0(\theta)$ and $\mathcal{C}_0(\theta)$ form semialgebraic families of nonempty compact sets. The coefficient images of the radius- t observation ball, scaled by $c_{r,a}/t^{\alpha k_{r,a}}$, converge in Hausdorff distance to $\mathcal{C}_0(\theta)$ for every fixed θ . If*

$$Z_r(V) = Z \left(z^{m_r} + \sum_{a < m_r} V_{r,a} z^a \right), \quad \mathcal{L}_0(\theta) = \{(Z_r(V))_r : V \in \mathcal{C}_0(\theta)\},$$

then the correspondingly scaled cluster-root images converge to $\mathcal{L}_0(\theta)$. Moreover

$$\omega_\theta(t) = C(\theta)t^\alpha + o(t^\alpha), \quad C(\theta) = \max_{V, V' \in \mathcal{C}_0(\theta)} \min_r d_\infty(Z_r(V), Z_r(V')) \in (0, \infty). \quad (4.3)$$

This is a joint feasible image, not a Cartesian product of separate coefficient envelopes.

Proof. Corollary 3.6 bounds all scaled coefficient coordinates for sufficiently small t ; U is already bounded. The defining closure is therefore compact and nonempty. First-order elimination gives its semialgebraicity with θ as a free parameter. Bounded graph coordinates can be eliminated after closure: every convergent sequence of scaled coordinates has a subsequence of graph coordinates converging in the compact fibre. Thus projection does not lose an initial point.

Fix θ and a point V of the outer limit. The distance from V to the positive-radius coefficient section is a bounded nonnegative semialgebraic function of t . It has a subsequence tending to zero. One-variable monotonicity gives limit zero. A finite net of the compact outer limit gives the inner Hausdorff inclusion. If the outer Hausdorff inclusion failed, bounded scaled coordinates would give an accumulation point outside the defining outer limit, a contradiction. This proves convergence for the entire ball, not just convergence along chosen arcs. Continuity of monic root multisets on bounded coefficient sets gives the asserted root-set limit.

Let Δ be the least distance between distinct base roots. For small t all roots in a cluster lie in its $\Delta/4$ disc. A matching within clusters has error tending to zero; a matching across clusters has an edge of length at least $\Delta/2$. An optimal matching therefore stays within clusters. Translation and scaling within each cluster are exact. Taking diameter in the maximum cluster metric proves (4.3). A minimizing coefficient transversal in Corollary 3.6 prevents the limiting diameter from being zero. $\square$

4.2. An explicit uniform asymptotics lemma.

Proposition 4.2 (Semialgebraic decay on strata). *Let $e(\theta, t) \geq 0$ be semialgebraic for $\theta \in S$ and $0 < t < \tau(\theta)$, where τ is positive and semialgebraic. Suppose $e(\theta, t) \rightarrow 0$ as $t \downarrow 0$ for each fixed θ . There is a finite Nash stratification of S such that, on every stratum, some rational $\nu > 0$ and positive continuous semialgebraic $\tau_0(\theta)$ satisfy*

$$0 < t < \tau_0(\theta) \implies e(\theta, t) \leq t^\nu.$$

In particular the estimate is uniform on compact subsets of the stratum. The exponent need not be the sharp decay exponent.

Proof. Eliminate quantifiers in a formula for the graph of e . It uses a finite list of polynomials in (θ, t, y) . For fixed θ , discard the polynomials that specialize identically to zero. A one-dimensional single-valued graph in the (t, y) plane must, near each of its points, lie on a zero of one of the remaining polynomials: otherwise all signs are locally constant in two dimensions and the formula cannot define a single-valued graph. Near $t = 0$ finitely many semialgebraic pieces suffice. The eventual-zero case is separated by the formula $\exists a > 0 \forall t (0 < t < a \implies e(\theta, t) = 0)$.

In the other case, one-variable algebraic branch expansion gives $e(\theta, t) = a_\theta t^{r_\theta} + o(t^{r_\theta})$ with $a_\theta > 0$ and $r_\theta > 0$ rational. If a polynomial $\sum p_{ij}(\theta) t^i y^j$ vanishes on that branch, the smallest number $i + r_\theta j$ occurs at least twice. Hence r_θ is among the positive ratios $(i - i')/(j' - j)$ formed from two distinct monomials of the fixed finite polynomial list. Specialization may remove monomials but cannot create a new support. There are consequently only finitely many candidate positive orders, uniformly over all parameters. This argument supplies the needed uniform lower bound on orders; pointwise convergence alone would not supply it.

Choose rational ν smaller than the least candidate order; use $\nu = 1$ on an eventual-zero piece if there is no candidate. Then for each parameter $e(\theta, t) \leq t^\nu$ eventually. Since ν is rational, this eventual inequality, with positive-power auxiliary variables, is a first-order real formula. Semialgebraic choice supplies a positive $\tau_0 \leq \min(\tau, 1)$ for which it holds. A finite stratification makes τ_0 continuous and the strata Nash. Its minimum on a compact subset is positive, which proves the uniform assertion. $\square$

Theorem 4.3 (Uniform spectral power law). *The identified nonrigid parameter locus has a finite Nash stratification on which α is fixed, $\mathcal{L}_0$ has constant semialgebraic topological type, and $C(\theta)$ is positive and Nash. On each compact H of a stratum, for some $\nu > 0$,*

$$d_H(\mathcal{L}_t(\theta), \mathcal{L}_0(\theta)) = O_H(t^\nu), \quad \omega_\theta(t) = C(\theta)t^\alpha + O_H(t^{\alpha+\nu}). \quad (4.4)$$

Here $\mathcal{L}_t$ is the exact clustered root image scaled by $t^{-\alpha}$, and d_H is Hausdorff distance for the maximum of the cluster bottleneck metrics.

Proof. We specify the set-valued distance used in applying Proposition 4.2. Represent each cluster by all its labelled root arrays through the elementary symmetric identities. For two arrays,

$$M_d(z, z') = \bigvee_{\sigma \in \mathfrak{S}_m} \bigwedge_{i=1}^m |z_i - z'_{\sigma(i)}|^2 \leq d^2 \quad (4.5)$$

is a polynomial formula for bottleneck distance at most d . For complex roots use real and imaginary parts. Taking the conjunction over clusters represents the maximum metric. An upper bound d on Hausdorff distance is the conjunction of

$$(\forall z \in \mathcal{L}_t)(\exists z' \in \mathcal{L}_0)M_d(z, z'), \quad (\forall z' \in \mathcal{L}_0)(\exists z \in \mathcal{L}_t)M_d(z, z').$$

Together with minimality over $d \geq 0$, this is a first-order formula for a single nonnegative number. Compactness ensures the minimum exists. Thus the Hausdorff error is semialgebraic in (θ, t) . Fractional powers of t in $\mathcal{L}_t$ are encoded by their unique positive algebraic roots. Theorem 4.1 says that this error tends to zero pointwise. Proposition 4.2 gives its compact-uniform power bound after refinement.

The family of leading coefficient sets and their root arrays is semialgebraic, so Hardt triviality separately supplies constant topological types. The diameter is specified by all pairs having distance at most C and some pair attaining C . Hence C is semialgebraic and positive. A semialgebraic function is Nash on a finite Nash stratification: stratify its graph so that its projection has constant full rank, use the analytic inverse function theorem, and repeat on the lower-dimensional exceptional sets. Refine by this stratification and by the continuous positive cluster-separation and observation-radius functions. On compact subsets their positive minima exclude cross-cluster matchings uniformly. Finally diameters of compact sets differ by at most twice their Hausdorff distance. This gives the second assertion of (4.4). Neither a rate nor Nash dependence has been inferred from topological triviality alone. $\square$

Theorem 4.4 (A finite real spectral atlas). *A fixed-format family has a finite semialgebraic stratification retaining these four distinct kinds of data:*

- (1) *the semialgebraic homeomorphism type and dimension of the exact target coefficient fibre, and of its root-multiset image;*
- (2) *on the identified locus, rigidity or the positive rational order (3.3), the cluster multiplicities, and a finite real monomial order list with fixed accessibility and minimizing indices;*
- (3) *on the nonrigid identified locus, the joint initial coefficient and root-set types and the Nash leading diameter with the compact-uniform remainder (4.4);*
- (4) *finite algebraic descriptions of these objects. On a subfamily where a finite Fisher-cone description is proved, its active supports and leading-constant choices may be retained in the same refinement.*

On a nonidentified stratum, the modulus tends to the positive diameter of the exact target root image. No general leading set is asserted to be quadratic, and no local leading constant is asserted to exclude a remote exact component.

Proof. The exact target coefficient fibre is the compact image $\{A_x : F_\theta(x) = P_\theta\}$, not the parameter fibre itself. Apply Hardt triviality to this family and its root graph. Singleton fibres are selected by a first-order formula. At fixed θ , decreasing compact observation balls converge in target image to the exact image, by subsequence compactness; their diameters therefore converge to its diameter. Apply Theorem 3.4 on the singleton locus, Corollary 3.6 for the orders, and Theorems 4.1 and 4.3 for the joint data and uniform remainder. These are separate inputs to a common finite refinement. On a proved Fisher subfamily there are only finitely many support and comparison conditions, which can be added to that refinement. $\square$

5. EFFECTIVE OUTPUT AND THE ROLE OF COMPUTATION

Theorem 5.1 (Effective algebraic specialization). *For an algebraic input, finite algorithms compute the rigidity decision, the exponent α in the nonrigid case, quantifier-free presentations of $\mathcal{J}_0, \mathcal{C}_0, \mathcal{L}_0$, and an integer polynomial with a rational isolating interval for C . In parameter families the corresponding strata and formulas are effectively describable. This assertion is separate*

from the existence of the real cover and from any executable finite regression in the accompanying material.

Proof. There are two finite routes to the exponent. The geometric route uses the algebraic interface of Lemma 3.3, constructive principalization, recursion on base dimension, and real accessibility elimination, followed by the rational minima (3.3). The following second route avoids treating the implementation of a resolution algorithm as a prerequisite for numerical spectral output.

For each squared coefficient H construct, by elimination, the graph of the compact maximum

$$m_H(s) = \max\{H(x) : x \in \Gamma, G(x) \leq s\}, \quad s \geq 0.$$

Its graph is the explicit formula

$$s \geq 0, y \geq 0, \quad (\forall x \in \Gamma)(G(x) \leq s \Rightarrow H(x) \leq y), \quad (\exists x \in \Gamma)(G(x) \leq s \wedge H(x) = y).$$

The maximum is an output of quantifier elimination, not an oracle or an input assumption. Detect eventual vanishing by a quantified formula. Otherwise isolate its positive real algebraic branch at $s = 0$ and use Newton polygon and real branch selection to obtain its first Puiseux order β_H . Since $G = h_w^2$ and $H = |c|^2$, this is exactly the optimal coefficient exponent β in Corollary 3.6. Take the finite minimum β_H/k_H . In a family, the finite candidate orders come from the polynomial supports as in Proposition 4.2. For each rational candidate $b > 0$, the formula

$$\exists c, C, \tau > 0 \forall s \in (0, \tau) : cs^b \leq m_H(s) \leq Cs^b$$

selects precisely the parameters with that order. Rational powers are encoded by their positive roots. This gives an explicit elimination test, including zero specializations, without quantifying over a real exponent. All operations are over the encoded real algebraic field.

Substitute the resulting rational p/q into (4.2). Real quantifier elimination returns $\mathcal{J}_0$ and its coefficient projection. Introduce the labelled roots by their symmetric identities and use (4.5). The assertion that all pairs have distance at most C and that a pair attains C defines a unique real number. Elimination and univariate real-root isolation give its polynomial over the algebraic input field. A primitive-element representation, field norm and renewed root isolation give an integer polynomial and rational isolating interval selecting the same number. These changes of representation use the arithmetic, elimination and real-root algorithms of [1, 6].

For completeness, the geometric route passes finite equations to reduced decomposition, to characteristic-zero principalization, to the finite spreading charts, and then to first-order real tests. Algebraic factorization and ideal membership determine the identically zero and vertical flags. Positive coordinate neighbourhoods, covering assertions, and unit bounds are first-order conditions after algebraic chart coordinates are adjoined. Finite branch selection and projection give the base refinement. Thus this route, too, has specified input and output representations. The semialgebraic-max route shows in addition that the spectral-output algorithm does not depend on an undocumented software implementation of this geometric route. $\square$

The finite symbolic calculations for the polynomial examples below test identities and specified quadratic programs. They do not execute the universal construction in Theorem 5.1, and they are not proof verification for Theorems 3.4 or 4.3.

6. AN EXPLICIT MULTIPLICITY AND BOUNDARY ATLAS

Fix d, D, π and $\ell \geq 2d + 1$. Let $\mathfrak{M}_d$ be the centres of (2.1) with strict weights, strictly positive invertible U, V , and coprime component polynomials. No restriction is imposed on root multiplicities or on contact with $0, D$. Competitors still range over the entire closed model. Partition $\mathfrak{M}_d$ by the component assignment, multiplicity and endpoint/interior status of each

distinct root. At fixed d this is a finite semialgebraic partition. On a pattern stratum write r_j, m_j for its distinct roots and multiplicities, and $a(j) \in \{1, 2\}$ for their component assignments.

For every cluster let s_j denote its first-order sum displacement. It is free at an interior root, nonnegative at 0, and nonpositive at D . For every repeated interior cluster let $X_j \geq 0$. Define

$$p_a(z) = - \sum_{j:a(j)=a} s_j \frac{f_a(z)}{z - r_j} - \sum_{\substack{j:a(j)=a \\ r_j \in (0,D), m_j \geq 2}} X_j \frac{f_a(z)}{(z - r_j)^2}. \quad (6.1)$$

Insert these polynomials into (2.3). Let L contain the free interior sum and stochastic columns. Let D_0 contain the nonnegative endpoint sum columns (reverse the sign at D) and all X_j columns. With $W = \text{diag}(w_j/P_{je})$, define the projected Gram matrix

$$Q = D_0^\top W D_0 - D_0^\top W L (L^\top W L)^{-1} L^\top W D_0. \quad (6.2)$$

For a repeated interior cluster j , let

$$\kappa_j = \min\{v^\top Q v : v \geq 0, v_{X_j} = 1\}. \quad (6.3)$$

All matrix entries are rational in the root, channel, weight and design coordinates. Empty collections of free or cone columns have their usual zero-dimensional meanings.

Lemma 6.1 (Ordered cluster diameter). *For $m \geq 2$, the set*

$$\{y_1 \leq \dots \leq y_m : \sum_i y_i = 0, \frac{1}{2} \sum_i y_i^2 \leq X\}$$

has diameter $\sqrt{2X(m-1)/m}$ in the maximum coordinate metric.

Proof. Fix a coordinate j . Minimizing the squared norm under its value, ordering and zero sum gives

$$\max y_j = \sqrt{\frac{2X(j-1)}{m(m-j+1)}}, \quad \min y_j = -\sqrt{\frac{2X(m-j)}{mj}}.$$

For the maximum, the last $m-j+1$ coordinates are equal and the first $j-1$ are equal; the minimum is analogous. This also covers the zero endpoint extrema when $j=1$ or $j=m$. Put $a=j-1, b=m-j$. For $a, b \geq 1$, the assertion that the resulting coordinate width is at most $\sqrt{2X(m-1)/m}$ is equivalent to

$$ab(a+b+2)^2 \geq 4(a+1)(b+1).$$

Writing $S = a+b \geq 2$ and $ab \geq 1$, its difference is $ab(S^2 + 4S) - 4S - 4 \geq S^2 - 4 \geq 0$. For $a=0$ or $b=0$ equality holds directly. An extreme vector with one coordinate of absolute value $\sqrt{2X(m-1)/m}$ and the other $m-1$ coordinates equal attains the endpoint width against the zero vector. The diameter of a subset in the maximum metric is its largest coordinate width. This proves the formula, including possible simultaneous constraints on other clusters when it is used below. $\square$

Theorem 6.2 (Multiplicity and boundary laws). *Every centre in $\mathfrak{M}_d$ is exactly identifying. On each root pattern stratum the following whole-model assertions hold uniformly on compact subsets H .*

(a) *If there is a repeated interior cluster, the exponent is $1/2$. The complete leading root set consists of zero clusters at the simple and boundary roots and ordered vectors $y_j \in \mathbb{R}^{m_j}$ at the repeated interior roots satisfying*

$$\sum_i y_{j,i} = 0, \quad X_j = \frac{1}{2} \sum_i y_{j,i}^2, \quad \min_{\text{endpoint sums} \geq 0} v^\top Q v \leq 4. \quad (6.4)$$

The X_j in v are the displayed variances. After grouping and scaling roots by $\sqrt{t}$, the observation ball is within $O_H(\sqrt{t})$ of this set in Hausdorff distance. In particular

$$\omega_\theta(t) = C(\theta)\sqrt{t} + O_H(t), \quad C(\theta) = \max_j 2\sqrt{\frac{m_j - 1}{m_j}} \kappa_j^{-1/4}. \quad (6.5)$$

Every leading point, including diameter-extremizing pairs, is realized by stochastic alternatives.

(b) If no interior root is repeated, the exponent is 1. For each simple interior cluster take a free real root increment, and for each boundary cluster take an ordered vector of increments with all entries nonnegative at 0 or nonpositive at D . Let s_j be their sums and use (6.1) without X terms. The compact set of these root increments with

$$\min_\eta \|S_\theta(p_1, p_2, \eta)\|_{L, \theta}^2 \leq 4 \quad (6.6)$$

is the complete leading set $\mathcal{L}_{\text{lin}, \theta}$. Its diameter $C_{\text{lin}}(\theta)$ is positive, and

$$\omega_\theta(t) = C_{\text{lin}}(\theta)t + O_H(t^2).$$

The scaled-root Hausdorff error is $O_H(t)$.

(c) The optimal coefficient orders of a cluster of multiplicity m , with coefficient weight k , are

$$\beta_k = \begin{cases} 1, & r \in (0, D), \ k = 1, \\ k/2, & r \in (0, D), \ 2 \leq k \leq m, \\ k, & r \in \{0, D\}, \ 1 \leq k \leq m. \end{cases} \quad (6.7)$$

The real order list, boundary and multiplicity pattern, and Fisher active supports admit a common finite semialgebraic refinement. On its square-root strata C^4 is a selected rational function of the root and experiment coordinates; the positive constants on the linear strata are algebraic semialgebraic functions. Both are Nash after refinement. Exact algebraic evaluation uses finite programs.

Proof. Lemma 2.2 gives exact identification and a uniform inverse in coefficients and stochastic parameters. The partial fractions $1/(z - r_j)$ and $1/(z - r_j)^2$ occurring in (6.1) are linearly independent at distinct roots, with their indicated pole orders. The coefficient derivative, and hence its score, is injective on these variables and the free stochastic coordinates. Thus L has full column rank and Q is positive definite whenever it is nonempty. The constrained minima exist, are positive, and have uniformly bounded minimizers on H .

We first obtain coordinates for *all* nearby competitors. Coefficient recovery and analytic coprime cluster factorization give an $O_H(t)$ error in each cluster coefficient and in the stochastic parameters. For an interior cluster, write its exact roots as

$$r_j + ts_j/m_j + \sqrt{t}y_{j,i}, \quad \sum_i y_{j,i} = 0.$$

The first coefficient bounds s_j , and the second bounds $\sum_i y_{j,i}^2$, since for centred real roots their second elementary symmetric function equals $-\frac{1}{2}\sum_i y_{j,i}^2$. At an endpoint all root displacements have the same sign. Their sum is $O_H(t)$, so each is $O_H(t)$. These observations prove the upper bounds in (6.7). For the lower bounds, translate an interior cluster by a nonzero order- t amount for $k = 1$; for $k \geq 2$ split it by $\sqrt{t}$ times a zero-sum real vector with nonzero k th elementary symmetric function. The vector with $m - 1$ equal nonzero coordinates and the last their negative sum has this property for every $k \geq 2$. Its first nonzero coefficient is of order t and the coefficient inverse makes the observation distance comparable to t . At an endpoint displace all roots by the same allowed order- t amount. This proves all lower orders, including the one-sided boundary case. In root-arc parameter ε , the interior split has $\text{ord } G = 4$, $\text{ord } H_k = 2k$, whereas translation or an

endpoint arc has orders $2, 2k$. These are admissible witnesses for the real order data, and (6.7) rules out any smaller candidate.

For part (a), expansion of the exact root coordinates gives precisely (6.1) at order t . Terms involving three centred root displacements are $O_H(t^{3/2})$; all higher ones and the bounded mean and endpoint terms satisfy the same bound. Consequently

$$F_T(\xi) = P_\theta + tS_\theta + O_H(t^{3/2}), \quad h_w(F_T(\xi), P_\theta) = \frac{t}{2}\|S_\theta\|_{L,\theta} + O_H(t^{3/2}). \quad (6.8)$$

This follows by Taylor expansion of the square-root probability vector and the reverse triangle inequality; it does not require a nonzero score. Membership in the ball implies $\|S_\theta\| \leq 2 + M\sqrt{t}$. Multiply all sum, variance and stochastic first-order variables by $\rho = 2/(2 + M\sqrt{t})$ and each centred root vector by $\sqrt{\rho}$. This gives a point of (6.4) within $O_H(\sqrt{t})$ of the scaled competitor. Endpoint and simple clusters contribute only $O_H(\sqrt{t})$ in these scaled coordinates.

Conversely take a point of (6.4) and bounded minimizing endpoint and free variables. Use the displayed interior root arcs; at an endpoint realize its sum by equal one-sided displacements, and use the prescribed order- t stochastic increments. Scale first-order variables by $\rho = 1 - M'\sqrt{t}$ and centred roots by $\sqrt{\rho}$. Strict weight and channel margins and cluster separation on H make these genuine closed-model alternatives. For a sufficiently large M' , (6.8) puts them inside the observation ball. This is the reverse Hausdorff inclusion, with the same uniform error.

Homogeneity gives $X_{j,\max} = 2/\sqrt{\kappa_j}$. The marginal projection of the leading set at cluster j includes every ordered zero-sum vector with squared norm at most $2X_{j,\max}$: scale a minimizing feasible vector for $X_j = 1$ and choose the desired shape in that cluster. Lemma 6.1 therefore gives its diameter $2\sqrt{(m_j - 1)/m_j} \kappa_j^{-1/4}$. The diameter in the maximum cluster metric is the largest marginal diameter; no independence of the clusters is used. Cluster separation excludes cross-cluster optimal matchings. Diameters differ by at most twice Hausdorff distance, proving (6.5).

For part (b), every root displacement is $O_H(t)$ by the same sum argument at the endpoints and the simple-root inverse in the interior. Divide the exact displacements by t . They belong to the stated polyhedral sign and ordering cone. Taylor expansion now has remainder $O_H(t^2)$. The first-order score controls their sums and all stochastic variables; on the one-sided endpoint cones the sums control the individual displacements. Hence the root set in (6.6) is compact. The radial argument with $\rho = 1 - Mt$ proves both Hausdorff inclusions with error $O_H(t)$. An allowed nonzero displacement gives positive diameter. This proves (b).

For (c), enumerate the subsets of the cone coordinates other than a fixed X_j . On a support A , put $v_A = -Q_{AA}^{-1}Q_{Aj}$ and require $v_A \geq 0$ and $Q_{bj} + Q_{bA}v_A \geq 0$ off the support. Its value is $Q_{jj} - Q_{jA}Q_{AA}^{-1}Q_{Aj}$. Convexity proves necessity and sufficiency. All tests and selected values are rational, and

$$C^4 = \max_j \frac{16(m_j - 1)^2}{m_j^2 \kappa_j}.$$

There are finitely many patterns and supports. For the linear law, (6.6) is an explicit quadratic inequality on a polyhedral root cone after projecting the free stochastic variables. Its coordinate maxima and minima, and hence its diameter, have finite real algebraic descriptions by (4.5) or by face optimization. Quantifier elimination gives the algebraic value at algebraic input. Stratify the resulting semialgebraic functions and refine with the order lists of Theorem 4.4. Positivity and compact-uniform coercivity ensure continuity within each fixed pattern; Nash refinement gives the final assertion. No continuity of C is asserted across a change of scaling exponent. $\square$

Example 6.3 (A multiplicity crossover). Let $x \in [0, b]$, $b < 1/2$, take $F(x) = (1/2 + x, 1/2 - x)$, and target $A_x(z) = (z - r)^2 - x$. At a fixed centre $a \in (0, b)$, differentiation of the square-root

probability vector gives

$$h(F(a+u), F(a)) = \frac{|u|}{\sqrt{1-4a^2}} + O_a(u^2).$$

The admissible x interval therefore has endpoints $a \pm t\sqrt{1-4a^2} + O_a(t^2)$, and its spectral diameter is

$$\omega_a(t) = \frac{\sqrt{1-4a^2}}{\sqrt{a}} t + O_a(t^2).$$

At $a = 0$ it instead has leading law $\sqrt{t}$. More precisely, uniformly for λ in compact subsets of $[0, \infty)$,

$$\frac{\omega_{\lambda t}(t)}{\sqrt{t}} \rightarrow \sqrt{\lambda+1} - \sqrt{\max(0, \lambda-1)}. \quad (6.9)$$

Indeed the feasible interval endpoints are $t \max(0, \lambda-1) + O(t^2)$ and $t(\lambda+1) + O(t^2)$, with truncation at zero, and $x \mapsto Z(A_x)$ has ordered roots $r \pm \sqrt{x}$. This example belongs to the general real-graph class of Section 3; it is not a claim about an unobserved restriction of the closed binary model. Here the linear constant really diverges at a multiplicity wall. Section 8 gives a different mechanism.

7. A RANK-DEFICIENT CUBIC STRATUM

Let $\mathfrak{F}$ denote the degree-three centres with

$$f = z(z-r)^2, \quad g = (z-s)^3, \quad 0 < s < r < D,$$

strict weights $\pi < \alpha < 1 - \pi$, strictly positive U, V , $\det V \neq 0$, and at least seven clocks. The first channel may have either rank. Put $\gamma = 1 - \alpha$ and

$$A = (r+3s)(2r-3s)^2, \quad B = (3D-r)s - 2rD, \quad E = \gamma A - 4\pi r^3. \quad (7.1)$$

Here A is a scalar, not the target polynomial.

Theorem 7.1 (The entire cubic fibre). *The exact spectral fibre is a singleton precisely on*

$$\mathfrak{I} = \{\det U \neq 0\} \cup \{B < 0\} \cup \{E < 0\}. \quad (7.2)$$

On its complement, besides (f, g) all component-polynomial pairs are, up to permutation,

$$(f, h_c), \quad h_c = (1-c)f + cg, \quad c_* \leq c \leq \min(c_D, \gamma/\pi), \quad (7.3)$$

where

$$c_* = \frac{4r^3}{A}, \quad b = \frac{rs}{3s-2r}, \quad R(z) = \frac{f(z)}{g(z)}, \quad c_D = \frac{R(D)}{R(D)-1}.$$

The interval is nonempty even at equality in either constraint, and every one of its pairs has a positive stochastic realization. Conditional on the component pair, that realization is unique.

Proof. Lemma 2.1 fixes q, K for every competitor. If U is invertible, Lemma 2.2 applies; in particular $\det K / \det K_3 = fg$ gives spectral identification.

If $\det U = 0$, stochasticity gives $u_1 = u_2 = u$. The second marginal $V \operatorname{diag}(\alpha, \gamma)(f, g)^\top$ has coefficient rank two. Every competitor therefore has invertible second channel and its two monic polynomials lie in the affine pencil h_c . Expansion gives

$$\operatorname{disc}(h_c) = c(c-1)^2(r-s)^3(Ac-4r^3), \quad h_c(0) = -cs^3, \quad (7.4)$$

and

$$\frac{R'(z)}{R(z)} = \frac{(2r-3s)z + rs}{z(z-r)(z-s)}, \quad A = 4r^3 - 27s^2(r-s) < 4r^3.$$

Nonnegative roots exclude $c < 0$, and the discriminant excludes $0 < c < 1$. For $c > 1$ there is a root in (s, r) and no root in $[0, s]$. The other two admissible roots must be above r , solving $R(z) = c/(c-1) > 1$. They exist only for $s > 2r/3$. In that case R increases to its maximum at b and then decreases to one. The double-root value is $c_* > 1$; the larger root increases from b to D as c increases from c_* to c_D . Thus the whole real-rooted pencil in $[0, D]$ is $\{0, 1\}$, with $[c_*, c_D]$ added exactly when $b \leq D$.

For a competing pair with pencil coordinates c_1, c_2 , the weights obey $\alpha'_1 c_1 + \alpha'_2 c_2 = \gamma \in (0, 1)$. One coordinate must be zero and the other is one or a value $c \geq c_*$. The latter weight is γ/c , so the floor imposes $c \leq \gamma/\pi$; the other weight is automatically at least $\alpha > \pi$. Conversely put

$$\beta_f = \alpha v_1 + \gamma(1 - 1/c)v_2, \quad \beta_h = \gamma v_2/c.$$

Both vectors are positive and $\beta_f f + \beta_h h_c = \alpha v_1 f + \gamma v_2 g$. Their entry sums are the stated weights; normalizing them and keeping first channel (u, u) gives the realization. Coefficient functionals dual to the independent polynomials recover the two rank-one coefficient matrices uniquely, hence their weights and normalized marginals. Finally $b \leq D$ is equivalent to $B \geq 0$ and, on this set, $c_* \leq \gamma/\pi$ is equivalent to $E \geq 0$. $\square$

7.1. Quantitative inversion across the rank locus. All coefficient norms below are Euclidean, including the leading coefficient when a monic polynomial is represented as a vector. Constants are specified from a fixed compact $K_0 \subset \mathfrak{F}$. Norm comparisons between these finite-dimensional spaces are included in these constants. Put

$$\delta = \max_j \|P'_j - P_{\theta,j}\|_F.$$

The coefficient matrix of the second marginal is denoted by B_θ ; its rows are the coefficients of that two-vector polynomial. Set

$$\beta_0 = \min_{K_0} \sigma_2(B_\theta) > 0, \quad d_{fg} = \min_{K_0} \|g - f\|_{\text{coef}} > 0.$$

Normalizer recovery supplies C_B with $\|B' - B_\theta\| \leq C_B \delta$. Let K_d bound $\|\text{diag}(\alpha') F'_{\text{coef}}\|$ on all closed-model competitors for the designs in K_0 , and set

$$W_0 = \frac{2K_d}{\pi\beta_0}, \quad C_f = \max_{K_0} \|f\|_{\text{coef}}, \quad K_{\text{aff}} = W_0 C_B (1 + C_f). \quad (7.5)$$

Proposition 7.2 (From the second marginal to the pencil). *If $\delta \leq \beta_0/(2C_B)$, every closed competitor has invertible V' and its component polynomials satisfy*

$$f'_b = h_{c_b} + e_b, \quad \deg e_b < 3, \quad \|e_b\|_{\text{coef}} \leq K_{\text{aff}} \delta. \quad (7.6)$$

Moreover $\|\text{diag}(\alpha')^{-1} V'^{-1}\| \leq W_0$. If the ordered component pair is within ρ of (f, g) , then

$$|c_1|, |c_2 - 1| \leq (\rho + K_{\text{aff}} \delta)/d_{fg}. \quad (7.7)$$

No lower bound on $|\det U|$ is used.

Proof. Since $B' = V' \text{diag}(\alpha') F'_{\text{coef}}$,

$$\beta_0/2 \leq \sigma_2(B') \leq \sigma_{\min}(V') K_d.$$

This proves the inverse bound, using $\alpha'_a \geq \pi$. Write C_θ for the matrix with coefficient rows f, g and put

$$T = \text{diag}(\alpha')^{-1} V'^{-1} V \text{diag}(\alpha, \gamma), \quad E_0 = \text{diag}(\alpha')^{-1} V'^{-1} (B' - B_\theta).$$

Then $F'_{\text{coef}} = T C_\theta + E_0$ and $\|E_0\| \leq W_0 C_B \delta$. If a row of T is (l, c) , monicity says $l + c + (E_0)_{\text{lead}} = 1$. Subtracting $(E_0)_{\text{lead}} f$ from that error row gives the exact decomposition $f'_b = (1 - c)f + cg + e_b$, with the bound in (7.5) and zero leading coefficient in e_b . Finally $h_c - f = c(g - f)$ and $h_c - g = (c - 1)(g - f)$ imply (7.7) by the triangle inequality. $\square$

Lemma 7.3 (Two robust isolation estimates). *There are explicit positive $r_f, r_g, \varepsilon_0, C_0, C_1$, depending only on K_0 , with the following property. If $h_c + e$ is monic with all roots in $[0, D]$, $\deg e < 3$ and $\|e\|_{\text{coef}} \leq \varepsilon$, then*

$$|c| \leq r_f, \varepsilon \leq \varepsilon_0 \implies |c| \leq C_0 \varepsilon, \quad |1 - c| \leq r_g, \varepsilon \leq \varepsilon_0 \implies |1 - c| \leq C_1 \varepsilon.$$

The constants require only root-separation and coefficient bounds on K_0 , not a sign or a nonzero margin for B, E .

Proof. Let $s_0 = \min_{K_0} s > 0$ and $d_0 = \min_{K_0} 4r^3(r - s)^3 > 0$. Choose L_D as a Lipschitz bound for the discriminant on a coefficient neighbourhood of $\{h_c : \theta \in K_0, |c| \leq 1\}$, and R_D such that

$$|\text{disc}(h_c) + 4r^3(r - s)^3 c| \leq R_D c^2 \quad (|c| \leq 1).$$

These constants are finite because the discriminant is polynomial. Take $r_f \leq \min(1, d_0/(2R_D))$, omitting the second bound if $R_D = 0$. The endpoint condition gives $-cs^3 + e(0) \leq 0$, hence $c \geq -\varepsilon/s_0^3$. For $0 \leq c \leq r_f$, nonnegative discriminant gives $0 \leq -d_0 c + R_D c^2 + L_D \varepsilon$, hence $c \leq 2L_D \varepsilon/d_0$. One may take $C_0 = \max(s_0^{-3}, 2L_D/d_0)$.

Near $c = 1$, put $v = 1 - c$ and translate $z = s + x$. There are exact identities

$$h_c(s + x) = x^3 + vA_2x^2 + vA_1x + vA_0, \quad A_2 = 3s - 2r, \quad A_1 = f'(s), \quad A_0 = f(s) = s(r - s)^2.$$

Let L_s bound coefficient translation of e on K_0 , set $E_s = L_s \varepsilon$, and write the perturbed coefficients as $u_i = vA_i + e_i$, $|e_i| \leq E_s$. Choose $M \geq \max_{K_0} (|A_1|, |A_2|)$ and $a_* = \min_{K_0} A_0 > 0$. Depressing the perturbed cubic by its own quadratic coefficient gives

$$p = u_1 - u_2^2/3, \quad q = u_0 - u_1u_2/3 + 2u_2^3/27.$$

For $u = |v| + E_s \leq 1$,

$$|p| \leq C_p u, \quad |q - A_0 v| \leq E_s + C_q u^2, \quad C_p = M + 1 + (M + 1)^2/3, \quad C_q = (M + 1)^2/3 + 2(M + 1)^3/27.$$

Real-rootedness is exactly $p \leq 0$ and $|q| \leq c_{\text{cub}}(-p)^{3/2}$ with $c_{\text{cub}} = 2/(3\sqrt{3})$. Thus

$$a_* |v| \leq E_s + C_q u^2 + c_{\text{cub}} C_p^{3/2} u^{3/2}. \quad (7.8)$$

Choose

$$\eta = \min\{1, a_*/(4C_q), (a_*/(4c_{\text{cub}} C_p^{3/2}))^2\}, \quad r_g = \eta/2, \quad \varepsilon_0 \leq \eta/(2L_s).$$

Then $u \leq \eta$, the last two terms in (7.8) are at most $a_* u/2$, and $|v| \leq (2/a_* + 1)E_s$. Take $C_1 = (2/a_* + 1)L_s$. Decrease ε_0 also to stay in the discriminant neighbourhood used in the first argument. Every threshold is determined by the stated positive lower margins and finite polynomial upper bounds. $\square$

Let d_{par} be Euclidean distance in monic coefficients and stochastic entries, minimized over the component permutation.

Lemma 7.4 (A local inverse through the walls). *For every compact $K_0 \subset \mathfrak{F}$, there exist $\rho, M, t_0 > 0$ such that a closed-model competitor whose component pair is within ρ of (f, g) , up to permutation, satisfies*

$$d_{\text{par}}(\xi, \theta) \leq M \max_j \|F_T(\xi)_j - P_{\theta, j}\|_F \quad (7.9)$$

whenever the observation error is below t_0 . These constants require no lower bound on $|\det U|$, $-B$, or $-E$. For a compact $H \subset \mathfrak{I}$ and small fixed ρ , the remote separation

$$\eta_H(\rho) = \min_{\theta \in H, \xi \text{ in the closed model}} h_w(F_T(\xi), P_\theta) \quad (7.10)$$

is positive if the minimizing set is nonempty. The whole-model inverse therefore holds uniformly on H .

Proof. Let $r_* = \min(r_f, r_g)$ from Lemma 7.3 and choose

$$\rho \leq d_{fg} r_*/4, \quad t_0 \leq \min\{\beta_0/(2C_B), \rho/K_{\text{aff}}, \varepsilon_0/K_{\text{aff}}\}.$$

Proposition 7.2 gives an error at most $\varepsilon = K_{\text{aff}}\delta$ in the affine pencil and pencil coordinates in the required neighbourhoods. The two isolation estimates then give $|c_1| + |c_2 - 1| \leq (C_0 + C_1)K_{\text{aff}}\delta$. Together with (7.6), this controls both polynomial coefficient vectors by a common multiple of δ .

To recover the stochastic parameters without dividing by $\det U$, choose coefficient functionals ℓ_1, ℓ_2 dual to f, g . Their norms are bounded by the inverse Gram matrix of this pair, uniformly on K_0 . The matrix $T'_{ij} = \ell_i(f'_j)$ is $I + O_{K_0}(\delta)$. Decrease t_0 so that $\|T' - I\| \leq 1/2$; then $\|(T')^{-1}\| \leq 2$. Apply the two functionals entrywise to K' . Inverting T' recovers $M'_a = \alpha'_a u'_a v'_a{}^\top$ with $O_{K_0}(\delta)$ error from M_a . Entry sums give the weights, and row and column sums divided by those weights give the two channels. These divisions have lower bound π . This proves (7.9) with constants constructed from normalizer, second-marginal, Gram, root and weight margins only. In particular no full-rank inverse has been continued through its singular determinant.

For the last assertion the feasible set in (7.10) is compact, allowing the centre's design to vary in H . A zero minimum would give an exact competitor separated from that centre. The entire fibre and conditional uniqueness in Theorem 7.1 exclude this on H . Small whole-model observation balls thus lie in the local neighbourhood. It is this exclusion radius, rather than a local coefficient constant, that may collapse at a remote wall. $\square$

7.2. The complete local Fisher fibre. Use $a, X, Y \geq 0$, free b, c , and five free stochastic coordinates η . The component jets are

$$p_f = -a(z - r)^2 - bz(z - r) - Xz, \quad p_g = -c(z - s)^2 - Y(z - s). \quad (7.11)$$

Let L be the seven free score columns and D_0 the three columns indexed by $0, x, y$ for (a, X, Y) . Define Q by (6.2) and

$$J(X, Y) = \min_{a \geq 0} (a, X, Y) Q(a, X, Y)^\top, \quad \kappa_x = \min_{Y \geq 0} J(1, Y), \quad \kappa_y = \min_{X \geq 0} J(X, 1). \quad (7.12)$$

Proposition 7.5 (Coercivity and finite supports). *Throughout $\mathfrak{F}$, L has rank seven, $Q_{00} > 0$, and κ_x, κ_y are positive and locally Lipschitz. For every compact $K_0 \subset \mathfrak{F}$,*

$$\|S_\theta(a, b, c, \eta, X, Y)\|_{I, \theta} \geq c_{K_0} \|(a, b, c, \eta, X, Y)\| \quad (a, X, Y \geq 0).$$

Each κ_i is given by four finite support tests. For $A \subseteq \{0, x, y\} \setminus \{i\}$ with Q_{AA} positive definite, put

$$v_A = -Q_{AA}^{-1} Q_{Ai}, \quad k_{i,A} = Q_{ii} - Q_{iA} Q_{AA}^{-1} Q_{Ai}.$$

The test is $v_A \geq 0$ and $Q_{ji} + Q_{jA} v_A \geq 0$ for inactive j . The empty support has value Q_{ii} . At least one test succeeds and all successful tests give κ_i .

Proof. Every feasible jet is realized by roots

$$ta, \quad r + tb/2 \pm \sqrt{tX}, \quad s + tc/3 + \sqrt{t}y_i, \quad \sum_i y_i = 0, \quad \frac{1}{2} \sum_i y_i^2 = Y,$$

and order- t stochastic increments. Their observation increment is $tS + O(t^{3/2})$. A zero score, combined with Lemma 7.4, forces the component jets and η to vanish. Coefficient comparison in (7.11) then forces every variable to vanish. Compactness of the feasible unit sphere gives coercivity, uniformly on K_0 . This also proves independence of the free columns, $Q_{00} > 0$, and existence, positivity and boundedness of the constrained minimizers.

Choose a minimizer with minimal positive support in the projected score columns. Those supported columns are independent: a nonzero dependence would allow a signed change of positive coefficients, preserving the represented vector, until one coefficient vanished. On an independent support, stationarity and the inactive inequalities are exactly the stated tests. Conversely their

KKT conditions prove optimality by convexity. This handles a singular full Q without inverting it. Uniformly bounded minimizers and smooth score dependence give local Lipschitz dependence by evaluating each problem at a minimizer of the other problem. $\square$

Theorem 7.6 (Local normal form and whole-model restriction). *At every centre of $\mathfrak{F}$ the local leading root set is*

$$(\{0\}, \{-\sqrt{X}, \sqrt{X}\}, Z(z^3 - Yz - Z)), \quad X, Y \geq 0, \quad J(X, Y) \leq 4, \quad 27Z^2 \leq 4Y^3. \quad (7.13)$$

On compact subsets of $\mathfrak{F}$, scaled local observation balls converge to this set with Hausdorff error $O(\sqrt{t})$. Hence

$$\omega_{\theta, \text{loc}}(t) = C_{\text{loc}}(\theta)\sqrt{t} + O(t), \quad C_{\text{loc}} = \max\{\sqrt{2}\kappa_x^{-1/4}, 2\sqrt{2/3}\kappa_y^{-1/4}\}. \quad (7.14)$$

The error is compact-uniform, and C_{loc} is positive and locally Lipschitz on all of $\mathfrak{F}$. On compact subsets of $\mathfrak{J}$ these statements hold for the whole-model modulus. The support tests, (7.2), and $9\kappa_y - 16\kappa_x$ give a finite semialgebraic partition. On its constant choices,

$$C_{\text{loc}}^4 = \max\{4/\kappa_x, 64/(9\kappa_y)\}$$

is one rational function, and C_{loc} is Nash after refinement.

Proof. The local inverse bounds coefficients and stochastic increments by $O(t)$ for every local competitor. Its first component has exact roots $ta, r + tb/2 \pm \sqrt{tX}$ with bounded a, b, X and $a, X \geq 0$. Its second component has exact roots $s + tc/3 + \sqrt{t}y_i$ with $\sum_i y_i = 0$ and bounded squared sum. Put $Y = \frac{1}{2}\sum_i y_i^2$ and $Z = \prod_i y_i$. The real-cubic condition is exactly $Y \geq 0, 27Z^2 \leq 4Y^3$. Expansion gives $F = P + tS + O(t^{3/2})$ and $h_w = t\|S\|_I/2 + O(t^{3/2})$ uniformly on these bounded coordinates. Radial contraction of all first-order variables, together with $Z \mapsto \rho^{3/2}Z$, proves the outer Hausdorff inclusion. Conversely choose bounded minimizing free and endpoint variables and realize the displayed root arcs with $\rho = 1 - M\sqrt{t}$. They lie in the observation ball and give the inner inclusion. The inequality boundary and all joint cubic coefficient relations are therefore realized.

Homogeneity gives $X_{\max} = 2/\sqrt{\kappa_x}$ and $Y_{\max} = 2/\sqrt{\kappa_y}$. The double-cluster diameter is $\sqrt{X_{\max}}$. Lemma 6.1 with $m = 3$ gives the triple diameter $2\sqrt{Y_{\max}/3}$, attained also by $(-2h, h, h)$ and $(-h, -h, 2h)$ with $h = \sqrt{Y_{\max}/3}$. The maximum of marginal diameters gives (7.14). Proposition 7.5, cluster separation and the positive remote exclusion in Lemma 7.4 give the remaining claims. The local formula remains valid at $B = 0$ or $E = 0$, but its identification with the whole-model modulus does not. $\square$

Example 7.7 (The earlier rational phase values). For $D = 1, \pi = 1/10, r = 3/5, s = 1/5, \alpha = 2/5, u_1 = u_2 = (1/3, 2/3)^T, v_1 = (1/4, 3/4)^T, v_2 = (3/4, 1/4)^T$, clocks $(6/5, 7/5, 8/5, 9/5, 2, 5/2, 3)$ and equal weights, $B = -18/25$. The empty supports are admissible, the double cluster dominates, and exact rational evaluation gives

$$C^4 = \frac{78550423766145732273576408862601811503774391446}{385279214348552940073664077156875}, \quad 3778.707 < C < 3778.708.$$

Changing only u_2 to $(2/5, 3/5)^T$ gives the triple-dominant phase, $59.970 < C < 59.971$. The exact arithmetic records for both values are retained in the archival companion. These large values are not evidence for an unspecified wall blow-up.

Corollary 7.8 (Centre-known local risk). *Let $H \subset \mathfrak{J}$ be compact, fix $0 < c \leq 1/(8\sqrt{2})$, and observe independent categorical samples at the design clocks with $n_j/N \rightarrow w_j$ uniformly on H (integer*

rounding suffices). For squared spectral loss on the specified neighbourhood $h_w(F_T(\xi), P_\theta) \leq cN^{-1/2}$, with the centre known, the minimax risk $\mathcal{R}_N(\theta, c)$ satisfies uniformly on H

$$\frac{3c}{32}C_{\text{loc}}(\theta)^2 - O_H(N^{-1/4}) \leq \sqrt{N} \mathcal{R}_N(\theta, c) \leq cC_{\text{loc}}(\theta)^2 + O_H(N^{-1/4}).$$

Proof. A constant decision in the specified spectral ball has loss at most its squared diameter. A diameter pair in that ball has weighted Hellinger distance at most $2cN^{-1/2}$. Tensorization bounds its product Hellinger distance by $2c\sqrt{\max_j n_j/(Nw_j)}$. Uniform convergence and the positive minimum of the design weights on H make the maximum ratio at most 2 for all sufficiently large N , uniformly on H . Thus the product distance is at most $2c\sqrt{2} \leq 1/4$. Total variation is bounded by this Hellinger distance in the present convention. The two-point testing bound for squared metric loss is $d^2(1 - \text{TV})/8$, hence at least $3d^2/32$. Substitute (7.14). The centre and its neighbourhood are known; unknown-centre adaptation and honest confidence sets across a wall are not asserted. $\square$

8. A REMOTE BRANCH AT THE STOCHASTIC WEIGHT WALL

Fix r, s, D with $B > 0$, positive u, v_1, v_2 with $v_1 \neq v_2$, and $U = (u, u)$, $V = (v_1, v_2)$. Assume

$$\gamma_0 = \pi c_*, \quad \alpha_0 = 1 - \pi c_*, \quad \pi < \alpha_0 < 1 - \pi. \quad (8.1)$$

The design is fixed. Let θ_δ be the original (f, g) centre with weight $\alpha = \alpha_0 + \delta$. Then $E = -A\delta$, so $\delta > 0$ is the identified side. Put

$$a = \frac{4rs}{r+3s}, \quad b = \frac{rs}{3s-2r}, \quad h_* = (1 - c_*)f + c_*g = (z - a)(z - b)^2. \quad (8.2)$$

Substitution gives $s < a < r < b < D$. At $\delta = 0$ the two exact component pairs are (f, g) and (f, h_*) . The remote pair has weight π on h_* and second-channel columns

$$v_f^* = \frac{\alpha_0 v_1 + (\gamma_0 - \pi)v_2}{1 - \pi}, \quad v_h^* = v_2. \quad (8.3)$$

Let ξ_* be this point; its observation is $P_0 = P_{\theta_0}$. Its second channel is invertible because $v_f^* - v_2 = \alpha_0(v_1 - v_2)/(1 - \pi) \neq 0$. The sorted base and remote spectra are respectively

$$(0, s, s, s, r, r), \quad (0, a, r, r, b, b),$$

whose distance is

$$D_* = \max(r - s, b - r) > 0. \quad (8.4)$$

Thus a zero-length additional pencil interval already has a positive exact-fibre spectral diameter.

8.1. A complete real corner chart at the remote point. Use the eleven coordinates $v = (a_0, b_0, X, c_0, d_0, Y, k, \dot{u}_f, \dot{u}_h, \dot{v}_f, \dot{v}_h)$, where a binary channel increment is represented by its one free zero-sum coordinate. Set

$$\begin{aligned} p_f &= -a_0(z - r)^2 - b_0z(z - r) - Xz, \\ p_h &= -c_0(z - b)^2 - d_0(z - a)(z - b) - Y(z - a), \\ \mathcal{T}_* &= \{v : a_0, X, Y, k \geq 0\}. \end{aligned} \quad (8.5)$$

The remaining seven coordinates are free. Here k is the increment of the weight on h_* ; the weight on f has increment $-k$. Let J_*v be the full coefficient and stochastic vector with polynomial increments p_f, p_h and these stochastic increments.

Proposition 8.1 (Remote tangent cone and converse realization). *In a neighbourhood of ξ_* , modulo the one component permutation, the closed model has a Nash coordinate chart from a neighbourhood of zero in $\mathcal{T}_*$. Its derivative is the invertible linear map J_* . Consequently its coefficient-stochastic tangent cone is exactly $J_*(\mathcal{T}_*)$. Every normalized secant has a subsequential limit in this cone, and every vector of the cone is realized by a feasible arc. For bounded $v \in \mathcal{T}_*$ the arcs can be chosen with*

$$\xi(\tau, v) = \xi_* + \tau J_* v + O(\tau^2), \quad F_T(\xi(\tau, v)) = P_0 + \tau S_* v + O(\tau^2), \quad (8.6)$$

uniformly on bounded coordinate sets. Here S_ is the score (2.3) at the remote point. All assertions are uniform on compact subsets of the wall family with its stated strict margins.*

Proof. Factor each nearby polynomial into its separated simple and quadratic factors. Coprime factorization is analytic in coefficients. Write the simple root of the first component as u , the mean of its quadratic roots as $r + v_f/2$, and their squared half-separation as x . Write the corresponding coordinates of the second component as $a + w$, $b + v_h/2$ and y . Thus the exact polynomials are

$$(z - u)((z - r - v_f/2)^2 - x), \quad (z - a - w)((z - b - v_h/2)^2 - y).$$

These six coordinates are Nash functions of the coefficients near the remote pair. Conversely the displayed formulas are polynomial in the six coordinates. Their derivative has the three columns in (8.5) for each component. They are independent: for the first polynomial, evaluation at 0 and r first determines a_0 and X , and then the remaining coefficient determines b_0 ; for the second, evaluation at a and b determines c_0 and Y , and then d_0 . The inverse function theorem gives the asserted local coefficient chart.

The first simple root is feasible exactly when $u \geq 0$, locally. Its upper endpoint is inactive. All other cluster centres are strictly inside $(0, D)$, so their interval constraints are inactive in this neighbourhood. Real-rootedness of the two quadratic factors is exactly $x, y \geq 0$. The weight on h_* is at its floor, giving $k \geq 0$; the other floor has strict margin $1 - 2\pi > 0$. Channel entries are strictly positive, so only their linear column-sum equations remain active. This proves that the complete feasible chart is the indicated orthant times seven free coordinates, not merely an outer cone. Separated factors are unique. The alternative component labelling is a second, identical chart under the permutation quotient and introduces no extra directions.

For the secant assertion, let τ be the norm of a coefficient and stochastic displacement. Analytic factorization bounds the simple and mean shifts by $O(\tau)$. The quadratic coefficients bound x, y by $O(\tau)$ as well, since variance is mean squared minus the constant coefficient. Dividing the exact chart coordinates by τ gives a bounded vector in $\mathcal{T}_*$. Its subsequential limits give all normalized tangent secants. In the norm-normalized case $J_* v$ has norm one in the limit, so no nonzero secant disappears in the chart.

Conversely set $u = \tau a_0$, $v_f = \tau b_0$, $x = \tau X$, $w = \tau c_0$, $v_h = \tau d_0$, $y = \tau Y$, choose weight $\pi + \tau k$ on h , and take the specified order- τ channel increments. The roots are

$$\tau a_0, \quad r + \tau b_0/2 \pm \sqrt{\tau X}, \quad a + \tau c_0, \quad b + \tau d_0/2 \pm \sqrt{\tau Y}.$$

They remain feasible for small τ , uniformly for bounded coordinates. They are analytic in $\sqrt{\tau}$; their polynomial coefficients are analytic in τ . In fact the exact identities are

$$f_\tau - f - \tau p_f = \tau^2 \{a_0 b_0 (z - r) + a_0 X + b_0^2 z/4\} - \tau^3 a_0 b_0^2/4, \quad (8.7)$$

$$h_\tau - h_* - \tau p_h = \tau^2 \{c_0 d_0 (z - b) + c_0 Y + d_0^2 (z - a)/4\} - \tau^3 c_0 d_0^2/4. \quad (8.8)$$

There is no order- $\tau^{3/2}$ coefficient term at these simple/double clusters. Taylor expansion of the rational observation map, whose normalizers stay positive, proves (8.6). Compact strict root, channel, weight and clock margins make all inverse chart and derivative bounds uniform as stated. $\square$

8.2. The constrained entrance program. Put $B_0 = \frac{d}{d\delta} P_{\theta_\delta} \Big|_{\delta=0}$ and use the Fisher norm at P_0 . Define

$$\mu_E = \frac{1}{2} \min_{v \in \mathcal{T}_*} \|S_* v - B_0\|_{I,0}. \quad (8.9)$$

This is an explicit finite-dimensional program on the cone just proved to be the full feasible tangent cone.

Lemma 8.2 (Positive entrance separation). *The score has no nonzero zero on $\mathcal{T}_*$, and B_0 is not in its image. Thus $\mu_E > 0$ and its minimum is attained. Eliminating seven free variables gives sixteen independent-support KKT tests on four nonnegative variables. For algebraic input μ_E is real algebraic.*

Proof. Suppose first that $S_* v = B_0$. The differentiated normalizer equations, using Lemma 2.1 at the coprime remote pair with invertible second channel, imply $\dot{q} = f - g$ and $\dot{K} = \frac{d}{d\delta} K_{\theta_\delta} \Big|_0$. The derivative of the second marginal belongs to $\text{span}(f, g)$. Its stochastic derivative terms already belong to that span; the remote second-channel weight matrix is invertible. Hence $p_f, p_h \in \text{span}(f, g)$. Since their degrees are below three, write $p_f = t_f(g - f)$ and $p_h = t_h(g - f)$.

Evaluation at 0 gives $t_f = a_0 r^2 / s^3 \geq 0$, while evaluation at r gives $t_f = -Xr / (r - s)^3 \leq 0$. Therefore $t_f = 0$ and $p_f = 0$. At b the second polynomial gives

$$t_h = \frac{Y(b - a)}{f(b) - g(b)} \geq 0,$$

because $R(b) > 1$. Differentiation of the monic normalizer now gives

$$\pi t_h + c_* k = -1, \quad (8.10)$$

an impossibility. For a zero score, the same argument has right-hand side zero and implies $t_h = k = 0$. All polynomial jets vanish. Coefficient functionals dual to f, h_* then imply that each coefficient matrix has zero derivative; taking its row and column sums forces all four channel derivatives to vanish. The independent chart columns in Proposition 8.1 force $v = 0$.

The intersection of $\mathcal{T}_*$ with the unit sphere is compact, so $\|S_* v\|_{I,0} \geq c\|v\|$ on the cone. Thus the objective in (8.9) tends to infinity with $\|v\|$; its minimum is attained and positive, since equality $S_* v = B_0$ was excluded.

Let L_* be the seven free columns, D_* the four nonnegative columns, $W = \text{diag}(w_j / P_{0,j\epsilon})$, and $\mathcal{P}$ the Fisher-orthogonal projection off $\text{im } L_*$. Put

$$Q_* = D_*^\top W \mathcal{P} D_*, \quad g_* = D_*^\top W \mathcal{P} B_0, \quad b_* = B_0^\top W \mathcal{P} B_0.$$

Coercivity implies full rank of the free columns, and

$$4\mu_E^2 = \min_{z \geq 0} \{z^\top Q_* z - 2g_*^\top z + b_*\}. \quad (8.11)$$

For each of the sixteen supports A with $Q_{*,AA}$ positive definite, set $z_A = Q_{*,AA}^{-1} g_{*,A}$, require $z_A \geq 0$ and $(Q_* z - g_*)_j \geq 0$ off A , and take value $b_* - g_{*,A}^\top Q_{*,AA}^{-1} g_{*,A}$. Include the empty support. The minimal-positive-support argument of Proposition 7.5 proves completeness, including singular Q_* . All entries are rational in the wall coordinates; take the positive square root after selecting the exact minimum. $\square$

Lemma 8.3 (Local coefficient control at the remote point). *In a coefficient and stochastic neighbourhood of ξ_* in the closed model,*

$$d_{\text{par}}(\xi, \xi_*) \leq C \|F_T(\xi) - P_0\|.$$

Every competitor at parameter distance $O(\delta)$ has bounded exact chart coordinates at scale δ and observation expansion $P_0 + \delta S_* v + O(\delta^2)$. Conversely each bounded cone vector has such a feasible realization.

Proof. If the inverse failed, normalize a sequence of displacements by their norm. Proposition 8.1 gives a nonzero limit $J_* v$ with $v \in \mathcal{T}_*$ and zero score, contradicting Lemma 8.2. Equivalently in its chart, Taylor's theorem and cone coercivity give $\|F_T(\Phi(w)) - P_0\| \geq c\|w\| - C\|w\|^2 \geq c\|w\|/2$ for small feasible w . The chart is locally bi-Lipschitz in coefficients and stochastic entries, which proves the inverse and the stated bounded coordinate assertion. The expansions and converse are (8.6). This proof does not assume the inverse in order to identify the tangent cone. $\square$

8.3. The two-scale law with a quadratic entrance remainder. Choose disjoint small closed coefficient and stochastic neighbourhoods N_0, N_* of the two exact representatives at the wall, modulo component permutation, with ξ_* in the relative interior of N_* . Put

$$d_{\text{rem}}(\delta) = \min_{\xi \in N_*} h_w(F_T(\xi), P_{\theta_\delta}). \quad (8.12)$$

Their boundaries have a positive observation separation from P_0 .

Theorem 8.4 (Weight-wall transition). *Under (8.1) and $B > 0$,*

$$d_{\text{rem}}(\delta) = \mu_E \delta + O(\delta^2) \quad (\delta \downarrow 0). \quad (8.13)$$

The minimum is independent of the sufficiently small neighbourhood N_ for all sufficiently small δ . The base local constant $C_{\text{loc}}(\theta_\delta)$ has the finite positive limit $C_{\text{loc}}(\theta_0)$; neither κ_x nor κ_y vanishes at this wall. For fixed $\lambda > 0$ the whole-model modulus satisfies*

$$0 < \lambda < \mu_E : \quad \omega_{\theta_\delta}(\lambda\delta) = C_{\text{loc}}(\theta_0)\sqrt{\lambda\delta} + O_\lambda(\delta), \quad (8.14)$$

$$\lambda > \mu_E : \quad \omega_{\theta_\delta}(\lambda\delta) = D_* + O_\lambda(\sqrt{\delta}). \quad (8.15)$$

The exact entrance curve for the closed ball is $t = d_{\text{rem}}(\delta)$, including equality. The layer not decided by the first-order threshold alone has width $O(\delta^2)$ in t .

On the nonidentified side, for small $\varepsilon > 0$,

$$\omega_{\theta_{-\varepsilon}}(0) = \max \left\{ r - s, \quad b - r + \sqrt{\frac{\chi\varepsilon}{\pi}} + O(\varepsilon) \right\}, \quad \chi = \frac{f(b) - g(b)}{b - a} > 0. \quad (8.16)$$

If $b - r < r - s$, the exact diameter stays equal to $r - s$ initially; if $b - r \geq r - s$, it opens above D_ at the displayed square-root rate.*

Proof. Theorem 7.1 gives exactly the two representatives at $\delta = 0$. Compactness gives a positive observation gap outside $N_0 \cup N_*$. Since ξ_* itself is at distance $O(\delta)$ from P_{θ_δ} , a minimizer in (8.12) has observation $O(\delta)$ from P_0 and lies away from the boundary of N_* . Lemma 8.3 bounds its coefficient and stochastic displacement by $O(\delta)$. Its exact chart coordinates are therefore δv for bounded $v \in \mathcal{T}_*$. Expansion at the common positive probability array and the reverse triangle inequality give

$$h_w(F_T(\xi), P_{\theta_\delta}) = \frac{\delta}{2} \|S_* v - B_0\|_{I,0} + O(\delta^2), \quad (8.17)$$

uniformly on those bounded coordinates. The lower bound in (8.13) follows by minimizing the leading norm. Conversely a minimizer of (8.9), realized by the exact chart, gives the upper bound. The error is quadratic because of (8.7)–(8.8), not merely an appeal to a generic root-continuity bound. This also implies the weaker $O(\delta^{3/2})$ entrance estimate. The same localization puts all minimizers inside any fixed smaller remote neighbourhood for small δ , proving independence of its choice.

Proposition 7.5 and Theorem 7.6 apply on a compact base path segment including θ_0 ; this pair stays locally coercive there. Thus C_{loc} is positive and locally Lipschitz through the wall. If

$\lambda < \mu_E$, the remote neighbourhood is excluded from the radius- $\lambda\delta$ ball for small δ . The exterior gap excludes every other competitor, so the whole ball is the base local ball. Formula (7.14) gives (8.14). Replacing $C_{\text{loc}}(\theta_\delta)$ by $C_{\text{loc}}(\theta_0)$ costs only $O(\delta^{3/2})$.

If $\lambda > \mu_E$, a remote minimizer and the base centre belong to the ball. Every competitor belongs to one of the two neighbourhoods. The two coefficient inverses put its roots within $O_\lambda(\sqrt{\delta})$ of the corresponding exact spectrum at the wall. The bottleneck triangle inequality gives both bounds in (8.15). Compact attainment makes remote membership exactly $t \geq d_{\text{rem}}(\delta)$, including equality. The width of the unresolved first-order layer follows from (8.13); it does not decide the sign of the difference on the critical ray.

For $\theta_{-\varepsilon}$ the extra exact pencil interval is $[c_*, c_* + \varepsilon/\pi]$, since $c_D > c_*$ is fixed. At $c = c_* + \sigma$, expansion at $z = b + u$ gives

$$h_c(b + u) = (b - a)u^2 + u^3 + \sigma(g(b) - f(b)) + O(\sigma|u|).$$

Since $R(b) > 1$, $\chi > 0$ and the two roots near b are $b \pm \sqrt{\chi\sigma} + O(\sigma)$. The third root remains in (s, r) . The larger root increases with c by the monotonic branch of R above its maximum. Comparing sorted lists, the distance to the base spectrum is exactly $\max(r - s, b_+(c) - r)$. Two remote spectra are only $O(\sqrt{\varepsilon})$ apart, less than D_* for small ε . The exact-fibre diameter is therefore attained against the base at the upper pencil endpoint. This proves (8.16) and its two cases. $\square$

Remark 8.5 (The critical ray). At $t = \mu_E\delta$, remote membership is decided by the sign of $d_{\text{rem}}(\delta) - \mu_E\delta$. For algebraic input, (8.12) has a compact first-order minimum formula. Elimination and real algebraic branch isolation determine its Puiseux expansion. The first nonzero higher coefficient decides the critical ray; if the difference is identically zero, the closed ball includes the remote branch. No strict regime is assigned to that ray by convention.

Remark 8.6 (The geometric wall). At $B = 0$ with $E > 0$, the remote double root is $b = D$, and the additional pencil interval again consists of one point at positive spectral distance. On the nonidentified side put $\ell_0 = D - b$. Expansion at the strict maximum of R gives

$$c_D - c_* = \frac{-R''(b)}{2(R(b) - 1)^2} \ell_0^2 + O(\ell_0^3), \quad \ell_0 = \frac{B}{3s - 2r}.$$

The coefficient is positive and the weight floor is inactive. This interval is born with quadratic length in geometric clearance, rather than linear length in the weight parameter. The base local constants remain finite on compact subsets of $\mathfrak{F}$. A two-scale entrance law at that endpoint requires a different cone with one-sided root coordinates. The complete entrance theorem here is for the weight wall.

Example 8.7 (An exact wall datum). Take $D = 1$, $r = 3/5$, $s = 11/20$, $\pi = 1/10$, the rank-one columns and clocks in Section 7, and equal design weights. Then

$$\alpha_0 = \frac{547}{675}, \quad c_* = \frac{256}{135}, \quad a = \frac{44}{75}, \quad b = \frac{11}{15}, \quad B = \frac{3}{25}, \quad D_* = \frac{2}{15}, \quad \chi = \frac{3}{64},$$

and $v_f^* = (1457, 3403)^T/4860$. Exact rational evaluation gives

$$5975.429249985072 < C_{\text{loc}}(\theta_0) < 5975.429249985073, \\ 2.131024914 \cdot 10^{-6} < \mu_E < 2.131024915 \cdot 10^{-6}.$$

The empty positive support is admissible in each base program and in the remote program. The ancillary exact records retain the rational Gram matrices, squared entrance constant, fourth power of the local constant and isolating intervals. The decimals follow from rational power comparisons, not optimization tolerances. They illustrate two different, poorly conditioned scales; they do not test the universal quantifiers of the family theorems.

Three stability questions. Multiplicity and one-sided boundary constraints determine the weights in Theorem 6.2. Exact identification in the cubic family is instead determined by (7.2). Conditioning of a fixed local law is measured by the positive Fisher programs. At the weight wall, identification fails while those programs stay coercive and the local square-root exponent persists. In Example 6.3, by contrast, the exponent changes and the linear constant diverges. The joint atlas retains these distinct limiting operations rather than identifying their strata.

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PROJECTIVE POLYNOMIAL OBSERVATIONS: REAL SPECTRAL ATLASES AND IDENTIFICATION WALLS

QIAN QI

ABSTRACT. We study the diameter of polynomial spectra compatible with a small neighbourhood of a normalized stochastic observation. A joint real weighted specialization, rather than separate coefficient bounds, gives the complete leading spectral set. We specify its effective construction and a finite parameter stratification retaining exact fibres, accessible vanishing orders, cluster weights, and leading diameters. For binary polynomial experiments of arbitrary fixed degree with invertible channels, the resulting atlas is explicit across all coprime root-multiplicity and boundary patterns: the exponent is one half precisely when an interior root is repeated, and is one otherwise. The leading constants are finite constrained Fisher programs; for an interior cluster of multiplicity m its diameter factor is $2\sqrt{(m-1)/m}$. A binary cubic family adds an explicit rank-deficient stratum. At its stochastic weight wall, identification is lost by the entry of a remote spectral branch while the local Fisher constant stays finite. We prove the linear entrance law for that branch, the resulting two-scale transition from a vanishing modulus to a positive diameter, and the square-root opening of the nonidentified exact fibre. These results separate root-multiplicity scaling, exact identification, and local conditioning.

1. INTRODUCTION

An observation may determine a spectrum without determining unrestricted coefficient directions. Conversely, a locally well-conditioned spectral inverse need not remain globally identifying. The first distinction is caused by real feasibility; the second can be caused by a remote component of an exact fibre. A theory of singular spectral observations must keep both distinctions, as well as the relations between different root clusters.

Let $F : X \rightarrow \mathcal{P}$ be a continuous semialgebraic observation on a compact model, and let A_x be a monic target polynomial. We write

$$\omega_P(t) = \text{diam}\{Z(A_x) : h_w(F(x), P) \leq t\},$$

where Z counts roots with multiplicity and the metric is bottleneck matching. The principal object is the simultaneous weighted specialization of the real observation-coefficient graph. If the exact fibre has one spectral value, Theorem 3.5 expresses the leading constant as the diameter of its root image. Proposition 3.3 specifies the real-accessible monomial data used to compute the exponent, including components entirely above the exact observation fibre. Theorem 3.7 places these objects in a finite semialgebraic parameter atlas. The scalar valuation formula is a classical ingredient, not the novelty claim of the construction.

The explicit part of the article begins with a class larger than a fixed cubic singularity. In the binary degree- d model, take coprime component polynomials and positive invertible channels. Roots may have arbitrary multiplicities and may lie at either endpoint of the root interval. There are finitely many such patterns at fixed d . Theorem 4.2 gives the full leading set for every pattern.

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An interior cluster has coefficient orders $1, k/2$ in weights $1, k \geq 2$, whereas a boundary cluster has orders k . Thus

$$\alpha = \begin{cases} 1/2, & \text{some interior root is repeated,} \\ 1, & \text{otherwise.} \end{cases}$$

In the first case the exact constant is

$$C = \max_{j: m_j \geq 2, r_j \in (0, D)} 2 \sqrt{\frac{m_j - 1}{m_j}} \kappa_j^{-1/4},$$

with κ_j given by a finite nonnegative least-squares problem. This formula includes every interior multiplicity, not just two and three. The proof is a whole-model Hausdorff comparison with realizing root arcs. It does not replace the joint fibre by a product of marginal bounds.

The rank-deficient cubic class has an additional global phenomenon. For $f = z(z - r)^2$, $g = (z - s)^3$, the exact spectral fibre is classified by three polynomial signs (Theorem 5.1). The local coefficient inverse extends through the identification walls, although its *whole-model exclusion radius* need not do so. At a transverse weight-wall crossing, put $\delta = \alpha - \alpha_0$. Theorem 6.3 proves that the distance of the remote branch from the observed centre is

$$d_{\text{rem}}(\delta) = \mu_E \delta + O(\delta^{3/2}), \quad \mu_E > 0.$$

For $t = \lambda \delta$, the modulus is of order $\sqrt{\delta}$ when $0 < \lambda < \mu_E$, but tends to a positive exact-fibre diameter when $\lambda > \mu_E$. In particular, a blow-up of the local Fisher constant is not necessary for loss of identification. The constant μ_E is another finite constrained Fisher calculation, now at the remote branch. An exact rational example evaluates both constants.

Classical inputs and the additional outputs. The distinctions in Table 1 are part of the statements of the results. In particular, no assertion of a new resolution algorithm, a universal ordinary Taylor-jet bound, or a quadratic description of every semialgebraic leading fibre is made.

Table 1: Inputs and outputs of the theorem graph.

Object	Closest input	Output retained here
Divisorial exponent	Principalization and the scalar order formulas of [4, Proposition 4.3, Theorem 4.8].	A classical scalar consequence; cluster weights convert it to the root exponent.
Real accessibility	Real uniformization and constructive resolution [2, 3].	A cover of the actual inequality-constrained graph, including its lower-dimensional real loci; no complex-only divisor enters.
Joint initial set	Real closure and elimination [5, 1]; set-valued limit constructions.	Simultaneous observation and cluster variables, with the centre fixed during closure, followed by the root image and its exact matching diameter.
Parameter strata	Semialgebraic triviality [7] and resolution on generic fibres followed by specialization.	One refinement retains exact-fibre type, accessible order lists and joint leading-set type; Fisher faces are added where a Fisher description is proved.

Object	Closest input	Output retained here
Current valuative theory	Hà's finite-test-set theorem for graded ideals and filtrations, and his fixed-toric-model chamber theorem [6, Theorems 5.8 and 12.3].	We do not claim scalar finiteness or wall-chamber phenomena absent from that work. Our target is a real feasible root-set diameter, including stochastic inequalities and remote-fibre entry.
Explicit spectral atlas	Polynomial factorization and finite convex quadratic minimization.	Arbitrary interior/boundary multiplicities, realizing alternatives, and uniform spectral remainders (Theorem 4.2).
Cubic fibre and its wall	The affine cubic pencil and its discriminant.	The full stochastic fibre, cross-rank local constants, and the remote entrance law (Theorems 5.1 and 6.3).

For clarity about the comparison with [6], its finite-test-set hypotheses concern integral-closure tests and positive asymptotic valuations. Its family theorem assumes a common toric model, constant Newton polyhedra and uniform nondegeneracy, with piecewise-linear weights. Neither hypothesis package is identified here with a real stochastic feasibility condition. A real-valued valuation is also not the same notion as an admissible real transversal. Granting the scalar finite-max results leaves the joint specialization, its root image, and the stochastic wall calculation to be established.

Organization. Sections 2–3 fix the experiment and the real-algebraic construction. Section 4 computes a multiplicity and boundary atlas. Sections 5–6 add a rank-deficient family and analyze its identification wall. The statistical consequence remains a centre-known local corollary. The accompanying archival volume retains the earlier global normalization, clock, quotient, residue, inference and perturbation results in full; none is needed as an unstated premise of the principal proofs below.

2. THE EXPERIMENT AND THE SPECTRAL METRIC

Fix $D > 0$, a degree $d \geq 1$, and $0 < \pi < 1/2$. A closed binary model point consists of two monic real polynomials f_1, f_2 of degree d , all of whose roots lie in $[0, D]$, weights $\alpha_1 + \alpha_2 = 1$ with $\alpha_a \geq \pi$, and column-stochastic matrices $U = (u_1, u_2)$ and $V = (v_1, v_2)$ with nonnegative entries. Set

$$K(z) = \sum_{a=1}^2 \alpha_a f_a(z) u_a v_a^T, \quad q(z) = \mathbf{1}^T K(z) \mathbf{1} = \sum_{a=1}^2 \alpha_a f_a(z), \quad P_j = \frac{K(T_j)}{q(T_j)}. \quad (2.1)$$

The clocks satisfy $D < T_1 < \dots < T_\ell$, and $w_j > 0$, $\sum_j w_j = 1$. Consequently $q(T_j) > 0$, including at every closed-model competitor. The target polynomial is $A = f_1 f_2$. The model is compact in monic coefficients and stochastic entries. Component permutations are identified. The matrices, roots, clocks and design weights are allowed to vary when a family of centres is considered, but every competitor uses its centre's fixed design.

For probability arrays define

$$h_w(P, Q)^2 = \sum_j w_j \sum_e (\sqrt{P_{je}} - \sqrt{Q_{je}})^2.$$

For two multisets $Z = \{z_i\}_{i=1}^n$ and $Z' = \{z'_i\}_{i=1}^n$, put

$$d_\infty(Z, Z') = \min_{\sigma \in \mathfrak{S}_n} \max_i |z_i - z'_{\sigma(i)}|.$$

For real roots this is the maximum discrepancy of their increasing orderings. All diameters in this article are diameters of sets of multisets, not maximal distances from the centre. Write

$$\omega_\theta(t) = \text{diam}\{Z(A_\xi) : h_w(F_T(\xi), P_\theta) \leq t\}. \quad (2.2)$$

A local modulus $\omega_{\theta, \text{loc}}$ restricts the competitors to a specified coefficient neighbourhood of the centre's component pair. This restriction is never implicit.

At a strictly positive datum the Fisher inner product on zero-sum arrays is

$$\langle H, H' \rangle_{I, \theta} = \sum_{j, e} \frac{w_j}{P_{\theta, je}} H_{je} H'_{je}.$$

For component increments p_a of degree below d and stochastic increments $\eta = (\dot{\alpha}_1, \dot{u}_1, \dot{u}_2, \dot{v}_1, \dot{v}_2)$, where channel increments have entry sum zero, define

$$\begin{aligned} \dot{K} &= \sum_a \{\alpha_a p_a u_a v_a^\top + f_a(\dot{\alpha}_a u_a v_a^\top + \alpha_a \dot{u}_a v_a^\top + \alpha_a u_a \dot{v}_a^\top)\}, \\ \dot{q} &= \sum_a (\alpha_a p_a + \dot{\alpha}_a f_a), \quad S_\theta(p_1, p_2, \eta)_j = \frac{\dot{K}(T_j) - P_{\theta, j} \dot{q}(T_j)}{q(T_j)}. \end{aligned} \quad (2.3)$$

Here $\dot{\alpha}_2 = -\dot{\alpha}_1$. These rational formulas define every Fisher program below in the original experiment coordinates.

Lemma 2.1 (Normalizer recovery). *Suppose f_1, f_2 are coprime, V is invertible, the weights are positive, and $\ell \geq 2d + 1$. The observed array determines q and K uniquely among all closed-model competitors. Their coefficients are locally Lipschitz functions of the observations, with uniform constants on compact sets of such centres. No rank assumption on U is required.*

Proof. Let e have degree below d , and let H be a matrix polynomial of degree at most d satisfying $H(T_j) = e(T_j)P_j$. Interpolation of the polynomial $qH - eK$, of degree at most $2d$, gives $qH = eK$. Its second marginal, multiplied by $\text{diag}(\alpha_1, \alpha_2)^{-1}V^{-1}$, gives $q \mid ef_1$ and $q \mid ef_2$. Coprimality implies $q \mid e$, hence $e = 0$ and $H = 0$.

More explicitly, let Π_T project onto the orthogonal complement of the degree- d Vandermonde column space. The map

$$N_\theta : e \longmapsto \Pi_T(e(T_j)P_{\theta, j})_j, \quad \deg e < d,$$

is injective. For a competitor, subtraction of the normalization equations yields $\|N_\theta(q' - q)\| \leq C \max_j \|P'_j - P_j\|$; the competitor normalizer coefficients are bounded by compactness of the root interval. The smallest singular value of N_θ is positive. A bounded left inverse recovers $q' - q$, and Vandermonde interpolation recovers $K' - K$. The singular values and the coefficient bounds have positive uniform margins on compact sets of the stated centres. $\square$

Lemma 2.2 (Full-rank coefficient inverse). *Under Lemma 2.1, suppose also that U is invertible. Then every sufficiently close observation has, up to one component permutation, coefficient and stochastic parameter error bounded by a constant times its observation error. Every exact competitor has the same component polynomials and stochastic parameters. The constants are uniform on compact subsets of these centres.*

Proof. Choose two clocks such that $f_1(T_j)/f_1(T_i) \neq f_2(T_j)/f_2(T_i)$. Otherwise the nonzero polynomial $f_2(T_i)f_1 - f_1(T_i)f_2$, of degree at most d , would vanish at all $2d + 1$ clocks. The matrix $K(T_j)K(T_i)^{-1}$ has those two distinct eigenvalues and eigenvectors u_1, u_2 . Normalize the eigenvectors to have entry sum one. Simple-eigenvalue inversion is analytic in a neighbourhood

of the centre. Applying the recovered U^{-1} to $K(z)$ and taking row sums recovers $\alpha_a f_a$; leading coefficients and normalized rows recover α_a, V . This construction applies to arbitrary closed-model competitors because their $K'(T_i)$ stays invertible. It also proves exact uniqueness. A finite neighbourhood cover gives compact-uniform constants. Equivalently, the derivative in monic coefficient and stochastic coordinates has a bounded left inverse. $\square$

3. JOINT REAL SPECIALIZATION AND FINITE PARAMETER DATA

This section applies to a compact semialgebraic model X , a continuous semialgebraic probability map F , and a continuous semialgebraic monic polynomial A_x of fixed degree n . The coefficients may be complex; real and imaginary parts are always separate real coordinates. Fix a datum $P \in F(X)$ and assume first that $A_x = A_0$ on $F^{-1}(P)$. Write $A_0(z) = \prod_r (z - r)^{m_r}$. Choose disjoint isolating discs for these roots. Compactness implies that all sufficiently small observation balls have exactly m_r target roots in the disc at r . Their monic factors satisfy

$$A_{r,x}(r + z) = z^{m_r} + \sum_{a < m_r} c_{r,a}(x) z^a, \quad k_{r,a} = m_r - a. \quad (3.1)$$

Polynomial multiplication and the isolating conditions define the unique cluster factors semialgebraically. The differential of multiplication is invertible for coprime factors, so the factors are analytic in the total polynomial coefficients on this neighbourhood.

3.1. Input representation and real-accessible covers.

Definition 3.1 (Effective input). Real algebraic input consists of finite first-order polynomial formulas for X and the graphs of F, A , together with P , the design weights, and cluster-isolating data. Every constant is represented by an integer polynomial and a rational isolating interval selecting a real root. Arithmetic in several such constants can equivalently be performed after passing to a primitive element. A complex constant is a pair of represented real algebraic numbers. Endpoints, clocks, polynomial coefficients, datum probabilities, disc centres and squared radii are included in this rule. All sign conditions, denominator exclusions and branch selections are part of the formulas, not information supplied to a numerical oracle. For a parameter family these are finite formulas in additional real parameter coordinates; the numerical output assertion applies to algebraic parameter values. No complexity bound is imposed.

Adjoin the graph variables

$$R_{je} = \sqrt{w_j}(\sqrt{F(x)_{je}} - \sqrt{P_{je}}), \quad G = \sum_{j,e} R_{je}^2, \quad H_{r,a} = (\Re c_{r,a})^2 + (\Im c_{r,a})^2.$$

The nonnegative square-root choices are equations and inequalities on a compact real graph Γ . Thus G, H are polynomial coordinate functions on Γ , even at zero observed cells.

Definition 3.2 (Monomial and exact pieces). Real monomial data near $G = 0$ are finitely many relatively compact coordinate boxes on smooth real sources mapping properly, collectively, onto a neighbourhood of $G = 0$ in Γ . Each source component is of one of the following two kinds. On an *exact piece*, $G \equiv 0$; then all $H_{r,a} \equiv 0$, and the piece is recorded separately without entering any quotient of orders. On every other piece,

$$G = u \prod_i |y_i|^{a_i}, \quad H_{r,a} \equiv 0 \quad \text{or} \quad H_{r,a} = v_{r,a} \prod_i |y_i|^{b_{r,a,i}}. \quad (3.2)$$

The nonzero units are positive and bounded above and below on smaller closed boxes. All exponents are nonnegative integers. A divisor is recorded only if $a_i > 0$ and it has a real point in the source box away from the other divisors, with an admissible transversal on which $G > 0$. Identically zero H contributes $+\infty$; $G \equiv 0$ never contributes $0/0$. Lower-dimensional exact and nonexact pieces are both retained.

Proposition 3.3 (A finite real cover). *The graph Γ has data as in Definition 3.2. The data cover every sufficiently small admissible observation neighbourhood, and each recorded divisor supplies a real admissible arc. For the input of Definition 3.1, all pieces, integer orders, and accessibility tests admit a finite effective construction.*

Proof. We give the interfaces of the construction. First decompose the compact semialgebraic graph into finitely many basic closed sets, using real elimination [5, 1]. On each piece replace an inequality $p(x) \geq 0$ by $p(x) = s^2$. The original graph coordinates range over a compact set and each slack is bounded by the maximum of $|p|^{1/2}$. The resulting real algebraic lift is compact and its real projection is exactly the original piece. This equality is the reason for using square slacks before, rather than inequalities after, resolution.

Resolve each reduced algebraic lift in characteristic zero [3]. Over its nonsingular locus the resolution is an isomorphism, so it covers every real point there. A real point not covered in this way lies in the algebraic singular locus. Resolve that locus separately and repeat. The dimension strictly decreases in each such recursion; after at most the original dimension plus one levels every real point is covered. This extra recursion is essential, for example at an isolated real singular point whose complex branches have no real generic point. Each map used is proper. The finite disjoint union of the resolved sources is therefore a proper cover of the real lift.

On each smooth source separate identically zero functions first, including G . Principalize the product of all remaining nonzero G, H by blow-ups with smooth centres. Since the factors are regular and their product has normal crossings, each factor is a monomial times a unit in the same coordinates. These blow-ups do not lose real points: the fibre above a real point of a smooth centre is a real projective space, hence nonempty. All source points still map into the square-slack lift and hence into Γ . Take finitely many coordinate neighbourhoods of the compact inverse image of $G = 0$ and shrink them to smaller boxes. Properness shows that their images contain $G \leq \varepsilon_0$ in Γ for some $\varepsilon_0 > 0$: otherwise a sequence outside the images with $G \rightarrow 0$ would have a lifted accumulation point above $G = 0$. The units have the stated bounds on these smaller boxes.

For each proposed divisor, test for a real point satisfying its equation, the strict nonvanishing of the other divisor coordinates and units, and the chart conditions. A transversal $y_i = s > 0$ with the other coordinates fixed then maps into Γ and has $G > 0$. If necessary reverse the coordinate sign. Charts with no such point are omitted; their lower-dimensional intersections have already been included in the cover. All tests are first-order real formulas. Characteristic-zero constructive resolution supplies finite algebraic equations for the centres and charts; real quantifier elimination decides the tests and selects smaller algebraic coordinate neighbourhoods. The recursion terminates by dimension, and each resolution and principalization terminates by its classical invariant. This proves both finite termination and coverage. The same argument using real uniformization [2] proves the existence statement without choosing an effective presentation. $\square$

Corollary 3.4 (The scalar order data). *For each nonrigid cluster coefficient define*

$$\beta_{r,a} = \min_{\text{recorded real divisors}} \frac{b_{r,a,i}}{a_i}, \quad \alpha = \min_{r,a} \frac{\beta_{r,a}}{k_{r,a}}. \quad (3.3)$$

Either all coefficients vanish on an observation neighbourhood, or the finite minimum α is positive rational. In the latter case $\omega_P(t) \asymp t^\alpha$. Every finite coefficient minimum is attained by an admissible transversal. The minima are independent of the chosen real cover.

Proof. For a fixed H , let β be the minimum in (3.3). On each nonexact box, $b_i \geq \beta a_i$ for $a_i > 0$, and $b_i \geq 0$ otherwise. Bounded units give $H \leq CG^\beta$. A finite cover makes this uniform, so $|c_{r,a}| \leq Ch_w^\beta$. On a minimizing real transversal, $G \asymp s^{a_i}$ and $H \asymp s^{b_i}$; inversion of the positive analytic branch gives the reverse coefficient order. Since $H = 0$ on $G = 0$, every relevant nonzero

H has $b_i > 0$. If no nonzero H meets an observation divisor, the proper cover gives an observation neighbourhood on which it vanishes.

For a monic polynomial, its maximal root modulus is comparable to $\max_{a < m} |c_a|^{1/(m-a)}$. One direction follows from the elementary symmetric functions; the other follows by summing a geometric series outside a sufficiently large multiple of that maximum. The centre belongs to every observation ball, so the diameter lies between the maximal displacement and twice that displacement. This proves the root order. Finally β is the optimal exponent in a coefficient-versus-observation inequality, an intrinsic property of the real graph. Two covers therefore give the same number. This is the usual divisorial presentation of a real Łojasiewicz order, not a bound on ordinary Taylor jets in the original coordinates. $\square$

3.2. The joint fibre and its exact diameter. Assume the nonrigid case and write $\alpha = p/q$ in lowest positive terms. For $\varepsilon > 0$ let $\Psi_P(\varepsilon, x, U, V)$ assert membership in Γ together with

$$R_{je} = \varepsilon^q U_{je}, \quad c_{r,a} = \varepsilon^{pk_{r,a}} V_{r,a}, \quad \sum_{j,e} U_{je}^2 \leq 1. \quad (3.4)$$

For complex coefficients the second equality is two real equations. The leading joint set is specified without suppressing a quantifier:

$$(U, V) \in \mathcal{J}_0(P) \iff \forall \rho > 0 \exists \varepsilon, x, U', V' : \\ 0 < \varepsilon < \rho, \quad \Psi_P(\varepsilon, x, U', V'), \quad \|(U', V') - (U, V)\|^2 < \rho^2. \quad (3.5)$$

Here the centre P is held fixed. Put $\mathcal{C}_0(P) = \text{pr}_V \mathcal{J}_0(P)$. Bounded model coordinates may be projected after closure: a convergent subsequence of them exists, so this projection loses no limiting point. The formula retains both the original inequalities and all joint coefficient relations. Algebraic saturation by ε is not a substitute for (3.5).

Theorem 3.5 (Joint specialization and root diameter). *The set $\mathcal{C}_0(P)$ is compact and is the Hausdorff limit of the rescaled joint coefficient images of the radius- t observation balls, with $t = \varepsilon^q$. Define*

$$Z_r(V) = Z \left(z^{m_r} + \sum_{a < m_r} V_{r,a} z^a \right).$$

Then

$$\omega_P(t) = C_P t^\alpha + o(t^\alpha), \quad C_P = \max_{V, V' \in \mathcal{C}_0(P)} \max_r d_\infty(Z_r(V), Z_r(V')), \quad 0 < C_P < \infty. \quad (3.6)$$

For real algebraic input there is a finite algorithm outputting the rational exponent, quantifier-free formulas for $\mathcal{J}_0, \mathcal{C}_0$, and an integer polynomial and rational isolating interval for C_P .

Proof. Corollary 3.4 bounds every V in (3.4); U is already bounded. For fixed positive ε the projected V -section is exactly the rescaled coefficient image. Its outer limit is the closed compact set in (3.5). For a point of that limit, distance to the positive section is a bounded semialgebraic function of ε with a zero subsequential limit. One-variable semialgebraic monotonicity gives limit zero. A finite net of the limit set proves the inner Hausdorff inclusion; boundedness and compactness prove the outer inclusion. This proves Hausdorff convergence, not merely the existence of individual arcs.

Continuity of monic root multisets on bounded coefficient sets now gives the limiting root set. To justify the metric, let Δ be the least separation of distinct base roots. For small t each cluster lies in its $\Delta/4$ disc. There is a within-cluster matching with error tending to zero; any cross-cluster matching has an edge of length at least $\Delta/2$. Thus optimal matchings cannot cross clusters. Inside cluster r , translation by r and dilation by t^α are exact in (3.4). The diameter

of the joint root set therefore has exactly the right-hand side of (3.6) as its limit. Its positivity follows from an attaining coefficient transversal and the root bound used in Corollary 3.4.

For completeness we specify the last algorithmic interface. Quantifier elimination applied to (3.5) produces $\mathcal{J}_0$ and then its V -projection. Introduce labelled root coordinates through all elementary symmetric identities of each monic cluster polynomial. For two such root arrays z, z' and $d \geq 0$, the condition

$$M_d(z, z') := \bigvee_{\sigma \in \mathfrak{S}_m} \bigwedge_{i=1}^m |z_i - z'_{\sigma(i)}|^2 \leq d^2 \quad (3.7)$$

is exactly $d_\infty \leq d$. Real and imaginary parts make these polynomial inequalities. Exact equality to d is M_d together with $\forall e (0 \leq e < d \Rightarrow \neg M_e)$. The maximum over the finitely many clusters is encoded by conjunction for an upper bound and by an attained equality in at least one cluster. Finally $C \geq 0$ is the diameter if every pair of coefficient/root arrays has distance at most C and some pair has distance exactly C . Compactness gives attainment. This is a first-order formula defining one real number over the real algebraic input field. Elimination followed by real-root isolation gives a polynomial over that field and an isolating interval; a norm to $\mathbb{Q}$ and further isolation give the stated integer polynomial. Ties between permutations or optimizing pairs do not require genericity. No observation-ball maximum is an input to the order construction. $\square$

3.3. Parameter dependence. A fixed-format family means a semialgebraic parameter set Θ , a semialgebraic $X \subset \Theta \times K$ with K compact and each X_θ compact, continuous semialgebraic graphs F_θ, A_θ on X , and a semialgebraic choice $P_\theta \in F_\theta(X_\theta)$. The degree and the number of observation coordinates are fixed. The graphs, including all inequalities, have finite real algebraic presentations. Statements on a stratum are uniform only on its compact subsets unless otherwise stated.

Lemma 3.6 (Relative finite order lists). *On the locus where the exact target fibre is a singleton, parameter space has a finite semialgebraic refinement with fixed cluster multiplicities and a fixed finite list of integer monomial orders and real-accessibility flags as in Definition 3.2. The source charts and units may vary. Neither the divisors nor their labels are asserted to be canonical.*

Proof. First apply semialgebraic triviality to the compact graph fibres. On each resulting parameter cell the graph projection is proper: its trivialization is a product with a compact fibre. Thus the graph is relatively closed over that cell. Next partition by the degree, multiplicities and an isolating choice for the base clusters. The distinct roots and the selected factors are Nash functions after a finite semialgebraic refinement; alternatively include their real and imaginary coordinates in a finite algebraic cover and select its branches on cells. Real semialgebraic choice supplies positive cluster-separation and observation radii; use smaller closed observation neighbourhoods so that no root meets an isolating boundary. The lifted cluster graph then has compact fibres. Apply elimination to this family graph and its square-slack lifts.

We detail why resolution can be made a finite family procedure. On an irreducible algebraic closure of a parameter cell, pass to its function field. Resolve every dominating component of the generic lifted fibre, then its successively singular subvarieties as in Proposition 3.3, and principalize the nonzero functions. This uses finitely many equations and blow-ups. Spread these equations out after clearing denominators. Delete a proper algebraic subset of the base so that the centres are smooth relative to the base, the source components and their relative singular loci have their generic dimensions, and every displayed divisor has relative normal crossings. These conditions are algebraic rank and nonvanishing conditions on finitely many charts. Their bad-fibre images are constructible by elimination and omit the generic point. Replacing those images by their proper algebraic closures gives effective exceptional base loci. The maps remain proper because the blow-ups are projective. The generic regular-locus isomorphisms and the decompositions into

regular and singular loci also spread out after this deletion. Thus the recursive coverage proof applies to each remaining fibre, not merely to the total space. Identically zero functions on a generic component are recorded before spreading; a nonzero one stays nonzero on the retained relative components after further shrinking.

Components not dominating this open base, and all excluded parameters, lie over a proper algebraic subset. Repeat the construction there. Noetherian induction on base dimension terminates. If a component or cluster label requires a finite algebraic base cover, perform the construction on that cover and then use real semialgebraic branch selection; the finite numerical order lists descend without requiring canonical labels. Finally the existence of a real generic divisor point is a first-order condition on the parameter and chart variables. Partition by these finitely many conditions. This last step discards complex-only divisors and retains lower-dimensional real specializations. It is not obtained by resolving the total space once and assuming that resolution commutes with every specialization. The resulting finite refinement has exactly the asserted lists and flags. $\square$

Theorem 3.7 (A finite real spectral atlas). *A fixed-format family admits a finite semialgebraic stratification with the following properties.*

- (i) *The exact target fibres have constant semialgebraic topological type and dimension on each stratum. On a nonidentified stratum, $\omega_\theta(t) \rightarrow \text{diam } Z(A(F_\theta^{-1}(P_\theta))) > 0$.*
- (ii) *On the identified locus, each stratum is either spectrally rigid on an observation neighbourhood or has a fixed positive rational exponent given by (3.3). Its cluster multiplicities, accessible integer order list and minimizing index set are fixed.*
- (iii) *On every nonrigid identified stratum, the joint coefficient leading sets and their root images are semialgebraic families with constant semialgebraic topological type after refinement. Their positive diameters $C(\theta)$ are Nash functions after refinement, and*

$$\omega_\theta(t) = C(\theta)t^\alpha + O_H(t^{\alpha+\nu})$$

on each compact H of a sufficiently refined stratum, for some $\nu > 0$. The exponent ν need not agree between strata.

- (iv) *All these objects have finite real-algebraic descriptions. On a subfamily for which a finite Fisher-cone normal form is proved, the stratification can additionally retain its active supports and leading-constant choices. Outside such a subfamily the constant program is semialgebraic, not asserted to be quadratic.*

Proof. The graph of the exact target fibre is a semialgebraic projection with compact fibres. Semialgebraic triviality [7] gives (i)'s partition; marking its singleton fibres is a first-order condition. For a fixed centre, decreasing compact observation balls converge to the exact fibre in Hausdorff distance by subsequence compactness. This proves the asserted nonidentified limit.

On the singleton locus apply Lemma 3.6. The finitely many integer lists and fixed multiplicities give (ii), including rigidity and fixed minimizing indices. On a stratum with $\alpha = p/q$, use (3.5) with θ held fixed throughout its quantifiers. Quantifier elimination gives semialgebraic families $\mathcal{S}_0(\theta), \mathcal{C}_0(\theta)$. In particular one must not take closure while letting θ vary: that different construction would insert limits from neighbouring strata. Apply triviality to these sets and their root graphs. The compact matching formula gives a semialgebraic positive function C , which is Nash on a finite stratification.

We explain the uniform remainder assertion. For each fixed parameter, Theorem 3.5 gives Hausdorff convergence of bounded rescaled root images to the leading root image. Its error is a semialgebraic function of (θ, t) . A finite cylindrical decomposition of its graph gives finitely many polynomial relations for its nonzero one-variable branches. The possible positive leading orders are among the finitely many slopes determined by pairs of monomials in these relations;

vanishing coefficient specializations are separated by further sign conditions. Choose ν smaller than the least positive order in each piece. The existence of $M < \infty$ and $t_0 > 0$ with error at most Mt^ν for $0 < t < t_0$ is first-order and true pointwise. Semialgebraic choice supplies such $M(\theta), t_0(\theta)$; refine until they are continuous. On compact subsets they have the required upper and positive lower bounds. The same refinement bounds rescaled root sets and separates base clusters. Diameters differ by at most twice Hausdorff distance, proving (iii). Finally a finite list of rational Fisher/KKT sign conditions can be added to any of these strata, proving (iv). This refinement connects the valuation and Fisher partitions without identifying two different kinds of walls. $\square$

Remark 3.8. The scalar part of Theorem 3.7 uses classical resolution and semialgebraic geometry. Its formulation records the joint target data needed by a spectral inverse problem. It is not a claim that exact-fibre homeomorphism type determines the exponent, that the exponent determines the leading set, or that the local leading set detects a remote fibre. Sections 4 and 6 compute these distinctions rather than suppressing them.

4. AN EXPLICIT MULTIPLICITY AND BOUNDARY ATLAS

Fix d, D, π and $\ell \geq 2d + 1$. Let $\mathfrak{M}_d$ be the centres of (2.1) with strict weights, strictly positive invertible U, V , and coprime component polynomials. No restriction is imposed on root multiplicities or on contact with $0, D$. Competitors still range over the entire closed model. Partition $\mathfrak{M}_d$ by the component assignment, multiplicity and endpoint/interior status of each distinct root. At fixed d this is a finite semialgebraic partition. On a pattern stratum write r_j, m_j for its distinct roots and multiplicities, and $a(j) \in \{1, 2\}$ for their component assignments.

For every cluster let s_j denote its first-order sum displacement. It is free at an interior root, nonnegative at 0 , and nonpositive at D . For every repeated interior cluster let $X_j \geq 0$. Define

$$p_a(z) = - \sum_{j:a(j)=a} s_j \frac{f_a(z)}{z - r_j} - \sum_{\substack{j:a(j)=a \\ r_j \in (0, D), m_j \geq 2}} X_j \frac{f_a(z)}{(z - r_j)^2}. \quad (4.1)$$

Insert these polynomials into (2.3). Let L contain the free interior sum and stochastic columns. Let D_0 contain the nonnegative endpoint sum columns (reverse the sign at D) and all X_j columns. With $W = \text{diag}(w_j/P_{je})$, define the projected Gram matrix

$$Q = D_0^\top W D_0 - D_0^\top W L (L^\top W L)^{-1} L^\top W D_0. \quad (4.2)$$

For a repeated interior cluster j , let

$$\kappa_j = \min\{v^\top Q v : v \geq 0, v_{X_j} = 1\}. \quad (4.3)$$

All matrix entries are rational in the root, channel, weight and design coordinates. Empty collections of free or cone columns have their usual zero-dimensional meanings.

Lemma 4.1 (Ordered cluster diameter). *For $m \geq 2$, the set*

$$\{y_1 \leq \dots \leq y_m : \sum_i y_i = 0, \frac{1}{2} \sum_i y_i^2 \leq X\}$$

has diameter $\sqrt{2X(m-1)/m}$ in the maximum coordinate metric.

Proof. Fix a coordinate j . Minimizing the squared norm under its value, ordering and zero sum gives

$$\max y_j = \sqrt{\frac{2X(j-1)}{m(m-j+1)}}, \quad \min y_j = -\sqrt{\frac{2X(m-j)}{mj}}.$$

For the maximum, the last $m - j + 1$ coordinates are equal and the first $j - 1$ are equal; the minimum is analogous. This also covers the zero endpoint extrema when $j = 1$ or $j = m$. Put $a = j - 1, b = m - j$. For $a, b \geq 1$, the assertion that the resulting coordinate width is at most $\sqrt{2X(m-1)/m}$ is equivalent to

$$ab(a + b + 2)^2 \geq 4(a + 1)(b + 1).$$

Writing $S = a + b \geq 2$ and $ab \geq 1$, its difference is $ab(S^2 + 4S) - 4S - 4 \geq S^2 - 4 \geq 0$. For $a = 0$ or $b = 0$ equality holds directly. An extreme vector with one coordinate of absolute value $\sqrt{2X(m-1)/m}$ and the other $m - 1$ coordinates equal attains the endpoint width against the zero vector. The diameter of a subset in the maximum metric is its largest coordinate width. This proves the formula, including possible simultaneous constraints on other clusters when it is used below. $\square$

Theorem 4.2 (Multiplicity and boundary laws). *Every centre in $\mathfrak{M}_d$ is exactly identifying. On each root pattern stratum the following whole-model assertions hold uniformly on compact subsets H .*

(a) *If there is a repeated interior cluster, the exponent is $1/2$. The complete leading root set consists of zero clusters at the simple and boundary roots and ordered vectors $y_j \in \mathbb{R}^{m_j}$ at the repeated interior roots satisfying*

$$\sum_i y_{j,i} = 0, \quad X_j = \frac{1}{2} \sum_i y_{j,i}^2, \quad \min_{\text{endpoint sums} \geq 0} v^\top Q v \leq 4. \quad (4.4)$$

The X_j in v are the displayed variances. After grouping and scaling roots by $\sqrt{t}$, the observation ball is within $O_H(\sqrt{t})$ of this set in Hausdorff distance. In particular

$$\omega_\theta(t) = C(\theta)\sqrt{t} + O_H(t), \quad C(\theta) = \max_j 2\sqrt{\frac{m_j - 1}{m_j}} \kappa_j^{-1/4}. \quad (4.5)$$

Every leading point, including diameter-extremizing pairs, is realized by stochastic alternatives.

(b) *If no interior root is repeated, the exponent is 1. For each simple interior cluster take a free real root increment, and for each boundary cluster take an ordered vector of increments with all entries nonnegative at 0 or nonpositive at D . Let s_j be their sums and use (4.1) without X terms. The compact set of these root increments with*

$$\min_\eta \|S_\theta(p_1, p_2, \eta)\|_{I, \theta}^2 \leq 4 \quad (4.6)$$

is the complete leading set $\mathcal{L}_{\text{lin}, \theta}$. Its diameter $C_{\text{lin}}(\theta)$ is positive, and

$$\omega_\theta(t) = C_{\text{lin}}(\theta)t + O_H(t^2).$$

The scaled-root Hausdorff error is $O_H(t)$.

(c) *The optimal coefficient orders of a cluster of multiplicity m , with coefficient weight k , are*

$$\beta_k = \begin{cases} 1, & r \in (0, D), \quad k = 1, \\ k/2, & r \in (0, D), \quad 2 \leq k \leq m, \\ k, & r \in \{0, D\}, \quad 1 \leq k \leq m. \end{cases} \quad (4.7)$$

The real order list, boundary and multiplicity pattern, and Fisher active supports admit a common finite semialgebraic refinement. On its square-root strata C^4 is a selected rational function of the root and experiment coordinates; the positive constants on the linear strata are algebraic semialgebraic functions. Both are Nash after refinement. Exact algebraic evaluation uses finite programs.

Proof. Lemma 2.2 gives exact identification and a uniform inverse in coefficients and stochastic parameters. The partial fractions $1/(z - r_j)$ and $1/(z - r_j)^2$ occurring in (4.1) are linearly independent at distinct roots, with their indicated pole orders. The coefficient derivative, and hence its score, is injective on these variables and the free stochastic coordinates. Thus L has full column rank and Q is positive definite whenever it is nonempty. The constrained minima exist, are positive, and have uniformly bounded minimizers on H .

We first obtain coordinates for *all* nearby competitors. Coefficient recovery and analytic coprime cluster factorization give an $O_H(t)$ error in each cluster coefficient and in the stochastic parameters. For an interior cluster, write its exact roots as

$$r_j + ts_j/m_j + \sqrt{t}y_{j,i}, \quad \sum_i y_{j,i} = 0.$$

The first coefficient bounds s_j , and the second bounds $\sum_i y_{j,i}^2$, since for centred real roots their second elementary symmetric function equals $-\frac{1}{2}\sum_i y_{j,i}^2$. At an endpoint all root displacements have the same sign. Their sum is $O_H(t)$, so each is $O_H(t)$. These observations prove the upper bounds in (4.7). For the lower bounds, translate an interior cluster by a nonzero order- t amount for $k = 1$; for $k \geq 2$ split it by $\sqrt{t}$ times a zero-sum real vector with nonzero k th elementary symmetric function. The vector with $m - 1$ equal nonzero coordinates and the last their negative sum has this property for every $k \geq 2$. Its first nonzero coefficient is of order t and the coefficient inverse makes the observation distance comparable to t . At an endpoint displace all roots by the same allowed order- t amount. This proves all lower orders, including the one-sided boundary case. In root-arc parameter ε , the interior split has $\text{ord } G = 4$, $\text{ord } H_k = 2k$, whereas translation or an endpoint arc has orders $2, 2k$. These are admissible witnesses for the real order data, and (4.7) rules out any smaller candidate.

For part (a), expansion of the exact root coordinates gives precisely (4.1) at order t . Terms involving three centred root displacements are $O_H(t^{3/2})$; all higher ones and the bounded mean and endpoint terms satisfy the same bound. Consequently

$$F_T(\xi) = P_\theta + tS_\theta + O_H(t^{3/2}), \quad h_w(F_T(\xi), P_\theta) = \frac{t}{2}\|S_\theta\|_{I,\theta} + O_H(t^{3/2}). \quad (4.8)$$

This follows by Taylor expansion of the square-root probability vector and the reverse triangle inequality; it does not require a nonzero score. Membership in the ball implies $\|S_\theta\| \leq 2 + M\sqrt{t}$. Multiply all sum, variance and stochastic first-order variables by $\rho = 2/(2 + M\sqrt{t})$ and each centred root vector by $\sqrt{\rho}$. This gives a point of (4.4) within $O_H(\sqrt{t})$ of the scaled competitor. Endpoint and simple clusters contribute only $O_H(\sqrt{t})$ in these scaled coordinates.

Conversely take a point of (4.4) and bounded minimizing endpoint and free variables. Use the displayed interior root arcs; at an endpoint realize its sum by equal one-sided displacements, and use the prescribed order- t stochastic increments. Scale first-order variables by $\rho = 1 - M'\sqrt{t}$ and centred roots by $\sqrt{\rho}$. Strict weight and channel margins and cluster separation on H make these genuine closed-model alternatives. For a sufficiently large M' , (4.8) puts them inside the observation ball. This is the reverse Hausdorff inclusion, with the same uniform error.

Homogeneity gives $X_{j,\max} = 2/\sqrt{\kappa_j}$. The marginal projection of the leading set at cluster j includes every ordered zero-sum vector with squared norm at most $2X_{j,\max}$: scale a minimizing feasible vector for $X_j = 1$ and choose the desired shape in that cluster. Lemma 4.1 therefore gives its diameter $2\sqrt{(m_j - 1)/m_j \kappa_j^{-1/4}}$. The diameter in the maximum cluster metric is the largest marginal diameter; no independence of the clusters is used. Cluster separation excludes cross-cluster optimal matchings. Diameters differ by at most twice Hausdorff distance, proving (4.5).

For part (b), every root displacement is $O_H(t)$ by the same sum argument at the endpoints and the simple-root inverse in the interior. Divide the exact displacements by t . They belong

to the stated polyhedral sign and ordering cone. Taylor expansion now has remainder $O_H(t^2)$. The first-order score controls their sums and all stochastic variables; on the one-sided endpoint cones the sums control the individual displacements. Hence the root set in (4.6) is compact. The radial argument with $\rho = 1 - Mt$ proves both Hausdorff inclusions with error $O_H(t)$. An allowed nonzero displacement gives positive diameter. This proves (b).

For (c), enumerate the subsets of the cone coordinates other than a fixed X_j . On a support A , put $v_A = -Q_{AA}^{-1}Q_{Aj}$ and require $v_A \geq 0$ and $Q_{bj} + Q_{bA}v_A \geq 0$ off the support. Its value is $Q_{jj} - Q_{jA}Q_{AA}^{-1}Q_{Aj}$. Convexity proves necessity and sufficiency. All tests and selected values are rational, and

$$C^4 = \max_j \frac{16(m_j - 1)^2}{m_j^2 \kappa_j}.$$

There are finitely many patterns and supports. For the linear law, (4.6) is an explicit quadratic inequality on a polyhedral root cone after projecting the free stochastic variables. Its coordinate maxima and minima, and hence its diameter, have finite real algebraic descriptions by (3.7) or by face optimization. Quantifier elimination gives the algebraic value at algebraic input. Stratify the resulting semialgebraic functions and refine with the order lists of Theorem 3.7. Positivity and compact-uniform coercivity ensure continuity within each fixed pattern; Nash refinement gives the final assertion. No continuity of C is asserted across a change of scaling exponent. $\square$

Example 4.3 (A multiplicity crossover). Let $x \in [0, b]$, $b < 1/2$, take $F(x) = (1/2 + x, 1/2 - x)$, and target $A_x(z) = (z - r)^2 - x$. At a fixed centre $a \in (0, b)$, differentiation of the square-root probability vector gives

$$h(F(a + u), F(a)) = \frac{|u|}{\sqrt{1 - 4a^2}} + O_a(u^2).$$

The admissible x interval therefore has endpoints $a \pm t\sqrt{1 - 4a^2} + O_a(t^2)$, and its spectral diameter is

$$\omega_a(t) = \frac{\sqrt{1 - 4a^2}}{\sqrt{a}} t + O_a(t^2).$$

At $a = 0$ it instead has leading law $\sqrt{t}$. More precisely, uniformly for λ in compact subsets of $[0, \infty)$,

$$\frac{\omega_{\lambda t}(t)}{\sqrt{t}} \longrightarrow \sqrt{\lambda + 1} - \sqrt{\max(0, \lambda - 1)}. \quad (4.9)$$

Indeed the feasible interval endpoints are $t \max(0, \lambda - 1) + O(t^2)$ and $t(\lambda + 1) + O(t^2)$, with truncation at zero, and $x \mapsto Z(A_x)$ has ordered roots $r \pm \sqrt{x}$. This example belongs to the general real-graph class of Section 3; it is not a claim about an unobserved restriction of the closed binary model. Here the linear constant really diverges at a multiplicity wall. Section 6 gives a different mechanism.

5. A RANK-DEFICIENT CUBIC STRATUM

Let $\mathfrak{F}$ denote the degree-three centres with

$$f = z(z - r)^2, \quad g = (z - s)^3, \quad 0 < s < r < D,$$

strict weights $\pi < \alpha < 1 - \pi$, strictly positive U, V , $\det V \neq 0$, and at least seven clocks. The first channel may have either rank. Put $\gamma = 1 - \alpha$ and

$$A = (r + 3s)(2r - 3s)^2, \quad B = (3D - r)s - 2rD, \quad E = \gamma A - 4\pi r^3. \quad (5.1)$$

The letter A in (5.1) is a scalar, not the target polynomial.

Theorem 5.1 (The entire cubic fibre). *The exact spectral fibre is a singleton precisely on*

$$\mathfrak{I} = \{\det U \neq 0\} \cup \{B < 0\} \cup \{E < 0\}. \quad (5.2)$$

On its complement, besides (f, g) all component-polynomial pairs are, up to permutation,

$$(f, h_c), \quad h_c = (1 - c)f + cg, \quad c_* \leq c \leq \min(c_D, \gamma/\pi), \quad (5.3)$$

where

$$c_* = \frac{4r^3}{A}, \quad b = \frac{rs}{3s - 2r}, \quad R(z) = \frac{f(z)}{g(z)}, \quad c_D = \frac{R(D)}{R(D) - 1}.$$

The interval is nonempty even at equality in either constraint, and every one of its pairs has a positive stochastic realization. Conditional on the component pair, that realization is unique.

Proof. Lemma 2.1 fixes q, K for every competitor. If U is invertible, Lemma 2.2 applies; in particular $\det K / \det K_3 = fg$ gives spectral identification directly.

If $\det U = 0$, stochasticity gives $u_1 = u_2 = u$. The second marginal $V \operatorname{diag}(\alpha, \gamma)(f, g)^\top$ has coefficient rank two. Every competitor therefore has invertible second channel and its two monic polynomials lie in the affine pencil h_c . Direct expansion gives

$$\operatorname{disc}(h_c) = c(c - 1)^2(r - s)^3(4r^3 - Ac), \quad h_c(0) = -cs^3, \quad (5.4)$$

and

$$\frac{R'(z)}{R(z)} = \frac{(2r - 3s)z + rs}{z(z - r)(z - s)}, \quad A = 4r^3 - 27s^2(r - s) < 4r^3.$$

Nonnegative roots exclude $c < 0$, and the discriminant excludes $0 < c < 1$. For $c > 1$ there is a root in (s, r) and no root in $[0, s]$. The two other admissible roots must be above r , solving $R(z) = c/(c - 1) > 1$. They exist only for $s > 2r/3$. In that case R increases to its maximum at b and then decreases to one. The double-root value is $c_* > 1$; the larger root increases from b to D as c increases from c_* to c_D . Thus the whole real-rooted pencil in $[0, D]$ is $\{0, 1\}$, with $[c_*, c_D]$ added exactly when $b \leq D$.

If a competing pair has pencil coordinates c_1, c_2 , its weights obey $\alpha'_1 c_1 + \alpha'_2 c_2 = \gamma \in (0, 1)$. One coordinate must therefore be zero, and the other is one or a value $c \geq c_*$. The latter weight is γ/c , so the floor imposes $c \leq \gamma/\pi$; the other weight is automatically at least $\alpha > \pi$. Conversely define

$$\beta_f = \alpha v_1 + \gamma(1 - 1/c)v_2, \quad \beta_h = \gamma v_2/c.$$

Both vectors are positive and $\beta_f f + \beta_h h_c = \alpha v_1 f + \gamma v_2 g$. Their entry sums are the stated weights; normalizing them and keeping first channel (u, u) gives the realization. Since the polynomials are linearly independent, coefficient functionals dual to them recover the two rank-one coefficient matrices uniquely, hence their weights and normalized marginals. Finally $b \leq D$ is equivalent to $B \geq 0$ and, on this set, $c_* \leq \gamma/\pi$ is equivalent to $E \geq 0$. This proves the entire classification, not only its open rank-one part. $\square$

5.1. Local constants and remote separation. Let d_{par} be Euclidean distance in monic coefficients and stochastic entries, minimized over the component permutation.

Lemma 5.2 (A local inverse through the walls). *For every compact $K_0 \subset \mathfrak{F}$, there exist $\rho, M, t_0 > 0$ such that a closed-model competitor whose component pair is within ρ of (f, g) , up to permutation, satisfies*

$$d_{\text{par}}(\xi, \theta) \leq M \max_j \|F_T(\xi)_j - P_{\theta, j}\|_F \quad (5.5)$$

whenever the observation error is below t_0 . These constants require no lower bound on $|\det U|$, $-B$, or $-E$. For a compact $H \subset \mathfrak{I}$ and a sufficiently small fixed ρ , the remote separation

$$\eta_H(\rho) = \min_{\theta \in H, \xi \text{ in the closed model } d_{\text{par}}(\xi, \theta) \geq \rho} h_w(F_T(\xi), P_\theta) \quad (5.6)$$

is positive whenever the minimizing set is nonempty. In particular the whole-model inverse holds uniformly on H .

Proof. We make the bounds independent of the rank of U . Normalizer recovery controls $q' - q, K' - K$ by $C\delta$, where δ is the maximum observation error. Let B_θ be the coefficient matrix of the second marginal and set $\beta = \sigma_2(B_\theta) > 0$. Writing $B' = V' \text{diag}(\alpha') F'_{\text{coef}}$, compactness gives a bound $M_0 \geq \|\text{diag}(\alpha') F'_{\text{coef}}\|$. Consequently

$$\beta/2 \leq \sigma_2(B') \leq \sigma_{\min}(V') M_0, \quad \|\text{diag}(\alpha')^{-1} V'^{-1}\| \leq \frac{2M_0}{\pi\beta}.$$

Multiplication by this bounded inverse expresses each competitor polynomial as a bounded linear combination of f, g with error $O(\delta)$. Monicity makes its coefficient sum $1 + O(\delta)$, giving $f'_b = h_{c_b} + O_{\text{coef}}(\delta)$. The prescribed small coefficient neighbourhood makes c_b close to zero and one respectively.

Near zero the endpoint inequality $f'_b(0) \leq 0$ gives $c_b \geq -C\delta$. The derivative of (5.4) at zero is $-4r^3(r-s)^3 < 0$; nonnegative discriminant gives $c_b \leq C\delta$ after absorbing a quadratic term. Near one put $v = 1 - c_b$ and depress the competing cubic using its own mean. Its two depressed coefficients satisfy

$$a = f'(s)v + O(v^2 + \delta), \quad b_0 = f(s)v + O(v^2 + \delta).$$

Real-rootedness gives $a \leq 0$ and $|b_0| \leq 2(-a)^{3/2}/(3\sqrt{3})$. Since $f(s) = s(r-s)^2 > 0$,

$$|v| \leq C\delta + Cv^2 + C(|v| + \delta)^{3/2}.$$

Choose the coefficient radius small enough to absorb the last terms. This yields $|v| \leq C\delta$. Fixed coefficient functionals dual to f, g now have an $I + O(\delta)$ value matrix on the competing pair. Applied to K' , they recover the two coefficient matrices with $O(\delta)$ error. Entry sums recover weights and normalized row and column sums recover both channels. This proves (5.5).

Here is a precise list of the uniform margins used in this argument: the smallest singular value of the normalization operator N_θ , $\sigma_2(B_\theta)$, the Gram determinant of the coefficient pair (f, g) , $4r^3(r-s)^3$, $s(r-s)^2$, the positive weight floor, and the root and positive-channel margins on K_0 . Their minima are positive and all upper coefficient and Taylor bounds are finite. Choose ρ so that the displayed absorption constants work with these common bounds, then t_0 so that $\sigma_2(B') \geq \beta/2$. This supplies a single M, ρ, t_0 , rather than a rank-two inverse whose constant could diverge at $\det U = 0$.

For (5.6), the feasible pairs (θ, ξ) form a compact set. A zero minimum would yield an exact competitor separated from the centre in parameter distance. Theorem 5.1 and its uniqueness statement exclude this on H . A sufficiently small observation ball therefore lies in the local coefficient neighbourhood, where the first part applies. This last exclusion uses identification; the local inverse does not. In particular $\eta_H(\rho)$, not the local coefficient constant, is the quantity allowed to degenerate at a remote identification wall. $\square$

5.2. The complete local Fisher fibre. Use variables $a, X, Y \geq 0$, free b, c , and five free stochastic coordinates η . The component jets are

$$p_f = -a(z-r)^2 - bz(z-r) - Xz, \quad p_g = -c(z-s)^2 - Y(z-s). \quad (5.7)$$

Let L be the seven free score columns and D_0 the three columns indexed by $0, x, y$ for (a, X, Y) . Define Q by (4.2) and

$$J(X, Y) = \min_{a \geq 0} (a, X, Y)Q(a, X, Y)^\top, \quad \kappa_x = \min_{Y \geq 0} J(1, Y), \quad \kappa_y = \min_{X \geq 0} J(X, 1). \quad (5.8)$$

Proposition 5.3 (Coercivity and finite supports). *Throughout $\mathfrak{F}$, L has rank seven, $Q_{00} > 0$, and κ_x, κ_y are positive and locally Lipschitz. For every compact $K_0 \subset \mathfrak{F}$,*

$$\|S_\theta(a, b, c, \eta, X, Y)\|_{L, \theta} \geq c_{K_0} \|(a, b, c, \eta, X, Y)\| \quad (a, X, Y \geq 0).$$

Each κ_i is given by four finite support tests. For $A \subseteq \{0, x, y\} \setminus \{i\}$ with Q_{AA} positive definite, put

$$v_A = -Q_{AA}^{-1}Q_{Ai}, \quad k_{i,A} = Q_{ii} - Q_{iA}Q_{AA}^{-1}Q_{Ai}.$$

The test is $v_A \geq 0$ and $Q_{ji} + Q_{jA}v_A \geq 0$ for inactive j . The empty support has value Q_{ii} . At least one test succeeds, and all successful tests give κ_i .

Proof. Every feasible jet is realized by roots

$$ta, \quad r + tb/2 \pm \sqrt{tX}, \quad s + tc/3 + \sqrt{t}y_i, \quad \sum_i y_i = 0, \quad \frac{1}{2} \sum_i y_i^2 = Y,$$

and order- t stochastic increments. Their observation increment is $tS + O(t^{3/2})$. A zero score, combined with the local inverse of Lemma 5.2, forces both component jets and η to vanish. Evaluation and coefficient comparison in (5.7) then force every variable to be zero. Compactness of the feasible unit sphere gives coercivity, uniformly on K_0 . It also proves independence of the free columns, $Q_{00} > 0$, and existence, positivity and boundedness of the constrained minimizers.

Choose a minimizer with minimal positive support in the projected score columns. Those supported columns are independent: a nonzero linear dependence would allow a signed change of the positive coefficients, preserving the represented vector, until one coefficient vanished. On an independent support the stationarity and inactive inequalities are exactly the stated tests. Conversely their KKT conditions prove optimality by convexity. This handles a singular full Q without inverting it. Uniformly bounded minimizing variables and smooth dependence of the score show local Lipschitz dependence of the minimum by evaluating each problem at a minimizer of the other. $\square$

Theorem 5.4 (Local normal form and whole-model restriction). *For every centre of $\mathfrak{F}$ the local leading root set is*

$$(\{0\}, \{-\sqrt{X}, \sqrt{X}\}, Z(z^3 - Yz - Z)), \quad X, Y \geq 0, \quad J(X, Y) \leq 4, \quad 27Z^2 \leq 4Y^3. \quad (5.9)$$

On compact subsets of $\mathfrak{F}$, scaled local observation balls converge to this set with Hausdorff error $O(\sqrt{t})$. Hence

$$\omega_{\theta, \text{loc}}(t) = C_{\text{loc}}(\theta)\sqrt{t} + O(t), \quad C_{\text{loc}} = \max\{\sqrt{2}\kappa_x^{-1/4}, 2\sqrt{2/3}\kappa_y^{-1/4}\}. \quad (5.10)$$

The error is compact-uniform, and C_{loc} is positive and locally Lipschitz on all of $\mathfrak{F}$. On compact subsets of $\mathfrak{J}$ these statements hold for the whole-model modulus. The support tests, (5.2), and $9\kappa_y - 16\kappa_x$ give a finite semialgebraic partition. On its constant choices,

$$C_{\text{loc}}^4 = \max\{4/\kappa_x, 64/(9\kappa_y)\}$$

is one rational function, and C_{loc} is Nash after refinement.

Proof. The local inverse bounds coefficients and stochastic increments by $O(t)$ for every local competitor. Its first component has exact roots $ta, r + tb/2 \pm \sqrt{tX}$ with bounded a, b, X and $a, X \geq 0$. Its second component has exact roots $s + tc/3 + \sqrt{t}y_i$ with $\sum_i y_i = 0$ and bounded squared sum. Put $Y = \frac{1}{2} \sum_i y_i^2$ and $Z = \prod_i y_i$. The real-cubic condition is exactly $Y \geq 0, 27Z^2 \leq 4Y^3$.

Expansion gives $F = P + tS + O(t^{3/2})$ and $h_w = t\|S\|_I/2 + O(t^{3/2})$ uniformly on these bounded coordinates. The radial contraction of all first-order variables, together with $Z \mapsto \rho^{3/2}Z$, proves the outer Hausdorff inclusion. Conversely choose bounded minimizing free and endpoint variables and realize the root arcs in the proof of Proposition 5.3, using $\rho = 1 - M\sqrt{t}$. They lie in the observation ball and give the inner inclusion. Thus the inequality boundary and all joint cubic coefficient relations are realized, not only contained in an outer tangent cone.

Homogeneity gives $X_{\max} = 2/\sqrt{\kappa_x}$ and $Y_{\max} = 2/\sqrt{\kappa_y}$. The double-cluster diameter is $\sqrt{X_{\max}}$. Lemma 4.1 with $m = 3$ gives the triple diameter $2\sqrt{Y_{\max}/3}$, also attained by $(-2h, h, h)$ and $(-h, -h, 2h)$ with $h = \sqrt{Y_{\max}/3}$. The maximum of marginal diameters yields (5.10). The remaining assertions follow from Proposition 5.3, cluster separation, and the positive remote exclusion in Lemma 5.2. In particular the local formula remains valid at $B = 0$ or $E = 0$, but its identification with the whole-model modulus does not. $\square$

Example 5.5 (The earlier rational phase values). For $D = 1$, $\pi = 1/10$, $r = 3/5$, $s = 1/5$, $\alpha = 2/5$, $u_1 = u_2 = (1/3, 2/3)^\top$, $v_1 = (1/4, 3/4)^\top$, $v_2 = (3/4, 1/4)^\top$, clocks $(6/5, 7/5, 8/5, 9/5, 2, 5/2, 3)$ and equal weights, $B = -18/25$. The empty supports are admissible, the double cluster dominates, and exact rational evaluation gives

$$C^4 = \frac{78550423766145732273576408862601811503774391446}{385279214348552940073664077156875}, \quad 3778.707 < C < 3778.708.$$

Changing only u_2 to $(2/5, 3/5)^\top$ gives the triple-dominant phase, $59.970 < C < 59.971$. The exact arithmetic records for both values are retained in the archival companion. The next section evaluates a wall datum and its different entrance constant, rather than interpreting these large values as evidence of an unspecified wall blow-up.

Corollary 5.6 (Centre-known local risk). *Let $H \subset \mathfrak{I}$ be compact, fix $0 < c \leq 1/(8\sqrt{2})$, and observe independent categorical samples at the design clocks with $n_j/N \rightarrow w_j$ uniformly on H (integer rounding suffices). For squared spectral loss on the specified neighbourhood $h_w(F_T(\xi), P_\theta) \leq cN^{-1/2}$, with the centre known, the minimax risk $\mathcal{R}_N(\theta, c)$ satisfies uniformly on H*

$$\frac{3c}{32}C_{\text{loc}}(\theta)^2 - O_H(N^{-1/4}) \leq \sqrt{N} \mathcal{R}_N(\theta, c) \leq cC_{\text{loc}}(\theta)^2 + O_H(N^{-1/4}).$$

Proof. A constant decision in the specified spectral ball has loss at most its squared diameter. For a diameter pair in the ball, product Hellinger distance is at most $1/4 + O_H(N^{-1})$ with integer rounding. Total variation is at most this distance. A two-point test gives squared-risk lower bound $(3/32 - O_H(N^{-1}))$ times the squared diameter. Substitute (5.10). This is an oracle local statement; it makes no assertion of unknown-centre adaptation or honest confidence sets across an identification wall. $\square$

6. A REMOTE BRANCH AT THE STOCHASTIC WEIGHT WALL

Fix r, s, D with $B > 0$, fix positive u, v_1, v_2 with $v_1 \neq v_2$, and set $U = (u, u)$, $V = (v_1, v_2)$. Assume

$$\gamma_0 = \pi c_*, \quad \alpha_0 = 1 - \pi c_*, \quad \pi < \alpha_0 < 1 - \pi. \quad (6.1)$$

The design is fixed. Write θ_δ for the original (f, g) centre with weight $\alpha = \alpha_0 + \delta$. Then $E = -A\delta$, so $\delta > 0$ is the identified side. Set

$$a = \frac{4rs}{r + 3s}, \quad b = \frac{rs}{3s - 2r}, \quad h_* = (1 - c_*)f + c_*g = (z - a)(z - b)^2. \quad (6.2)$$

Direct substitution gives $s < a < r < b < D$. At $\delta = 0$ the exact component pairs are precisely (f, g) and (f, h_*) . The remote pair has weight π on h_* and second-channel columns

$$v_f^* = \frac{\alpha_0 v_1 + (\gamma_0 - \pi)v_2}{1 - \pi}, \quad v_h^* = v_2. \quad (6.3)$$

Let ξ_* denote this parameter point. Its observed array equals $P_0 = P_{\theta_0}$. The sorted base and remote spectra are

$$(0, s, s, s, r, r), \quad (0, a, r, r, b, b),$$

and their distance is

$$D_* = \max(r - s, b - r) > 0. \quad (6.4)$$

In particular, a zero-length additional pencil interval at the wall already produces a positive exact-fibre diameter.

6.1. The constrained entrance program. At ξ_* the full first-order feasible coefficient and stochastic cone is described by

$$\begin{aligned} p_f &= -a_0(z - r)^2 - b_0z(z - r) - Xz, \\ p_h &= -c_0(z - b)^2 - d_0(z - a)(z - b) - Y(z - a), \\ a_0, X, Y, k &\geq 0. \end{aligned} \quad (6.5)$$

Here k is the derivative of the weight on h_* ; b_0, c_0, d_0 and four channel increments are free. Denote the score by S_* , with weights $(1 - \pi, \pi)$ and columns (6.3), and put $B_0 = \frac{d}{d\delta} P_{\theta_\delta} \Big|_{\delta=0}$. The norm is Fisher at P_0 . Define

$$\mu_E = \frac{1}{2} \min_{v \in \mathcal{T}_*} \|S_* v - B_0\|_{I,0}, \quad (6.6)$$

where $\mathcal{T}_*$ is the cone (6.5) with its free variables. It is a finite-dimensional cone, not a closure of unspecified observation-ball optimizers.

Lemma 6.1 (Positive entrance separation). *The score has no nonzero zero on $\mathcal{T}_*$, and B_0 is not in its image. Thus $\mu_E > 0$ and its minimum is attained. After eliminating seven free variables, μ_E^2 is given by sixteen independent-support KKT tests on four nonnegative variables. For algebraic input this is an exact real algebraic number.*

Proof. Suppose $S_* v = B_0$. Normalizer injectivity (Lemma 2.1, or its differentiated linear equations) gives $\dot{q} = f - g$ and $\dot{K} = \dot{K}_{\theta_\delta}|_0$. The second marginal of this derivative lies in $\text{span}(f, g)$. Since the coefficient vectors of the remote two components and their second-channel weights are independent, differentiating their marginal factorization gives $p_f, p_h \in \text{span}(f, g)$. Both have degree below three, hence are multiples of $g - f$.

At the f endpoint of the pencil, the endpoint and discriminant inequalities in the proof of Lemma 5.2 force the first-order pencil displacement to be zero. Thus $p_f = 0$. At h_* , (5.4) has positive derivative with respect to c at c_* , so feasible $p_h = \dot{c}(g - f)$ requires $\dot{c} \geq 0$. Differentiating the monic normalizer now gives

$$\pi \dot{c} + c_* k = -1, \quad (6.7)$$

which is impossible because $k, \dot{c} \geq 0$. For a zero score the same argument gives right-hand side zero, so $k = \dot{c} = 0$. All component jets vanish, and dual coefficient functionals then force the four channel derivatives to vanish. The basis in (6.5) is independent within each component, so all variables are zero. Compactness of the intersection of $\mathcal{T}_*$ with the unit sphere gives coercivity. It implies boundedness of minimizing sequences in (6.6), hence attainment and strict positivity.

For an explicit program, let $\mathbf{L}_*$ be the seven free columns, $\mathbf{D}_*$ the four nonnegative columns, and let $\mathcal{P}$ be the Fisher-orthogonal projection off $\text{im } \mathbf{L}_*$. Put

$$Q_* = \mathbf{D}_*^\top W \mathcal{P} \mathbf{D}_*, \quad g_* = \mathbf{D}_*^\top W \mathcal{P} B_0, \quad b_* = B_0^\top W \mathcal{P} B_0.$$

The already proved coercivity implies that L_* has full rank. Then

$$4\mu_E^2 = \min_{v \geq 0} \{v^\top Q_* v - 2g_*^\top v + b_*\}. \quad (6.8)$$

For each of the sixteen supports A , with $Q_{*,AA}$ positive definite, set $v_A = Q_{*,AA}^{-1}g_{*,A}$, require $v_A \geq 0$ and $(Q_* v - g_*)_j \geq 0$ off A , and take value $b_* - g_{*,A}^\top Q_{*,AA}^{-1}g_{*,A}$. The empty support is included. The minimal-positive-support argument of Proposition 5.3 proves completeness even if Q_* is singular. All entries are rational in the wall coordinates; selecting the positive square root gives the exact algebraic output. $\square$

Lemma 6.2 (Local coefficient control at the remote point). *There is a coefficient and stochastic neighbourhood of ξ_* in the closed model on which*

$$d_{\text{par}}(\xi, \xi_*) \leq C \|F_T(\xi) - P_0\|.$$

Every competitor at coefficient distance $O(\delta)$ has bounded coordinates of (6.5) at scale δ , with observation expansion $P_0 + \delta S_ v + O(\delta^{3/2})$. Conversely each bounded cone vector has a realizing stochastic root arc with that expansion.*

Proof. The coefficient tangent cone is exactly (6.5). Indeed, at the endpoint 0 a simple root has nonnegative displacement; at either interior double root its centred squared separation is nonnegative; the two simple/mean coordinates are unrestricted. The weight on h_* is at its lower floor, so its derivative is nonnegative. All channels and the other weight are interior. Analytic factorization into the separated simple and double clusters gives these coordinates for normalized coefficient secants. Conversely the root arcs with simple and mean shifts of order δ , double splits $\pm\sqrt{\delta X}$ and $\pm\sqrt{\delta Y}$, and weight $\pi + \delta k$, realize every cone vector.

If the asserted inverse failed, normalize a sequence of coefficient and stochastic displacements by their norm. A subsequential limit is a nonzero vector of this tangent cone with zero score, contrary to Lemma 6.1. For the stronger expansion, the coefficient bound controls simple and mean shifts by $O(\delta)$ and the squared double separations by $O(\delta)$. Extract them as exact coordinates. Expansion of their cubic products gives (6.5) and a bounded $O(\delta^{3/2})$ remainder. The rational observation map and the square-root probabilities have bounded derivatives in this neighbourhood. The converse follows from the displayed realizing arcs; all roots remain in $[0, D]$ because a, r, b are strictly interior. $\square$

6.2. The two-scale law. Choose disjoint small closed coefficient and parameter neighbourhoods N_0, N_* of the two exact representatives at the wall, modulo component permutation, with ξ_* in the relative interior of N_* . Define

$$d_{\text{rem}}(\delta) = \min_{\xi \in N_*} h_w(F_T(\xi), P_{\theta_\delta}). \quad (6.9)$$

Different sufficiently small choices give the same minimum for all sufficiently small $\delta > 0$; their boundaries have a positive observation separation from P_0 .

Theorem 6.3 (Weight-wall transition). *Under (6.1) and $B > 0$,*

$$d_{\text{rem}}(\delta) = \mu_E \delta + O(\delta^{3/2}) \quad (\delta \downarrow 0). \quad (6.10)$$

The local constant $C_{\text{loc}}(\theta_\delta)$ has the finite positive limit $C_{\text{loc}}(\theta_0)$; neither κ_x nor κ_y vanishes at this wall. For a fixed $\lambda > 0$ the whole-model modulus has the following two distinct regimes:

$$0 < \lambda < \mu_E : \quad \omega_{\theta_\delta}(\lambda\delta) = C_{\text{loc}}(\theta_0)\sqrt{\lambda\delta} + O_\lambda(\delta), \quad (6.11)$$

$$\lambda > \mu_E : \quad \omega_{\theta_\delta}(\lambda\delta) = D_* + O_\lambda(\sqrt{\delta}). \quad (6.12)$$

The exact entrance curve, including equality for the closed observation ball, is $t = d_{\text{rem}}(\delta)$. The transition layer left unresolved by the first-order threshold alone has width $O(\delta^{3/2})$ in t .

On the nonidentified side, for $\varepsilon > 0$ sufficiently small,

$$\omega_{\theta_{-\varepsilon}}(0) = \max \left\{ r - s, b - r + \sqrt{\frac{\chi \varepsilon}{\pi}} + O(\varepsilon) \right\}, \quad \chi = \frac{f(b) - g(b)}{b - a} > 0. \quad (6.13)$$

If $b - r < r - s$, the exact diameter remains equal to $r - s$ initially; if $b - r \geq r - s$, it opens above D_* at the displayed square-root rate.

Proof. At $\delta = 0$, Theorem 5.1 gives exactly the two representatives and no others. Compactness gives a positive observation gap outside $N_0 \cup N_*$. Since ξ_* itself has distance $O(\delta)$ from P_{θ_δ} , a minimizer in (6.9) has observation $O(\delta)$ from P_0 , lies away from the boundary of N_* , and has coefficient distance $O(\delta)$ by Lemma 6.2. Its exact scaled root and stochastic coordinates are bounded in $\mathcal{T}_*$. Taylor expansion at the common positive probability array gives, uniformly on these bounded coordinates,

$$h_w(F_T(\xi), P_{\theta_\delta}) = \frac{\delta}{2} \|S_* v - B_0\|_{I,0} + O(\delta^{3/2}).$$

This is the lower bound in (6.10). A minimizing cone vector in (6.6), realized by the arcs of Lemma 6.2, gives the upper bound. This also proves independence from the selected neighbourhood for small δ .

Proposition 5.3 and Theorem 5.4 apply on a compact path segment including θ_0 ; the base pair stays locally coercive there. Hence C_{loc} is positive and locally Lipschitz through the wall. If $\lambda < \mu_E$, the remote neighbourhood is excluded from the radius- $\lambda\delta$ ball for small δ . The positive exterior gap excludes every other competitor. The entire ball is therefore the base local ball, and (5.10) gives (6.11). Replacing $C_{\text{loc}}(\theta_\delta)$ by $C_{\text{loc}}(\theta_0)$ changes the expression by only $O(\delta^{3/2})$.

If $\lambda > \mu_E$, a remote minimizer belongs to the ball, as does the base centre. Every competitor lies in one of the two neighbourhoods. The respective coefficient inverses put its roots within $O_\lambda(\sqrt{\delta})$ of the corresponding exact spectrum at the wall. The triangle inequality for bottleneck distance then gives both bounds in (6.12). Compact attainment makes remote membership exactly the inequality $t \geq d_{\text{rem}}(\delta)$, including equality. The stated layer width follows from (6.10); no universal answer at $t = \mu_E \delta$ is inferred from a first-order expansion.

For $\theta_{-\varepsilon}$, the extra exact pencil interval is $[c_*, c_* + \varepsilon/\pi]$, since $c_D > c_*$ is fixed. At $c = c_* + \sigma$, expand the polynomial at $z = b + u$:

$$h_c(b + u) = (b - a)u^2 + u^3 + \sigma(g(b) - f(b)) + O(\sigma|u|).$$

Since $R(b) > 1$, the coefficient χ is positive, and the two roots near b are $b \pm \sqrt{\chi\sigma} + O(\sigma)$. The third root stays in (s, r) . The larger root increases with c , by the monotonic branch of R above its maximum. Comparing the increasing root lists shows that the distance from the base spectrum is exactly $\max(r - s, b_+(c) - r)$. Distances between two remote spectra are $O(\sqrt{\varepsilon})$ and hence smaller than D_* for sufficiently small ε . The whole exact-fibre diameter is therefore attained against the base at the upper pencil endpoint. This proves (6.13) and the last two assertions. $\square$

Remark 6.4 (The critical ray and effective higher orders). At $t = \mu_E \delta$, remote membership is decided by the sign of $d_{\text{rem}}(\delta) - \mu_E \delta$, not by choosing either strict regime by convention. For algebraic input, (6.9) has a compact first-order minimum formula. Elimination and one-variable real algebraic branch isolation determine its Puiseux expansion. If the difference is not identically zero, its first nonzero coefficient decides membership for all small positive δ ; if it is identically zero the remote branch belongs to the closed ball. This gives an effective determination of the critical ray while keeping the universal first-order statement honest about its transition layer.

Remark 6.5 (The geometric wall). At $B = 0$ with $E > 0$, the remote double root is $b = D$ and the additional pencil interval again consists of one point with a positive spectral distance from

the base. From the nonidentified side, put $\ell_0 = D - b$. Taylor expansion at the strict maximum of R gives

$$c_D - c_* = \frac{-R''(b)}{2(R(b) - 1)^2} \ell_0^2 + O(\ell_0^3), \quad \ell_0 = \frac{B}{3s - 2r}.$$

The coefficient is positive and the weight floor is inactive near this wall. Thus this pencil interval is born with quadratic length in the geometric clearance, unlike the linear length at the weight wall. The local constants in (5.10) remain finite on compact subsets of $\mathfrak{F}$ here as well. A two-scale entrance program at the endpoint has different one-sided root coordinates from (6.5); the explicit entrance theorem above concerns the weight wall, not that different cone.

Example 6.6 (An exact wall datum). Take $D = 1$, $r = 3/5$, $s = 11/20$, $\pi = 1/10$, the columns and clocks of the rank-one rational example in Section 5, and equal design weights. Then

$$\alpha_0 = \frac{547}{675}, \quad c_* = \frac{256}{135}, \quad a = \frac{44}{75}, \quad b = \frac{11}{15}, \quad B = \frac{3}{25}, \quad D_* = \frac{2}{15}, \quad \chi = \frac{3}{64},$$

and $v_f^* = (1457, 3403)^\top / 4860$. Exact rational evaluation of the two base programs and the remote program gives

$$5975.429249985072 < C_{\text{loc}}(\theta_0) < 5975.429249985073, \\ 2.131024914 \cdot 10^{-6} < \mu_E < 2.131024915 \cdot 10^{-6}.$$

In both base programs, and in the four-variable remote program, the empty positive support is admissible. The ancillary exact record contains the rational Gram matrix, positive squared entrance constant, fourth power of the local constant, and rational isolating intervals. The decimal displays are consequences of rational power comparisons, not optimization tolerances. They illustrate a well-defined but poorly conditioned local law and a distinct, very small remote entrance scale. They do not certify the universal quantifiers in the preceding theorems.

Three stability questions. Root multiplicity and one-sided boundary constraints determine the valuation weights in Theorem 4.2. Exact identification in the cubic family is determined instead by (5.2). Conditioning of a fixed local law is measured by the positive Fisher programs. The weight-wall theorem shows that the second can fail while the third stays finite and the local square-root exponent persists. In contrast, Example 4.3 changes the exponent itself and makes the linear constant diverge. These are different strata and different limiting operations in the real spectral atlas.

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PROJECTIVE POLYNOMIAL OBSERVATIONS: REAL VALUATIONS AND STRATIFIED SPECTRAL LAWS

QIAN QI

ABSTRACT. We study polynomial spectra observed through stochastic channels and an unknown scalar normalization. Real monomialization of the observation–spectral graph gives finite valuative data for the intrinsic root exponent; real weighted specialization retains the joint leading spectral set and its inequality constraints. For the binary boundary cubic family with an interior double root and an interior triple root, we classify the entire exact spectral fibre across both ranks of the first channel. The identifiable set is an explicit open semialgebraic set, including regions isolated by the weight constraint. On each of its compact subsets the whole-model modulus has the uniform expansion $C(\theta)\sqrt{t} + O(t)$. A finite active-set calculation gives a semialgebraic partition on which C^4 is rational, and constructs extremizing stochastic alternatives. The constant is continuous across identified rank changes. The same expansion yields uniform squared-risk bounds on specified shrinking oracle neighbourhoods. Additive normalization, clock geometry, quotient recovery, and structured singular perturbation results provide the complementary global theory.

1. INTRODUCTION

A singular observation map can identify an unordered polynomial spectrum even when it fails to identify unrestricted coefficient directions. Stochastic inequalities distinguish these two assertions: an ambient invisible direction need not have an admissible nonconstant representative. The intrinsic spectral law must consequently be computed on the real observation–spectral graph. This article passes from finite algebraic data for that graph to an explicit family classification, a uniform normal form, and a finite formula for its leading constant.

For a compact semialgebraic model with observation map F and monic target polynomial A_x , fix a datum P whose exact fibre has a single spectral value. Let $c_{r,a}$ be the coefficient of degree a in the monic cluster at a root r of multiplicity m_r . On a compact real lift put $G = h_w(F(x), P)^2$ and $H_{r,a} = |c_{r,a}|^2$. Theorem 3.3 computes the nonrigid exponent as

$$\alpha_P = \min_{r,a,i} \frac{b_{r,a,i}}{a_i(m_r - a)},$$

where $a_i, b_{r,a,i}$ are integer orders on accessible real divisors of a simultaneous monomialization. The input is the defining graph, not optimized coefficient envelopes. Proposition 3.4 retains the simultaneous weighted coefficient specialization and outputs its exact variational diameter. This construction includes singular observation charts and positive-dimensional parameter fibres.

The principal family result is more explicit. Take $f = z(z-r)^2$, $g = (z-s)^3$, with $0 < s < r < D$, strict weights and positive binary channels, the second channel invertible and the first of either rank. Define

$$B = (3D - r)s - 2rD, \quad E = (1 - \alpha)(r + 3s)(2r - 3s)^2 - 4\alpha_* r^3.$$

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Theorem 4.1 shows that the entire closed-model spectral fibre is a singleton exactly on

$$\mathfrak{I} = \{\det U \neq 0\} \cup \{B < 0\} \cup \{E < 0\}.$$

It describes all additional component polynomials on the complement. Thus channel rank, the active root boundary and the weight floor enter one necessary-and-sufficient classification. The earlier open root region $s < 2rD/(3D - r)$ is only one part of $\mathfrak{I}$; the region $B \geq 0, E < 0$ is identified by the weight constraint instead.

On every compact $H \in \mathfrak{I}$, Theorem 5.2 identifies the complete leading spectral set and proves

$$\sup_{\theta \in H} |\omega_\theta(t) - C(\theta)\sqrt{t}| \leq C_H t.$$

Here competitors range over the whole closed stochastic model, not merely over the displayed family. The proof gives a quantitative Hausdorff comparison with the cluster polynomials

$$z, \quad z^2 - X, \quad z^3 - Yz - Z, \quad X, Y \geq 0, \quad J_\theta(X, Y) \leq 4, \quad 27Z^2 \leq 4Y^3.$$

The constrained Fisher cost J_θ retains all seven free nuisance directions and the one-sided endpoint motion. Necessity follows from a uniform coefficient inverse; sufficiency follows from actual root arcs with a controlled radial correction. Neither part is inferred from a pointwise Puiseux expansion. In particular the error remains uniform when the first channel changes rank inside H .

Theorem 6.1 eliminates the two remaining nonnegative least-squares problems by four explicit support tests apiece. On a finite semialgebraic partition, the fourth power of the leading constant is the rational function

$$C^4 = \max\{4/\kappa_x, 64/(9\kappa_y)\},$$

with the switching sign $9\kappa_y - 16\kappa_x$. The constant is continuous across the partition and analytic on a finite refinement. This is a family-level formula, rather than an existence statement for the coefficient-envelope orders of a single datum.

Relation to classical results. Uniformization and resolution are classical constructions [1, 2]; so is the calculation of a Łojasiewicz order from ratios of divisor orders [3]. Semialgebraic elimination and stratification [13, 9], and general definable metric regularity [4], provide the appropriate setting. The broader filtration perspective in [5] is contextual rather than a required technical input. The contributions specific to the present experiment are the real probability graph with its joint cluster weights, the exact stochastic fibre classification, and the uniform spectral normal form with the finite Fisher phase calculation. The active-set calculation itself is elementary convex quadratic minimization; its role is to make the complete leading invariant explicit in the original statistical coordinates. We do not claim a new resolution algorithm, a universal ordinary-jet cutoff, or a classification of unrelated root-multiplicity patterns.

Structure and scope. The main text proceeds from the experiment to finite real-valuative data, then to the family classification, uniform normal form and finite phase formulas. Direct rank-one calculations are preserved as complementary proofs, followed by the global additive and clock results, general intrinsic moduli, quotient and inference theory, and structured perturbation results. These are organized by mathematical dependence rather than by the dates of their derivation. Every previously active mathematical and bibliography module remains in the complete manuscript. Fixed-datum constants, uniform constants on compact identifiable families, and boundary-uniform global bounds have distinct quantifiers throughout. The risk statements concern specified shrinking oracle neighbourhoods, not an unknown-centre adaptive procedure.

2. THE POLYNOMIAL EXPERIMENT

Fix $d \geq 1$, $2 \leq k \leq \min(m_1, m_2)$, an interval $[L, D]$, distinct clocks $D < T_1 < \dots < T_\ell$ with $\ell \geq d+1$, and $0 < \alpha_* < 1/k$. The columns of $U \in \mathbb{R}^{m_1 \times k}$ and $V \in \mathbb{R}^{m_2 \times k}$ are probability vectors. Let $\alpha_b \geq \alpha_*$ and $\sum_b \alpha_b = 1$. Put

$$\begin{aligned} f_b(z) &= \prod_{a=1}^d (z - \lambda_{ba}), \quad \lambda_{ba} \in [L, D], & q(z) &= \sum_b \alpha_b f_b(z), \\ K(z) &= U \operatorname{diag}(\alpha_b f_b(z)) V^\top, & P(T) &= K(T)/q(T). \end{aligned} \quad (2.1)$$

The closed parameter set $\mathcal{K}_d$ is compact; singular channels, boundary roots, and repeated roots are included. Every $q(T_j)$ and every component probability $\alpha_b f_b(T_j)/q(T_j)$ has uniform positive bounds. A common positive scalar detector disappears on normalization; a component-dependent detector defines a different experiment. Write $F(\theta) = (\operatorname{vec} P_\theta(T_j))_j$. The target is the unordered multiset Λ of all kd roots, with no component partition in the output. Set

$$d_\infty(\Lambda, \Lambda') = \min_\pi \max_a |\lambda_a - \lambda'_{\pi(a)}|, \quad \delta = \max_j \|P(T_j) - P'(T_j)\|_F.$$

For real multisets this is the maximum discrepancy of increasing orders. Unless otherwise stated, uniform constants depend only on the displayed experiment constants. Internal separation is charged only when specified.

Let $W = (T_j^r)_{j=1, \dots, \ell; r=0, \dots, d}$ and let Ω project orthogonally onto $\operatorname{col}(W)^\perp$. Write $q(z) = z^d + \sum_{r < d} a_r z^r$. For $c \in \mathbb{R}^d$, put $r_c(z) = \sum_{r < d} c_r z^r$ and define

$$\begin{aligned} \mathcal{N}_{PC} &= (\Omega \otimes I)(r_c(T_j) \operatorname{vec} P(T_j))_j, & \tau(P) &= \sigma_{\min}(\mathcal{N}_P), \\ b_P &= -(\Omega \otimes I)(T_j^d \operatorname{vec} P(T_j))_j, & \kappa &= \sigma_k(U) \sigma_k(V). \end{aligned} \quad (2.2)$$

Here I has size $m_1 m_2$, and the norms are Euclidean coefficient and product Frobenius norms. The equation is $\mathcal{N}_P a = b_P$. Thus τ is observable; κ is a representation quantity.

For positive w_j summing to one, use the Hellinger convention

$$h_w(P, Q)^2 = \sum_j w_j \sum_e (\sqrt{P_{je}} - \sqrt{Q_{je}})^2.$$

For a compact admissible model X , with monic target polynomial A_x , write $B_{P,X}(t) = \{x \in X : h_w(F(x), P) \leq t\}$ and $\omega_{P,X}(t) = \operatorname{diam}\{Z(A_x) : x \in B_{P,X}(t)\}$. In the stochastic experiment $X = \mathcal{K}_d$ and $A_x = \prod_b f_b$. The omission of X denotes this whole-model modulus, never a selected parametric slice. For complex targets $|z|$ in the bottleneck metric means Euclidean modulus. The squared local minimax risk discussed below is over a ball of radius $cN^{-1/2}$ centred at a specified datum P ; a decision rule for this local experiment may depend on the centre.

3. FINITE NEWTON DATA ON THE REAL OBSERVATION GRAPH

Let X be compact semialgebraic, let $F : X \rightarrow \mathcal{P}$ be continuous semialgebraic, and let A_x be a continuous semialgebraic monic polynomial of degree n . Fix $P \in F(X)$ and suppose $A_x = A_0$ on $F^{-1}(P)$. The competitors in this section range over all of X . A restriction to a compact local model will always be denoted X_{loc} instead. Write $A_0(z) = \prod_r (z - r)^{m_r}$. Near the exact fibre, unique monic cluster factors give

$$A_{r,x}(r+z) = z^{m_r} + \sum_{a < m_r} c_{r,a}(x) z^a, \quad k_{r,a} = m_r - a.$$

The factor equations, with isolating discs fixed, define semialgebraic functions by unique choice. Their analyticity in the original polynomial coefficients follows from the invertible multiplication differential. These are separate assertions; analyticity alone would not imply semialgebraicity.

Put $R_{je} = \sqrt{w_j}(\sqrt{F(x)_{je}} - \sqrt{P_{je}})$ and introduce the compact graph of $x, F, A, (c_{r,a}), R$ over a sufficiently small closed observation neighbourhood. On this graph set

$$G = \sum_{j,e} R_{je}^2 = h_w(F(x), P)^2, \quad H_{r,a} = (\Re c_{r,a})^2 + (\Im c_{r,a})^2. \quad (3.1)$$

Thus G and $H_{r,a}$ are polynomial functions on the graph coordinates, even at zero observed cells. We use their real graph, not its unrestricted complexification.

Definition 3.1 (Real monomial data). A real monomial cover of the graph near $G = 0$ is a finite collection of relatively compact coordinate boxes on smooth real manifolds, with maps into the graph, which cover an observation neighbourhood and in which

$$G = u \prod_i |y_i|^{a_i}, \quad H_{r,a} = v_{r,a} \prod_i |y_i|^{b_{r,a,i}}. \quad (3.2)$$

A displayed H may instead be identically zero on a box. Nonzero units are positive and bounded above and below on the closures of smaller boxes. The exponents are nonnegative integers. Only coordinate divisors with $a_i > 0$ and a real generic point in a box are recorded. In particular every recorded transversal is an admissible real arc. The maps need not be injective, and all lower-dimensional pieces are included.

Lemma 3.2 (Construction of the data). *The graph in (3.1) admits real monomial data. With real algebraic input it admits an effective finite construction. The input consists of a polynomial formula for X and the graphs of F, A , the algebraic datum P , the weights, and an isolating description of the base clusters. No observation-ball maximum is an input to the construction.*

Proof. We spell out the real-locus issue. Uniformize the compact real graph by a proper map from smooth real manifolds. Simultaneously make the nonzero functions $G, H_{r,a}$ normal crossings on each component. A finite cover of the inverse image of $G = 0$, followed by shrinking boxes, gives (3.2). Components on which a function vanishes identically are treated separately, not included in a product of nonzero functions. Properness and compactness ensure that the images of the boxes contain every sufficiently small observation ball. All source points map into the original graph, including its inequalities. This is the uniformization and normal-crossing construction of [1], not an additional resolution theorem.

For effective algebraic input one can avoid an arbitrary analytic uniformization. Decompose the compact semialgebraic graph into finitely many basic closed sets. Replace each inequality $p \geq 0$ by $p = s^2$. The slack variables are bounded over the compact graph, so this produces compact real algebraic lifts. Resolve the lifts over the real algebraic numbers, and resolve their singular algebraic subsets recursively when real points are not covered by the resolved regular locus. The uncovered locus has strictly smaller dimension at each such step. The real images remain in the lift throughout. This gives a finite proper cover of its real points. Simultaneous principalization on each smooth source, applied to the product of the nonzero $G, H_{r,a}$, gives the required monomials. Constructive characteristic-zero resolution [2] supplies the algebraic operations. Real feasibility of a chart and its divisors is decided by quantifier elimination. One must retain this test: a divisor seen only on the complexification supplies no admissible transversal. $\square$

Theorem 3.3 (Finite real-valuative Newton formula). *For each cluster coefficient take the minimum over all real monomial boxes and their recorded divisors:*

$$\beta_{r,a} = \min \frac{b_{r,a,i}}{a_i}, \quad \alpha = \min_{r,a} \frac{\beta_{r,a}}{k_{r,a}}. \quad (3.3)$$

Identically zero functions contribute $+\infty$. Then either all coefficients vanish on an observation neighbourhood, or $\alpha \in \mathbb{Q}_{>0}$ and

$$\omega_{P,X}(t) = C_{P,X}t^\alpha + o(t^\alpha), \quad 0 < C_{P,X} < \infty. \quad (3.4)$$

The numbers $\beta_{r,a}$ are the orders of the corresponding coefficient envelopes, but formula (3.3) computes them from finite integer data without constructing those envelopes. Each finite minimum is attained by an admissible transversal on the resolved real graph. The formula is independent of the chosen cover. It applies to singular observation maps, positive-dimensional exact parameter fibres, and inequality faces of the entire closed stochastic model.

Proof. Fix a nonzero $H = H_{r,a}$ and let β be the first minimum in (3.3). On every box, $b_i \geq \beta a_i$ for all $a_i > 0$; for $a_i = 0$ one has $b_i \geq 0$. Bounded units and $|y_i| \leq 1$ therefore give $H \leq CG^\beta$. A finite cover makes the constant uniform over the observation neighbourhood. Hence $|c_{r,a}| \leq Ch_w^\beta$.

Choose a divisor realizing the minimum and a real point on it away from all other divisors. Fix the other coordinates at this point and let $y_i = s > 0$. Its image is an admissible arc with $G \asymp s^{a_i}$ and $H \asymp s^{b_i}$. Inverting the first relation (or restricting its analytic branch) gives a lower bound of order t^β for the coefficient envelope. This proves the first equality of orders without optimizing over arcs or balls. Since $H = 0$ on $G = 0$, a nonzero H has $b_i > 0$ on every recorded observation divisor. Thus all finite orders are positive. If no nonzero H occurs over such a divisor, properness gives an observation neighbourhood on which $H = 0$.

For a monic cluster polynomial with coefficients c_a , its largest root modulus is comparable, with constants depending only on the degree, to $\max_a |c_a|^{1/(m-a)}$. This follows from elementary symmetric functions and the geometric-series root bound. The base spectrum is in every ball; therefore the diameter is between its maximal displacement and twice that displacement. This gives the exponent in (3.3). The joint coefficient specialization below, as proved in Proposition 3.4, supplies the leading constant and the little-oh term. Since the envelope orders are intrinsic to the real observation-spectral graph, two covers give the same minimum. $\square$

What the valuation formula does and does not assert. Ratios of vanishing orders on a principalization are classical Łojasiewicz data; compare [3, Proposition 4.3 and Theorem 4.8] and the broader filtration framework of [5]. The additional steps here are the real probability-graph lift, the cluster weights $k_{r,a}$, and the joint leading spectral set. Neither bare channel ranks nor root multiplicities replace the exponents $a_i, b_{r,a,i}$. The finite cutoff is the output of a resolution of the given graph, not a universal bound on ordinary Taylor jets.

3.1. The joint leading set as a real weighted specialization. Write $\alpha = p/q$ in lowest positive terms. Use $\varepsilon > 0$ and $t = \varepsilon^q$. In the same graph impose

$$R_{je} = \varepsilon^q U_{je}, \quad c_{r,a} = \varepsilon^{pk_{r,a}} V_{r,a}, \quad \sum_{j,e} U_{je}^2 \leq 1. \quad (3.5)$$

Let $\mathcal{J}_0$ be the (U, V) -projection of the fibre $\varepsilon = 0$ of the real closure of this set. The model coordinates are retained until the closure is taken. They are bounded, so projecting them afterwards loses no limiting point. Set $\mathcal{C}_0 = \text{pr}_V \mathcal{J}_0$ and

$$Z_r(V) = Z \left(z^{m_r} + \sum_{a < m_r} V_{r,a} z^a \right).$$

The simultaneous observation variables in (3.5) retain the metric; the simultaneous coefficient variables retain all relations within and between clusters.

Proposition 3.4 (Effective leading fibre and diameter). *The set $\mathcal{C}_0$ is compact and equals the joint limiting coefficient set of the observation balls. In particular*

$$C_{P,X} = \max_{V,V' \in \mathcal{C}_0} \max_r d_\infty(Z_r(V), Z_r(V')). \quad (3.6)$$

For real algebraic input, a finite algorithm outputs a quantifier-free formula for $\mathcal{I}_0$, hence for $\mathcal{C}_0$, and an isolating polynomial and rational interval for the positive real algebraic number $C_{P,X}$. It also outputs the rational exponent from (3.3). These outputs require no coefficient-envelope maximization or Puiseux analysis of such a maximum.

Proof. The monomial bound makes V uniformly bounded in (3.5); U already lies in the unit ball. For every $t > 0$ the projected V -section is exactly the rescaled coefficient image of $B_P(t)$. Thus its closure fibre is the joint coefficient limit. Definability rules out oscillatory subsequential limits: for a point of that fibre, its distance to the section is a bounded semialgebraic function with a zero subsequential limit, hence has limit zero. A finite net gives Hausdorff convergence. Continuity of monic root multisets on compact coefficient sets gives (3.6). Zero belongs to $\mathcal{C}_0$ because an exact-fibre representative belongs to every observation ball.

We specify the output procedure. Encode each algebraic constant by its minimal polynomial and isolating interval. Represent complex quantities by real and imaginary coordinates. Isolate each cluster using disjoint algebraic discs and the polynomial multiplication equations; their unique near-base branch is selected by strict distance inequalities. All powers in (3.5) are integral. Closure is encoded by the first-order condition that every positive-radius ball meets the set with $\varepsilon > 0$. Real quantifier elimination removes these quantifiers and the bounded model coordinates. Encode the roots of a monic polynomial by their elementary symmetric identities. For each of the finitely many permutations, squared Euclidean distances encode the bottleneck metric using finite maxima and minima. A final compact maximization formula isolates the unique value in (3.6). A singleton semialgebraic over the real algebraic numbers is real algebraic, giving the stated number output. There is no complexity claim. Algebraic saturation by ε alone is not this algorithm: its special fibre can contain inadmissible real points. The real closure and the original inequalities are essential. $\square$

Example 3.5. For $x \in [-1/2, 1/2]$, take $F(x) = (1/2 + x^4, 1/2 - x^4)$ and $A_x(z) = z^2 - x^6$, at $x = 0$. The observation derivative vanishes. Here G has order eight and the only nonzero H has order twelve. Thus $\beta = 3/2$ and $\alpha = 3/4$. This is a finite singular-chart calculation, not the immersed-jet formula. It also illustrates why observation orders, rather than parameter orders, appear in (3.3).

4. A FINITE CLASSIFICATION ACROSS CHANNEL RANKS

The boundary family admits a classification which includes both channel ranks and the weight constraint. We fix $D > 0$, $0 < \alpha_* < 1/2$, and an integer $\ell \geq 7$. Let $\mathfrak{F}$ consist of the parameters

$$\theta = (r, s, \alpha, U, V, T, w), \quad 0 < s < r < D, \quad \alpha_* < \alpha < 1 - \alpha_*,$$

where $U = (u_1, u_2)$ and $V = (v_1, v_2)$ are strictly positive column-stochastic 2×2 matrices, $\det V \neq 0$, $D < T_1 < \dots < T_\ell$, and $w_j > 0$, $\sum_j w_j = 1$. No rank assumption is imposed on U . Write $\gamma = 1 - \alpha$ and set

$$\begin{aligned} f(z) &= z(z - r)^2, & g(z) &= (z - s)^3, \\ K(z) &= \alpha f(z) u_1 v_1^\top + \gamma g(z) u_2 v_2^\top, & q(z) &= \alpha f(z) + \gamma g(z), \\ P_{\theta,j} &= K(T_j)/q(T_j). \end{aligned} \quad (4.1)$$

Competitors always range over the entire closed binary degree-three model: both channels may be singular or have zero entries, the weights are at least α_* , and all six roots belong to $[0, D]$. They use the same T, w as the centre. The target is the unordered multiset of six roots. Denote its diameter over the radius- t observation ball by $\omega_\theta(t)$. Our convention is

$$h_w(P, Q)^2 = \sum_j w_j \sum_e (\sqrt{P_{je}} - \sqrt{Q_{je}})^2.$$

Coefficient norms are Euclidean; matrix coefficients are stacked with Frobenius norm. The metric on real root multisets is the maximum discrepancy between their increasing orderings.

Introduce the following two polynomials in the family parameters:

$$\begin{aligned} A(r, s) &= (r + 3s)(2r - 3s)^2, \\ B(r, s) &= (3D - r)s - 2rD, \\ E(r, s, \alpha) &= (1 - \alpha)A(r, s) - 4\alpha_*r^3. \end{aligned} \tag{4.2}$$

They involve neither an observation-ball optimization nor a selected parameter tangent space.

Theorem 4.1 (Classification of the family). *The exact spectral fibre of a datum in $\mathfrak{F}$ is a singleton if and only if*

$$\theta \in \mathfrak{I} := \{\det U \neq 0\} \cup \{B < 0\} \cup \{E < 0\}. \tag{4.3}$$

In particular $\mathfrak{I}$ is an open semialgebraic subset of $\mathfrak{F}$, containing identifiable rank-one and rank-two data. On its complement put

$$z_* = \frac{rs}{3s - 2r}, \quad c_* = \frac{4r^3}{A(r, s)}, \quad R(z) = \frac{z(z - r)^2}{(z - s)^3}, \quad c_D = \frac{R(D)}{R(D) - 1}.$$

Besides (f, g) , all exact component-polynomial pairs, up to permutation, are precisely

$$(f, (1 - c)f + cg), \quad c_* \leq c \leq \min\{c_D, \gamma/\alpha_*\}. \tag{4.4}$$

This interval is nonempty, including at equality in either boundary condition. Every pair in it is realized by positive stochastic channels and admissible weights.

Proof. We first record the normalization argument in a form which is independent of the rank of U . If e has degree below three and a matrix polynomial H of degree at most three satisfies $H(T_j) = e(T_j)P_{\theta, j}$, then seven-point interpolation gives $qH = eK$. Its second marginal is

$$qH^\top \mathbf{1} = eV \operatorname{diag}(\alpha, \gamma)(f, g)^\top.$$

Multiplication by $\operatorname{diag}(\alpha, \gamma)^{-1}V^{-1}$ shows that $q \mid ef$ and $q \mid eg$. Since $\gcd(f, g) = 1$, Bezout's identity implies $q \mid e$, and consequently $e = 0$. Thus the monic normalizer and the matrix polynomial K are determined by the observations. The same argument proves injectivity of the finite normalization operator, not merely uniqueness within the displayed family.

When $\det U \neq 0$, the leading coefficient K_3 is invertible and

$$\frac{\det K(z)}{\det K_3} = f(z)g(z).$$

Every exact closed-model competitor has this same K and therefore this same monic spectral polynomial. This proves identification on the first set in (4.3).

Suppose next that $\det U = 0$. Column stochasticity gives $u_1 = u_2 = u$. The coefficient matrix of the second marginal $V \operatorname{diag}(\alpha, \gamma)(f, g)^\top$ has rank two. For any exact competitor its second

channel is therefore invertible and its two monic component polynomials belong to the affine pencil $h_c = (1 - c)f + cg$. Direct calculation gives

$$\begin{aligned} \text{disc}(h_c) &= c(c-1)^2(r-s)^3\{A(r,s)c - 4r^3\}, \\ h_c(0) &= -cs^3, \quad \frac{R'(z)}{R(z)} = \frac{(2r-3s)z + rs}{z(z-r)(z-s)}. \end{aligned}$$

Nonnegative roots exclude $c < 0$; the discriminant excludes $0 < c < 1$. For $c > 1$ there is one root in (s, r) . The other two must exceed r and solve $R(z) = c/(c-1) > 1$. They exist only when $s > 2r/3$. In this case R increases to its maximum at z_* and then decreases to one. The double-root value is c_* , and the larger root increases from z_* to D as c increases from c_* to c_D . Thus the complete admissible pencil is $\{0, 1\}$, together with $[c_*, c_D]$ exactly when $z_* \leq D$. At $z_* = D$ this interval is a singleton. Notice that $A = 4r^3 - 27s^2(r-s) < 4r^3$, so $c_* > 1$.

If the two pencil parameters are c_1, c_2 , the monic normalizer gives $\alpha'_1 c_1 + \alpha'_2 c_2 = \gamma \in (0, 1)$. Hence one is zero and the other is either one or a c in the additional interval. Its weight is γ/c , admissible exactly when $c \leq \gamma/\alpha_*$; the other weight is $1 - \gamma/c \geq \alpha > \alpha_*$. Conversely set

$$\beta'_f = \alpha v_1 + \gamma(1 - 1/c)v_2, \quad \beta'_h = \gamma v_2/c.$$

These vectors are positive, have the stated weights as entry sums, and satisfy $\beta'_f f + \beta'_h h_c = \alpha v_1 f + \gamma v_2 g$. Normalizing them and keeping the first channel (u, u) realizes the whole interval. Since $h_c \neq g$ for $c > 1$, each is a second spectral value. Finally $z_* \leq D$ is equivalent to $B \geq 0$; this already implies $s > 2r/3$ and $A > 0$. On this set $c_* \leq \gamma/\alpha_*$ is equivalent to $E \geq 0$. This proves all assertions, including the strict inequalities describing the open identifiable set. $\square$

Lemma 4.2 (Uniform coefficient inverse on identifiable compacta). *Let $H \subseteq \mathfrak{I}$ be compact. There exist $M, \delta_0 > 0$ such that, for every $\theta \in H$ and every closed-model competitor ξ with $\delta = \max_j \|F_T(\xi)_j - P_{\theta,j}\|_F < \delta_0$, one component permutation gives*

$$\|f'_1 - f\|_{\text{coef}} + \|f'_2 - g\|_{\text{coef}} + |\alpha'_1 - \alpha| + \|U' - U\|_F + \|V' - V\|_F \leq M\delta. \quad (4.5)$$

The constants are uniform even when H meets both channel ranks.

Proof. Injectivity of the normalization operator in the preceding proof, subtraction of its linear equations, and the uniform boundedness of all competitor normalizer coefficients imply

$$\|q' - q\|_{\text{coef}} + \|K' - K\|_{\text{coef}} + \|K'^T \mathbf{1} - K^T \mathbf{1}\|_{\text{coef}} \leq C\delta.$$

For completeness, the normalization equation projects the array $(q(T_j)P_j)_j$ onto the orthogonal complement of the degree-three Vandermonde column space. Subtracting the two equations gives the fixed injective operator at P_θ , applied to the degree-below-three polynomial $q' - q$, with right side bounded by $C\delta$. A bounded left inverse gives the first estimate; interpolation gives the others. All these bounds are locally uniform in the centre.

At a rank-two centre choose two clocks T_i, T_j for which $f(T_j)/f(T_i) \neq g(T_j)/g(T_i)$. Such clocks exist: otherwise the nonzero degree-at-most-three polynomial $g(T_i)f - f(T_i)g$ would vanish at all seven clocks. The matrix $K(T_j)K(T_i)^{-1}$ has distinct real eigenvalues and eigenvectors u_1, u_2 . Normalize each eigenvector to have entry sum one. The simple-eigenvalue inverse is analytic near the centre. Applying U^{-1} to $K(z)$, and taking row sums, then recovers αf and γg ; leading coefficients and normalized rows recover weights and V . This gives (4.5) for arbitrary nearby competitors, since $K'(T_i)$ remains invertible.

At an identifiable rank-one centre, Theorem 4.1 and compactness first confine all sufficiently close competitors to small coefficient neighbourhoods of (f, g) , after permutation. There is no hidden non-root fibre here: fixed functionals dual to f, g recover their coefficient matrices, whose entry sums and normalized marginals recover the weights and both channels uniquely. Put

$B_{\text{coef}} = (K^\top \mathbf{1})_{\text{coef}}$ and $\beta = \sigma_2(B_{\text{coef}}) > 0$. In $B'_{\text{coef}} = V' \text{diag}(\alpha') F'_{\text{coef}}$, compactness supplies $M_0 \geq \|\text{diag}(\alpha') F'_{\text{coef}}\|$. Thus

$$\beta/2 \leq \sigma_2(B'_{\text{coef}}) \leq \sigma_{\min}(V') M_0, \quad \|\text{diag}(\alpha')^{-1} V'^{-1}\| \leq \frac{2M_0}{\alpha_* \beta}.$$

Multiplication by this bounded inverse and monicity yield $f'_b = h_{c_b} + O_{\text{coef}}(\delta)$, with bounded c_b respectively near zero and one. Only *local* pencil isolation at these two points is now needed; additional remote pencil polynomials have already been excluded by whole-model fibre compactness.

Near zero, $f'_b(0) \leq 0$ gives $c_b \geq -C\delta$, and the discriminant formula, whose derivative at zero is $-4r^3(r-s)^3 < 0$, gives $c_b \leq C\delta$. Near one put $v = 1 - c_b$ and depress f'_b using its own root mean. Its depressed coefficients satisfy

$$a = f'(s)v + O(v^2 + \delta), \quad b = f(s)v + O(v^2 + \delta).$$

Real-rootedness implies $a \leq 0$ and $|b| \leq 2(-a)^{3/2}/(3\sqrt{3})$. Since $f(s) > 0$, these inequalities give $|v| \leq C\delta + Cv^2 + C(|v| + \delta)^{3/2}$, and the last two terms are absorbed locally. The component coefficients are therefore recovered with error $O(\delta)$.

Choose coefficient functionals dual to f, g . Their values on the nearby components form $I + O(\delta)$. Applying them to K' recovers $\alpha'_b u'_b (v'_b)^\top$ with error $O(\delta)$. Entry sums give weights; normalized row and column sums give channels. This proves the desired estimate at rank one, including cases isolated by the weight constraint rather than by the root interval alone.

All confinement and displayed estimates hold on a neighbourhood of each such centre. To see the first assertion uniformly, a contrary sequence of centres tending to it and competitors with observation error tending to zero would have a limiting exact competitor, which has just been classified. Positivity, coefficient independence and the local pencil inequalities have strict margins on a smaller neighbourhood. A finite cover of H completes the proof. $\square$

5. A UNIFORM SPECTRAL NORMAL FORM

We compute the whole-model observation ball, not just the image of a chosen root-splitting path. For a fixed centre in $\mathfrak{J}$ define

$$\langle H, H' \rangle_{I, \theta} = \sum_j w_j \sum_e \frac{H_{je} H'_{je}}{P_{\theta, je}}.$$

The column derivatives in $\eta = (\dot{\alpha}, \dot{u}_1, \dot{u}_2, \dot{v}_1, \dot{v}_2)$ have entry sum zero, so η has five free scalar coordinates. Define

$$\begin{aligned} p_f(z) &= -a(z-r)^2 - bz(z-r) - Xz, \\ p_g(z) &= -c(z-s)^2 - Y(z-s), \end{aligned} \tag{5.1}$$

$$\begin{aligned} \dot{K} &= \alpha u_1 v_1^\top p_f + \gamma u_2 v_2^\top p_g \\ &\quad + f\{\dot{\alpha} u_1 v_1^\top + \dot{\alpha} u_1 v_1^\top + \alpha u_1 \dot{v}_1^\top\} + g\{-\dot{\alpha} u_2 v_2^\top + \gamma \dot{u}_2 v_2^\top + \gamma u_2 \dot{v}_2^\top\}, \\ \dot{q} &= \alpha p_f + \gamma p_g + \dot{\alpha}(f-g), \end{aligned}$$

$$S_\theta(a, b, c, \eta, X, Y)_j = \frac{\dot{K}(T_j) - P_{\theta, j} \dot{q}(T_j)}{q(T_j)}. \tag{5.2}$$

Here $a, X, Y \geq 0$, whereas b, c, η are free. The first inequality is the endpoint condition; the other two describe real root splitting. These formulas allow $u_1 \neq u_2$.

Let L_θ be the seven-column matrix of the free directions (b, c, η) , and let D_θ be the three-column matrix of the directions (a, X, Y) . Let W_θ be the diagonal matrix with entries $w_j/P_{\theta,je}$. Define

$$Q_\theta = D_\theta^\top W_\theta D_\theta - D_\theta^\top W_\theta L_\theta (L_\theta^\top W_\theta L_\theta)^{-1} L_\theta^\top W_\theta D_\theta, \quad (5.3)$$

$$J_\theta(X, Y) = \min_{a \geq 0} (a, X, Y) Q_\theta (a, X, Y)^\top, \quad (5.4)$$

$$\kappa_x(\theta) = \min_{Y \geq 0} J_\theta(1, Y), \quad \kappa_y(\theta) = \min_{X \geq 0} J_\theta(X, 1).$$

The following lemma proves, in particular, that the inverse in (5.3) exists even at rank-one observations.

Lemma 5.1 (Uniform feasible coercivity). *The matrix L_θ has rank seven, $Q_{\theta,00} > 0$, and κ_x, κ_y are positive and continuous on $\mathfrak{J}$. For every compact $H \subseteq \mathfrak{J}$ there is $c_H > 0$ such that*

$$\|S_\theta(a, b, c, \eta, X, Y)\|_{L, \theta} \geq c_H \|(a, b, c, \eta, X, Y)\| \quad (a, X, Y \geq 0, \theta \in H). \quad (5.5)$$

In particular $J_\theta(X, Y) \geq c'_H (X + Y)^2$ on the nonnegative quadrant. Every minimum in (5.4) is attained, with minimizing jet variables uniformly bounded on H after fixing $X = 1$ or $Y = 1$.

Proof. Every feasible jet is realized by admissible root and parameter arcs. For $t > 0$ choose a real triple y with $\sum_i y_i = 0$ and $\frac{1}{2} \sum_i y_i^2 = Y$, and use roots

$$ta, \quad r + tb/2 \pm \sqrt{tX}, \quad s + tc/3 + \sqrt{t} y_i \quad (1 \leq i \leq 3), \quad (5.6)$$

together with parameter increment $t\eta$. Positivity and the interior margins at r, s make these genuine closed-model alternatives, for small t , including when some roots coincide. Their observation increment is $tS_\theta + O(t^{3/2})$.

If a feasible jet has zero score, Lemma 4.2 applied to these arcs makes all polynomial and non-root derivatives zero. Thus $p_f = p_g = 0$ and $\eta = 0$. Evaluation at $z = 0$, followed by coefficient comparison in (5.1), gives $a = b = c = X = Y = 0$. The score therefore has no nonzero zero on the closed feasible cone. Compactness of its intersection with the unit sphere, and then of H , proves (5.5). This also proves that the free columns are independent. If the endpoint column were in their span, a feasible jet with $a = 1$ would have zero score; hence $Q_{00} > 0$.

Coercivity proves existence and boundedness of minimizing jets. The upper bound is obtained by setting all other variables to zero; the lower bound follows from (5.5). A convergent sequence of centres has bounded minimizing jets, so a subsequence and continuity of the score prove lower semicontinuity of either minimum. Evaluation at a fixed minimizing jet proves upper semicontinuity. This proves continuity and all remaining assertions. $\square$

Let $\mathcal{L}_\theta$ be the compact set of triples of ordered root clusters

$$(\{0\}, \{-\sqrt{X}, \sqrt{X}\}, Z(z^3 - Yz - Z)), \quad X, Y \geq 0, \quad J_\theta(X, Y) \leq 4, \quad 27Z^2 \leq 4Y^3. \quad (5.7)$$

Its metric is the maximum of the three ordered-cluster metrics. For small t , define $\mathcal{R}_\theta(t)$ by taking every competitor in the observation ball, grouping its roots near $0, r, s$, subtracting these centres, and dividing by $\sqrt{t}$. Lemma 4.2 ensures that these groups have sizes one, two and three, uniformly on compact subsets of $\mathfrak{J}$.

Theorem 5.2 (Uniform leading set and linear remainder). *For every compact $H \subseteq \mathfrak{J}$ there exist $C_H, t_H > 0$ such that, for $\theta \in H$ and $0 < t < t_H$,*

$$d_H(\mathcal{R}_\theta(t), \mathcal{L}_\theta) \leq C_H \sqrt{t}. \quad (5.8)$$

Consequently the entire closed-model modulus satisfies

$$\omega_\theta(t) = C(\theta)\sqrt{t} + O_H(t), \quad (5.9)$$

$$C(\theta) = \max \left\{ \sqrt{2} \kappa_x(\theta)^{-1/4}, 2\sqrt{2/3} \kappa_y(\theta)^{-1/4} \right\}. \quad (5.10)$$

The error is uniform across channel-rank changes within H . All points of the leading set, including diameter-extremizing pairs, are realized by actual stochastic alternatives. The function C is positive and continuous on $\mathfrak{I}$.

Proof. We give quantitative necessity and sufficiency. Positivity on H and (4.5) imply that every radius- t competitor has coefficient and non-root parameter increments $O_H(t)$. Coprime factorization of its first component, at 0 and r , gives a simple root ta and pair sum $2r + tb$, with $a \geq 0$ and bounded a, b . Its pair is exactly $r + tb/2 \pm \sqrt{tX}$ with $X \geq 0$ bounded. The second component has mean $s + tc/3$ and roots $s + tc/3 + \sqrt{t}y_i$, where $\sum_i y_i = 0$. The first two symmetric coefficients give $\sum_i y_i^2 = O_H(1)$. Thus $Y = \frac{1}{2} \sum_i y_i^2$ and $Z = \prod_i y_i$ are bounded and satisfy $Y \geq 0$, $27Z^2 \leq 4Y^3$. The parameter increment is $t\eta$ with η uniformly bounded. These are exact coordinates for the competitor, not a choice of an approximating arc.

Expanding its component polynomials gives (5.1) at order t . The only possible order- $t^{3/2}$ term needed here is the centred triple product; all remaining errors are bounded by that order. The rational probability map has uniformly bounded derivatives on these coefficient neighbourhoods. Hence, uniformly in all the competitors under consideration,

$$F_T(\xi) = P_\theta + tS_\theta + O_H(t^{3/2}), \quad h_w(F_T(\xi), P_\theta) = \frac{t}{2} \|S_\theta\|_{I,\theta} + O_H(t^{3/2}). \quad (5.11)$$

The second assertion follows by Taylor expansion of the vector of square roots and the reverse triangle inequality for its Euclidean norm; it does not require a nonzero score.

For necessity, ball membership gives $\|S_\theta\|_{I,\theta} \leq 2 + M_H\sqrt{t}$. Let $\rho = 2/(2 + M_H\sqrt{t})$. Scaling all first-order jet variables by ρ , and Z by $\rho^{3/2}$, produces a point of (5.7). Its root clusters differ from the competitor's rescaled clusters by $O_H(\sqrt{t})$: the centres contribute $\sqrt{t}(a, b/2, c/3)$ and the radial change multiplies bounded centred roots by $\sqrt{\rho} = 1 + O_H(\sqrt{t})$. This proves one directed Hausdorff estimate.

For sufficiency, take any point of (5.7) and choose minimizing nuisance variables and $a \geq 0$. By Lemma 5.1 these are uniformly bounded, because the score is at most two. For a fixed sufficiently large M'_H , put $\rho = 1 - M'_H\sqrt{t} > 0$ and use the actual arcs (5.6) with first-order variables multiplied by ρ , and centred triple roots by $\sqrt{\rho}$. Their score is multiplied by ρ . The uniform expansion (5.11) then gives $h_w \leq t(\rho + O_H(\sqrt{t})) \leq t$. Their rescaled roots differ from the prescribed point by $O_H(\sqrt{t})$. All stochastic and root constraints hold uniformly for small t . This proves the reverse Hausdorff estimate, including boundary points of the Fisher ball, without exchanging a pointwise limit with a family supremum.

It remains to compute the diameter. Homogeneity gives $X_{\max} = 2/\sqrt{\kappa_x}$ and $Y_{\max} = 2/\sqrt{\kappa_y}$. The double-cluster diameter is $\sqrt{X_{\max}}$. A sorted zero-sum triple with $\sum_i y_i^2 \leq 2Y_{\max}$ satisfies

$$-2\sqrt{Y_{\max}/3} \leq y_1 \leq 0, \quad |y_2| \leq \sqrt{Y_{\max}/3}, \quad 0 \leq y_3 \leq 2\sqrt{Y_{\max}/3}.$$

The outer-coordinate bounds follow from Cauchy-Schwarz and zero sum; the middle bound follows by minimizing the norm subject to ordering. The resulting diameter $2\sqrt{Y_{\max}/3}$ is attained by $(-2h, h, h)$ and $(-h, -h, 2h)$, $h = \sqrt{Y_{\max}/3}$. They have the same Y and opposite Z , hence the same leading score. The maximum-cluster metric has diameter equal to its largest marginal diameter; no independence of the cluster coordinates is assumed. This proves (5.10). Diameters of two compact sets differ by at most twice their Hausdorff distance. Multiplying (5.8) by $\sqrt{t}$

proves (5.9). Separation of the base clusters prevents cross-cluster optimal matchings, uniformly on H . Continuity and positivity follow from Lemma 5.1. $\square$

Corollary 5.3 (Uniform local oracle risk). *Fix $0 < c \leq 1/(8\sqrt{2})$ and $H \in \mathfrak{I}$. At the design of each centre take independent categorical samples with $n_j/N \rightarrow w_j$ uniformly on H ; integer rounding of Nw_j is sufficient. For the squared minimax risk $\mathcal{R}_N(\theta, c)$ on the specified ball $h_w(F_T(\xi), P_\theta) \leq cN^{-1/2}$, one has, uniformly on H ,*

$$\frac{3c}{32}C(\theta)^2 - O_H(N^{-1/4}) \leq \sqrt{N}\mathcal{R}_N(\theta, c) \leq cC(\theta)^2 + O_H(N^{-1/4}).$$

The centre is part of this local decision problem; no unknown-centre or global adaptation is asserted.

Proof. The constant-decision upper bound is $\omega_\theta(t_N)^2$. For a diameter pair in the ball, the product Hellinger distance is at most $\sqrt{8N}t_N \leq 1/4$, eventually uniformly on H . Total variation is at most this distance. Testing the two targets therefore gives the lower bound $3\omega_\theta(t_N)^2/32$. Substitute $t_N = cN^{-1/2}$ into (5.9); squaring its uniform $O(t_N)$ remainder gives the displayed $O(N^{-1/4})$ error after multiplication by $\sqrt{N}$. $\square$

6. FINITE ACTIVE-SET FORMULAS AND THE SPECTRAL PHASE PARTITION

The leading constant can be evaluated without a numerical optimizer. Index the rows and columns of Q_θ by $0, x, y$. For $i = x$ or y let $M_i = \{0, x, y\} \setminus \{i\}$. For a subset $A \subseteq M_i$ with $\det Q_{AA} > 0$, set

$$v_A = -Q_{AA}^{-1}Q_{Ai}, \quad k_{i,A} = Q_{ii} - Q_{iA}Q_{AA}^{-1}Q_{Ai}. \quad (6.1)$$

For $A = \emptyset$ use $k_{i,\emptyset} = Q_{ii}$. Say that a candidate is admissible when

$$v_A \geq 0, \quad Q_{ji} + Q_{jA}v_A \geq 0 \quad (j \in M_i \setminus A). \quad (6.2)$$

For the empty set only the second inequalities are imposed. All these tests are explicit signs of rational functions in the original root, weight, channel and design parameters. There are four candidates for each of κ_x and κ_y .

Theorem 6.1 (Finite formulas and a stratified law). *At every $\theta \in \mathfrak{I}$ at least one of the four candidates is admissible, and every admissible candidate has value*

$$\kappa_i(\theta) = k_{i,A}(\theta). \quad (6.3)$$

Together with (4.3), these tests give a finite semialgebraic partition of the entire named family into exact fibre classes and, on its identifiable part, leading-constant classes. On each identifiable class, $C(\theta)^4$ is one explicit rational function. More precisely,

$$C(\theta)^4 = \begin{cases} 4/\kappa_x(\theta), & 9\kappa_y(\theta) \geq 16\kappa_x(\theta), \\ 64/(9\kappa_y(\theta)), & 9\kappa_y(\theta) \leq 16\kappa_x(\theta). \end{cases} \quad (6.4)$$

The two formulas agree on equality. The partition has a finite semialgebraic real-analytic stratification on which C is analytic. Its formula extends continuously across active-set boundaries and across channel-rank changes contained in $\mathfrak{I}$. The uniform remainder in Theorem 5.2 holds on every compact subset of $\mathfrak{I}$, not just on a compact subset of one stratum. For real algebraic input the tests produce algebraic κ_i and a polynomial and isolating interval for C , using finite exact arithmetic and real-root isolation.

Proof. Write the three projected score columns as p_0, p_x, p_y in the Fisher inner product, so $Q_{ij} = \langle p_i, p_j \rangle$. For fixed i the required problem is

$$\min_{v \in \mathbb{R}_+^{M_i}} \left\| p_i + \sum_{j \in M_i} v_j p_j \right\|^2.$$

A minimizer exists by Lemma 5.1. Choose one with minimal positive support. Its supported columns are linearly independent: a nonzero linear dependence permits a signed variation of its positive coefficients, preserving the represented vector, until one coefficient becomes zero. This contradicts minimality. Its supported Gram matrix is consequently positive definite. The derivative vanishes on its support and is nonnegative on inactive coordinates. These are exactly (6.1) and (6.2). Conversely these conditions imply optimality: expanding the quadratic at the candidate gives a nonnegative linear term on every feasible displacement and a nonnegative quadratic term. The value is (6.3). Singular two-column Gram matrices cause no loss of candidates; an independent minimal support still exists. Zero active coefficients and ties are also allowed.

Each entry of the score is rational in θ , because the clocks are above D and every base probability is positive. The Fisher weights are $w_j/P_{\theta,je}$, not their square roots. The seven-column Gram matrix is invertible by Lemma 5.1. Thus (5.3) is rational throughout $\mathfrak{J}$. Select, for example, the first admissible support in a fixed ordering of the four subsets. This yields a finite semialgebraic partition with a rational formula for each κ_i . Taking fourth powers in (5.10) gives (6.4); the switching test is $9\kappa_y - 16\kappa_x$, with no fractional-power comparison. Refine these finitely many sign conditions, the rank condition, and the identification signs into a semialgebraic analytic stratification [13]. The positive fourth root of the selected positive rational function is analytic on every such stratum. Continuity at boundaries follows from Lemma 5.1, not from an assumption that the support remains fixed.

With algebraic input, all matrix inversions, sign tests and selected rational operations take place in a real algebraic number field. The positive solution of $z^4 = C(\theta)^4$ is then isolated in the usual real algebraic representation. No maximization over shrinking observation balls, no fitted Puiseux exponent, and no inversion of a singular ambient Fisher matrix enters this calculation. $\square$

Scope of the classification. The family has one boundary simple root, one interior double root and one interior triple root, with $0 < s < r < D$, an invertible second channel, and an arbitrary positive first channel. It includes the full identifiable portion of that family, including the weight-isolated rank-one region $B \geq 0$, $E < 0$. It is not a classification of unrelated multiplicity patterns. Within its stated scope, the exact fibre is classified by three polynomial signs, the leading constant by eight finite support tests and one comparison, and its approximation error is uniform with order t on identifiable compacta. The real-valuative construction supplies the general singular-germ framework; the results of these sections give a finite parameter-family computation inside that framework.

Example 6.2 (A rational datum and an exact constant). Take $D = 1$, $\alpha_* = 1/10$, $r = 3/5$, $s = 1/5$, $\alpha = 2/5$,

$$u_1 = u_2 = (1/3, 2/3)^\top, \quad v_1 = (1/4, 3/4)^\top, \quad v_2 = (3/4, 1/4)^\top,$$

with clocks $(6/5, 7/5, 8/5, 9/5, 2, 5/2, 3)$ and weights $1/7$. Here $B = -18/25 < 0$. The projected Gram matrix has rank two; the empty support is admissible for both programs, and $9\kappa_y > 16\kappa_x$. Exact substitution in (5.3)–(6.4) gives

$$C^4 = \frac{78550423766145732273576408862601811503774391446}{385279214348552940073664077156875}, \quad 3778.707 < C < 3778.708.$$

Replacing only u_2 by $(2/5, 3/5)^\top$ gives a rank-two datum, a rank-three projected Gram matrix, and the triple-dominant phase, with $59.970 < C < 59.971$. Both intervals are rational isolating intervals, verified by fourth powers, not error bars from floating-point fitting. The ancillary exact-arithmetic evaluator records the rational Gram matrices, support inequalities and fourth powers for both data. These examples also show why positivity and fixed multiplicities alone do not supply a useful universal leading constant.

APPENDIX A. CLASSIFICATION OF A BOUNDARY CUBIC FAMILY

We now compute on a family for which the observation derivative need not be injective. Throughout this section $k = m_1 = m_2 = 2$, $d = 3$, the root interval is $[0, D]$, and at least seven distinct clocks exceed D . The weights satisfy $\alpha_b \geq \alpha_* > 0$. Fix

$$0 < s < r < D, \quad f = z(z-r)^2, \quad g = (z-s)^3, \quad \gamma = 1 - \alpha, \quad \alpha \in (\alpha_*, 1 - \alpha_*). \quad (\text{A.1})$$

Let u be a positive probability column, $U = (u, u)$, and let $V = (v_1, v_2)$ be positive, invertible, and column stochastic. Put

$$K = u\{\alpha f v_1^\top + \gamma g v_2^\top\}, \quad q = \alpha f + \gamma g, \quad P_j = K(T_j)/q(T_j).$$

All coefficient norms in this and the next section are Euclidean norms of coefficients in increasing powers of z ; matrix coefficient norms are Frobenius norms of the stacked coefficients. Constants may depend on the fixed datum and design, but never on a nearby competitor.

Lemma A.1 (The complete admissible pencil). *Set $h_c = (1-c)f + cg$. If $s \leq 2r/3$, the set of $c \in \mathbb{R}$ for which all roots of h_c lie in $[0, D]$ is $\{0, 1\}$. If $s > 2r/3$, put*

$$z_* = \frac{rs}{3s-2r}, \quad c_* = \frac{4r^3}{(r+3s)(2r-3s)^2}, \quad R(z) = \frac{z(z-r)^2}{(z-s)^3}. \quad (\text{A.2})$$

For $z_* > D$ the same conclusion holds. For $z_* \leq D$ the set is

$$\{0, 1\} \cup [c_*, c_D], \quad c_D = \frac{R(D)}{R(D)-1}. \quad (\text{A.3})$$

The interval is a singleton when $z_* = D$.

Proof. Direct expansion gives

$$\text{disc}(h_c) = c(c-1)^2(r-s)^3\{(r+3s)(2r-3s)^2c - 4r^3\}. \quad (\text{A.4})$$

Also $h_c(0) = -cs^3$. Nonnegative roots exclude $c < 0$. Since $(r+3s)(2r-3s)^2 = 4r^3 - 27s^2(r-s) < 4r^3$, the discriminant excludes $0 < c < 1$. For $c > 1$, h_c is negative on $[0, s]$ and has a root between s and r . Any remaining nonnegative roots must exceed r . On this last interval their equation is $R(z) = c/(c-1) > 1$, and

$$\frac{R'(z)}{R(z)} = \frac{(2r-3s)z + rs}{z(z-r)(z-s)}.$$

If $s \leq 2r/3$, R increases from zero to one and there are no such roots. If $s > 2r/3$, it increases to its unique maximum at z_* and then decreases to one from above. At the maximum the double root condition gives $c = c_*$; substitution into h_c gives its third root $4rs/(r+3s) \in (s, r)$. For $c > c_*$ the larger of the two roots above r increases strictly from z_* to infinity, because $c/(c-1)$ decreases strictly to one. It reaches D precisely when $c = c_D$. These three roots exhaust the cubic. This proves all cases, including the endpoint equality. $\square$

Theorem A.2 (Exact-fibre classification). *For every datum (A.1), $\tau(P) > 0$. Its spectral fibre in the entire closed stochastic model is a singleton unless all three inequalities*

$$s > 2r/3, \quad z_* \leq D, \quad c_* \leq \frac{\gamma}{\alpha_*} \quad (\text{A.5})$$

hold. A second spectral value exists if and only if all three hold. In that event the complete additional component pairs, up to permutation, are

$$(f, h_c), \quad c_* \leq c \leq \min\{c_D, \gamma/\alpha_*\}.$$

In particular, on the open parameter region

$$0 < s < \frac{2rD}{3D-r}, \quad 0 < r < D, \quad (\text{A.6})$$

the exact spectral fibre is $\{Z(fg)\}$. This region contains $s = 2r/3$ in its interior, not on its boundary.

Proof. Let e have degree below three and belong to the normalization kernel. Interpolation yields a matrix polynomial E of degree at most three with $E(T_j) = e(T_j)P_j$. The polynomial $qE - eK$ has degree at most six and vanishes at seven clocks. Hence $qE = eK$. For its second marginal,

$$qE^\top \mathbf{1} = eB, \quad B = K^\top \mathbf{1} = V \operatorname{diag}(\alpha, \gamma) \begin{pmatrix} f \\ g \end{pmatrix}.$$

Multiplying by $\operatorname{diag}(\alpha, \gamma)^{-1}V^{-1}$ proves that q divides both ef and eg . Bezout's identity for the coprime polynomials f, g then implies $q \mid e$, so $e = 0$. This proves $\tau(P) > 0$ without a full-rank hypothesis on U .

Every exact competitor has the same monic normalizer, by the injective normalization equation, and the same K , by interpolation. The 2×4 coefficient matrix of B has rank two. Thus in $B = V' \operatorname{diag}(\alpha') F'_{\text{coef}}$ both V' and the row span of F'_{coef} have rank two. Each monic component is h_{c_b} . The two c_b are distinct and, by summing the entries of B , satisfy $\alpha'_1 c_1 + \alpha'_2 c_2 = \gamma \in (0, 1)$. Lemma A.1 implies that one coefficient is zero and the other is either one or belongs to $[c_*, c_D]$. In the latter case the weight on h_c must be γ/c . It is admissible precisely when $c \leq \gamma/\alpha_*$; the other weight is $1 - \gamma/c \geq \alpha > \alpha_*$. Conversely write

$$\beta'_f = \alpha v_1 + \gamma(1 - 1/c)v_2, \quad \beta'_h = \gamma v_2/c.$$

Both columns are positive, their sums are the required weights, and $\beta'_f f + \beta'_h h_c = \alpha v_1 f + \gamma v_2 g$. Normalizing the columns and retaining $U = (u, u)$ realizes an exact stochastic competitor. This proves necessity and sufficiency over the full model. Finally, for $s > 2r/3$, $z_* > D$ is equivalent to $s < 2rD/(3D - r)$. The latter upper endpoint strictly exceeds $2r/3$ when $r < D$. $\square$

Lemma A.3 (Quantitative pencil isolation). *On (A.6), for each fixed bounded range of c there are $C, \epsilon_0 > 0$ such that every monic p with roots in $[0, D]$ and $\|p - h_c\|_{\text{coef}} \leq \epsilon < \epsilon_0$ satisfies $\min\{|c|, |1 - c|\} \leq C\epsilon$.*

Proof. Compactness and Lemma A.1 first confine c to small neighbourhoods of zero and one. Near zero, $p(0) \leq 0$ gives $c \geq -C\epsilon$. The discriminant is locally Lipschitz on bounded coefficients and its derivative at zero in (A.4) is $-4r^3(r - s)^3 < 0$. Nonnegative discriminant therefore gives $c \leq C\epsilon$. Near one set $v = 1 - c$ and depress p using its own root mean. Its depressed coefficients a, b satisfy

$$a = f'(s)v + O(v^2 + \epsilon), \quad b = f(s)v + O(v^2 + \epsilon), \quad f(s) > 0.$$

For a real-rooted depressed cubic, $a \leq 0$ and $|b| \leq 2(-a)^{3/2}/(3\sqrt{3})$. Hence $|v| \leq C\epsilon + Cv^2 + C(|v| + \epsilon)^{3/2}$. Shrinking the fixed neighbourhood absorbs the terms in $|v|$ on the right. Thus $|v| \leq C\epsilon$. This argument does not assume that $f'(s)$ is nonzero. $\square$

Theorem A.4 (Whole-model coefficient recovery and stability). *At every datum in (A.6), every sufficiently nearby closed-model competitor can be permuted so that, with $\delta = \max_j \|P'_j - P_j\|_F$,*

$$\|f'_1 - f\|_{\text{coef}} + \|f'_2 - g\|_{\text{coef}} + |\alpha'_1 - \alpha| + \|U' - U\|_F + \|V' - V\|_F \leq C\delta. \quad (\text{A.7})$$

Consequently $\omega_P(t) \asymp \sqrt{t}$ for the entire closed model. The constants can be chosen uniformly on compact subsets of the strict region, with positive channel, weight, clock-separation, and design-weight margins. The coefficient bound and the square-root order persist when U is replaced by any sufficiently close positive column-stochastic matrix, including matrices of rank two.

Proof. Subtract the normalization equations, substitute the uniformly bounded coefficients of q' , and apply the left inverse of $\mathcal{N}_P$. Interpolation then gives

$$\|q' - q\|_{\text{coef}} + \|K' - K\|_{\text{coef}} + \|B' - B\|_{\text{coef}} \leq C\delta.$$

Here $B' = K'^T \mathbf{1}$. Put $\beta = \sigma_2(B_{\text{coef}}) > 0$ and choose M bounding $\|\text{diag}(\alpha')F'_{\text{coef}}\|_{\text{op}}$ on the compact closed model. For small δ ,

$$\frac{\beta}{2} \leq \sigma_2(B'_{\text{coef}}) \leq \sigma_{\min}(V') \|\text{diag}(\alpha')F'_{\text{coef}}\|_{\text{op}}, \quad \|(V')^{-1}\|_{\text{op}} \leq \frac{2M}{\beta}.$$

The singular-value inequality follows by testing B'^T at a unit left singular vector of V' for its least singular value. Set $L' = \text{diag}(\alpha')^{-1}(V')^{-1}$. Then $\|L'\| \leq 2M/(\alpha_*\beta)$ and

$$F'_{\text{coef}} = L'B_{\text{coef}} + L'(B'_{\text{coef}} - B_{\text{coef}}).$$

The exact rows of $L'B$ belong to $\text{span}\{f, g\}$, with bounded coefficients. Since the rows of F' are monic, these two coefficients sum to $1 + O(\delta)$. Absorbing their excess into the remainder yields $f'_b = h_{c_b} + O_{\text{coef}}(\delta)$ with bounded c_b . Lemma A.3 places each row within $C\delta$ of f or g . If both rows approached the same polynomial, $\sigma_2(F'_{\text{coef}}) \leq C\delta$, and the uniform bound on $\|V' \text{diag}(\alpha')\|$ would contradict $\sigma_2(B'_{\text{coef}}) \geq \beta/2$. This proves the polynomial part of (A.7) after permutation.

For the remaining parameters put $A'_b = \alpha'_b u'_b (v'_b)^T$ and $A_1 = \alpha u v_1^T, A_2 = \gamma u v_2^T$. Choose fixed bounded coefficient functionals ℓ_1, ℓ_2 dual to f, g . The matrix $(\ell_b(f'_c))_{bc}$ is $I + O(\delta)$. Applying ℓ_b to $K' = \sum_c A'_c f'_c$ and inverting this matrix gives $A'_b = A_b + O(\delta)$. Entry sums give the weights; normalized row and column sums give u'_b, v'_b , since the weights are bounded below. This proves the full bound, and in particular rules out an uncontrolled nuisance drift in the calculation of a leading constant.

For a real-rooted cluster whose coefficient displacement is $O(\delta)$, its first two symmetric functions give $\sum_i y_i^2 = (\sum_i y_i)^2 - 2 \sum_{i < j} y_i y_j = O(\delta)$. Coprime cluster factorization transfers the coefficient bound to each cluster. Simple clusters move by $O(\delta)$, repeated clusters by $O(\sqrt{\delta})$. Since $\delta \leq 2t/\sqrt{\min_j w_j}$ in the Hellinger ball, this proves the upper modulus bound. For a lower bound replace g by $(z-s)((z-s)^2 - v^2)$ with actual roots $s-v, s, s+v$. The observation derivative in v^2 is nonzero: a zero derivative would produce a normalization-kernel polynomial, force the normalizer derivative to vanish, and then force the nonzero numerator derivative to vanish. Thus $h_w(P_v, P) \asymp v^2$ while the spectral displacement is $|v|$.

All inequalities used have strict margins on a compact subregion. The second marginal B is independent of the first channel; its rank, the normalization argument, and the pencil isolation survive a change of U . Recovery of A_b by the same functionals still applies. Positivity keeps the root-splitting alternatives admissible. This proves the last two assertions without asserting that the rank-one locus itself is open. $\square$

The role of the endpoint. There is a genuine change when the simple root leaves the endpoint. Take $f_a = (z-a)(z-r)^2$ with $0 < a < s < r < D$. For negative sufficiently small c , the two roots near r of $(1-c)f_a + cg$ split into real roots: the leading equation is $(r-a)(z-r)^2 + c(r-s)^3 = 0$. The simple root near a stays in the interior. All three roots therefore remain admissible. The

changes $\beta_1 \mapsto \beta_1/(1-c)$ and $\beta_2 \mapsto \beta_2 - c\beta_1/(1-c)$ preserve the marginal polynomial and, with $U = (u, u)$, the entire observation. Strict channel entries and weights remain strict. This explains precisely why the boundary face, not the relation $s = 2r/3$, supplies the isolation mechanism. An affine change of the root coordinate gives the corresponding family on any interval $[L, D]$; clocks and spectral distances transform with it.

APPENDIX B. THE COMPLETE LEADING FIBRE AND ITS FISHER PROGRAM

Continue at a rank-one datum in the strict region (A.6). The conclusion of Theorem A.4 is stronger than a root bound: it controls every coefficient and every non-root parameter to first order in the observation distance. It therefore identifies all nuisance directions which can enter the leading spectral set.

For arrays $H = (H_j)$ define the positive definite Fisher inner product

$$\langle H, H' \rangle_I = \sum_j w_j \sum_e \frac{H_{je} H'_{je}}{P_{je}}, \quad \|H\|_I^2 = \langle H, H \rangle_I.$$

This is the metric at the fixed datum and in the original probability coordinates; no untransported change of matrix basis is made. Let $\eta = (\dot{\alpha}, \dot{u}_1, \dot{u}_2, \dot{v}_1, \dot{v}_2)$, where each column derivative has entry sum zero. These five scalar degrees of freedom are unconstrained, since all the corresponding inequalities are strict. For $a \geq 0$, $X, Y \geq 0$, and $b, c \in \mathbb{R}$, set

$$\begin{aligned} p_f(z) &= -a(z-r)^2 - bz(z-r) - Xz, \\ p_g(z) &= -c(z-s)^2 - Y(z-s), \end{aligned} \tag{B.1}$$

$$\begin{aligned} \dot{K} &= \alpha u v_1^\top p_f + \gamma u v_2^\top p_g \\ &\quad + f\{\dot{\alpha} u v_1^\top + \alpha \dot{u}_1 v_1^\top + \alpha u \dot{v}_1^\top\} + g\{-\dot{\alpha} u v_2^\top + \gamma \dot{u}_2 v_2^\top + \gamma u \dot{v}_2^\top\}, \\ \dot{q} &= \alpha p_f + \gamma p_g + \dot{\alpha}(f-g), \\ S(a, b, c, \eta, X, Y)_j &= \frac{\dot{K}(T_j) - P_j \dot{q}(T_j)}{q(T_j)}. \end{aligned} \tag{B.2}$$

Thus S is an explicitly specified linear map to a finite-dimensional space of arrays. The variable a is the one-sided motion of the endpoint root. The variables b, c describe motions of cluster sums; X, Y describe the leading splitting of the double and triple roots.

Let $\mathcal{L}$ be the span of $S(0, b, c, \eta, 0, 0)$, and let Π be the orthogonal projection onto $\mathcal{L}^\perp$ for $\langle \cdot, \cdot \rangle_I$. Put

$$p_0 = \Pi S(1, 0, 0, 0, 0, 0), \quad p_x = \Pi S(0, 0, 0, 0, 1, 0), \quad p_y = \Pi S(0, 0, 0, 0, 0, 1).$$

Only finite linear algebra is needed for this projection; it may be computed by a Gram matrix or a Moore–Penrose inverse. Define

$$J(X, Y) = \min_{a \geq 0} \|ap_0 + Xp_x + Yp_y\|_I^2. \tag{B.3}$$

In the present family $p_0 \neq 0$, as proved below. Consequently, with $z = Xp_x + Yp_y$, this is the explicit piecewise quadratic function

$$J(X, Y) = \|z\|_I^2 - \frac{\min\{0, \langle p_0, z \rangle_I\}^2}{\|p_0\|_I^2}. \tag{B.4}$$

Theorem B.1 (The leading coefficient fibre). *For the entire closed stochastic model, the joint leading cluster polynomials at scale $\sqrt{t}$ are exactly*

$$z, \quad z^2 - X, \quad z^3 - Yz - Z, \quad X, Y \geq 0, \quad J(X, Y) \leq 4, \quad 27Z^2 \leq 4Y^3. \tag{B.5}$$

The clusters are respectively based at $0, r, s$. The displayed set of (X, Y, Z) is compact. Every point is realized as a limit of actual stochastic alternatives, not just as a solution of tangent equations. In particular this is an explicit finite description of $\mathcal{C}_0$ for the binary rank-loss intersection.

Proof. We first describe admissible arcs for arbitrary finite jet variables. Choose real y_1, y_2, y_3 with

$$\sum_i y_i = 0, \quad \frac{1}{2} \sum_i y_i^2 = Y, \quad \prod_i y_i = Z. \quad (\text{B.6})$$

Such a triple exists exactly when $Y \geq 0$ and $27Z^2 \leq 4Y^3$, by the discriminant of the real depressed cubic. For $\varepsilon > 0$ use roots

$$\varepsilon^2 a, \quad r - \varepsilon\sqrt{X} + \varepsilon^2 b/2, \quad r + \varepsilon\sqrt{X} + \varepsilon^2 b/2, \quad s + \varepsilon y_i + \varepsilon^2 c/3 \quad (1 \leq i \leq 3). \quad (\text{B.7})$$

Vary each non-root parameter by ε^2 times its component of η . All constraints hold for small ε , including when some of the y_i coincide or $X = 0$; the endpoint condition is exactly $a \geq 0$. The polynomial derivatives in $t = \varepsilon^2$ are (B.1), and

$$P_t = P + tS + o(t), \quad h_w(P_t, P) = \frac{1}{2}t\|S\|_I + o(t). \quad (\text{B.8})$$

The terms containing Z have order $t^{3/2}$ in the coefficients and do not enter this score.

There is no nonzero feasible coefficient direction with zero score. Indeed apply (A.7) to the arcs just constructed. If $S = 0$, their observation error is $o(t)$, so $p_f = p_g = 0$. Evaluation of p_f at zero gives $a = 0$, and then its quadratic and linear coefficients give $b = X = 0$. Likewise $p_g = 0$ gives $c = Y = 0$. This also shows that $p_0 \neq 0$: otherwise a suitable nuisance direction would cancel $S(1, 0, 0, 0, 0, 0)$. The sum of a linear subspace and one closed ray is closed, so all minima defining J are attained. Continuity of (B.4) and the zero-score assertion on the compact segment $X, Y \geq 0, X + Y = 1$, yield

$$J(X, Y) \geq c_0(X + Y)^2 \quad (X, Y \geq 0), \quad c_0 > 0. \quad (\text{B.9})$$

For clarity, if J vanished at a point of this segment, its minimizing a and the nuisance least-squares solution would give the forbidden zero-score arc. This proves coercivity without assuming invertibility of the ambient coefficient Jacobian.

Now take any sequence in the observation ball of radius $t \downarrow 0$. The whole-model bound (A.7) gives $O(t)$ changes of both component coefficients and all non-root parameters, after the unique nearby component permutation. Coprime factorization of the first component at 0 and r shows that its endpoint root is ta_t with bounded $a_t \geq 0$, and its pair sum is $2r + tb_t$ with bounded b_t . Its pair splitting, divided by $\sqrt{t}$, has a subsequential limit $\{-\sqrt{X}, \sqrt{X}\}$, $X \geq 0$. The triple-root component has sum $3s + tc_t$ with bounded c_t and rescaled roots y_i with zero sum. The real-rooted coefficient bound gives bounded y_i . Taking a common subsequence supplies bounded limiting a, b, c, η, X, Y, Z with (B.1)–(B.6). Taylor expansion of the smooth probability map in coefficients and non-root parameters gives (B.8), hence $\|S\|_I \leq 2$. Minimizing over the nuisance variables and $a \geq 0$ proves $J(X, Y) \leq 4$. The simple cluster has zero rescaled coefficient; the pair and triple give precisely (B.5). This proves necessity over all closed-model competitors.

Conversely, take a point in (B.5), choose the least-squares nuisance variables and the minimizing a , and use (B.7). For $J(X, Y) < 4$ these arcs lie in the radius- t ball for all small t . At the boundary replace (a, b, c, η, X, Y, Z) by $(\rho a, \rho b, \rho c, \rho \eta, \rho X, \rho Y, \rho^{3/2} Z)$, $0 < \rho < 1$, and then let $\rho \uparrow 1$. This realizes the boundary as a closure limit. If all jet variables vanish, the exact datum itself suffices. Coercivity and the cubic discriminant bound prove compactness. The joint Hausdorff convergence established in Proposition 3.4 now identifies this complete set with $\mathcal{C}_0$. $\square$

Define two nonnegative least-squares values, both on fixed finite spaces:

$$\begin{aligned}\kappa_x &= \min_{a, Y \geq 0; b, c, \eta} \|S(a, b, c, \eta, 1, Y)\|_I^2 = \min_{Y \geq 0} J(1, Y), \\ \kappa_y &= \min_{a, X \geq 0; b, c, \eta} \|S(a, b, c, \eta, X, 1)\|_I^2 = \min_{X \geq 0} J(X, 1).\end{aligned}\tag{B.10}$$

They are strictly positive by (B.9). A Gram matrix for the seven free directions b, c, η , followed by a quadratic program in two nonnegative variables, computes each value.

Theorem B.2 (An explicit leading constant). *At every rank-one datum in (A.6),*

$$\omega_P(t) = C_P \sqrt{t} + o(\sqrt{t}), \quad C_P = \max \left\{ \sqrt{2} \kappa_x^{-1/4}, 2\sqrt{2/3} \kappa_y^{-1/4} \right\}.\tag{B.11}$$

Thus the leading constant is determined by the displayed finite Fisher programs in the stated root, channel, weight, and design parameters. The programs do not contain an inverse observation-ball optimization. Their minimizers give admissible arcs attaining the limiting diameter. On the shrinking oracle neighbourhood $h_w(F(\theta), P) \leq cN^{-1/2}$, take independent categorical samples with $n_j/N \rightarrow w_j > 0$ and $0 < c \leq 1/(8\sqrt{2})$. If $\mathcal{R}_N(P, c)$ denotes the squared local minimax risk, then

$$\frac{3cC_P^2}{32} \leq \liminf_N \sqrt{N} \mathcal{R}_N(P, c) \leq \limsup_N \sqrt{N} \mathcal{R}_N(P, c) \leq cC_P^2.\tag{B.12}$$

No unknown-centre adaptation is asserted.

Proof. The feasible set of (X, Y) is star-shaped by homogeneity of J . Its projections are intervals $[0, X_{\max}]$, $[0, Y_{\max}]$, where

$$X_{\max} = 2/\sqrt{\kappa_x}, \quad Y_{\max} = 2/\sqrt{\kappa_y}.$$

For the double cluster the diameter of the sorted pairs $\{-\sqrt{X}, \sqrt{X}\}$ is $\sqrt{X_{\max}}$. For a sorted real triple $y_1 \leq y_2 \leq y_3$ with zero sum and $\sum y_i^2 = 2Y \leq 2Y_{\max}$, elementary optimization gives

$$-2\sqrt{Y_{\max}/3} \leq y_1 \leq 0, \quad |y_2| \leq \sqrt{Y_{\max}/3}, \quad 0 \leq y_3 \leq 2\sqrt{Y_{\max}/3}.$$

For the outer coordinates use $y_2 + y_3 = -y_1$ and $y_2^2 + y_3^2 \geq y_1^2/2$; for the middle coordinate use the ordering and zero sum, with the minimum norm at $(y_2, y_2, -2y_2)$ or its reflection. These bounds show that the triple diameter is at most $2\sqrt{Y_{\max}/3}$. Equality is attained by the two sorted triples $(-2h, h, h)$ and $(-h, -h, 2h)$ with $h = \sqrt{Y_{\max}/3}$. Both have $Y = Y_{\max}$ and differ only in the sign of $Z = 2h^3$; they therefore have the same score and are simultaneously feasible.

For sufficiently small t , optimal spectral matchings cannot cross the three separated base clusters. The diameter in the maximum of these three cluster metrics equals the largest marginal diameter: for finitely many coordinates, supremum and maximum commute. This argument does not replace the joint feasible set by a product. The simple cluster contributes zero. Substitution of $X_{\max}, Y_{\max}$ proves (B.11) by Theorem B.1 and weighted coefficient convergence. The root arcs (B.7), with minimizing nuisance variables and radial approximation on the unit boundary, realize the pair and triple extrema.

For completeness put $t_N = cN^{-1/2}$ and $D_N = \omega_P(t_N)$. A constant decision chosen in the local target set has risk at most D_N^2 . A maximizing pair in that set has one-sample weighted Hellinger distance at most $2t_N$. Eventually $n_j/N \leq 2w_j$, so the product Hellinger distance squared is at most $8Nt_N^2$. Its total variation is at most $\sqrt{8c} \leq 1/4$. A decision rule induces a test by choosing the nearer of the two targets; a testing error entails squared loss at least $D_N^2/4$. The two-point testing bound therefore gives risk at least $3D_N^2/32$. Substituting $D_N^2 = C_P^2 cN^{-1/2} + o(N^{-1/2})$ proves (B.12). The constant upper decision depends on the specified centre, as the stated local experiment permits. $\square$

Which split is extremal? The symmetric triple split $(-v, 0, v)$ has $Y = v^2$ and largest root modulus $\sqrt{Y}$. At exactly the same first-order observation score, $(-2h, h, h)$ with $h = \sqrt{Y/3}$ has largest root modulus $2\sqrt{Y/3} > \sqrt{Y}$. The symmetric split establishes the exponent but does not exhaust the leading triple fibre. In addition, optimizing the nuisance directions and the other cluster can reduce its Fisher cost. Formula (B.10) includes both effects. This distinction is why a one-path lower bound cannot evaluate the leading diameter of the whole model.

APPENDIX C. GLOBAL ADDITIVE NORMALIZATION AND OBSERVABLE CONDITIONING

Proposition E.2 gives explicit degree and clock dependence without claiming stable monomial interpolation in growing degree.

Theorem C.1 (Normalization, clock design, and spectral recovery). *Suppose the true channels have full column rank. If $\ell \geq 2d + 1$ and $g = \gcd(f_1, \dots, f_k)$ is monic, then*

$$\ker \mathcal{N}_P = \{(q/g)s : \deg s < \deg g\}, \quad \dim \ker \mathcal{N}_P = \deg g. \quad (\text{C.1})$$

For any design with $\ell \geq d + 1$ and $\tau(P) > 0$, every competitor in $\mathcal{K}_d$ satisfies

$$\|a - a'\|_2 \leq C \min\{1, \delta/\tau(P)\}, \quad (\text{C.2})$$

$$d_\infty(\Lambda, \Lambda') \leq C \min\{1, [\delta(\kappa^{-1} + \tau(P)^{-1})]^{1/d}\}. \quad (\text{C.3})$$

For simple roots within each true component, separated by $\nu > 0$, the spectral exponent is one, with the constant also depending on ν . Cross-component collisions are allowed. No rank or separation hypothesis is imposed on a competitor.

At an interior parameter with internally simple roots, $\tau(P) = 0$ gives a nonconstant local exact fibre with fixed channels and changing aggregate roots. For every fixed admissible k , relatively prime interior examples fail identification at $2d$ clocks. This worst-case statement is realized by an affine-line matrix curve. If the component polynomials span all polynomials of degree at most d , then $d + 2$ clocks suffice and $d + 1$ do not.

The proof, including the clock-complexity alternatives, is retained in the appendices. To state its observable strengthening, reduce K in orthonormal frames of its rank- k row and column spaces, write $M(z) = \sum_{j=0}^d M_j z^j$, and put $A_0 = M_d$. For each distinct complete component polynomial f , let E_f be the joint spectral projector of the commuting $M_j A_0^{-1}$, and set

$$B_d(P) = \sum_f \|A_0^{-1} E_f\|_{\text{op}}, \quad \eta_d(P) = (B_d(P) + \tau(P)^{-1})^{-1}.$$

Theorem E.1 proves observability, frame independence, and $d_\infty(\Lambda, \Lambda') \leq C \min\{1, [\delta/\eta_d(P)]^{1/d}\}$. Separated clusters of at most m roots per component have exponent $1/m$. The two losses add; they are not multiplied.

In degree one, $f_b = z - \xi_b$, put

$$h = \sum_b \alpha_b \xi_b, \quad A = U \text{diag}(\alpha) V^\top, \quad C = U \text{diag}(\alpha_b \xi_b) V^\top, \quad B = hA - C. \quad (\text{C.4})$$

Then $P(T) = A + B/(T - h)$, $K(z) = zA - C$, and at three clocks $\tau(P) \asymp \gamma(P) = \|B\|_F$. The affine condition is $\vartheta = \kappa\gamma/(\kappa + \gamma)$, zero on either boundary, with exact-fibre envelope

$$\vartheta_*(P) = \max\{\vartheta(\omega) : \mathcal{P}(\omega) = P\}. \quad (\text{C.5})$$

The residue condition is developed in Appendix O; B_1 is its residue sum and $\eta_1 \asymp \eta$. Appendix P gives the least-squares and polynomial-root estimator without a supplied signal floor. This is exact-arithmetic stability, not a bit-complexity or confidence-optimization assertion.

APPENDIX D. NORMALIZATION AND THE FINITE-CLOCK FIBRE

The normalizing polynomial is constrained by the requirement that $q(T_j)P(T_j)$ interpolate a matrix polynomial of degree at most d . The operator in (2.2) records precisely this constraint.

Lemma D.1 (The normalization defect). *Under the rank assumption of Theorem C.1, the greatest common divisor of all entries of K is $g = \gcd(f_1, \dots, f_k)$. If $\ell \geq 2d + 1$, then (C.1) holds.*

Proof. Every entry of K is a constant linear combination of the f_b , so g divides those entries. Conversely, constant left and right channel inverses recover $\text{diag}(\alpha_b f_b)$. Every common divisor of the entries therefore divides every f_b .

Let $r \in \ker \mathcal{N}_P$ have degree less than d . There is a matrix polynomial H of degree at most d with $H(T_j) = r(T_j)P(T_j)$. Every entry of $qH - rK$ has degree at most $2d$ and vanishes at $2d + 1$ nodes. Hence $qH = rK$ identically. Write $K = g\tilde{K}$ and $q = g\tilde{q}$. The entries of $\tilde{K}$ have greatest common divisor one. Their Bezout identity implies $\tilde{q} \mid r$. Thus $r = \tilde{q}s$ with $\deg s < \deg g$. Conversely each such s gives $H = \tilde{K}s$ of degree at most $d - 1$ and satisfies the identity. This includes the case $\deg g = d$. $\square$

Cancelling g lowers the visible degree to $d - \deg g$. The zeros of the cancelled factor cannot be read from the normalized curve. A shorter design can have a normalization defect even when $g = 1$.

Lemma D.2 (Positive local fibres). *Suppose $\ell \geq d + 1$, the true channels have full column rank, $\alpha_b > \alpha_*$, and all roots of each f_b are simple in (L, D) . If $0 \neq r \in \ker \mathcal{N}_P$, there is a smooth parameter curve with the same U, V , unchanged observations at every sampled clock, and a nonconstant root multiset.*

Proof. Let H interpolate $r(T_j)P(T_j)$ with degree at most d . Interpolation at the first $d + 1$ nodes shows that $H = U \text{diag}(h_b) V^\top$, with each h_b of degree at most d . Since every $P(T_j)$ has entry sum one, $\sum_b h_b - r$ vanishes at $d + 1$ nodes and is identically zero. For small real u put

$$q_u = q + ur, \quad K_u = K + uH, \quad w_{b,u} = \alpha_b f_b + uh_b.$$

Let $\alpha_{b,u}$ be the leading coefficient of $w_{b,u}$ and set $f_{b,u} = w_{b,u}/\alpha_{b,u}$. Then $\sum_b \alpha_{b,u} = 1$, $q_u = \sum_b \alpha_{b,u} f_{b,u}$, and the strict weight inequalities persist. Around each root of each component choose disjoint brackets with opposite signs at their endpoints. Small coefficient perturbations preserve a root in each bracket, accounting for all d roots. These real roots are smooth in u by the implicit function theorem. The identity $K_u(T_j) = q_u(T_j)P(T_j)$ proves exact agreement.

Suppose the aggregate multiset were constant on a neighbourhood of $u = 0$. Each continuously tracked component root would then take values in a fixed finite set. A continuous map from a connected interval to a finite set is constant. This argument applies also when roots of different components coincide. Hence every $f_{b,u} = f_b$. Applying the true channel inverses at a clock gives $\alpha_{b,u}/\alpha_b = q_u(T_j)/q(T_j)$ for every b, j . Summing the weights makes this ratio one. Thus r vanishes at $d + 1$ nodes, contradicting $r \neq 0$ and $\deg r < d$. $\square$

Proposition D.3 (Sharp worst-case clock count). *For every admissible fixed k, m_1, m_2 , there are positive full-rank channels and interior, relatively prime component polynomials for which any specified $2d$ distinct clocks admit the fibre of Lemma D.2.*

Proof. For $k = 2$, write $E_b = U_{:b} V_{:b}^\top$ and $P(T) = E_2 + (E_1 - E_2)w(T)$, where $w = \alpha_1 f_1/q$. Since $\Omega(r(T_j))_j = 0$ for $\deg r < d$,

$$\mathcal{N}_P r = [\Omega(r(T_j)w(T_j))_j] \otimes \text{vec}(E_1 - E_2).$$

At $2d$ nodes the scalar residual space has dimension $d - 1$, so the map from the d coefficients of r has nonzero kernel, even for relatively prime f_1, f_2 .

For arbitrary k , choose close relatively prime monic polynomials f, g with simple roots in (L, D) , so that every convex combination $(1 - s)f + sg$ retains that property by the bracket argument. Choose $s_b \in [0, 1]$ including both endpoints and set $f_b = (1 - s_b)f + s_b g$, with $\alpha_b = 1/k$ and any strictly positive full-rank stochastic channels. The collection is relatively prime. Put $A = \sum_b \alpha_b E_b$, $D_0 = \sum_b \alpha_b s_b E_b$, and $\bar{s} = \sum_b \alpha_b s_b$. Then

$$P(T) = A + (D_0 - \bar{s}A) \frac{g(T) - f(T)}{q(T)}.$$

This is a genuinely nonconstant affine-line curve. Indeed the matrices E_b are linearly independent, as follows by applying the channel inverses. Thus $D_0 - \bar{s}A \neq 0$, since the s_b are not all equal. Moreover $g - f$ has degree less than d and is nonzero, so $(g - f)/q$ is nonconstant. The same scalar residual-dimension argument gives a nonzero kernel. Lemma D.2 completes the proof. $\square$

D.1. Root matching without cross-component gaps. Write $\|F\|_{\text{coef}} = \sum_{r=0}^d \|F_r\|_{\text{op}}$ for a matrix polynomial $F(z) = \sum_{r=0}^d F_r z^r$.

Lemma D.4 (Diagonal polynomial perturbations). *Let $D_f(z) = \text{diag}(\alpha_b f_b(z))$ be as in (2.1). If $\|E\|_{\text{coef}} \leq \varepsilon$ is sufficiently small, then $\det(D_f + E)$ has degree kd and its root multiset Z_E satisfies $d_\infty(\Lambda, Z_E) \leq C\varepsilon^{1/d}$. The roots of Z_E may be complex. If roots within each f_b are simple and ν -separated, the bound is $C_\nu \varepsilon$. No separation between roots of different components is required.*

Proof. For $\varepsilon < \alpha_*/2$, the leading coefficient of $D_f + sE$ is invertible for every $0 \leq s \leq 1$. For sufficiently large $|z|$, $|f_b(z)| \geq c|z|^d$ and $\|E(z)\| \leq C\varepsilon|z|^d$. A Neumann series excludes zeros outside a fixed disk uniformly in s . Inside this disk, with $\rho = \text{dist}(z, \Lambda)$,

$$\|D_f(z)^{-1}\| \leq \alpha_*^{-1} \rho^{-d}, \quad \|E(z)\| \leq C\varepsilon.$$

Thus every zero throughout the homotopy lies in the union of disks of radius $r = (C_1 \varepsilon)^{1/d}$ about the true roots. Choose C_1 large enough that exclusion is strict outside these disks.

Fix $r' > r$ avoiding the finitely many radii at which components of the disk union first touch. Distinct components of the closed r' -disk union have positive distance from each other. Surround each by piecewise smooth contours in the complement of the closed r -disks, enclosing that component and no other. Such contours follow from a slight outer perturbation of the disk-union boundary; holes may be filled because all zeros are already in the r -disk union. The homotopy determinant is nonzero on each contour. The argument principle therefore preserves the enclosed number of roots with algebraic multiplicity, equal to the number of true centres in that component.

If this number is n , a chain of overlapping disks joins centres in at most $n - 1$ steps. Their mutual distance is at most $2(n - 1)r'$. Every perturbed root is within r of a centre, and there are exactly n such roots. Any bijection within the component has distance at most $(2n - 1)r'$. Let $r' \downarrow r$ through nonexceptional radii. Since there are finitely many permutations, a subsequence has one fixed matching. This proves the claim even at touching disks and repeated centres. For real spectra, uncrossing inverted pairs gives ordered matching.

Under the within-component separation assumption, the other factors near a root of f_b have magnitude bounded below in terms of ν . Away from these neighbourhoods $|f_b|$ is uniformly positive. On the fixed disk, $|f_b(z)| \geq c_\nu \min\{1, \text{dist}(z, Z(f_b))\}$. The same argument uses radius $C_\nu \varepsilon$. A repeated aggregate root from different simple scalar components is a semisimple collision; no eigenvector labeling or cross-component separation is needed. $\square$

Proof of Theorem C.1. The defect, fibre and worst-case clock assertions have been proved. The full-span assertion is proved in Proposition D.6 below. A competitor satisfies $\mathcal{N}_{P'}a' = b_{P'}$. Projection gives

$$\mathcal{N}_P(a' - a) = (\Omega \otimes I)(q'(T_j) \text{vec}(P(T_j) - P'(T_j)))_j. \quad (\text{D.1})$$

Uniform clock-value bounds for q' yield $\|a' - a\| \leq C\delta/\tau$; compactness gives the cap in (C.2).

Let $L_U = U^\dagger$ and $R_V = (V^\dagger)^\top$, so $\|L_U\| \|R_V\| = \kappa^{-1}$. Set $M = L_U K' R_V$. At the first $d+1$ clocks,

$$\begin{aligned} M(T_j) - \text{diag}(\alpha_b f_b(T_j)) &= q'(T_j) L_U (P'(T_j) - P(T_j)) R_V \\ &\quad + (q'(T_j) - q(T_j)) \text{diag}(p_b(T_j)). \end{aligned} \quad (\text{D.2})$$

The second term is diagonal. Interpolation gives $\|M - D_f\|_{\text{cof}} \leq C\delta(\kappa^{-1} + \tau^{-1}) =: C\epsilon$. In particular the normalization error has not acquired another channel factor. For small ϵ the leading coefficient of M is invertible. It equals $(L_U U') \text{diag}(\alpha')(V'^\top R_V)$, so both square outer factors are invertible. Hence

$$\det M(z) = \det(L_U U') \det(V'^\top R_V) \prod_b \alpha'_b f'_b(z).$$

Its roots are precisely the competitor multiset; no initial rank hypothesis on that competitor was used. Lemma D.4 gives both small-error bounds. Bounded target range handles larger errors. $\square$

Remark D.5. Common-factor cancellation and multiplicity within a scalar component are different mechanisms. The exponent $1/d$ is a uniform bound, not a claim of the optimal exponent at every real-rooted singularity. Theorem R.3 supplies matching neighbourhood rates on explicitly stated multiplicity strata. Cross-component collisions of simple scalar roots remain in the linear regime.

D.2. Functional complexity and a second sharp clock count. The worst-case obstruction in Proposition D.3 is a low-functional-rank obstruction, even when the latent capacity is large. The following result determines a different sharp count for curves whose numerator uses the whole scalar polynomial space.

Proposition D.6 (Full polynomial span). *Let the channels have full rank, and let $s = \dim \text{span}\{f_1, \dots, f_k\}$. At any design,*

$$\text{rank } \mathcal{N}_P \leq \min\{d, (s-1)(\ell-d-1)\}.$$

If $s = d+1$, every $d+2$ distinct admissible clocks give $\tau(P) > 0$. This count is necessary and sufficient on that full-span class. For every $k \geq d+1$ the class contains interior real-rooted examples with strictly positive full-rank stochastic channels.

Proof. The curve belongs to an affine matrix space of dimension at most $s-1$, since its scalar functions are f_b/q and their weighted sum is one. Write it as $P_0 + \sum_{i=1}^{s-1} w_i(T) D_i$. The residual annihilates $r(T)P_0$, and each remaining scalar residual belongs to an $(\ell-d-1)$ -dimensional space. This proves the rank bound.

Now let $\ell = d+2$ and let v span the one-dimensional orthogonal complement of the degree- d Vandermonde space. Each v_j is nonzero: otherwise its other $d+1$ coordinates would annihilate an invertible square Vandermonde matrix. If $r \in \ker \mathcal{N}_P$, applying the fixed channel inverses shows that $\sum_j v_j r(T_j) f(T_j)/q(T_j) = 0$ for every $f \in \text{span}\{f_b\}$. When that span is the whole $\mathbb{R}[z]_{\leq d}$, the vector $(v_j r(T_j)/q(T_j))_j$ therefore lies in the same one-dimensional complement and equals cv . Thus $r(T_j) = cq(T_j)$ at $d+2$ nodes. Polynomial identity and monicity give $c = 0$ and $r = 0$. At $d+1$ clocks the residual projector is zero, and Lemma D.2 gives actual nonidentification at interior simple-root parameters. Finally choose a monic F with simple roots in (L, D) and the $d+1$ polynomials $F, F+\epsilon, F+\epsilon z, \dots, F+\epsilon z^{d-1}$. For sufficiently small nonzero ϵ they retain

simple interior roots and span the whole space. Extra components may repeat F . Choose any positive full-rank stochastic channels and strict weights. $\square$

APPENDIX E. THE JOINT COEFFICIENT ALGEBRA AND OBSERVABLE CONDITIONING

The affine residues have a polynomial counterpart, but residues at individual roots are not the appropriate first decomposition. Distinct component polynomials may share a root. We separate the polynomials by their complete coefficient tuples before estimating their scalar zeros. This distinction is what preserves the additive normalization loss.

Let $\mathcal{R}_d$ be the model image on which $\tau(P) > 0$ and $\text{rank } P(T_1) = k$. Recover q and K by the normalization equation and interpolation. The leading coefficient is $A = K_d = U \text{diag}(\alpha) V^\top$. Choose orthonormal frames Q_1, Q_2 for its column and row spaces, and put

$$M(z) = Q_1^\top K(z) Q_2 = \sum_{j=0}^d M_j z^j, \quad A_0 = M_d, \quad C_j = M_j A_0^{-1} \quad (j < d).$$

These matrices are determined up to orthogonal changes of frames. If $S = Q_1^\top U$ and $T = Q_2^\top V$, then

$$C_j = S \text{diag}(f_{bj}) S^{-1}, \quad M(z) A_0^{-1} = S \text{diag}(f_b(z)) S^{-1}. \quad (\text{E.1})$$

The C_j therefore commute and generate a semisimple real algebra. For each distinct monic polynomial f occurring as a joint eigenvalue tuple, let E_f denote its joint spectral projector. Concretely, choose real c_j for which the numbers $\sum_{j < d} c_j f_j$ are distinct for these finitely many tuples, and take the spectral projectors of $\sum_{j < d} c_j C_j$. The forbidden choices form finitely many proper hyperplanes. The resulting projectors are independent of this choice. Set

$$W_f = A_0^{-1} E_f, \quad B_d(P) = \sum_{f \text{ distinct}} \|W_f\|_{\text{op}}, \quad \eta_d(P) = (B_d(P) + \tau(P)^{-1})^{-1}. \quad (\text{E.2})$$

Set $\eta_d = 0$ off $\mathcal{R}_d$. This is a finite construction from the data. No optimization over latent factorizations enters it. The rank of E_f records the number of components with polynomial f .

Theorem E.1 (Observable polynomial inverse). *The quantities in (E.2) are independent of the orthonormal frames and of the latent representation. On $\mathcal{R}_d$,*

$$\|A_0^{-1}\| \leq B_d(P) \leq \frac{k}{\alpha_* \kappa}, \quad \eta_d(P) \geq c(\kappa^{-1} + \tau(P)^{-1})^{-1}. \quad (\text{E.3})$$

Every competitor in the entire closed model $\mathcal{K}_d$ satisfies

$$d_\infty(\Lambda, \Lambda') \leq C \min\{1, [\delta/\eta_d(P)]^{1/d}\}. \quad (\text{E.4})$$

If the distinct roots within each true component polynomial have separation at least $\nu > 0$ and multiplicity at most m , the exponent is $1/m$, with an additional dependence on ν . In particular, within-component simple roots give a linear estimate through arbitrary cross-component collisions. The same estimates hold for the direct estimator of Section P, with its error e in place of δ and no conditioning floor supplied to the estimator.

On $\mathcal{R}_d$, B_d is lower semicontinuous and is continuous on strata with a fixed partition into equal component polynomials. The zero extension of η_d is upper semicontinuous on the whole model image, and

$$\eta_d(P) > 0 \iff \tau(P) > 0 \text{ and } \text{rank } P(T_1) = k.$$

For $d = 1$, $B_1 = \beta$ and $\eta_1 \asymp \eta$, with the notation of Section O.

Proof. Formula (E.1) gives

$$E_f = S \operatorname{diag}(\mathbf{1}_{f_b=f}) S^{-1}, \quad W_f = T^{-\top} \operatorname{diag}(\alpha_b^{-1} \mathbf{1}_{f_b=f}) S^{-1}, \quad M(z)^{-1} = \sum_f \frac{W_f}{f(z)}. \quad (\text{E.5})$$

The algebra, its joint eigenvalue tuples and its joint projectors are observable. An orthogonal change $Q_1 \mapsto Q_1 O_1$, $Q_2 \mapsto Q_2 O_2$ sends W_f to $O_2^\top W_f O_1$. The norm is unchanged. Since $\sum_f W_f = A_0^{-1}$, the triangle inequality gives the lower bound in (E.3). Each summand is bounded by $(\alpha_* \kappa)^{-1}$; there are at most k of them. This proves both comparisons.

Write $r = q' - q$. The normalization identity (D.1) gives $\|r\|_{\text{coef}} \leq C\delta/\tau$. Here and below the scalar coefficient norm is Euclidean or the sum norm; their fixed dimensional comparison is included in C . Let I_d be interpolation at the first $d+1$ clocks and define the scalar polynomial

$$h_f[r](z) = I_d(r(T_j)f(T_j)/q(T_j))(z).$$

It is linear in r , and $\|h_f[r]\|_{\text{coef}} \leq C\|r\|_{\text{coef}}$ uniformly over the model. The bounds use only clock geometry and the positive bounds for $f(T_j)/q(T_j)$, not channel inverses. Form

$$\overline{M}_r(z) = S \operatorname{diag}(\alpha_b(f_b(z) + h_{f_b}[r](z))) T^\top. \quad (\text{E.6})$$

This is the reduction of $K + I_d(rP)$. It is used algebraically; its scalar roots need not remain real. It satisfies the exact identity

$$\overline{M}_r(z)^{-1} = \sum_f \frac{W_f}{f(z) + h_f[r](z)}. \quad (\text{E.7})$$

In particular equal complete polynomials undergo the same gauge. There is no need to split a residue at a root shared by different polynomials.

The reduction of the competitor has the form $Q_1^\top K'(z) Q_2 = \overline{M}_r(z) + H(z)$, where H interpolates $q'(T_j)Q_1^\top(P'_j - P_j)Q_2$. Hence $\|H\|_{\text{coef}} \leq C\delta$. On a fixed disk containing all roots, put $\rho = \operatorname{dist}(z, \Lambda)$. Since $|f(z)| \geq \rho^d$, whenever $\rho^d \geq C_1\|r\|_{\text{coef}}$ the scalar denominators in (E.7) have modulus at least $\rho^d/2$. Thus

$$\|\overline{M}_r(z)^{-1} H(z)\| \leq C B_d \delta \rho^{-d}.$$

Choose $a^d = C_2(\|r\|_{\text{coef}} + B_d \delta)$, with C_2 large. Every determinant on the homotopy $\overline{M}_r + sH$ is nonzero outside the a -disks about the true roots. The same is true on the preliminary homotopy $M + u(\overline{M}_r - M)$, since its scalar denominators obey the same estimate. Leading coefficients stay invertible: the scalar leading coefficients in (E.6) are close to one, and their reduced inverse has norm at most $2B_d$; then H_d is a small relative perturbation. For large $|z|$, $|f(z)| \geq c|z|^d$ and both errors are bounded by their coefficient norms times $C|z|^d$. The same Neumann argument excludes roots there. There are therefore exactly kd roots through both homotopies. The contour and fixed-permutation limiting argument of Lemma D.4 gives the matching in (E.4). An invertible leading coefficient of the reduced competitor forces its two reduced channel factors to be invertible, so these are exactly its kd component roots. Larger errors are covered by the bounded target interval.

For the refined exponent, near a root cluster of a given f , all factors outside that cluster are bounded below in terms of ν . There are at most m factors in the cluster. On a fixed disk this gives $|f(z)| \geq c_\nu \min\{1, \rho^m\}$. Repeat the two homotopies with $a^m = C_\nu(\|r\|_{\text{coef}} + B_d \delta)$. This argument is also uniform on a fixed system of separated clusters containing at most m roots of each component: gaps inside the clusters may vanish.

For the estimator, $\|\hat{q} - q\|_{\text{coef}} \leq Ce/\tau$ as in (P.3). Formula (E.5) at T_1 gives $\sigma_k(P_1)^{-1} \leq C B_d$. The estimated row and column frames therefore have invertible overlaps with the true frames, with inverse norms at most two, when eB_d is small. In those frames each W_f is multiplied on the

left and right by these inverse overlaps, and their norm sum is at most $4B_d$. The preceding proof applies verbatim with $r = \hat{q} - q$ and interpolation error Ce . All fallback conditions are then excluded. Projecting complex roots to real parts and to $[L, D]$ cannot increase their matched distances.

For semicontinuity, take a sequence of data converging inside $\mathcal{R}_d$ and choose continuous local orthonormal frames. The normalizations, coefficients and the matrices C_j converge. Choose one linear combination separating the distinct joint tuples of the limit. Its contour projectors give, for each limiting polynomial f ,

$$\sum_{g \text{ in the cluster of } f} W_g^{(n)} \longrightarrow W_f.$$

To justify clustering of the full tuples, one may also take a compact subsequence of the original model parameters. The rank- k limiting observation forces both limiting channels to have full rank; their component coefficients consequently give exactly those joint tuples. The triangle inequality proves lower semicontinuity. If groups neither split nor merge, every summand converges, giving continuity. Since τ is continuous and positive here, the assertions for η_d follow locally. At a limit with $\tau = 0$, use $\eta_d \leq \tau$. At a limit with $\tau > 0$ and rank loss, the normalizations converge and (E.5) gives $B_d \geq c\sigma_k(P_1)^{-1} \rightarrow \infty$. This proves the zero-extension statement. Finally, in degree one the distinct polynomials are precisely $z - x$, so W_{z-x} is the residue R_x and $\tau \asymp \gamma$. $\square$

E.1. Explicit dependence on degree and clock geometry. The following estimate makes the constants in the uniform polynomial bound quantitative. It is not a claim that the monomial interpolation problem stays well conditioned as degree grows. Let ℓ_j be the Lagrange polynomials at $T_1, \dots, T_{d+1}$ and put

$$\begin{aligned} R_0 &= \max\{1, |L|, |D|\}, & R &= 2R_0 + 2, & \mathcal{L} &= \sum_{j=1}^{d+1} \|\ell_j\|_{\text{coef}}, \\ Q &= \max_{j \leq \ell} (T_j - L)^d, & V_d &= \max_{j \leq \ell} \left(\sum_{i=0}^{d-1} |T_j|^{2i} \right)^{1/2}, & \chi &= \max_{j \leq \ell} \left(\frac{T_j - L}{T_j - D} \right)^d, \\ \mathcal{A}(P) &= \mathcal{L}Q \left(B_d + \chi V_d \sqrt{\ell}/\tau \right). \end{aligned}$$

Here the scalar coefficient norm is the sum of absolute coefficients.

Proposition E.2 (A quantitative bound). *If $\delta\mathcal{A}(P) \leq 2^{-d-4}$, then*

$$d_\infty(\Lambda, \Lambda') \leq (2kd - 1)(4R^d \delta\mathcal{A}(P))^{1/d}. \quad (\text{E.8})$$

There is no hidden dependence on m_1, m_2 in this deterministic formula: the data norm is Frobenius, and B_d, τ already use that observation.

Proof. The Euclidean coefficient norm of r is at most $\sqrt{\ell}Q\delta/\tau$. Hence $\|h_f[r]\|_{\text{coef}} \leq \mathcal{L}\chi V_d \sqrt{\ell}Q\delta/\tau$, whereas $\|H\|_{\text{coef}} \leq \mathcal{L}Q\delta$. Their sum after weighting the latter by B_d is at most $\delta\mathcal{A}$. On $|z| \leq R$, evaluation costs at most R^d . With $a^d = 4R^d \delta\mathcal{A}$, both homotopy exclusions above are strict outside the a -disks. On $|z| \geq R$ one has $|f(z)| \geq 2^{-d}|z|^d$; the imposed smallness makes the scalar and matrix relative errors less than $1/4$. It also ensures invertibility of every leading coefficient. A disk component with n centres, counted with multiplicity, admits matching distance at most $(2n - 1)a$. Use $n \leq kd$ and pass to nonexceptional radii as in Lemma D.4. $\square$

E.2. Degree preservation, closed-model competitors, and grouping. The following lemmas make explicit the quantitative and algebraic steps in the proof of Theorem E.1. In particular, regularity of the competing channels is a conclusion of the small-error argument, not an additional hypothesis on the competitor.

Lemma E.3 (The two leading-coefficient homotopies). *In the notation of (E.6), put*

$$h = \max_f \|h_f[r]\|_{\text{coef}}, \quad e = \|H\|_{\text{coef}}, \quad B = B_d(P).$$

If $h \leq 1/4$ and $Be \leq 1/4$, the leading coefficients of

$$M + u(\overline{M}_r - M), \quad 0 \leq u \leq 1, \quad \text{and} \quad \overline{M}_r + sH, \quad 0 \leq s \leq 1,$$

are invertible. Their inverse norms are at most $2B$ and $4B$, respectively. Each determinant has degree exactly kd . In the observation comparison it suffices that

$$C_0\delta/\tau(P) \leq 1/4, \quad C_1B_d(P)\delta \leq 1/4, \quad (\text{E.9})$$

where $\|h_f[r]\|_{\text{coef}} \leq C_0\delta/\tau(P)$ and $\|H\|_{\text{coef}} \leq C_1\delta$ are the interpolation bounds in that proof. The constants involve only the fixed scalar experiment.

Proof. Write $h_{f,d}$ for the leading coefficient of $h_f[r]$. The inverse of the leading matrix in the first homotopy is

$$\sum_f \frac{W_f}{1 + uh_{f,d}}.$$

All denominators have modulus at least $3/4$, so its norm is at most $4B/3$. If L is the leading matrix at $u = 1$, then $\|L^{-1}sH_d\| \leq (4/3)Be \leq 1/3$. A Neumann series proves the second assertion, with the stated bounds. The coefficient of z^{kd} in the determinant of any $k \times k$ degree- d matrix polynomial is the determinant of its leading coefficient. It is nonzero throughout both homotopies. Substitution gives (E.9). $\square$

Lemma E.4 (The reduced competitor is the target). *Let $Q_1 \in \mathbb{R}^{m_1 \times k}$ and $Q_2 \in \mathbb{R}^{m_2 \times k}$ have orthonormal columns, and let a competitor in $\mathcal{K}_d$ have numerator $K' = U' \text{diag}(\alpha'_b f'_b) V'^T$. If the leading coefficient of $M' = Q_1^T K' Q_2$ is invertible, then both $S' = Q_1^T U'$ and $T' = Q_2^T V'$ are invertible, and*

$$\det M'(z) = \det S' \det T' \prod_{b=1}^k \alpha'_b f'_b(z). \quad (\text{E.10})$$

In particular the kd roots counted by the homotopy argument are exactly the competitor's component-root multiset, including all multiplicities. Its original channels necessarily have full column rank, even if this was not assumed when introducing the competitor.

Proof. The leading matrix is the product of the three square matrices S' , $\text{diag}(\alpha')$, and T'^T . All weights are positive. Invertibility of the product forces both outer factors to be invertible. Taking determinants of the polynomial factorization proves (E.10). Since the f'_b are monic of degree d and the scalar prefactor is nonzero, no root is cancelled and no leading degree is lost. A rank-deficient U' or V' could not have an invertible $k \times k$ compression. $\square$

Lemma E.5 (Componentwise clusters). *For each scalar component polynomial f , partition its roots, counted with multiplicity, into clusters of size at most m . Assume roots in different clusters of the same polynomial are separated by at least $\nu > 0$. No separation or common partition is imposed between different component polynomials. On a fixed disk containing the roots,*

$$|f(z)| \geq c_\nu \min\{1, \text{dist}(z, Z(f))^m\} \geq c_\nu \min\{1, \text{dist}(z, \Lambda)^m\}. \quad (\text{E.11})$$

The constant is uniform when gaps inside a cluster vanish. Accordingly the two exclusions in Theorem E.1 use radius $C_\nu(\|r\|_{\text{coef}} + B_d\delta)^{1/m}$, with no cross-component gap in the constant.

Proof. Put $a = \min\{1, \nu/4\}$. If the distance from z to every root of f is at least a , then $|f(z)| \geq a^d$. Otherwise select a nearest root and its cluster, containing $s \leq m$ roots. Every root in another cluster has distance at least $\nu - a \geq 3\nu/4$ from z . Each of the s factors inside the chosen cluster has modulus at least $\text{dist}(z, Z(f))$. This gives $|f(z)| \geq (3\nu/4)^{d-s} \text{dist}(z, Z(f))^s$. Since the last distance is less than one, its s th power is at least its m th power. Taking the minimum of these finitely many constants proves the first bound; $Z(f) \subset \Lambda$ gives the second. The scalar gauge error is bounded by $C\|r\|_{\text{coef}}$ and the matrix error by $C\delta$. Applying (E.7) with this lower bound gives both homotopy exclusions. The leading-degree lemma and the original contour matching argument then apply unchanged. For simple roots take singleton clusters, so $m = 1$. Disjoint root brackets in Lemma D.2 are likewise required only within each scalar polynomial, never between components. $\square$

Lemma E.6 (Observable grouping semicontinuity). *The lower semicontinuity and fixed-partition continuity of B_d in Theorem E.1 follow from the recovered coefficient algebra alone. No choice or limiting subsequence of latent representations is needed.*

Proof. Let $P_n \rightarrow P$ inside $\mathcal{R}_d$. Normalization recovery and locally continuous orthonormal frames give $A_{0,n} \rightarrow A_0$ and $C_{j,n} \rightarrow C_j$. Every joint tuple of the $C_{j,n}$ is bounded, since each coordinate is an eigenvalue of a bounded matrix. Choose a unit common eigenvector for any sequence of such tuples. Along a subsequence both the vector and its tuple converge; the limit vector is nonzero and is a common eigenvector for the C_j with the limiting tuple. Thus every nearby joint tuple lies near one of the finitely many joint tuples at P . This argument is entirely within observable matrix space.

Choose a fixed real linear combination $C(c)$ separating the distinct limit tuples, and disjoint contours around its eigenvalues. The Riesz projector of $C_n(c)$ on a contour converges to E_f . It equals the sum of all joint projectors whose full tuples approach f , even if some of their values under c coincide. Consequently

$$\sum_{g \text{ approaching } f} A_{0,n}^{-1} E_{g,n} \longrightarrow A_0^{-1} E_f = W_f.$$

The triangle inequality, summed over limit tuples, gives $B_d(P) \leq \liminf_n B_d(P_n)$. With a fixed equality partition no cluster splits or merges, and each summand converges separately. Finally $\eta_d \leq \tau$ handles limits with $\tau = 0$; at rank-loss limits with $\tau > 0$, the resolved inverse at T_1 gives $\sigma_k(P_n(T_1))^{-1} \leq C B_d(P_n)$, forcing $B_d(P_n) \rightarrow \infty$. This also proves the asserted upper semicontinuity of the zero extension of η_d . $\square$

APPENDIX F. AN INTRINSIC MODULUS AT ARBITRARY MODEL POINTS

A local neighbourhood must be specified before one assigns a statistical exponent to a singular point. A fixed coefficient neighbourhood, an expanding tangent ball, and a shrinking observation ball are different experiments. The preceding sections address the first two. We now give a representation-free characterization for the third, allowing all intersections of the singular sets of the closed stochastic model.

Fix positive design weights w_j , with $\sum_j w_j = 1$. For probability arrays define the Hellinger metric

$$h_w(P, Q)^2 = \sum_j w_j \sum_e (\sqrt{P_{je}} - \sqrt{Q_{je}})^2.$$

The cell index e ranges over the $m_1 m_2$ observed outcomes. This normalization has no factor $1/2$. It remains meaningful at zero cells. For a datum P in the closed-model image put

$$\begin{aligned}\mathcal{B}_P(t) &= \{\theta \in \mathcal{K}_d : h_w(F(\theta), P) \leq t\}, \\ \omega_P(t) &= \max_{\theta, \theta' \in \mathcal{B}_P(t)} d_\infty(\Lambda(\theta), \Lambda(\theta')).\end{aligned}\tag{F.1}$$

Both the maximum and the set are intrinsic to P . In particular, $\omega_P(0)$ is the diameter of the spectral image of the exact fibre, rather than the diameter of a chosen latent parametrization. The function is nondecreasing, and compactness implies $\omega_P(t) \downarrow \omega_P(0)$ as $t \downarrow 0$.

Theorem F.1 (Observation-neighbourhood minimax equivalence). *Take independent categorical samples at the fixed clocks, with $n_j/N \rightarrow w_j > 0$. Choose a fixed $0 < c \leq 1/(8\sqrt{2})$ and set $t_N = c/\sqrt{N}$. For all sufficiently large N ,*

$$\frac{3}{32} \omega_P(t_N)^2 \leq \inf_{\hat{\Lambda}} \sup_{\theta \in \mathcal{B}_P(t_N)} \mathbb{E}_\theta d_\infty(\hat{\Lambda}, \Lambda(\theta))^2 \leq \omega_P(t_N)^2.\tag{F.2}$$

No rank, normalization, positivity-of-cells, or component-separation hypothesis is imposed on P . The local decision rule may depend on P, t_N , as usual for a minimax problem on a specified neighbourhood, but not on its unknown member.

If $\omega_P(0) = 0$, then either ω_P vanishes on a neighbourhood of zero, or there are $a_P \in \mathbb{Q}_{>0}$ and $b_P > 0$ such that

$$\omega_P(t) = b_P t^{a_P} + o(t^{a_P}) \quad (t \downarrow 0).\tag{F.3}$$

In the latter case the risk in (F.2) has order N^{-a_P} . If $\omega_P(0) > 0$, the same equivalence retains that exact-fibre diameter as a nonvanishing term.

Proof. The compactness assertion follows from the compact root and stochastic parameter ranges and the positive clock denominators. For the upper bound select any target in $\Lambda(\mathcal{B}_P(t_N))$ and use it as a constant decision. Every target in this local experiment is within $\omega_P(t_N)$ of that decision. This is a local information-theoretic upper bound, not a proposed globally adaptive estimator.

For the lower bound choose a maximizing pair in (F.1), and write $D = \omega_P(t_N)$. The triangle inequality gives $h_w(F(\theta), F(\theta')) \leq 2t_N$. Eventually $n_j/N \leq 2w_j$ for all j . The affinity of independent product laws is the product of their affinities. Consequently their squared Hellinger distance is at most the sum of the individual squared distances, and

$$h(\mathbb{P}_\theta^{(N)}, \mathbb{P}_{\theta'}^{(N)})^2 \leq \sum_j n_j h(F(\theta)_j, F(\theta')_j)^2 \leq 8N t_N^2.$$

Total variation is at most the Hellinger distance in our normalization, so it is at most $1/4$. Given any target estimate, choose the nearer of the two targets as a test. A wrong choice has loss at least $D^2/4$. The sum of the testing errors is at least $1 - \text{TV}$, whence the larger of the two risks is at least $D^2(1 - \text{TV})/8 \geq 3D^2/32$. This proof also covers zero cells, since it uses no likelihood-ratio denominator.

We justify the exponent statement in the actual closed model. Work first in labelled root coordinates, with the finite permutation quotient imposed only by the target distance. The constraints on U, V, α and the roots are polynomial inequalities and equalities. The map F is rational with denominators uniformly positive at the clocks. Introduce nonnegative variables for the square roots of cell probabilities. The Hellinger-ball constraints are then semialgebraic. The bottleneck distance is the minimum, over finitely many permutations, of a maximum of absolute coordinate differences, and is semialgebraic too. Quantifier elimination shows that the graph of the maximum in (F.1) is semialgebraic, with the entries of P and the fixed experiment

treated as real parameters. A one-variable semialgebraic function which is positive near zero and tends to zero has a positive leading Puiseux monomial. This standard consequence of real algebraic geometry [13, 9] gives (F.3). A semialgebraic function cannot alternate between zero and positive values infinitely often near zero, so these cases exhaust the possibilities. Substituting $t_N = cN^{-1/2}$ in the already proved two-sided bound proves the risk assertion. $\square$

The exponent has an exact arc description. This is useful when several singular mechanisms meet, because no singularity is suppressed before forming the admissible alternatives. For a positive semialgebraic function $u(s)$ tending to zero, let $\text{ord}_0 u$ denote its leading Puiseux exponent.

Corollary F.2 (Admissible-arc characterization). *In the nonzero case of (F.3),*

$$a_P = \min \frac{\text{ord}_0 d_\infty(\Lambda(\theta(s)), \Lambda(\theta'(s)))}{\text{ord}_0 \max\{h_w(F(\theta(s)), P), h_w(F(\theta'(s)), P)\}}, \quad (\text{F.4})$$

where the minimum ranges over semialgebraic pairs of admissible arcs whose observation distances tend to zero and whose displayed numerator and denominator are not identically zero. The minimum is attained. With real algebraic input data and experiment constants, its value is computable in principle by quantifier elimination and one-variable algebraic branch analysis. No polynomial-time bound is asserted.

Proof. For any such pair set the maximum observation distance equal to $t(s)$. The definition gives $d_\infty \leq \omega_P(t(s))$. Comparing positive leading powers yields a ratio at least a_P . Semialgebraic choice selects a maximizing pair for every sufficiently small $t > 0$; the selection can be restricted to an interval on which its coordinates are continuous and have Puiseux expansions. Parameterize it by $s = t$. If its actual maximal observation distance has order $r \geq 1$, its positive target distance has order a_P . The preceding inequality forces $a_P/r \geq a_P$, hence $r = 1$. This pair attains the minimum. Compactness ensures that its limits belong to the exact observation fibre. The final assertion follows by constructing the semialgebraic graph in the proof of the theorem and extracting its leading algebraic branch; it is an effective description, not a numerical algorithm for large instances. $\square$

This theorem specifies the interaction through the admissible observation geometry. It is not a claim that kernel dimension, rank and multiplicity integers alone determine a_P , nor a smooth coordinate normal form at every singular point. The semialgebraic and two-point ingredients are classical; their role here is to give one exact modulus, in the same experiment, on all intersecting faces of the model. The next result computes that modulus on a substantial singular stratum rather than leaving it as a variational definition.

F.1. Anchored real-rooted multiplicities.

Lemma F.3 (Anchored real-rooted root displacement). *Let f_0 be a monic real-rooted polynomial. On a sufficiently small coefficient neighbourhood restricted to monic real-rooted polynomials,*

$$d_\infty(Z(f), Z(f_0)) \leq C \|f - f_0\|_{\text{coef}}^{1/2}.$$

If all roots of f_0 are simple, the exponent is 1. If f_0 has a multiple root, the exponent 1/2 cannot be improved for this anchored comparison, irrespective of its multiplicity.

Proof. Factor f_0 into its powers of distinct linear factors. These cluster factors are pairwise coprime. The derivative of their multiplication map, with monic leading terms fixed, is invertible: a vanishing linear combination of the varied factors reduces modulo each cluster factor to its own variation times an invertible coprime product. Each variation must therefore vanish; domain and codomain dimensions agree. The implicit function theorem gives analytic cluster factors for f

near f_0 , with coefficient errors $O(\epsilon)$, where $\epsilon = \|f - f_0\|_{\text{coef}}$. Root continuity identifies the roots of each factor with those near its corresponding base root.

Translate a cluster of multiplicity $r \geq 2$ to zero and denote its real perturbed roots by $y_1, \dots, y_r$. Its first two elementary symmetric coefficients satisfy $e_1 = O(\epsilon)$ and $e_2 = O(\epsilon)$. Hence

$$0 \leq \sum_i y_i^2 = e_1^2 - 2e_2 \leq C\epsilon,$$

so every $|y_i| \leq C\sqrt{\epsilon}$. A singleton cluster moves by $O(\epsilon)$ instead. Matching within the separated base clusters proves both upper bounds. For sharpness at a multiple root x , replace two copies of x by $x - s, x + s$, leaving all other roots fixed. The polynomial changes by $-s^2 f_0(z)/(z - x)^2$, whereas its anchored bottleneck displacement is exactly s . $\square$

Theorem F.4 (The anchored multiplicity exponent). *Suppose $P = F(\theta_0)$ has strictly positive full-rank channels, strict weights, $\tau(P) > 0$, pairwise distinct complete component polynomials, and component roots in (L, D) . Repeated roots within a component and cross-component collisions are allowed. Then*

$$\omega_P(t) \asymp \begin{cases} t, & \text{every component root is simple,} \\ t^{1/2}, & \text{some component root is multiple.} \end{cases} \quad (\text{F.5})$$

Accordingly, the shrinking-observation-neighbourhood minimax squared risk in Theorem F.1 has order N^{-1} or $N^{-1/2}$, respectively. The constants may depend on the base datum.

Proof. Proposition K.3 supplies an ambient analytic coefficient inverse under exactly the hypotheses stated here, without any simple-root assumption. It identifies all nearby closed-model coefficient representatives up to component permutation, even though the real-rooted coefficient set at the base need not be a manifold. Its local derivative bound gives coefficient distance at most $C\|F(\theta) - P\|_2$ from the recovered base representative. Positive cells imply local equivalence of this norm and $h_w(F(\theta), P)$. The preceding scalar lemma, component by component, therefore bounds the target distance from the base by Ct or $C\sqrt{t}$. The triangle inequality gives the asserted upper bound for two targets in $\mathcal{B}_P(t)$.

At an internally simple base, move one component root by s and keep all other parameters fixed. The observation distance is at most Cs , and the aggregate anchored displacement is s for sufficiently small s , including when the original root is shared by another component. At a multiple root split two copies symmetrically as in Lemma F.3. The component coefficient change and thus the observation distance are $O(s^2)$, while the target moves by s . All roots remain real and interior, and the strict normalization, rank and tuple conditions persist. Choose s a fixed small multiple of t or $\sqrt{t}$, respectively, and use the base point as the other member of (F.1). This proves the lower bounds. The risk conclusion is (F.2). $\square$

There is no conflict with Theorem R.3. That theorem takes a supremum over a *fixed coefficient neighbourhood* and compares two moving alternatives. Here both alternatives must remain within $O(N^{-1/2})$ Hellinger distance of the *fixed base datum*. For example, choose the polynomial H in (R.9) with distinct real roots of sum zero. When $m > 2$, both H and $H + c$ have the same nonzero second elementary symmetric coefficient, namely minus half the sum of squared roots. Their dilations are each at coefficient distance of order u^2 from the multiple-root base, but their mutual coefficient difference is of order u^m . Analytic coefficient recovery and positive cells make these also the corresponding local observation orders. With $u = N^{-1/(2m)}$, their distance from the base is of order $N^{-1/m}$, larger than $N^{-1/2}$. They belong to every sufficiently small fixed neighbourhood eventually, but not to the shrinking ball used here. Thus the two exponents describe different geometries of the same real-rooted singularity; neither theorem replaces the other.

APPENDIX G. THE EXACT BRIDGE BETWEEN METRIC CONDITIONING AND OBSERVATION BALLS

We now relate the three condition objects used in the paper. The certificate η_d is observable and controls all closed-model competitors. The matrix V_H is invariant under coefficient-chart changes. The function ω_P is an exact target diameter. These are three precise meanings, respectively, of representation-free, coordinate-invariant, and decision-theoretic conditioning.

At a point of Θ_d° , partition the kd locally labelled roots into their equal-base-value blocks B . Let $\Gamma = \prod_B \mathfrak{S}_B$, acting only within those blocks, and write

$$d_\Gamma(x, y) = \min_{\pi \in \Gamma} \|x - \pi y\|_\infty, \quad \mathcal{E}_V = \{V^{1/2}z : \|z\|_2 \leq 1\}, \quad D_\Gamma(V) = \max_{x, y \in \mathcal{E}_V} d_\Gamma(x, y).$$

The metric compares two perturbed targets, rather than one target with the base point. This distinction is essential at collisions.

Theorem G.1 (Exact regular observation-ball modulus). *Let $P = F(\theta_0)$, $\theta_0 \in \Theta_d^\circ$, and let $H = H_{P,w} = \text{diag}(w_j/P_{je})$ be the categorical Fisher form. With the Hellinger normalization of Section F,*

$$\omega_P(t) = 2D_\Gamma(V_H(P))t + o(t). \quad (\text{G.1})$$

In particular, writing $\mathbf{c}_H = \max_a \sqrt{(V_H)_{aa}}$,

$$2\mathbf{c}_H \leq \lim_{t \downarrow 0} \frac{\omega_P(t)}{t} \leq 4\mathbf{c}_H.$$

If all aggregate base roots are distinct, the limit is exactly $4\mathbf{c}_H$. At a collision the constant is the stated finite quotient functional, not an assumed multiple of a radial condition.

Proof. Theorem K.1 accounts for the whole closed-model observation neighbourhood. Write $I = J^\top H J$. Uniformly in a sufficiently small coefficient chart,

$$h_w(F(\theta_0 + u), P) = \frac{1}{2}\|Ju\|_H + O(\|u\|^2), \quad g(\theta_0 + u) = g(\theta_0) + Gu + O(\|u\|^2).$$

The first expansion follows by Taylor expansion of every square root; all base cells are positive. Injectivity of J implies $\|u\| = O(t)$ for points in the Hellinger ball. Thus the rescaled coefficient balls $t^{-1}(R(\mathcal{B}_P(t)) - \theta_0)$ converge in Hausdorff distance to $\{u : u^\top I u \leq 4\}$. For the inner inclusion, first use a strict ellipsoid and then let its radius increase to 2; the outer inclusion follows from the uniform expansion and the lower singular value of J .

The image of $\{u : u^\top I u \leq 1\}$ under G is precisely $\mathcal{E}_{V_H}$: the minimum of $u^\top I u$ subject to $Gu = x$ is $x^\top V_H^{-1}x$. Distinct base-value blocks have a fixed positive gap. Hence for two sufficiently close representatives, matching takes place within the blocks, and their target distance is $d_\Gamma(Gu, Gv) + O(\|u\|^2 + \|v\|^2)$. The quotient metric is Lipschitz in both arguments. Hausdorff convergence and compactness therefore give (G.1). Since 0 belongs to the centrally symmetric ellipsoid, its quotient diameter is at least its sup-norm radius $\mathbf{c}_H$, and the triangle inequality gives at most twice that radius. Without collisions the pair $x, -x$ in a maximal-radius direction attains the latter bound. $\square$

The constant in this theorem can be evaluated by finitely many convex programs using only V_H . If B has size b , let $x_{B,(i)}$ denote its i th increasing coordinate and put

$$M_{B,i} = \max_{x \in \mathcal{E}_{V_H}} x_{B,(i)}.$$

Then

$$D_\Gamma(V_H) = \max_{B, 1 \leq i \leq b} \{M_{B,i} + M_{B,b+1-i}\}. \quad (\text{G.2})$$

Indeed sorting is an isometry for the real bottleneck metric, and central symmetry changes the minimum of the i th order statistic into the negative maximum of the $(b+1-i)$ th. More explicitly, $M_{B,i}$ is the maximum, over subsets $S \subset B$ of cardinality $b-i+1$, of

$$\max\{a : \|z\|_2 \leq 1, (V_H^{1/2}z)_r \geq a \ (r \in S)\}.$$

Each inner problem is a second-order cone problem. This is a finite observable formula, not quantifier elimination over latent factors. For example, for one two-coordinate collision block and $V = \text{diag}(1, \epsilon^2)$, $0 < \epsilon \leq 1$,

$$D_\Gamma(V) = 1 + \frac{\epsilon}{\sqrt{1 + \epsilon^2}} < 2.$$

This example concerns the covariance functional itself and explains why a universal collision-blind factor $4\mathbf{c}_H$ would be incorrect.

Proposition G.2 (Global certificate comparison and its limitation). *At a full-rank true datum with $\tau(P) > 0$, the global observable theorem implies, for every $t \geq 0$,*

$$\omega_P(t) \leq C \min\{1, (t/\eta_d(P))^{1/d}\}. \quad (\text{G.3})$$

On the within-component ν -separated class the exponent is 1; the cluster versions have exponent $1/m$ under their stated margins. Consequently $2D_\Gamma(V_{H_{P,w}}) \leq C_\nu/\eta_d(P)$ on the regular stratum. There is no uniform reverse comparison of $\eta_d(P)^{-1}$ with this Fisher-metric constant over all strictly positive regular data, even when the component roots are fixed.

Proof. For probability vectors p, q , factoring $p_e - q_e$ into the difference and sum of their square roots gives $\|p - q\|_2 \leq 2h(p, q)$. Hence a member of $\mathcal{B}_P(t)$ has observation error at most $2t/\sqrt{\min_j w_j}$ from P . Apply the global observable estimate from P to each member of a pair and use the triangle inequality. The separated and clustered versions follow from the same argument. The regular limit is Theorem G.1.

For the last assertion fix an interior regular binary, two-component, degree-one datum $\bar{P}$ with distinct roots, and replace its first channel by $D_\epsilon \bar{U}$, where

$$D_\epsilon = \begin{pmatrix} 1 & 1 - \epsilon \\ 0 & \epsilon \end{pmatrix}, \quad P_\epsilon = D_\epsilon \bar{P}, \quad 0 < \epsilon < 1.$$

Both channels remain strictly positive and full rank; the roots and weights are unchanged. The normalization residual has the same kernel because D_ϵ is invertible. In square coordinates the leading matrix is $A_\epsilon = D_\epsilon \bar{A}$, so $B_1(P_\epsilon) \geq \|A_\epsilon^{-1}\|_{\text{op}} \geq c/\epsilon$. Thus $\eta_1(P_\epsilon)^{-1} \geq c/\epsilon$.

The row transformation is a local diffeomorphism between the two parameter charts. For an observation tangent Z_ϵ , $\bar{Z} = D_\epsilon^{-1}Z_\epsilon$ has second row $Z_{\epsilon,2}/\epsilon$ and first row $Z_{\epsilon,1} - (1 - \epsilon)Z_{\epsilon,2}/\epsilon$. The second row probabilities of P_ϵ are $\epsilon \bar{P}_2$, whereas its first row probabilities have a fixed positive lower bound. It follows, clock by clock, that

$$\|\bar{Z}\|_{H_{\bar{P},w}} \leq C\epsilon^{-1/2}\|Z_\epsilon\|_{H_{P_\epsilon,w}}.$$

The target is unchanged by the row transformation, so $\mathbf{c}_{H_{P_\epsilon,w}}(P_\epsilon) \leq C\epsilon^{-1/2}$. The regular modulus constant is at most four times this quantity, while $\eta_1^{-1} \geq c/\epsilon$. Their ratio tends to zero. The failure occurs as a cell floor vanishes; no reverse comparison on a fixed positive-cell compact class is being disproved by this example. $\square$

APPENDIX H. THE NEWTON LAW OF AN ADMISSIBLE OBSERVATION GERM

The order of a determinant perturbation must be computed after its composition with the admissible model. This section gives such a construction without assuming that the observation map is an immersion. The construction uses the monic spectral polynomial. It therefore continues to make sense when the observed matrix loses rank and its determinant is identically zero.

Let X be a compact semialgebraic set, let $F : X \rightarrow \mathcal{P}$ be a continuous semialgebraic map into a finite product of probability simplices, and let $A_x(z)$ be a continuous semialgebraic family of monic polynomials of fixed degree n . Complex coefficients are identified with their real and imaginary parts. Put

$$h_w(Q, Q')^2 = \sum_j w_j \sum_e (\sqrt{Q_{je}} - \sqrt{Q'_{je}})^2, \quad w_j > 0, \quad \sum_j w_j = 1.$$

Fix $P \in F(X)$ and write $B_P(t) = \{x \in X : h_w(F(x), P) \leq t\}$. The target is $Z(A_x)$, including multiplicity. Its metric is

$$d_\infty(Z, Z') = \min_{\pi \in \mathfrak{S}_n} \max_i |z_i - z'_{\pi(i)}|, \quad |\cdot| \text{ is the Euclidean modulus on } \mathbb{C}.$$

Real polynomials impose conjugacy constraints; those constraints are part of X and are never discarded. Define

$$\omega_{P,X}(t) = \text{diam}\{Z(A_x) : x \in B_P(t)\}. \quad (\text{H.1})$$

The whole inverse image of the observation ball is used. A pointed version is obtained by specifying a compact local admissible set X instead; it is not identified with the whole-model version.

Assume that the spectral polynomial is constant on the exact fibre:

$$A_x = A_0 \quad \text{for every } x \in F^{-1}(P). \quad (\text{H.2})$$

This is identification of the target, not of the latent parameter. Write $A_0(z) = \prod_{r \in \mathcal{R}} (z - r)^{m_r}$. Compactness and (H.2) imply that $A_x \rightarrow A_0$ uniformly for $x \in B_P(t)$ as $t \downarrow 0$. In disjoint discs about the distinct r , the corresponding monic cluster factors are therefore uniquely defined for small t :

$$A_{r,x}(r + \zeta) = \zeta^{m_r} + \sum_{a=0}^{m_r-1} c_{r,a}(x) \zeta^a, \quad A_x = \prod_r A_{r,x}. \quad (\text{H.3})$$

These coefficients are analytic functions of the coefficients of A_x in a neighbourhood of A_0 , and are semialgebraic there. Indeed, the differential of multiplication of the coprime monic cluster factors is invertible: reducing a zero differential modulo any one factor forces its own variation to vanish. The analytic inverse function theorem and the locally unique polynomial factor equations prove the two assertions.

For each of the finitely many pairs (r, a) put

$$b_{r,a}(t) = \max_{x \in B_P(t)} |c_{r,a}(x)|, \quad \beta_{r,a} = \text{ord}_{t=0} b_{r,a}(t). \quad (\text{H.4})$$

The order is $+\infty$ when the envelope vanishes on a neighbourhood of zero. Otherwise it is the positive exponent of its leading Puiseux monomial. Define

$$\alpha_P = \min_{r,a < m_r} \frac{\beta_{r,a}}{m_r - a}. \quad (\text{H.5})$$

The order in (H.4) is an observation order. It includes all orders of the observation map, the spectral map, and the admissibility constraints. It is not the order of a coefficient on an affine tangent space.

Suppose $\alpha_P < \infty$. Consider the joint, rather than coordinatewise, rescaled coefficient sets

$$\mathcal{C}_t = \left\{ \left(t^{-\alpha_P(m_r-a)} c_{r,a}(x) \right)_{r,a} : x \in B_P(t) \right\}. \quad (\text{H.6})$$

Let $\mathcal{C}_0$ be the fibre at $t = 0$ of the closure of their graph for $0 < t < t_0$. For $v \in \mathcal{C}_0$ let

$$\mathcal{Z}_r(v) = Z \left(\zeta^{m_r} + \sum_{a < m_r} v_{r,a} \zeta^a \right), \quad C_P = \max_{v, v' \in \mathcal{C}_0} \max_r d_\infty(\mathcal{Z}_r(v), \mathcal{Z}_r(v')).$$

The common parameter in (H.6) retains all cross-cluster, conjugacy, and stochastic constraints. Replacing $\mathcal{C}_0$ by a product of its coordinate projections would in general give an incorrect leading constant.

Theorem H.1 (Intrinsic Newton law). *Under the preceding hypotheses, either $\alpha_P = \infty$ and the spectral target is constant on $B_P(t_0)$ for some $t_0 > 0$, or $\alpha_P \in \mathbb{Q}_{>0}$, $0 < C_P < \infty$, and*

$$\omega_{P,X}(t) = C_P t^{\alpha_P} + o(t^{\alpha_P}). \quad (\text{H.7})$$

Moreover $\mathcal{C}_t \rightarrow \mathcal{C}_0$ in Hausdorff distance. The exponent and the leading root-multiset set depend only on the germ of the observation-spectral image $(F, A)(X)$ at (P, A_0) , with its specified observation metric. They are unchanged by analytic or semialgebraic reparametrizations preserving that image germ, including its admissibility constraints.

For real algebraic input, the exponents, the limiting coefficient set, and its variational diameter admit a finite description by real quantifier elimination and algebraic branch analysis. No polynomial-time claim or model-independent jet cutoff is made. In particular the theorem applies to the entire closed stochastic model (2.1), with $A_\theta = \prod_b f_b$, without a channel-rank assumption, without $\tau(P) > 0$, and without positive observed-cell assumptions, whenever its exact spectral fibre is a singleton.

Proof. All maxima in (H.4) exist. The graph of each is semialgebraic: introduce nonnegative square-root variables for the finitely many cell probabilities, use the polynomial equations for the cluster factors, and quantify the condition of being a maximum. Each envelope tends to zero. One-variable semialgebraic preparation therefore gives either local zero or a positive leading Puiseux monomial. These are the standard real-algebraic ingredients [13, 9]; the remaining argument identifies what their orders and leading sets mean for the observed spectrum.

For a monic polynomial $p(\zeta) = \zeta^m + \sum_{a < m} u_a \zeta^a$, let $R(p)$ be its largest root modulus and put $H(p) = \max_{a < m} |u_a|^{1/(m-a)}$. Elementary symmetric functions and a geometric-series root bound give constants depending only on m such that

$$c_m H(p) \leq R(p) \leq 2H(p). \quad (\text{H.8})$$

For the first inequality use $|u_a| \leq \binom{m}{a} R(p)^{m-a}$. For the second, if $|\zeta| > 2H(p)$, then $\sum_{a < m} |u_a \zeta^a| < |\zeta|^m$. Thus the maximal displacement from the base spectrum has order (H.5). Its diameter is between that maximal displacement and twice it, since the base spectrum belongs to every ball. This proves the exponent, but not yet the exact constant.

By the definition of α_P , every coordinate of (H.6) is uniformly bounded. The sets $\mathcal{C}_t$ are nonempty compact semialgebraic sets in a fixed compact box. Their set of cluster limits $\mathcal{C}_0$ is nonempty, compact, and semialgebraic. We verify Hausdorff convergence, rather than replacing a subsequential limit by a limit without justification. The outer inclusion follows from compactness: any sequence violating it has a convergent subsequence belonging to $\mathcal{C}_0$. For fixed $v \in \mathcal{C}_0$, the bounded semialgebraic function $t \mapsto \text{dist}(v, \mathcal{C}_t)$ has a limit at zero. Its definition as a cluster point supplies a subsequence along which this distance tends to zero, so the limit is zero. A finite net in $\mathcal{C}_0$, and the fact that distance to a set is 1-Lipschitz, make this convergence uniform in v . This proves the inner inclusion.

After the exact change $z = r + t^{\alpha_P} \zeta$, the monic cluster polynomial divided by $t^{\alpha_P m_r}$ has precisely the coefficients in (H.6). Root multisets are continuous functions of monic coefficients in the bottleneck metric, uniformly on compact coefficient sets. Hence their joint rescaled sets converge to $(\mathcal{Z}_r(v))_r$, $v \in \mathcal{C}_0$. Small disjoint base discs prevent a minimizing matching from crossing between different base clusters. Taking diameters proves (H.7). The set $\mathcal{C}_0$ contains zero. At least one minimizing envelope has a positive leading coefficient, so it also contains a nonzero vector. The corresponding collection of monic polynomials cannot all be pure powers. Its distance from the zero-root collection is positive. Thus $C_P > 0$.

If all orders are infinite, all finitely many cluster coefficients vanish on a common observation neighbourhood. This proves the rigidity alternative. Conversely, a constant spectral target makes all these coefficients zero. The entire construction takes place in $(F, A)(X)$, proving the asserted germ invariance, not merely basis invariance of a tangent space.

For algebraic input, factor isolation, the graphs of the envelopes, and the closure defining $\mathcal{C}_0$ have finite formulas over real algebraic numbers. Quantifier elimination constructs their semialgebraic descriptions, and Puiseux branch analysis gives the orders. Polynomial root equations and the finitely many matchings then describe C_P by a further compact optimization formula. These are effective finite procedures, not claims of efficient computation. In the stochastic model, compactness, semialgebraicity, and positivity of the clock denominators were established in the model definition; $\prod_b f_b$ remains monic even when K loses rank. All hypotheses are therefore verified as stated. $\square$

H.1. Composed analytic jets. The preceding theorem handles singular observation maps and inequality faces directly. When an admissible germ is an immersed analytic chart, its coefficient envelopes can instead be calculated from ordinary homogeneous jets. The following statement makes the required composition explicit.

Let $x \mapsto M(x, z)$ be real analytic for x in a neighbourhood of zero in $\mathbb{R}^N$, polynomial of fixed degree in z , with invertible leading matrix. Suppose its probability-normalized observation map $F(x)$ has positive cells and injective derivative $J = DF(0)$. Choose $\rho > 0$ so small that F is an embedding on a neighbourhood of the closed ball $\|x\| \leq \rho$. All competitors in this subsection are explicitly restricted to that ball. Put

$$I(v) = \sum_j w_j \sum_e \frac{(Jv)_{je}^2}{P_{je}}, \quad \mathcal{E} = \{v : I(v) \leq 4\}.$$

At a root r of $D_0(z) = \det M(0, z)$ of multiplicity m_r write

$$\det M(x, r + \zeta) = D_0(r + \zeta) + \sum_{j \geq 1} \sum_a H_{r,j,a}(x) \zeta^a, \quad H_{r,j,a} \text{ homogeneous of degree } j. \quad (\text{H.9})$$

The determinant in (H.9) is taken after substitution of the complete map $M(x)$. Set

$$\alpha = \min_{r,j,a < m_r : H_{r,j,a} \neq 0} \frac{j}{m_r - a}, \quad \Pi_{r,v}(\zeta) = b_r \zeta^{m_r} + \sum_{j=\alpha(m_r-a)} H_{r,j,a}(v) \zeta^a, \quad b_r = \frac{D_0^{(m_r)}(r)}{m_r!}.$$

Empty sums are zero; an empty minimum is infinite.

Theorem H.2 (Composed-jet formula). *If $\alpha < \infty$, the local observation-ball diameter equals*

$$t^\alpha \max_{v,v' \in \mathcal{E}} \max_r d_\infty(Z(\Pi_{r,v}), Z(\Pi_{r,v'})) + o(t^\alpha).$$

The displayed constant is finite and positive. Once any nonzero coefficient below a base multiplicity has supplied a provisional upper bound α_0 on α , only parameter jets of orders $j \leq \lfloor \alpha_0 \max_r m_r \rfloor$ can lower that bound or contribute to a minimizing edge. The all-zero case is local spectral rigidity

and, for a general analytic germ, requires an identity argument rather than any fixed finite test of Taylor coefficients. The formula is covariant under analytic changes of parameter with invertible derivative, provided the complete admissible and observed maps are composed with that change.

Proof. The square-root expansion gives $h_w(F(tv), P) = \frac{1}{2}t\sqrt{I(v)} + O(t^2)$ uniformly on bounded sets. Injectivity of J bounds $\|x\|$ by a constant times the observation distance in the chosen neighbourhood. The rescaled parameter balls converge to $\mathcal{E}$, including its boundary by radial approximation.

For $x = tv$ and $z = r + t^\alpha \zeta$, division by $t^{\alpha m_r}$ leaves precisely $\Pi_{r,v}$ in the limit. The convergence is uniform on bounded v, ζ sets. To justify this assertion for an infinite analytic expansion, truncate at an integer $J_0 > \alpha_0 \max_r m_r$. Analyticity bounds the remaining coefficient functions by $O(t^{J_0+1})$ uniformly on a smaller parameter ball; after the division their contribution tends to zero. All retained nonedge terms have strictly larger weight. On $|\zeta| = R$, first choosing R sufficiently large and then t small makes the perturbation smaller than the base leading monomial. Rouché's theorem, applied also on small fixed circles around the base clusters, localizes exactly m_r roots and excludes loss at infinity. Continuity of root multisets now gives uniform convergence, and taking diameters proves the formula. A nonzero edge polynomial at an interior point of $\mathcal{E}$ proves positivity by comparison with $v = 0$.

Terms of degree $j > \alpha_0 \max_r m_r$ have $j/(m_r - a) > \alpha_0$ and cannot contribute. If every lower coefficient vanishes identically, each determinant retains all base roots with their full multiplicities and the degree is unchanged; it is a nonzero scalar multiple of the base determinant. Finally, an analytic parameter diffeomorphism is a bijection of local admissible alternatives. It transports the observation balls exactly, and its derivative transports their limiting ellipsoids. Equivalently, the lowest nonzero homogeneous part of each composed coefficient transforms by that derivative. This proves the stated covariance while retaining, rather than discarding, higher model jets. $\square$

For $M(x) = M_0 + E_1(x) + E_2(x) + \dots$, its quadratic determinant jet is

$$D \det_{M_0}[E_2(x)] + \frac{1}{2} D^2 \det_{M_0}[E_1(x), E_1(x)].$$

Both summands are necessary. The affine result Theorem V.2 is the specialization in which $E_j = 0$ for $j \geq 2$. For a singular observation derivative one must use Theorem H.1, not a degenerate Fisher ellipsoid in the last theorem.

H.2. A curved cancellation with the same affine tangent.

Proposition H.3 (Arbitrarily delayed composed exposure). *For every $m \geq 3$ and integer $q > 2$, a positive rational polynomial experiment has an analytic one-parameter admissible germ whose affine tangent predicts exponent $2/m$, whereas its actual exponent is q/m . A corresponding germ with the same affine tangent can also be spectrally constant.*

Proof. Take $f = z^m$ and $g = \prod_{i=1}^m (z - a_i)$ with distinct $a_i > 0$, and at least $2m + 1$ clocks greater than all a_i . Let $O = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and define $M_x = \frac{1}{2} O B_x O^T$, where

$$B_x = \begin{pmatrix} f + x^2 + x^q & xg \\ x & g \end{pmatrix}.$$

The entry sum of M_x is $f + x^2 + x^q$, monic of degree m . At $x = 0$ all observed cells are positive since $0 < g(T) < f(T)$; positivity persists locally. The coprimality argument in Proposition V.6 gives $\tau(P) > 0$. The first derivative of the observation is nonzero by Lemma V.1 applied to the nonzero tangent. But the exact identities are

$$\det B_x = g(f + x^q), \quad \det(B_0 + xB'_0) = g(f - x^2).$$

At zero these give exponents q/m and $2/m$, respectively. The roots of g stay fixed; the roots of $f + x^q$ have modulus $|x|^{q/m}$. Taking the exact local Hellinger diameter therefore has the claimed exponent even when it exceeds one. Dropping x^q makes the determinant fg identically. These are germs in the positive rational extension, not asserted stochastic factorizations. They show why no jet cutoff depending only on matrix size and polynomial degree can cover arbitrary analytic parametrizations. $\square$

H.3. Statistical consequence.

Corollary H.4 (The intrinsic local risk exponent). *In the nonrigid case of Theorem H.1, take independent categorical samples with $n_j/N \rightarrow w_j > 0$, and fix $0 < c \leq 1/(8\sqrt{2})$. On the anchored observation neighbourhood $B_P(cN^{-1/2})$, the minimax squared bottleneck risk R_N satisfies*

$$\frac{3}{32}C_P^2c^{2\alpha_P} \leq \liminf_{N \rightarrow \infty} N^{\alpha_P} R_N \leq \limsup_{N \rightarrow \infty} N^{\alpha_P} R_N \leq C_P^2c^{2\alpha_P}.$$

The same assertion holds for the compact local positive rational experiment of Theorem H.2. The diameter constant is exact; this sandwich is not an assertion of an exact minimax decision constant.

Proof. The two-point proof of Theorem F.1 uses only the triangle inequality, the compact target diameter, and categorical Hellinger product bounds, so applies to both specified experiments. Substitute (H.7), or the composed-jet diameter, into its bounds $3\omega^2/32 \leq R_N \leq \omega^2$. $\square$

APPENDIX I. AN IDENTIFIABLE RANK-LOSS INTERSECTION IN THE STOCHASTIC MODEL

The full-collapse binary fibre is not an identifiable singularity. We give instead a closed-model datum at which one channel loses rank, one component has a double root, another has a triple root, and the spectral image of the exact fibre is a singleton. All probability cells, channel entries, and weight inequalities are strict. The root constraint has one active endpoint. The upper bound below is over the whole closed model, not over the displayed family alone.

Take $k = m_1 = m_2 = 2$, $d = 3$, $[L, D] = [0, 1]$, and at least seven distinct clocks greater than one. Fix $0 < r < 1$, put $s = 2r/3$, and set

$$f(z) = z(z-r)^2, \quad g(z) = (z-s)^3. \quad (\text{I.1})$$

Let u be a strictly positive probability vector, let $U = (u, u)$, and let $V = (v_1, v_2)$ be a strictly positive invertible column-stochastic matrix. Choose $\alpha \in (\alpha_*, 1 - \alpha_*)$ and define

$$K(z) = u\{\alpha f(z)v_1^\top + (1-\alpha)g(z)v_2^\top\}, \quad q(z) = \alpha f(z) + (1-\alpha)g(z).$$

Denote the resulting observation by P_* .

Lemma I.1 (An isolated real-rooted pencil). *For the monic affine pencil $h_c = (1-c)f + cg$, the only polynomials whose three roots all belong to $[0, 1]$ are $h_0 = f$ and $h_1 = g$. Furthermore, on every bounded c -range there are constants C and $\epsilon_0 > 0$ such that, if a monic polynomial p has all roots in $[0, 1]$ and $\|p - h_c\|_{\text{coef}} \leq \epsilon < \epsilon_0$, then*

$$\min\{|c|, |1-c|\} \leq C\epsilon.$$

Proof. All polynomials h_c have root sum $2r$. Their exact identities are

$$\begin{aligned} h_c(0) &= -\frac{8r^3}{27}c, \\ h_c(s+y) &= y^3 - \frac{r^2}{3}(1-c)y + \frac{2r^3}{27}(1-c), \\ \text{disc}(h_c) &= -\frac{4r^6}{27}c(1-c)^2. \end{aligned} \tag{I.2}$$

Nonnegative real roots imply $h_c(0) \leq 0$, hence $c \geq 0$. Their centered sum of squares is $2r^2(1-c)/3$, hence $c \leq 1$. A real-rooted cubic has nonnegative discriminant, which on $[0, 1]$ forces $c = 0$ or $c = 1$.

For the quantitative assertion, compactness first places c in arbitrarily small neighbourhoods of zero and one as ϵ tends to zero. Near zero, $p(0) \leq 0$ and the first identity give $c \geq -C\epsilon$. The discriminant is locally Lipschitz in bounded monic coefficients; its nonnegativity and the third identity give $c \leq C\epsilon$ there.

Near one put $t = 1 - c$ and depress the actual polynomial p by its own root mean. Since that mean is $s + O(\epsilon)$, its depressed coefficients a, b satisfy

$$a = -\frac{r^2}{3}t + O(\epsilon), \quad b = \frac{2r^3}{27}t + O(\epsilon).$$

Real-rootedness gives $a \leq 0$ and $|b| \leq 2(-a)^{3/2}/(3\sqrt{3})$. Consequently

$$|t| \leq C\epsilon + C(|t| + \epsilon)^{3/2}.$$

In a sufficiently small fixed neighbourhood the last term involving $|t|$ is absorbed in the left side, proving $|t| \leq C\epsilon$. This argument is important: a Lipschitz bound for the discriminant alone would only give $|1 - c| = O(\sqrt{\epsilon})$ at its double zero. The real-rooted cubic inequality supplies the sharper estimate. $\square$

Theorem I.2 (Rank loss, multiple roots, and a vanishing intrinsic modulus). *At P_* , all observed cells are positive, $\tau(P_*) > 0$, $\text{rank } U = 1$, and*

$$\Lambda(F^{-1}(P_*)) = \{\{0, r, r, s, s, s\}\}. \tag{I.3}$$

For the entire closed stochastic model with these fixed experiment constants, not merely a pointed subfamily, there is $C_ > 0$ such that*

$$\omega_{P_*}(t) = C_*\sqrt{t} + o(\sqrt{t}). \tag{I.4}$$

The constant is the joint coefficient-fibre diameter in Theorem H.1. Its observation-neighbourhood minimax squared risk has order $N^{-1/2}$ and satisfies the constant bounds of Corollary H.4.

Proof. Positivity at the clocks is immediate. We first prove $\tau > 0$ without applying any full-rank-channel theorem. If a polynomial e of degree below three lies in the normalization kernel, interpolation gives a matrix polynomial E of degree at most three with $E(T_j) = e(T_j)P_*(T_j)$. The polynomial $qE - eK$ has degree at most six and vanishes at seven clocks, so it is identically zero. Take the second marginal and use invertibility of V and positivity of the weights. It follows that q divides both ef and eg . Since f and g are coprime, q divides e , and $e = 0$.

For any exact competitor, its monic normalizer solves the same injective normalization equation. Thus $q' = q$, and interpolation gives $K' = K$. Write the second marginal as a column of polynomials

$$B(z) = K(z)^T \mathbf{1} = V \text{diag}(\alpha, 1 - \alpha) \begin{pmatrix} f(z) \\ g(z) \end{pmatrix}.$$

Its two by four coefficient matrix has rank two. In any competitor representation $B = V' \text{diag}(\alpha') \begin{pmatrix} f'_1 \\ f'_2 \end{pmatrix}$, the matrix V' must therefore be invertible. Each f'_b is a linear combination of f, g , and monicity makes it h_{c_b} for a real c_b . Lemma I.1 forces $f'_b \in \{f, g\}$. The rank of B prevents both from being the same. This proves (I.3), for all closed-model competitors. In particular it is not a statement about the fibre of a selected rank-one parametrization.

We prove a quantitative upper bound before extracting its leading constant. Put $\delta = \max_j \|P'_j - P_{*,j}\|_F$ for an arbitrary closed-model competitor. The normalization equation at P_* gives

$$\|q' - q\|_{\text{coef}} \leq C\delta, \quad \|K' - K\|_{\text{coef}} \leq C\delta.$$

For completeness, substitute the bounded coefficients of q' into the difference of the two normalization equations, apply the left inverse of $\mathcal{N}_{P_*}$, and then interpolate $q'P' - qP_*$. This uses $\tau(P_*) > 0$, but not the rank of U or of a competitor. It also gives $\|B' - B\|_{\text{coef}} \leq C\delta$.

The coefficient matrix of B' has its second singular value bounded below. The component polynomial coefficients and weights range over a fixed compact set. In $B' = V' \text{diag}(\alpha') \begin{pmatrix} f'_1 \\ f'_2 \end{pmatrix}$ this forces the least singular value of V' to be bounded below as well: otherwise a unit left singular vector of V' would contradict the lower singular value of B' . Its inverse and the inverse weights are consequently uniformly bounded for small δ . Multiplying the coefficient error $B' - B$ by these inverses shows

$$f'_b = (1 - c_b)f + c_bg + O_{\text{coef}}(\delta), \quad |c_b| \leq C.$$

Here monicity makes the two combination coefficients sum to $1 + O(\delta)$; its $O(\delta)$ correction is included in the remainder. The quantitative part of Lemma I.1 places each f'_b within $C\delta$ of f or g . They cannot both approach the same polynomial, since B' retains rank two with a fixed singular-value lower bound. After a component permutation,

$$\|f'_1 - f\|_{\text{coef}} + \|f'_2 - g\|_{\text{coef}} \leq C\delta. \quad (\text{I.5})$$

The anchored real-rooted estimate Lemma F.3, whose proof uses the first two cluster symmetric functions and allows endpoint roots, gives $d_\infty(\Lambda', \Lambda_*) \leq C\sqrt{\delta}$. For the stated Hellinger convention, $\delta \leq 2t/\sqrt{\min_j w_j}$ on $B_{P_*}(t)$. The triangle inequality proves $\omega_{P_*}(t) \leq C\sqrt{t}$.

For a matching lower order, keep the channels, weights, and f fixed and replace g by

$$g_u(z) = (z - s)((z - s)^2 - u^2), \quad |u| \text{ small.}$$

These are actual stochastic alternatives: all three roots $s - u, s, s + u$ remain in $(0, 1)$, and every other constraint is unchanged. Their aggregate root displacement is $|u|$, whereas the polynomial and observation changes are $O(u^2)$. The derivative of the observation with respect to u^2 is nonzero. Indeed a zero derivative would give $E(T_j) = e(T_j)P_{*,j}$ for the nonzero numerator derivative E and its entry sum e . The already proved normalization-kernel argument would force $e = 0$ and then $E = 0$, a contradiction. Hence $h_w(P_u, P_*) = \gamma u^2 + o(u^2)$ with $\gamma > 0$. This proves the lower bound $\omega_{P_*}(t) \geq c\sqrt{t}$.

The entire exact spectral fibre is the singleton already established. Theorem H.1 applies to $A_\theta = f'_1 f'_2$ and supplies a positive exact leading constant. The matching orders force $\alpha_{P_*} = 1/2$, proving (I.4). The statistical assertion follows from Corollary H.4. $\square$

Why regular coefficient charts do not prove this theorem. There is a genuine kernel in the ambient normalized coefficient experiment at this datum. Put $\beta_1 = \alpha v_1$ and $\beta_2 = (1 - \alpha)v_2$. For small c , the analytic changes

$$f \mapsto h_c, \quad \beta_1 \mapsto \frac{\beta_1}{1 - c}, \quad \beta_2 \mapsto \beta_2 - \frac{c\beta_1}{1 - c}$$

leave B , K , and P_* exactly unchanged. The coefficient vectors remain positive, and their sums give strict normalized weights and stochastic columns. The curve therefore witnesses the failure of an injective ambient coefficient Jacobian. Its nonzero points, however, are excluded by the actual real-root and endpoint constraints, exactly as Lemma I.1 proves. Replacing the admissible germ by that ambient tangent would lose identification altogether.

The anchored coefficient bound (I.5) is a consequence of this particular inequality geometry. It is not an assertion of an analytic coefficient inverse on an open neighbourhood of the rank-loss datum. Nor is the example a new exponent different from $1/2$: its contribution is a whole-model, spectrally identifiable intersection for which the regular full-rank recovery hypotheses fail. The exact constant comes from the full admissible coefficient fibre, not from the ambient singular Fisher matrix.

APPENDIX J. COORDINATE TRANSPORT OF THE OBSERVATION METRIC

Algebraic covariance and statistical isometry are different statements. Throughout the rational-coordinate results in Appendix V, including Theorem V.2, a change of numerator coordinates transports the observation metric as specified here. It never declares the transformed matrix entries to be new probability cells. This convention is part of the hypotheses of the invariance assertion, including its leading diameter constant.

Proposition J.1 (Left–right coordinates and the metric). *Let $T(X) = AXB$ for fixed invertible real matrices A, B , and let $\mathfrak{l}$ be the original scalar normalization functional. Set $\tilde{\mathfrak{l}} = \mathfrak{l} \circ T^{-1}$ and $\tilde{M} = T(M)$. Then the determinant root multiset and its algebraic Newton orders are preserved. The exact observation moduli, the Fisher ellipsoids, and their leading diameter constants are preserved when the transformed observations are equipped with*

$$\tilde{h}(Q, Q') = h_w(T^{-1}Q, T^{-1}Q'). \quad (\text{J.1})$$

At one clock the vectorized map is $\mathsf{T}_0 = B^\top \otimes A$; on the stacked clock arrays it is $\mathsf{T} = I_\ell \otimes \mathsf{T}_0$. If H is the Fisher form on those arrays, the transported form is

$$\tilde{H} = \mathsf{T}^{-\top} H \mathsf{T}^{-1}.$$

Without this transport, a general invertible left–right map is not asserted to preserve an entrywise Hellinger metric or even the probability simplex.

Proof. The denominator is unchanged because $\tilde{\mathfrak{l}}(\tilde{M}) = \mathfrak{l}(M)$. Thus transformed rational observations are $T(P)$, and applying T^{-1} recovers the original statistical experiment exactly. Equation (J.1) therefore preserves every observation ball, not only its tangent approximation. The determinant is multiplied by the constant $\det A \det B \neq 0$, which does not change any roots or any nonzero Newton order. On differentiating, $\tilde{J} = \mathsf{T}J$ and $\tilde{J}^\top \tilde{H} \tilde{J} = J^\top H J$. This proves the assertion about ellipsoids and constants as well. For rectangular arrays the same proof applies to the observation coordinates wherever the invertible left–right map is defined. $\square$

Scalar interpolation notation. The operator denoted by $\mathcal{H}_e M$ in the retained affine proof is more explicitly $\mathcal{T}_P(e) = I_d((e(T_j)P_j)_j)$; it is an interpolation operator, not multiplication by a polynomial $\mathcal{H}_e$. The scalar shear is $E = \mathcal{T}_P(e) + H$, with $\mathbf{1}^\top H \mathbf{1} = 0$. It transports the complete numerator map and the same observation metric. Applied to a nonlinear germ, it must be applied to all of that germ’s jets. Theorem H.1 then gives germ invariance; mere basis invariance of an affine slice is its strictly narrower special case.

Explicit local convention for the affine appendix. In the definition of ω_P^S in Appendix V, fix $\rho > 0$ such that the rational observation map is an embedding on a neighbourhood of $\{x : \|x\| \leq \rho\}$, and use exactly

$$\{x : \|x\| \leq \rho, \quad h_w(F_S(x), P) \leq t\}.$$

All determinant-root multisets there are in $\mathbb{C}$ with the metric stated in Section H. The radius is fixed before t tends to zero. The algebraic rigidity alternative follows from the following elementary fact: a degree- n polynomial with nonzero leading coefficient which retains every base root with its full multiplicity is a nonzero scalar multiple of the base polynomial. These conventions also specify the empty-edge case of the affine theorem.

Coefficient representatives. The quotient statements use monic-coefficient coordinates, or roots ordered within each component. The internal root-permutation quotient has already been taken. Raw root labels add internal permutations, not new coefficient representatives; on the internally simple stratum the remaining component action is the stated free $\mathfrak{S}_k$ action. In the immersed determinant calculation an invertible leading matrix fixes the determinant degree locally and excludes roots escaping to infinity.

APPENDIX K. THE REGULAR QUOTIENT AND ITS INTRINSIC METRIC

The global certificate η_d and an intrinsic local condition answer different questions. The former controls competitors throughout the closed model. The latter is the norm of the differential of the target on the observation image. We first identify that image, including all its nearby closed-model representatives.

Let Θ_d° consist of the parameters in $\mathcal{R}_d$ with strictly positive channel entries, $\alpha_b > \alpha_*$, pairwise distinct complete component polynomials, and simple component roots in (L, D) . Use the first $m_i - 1$ entries of each stochastic column, the first $k - 1$ weights, and the nonleading coefficients of the f_b as coordinates. Thus Θ_d° is an open subset of $\mathbb{R}^p$, where

$$p = k(m_1 + m_2 - 2) + (k - 1) + kd.$$

The simple real-rooted coefficient region is open, by the implicit function theorem and the sign-bracket argument. All rank and strict inequality conditions above are open as well. The group $\mathfrak{S}_k$ acts by simultaneous permutation of components. Put $F(\theta) = (\text{vec } P_\theta(T_j))_j$.

Theorem K.1 (Local quotient identification). *The action of $\mathfrak{S}_k$ on Θ_d° is free. For every $\theta_0 \in \Theta_d^\circ$, with $P = F(\theta_0)$, there are an open observation neighbourhood $\mathcal{U}$ of P and a coordinate neighbourhood $\mathcal{V}$ of θ_0 such that:*

- (1) *every closed-model representative of a datum in $\mathcal{U} \cap F(\mathcal{K}_d)$ belongs to exactly one of the disjoint sets $\pi\mathcal{V}$, $\pi \in \mathfrak{S}_k$;*
- (2) *each such datum has exactly one representative in $\mathcal{V}$;*
- (3) *the inverse from this model image to $\mathcal{V}$ is real analytic and extends to a real-analytic map on an ambient neighbourhood of P .*

Consequently $\Theta_d^\circ/\mathfrak{S}_k$ is a manifold, not an orbifold with nontrivial stabilizers, and its observation image is locally an embedded analytic manifold. The derivative $J = DF(\theta_0)$ has full column rank. Collisions of roots belonging to different component polynomials do not alter any of these conclusions.

Proof. Exact fibres. A permutation fixing a parameter fixes its complete polynomial tuple. Distinctness of these tuples makes the permutation the identity. Suppose any parameter in the closed model represents P . Since $\tau(P) > 0$, its normalizing polynomial is the unique monic solution of $\mathcal{N}_P a = b_P$. Interpolation therefore recovers the same K . Its leading coefficient A has rank k . The factorization $A = U' \text{diag}(\alpha') V'^T$ forces both competing channels to have full

column rank. In fixed orthonormal frames the observable coefficient algebra has k distinct joint tuples, with rank-one joint projectors. Any competing diagonal realization must assign exactly one component to each tuple. For the projector E_b of that tuple,

$$Q_b = Q_1 E_b A_0 Q_2^\top = \alpha_b U_{:b} V_{:b}^\top. \quad (\text{K.1})$$

Its entry sum is α_b ; its row and column sums, divided by α_b , are the two stochastic columns. Thus the entire exact fibre in $\mathcal{K}_d$ is the single permutation orbit of θ_0 .

Analytic recovery. For an ambient datum Y near P , define

$$a(Y) = (\mathcal{N}_Y^\top \mathcal{N}_Y)^{-1} \mathcal{N}_Y^\top b_Y, \quad K(Y) = I_d(q_Y Y).$$

The least-squares inverse exists on a neighbourhood of P . Select k independent columns and k independent rows of the leading coefficient $A(P)$. The same selections in $A(Y)$, orthonormalized by the analytic positive definite inverse square root of their Gram matrices, give analytic frames $Q_1(Y), Q_2(Y)$. On rank- k model data they span the whole column and row spaces. The reduced leading matrix $A_0(Y)$ is invertible near P , so each $C_j(Y) = M_j(Y) A_0(Y)^{-1}$ is analytic. Choose fixed real coefficients c_j for which $\sum c_j C_j(P)$ has k distinct eigenvalues. Disjoint small circles about these eigenvalues define analytic rank-one Riesz projectors $E_b(Y)$; simple real eigenvalues remain real on this real neighbourhood. Recover the coefficient tuple by $f_{bj}(Y) = \text{tr}(C_j(Y) E_b(Y))$, and recover the factors from (K.1) and its row, column and total sums. These formulae are analytic ambient maps. For data on the model, they give exactly its parameters, with the labelling selected by the circles.

All nearby representatives. The recovered parameters tend to θ_0 as $Y \rightarrow P$. Shrink the neighbourhood so that the recovered weights, entries, ranks, distinct tuples and simple interior roots satisfy the strict conditions defining Θ_d^o . Every closed-model representative of such a datum has the recovered q, K . The rank argument and the singleton projectors just used apply to that datum itself, not only to its limit. Its representatives are therefore precisely the permutations of the recovered point. Choose $\mathcal{V}$ small enough that its finitely many permutation translates are disjoint, and shrink $\mathcal{U}$ once more. This proves the first two assertions without assuming in advance that a competitor is regular.

On $\mathcal{V}$ the recovery identity is $R \circ F = \text{id}$. Differentiation gives $DRJ = I_p$, so J is injective. The constant-rank theorem identifies the local image as an embedded manifold, and the finite free action gives its quotient charts. Internal root simplicity makes the continuously labelled component roots analytic. Equality of a root from one component with a root from another neither identifies the complete tuples nor introduces a permutation stabilizer. It is therefore immaterial to this recovery construction. $\square$

Let $g(\theta)$ list the component roots, increasingly within each component, and put $G = Dg(\theta_0)$ and $q_0 = kd$. This labelling is a local device, not an additional target supplied to an estimator. Differentiating $f_b(\lambda_{ba}) = 0$ gives

$$\frac{\partial \lambda_{ba}}{\partial f_{bj}} = -\frac{\lambda_{ba}^j}{f_b'(\lambda_{ba})}, \quad j < d. \quad (\text{K.2})$$

For each component this is a nonsingular diagonal matrix times a Vandermonde matrix. In particular G has full row rank q_0 .

Theorem K.2 (Two-sided local conditioning). *Let H be a positive definite form on the product probability-simplex tangent space*

$$\mathcal{T} = \{Z : \mathbf{1}^\top \text{vec } Z_j = 0 \text{ for every } j\}.$$

It may be extended arbitrarily as a positive definite form to the ambient space. Set

$$V_H(P) = G(J^\top H J)^{-1} G^\top. \quad (\text{K.3})$$

This positive definite $q_0 \times q_0$ matrix is determined by the observation and the specified metric, up to simultaneous permutation of its rows and columns. The exact local inverse condition is

$$\mathfrak{c}_H(P) := \lim_{\epsilon \downarrow 0} \sup_{\substack{P' \in F(\mathcal{K}_d) \\ 0 < \|P' - P\|_H \leq \epsilon}} \frac{d_\infty(\Lambda(P'), \Lambda(P))}{\|P' - P\|_H} = \max_{a \leq q_0} \sqrt{[V_H(P)]_{aa}}. \quad (\text{K.4})$$

The maximum first-order ratio is attained along a smooth admissible path. For $H = I$, $\mathfrak{c}_2(P) \leq C_\nu/\eta_d(P)$ on the within-component ν -separated class. The same comparison holds for a fixed uniformly equivalent metric.

Proof. Theorem K.1 accounts for every nearby model datum. Writing $v = \theta' - \theta_0$, Taylor expansion gives $F(\theta') - P = Jv + O(\|v\|^2)$ and $g(\theta') - g(\theta_0) = Gv + O(\|v\|^2)$. Distinct values in the base aggregate multiset have a positive gap. For sufficiently small v , matching cannot improve upon matching within the corresponding base-value groups: a crossing between groups has a fixed positive cost, whereas the within-group matching cost tends to zero. Within an equal-value group every base root is the same number. Hence its bottleneck distance from the perturbed roots is exactly their largest absolute displacement. Thus, even at a collision,

$$d_\infty(\Lambda(\theta'), \Lambda(\theta_0)) = \|Gv\|_\infty + O(\|v\|^2).$$

This is a radial metric identity at the base datum; it does not assert that sorting is differentiable. Injectivity of J makes the desired limit $\sup_{v \neq 0} \|Gv\|_\infty / \|Jv\|_H$. For each row of G , duality for the Gram matrix $J^\top H J$ gives (K.4). A maximizing row and its dual vector supply an admissible two-sided line in the open parameter chart. The global comparison follows by differentiating Theorem E.1. Under a coordinate change with invertible Jacobian D , the matrices become JD and GD ; substitution leaves (K.3) unchanged. Component relabelling only permutes its indices. Full row rank of G proves positive definiteness. Since columns of J belong to $\mathcal{T}$, the ambient extension of H has no effect. $\square$

For balanced categorical sampling the per-observation Fisher form on $\mathcal{T}$ is

$$H_P = \ell^{-1} \text{diag} (1/(P_j)_{ab})_{j,a,b}.$$

Section L will identify the exact local statistical meaning of the whole matrix V_{H_P} , including collision points.

An alternative derivative formula avoids differentiating an eigenvector normalization. For an admissible observation tangent Z , let

$$\begin{aligned} r_Z &= \mathcal{N}_P^\top [- (\Omega \otimes I)(q(T_j) \text{vec } Z_j)_j], \\ \dot{M}_Z &= Q_1^\top I_d(qZ + r_Z P) Q_2. \end{aligned} \quad (\text{K.5})$$

The first expression denotes the polynomial with that coefficient vector; the frames in the second are held fixed at the base point. For an aggregate simple root x ,

$$\dot{x} = -\text{tr}((\text{Res}_{z=x} M(z)^{-1}) \dot{M}_Z(x)). \quad (\text{K.6})$$

Differentiate $\mathcal{N}_{Pa} = b_P$ for the first formula and the simple zero of $\det M$ for the second. At a cross-component collision, one instead differentiates the distinct joint tuples and their internally simple roots using (K.2). The trace of the combined residue gives a sum of velocities, not the individual velocities. The exact local condition is not being identified with a distance to a discriminant or with the global certificate η_d .

K.1. Coefficient recovery across internal multiplicities. The coefficient inverse, the extraction of scalar roots, and the finite quotient of the target are separate operations. The first continues analytically through an internal root multiplicity.

Proposition K.3 (Ambient coefficient recovery). *Let $\theta_0 \in \mathcal{K}_d$ have strictly positive full-rank channels, strict weights, $\tau(F(\theta_0)) > 0$, and pairwise distinct complete monic component polynomials. Roots may repeat, both within and between components. In monic-coefficient coordinates there is an ambient real-analytic map R near $P = F(\theta_0)$ with the following properties. Every sufficiently nearby closed-model datum has precisely the component permutations of $R(Y)$ as its coefficient representatives; $R(P) = \theta_0$ for the selected labelling. Moreover, after removing only the real-rootedness restriction in a neighbourhood of θ_0 , $R \circ F$ is the identity on an open coefficient chart. In particular $J_c = DF(\theta_0)$ is injective on that ambient chart. No manifold assertion is made about its real-rooted subset at a multiple root.*

Proof. The coefficient coordinates consist of the stochastic-column and weight coordinates of Section K and the kd nonleading polynomial coefficients. Positivity of the finitely many clock values, strict stochastic inequalities, full channel ranks, $\tau > 0$, and distinct complete coefficient tuples persist on an open set of these coordinates. This set does not require real-rootedness.

For an ambient observation Y , recover the monic normalizer by $(\mathcal{N}_Y^\top \mathcal{N}_Y)^{-1} \mathcal{N}_Y^\top b_Y$ and the numerator by interpolation. Select the same independent leading-coefficient rows and columns as at P , and form their analytic orthonormal frames by positive definite Gram square roots. The reduced leading matrix is invertible. A fixed real linear combination of its normalized coefficient matrices has distinct eigenvalues at P , because the complete polynomial tuples are distinct. Its disjoint Riesz contours therefore give analytic rank-one projectors near P . The traces against these projectors recover every component coefficient. Formula (K.1), followed by total, row, and column sums, recovers the weight and stochastic columns. All these operations take place before scalar root extraction.

For data in the enlarged coefficient chart, the normalized coefficient matrices still have their exact simultaneous diagonal realization. The recovered projectors consequently give the original coefficients and factors, so $R \circ F = \text{id}$. Differentiating proves the assertion about J_c . For a nearby closed-model datum, uniqueness of the normalizer forces the same numerator; its rank- k leading coefficient forces full-rank competing channels. The k distinct rank-one joint projectors then force exactly one competing component per tuple and the same row, column, and total sums. Hence every competing coefficient representation is one of the finitely many component permutations of $R(Y)$. Choose disjoint neighbourhoods of these permutations and shrink the observation neighbourhood. This proves the assertion for all representatives, not just for a selected parametrization. Restriction of the ambient analytic map to the real-rooted coefficient set completes the proof. $\square$

APPENDIX L. INTRINSIC LOCAL ASYMPTOTIC MINIMAXITY

Fix a datum on the regular quotient of Theorem K.1. At clock T_j take n_j independent categorical observations, independently between clocks, where $N = \sum_j n_j$ and $n_j/N \rightarrow w_j > 0$. The design weights are known and $\sum_j w_j = 1$. All cell probabilities are positive on this stratum. Define, on the product simplex tangent space,

$$H_{P,w} = \text{diag}(w_j/(P_j)_{ab})_{j,a,b}, \quad I = J^\top H_{P,w} J, \quad V = G I^{-1} G^\top. \quad (\text{L.1})$$

The matrix V is an observable positive definite form on the locally labelled root coordinates. Its conjugacy class under component permutations is intrinsic. Put

$$\mathfrak{R}(P, w) = \mathbb{E} \|Z\|_\infty^2, \quad Z \sim \mathcal{N}(0, V). \quad (\text{L.2})$$

Thus the local deterministic condition and the statistical constant will be two functionals of the *same* observable matrix, rather than unrelated lower and upper certificates.

For $v \in \mathbb{R}^p$, write $\theta_N(v) = \theta_0 + v/\sqrt{N}$. For fixed $R < \infty$ these parameters are admissible when N is sufficiently large. Define the local minimax risk

$$\mathfrak{R}_N(R) = \inf_{\hat{\Lambda}_N} \sup_{v^\top I v \leq R^2} N \mathbb{E}_{\theta_N(v)} d_\infty(\hat{\Lambda}_N, \Lambda(\theta_N(v)))^2. \quad (\text{L.3})$$

The infimum is over all measurable multiset estimators. An estimator in this local problem may depend on the centre θ_0 and the neighbourhood radius, but not on the unknown local parameter v .

Theorem L.1 (Exact local statistical constant). *At every datum in the regular quotient, including cross-component root collisions,*

$$\lim_{R \rightarrow \infty} \liminf_{N \rightarrow \infty} \mathfrak{R}_N(R) = \lim_{R \rightarrow \infty} \limsup_{N \rightarrow \infty} \mathfrak{R}_N(R) = \mathfrak{R}(P, w). \quad (\text{L.4})$$

A locally efficient minimum-distance estimator followed by component root extraction and sorting attains the upper bound. Moreover,

$$\mathfrak{c}_{H_{P,w}}(P)^2 \leq \mathfrak{R}(P, w) \leq 2\{\log(2kd) + 1\} \mathfrak{c}_{H_{P,w}}(P)^2. \quad (\text{L.5})$$

For fixed dimensions the same observable local condition therefore controls both sides of local asymptotic minimax risk. Neither equality asserts a risk lower bound for every estimator at the single point θ_0 .

The non-differentiability of sorting at a collision requires a separate argument. We give it before passing to the categorical experiment. Let a partition of $\{1, \dots, q_0\}$ define a group Π of permutations within its blocks, and write

$$d_\Pi([u], [v]) = \min_{\pi \in \Pi} \|u - \pi v\|_\infty.$$

A quotient point can equivalently be represented by sorting within each block. The blocks here will be the groups of equal roots at the base datum, not the latent components.

Lemma L.2 (A Gaussian quotient experiment). *Let I be positive definite and let $G : \mathbb{R}^p \rightarrow \mathbb{R}^{q_0}$ be surjective. In the experiment $X \sim \mathcal{N}(v, I^{-1})$, $v \in \mathbb{R}^p$, with target $[Gv] \in \mathbb{R}^{q_0}/\Pi$, one has*

$$\inf_{\hat{t}} \sup_{v \in \mathbb{R}^p} \mathbb{E}_v d_\Pi(\hat{t}, [Gv])^2 = \mathbb{E} \|\mathcal{N}(0, GI^{-1}G^\top)\|_\infty^2. \quad (\text{L.6})$$

The same constant is approached from below by Bayes risks of finite priors supported in bounded subsets of $\mathbb{R}^p$.

Proof. The estimator $[GX]$ has loss at most $\|G(X - v)\|_\infty^2$, giving the upper bound. First consider the labelled linear target Gv . For the prior $v \sim \mathcal{N}(0, t^2 I^{-1})$, the posterior covariance is $t^2(1+t^2)^{-1}I^{-1}$. A centrally symmetric distribution and a convex even loss have a risk-minimizing centre: average the losses at a displacement and its negative and use convexity. The posterior mean is consequently Bayes for squared sup-norm loss, and its Bayes risk is

$$\frac{t^2}{1+t^2} \mathbb{E} \|\mathcal{N}(0, GI^{-1}G^\top)\|_\infty^2.$$

Letting $t \rightarrow \infty$ proves the labelled minimax lower bound. For each fixed t , truncate this prior to growing cubes, and then approximate it by finitely supported priors. Gaussian likelihoods are continuous in v in total variation; compactly supported targets allow actions to be clipped to a common coordinate box without increasing loss. The Bayes risks of the finite approximations therefore converge to the truncated Bayes risk. Removing truncation can be checked directly.

Clip an estimator for the truncated prior to the coordinate box containing $G[-r, r]^p$, whose radius is at most $C_G r$. Its contribution on the omitted prior tail is at most

$$2C_G^2 r^2 \Pr(\|v\|_\infty > r) + 2\mathbb{E}[\|Gv\|_\infty^2 \mathbf{1}_{\{\|v\|_\infty > r\}}],$$

which tends to zero for the fixed Gaussian prior. The Bayes risk of the untruncated prior is at most the truncated Bayes risk times its retained mass plus this tail term. This proves the needed lower approximation. Thus finite priors on growing cubes approach the labelled constant.

It remains to show that the finite permutation quotient cannot reduce this worst-case constant. Fix a cube $C = [-r, r]^p$ supporting one of these finite priors. Since G is onto, there is v_* such that the coordinates of Gv_* are pairwise distinct within every block. For large enough M , the coordinate ranges

$$\{(Gv)_a : v \in Mv_* + C\}, \quad a \text{ in a fixed block,}$$

are disjoint intervals in a fixed order. Translate the finite prior by Mv_* . On its support, sorting within each block is a fixed permutation of the labelled target. Given any quotient estimate, sort each block and clip its successive coordinates to these disjoint intervals. This cannot increase the error from any ordered target in the support. The resulting coordinates have a unique labelling, and their quotient loss is exactly their labelled sup-norm loss. Hence the quotient Bayes risk on this translated prior is at least the labelled Bayes risk on the original prior. Translation of a Gaussian location experiment does not change that latter risk. Allow r and the prior scale to grow. The translated supports are bounded at each stage, proving both assertions. $\square$

Proof of Theorem L.1. The limit experiment. Let e index the $m_1 m_2$ cells. Write J_{je} for the row derivative of their probability at clock j . The score sum, normalized by $\sqrt{N}$, is

$$\Delta_N = \frac{1}{\sqrt{N}} \sum_{j=1}^{\ell} \sum_{i=1}^{n_j} \frac{J_{j,X_{ji}}^\top}{P_{j,X_{ji}}}.$$

It has mean zero, since $\sum_e J_{je} = 0$, and covariance tending to I . Positive cell floors and the analytic chart make all derivatives bounded on a fixed neighbourhood. Taylor expansion of each log probability, the law of large numbers and the finite-dimensional central limit theorem give, uniformly for bounded v ,

$$\log \frac{d\mathbb{P}_{\theta_N(v)}}{d\mathbb{P}_{\theta_0}} = v^\top \Delta_N - \frac{1}{2} v^\top I v + o_{\mathbb{P}}(1), \quad \Delta_N \implies \mathcal{N}(0, I). \quad (\text{L.7})$$

For completeness, the sum of the third-order log-probability remainders is $O(N^{-1/2}\|v\|^3)$, and replacing the empirical second derivative by its expectation costs $o_{\mathbb{P}}(1)$ on each bounded set. Differentiating $\sum_e p_{je} = 1$ identifies its expected negative Hessian with $\sum_e J_{je}^\top J_{je} / p_{je}$. This is precisely (L.1). These are the usual local asymptotic normality calculations [21, 31], here in the specified observation geometry.

Upper bound and moments. On a small closed coordinate ball about θ_0 , minimize $\|\hat{P} - F(\theta)\|_{H_{P,w}}^2$. For $\hat{P}$ sufficiently close to P , its Hessian is uniformly positive definite on a suitably chosen ball; its unique minimizer is interior and depends smoothly on $\hat{P}$. Define the estimator arbitrarily in $\mathcal{K}_d$ on the complementary event. The implicit normal equations yield, uniformly for bounded v ,

$$\sqrt{N}(\hat{\theta} - \theta_N(v)) = I^{-1} J^\top H_{P,w} \sqrt{N}(\hat{P} - F(\theta_N(v))) + o_{L^2}(1). \quad (\text{L.8})$$

To justify the L^2 assertion, cell-frequency fourth moments are $O(N^{-2})$ and the local Taylor remainder is quadratic. The probability of leaving the fixed neighbourhood is exponentially small,

uniformly on each bounded local parameter set; bounded targets control the fallback contribution after multiplication by N . The block covariance of $\sqrt{N}\hat{P}$ tends to

$$\Sigma_w = \bigoplus_j w_j^{-1} \{\text{diag}(P_j) - \text{vec } P_j \text{ vec } P_j^\top\}.$$

On the simplex tangent, $J^\top H_{P,w} \Sigma_w H_{P,w} J = I$. Thus the labelled root error obtained from (L.8) converges, with second moments, to $\mathcal{N}(0, V)$. Matching the labelled roots and then sorting cannot increase its sup norm, proving the upper bound in (L.4) for every fixed R . The construction is a local statistical estimator, not a claim of global optimization complexity. A consistent minimum-distance selector and localization in the recovered quotient chart give the same expansion without supplying a latent labelling.

Lower bound at collisions. Partition the base aggregate roots into equal-value groups and let Π be their within-group permutation group. Taylor expansion gives local root velocities Gv . For every fixed finite set of local parameters, their scaled bottleneck losses converge uniformly, on bounded local action sets, to $d_\Pi([u], [Gv])^2$. Indeed the different base-value groups are separated by a fixed gap. Sort any estimated multiset and clip the coordinates in each base-value group to a common interval of radius $C/\sqrt{N}$ about that base value, containing all targets of the finite experiment. Coordinatewise clipping cannot increase the real ordered-matching loss. After centering and scaling this restricts actions to a compact set; within each group the limiting operation is exactly the quotient by Π , not differentiation of the sorting map.

For a finite prior on local parameters, (L.7) implies joint convergence of its likelihood ratios to those of the Gaussian shift with information I . The likelihood ratios are uniformly integrable: their nonnegative limits have expectation one, as do the ratios themselves. On the compact action set just described, losses converge uniformly and are bounded. The formula for posterior Bayes risk as the integral of the minimum of weighted losses therefore gives convergence of the finite-experiment Bayes risks. This is the finite-prior form of the local asymptotic minimax argument, and does not require an exchange of the limits N and R .

By (K.2), G is onto, so Lemma L.2 supplies finite priors with Gaussian Bayes risks arbitrarily close to (L.2). Each prior is contained in an information ball of some finite radius R . Use that prior first, let $N \rightarrow \infty$, and only then let $R \rightarrow \infty$. A minimax risk dominates every such Bayes risk. This proves the lower bound in (L.4), including at cross-component collisions.

Finally let $\sigma^2 = \max_a V_{aa} = \mathbf{c}_{H_{P,w}}(P)^2$. One coordinate gives $\mathbb{E}\|Z\|_\infty^2 \geq \sigma^2$. The union bound $\Pr(\|Z\|_\infty > t) \leq 2q_0 \exp(-t^2/(2\sigma^2))$, integrated above $t^2 = 2\sigma^2 \log(2q_0)$, gives the upper bound in (L.5) without any independence assumption on the coordinates of Z . $\square$

The theorem supplies intrinsic sharpness on the full open identifiable quotient stratum, not merely on a specially chosen testing family. It complements, and does not replace, the closed-model confidence sets, signal-class lower bounds, and internal-multiplicity neighbourhood rates below. It uses a fixed limiting allocation w ; no optimal adaptive allocation constant is being inferred from it. The adaptive lower bounds in the later sections retain their stated scope.

APPENDIX M. RATIONAL MATRIX QUOTIENTS WITHOUT SIMULTANEOUS DIAGONALIZATION

The normalization operator is not peculiar to the stochastic diagonal model. We state its exact relation to scalar-denominator rational interpolation, and then separate the general resolvent bound from the stronger additive law furnished by the joint coefficient algebra.

Theorem M.1 (The rational interpolation module). *Let K be any rectangular matrix polynomial of degree at most d , and let q be monic of degree d , with $q(T_j) \neq 0$. Form N_P from $P(T_j) =$*

$K(T_j)/q(T_j)$ at $\ell \geq 2d + 1$ distinct nodes. Then

$$\ker \mathcal{N}_P = \{(q/h)s : \deg s < \deg h\}, \quad h = \gcd(q, \text{all entries of } K). \quad (\text{M.1})$$

More generally, the pairs (H, r) with $\deg H \leq d$, $\deg r < d$ and $H(T_j) = r(T_j)P(T_j)$ are exactly $((K/h)s, (q/h)s)$ with $\deg s < \deg h$. No diagonalization, channel factorization or positivity is assumed.

Proof. Interpolation gives $qH - rK = 0$ at $2d + 1$ nodes. Each entry has degree at most $2d$, so this is an identity. The polynomial q/h and the entries of K/h have no common divisor. A Bezout identity for this collection, or prime-by-prime divisibility, shows that q/h divides r . Write $r = (q/h)s$ and then $H = (K/h)s$. The degree constraint on r is exactly $\deg s < \deg h$. Conversely every such pair satisfies the identity and the degree constraints. $\square$

In the probability model q is the sum of the entries of K , so the common divisor of those entries already divides q . Full-rank constant channels then identify it with $\gcd(f_b)$, giving Lemma D.1. The module itself is an instance of classical simultaneous rational interpolation, not a new interpolation algorithm. For any monic $h_u = h + us$ of the same degree, the pair $((K/h)h_u, (q/h)h_u)$ has exactly the same rational curve. In a square regular numerator the cancelled scalar factor contributes its zeros with multiplicity equal to the matrix size. This identifies the spectral information absent from a minimal realization.

Here is a local observable spectral theorem for general square numerators. It genuinely removes simultaneous diagonalization, though it does not impose the diagonal model's better normalization constant. Let $M = K$ have invertible leading coefficient, let $\tau(P) > 0$, and let X be its finite root set. On a fixed coefficient-bounded neighbourhood assume the clock denominators are bounded away from zero. Let m be the largest pole order of M^{-1} . Write

$$\begin{aligned} M(z)^{-1} &= \sum_{x \in X} \sum_{j=1}^m \frac{R_{x,j}}{(z-x)^j}, \\ \mathcal{T}(z)r &:= M(z)^{-1}I_d(rP)(z) = T_\infty r + \sum_{x \in X} \sum_{j=1}^m \frac{T_{x,j}r}{(z-x)^j}, \\ b &= \|M_d^{-1}\| + \sum_{x,j} \|R_{x,j}\|, \quad t = \|T_\infty\| + \sum_{x,j} \|T_{x,j}\|. \end{aligned}$$

The T coefficients are linear maps from the Euclidean coefficient space of r to matrices with operator norm. All coefficients are observable Laurent coefficients of the recovered rational matrix.

Proposition M.2 (General quotient resolvent bound). *For sufficiently close degree- d competitors with regular leading coefficient, their kd numerator roots satisfy*

$$d_\infty(Z(M), Z(M')) \leq C\{\delta(b + t/\tau(P))\}^{1/m}.$$

Complex roots and nondiagonalizable matrix coefficients are allowed.

Proof. Write $r = q' - q$. The normalization identity gives $\|r\| \leq C\delta/\tau$, and $M' = M + I_d(rP) + H$ with $\|H\|_{\text{coef}} \leq C\delta$. For $\rho = \text{dist}(z, X) \leq 1$ on a fixed disk, the displayed Laurent expansions imply

$$\|M(z)^{-1}(I_d(rP) + H)(z)\| \leq C\rho^{-m}(t\|r\| + b\delta).$$

The homotopy is nonzero outside disks of radius $C(t\|r\| + b\delta)^{1/m}$. Coefficient closeness preserves its leading invertibility and excludes roots outside a fixed disk. Argument-principle counting on each disk component, followed by the finite-permutation limit, proves the assertion including algebraic multiplicities. $\square$

M.1. Comparison with realization and matrix interpolation. Anderson–Antoulas [7] recover minimal state-variable realizations from rational matrix interpolation data through Loewner matrices. Beckermann–Labahn [11] compute matrix rational interpolants and matrix greatest common divisors, including singular interpolation subproblems, by fraction-free structured algorithms. Mayo–Antoulas [26] develop the generalized realization framework. These are direct predecessors of the normalization module, not merely neighbouring applications. In particular Theorem M.1 locates our linear system in their interpolation setting: it is the bounded-degree, scalar-denominator submodule obtained after eliminating numerator coefficients by an orthogonal Vandermonde residual.

The statistical problem asks for something different from the minimal realization alone. A common factor changes the latent numerator spectrum while leaving the entire rational datum unchanged. Specifying the original degree and stochastic realization does not undo that cancellation. Lemma D.2 shows that the ambiguity is realized by positive fixed-channel parameters, not just by formal rational representations. Proposition D.6 distinguishes the full-complexity clock count from the worst-case scalar obstruction.

For a given polynomial matrix, the exclusion and multiplicity argument is the classical mechanism of Bauer–Fike and its polynomial and pencil extensions [10, 16, 17, 30, 22, 28]. We use it explicitly in Lemma D.4 and Proposition M.2. The further assertion of Theorem E.1 is the conversion from normalized probability error to that spectral perturbation: the joint coefficient projectors keep the normalization term scalar on each complete-polynomial group, yielding $B_d + \tau^{-1}$ instead of a second multiplication by channel conditioning. The exact metric identity in Theorem K.2 and the positive testing families below then address the target and the observation experiment, not the numerical complexity of rational interpolation. No claim that Loewner or matrix-GCD methods fail on these data is needed or made.

APPENDIX N. AN ADDITIVE GAUGE ON NONCOMMUTING FLAG ALGEBRAS

The scalar normalization acts on a matrix polynomial by an entrywise interpolation operator. It consequently preserves every constant invariant flag of the normalized coefficient algebra. Commutativity and semisimplicity are not needed for this preservation. The resulting inverse estimate has two scales: scalar multiplicity governs the normalization term, while paths along the flag govern general matrix perturbations.

Consider square quotients $P = M/q$, with M of degree d , invertible leading coefficient A_0 , and q monic of degree d . Assume fixed coefficient bounds for q and its competitors, a positive lower bound for $|q(T_j)|$, and a fixed disk containing the zeros of $\det M$. The clock geometry and d, k are fixed. The data norm is again the maximum Frobenius norm at the clocks. Suppose $\tau(P) > 0$ and the coefficients of $C(z) = M(z)A_0^{-1}$ have a common complete invariant flag over $\mathbb{C}$. An orthonormal basis adapted to that flag gives

$$Q^*C(z)Q = D_f(z) + N(z), \quad D_f = \text{diag}(f_1, \dots, f_k), \quad (\text{N.1})$$

where Q is unitary, each f_i is monic of degree d , and N is strictly upper triangular of degree at most $d - 1$. The f_i need not be distinct, and the coefficients need not commute or be diagonalizable. Define

$$n_Q = \|N\|_{\text{coef}}, \quad B_Q = \|A_0^{-1}\|_{\text{op}} \sum_{h=0}^{k-1} (1 + n_Q)^h, \quad B_\Delta(P) = \min_Q B_Q, \quad (\text{N.2})$$

where the minimum is over unitary simultaneous triangularizers of the recovered coefficient matrices. This nonempty set is closed in the unitary group; the minimum is attained. It is an observable finite optimization over flags of a known matrix algebra, not over latent stochastic

factorizations. It is invariant under constant unitary changes of row and column coordinates. The assertions below are stability statements; no complexity bound for finding this flag is required.

Theorem N.1 (Two-scale additive flag gauge). *For every degree- d competitor M'/q' sufficiently close in the sense $\delta(\tau(P)^{-1} + B_\Delta(P)) \leq c$, its numerator has invertible leading coefficient and exactly kd roots, and*

$$d_\infty(Z(M), Z(M')) \leq C \left\{ \left(\frac{\delta}{\tau(P)} \right)^{1/d} + (B_\Delta(P)\delta)^{1/(kd)} \right\}. \quad (\text{N.3})$$

The competitor need not preserve a flag, commute, or be semisimple. The constants depend on the declared degree, dimension, clock and scalar-root bounds, not on τ , B_Δ , or a latent channel condition.

There is a refined version for separated clusters. Suppose the roots of all the f_i lie in a fixed collection of mutually separated disks, with a fixed margin between the roots and the disk boundaries. Let m_{is} be the number of roots of f_i in disk s , counted with multiplicity, and put $m = \max_{i,s} m_{is}$. In a chosen flag draw an edge $i \rightarrow j$, $i < j$, when the polynomial N_{ij} is nonzero, and set

$$\nu_Q = \max_s \max_{\text{directed paths } \mathcal{P}} \sum_{i \in \mathcal{P}} m_{is}. \quad (\text{N.4})$$

Single-vertex paths are included. Then $m \leq \nu_Q \leq kd$, and

$$d_\infty(Z(M), Z(M')) \leq C \left\{ (\delta/\tau(P))^{1/m} + (B_Q\delta)^{1/\nu_Q} \right\}, \quad (\text{N.5})$$

with additional dependence on the fixed cluster separation and margin. The path exponent is a valid observable upper certificate; cancellations between paths may improve it, so it is not asserted to be the minimal pole order in every flag algebra.

Proof. Write $r = q' - q$ and let $\mathcal{H}_r$ act on a scalar or matrix polynomial F by

$$\mathcal{H}_r F = I_d(r(T_j)F(T_j)/q(T_j)).$$

It is linear, commutes with constant left and right multiplication, and preserves every identically zero matrix entry. The normalization equation and interpolation give

$$\|r\|_{\text{coef}} \leq C\delta/\tau(P), \quad M' = \overline{M}_r + H, \quad \overline{M}_r = M + \mathcal{H}_r M, \quad \|H\|_{\text{coef}} \leq C\delta. \quad (\text{N.6})$$

In the chosen flag,

$$Q^* \overline{M}_r A_0^{-1} Q = D_f + \mathcal{H}_r D_f + N + \mathcal{H}_r N =: \overline{D} + \overline{N}. \quad (\text{N.7})$$

In particular the diagonal perturbations $h_i = \mathcal{H}_r f_i$ satisfy $\|h_i\|_{\text{coef}} \leq C\|r\|_{\text{coef}}$, whereas $\|\overline{N}\|_{\text{coef}} \leq Cn_Q$ for $\|r\|_{\text{coef}} \leq 1$. No inverse of A_0 or norm of N occurs in the first bound.

On a fixed disk containing the roots put $\rho = \text{dist}(z, Z(M))$. Since $|f_i(z)| \geq \rho^d$, the inequalities $\rho^d \geq C_1\|r\|_{\text{coef}}$ give $|f_i(z) + uh_i(z)| \geq \rho^d/2$ for $0 \leq u \leq 1$. The preliminary homotopy $M + u\mathcal{H}_r M$ is triangular after the constant transformations in (N.7); its determinant has no zeros outside these scalar disks. For the second homotopy use the terminating triangular inverse

$$(\overline{D} + \overline{N})^{-1} = \sum_{h=0}^{k-1} (-\overline{D}^{-1}\overline{N})^h \overline{D}^{-1}. \quad (\text{N.8})$$

It follows, for $0 < \rho \leq 1$, that

$$\|\overline{M}_r(z)^{-1}\| \leq CB_Q \rho^{-kd}.$$

Thus $\overline{M}_r + sH$ is invertible whenever

$$\rho \geq C_2 \{ \|r\|_{\text{coef}}^{1/d} + (B_Q\delta)^{1/(kd)} \}, \quad 0 \leq s \leq 1.$$

This is the decisive separation: the normalization error is used to control diagonal scalar denominators before the inverse series is applied to H . Inserting both errors into a single Neumann estimate would unnecessarily apply the longest flag path to r as well.

We verify that no degree or roots are lost at infinity. The leading diagonal entries of the first homotopy are $1 + u(h_i)_d$, bounded below by $1/2$ when $C\|r\|_{\text{coef}} \leq 1/2$. Its leading strictly upper part has norm at most $C\|r\|_{\text{coef}} n_Q$. The same finite series bounds the inverse of its leading coefficient by CB_Q . If $CB_Q\delta \leq 1/2$, adding sH_d preserves invertibility. Every homotopy determinant therefore has degree exactly kd . For $|z|$ larger than a fixed multiple of the scalar root bound, $|f_i(z)| \geq c|z|^d$, while $|h_i(z)| \leq C\|r\|_{\text{coef}}|z|^d$ and $\|\bar{N}(z)\| \leq Cn_Q|z|^d$. Equation (N.8) then gives $\|\bar{M}_r(z)^{-1}\| \leq CB_Q|z|^{-d}$. Since $\|H(z)\| \leq C\delta|z|^d$, the same smallness excludes zeros outside this fixed disk. Argument-principle counting on the connected components of the remaining root disks and the fixed-permutation limit of Lemma D.4 give the matching bound. Choose a minimizing flag in (N.2) to obtain (N.3). This also proves leading invertibility for a competitor that was not initially assumed regular.

For the refinement, near cluster s the factors outside that cluster are uniformly bounded below. Therefore

$$|f_i(z)| \geq c \min\{1, \rho^{m_{is}}\}.$$

The scalar gauge is excluded by radius $C\|r\|_{\text{coef}}^{1/m}$. Expand each entry of (N.8) along increasing index paths. A path has one diagonal denominator for each visited vertex and only the corresponding off-diagonal entries in its numerator. Zero edges remain zero under $\mathcal{H}_r$. On cluster s the denominator exponent along that path is $\sum_{i \in \mathcal{P}} m_{is}$, at most ν_Q . The finite sum of its numerator bounds is at most CB_Q . Consequently the second exclusion radius is $C(B_Q\delta)^{1/\nu_Q}$. The leading-coefficient and counting arguments are unchanged. This proves (N.5) uniformly even if roots within a cluster coalesce. $\square$

N.1. A noncommuting positive quotient and a Jordan intersection. The class in Theorem N.1 is strictly larger than the simultaneously semisimple stochastic model. Let $\epsilon > 0$ and put

$$C(z) = \begin{pmatrix} z^2 + 7z + 10 & \epsilon z \\ 0 & z^2 + 7z + 12 \end{pmatrix}, \quad L = R = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad K(z) = \frac{1}{2} LC(z)R^\top.$$

Its entry sum is $q(z) = z^2 + (7 + \epsilon/2)z + 11$, so $K(T)/q(T)$ is a strictly positive probability matrix for $T > 0$. The leading coefficient is $LR^\top/2$, and $K(z)K_2^{-1} = LC(z)L^{-1}$. Before this similarity the coefficient of z is $\begin{pmatrix} 7 & \epsilon \\ 0 & 7 \end{pmatrix}$, which is not diagonalizable, and its commutator with $\text{diag}(10, 12)$ has nonzero upper entry 2ϵ . Thus these coefficients neither commute nor admit simultaneous diagonalization. The diagonal polynomials have disjoint root sets, so their greatest common divisor, and that of the entries of K , is one. Theorem M.1 gives $\tau > 0$ at five clocks. All hypotheses are therefore realized by a genuine positive rational observation, not merely by a formal matrix example. It is a broader quotient experiment, not a new representation of (2.1).

The difference between the two exponents is already visible for

$$C(z) = \begin{pmatrix} z & \beta \\ 0 & z \end{pmatrix}, \quad \beta > 0.$$

With the same L, R and $K = LCR^\top/2$, the normalizer is $q = z + \beta/2$ and the observations are strictly positive for $T > 0$. A normalization-only gauge changes the two diagonal polynomials by the same scalar affine perturbation; its repeated root moves by $O(\|r\|)$. In contrast, replacing the lower zero entry of C by $t > 0$ produces roots $\pm\sqrt{\beta t}$. The corresponding normalizer changes to $z + (\beta + t)/2$, and the observation error at fixed positive clocks is $O(t)$. Here $m = 1$ but the flag path has $\nu_Q = 2$. Normalization and the nilpotent collision consequently contribute linearly and

by a square root, respectively. This establishes the mechanism behind the two-scale sum without claiming a classification of arbitrary intersections with normalization-kernel or channel-rank boundaries.

APPENDIX O. THE AFFINE QUOTIENT AND ITS OBSERVABLE RESIDUE CONDITION

Throughout this section $d = 1$ and $\ell = 3$. The parameters are as in (C.4). Write $P_j = P(T_j)$ and $\gamma = \|B\|_F$.

Lemma O.1 (The third clock and the pole-shift fibre). *If $\gamma > 0$, put $X = P_1 - P_2$, $Y = P_2 - P_3$, and $a_0 = (T_2 - T_1)/(T_3 - T_2)$. Then*

$$r := \frac{\langle X, Y \rangle_F}{\|Y\|_F^2} = a_0 \frac{T_3 - h}{T_1 - h}, \quad h = \frac{rT_1 - a_0T_3}{r - a_0}, \quad \|Y\|_F \asymp \gamma. \quad (\text{O.1})$$

The map from $h \in [L, D]$ to r is uniformly bi-Lipschitz. If two model observations are at distance $\delta \leq c\gamma$, then $|h - h'| \leq C\delta/\gamma$. Also $\tau(P) \asymp \gamma$.

Let $H_0 = (T_1 - h)(T_2 - h)$. For sufficiently small real u , define

$$\begin{aligned} a_x(u) &= 1 + u(h - x)/H_0, \\ x_u &= \frac{x + u\{1 + (h - x)(T_1 + T_2 - h)/H_0\}}{a_x(u)}, \\ \alpha_{b,u} &= \alpha_b a_{\xi_b}(u), \quad h_u = h + u. \end{aligned} \quad (\text{O.2})$$

With the same channels, these parameters give exactly the same two observations at T_1, T_2 , whenever they remain admissible. At interior parameters they give a nonconstant exact fibre, since

$$\left. \frac{dx_u}{du} \right|_{u=0} = \frac{(T_1 - x)(T_2 - x)}{H_0} > 0. \quad (\text{O.3})$$

Proof. Subtract $P(T) = A + B/(T - h)$ at consecutive clocks. This gives (O.1); the derivative of $r(h)$ is $a_0(T_3 - T_1)/(T_1 - h)^2$, bounded above and below by positive constants. At error $\delta \leq c\gamma$, both difference norms are bounded above by $C\gamma$ and the perturbed denominator norm is bounded below by $c\gamma$. Expanding the ratio gives error $C\delta/\gamma$, hence the pole estimate.

For $d = 1$, $\mathcal{N}_P$ has one column. Since Ω annihilates constant and affine clock vectors,

$$\tau(P) = \|B\|_F \left\| \Omega((T_j - h)^{-1})_{j=1}^3 \right\|_2.$$

The last norm is continuous and strictly positive for $h \in [L, D]$: if a degree-one polynomial agreed with $(T - h)^{-1}$ at three distinct points, multiplying by $T - h$ would give an impossible polynomial identity. Compactness proves the comparison.

The affine interpolant of $p_b(T)$ at T_1, T_2 is

$$Q_b(z) = \alpha_b \left[1 + (h - \xi_b) \frac{T_1 + T_2 - h - z}{H_0} \right].$$

The polynomial $\alpha_b(z - \xi_b) - uQ_b(z)$ has slope $\alpha_{b,u}$ and root $(\xi_b)_u$. Summing these polynomials gives $z - h - u$, so the weights sum to one and their weighted actions to $h + u$. At either clock the polynomial equals $(T_j - h - u)p_b(T_j)$. This proves the fibre assertion and differentiation gives (O.3). In the perturbation arguments below this gauge polynomial is used algebraically even if it leaves the parameter interval; its slopes stay positive and its roots remain in a fixed slightly larger interval. $\square$

Let $\mathcal{R}$ be the subset of the affine model image on which $\text{rank } P_1 = k$ and $P_2 \neq P_3$. On $\mathcal{R}$, the preceding lemma recovers h , then the two coefficients A, C of $K(z) = zA - C$. Choose orthonormal bases Q_1, Q_2 of the column and row spaces of A , and set

$$A_0 = Q_1^\top A Q_2, \quad C_0 = Q_1^\top C Q_2, \quad M(z) = zA_0 - C_0. \quad (\text{O.4})$$

For each distinct generalized eigenvalue x , define

$$R_x = \text{Res}_{z=x} M(z)^{-1}, \quad \beta(P) = \sum_x \|R_x\|_{\text{op}}, \quad \eta(P) = (\beta(P) + \gamma(P)^{-1})^{-1}. \quad (\text{O.5})$$

Set $\eta = 0$ off $\mathcal{R}$. The residues are residues of a matrix function, not rank-one factors chosen inside repeated eigenspaces.

Theorem O.2 (An observable affine singular invariant). *The quantities β and η do not depend on the chosen orthonormal bases or on a latent representation. On the affine model image,*

$$\eta(P) > 0 \iff \vartheta_*(P) > 0 \iff \text{rank } P_1 = k \text{ and } P_2 \neq P_3. \quad (\text{O.6})$$

For every representation of $P \in \mathcal{R}$,

$$\sigma_k(A)^{-1} \leq \beta(P) \leq \frac{k}{\alpha_* \kappa}, \quad \eta(P) \geq c \vartheta_*(P). \quad (\text{O.7})$$

Every competitor in the full closed affine model satisfies

$$d_\infty(\xi, \xi') + |h - h'| \leq C \min\{1, \delta/\eta(P)\}. \quad (\text{O.8})$$

All constants have precisely the fixed-parameter dependencies stated in Section 1; no individual action gap enters them.

On $\mathcal{R}$, β is lower semicontinuous and is continuous on each fixed action-multiplicity stratum. The extension of η by zero is upper semicontinuous on the whole model image, continuous on those strata, and tends to zero at its boundary zero set. The envelope ϑ_* is upper semicontinuous on the whole image.

Proof. For any representation, put $S = Q_1^\top U$ and $T = Q_2^\top V$. These are invertible, and

$$M(z)^{-1} = T^{-\top} \text{diag} \left(\frac{1}{\alpha_b(z - \xi_b)} \right) S^{-1} = \sum_x \frac{R_x}{z - x}, \quad R_x = T^{-\top} \text{diag}(\alpha_b^{-1} \mathbf{1}_{\xi_b=x}) S^{-1}. \quad (\text{O.9})$$

Thus every pole is simple, even when the corresponding action has higher multiplicity. An orthogonal change of bases multiplies each residue on its left and right by orthogonal matrices, preserving its operator norm. The construction otherwise uses only h, A, C . Taking the coefficient at infinity gives $\sum_x R_x = A_0^{-1}$. Each residue norm is at most $1/(\alpha_* \kappa)$, proving (O.7) before its last assertion. Every representation of a rank- k observation has full rank, since all diagonal probabilities are positive. Further, $P_2 = P_3$ is equivalent to $B = 0$. This proves (O.6). The bounds on β imply $\eta \geq c\vartheta$ for every representation, hence $\eta \geq c\vartheta_*$.

To prove the inverse, it suffices to consider $\delta(\beta + \gamma^{-1})$ small. The pole lemma gives $u = h' - h = O(\delta/\gamma)$. Let $Q(z)$ be the affine interpolant of P_1, P_2 , and form the true gauge polynomial $\bar{K}_u(z) = K(z) - uQ(z)$. Its reduced inverse is

$$\bar{M}_u(z)^{-1} = \sum_x \frac{R_x/a_x(u)}{z - x_u}. \quad (\text{O.10})$$

For sufficiently small $|u|$, uniformly in the actions, $1/2 \leq a_x(u) \leq 3/2$ and $|x_u - x| \leq C|u|$. The fractional-linear map $x \mapsto x_u$ has positive derivative close to one on $[L, D]$, so it preserves distinct groups and their multiplicities. Consequently

$$\|\bar{M}_u(z)^{-1}\| \leq \frac{2\beta(P)}{\text{dist}(z, \{x_u\})}.$$

On the other hand, $M'(z) = Q_1^\top K'(z) Q_2$ satisfies $M' = \overline{M}_u + E$, where E is the affine interpolant of

$$(T_j - h') Q_1^\top (P'_j - P_j) Q_2, \quad j = 1, 2.$$

Its coefficient norm is at most $C\delta$, with no channel inverse. The leading inverse of $\overline{M}_u$ has norm at most 2β , so the leading coefficient remains invertible throughout $\overline{M}_u + sE$, $0 \leq s \leq 1$. The same resolvent bound first excludes roots outside a fixed disk and then excludes them outside disks of radius $C\beta\delta$ about the x_u . The contour and multiplicity argument in Lemma D.4, now with degree k , gives a matching within $C\beta\delta$. The invertible leading coefficient of M' forces both reduced competitor channel factors to be invertible. Its roots are therefore precisely ξ' . Combining this matching with $|x_u - x| \leq C\delta/\gamma$ proves (O.8). Compactness covers larger errors.

For regularity, let $P^{(n)} \rightarrow P$ in $\mathcal{R}$. Locally one can choose continuous orthonormal frames for the rank- k column and row spaces, for example by projecting fixed independent columns and orthonormalizing. The recovered h, A, C , hence the reduced pencils, then converge. Choose disjoint circles enclosing the distinct actions of the limit pencil and no others. The inverse pencils converge uniformly on these circles. Integrating gives

$$\sum_{y \text{ in the cluster of } x} R_y^{(n)} \longrightarrow R_x.$$

The triangle inequality and summation imply $\beta(P) \leq \liminf_n \beta(P^{(n)})$. If the multiplicity pattern is fixed locally, each circle contains one distinct pole, so each residue converges and β is continuous. Since $\gamma > 0$ is continuous here, this gives the assertions for η on $\mathcal{R}$.

At a constant-curve limit, the difference formula in the pole lemma, uniformly for $h \in [L, D]$, gives $\gamma \rightarrow 0$, hence $\eta \rightarrow 0$. At a nonconstant rank-loss limit, h, A still converge and $\sigma_k(A) \rightarrow 0$. Since $\beta \geq \sigma_k(A)^{-1}$, again $\eta \rightarrow 0$. This proves the extension statement. Finally, choose maximizing representatives in (C.5) along a sequence realizing a limsup. A compact subsequence converges to a representation of the limit datum. Continuity of ϑ proves upper semicontinuity of ϑ_* . $\square$

Remark O.3 (Why semicontinuity is the appropriate statement). Take $k = m_1 = m_2 = 3$, $U = V = I$, $\alpha_b = 1/3$, and actions $(a - t, a + t, b)$ with $a < b$ and small $t \geq 0$. For $t > 0$ the three residue norms equal 3, so $\beta = 9$. At $t = 0$ the residue at a is $3 \operatorname{diag}(1, 1, 0)$, while the residue at b has norm 3; hence $\beta = 6$. The contrast stays positive. Thus η has an upward jump at this partial collision and is not globally continuous on $\mathcal{R}$. This example does not assert discontinuity of ϑ_* . The zero-set statement is a characterization of the certificate, not a claim that every rank-deficient datum loses every possible spectral feature, nor a distance-to-discriminant formula.

Proposition O.4 (Exact factors and the original affine modulus). *For a full-rank nonconstant affine datum, every exact representation has the same h and spectrum and has full-rank channels. If the spectrum is simple, the channels and weights are unique up to a common permutation. At a repeated action, its total matrix coefficient is unique, but its individual positive rank-one factors need not be. Moreover, if $\xi_k - \xi_1 \geq \Delta > 0$, then $\vartheta \geq c_\Delta \kappa$. The original one-sided estimate with ϑ_* and all its gap-free consequences follow from (O.8).*

Proof. The pole is determined by (O.1). Hence A, C are determined, and $\operatorname{rank} A = k$. In an alternative exact representation $A = U' \operatorname{diag}(\alpha') V'^\top$; its rank forces both alternative channels to have full column rank. This justifies applying spectral reconstruction to every point of the exact fibre, not just to a selected full-rank representative.

Choose row and column index sets I, J with A_{IJ} invertible. The spectral projector Π_x of $K(T_1)_{IJ} A_{IJ}^{-1}$ at $T_1 - x$ gives

$$E_x = A_{:J} A_{IJ}^{-1} \Pi_x A_{I:} = \sum_{\xi_b = x} \alpha_b U_{:b} V_{:b}^\top.$$

For a simple action, entry sum, row sums and column sums of E_x recover $\alpha_b, U_{\cdot b}, V_{\cdot b}$. At a repeated action only their displayed sum is determined. Finally,

$$\gamma \geq \kappa \left(\sum_b \alpha_b^2 (h - \xi_b)^2 \right)^{1/2} \geq \frac{1}{2} \alpha_* \Delta \kappa.$$

Together with $\vartheta = \kappa\gamma/(\kappa + \gamma)$ this proves the last claim. In particular, on every simple-spectrum exact fibre, $\vartheta_* = \vartheta$. $\square$

APPENDIX P. A DIRECT SPECTRAL ESTIMATOR

The inverse estimates concern the true experiment, but their proof does not require a global search over latent factors. We give an explicit estimator for the multiset alone. Computation of the exact residual confidence region in the next section is a separate problem.

Given matrices $\hat{P}_j$, form $\mathcal{N}_{\hat{P}}$ and $b_{\hat{P}}$ by (2.2). If the former has full column rank, let $\hat{a}$ be its least-squares solution and put $\hat{q}(z) = z^d + \sum_{r < d} \hat{a}_r z^r$. For $d = 1$, replace $-\hat{a}_0$ by its projection onto $[L, D]$; write the resulting polynomial as $z - \hat{h}$. Interpolate the matrices $\hat{q}(T_j)\hat{P}_j$ at the first $d + 1$ clocks to obtain $\hat{K}(z)$.

Take the leading k left and right singular subspaces of $\hat{P}_1$, with orthonormal bases $\hat{Q}_1, \hat{Q}_2$. Form the reduced polynomial

$$\hat{M}(z) = \hat{Q}_1^T \hat{K}(z) \hat{Q}_2.$$

Compute its kd roots, counted with algebraic multiplicity, project the real parts onto $[L, D]$, and sort them. This is the estimator $\hat{\Lambda}$. If the normalization matrix lacks full column rank, a sampled $\hat{q}$ is nonpositive, $\hat{P}_1$ has rank less than k , or the leading coefficient of $\hat{M}$ is singular, return a fixed multiset in $[L, D]$ instead. This convention defines the estimator everywhere. Numerical root tolerances can be added to the bounds below; the statements themselves refer to exact arithmetic.

Theorem P.1 (Constructive recovery). *Put $e = \max_j \|\hat{P}_j - P_j\|_F$. At every polynomial truth with $\kappa > 0$ and $\tau(P) > 0$, the estimator just defined satisfies*

$$d_\infty(\hat{\Lambda}, \Lambda) \leq C \min\{1, [e(\kappa^{-1} + \tau(P)^{-1})]^{1/d}\}. \quad (\text{P.1})$$

The within-component simple-root hypothesis of Theorem C.1 gives the corresponding linear bound. In the affine experiment the stronger observable estimate holds:

$$d_\infty(\hat{\xi}, \xi) \leq C \min\{1, e/\eta(P)\}, \quad P \in \mathcal{R}. \quad (\text{P.2})$$

The procedure uses a least-squares solve, a singular value decomposition, polynomial interpolation, and the roots of a regular matrix polynomial of size k and degree d . It does not optimize over U, V or enumerate a parameter net. No conditioning floor is an input to the estimator.

Proof. The normalization operator and its right-hand side depend linearly on the observed entries. Therefore $\|\mathcal{N}_{\hat{P}} - \mathcal{N}_P\| \leq Ce$, and the residual of the true coefficient vector in the perturbed equation is at most Ce . If $e \leq c\tau$, the least singular value is at least $\tau/2$. Least-squares optimality followed by the triangle inequality gives

$$\|\hat{a} - a\| \leq Ce/\tau. \quad (\text{P.3})$$

Projection of $\hat{h}$ cannot increase its distance from h . For sufficiently small e/τ , all sampled denominator values remain positive.

We record the subspace estimate used in both parts of the proof. Let Q_1, Q_2 span the true column and row spaces of P_1 . The best rank- k approximation property implies

$$\|(I - \hat{Q}_1 \hat{Q}_1^T) P_1\| \leq 2e, \quad \|P_1(I - \hat{Q}_2 \hat{Q}_2^T)\| \leq 2e.$$

If $2e/\sigma_k(P_1)$ is sufficiently small, the largest principal angle has sine at most that ratio. In particular both matrices $\widehat{Q}_1^\top Q_1$ and $Q_2^\top \widehat{Q}_2$ are invertible with inverse norms at most two. This assertion also follows directly by applying the preceding residual inequalities to a singular value decomposition of P_1 .

For the polynomial bound, $\sigma_k(P_1) \geq c\kappa$. Consequently $S = \widehat{Q}_1^\top U$ and $T = \widehat{Q}_2^\top V$ are invertible, with $\|S^{-1}\| \|T^{-1}\| \leq C/\kappa$, when e/κ is small. At the interpolation nodes, reduction by these inverses yields

$$\begin{aligned} S^{-1} \widehat{M}(T_j) T^{-\top} - D_f(T_j) &= \widehat{q}(T_j) S^{-1} \widehat{Q}_1^\top (\widehat{P}_j - P_j) \widehat{Q}_2 T^{-\top} \\ &\quad + (\widehat{q}(T_j) - q(T_j)) \text{diag}(p_b(T_j)). \end{aligned}$$

As in (D.2), interpolation bounds the coefficient error by $Ce(\kappa^{-1} + \tau^{-1})$. The leading coefficient is then invertible and Lemma D.4 applies. The real-part and interval projections cannot increase the distance to any matched true real root. Sorting gives the claimed multiset estimate. All failure conditions are excluded in this small-error regime; bounded range handles its complement.

For the stronger affine assertion, no comparison with κ is needed. The resolvent formula (O.9) gives

$$\sigma_k(P_1)^{-1} = \|(T_1 - h)M(T_1)^{-1}\| \leq C\beta(P).$$

Thus the two estimated subspaces have the preceding inverse bounds as soon as $e\beta$ is small. The true pencil, reduced into these estimated subspaces, has residues

$$(Q_2^\top \widehat{Q}_2)^{-1} R_x (\widehat{Q}_1^\top Q_1)^{-1},$$

whose total norm is at most 4β . By (P.3) and $\tau \asymp \gamma$, $u = \widehat{h} - h$ satisfies $|u| \leq Ce/\gamma$. Apply the exact gauge (O.2) to this true reduced pencil. The difference between the estimated pencil and that gauge polynomial is the affine interpolant of

$$(T_j - \widehat{h}) \widehat{Q}_1^\top (\widehat{P}_j - P_j) \widehat{Q}_2, \quad j = 1, 2,$$

and has coefficient norm at most Ce . The resolvent and root-counting argument of Theorem O.2 now gives root matching error $C\beta e$ about the shifted actions, plus Ce/γ for the shift itself. It applies to complex estimated roots as well. Projection and the bounded-range convention complete (P.2). $\square$

The theorem is an arithmetic and stability statement for this explicit procedure. It is not a bit-complexity theorem uniform in dimension, precision and conditioning. In degree one only a $k \times k$ generalized eigenvalue problem is required. Confidence-region optimization is not being declared tractable on the strength of this point-estimation result.

APPENDIX Q. HONEST INFERENCE AND SHARP OBSERVABLE SIGNAL CLASSES

We first record the compact-quotient principle behind residual confidence sets. It applies to the polynomial model as well as its affine specialization.

Lemma Q.1 (Residual inversion). *Let a compact parameter space have a continuous observation map F into a finite product of probability simplices and a continuous target g in a compact metric space. For a true parameter ω , put*

$$\mathbf{m}_\omega(t) = \sup\{d(g(\omega), g(\omega')) : \|F(\omega') - F(\omega)\| \leq t\}.$$

Then $\mathbf{m}_\omega(0) = 0$ exactly when the target is constant on the exact fibre. In that case $\mathbf{m}_\omega(t) \rightarrow 0$ as $t \downarrow 0$. If $e = \|\widehat{F} - F(\omega)\|$, a minimum-residual estimate has target error at most $\mathbf{m}_\omega(2e)$. The target image of a residual ball of radius ρ about $\widehat{F}$, when nonempty, has diameter at most $2\mathbf{m}_\omega(e + \rho)$ and contains the truth on $e \leq \rho$.

Proof. The zero-radius assertion is the definition. If continuity at zero failed, compactness would give a limiting competitor in the exact fibre with different target. A residual minimizer has residual at most e , because the truth is feasible. The triangle inequality gives observation error at most $2e$. Every point of the residual ball is at observation distance at most $e + \rho$ from the truth, proving the remaining assertions. $\square$

For the affine experiment, take n independent categorical draws at each of the three clocks and put $N = 3n$, $r = m_1 m_2$. Let $\hat{P}_j$ be the empirical matrices. For $0 < \zeta < 1$, define

$$\rho_{N,\zeta} = \sqrt{\frac{r}{2n} \log \frac{6r}{\zeta}}, \quad \mathcal{C} = \{\omega \in \mathcal{K}_1 : \max_j \|P_\omega(T_j) - \hat{P}_j\|_F \leq \rho_{N,\zeta}\}. \quad (\text{Q.1})$$

Its target image is $\mathcal{C}_\xi$. If it is empty, report model inconsistency and the empty set. No action is assigned by the confidence procedure in that case. Coordinate extrema of the ordered spectra give simultaneous bands whenever the set is nonempty.

Theorem Q.2 (Risk, honesty, and adaptation). *The set (Q.1) has coverage at least $1 - \zeta$ uniformly over the entire closed model, including singular boundaries. The empty set is reported explicitly, and its diameter is defined as zero solely for the expected-diameter loss. For either $\psi = \eta$ or $\psi = \vartheta_*$, and every true datum with $\psi(P) > 0$, the direct estimator of Theorem P.1 satisfies*

$$\begin{aligned} \mathbb{E}_P d_\infty(\hat{\xi}, \xi)^2 &\leq C \min\{1, (N\psi(P)^2)^{-1}\}, \\ \mathbb{E}_P \text{diam}_\infty \mathcal{C}_\xi &\leq C_\zeta \min\{1, (\sqrt{N}\psi(P))^{-1}\}. \end{aligned} \quad (\text{Q.2})$$

On the coverage event the diameter is at most $C \min\{1, \rho_{N,\zeta}/\psi(P)\}$. Neither signal nor either channel factor is supplied to the procedures. At $\psi = 0$ the statements are coverage and bounded-range loss, not shrinking diameter.

Proof. Hoeffding's inequality and a union bound over the $3r$ cell frequencies give

$$\Pr\{e > \rho_{N,\zeta}\} \leq 6r \exp(-2n\rho_{N,\zeta}^2/r) = \zeta.$$

An entrywise error bound t gives a Frobenius bound $\sqrt{r}t$; for binary readouts the factor is 2, not 1. Theorem O.2 and Lemma Q.1 give the eventwise assertions, and on every outcome with nonempty set give $\text{diam } \mathcal{C}_\xi \leq C \min\{1, (e + \rho_{N,\zeta})/\psi(P)\}$. Further,

$$\mathbb{E} e^2 \leq \sum_{j=1}^3 \mathbb{E} \|\hat{P}_j - P_j\|_F^2 = \frac{1}{n} \sum_{j=1}^3 \left(1 - \sum_{a,b} (P_j)_{ab}^2 \right) \leq 3/n.$$

No exceptional outcome is discarded. The empty-set convention preserves the diameter bound. Taking expectations proves its asserted order. Squaring (P.2) and using this second moment proves the risk assertion; $\eta \geq c\vartheta_*$ supplies the other signal. Compactness gives extrema of the residual region. For fixed n the categorical sample space is finite, so choosing minimizers, singular bases and tie conventions causes no measurable selection issue. $\square$

Exact computation of $\mathcal{C}_\xi$ is an information-theoretic construction; it is not implied to be efficient by the preceding point estimator. Finite forward nets with mesh r_0 provide an exhaustive approximation, with residual padding Cr_0 and action padding r_0 , by uniform forward Lipschitz continuity. This may be exponential in dimension and resolution. For degree $d > 1$, Lemma Q.1 and Theorem C.1 give the analogous polynomial confidence modulus. The following sharp statements concern the affine experiment.

Q.1. Two positive least-favourable families. We give all constants in one lemma, so that the parameter class and the testing displacement use the same admissible neighbourhood.

Let $u_0 = m_1^{-1}\mathbf{1}$, $v_0 = m_2^{-1}\mathbf{1}$. Choose unit zero-sum vectors e, f and, when $k > 2$, orthonormal zero-sum vectors $e_3, \dots, e_k$ perpendicular to e and $g_3, \dots, g_k$ perpendicular to f . The dimension assumption $m_i \geq k$ gives these vectors. Fix a small constant $c_0 > 0$ and interior distinct actions whose first pair is $a_0 - b_0, a_0 + b_0$, with $b_0 > 0$, all other actions being above this pair. Set $\alpha_b = 1/k$. For $0 \leq t \leq t_0 < b_0/2$, let $\lambda = 1 - t/b_0$ and put

$$\begin{aligned} \xi_{1,t} &= a_0 - b_0 + t, & \xi_{2,t} &= a_0 + b_0 - t, & \xi_{b,t} &= \xi_b \quad (b \geq 3), \\ U_{t,1} &= u_0 + \frac{\epsilon_u e}{2\lambda}, & U_{t,2} &= u_0 - \frac{\epsilon_u e}{2\lambda}, & U_{t,b} &= u_0 + c_0 e_b \quad (b \geq 3), \\ V_{t,1} &= v_0 + \frac{\epsilon_v f}{2\lambda}, & V_{t,2} &= v_0 - \frac{\epsilon_v f}{2\lambda}, & V_{t,b} &= v_0 + c_0 g_b \quad (b \geq 3). \end{aligned} \quad (\text{Q.3})$$

The pole h is fixed along this family.

The second family fixes positive full-rank channels and positive weights. Choose distinct real z_b with $\sum_b \alpha_b z_b = 0$ and put, for an interior a ,

$$\xi_b(t) = a + t + sz_b, \quad 0 \leq t \leq t_0, \quad B = -sU \operatorname{diag}(\alpha_b z_b) V^\top. \quad (\text{Q.4})$$

Here $h_t = a + t$, while A, B do not depend on t .

Lemma Q.3 (Uniform family constants). *There exist fixed $t_0 > 0$, $\epsilon_* > 0$, $s_0 > 0$, a cell floor $p_* > 0$, and positive constants c_1, C_1, C_2 , depending only on the fixed experiment and the fixed choices just made, with the following properties. Both families are admissible throughout these ranges, all joint cell probabilities are at least p_* for every $T \in [T_1, T_3]$, and $d_\infty(\xi(t), \xi(0)) = t$. For either signal $\psi \in \{\eta, \vartheta_*\}$ and positive $w = \epsilon_u \epsilon_v$ in (Q.3), or $w = s$ in (Q.4), uniformly on $[0, t_0]$,*

$$c_1 w \leq \psi(P_t) \leq C_1 w, \quad \sup_{T \in [T_1, T_3]} \|P_t(T) - P_0(T)\|_F \leq C_2 w t. \quad (\text{Q.5})$$

In the product family both observed marginals are exactly fixed, and

$$P_t(T) - P_0(T) = \frac{T - a_0}{2k(T - h)} \epsilon_u \epsilon_v (\lambda^{-2} - 1) e f^\top, \quad (\text{Q.6})$$

$$\kappa(\omega_t) = \frac{\epsilon_u \epsilon_v}{2\lambda^2}, \quad \sigma_k(A_t) = \kappa(\omega_t)/k. \quad (\text{Q.7})$$

In the collapse family,

$$P_t(T) - P_0(T) = \frac{tB}{(T - a - t)(T - a)}. \quad (\text{Q.8})$$

If $w = 0$, the respective family has exactly the same laws for all t and all clocks in the interval, despite its changing action multiset.

Proof. Choose c_0, ϵ_* small enough that every channel entry in the product family is at least half the corresponding uniform entry. Shrink t_0 and s_0 to keep every action in the declared interval and to preserve order. The collapse family also has a fixed positive cell floor, since its channels are strictly positive. These choices precede all later choices of N, w and the testing displacement.

In the first two terms of the product family the unnormalized sum is

$$\begin{aligned} & (T - a_0 + \lambda b_0)(u_0 + \epsilon_u e/(2\lambda))(v_0 + \epsilon_v f/(2\lambda))^\top \\ & + (T - a_0 - \lambda b_0)(u_0 - \epsilon_u e/(2\lambda))(v_0 - \epsilon_v f/(2\lambda))^\top. \end{aligned}$$

Its constant part is $2(T - a_0)u_0 v_0^\top$, its linear parts are $b_0 \epsilon_u \epsilon_v u_0^\top + b_0 \epsilon_v u_0 f^\top$, and its interaction part is $(T - a_0) \epsilon_u \epsilon_v e f^\top / (2\lambda^2)$. Division by $k(T - h)$ proves (Q.6). Zero sums of e, f prove exact marginal cancellation.

The latent vector $(1, -1, 0, \dots, 0)/\sqrt{2}$ is a right singular vector of U_t with singular value $\epsilon_u/(\sqrt{2}\lambda)$. Its orthogonal complement maps, independently of ϵ_u , into the independent directions $u_0, e_3, \dots, e_k$, all perpendicular to e . Its least singular value is a fixed positive number. Choosing ϵ_* smaller if necessary gives the stated smallest singular value, and the same proof applies to V_t . In $A_t = U_t V_t^T/k$ the pair direction maps f to $(\kappa/k)e$; the complementary rank- $(k-1)$ map is fixed and has its nonzero singular values bounded below. This proves (Q.7).

Taking the marginals of B gives components of magnitude $b_0\epsilon_u/k$ and $b_0\epsilon_v/k$ in the e and f directions. Thus $\gamma \geq c \max(\epsilon_u, \epsilon_v)$ and $\vartheta \asymp \epsilon_u \epsilon_v$. The actions are simple and both channels have full rank when the two contrasts are positive. Proposition O.4, including its rank argument for every alternative exact representation, gives $\vartheta_* = \vartheta$. Further, (O.7) and (Q.7) imply $\beta \asymp \kappa^{-1}$; hence $\eta \asymp \epsilon_u \epsilon_v$. Since λ stays in a fixed compact subinterval of $(0, 1]$, $\lambda^{-2} - 1 \asymp t$, uniformly on the chosen interval.

For the collapse family, subtraction of $A + B/(T - a - t)$ gives (Q.8). The fixed channel ranks imply $\gamma \asymp s$, while κ stays bounded below. Simple-spectrum uniqueness gives $\vartheta_* = \vartheta \asymp s$ for $s > 0$. The inequalities $c\vartheta_* \leq \eta \leq \gamma$ give $\eta \asymp s$. The denominators in both exact identities have fixed positive bounds, which gives (Q.5) with one common set of constants after taking minima and maxima. Both exact identities also prove the assertions at $w = 0$. $\square$

Q.2. Adaptive experiments and the lower bounds. An adaptive experiment consists of parameter-independent probability kernels $\pi_i(dT \mid \mathcal{H}_{i-1})$ on $[T_1, T_3]$. Here $\mathcal{H}_{i-1}$ contains all preceding clocks and categorical observations, and any external randomization independent of the parameter. At step i sample the clock with this kernel and then one categorical observation with law $P_\omega(T_i)$. Deterministic predictable clocks and fixed designs are included. The complete transcript, including clocks, is available to the estimator.

Theorem Q.4 (Sharp classes and honest-diameter lower bounds). *For either $\psi = \eta$ or $\psi = \vartheta_*$, there exists $s_* > 0$ such that for $0 < s \leq s_*$ and $N = 3n$,*

$$\inf_{\hat{\xi}} \sup_{\omega: \psi(\mathcal{P}(\omega)) \geq s} \mathbb{E}_\omega d_\infty(\hat{\xi}, \xi)^2 \asymp \min\{1, (Ns^2)^{-1}\}. \quad (\text{Q.9})$$

The direct estimator at the balanced three-clock design gives the upper bound. The lower bound remains valid when the infimum includes every adaptive experiment just defined.

If a random spectrum set J is honest at level $1 - \zeta$ over $\mathcal{K}_1$, with $0 < \zeta < 1/4$, then at the base point of either positive family,

$$\mathbb{E}_0 \text{diam}_\infty J \geq c_\zeta \min\{1, (\sqrt{N}\psi(P_0))^{-1}\}. \quad (\text{Q.10})$$

This is true under any such clock strategy. The exact singular limits of these families have nonvanishing risk and honest-diameter lower bounds. Equal channel contrasts ϵ give the transition $N\epsilon^4 \asymp 1$; full-spectrum spread s with fixed channels gives $Ns^2 \asymp 1$.

Proof. The upper bound follows from Theorem Q.2. Lemma Q.3 and its cell floor give, uniformly in the clock,

$$\text{KL}(P_t(T), P_0(T)) \leq \sum_{a,b} \frac{((P_t)_{ab} - (P_0)_{ab})^2}{(P_0)_{ab}} \leq Cw^2t^2.$$

The conditional clock kernel is the same under both parameters. The relative-entropy chain rule therefore gives

$$\text{KL}(\mathbb{P}_t^{(N)}, \mathbb{P}_0^{(N)}) = \sum_{i=1}^N \mathbb{E}_t \text{KL}(P_t(T_i), P_0(T_i)) \leq CNw^2t^2.$$

This equality includes the external randomization and the entire transcript, so no fixed-design assumption is hidden in the bound. Take $t = c \min\{t_0, (\sqrt{N}w)^{-1}\}$ with a sufficiently small fixed

c. Pinsker's inequality makes the total variation at most $1/2$. The spectra are distance t apart. Turning an estimate into the nearer of the two spectra gives maximal squared risk at least $t^2/16$.

Here is the class bookkeeping. Fix once and for all the constants in Lemma Q.3, choose $M \geq 1/c_1$, and set $w = Ms$, $\epsilon_u = \epsilon_v = \sqrt{Ms}$ in the product family. Take $s_* \leq \epsilon_*^2/M$. Then every $t \in [0, t_0]$, including the testing displacement just chosen, has positive cell floor p_* , admissible actions and weights, and $\psi(P_t) \geq c_1 Ms \geq s$. All constants in the entropy estimate remain unchanged. This proves the lower bound on the exact class in (Q.9); the collapse family gives the other attained singular order.

For an honest set, transfer coverage at the alternative to the base law. The union bound gives

$$\Pr\{\xi(0), \xi(t) \in J\} \geq 1 - 2\zeta - \text{TV}(\mathbb{P}_t^{(N)}, \mathbb{P}_0^{(N)}).$$

Decreasing c depending on ζ leaves a positive fixed lower bound. On this event the diameter is at least t . The comparison $\psi(P_0) \asymp w$ proves (Q.10). At $w = 0$ use a fixed nonzero t ; all conditional laws agree at every clock, so the transcript laws are identical for every policy. The same testing and coverage arguments give the stated nonvanishing bounds. $\square$

The two paths identify different losses of the observation map. The first removes both first-order marginal changes and leaves a tensor product of weak channel directions. The second keeps the channels fixed and removes the whole projective contrast. Neither construction charges a small gap between two otherwise visible actions. The results are class-sharp and pathwise sharp as stated, rather than a complete classification of all directional moduli at intersections of singular strata.

APPENDIX R. SHARP INFERENCE IN EVERY DEGREE

The observable inverse gives inference without a latent channel floor. We first prove sharp signal-class rates with simple component roots, and then prove sharp multiplicity-dependent rates on neighbourhoods of polynomial singularities. The latter are neighbourhood minimax rates, not a pointwise lower bound at a single fixed parameter.

Take n independent categorical draws at each of the ℓ fixed clocks, put $N = \ell n$, $r = m_1 m_2$, and define

$$\rho = \sqrt{\frac{r}{2n} \log \frac{2\ell r}{\zeta}}, \quad \mathcal{D} = \{\omega \in \mathcal{K}_d : \max_j \|P_\omega(T_j) - \hat{P}_j\|_F \leq \rho\}. \quad (\text{R.1})$$

Use the same empty-set convention as in (Q.1). Write $\mathcal{D}_\Lambda$ for its multiset image.

Theorem R.1 (Observable polynomial inference). *The set (R.1) has coverage at least $1 - \zeta$ uniformly over the entire closed model. If the true datum has $\eta_d(P) > 0$, the direct estimator satisfies*

$$\begin{aligned} \mathbb{E}_P d_\infty(\hat{\Lambda}, \Lambda)^2 &\leq C \min\{1, (N\eta_d(P)^2)^{-1/d}\}, \\ \mathbb{E}_P \text{diam}_\infty \mathcal{D}_\Lambda &\leq C_\zeta \min\{1, (\sqrt{N}\eta_d(P))^{-1/d}\}. \end{aligned} \quad (\text{R.2})$$

The exponent $1/d$ is replaced by $1/m$ under the separated-cluster hypothesis in Theorem E.1. In particular, simple component roots separated by ν give squared risk $C_\nu \min\{1, (N\eta_d(P)^2)^{-1}\}$ and expected diameter $C_{\nu, \zeta} \min\{1, (\sqrt{N}\eta_d(P))^{-1}\}$.

Proof. The union bound for the ℓr cell frequencies gives $\Pr(e > \rho) \leq 2\ell r \exp(-2n\rho^2/r) = \zeta$. Also $\mathbb{E}e^2 \leq \sum_j \mathbb{E}\|\hat{P}_j - P_j\|_F^2 \leq \ell/n = \ell^2/N$. Apply Theorem E.1 to the direct estimator, and use Jensen's inequality for the power $1/d$ of e^2 . For nonempty residual regions, Lemma Q.1 bounds their diameter by $C \min\{1, [(e + \rho)/\eta_d(P)]^{1/d}\}$ on every outcome. Taking expectations

gives the second assertion; the empty-set convention preserves it. The identical argument proves the cluster refinements. $\square$

Theorem R.2 (Sharp semisimple signal classes). *Fix d, k, m_1, m_2 and $\ell \geq 2d + 1$. There exist $\nu_0, s_0 > 0$ such that, for every fixed $0 < \nu \leq \nu_0$ and $0 < s \leq s_0$, the class*

$$\mathcal{S}_{\nu,s} = \{\omega \in \mathcal{K}_d : \eta_d(P_\omega) \geq s, \quad \text{roots within each } f_b \text{ are } \nu\text{-separated}\}$$

satisfies

$$\inf_{\hat{\Lambda}} \sup_{\omega \in \mathcal{S}_{\nu,s}} \mathbb{E}_\omega d_\infty(\hat{\Lambda}, \Lambda)^2 \asymp \min\{1, (Ns^2)^{-1}\}. \quad (\text{R.3})$$

The balanced fixed design attains the upper bound. The lower bound holds even if the infimum includes all parameter-independent adaptive clock strategies on $[T_1, T_\ell]$. For confidence sets honest at level $1 - \zeta$, $0 < \zeta < 1/2$, over the closed model, the corresponding worst-case expected diameter on this class has order $\min\{1, (\sqrt{N}s)^{-1}\}$.

Proof. Only the lower bound remains. Choose a monic F with d simple interior roots and a sufficiently small nonzero constant G . Choose $t_0 > 0$ so that, for $0 \leq t \leq t_0$, $\lambda = 1 - t$ and the polynomials

$$f_{1,t} = F + \lambda G, \quad f_{2,t} = F - \lambda G, \quad f_{b,t} = F \quad (b \geq 3) \quad (\text{R.4})$$

have simple interior roots with a common positive within-component separation. Their collection is relatively prime and, with equal weights, $q = F$. Use the positive stochastic channels of (Q.3), now with this λ . The extra directions are unchanged, and take sufficiently small contrasts $\epsilon_u, \epsilon_v > 0$. Put $w = \epsilon_u \epsilon_v$. Direct expansion gives

$$P_t(T) = A_t + \frac{G}{F(T)} D, \quad D = \frac{\epsilon_u \epsilon_v v_0^\top + \epsilon_v u_0 f^\top}{k}, \quad (\text{R.5})$$

$$P_t(T) - P_0(T) = \frac{w}{2k} (\lambda^{-2} - 1) e f^\top. \quad (\text{R.6})$$

Both observed marginals are fixed, and every cell has a uniform positive floor on the whole clock interval. The leading matrix is $A_t = U_t V_t^\top / k$, so the orthogonal pair-direction calculation in Lemma Q.3 gives $\sigma_k(A_t) = w / (2k\lambda^2)$ and $\kappa = w / (2\lambda^2)$. Consequently (E.3) gives $B_d(P_t) \asymp w^{-1}$.

The normalization operator in this family is the tensor product of $r \mapsto \Omega(r(T_j)G/F(T_j))_j$ and $\text{vec } D$. The scalar map is injective: an interpolating polynomial would give $FH - rG = 0$ at $2d + 1$ nodes, hence identically, and coprimality forces $F \mid r$, so $r = 0$. Its least singular value is a fixed positive number. The two summands in D are Frobenius orthogonal; hence $\tau(P_t) \asymp \max\{\epsilon_u, \epsilon_v\}$. Uniformly over the declared ranges, therefore,

$$c_1 w \leq \eta_d(P_t) \leq C_1 w, \quad \sup_T \|P_t(T) - P_0(T)\|_F \leq C_2 w t. \quad (\text{R.7})$$

No common polynomial factor was attached to an affine experiment: $\gcd(F, G) = 1$ and the visible denominator has degree d .

The roots of $F \pm \lambda G$ have nonzero derivative with respect to λ , since their derivatives are $\mp G/(F')$ at these roots. Shrinking t_0 preserves their order relative to the roots of F and gives $c_3 t \leq d_\infty(\Lambda_t, \Lambda_0) \leq C_3 t$. All constants here are fixed before choosing N, w . The positive cell floor and (R.7) imply $\sup_T \text{KL}(P_t(T), P_0(T)) \leq C w^2 t^2$. The adaptive transcript chain rule from Theorem Q.4 gives the bound $C N w^2 t^2$ for every allowed policy. Take $t = c \min\{t_0, (\sqrt{N}w)^{-1}\}$ to keep total variation bounded away from one. The two-point risk is at least ct^2 .

For the exact signal class choose $M \geq 1/c_1$, set $w = Ms$ and $\epsilon_u = \epsilon_v = \sqrt{Ms}$, and restrict s_0 so that all parameters remain in their fixed admissible ranges. Equation (R.7) puts both testing points, and the path between them, in $\mathcal{S}_{\nu,s}$. Choose ν_0 from the separation of (R.4). This proves (R.3). For honest sets, transfer coverage from the alternative to the base law and

use $\Pr_0\{\Lambda_0, \Lambda_t \in J\} \geq 1 - 2\zeta - \text{TV}$. Choosing c in terms of ζ proves the diameter lower bound. Theorem R.1 proves the matching upper bound. $\square$

R.1. Neighbourhoods of multiple component roots. The exponent $1/d$ is not merely an artefact of allowing complex roots. Real-rooted alternatives can force its multiplicity-refined version. The correct lower-bound construction uses two moving alternatives, not an arbitrary constant perturbation of $(z - x)^m$.

Theorem R.3 (Multiplicity-dependent local minimax exponents). *Let ω_0 have strictly positive full-rank channels, strict weights, $\tau(P_0) > 0$, and pairwise distinct component polynomials whose roots are in (L, D) . Let m be the largest multiplicity of a root within a single component. There is a sufficiently small fixed neighbourhood $\mathcal{U}$ of ω_0 , in channel, weight and monic-coefficient coordinates and restricted to the real-rooted model, such that*

$$\inf_{\hat{\Lambda}} \sup_{\omega \in \mathcal{U}} \mathbb{E}_{\omega} d_{\infty}(\hat{\Lambda}, \Lambda)^2 \asymp N^{-1/m} \quad (N \rightarrow \infty). \quad (\text{R.8})$$

The direct estimator at the balanced design attains the upper bound. The lower bound remains valid for all adaptive clock strategies on $[T_1, T_\ell]$. The corresponding worst-case expected diameter of sets honest on $\mathcal{U}$ has lower order $N^{-1/(2m)}$, attained by (R.1), which is honest even on the closed model.

Proof. Choose disjoint small intervals about the distinct roots of each base component polynomial. By continuity of roots, a sufficiently small coefficient neighbourhood has the same number of roots in each such interval. There are at most m roots per component interval; distances between its different intervals have a positive lower bound. Keep the k complete polynomial tuples distinct and the channels full rank by shrinking the neighbourhood. The joint projectors, B_d and τ are continuous there; in particular $\eta_d \geq s_* > 0$. The cluster version of Theorem E.1 is uniform on this neighbourhood, even when roots inside a cluster split or collide. Theorem R.1 gives the required risk and diameter upper bounds with exponent $1/m$.

Choose one base component $f_b(z) = (z - x)^m a(z)$ with $a(x) \neq 0$. Let $H(y)$ be a monic degree- m polynomial with distinct real roots. A sufficiently small nonzero constant c makes both H and $H + c$ have m distinct real roots, by disjoint sign-changing brackets. For small $u > 0$ replace only this component by the two alternatives

$$f_{b,u,0}(z) = u^m H((z - x)/u) a(z), \quad f_{b,u,1}(z) = u^m \{H((z - x)/u) + c\} a(z). \quad (\text{R.9})$$

All other parameters stay fixed. These are monic, real-rooted and converge in coefficients to the original component. Thus both lie in $\mathcal{U}$ for sufficiently small u , and their difference is exactly $cu^m a(z)$. Their aggregate root multisets are at distance $c_0 u$ for some $c_0 > 0$. Indeed the changed cluster is the dilation by u of two different fixed multisets; unchanged roots at x , if any, add the same finite number of zeros to both and cannot make those multisets equal. Distinct base clusters remain separate.

On the compact clock interval the denominator has a positive lower bound, so the rational observations differ by at most Cu^m uniformly in the clock. The positive channels give a common cell floor. Hence every adaptive transcript has relative entropy at most CNu^{2m} . Choose $u = c'N^{-1/(2m)}$. The two-point test gives squared risk at least $c''u^2$, proving (R.8). Coverage transfer between these same two moving alternatives gives expected diameter at least a fixed multiple of u . This proves the final assertion. $\square$

The local theorem is valid for every $1 \leq m \leq d$ for which such a regular-normalization base exists; for instance one component may have an m -fold root and another a polynomial with no common zero with it. Cross-component multiplicity alone does not change m . The sharp neighbourhood rate is therefore compatible with linear semisimple stability. Intersections with

$\tau = 0$ or channel-rank loss require their own signal geometry; the global certificate, the sharp semisimple classes, the retained affine collapse family, and the local multiplicity theorem state separately the conclusions proved here.

APPENDIX S. OBSERVATION FIBRES AND ADMISSIBLE COEFFICIENT PERTURBATIONS

A defect of a latent representative is not necessarily a defect of an observation. This distinction is needed when a singular fibre contains different component-equality or root-multiplicity patterns. For the stochastic map F in (2.1), write

$$\mathfrak{F}_P = \{\theta \in \mathcal{K}_d : F(\theta) = P\}.$$

The intrinsic observation germ uses the entire inverse image of shrinking observation balls, and hence includes $\mathfrak{F}_P$. A pointed latent germ (P, θ_0) instead restricts the parameters to neighbourhoods of a specified $\theta_0 \in \mathfrak{F}_P$. The stratified fibre records the subsets of $\mathfrak{F}_P$ having each specified rank, equality, or multiplicity pattern, modulo the finite component action. These are three different objects.

In the binary family of Theorem T.3, the intersection $a = b = x = 0$ always means the selected *pointed latent* intersection. Its representative has equal repeated component polynomials. The observation P_* itself has neither a well-defined component-equality pattern nor a well-defined internal multiplicity: both collapsed channels give the same P_* for every admissible choice of the component polynomials. Its intrinsic modulus is $D - L$. The ideal (ab, ax, bx) and the transverse arc orders describe that pointed family; they do not define a vanishing exponent of ω_{P_*} . The quartic theorem concerns a different stratum, where coefficient recovery is regular and scalar root extraction is singular. Its matrix S is reconstructed by first recovering the monic normalizer and numerator coefficients, then their joint projectors and component tuples, and finally the profiled Fisher form. The only remaining ambiguity is a finite relabelling of components. Whenever coefficient coordinates are changed in this paper, the actual observed-cell map and its Fisher or Hellinger metric are transported with them. Applying an entrywise Hellinger formula afresh to arbitrary left-right transforms of probability matrices would define a different experiment; invariance under that replacement is not asserted.

S.1. Positive refactorization of pencils. The following lemma makes explicit the admissibility of the binary whole-model perturbation used below. Its neighbourhood can depend on the base point; it is not a uniform lower bound for that neighbourhood as the channel ranks or root separation degenerate.

Lemma S.1 (Analytic stochastic refactorization of a binary pencil). *Let $K_\epsilon(z) = zA_\epsilon - C_\epsilon$ be a real analytic two by two pencil with A_0 invertible and*

$$\mathbf{1}^\top A_\epsilon \mathbf{1} = 1, \quad \mathbf{1}^\top C_\epsilon \mathbf{1} = h_\epsilon.$$

Suppose $K_0 = U \operatorname{diag}(\alpha_b(z - \xi_b))V^\top$, with strictly positive stochastic columns, strict weights $\alpha_b > \alpha_$, full-rank channels, and distinct interior roots. Then for all sufficiently small $|\epsilon|$ there is an analytic factorization of the same type, with roots remaining interior and normalizer exactly $z - h_\epsilon$. In particular a numerator perturbation of zero total preserves a fixed normalizer exactly.*

Proof. The relevant normalized coefficient is $X_\epsilon = C_\epsilon A_\epsilon^{-1}$. Its two eigenvalues $\xi_1(\epsilon), \xi_2(\epsilon)$ are distinct real analytic functions for small real ϵ , by the simple-root implicit function theorem. They stay real because the real quadratic has strictly positive discriminant at zero. They remain in the permitted interval after shrinking the neighbourhood. Its spectral projectors are explicitly

$$Q_b(\epsilon) = \frac{X_\epsilon - \xi_{3-b}(\epsilon)I}{\xi_b(\epsilon) - \xi_{3-b}(\epsilon)}, \quad W_b(\epsilon) = Q_b(\epsilon)A_\epsilon.$$

They give $K_\epsilon(z) = \sum_b (z - \xi_b(\epsilon)) W_b(\epsilon)$. Each W_b has rank one, depends analytically on ϵ , and at zero equals $\alpha_b U_{:b} V_{:b}^\top$. All its finitely many entries therefore remain strictly positive. Set

$$\alpha_b(\epsilon) = \mathbf{1}^\top W_b(\epsilon) \mathbf{1}, \quad u_b(\epsilon) = \frac{W_b(\epsilon) \mathbf{1}}{\alpha_b(\epsilon)}, \quad v_b(\epsilon) = \frac{W_b(\epsilon)^\top \mathbf{1}}{\alpha_b(\epsilon)}.$$

These are analytic; the weights remain above α_* and the columns are strictly positive and stochastic. Rank one gives $W_b = \alpha_b u_b v_b^\top$, since for a positive rank-one matrix the product of its row and column sums equals that matrix multiplied by its total sum. Also $\sum_b W_b = A_\epsilon$ and $\sum_b \xi_b W_b = C_\epsilon$. Thus $\sum_b \alpha_b = 1$ and $\sum_b \alpha_b \xi_b = h_\epsilon$, proving the exact normalizer identity. Full rank of the recovered channels persists by continuity. $\square$

For the lower tangent in Theorem T.3, take $A_\epsilon = A$ and $C_\epsilon = hA - B - \epsilon D$, where $D = \frac{1}{4} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. Then $\mathbf{1}^\top D \mathbf{1} = 0$, its binary moments are $(0, 0, 1)$, and the exact determinant is

$$\det((z - h)A + B + \epsilon D) = \frac{1}{4} \{ab(z - h)^2 + \epsilon(z - h) - abx^2\}.$$

At $\epsilon = 0$, both root derivatives are $-1/(2ab)$. The lemma verifies every stochastic constraint for this perturbation; an algebraic pencil perturbation is therefore an actual tangent of the original model in this use.

S.2. Comparison on fixed-margin regular classes.

Proposition S.2 (Compact regular equivalence). *Fix positive lower margins for the observed cells, the channel least singular values, $\tau(P)$, the weight inequalities, the distances of roots from the endpoints, the internal simple-root gaps, and the separation of distinct complete component coefficient tuples. On every nonempty resulting compact regular class there are constants $0 < c \leq C < \infty$, depending on these margins and the fixed design, such that*

$$c\eta_d(P)^{-1} \leq \mathbf{c}_{H_{P,w}}(P) \leq C\eta_d(P)^{-1}.$$

Cross-component scalar collisions are allowed.

Proof. Pass to ordered internal roots and the finite component quotient. The specified set is compact. The analytic coefficient recovery and quotient-chart theorems give local continuous matrices J, G ; the Fisher metric is continuous because the cells have a fixed positive floor. The injective Jacobians remain injective throughout this set, so $V_H = G(J^\top H J)^{-1} G^\top$ is continuous in each chart. Relabelling only permutes its root coordinates, and $\mathbf{c}_H = \max_a \sqrt{(V_H)_{aa}}$ is a continuous quotient function, also at cross-component scalar collisions. It is strictly positive: shifting every component root by the same small amount is an admissible two-sided tangent with nonzero target derivative; the injective observation derivative has nonzero finite norm on that tangent.

The joint projectors, the leading coefficient inverse, and τ are continuous on the same quotient under the stated tuple and rank margins. Hence $\eta_d^{-1} = B_d + \tau^{-1}$ is continuous, finite, and strictly positive there. The ratio $\mathbf{c}_H/\eta_d^{-1}$ therefore attains a positive minimum and a finite maximum on the compact set. This proves the assertion. The argument supplies a fixed-margin equivalence, not a universal sharp identity between the certificate and the Fisher condition. It does not extend the constants across the vanishing-cell example of Proposition G.2. $\square$

APPENDIX T. EXPLICIT NORMAL FORMS AT SINGULAR STRATA

The semialgebraic envelope does not by itself compute a singularity. We give two complementary computations. The first identifies the leading constant at real-rooted multiplicities by a finite quartic cone in the recovered coefficient algebra. The second treats an intersection of channel collapse, normalization failure, component equality, and internal multiplicity directly from the

model equations. At the latter intersection the modulus has a nonzero constant term; its transverse regular condition has an explicit two-regime law.

T.1. A profiled quartic cone for real-rooted multiplicities. Assume the hypotheses of Proposition K.3, with all roots interior, and at least one internal multiple root. For each component let r_g run over its distinct root values, and let m_g be their multiplicities. A group g includes its component index, even when another component has the same scalar root. Let $\mathcal{M} = \{g : m_g \geq 2\}$. In the ambient coefficient tangent space, define D_g , $g \in \mathcal{M}$, to have zero channel and weight coordinates and component-polynomial variation

$$(D_g)_b(z) = f_b(z)/(z - r_g)^2$$

in its own component, and zero variations in other components. Let $\mathcal{T}_0$ be the linear space generated by all channel and weight tangents and by the vectors with polynomial variation $f_b(z)/(z - r_g)$, one for every distinct root group. These last vectors are the cluster-centroid tangents, including the simple roots.

Put $\langle u, v \rangle_I = \langle J_c u, J_c v \rangle_{H_{P,w}}$. This is positive definite by Proposition K.3. Let Π_0 be its orthogonal projection onto $\mathcal{T}_0$, and set

$$S_{gh} = \langle (I - \Pi_0)D_g, (I - \Pi_0)D_h \rangle_I, \quad g, h \in \mathcal{M}. \quad (\text{T.1})$$

Here the identity in $I - \Pi_0$ is an operator, not the metric matrix. The matrix S is a Schur complement of the recovered coefficient Fisher information. In particular it profiles out the stochastic factors and the cluster centroids, rather than holding nuisance parameters fixed.

For a vector x with one coordinate per root, set $v_g(x) = \frac{1}{2} \sum_{i \in g} x_i^2$. Define the compact set

$$\mathcal{X}_S = \left\{ x : \begin{array}{l} x_i = 0 \text{ on simple groups,} \quad \sum_{i \in g} x_i = 0 \quad (g \in \mathcal{M}), \\ v(x)^\top S v(x) \leq 4 \end{array} \right\}. \quad (\text{T.2})$$

As before, Γ permutes all root coordinates with the same base scalar value, including coordinates belonging to different components.

Theorem T.1 (The real-rooted singular leading constant). *The matrix S in (T.1) is positive definite, and*

$$\omega_P(t) = C_S \sqrt{t} + o(\sqrt{t}), \quad C_S = \max_{x, y \in \mathcal{X}_S} d_\Gamma(x, y) > 0. \quad (\text{T.3})$$

All quantities defining $\mathcal{X}_S$ are recovered from P ; coefficient-coordinate changes leave the construction invariant, and component relabelling only permutes its groups. Thus the exponent and leading constant are determined by a finite profiled quartic problem, not by elimination over the original closed parameter space. If there is exactly one multiple group, it has multiplicity two, and its base value is not shared by any other component, then, writing $S = s > 0$, the constant is

$$C_S = \sqrt{2} s^{-1/4}. \quad (\text{T.4})$$

Proof. For each component the rational functions $1/(z - r_g)$ and $1/(z - r_g)^2$ have linearly independent principal parts at their distinct poles. Multiplication by f_b shows that the D_g are independent modulo $\mathcal{T}_0$. Injectivity of J_c then makes S positive definite. It follows that $v^\top S v \leq 4$ bounds every v_g and every root coordinate; hence (T.2) is compact.

We prove the limiting set assertion underlying (T.3). Proposition K.3 gives coefficient displacement $O(t)$ for every member of $\mathcal{B}_P(t)$. Analytic factorization into the pairwise coprime cluster factors, as in Lemma F.3, therefore gives $O(t)$ displacement of each cluster coefficient. Write the real root displacements in group g as y_{gi} . The first two elementary symmetric coefficients give

$$\sum_i y_{gi} = O(t), \quad \sum_i y_{gi}^2 = O(t).$$

Simple roots have displacement $O(t)$. After a subsequence, $x_{gi} = y_{gi}/\sqrt{t}$ converges, its sum is zero, and the first coefficient divided by t also converges. Elementary symmetric terms of degree at least three are $O(t^{3/2}) = o(t)$. Consequently every limiting ambient coefficient displacement divided by t has the form

$$a_0 - \sum_{g \in \mathcal{M}} v_g(x) D_g, \quad a_0 \in \mathcal{T}_0. \quad (\text{T.5})$$

The Hellinger expansion in the proof of Theorem G.1 is valid on this ambient coefficient chart. Profiling (T.5) over a_0 therefore gives $v(x)^\top S v(x) \leq 4$. This proves the outer limiting-set inclusion for the rescaled target.

Conversely take $x \in \mathcal{X}_S$ with strict inequality. Choose $a_0 \in \mathcal{T}_0$ attaining the orthogonal projection minimum. Realize its channel and weight coordinates by linear perturbations of size t . In group g place the roots at

$$r_g + \sqrt{t} x_{gi} + t b_g,$$

where the common centroid shift b_g is selected to give the corresponding $f_b/(z - r_g)$ coefficient of a_0 ; use the analogous linear shift for each simple root. The zero-sum condition gives exactly (T.5) at order t . Higher terms are $o(t)$. These are real admissible interior roots, and the strict stochastic, rank, normalization, and tuple conditions persist. Their Hellinger distance divided by t tends to $\frac{1}{2} \sqrt{v(x)^\top S v(x)} < 1$. Points on the boundary are approached by replacing x by $(1 - \varepsilon)x$ and then letting $\varepsilon \downarrow 0$. Thus the target sets, divided by $\sqrt{t}$ and taken modulo Γ , converge in Hausdorff distance to $\mathcal{X}_S/\Gamma$. Matching between distinct base values is excluded by their fixed gaps. Taking diameters proves (T.3).

A symmetric split in any multiple group gives a nonzero member of $\mathcal{X}_S$, while 0 also belongs to it, so $C_S > 0$. The coordinate invariance follows from the pullback metric and the intrinsic principal-part directions; permutations just relabel them. In the final case $x = (-a, a)$ in the double group and zero elsewhere, with $a \geq 0$ after quotienting. Its constraint is $sa^4 \leq 4$ and its quotient distance is $|a - a'|$. The diameter of this interval is $\sqrt{2} s^{-1/4}$, proving (T.4). $\square$

Theorem T.2 (Uniform crossover at a splitting double root). *Suppose just one component has a double root r , all other component roots are simple and separated from r , and the strict hypotheses of Proposition K.3 hold. Replace this double factor by $(z - r)^2 - \Delta^2$, leaving all other parameters fixed, and denote the datum by P_Δ . For sufficiently small fixed $\Delta_0, t_0 > 0$, uniformly for $0 \leq \Delta \leq \Delta_0$ and $0 < t \leq t_0$,*

$$\omega_{P_\Delta}(t) \asymp t + \frac{t}{\Delta + \sqrt{t}}. \quad (\text{T.6})$$

For $\Delta > 0$ its regular Fisher condition is of order $1 + \Delta^{-1}$. The linear regime $t \ll \Delta^2$ and the singular regime $t \gg \Delta^2$ therefore join at $t \asymp \Delta^2$. At $\Delta = 0$ the exact leading constant is (T.4).

Proof. The coefficient recovery map and its inverse differential are bounded uniformly on a small compact neighbourhood of P_0 . Coprime cluster factorization is uniform there as well. Write the exceptional monic quadratic as $(z - r - \mu)^2 - v$, with $v \geq 0$. In coefficient coordinates this is an analytic change of variables even at $v = 0$. A Hellinger ball of radius t about P_Δ consequently has $|\mu| \leq Ct$, $|v - \Delta^2| \leq Ct$, and $O(t)$ changes of all the other roots. Uniform exclusion of distant representatives follows from the recovery proposition and compactness. For two such quadratic factors, their sorted-root distance is at most the difference of centres plus $|\sqrt{v} - \sqrt{v'}|$. The latter is bounded by $Ct/(\Delta + \sqrt{t})$, by considering separately $t \leq c\Delta^2$ and its complement. This proves the upper bound.

For the lower bound increase v from Δ^2 to $\Delta^2 + ct$, with all other coordinates fixed. For sufficiently small fixed $c > 0$ this alternative has Hellinger distance at most t , and its target

displacement is

$$\sqrt{\Delta^2 + ct} - \Delta = \frac{ct}{\sqrt{\Delta^2 + ct} + \Delta} \asymp \frac{t}{\Delta + \sqrt{t}}.$$

A common shift of the quadratic roots by ct supplies the linear term. The larger of the two lower bounds controls their sum up to a factor two. Interior margins preserve admissibility. For $\Delta > 0$ root differentiation in (μ, v) gives derivatives 1 and $\pm(2\Delta)^{-1}$, whereas the coefficient Fisher form is uniformly positive and bounded. This proves the condition estimate and all the stated transitions. $\square$

T.2. A binary intersection of normalization and channel defects. Here $k = m_1 = m_2 = 2$, the weights are $1/2$, and $\ell \geq 2d + 1$. For $0 \leq a, b < 1$, put

$$U_a = \frac{1}{2} \begin{pmatrix} 1+a & 1-a \\ 1-a & 1+a \end{pmatrix}, \quad V_b = \frac{1}{2} \begin{pmatrix} 1+b & 1-b \\ 1-b & 1+b \end{pmatrix}, \quad f_{\pm}(z) = (z - h \mp x)^d,$$

with $h \pm x$ interior. Define the observable binary moments $u = P_{11} + P_{12} - P_{21} - P_{22}$, $v = P_{11} + P_{21} - P_{12} - P_{22}$, and $w = P_{11} + P_{22} - P_{12} - P_{21}$. Direct substitution gives the exact normal form

$$(u_j, v_j, w_j) = (a\chi_j, b\chi_j, ab), \quad \chi_j = \frac{(T_j - h - x)^d - (T_j - h + x)^d}{(T_j - h - x)^d + (T_j - h + x)^d}. \quad (\text{T.7})$$

Let P_* be the constant matrix with every entry $1/4$.

Theorem T.3 (Collapse normal form and transverse phase law). *Near $a = b = x = 0$, the observation ideal in analytic coordinates is*

$$(ab, ax, bx). \quad (\text{T.8})$$

In particular, uniformly for h in an interior compact interval,

$$h_w(P_{a,b,x,h}, P_*) \asymp |ab| + |ax| + |bx|.$$

The intersection $a = b = x = 0$ has two rank-one channels, equal complete component polynomials, $\tau(P_) = 0$, and internal multiplicity d when $d \geq 2$. Its whole-model modulus is exactly*

$$\omega_{P_*}(t) = D - L \quad (t \geq 0). \quad (\text{T.9})$$

For arcs $a = As^p + o(s^p)$, $b = Bs^q + o(s^q)$, $x = Xs^r + o(s^r)$ with $A, B, X > 0$, $p, q, r > 0$, and $h = h_0$, the observation order is $\min\{p+q, p+r, q+r\}$, whereas the target displacement from the selected repeated-root representative is of order s^r .

In degree one, at every $a, b, x > 0$ in a sufficiently small fixed parameter box, the regular condition for the entire binary closed model, not merely for the displayed subfamily, satisfies

$$\mathfrak{c}_{H_{P,w}}(P) \asymp \eta_1(P)^{-1} \asymp \frac{1}{ab} + \frac{1}{x\sqrt{a^2 + b^2}}. \quad (\text{T.10})$$

The constants are uniform in a, b, x . Along the preceding power arcs its divergence exponent is

$$\max\{p+q, r + \min(p, q)\}. \quad (\text{T.11})$$

The phase boundary is $r = \max(p, q)$. At each regular datum the leading observation-ball constant is four times this Fisher condition, by Theorem G.1.

Proof. The binary moment transformation is invertible, with $P_{ij} = \frac{1}{4}(1 + \varepsilon_i u + \varepsilon_j v + \varepsilon_i \varepsilon_j w)$, where $\varepsilon_1 = 1$ and $\varepsilon_2 = -1$. Equation (T.7) is obtained by averaging the two rank-one component matrices. At each clock,

$$\chi_j = -\frac{dx}{T_j - h} + O(x^3).$$

Thus χ_j/x is an analytic unit near the intersection. The ideal generated by all moment differences from P_* is exactly (T.8). Positive cell floors make Hellinger and moment norms uniformly equivalent. This proves the stated norm and arc orders without quantifier elimination. More precisely, if $s_0 = \min\{p+q, p+r, q+r\}$ and $C_h = \sum_j w_j(T_j - h_0)^{-2}$, the squared leading Hellinger coefficient is

$$\frac{1}{4} \left[\mathbf{1}_{p+q=s_0}(AB)^2 + d^2 C_h \{ \mathbf{1}_{p+r=s_0}(AX)^2 + \mathbf{1}_{q+r=s_0}(BX)^2 \} \right].$$

The three moment directions are orthogonal at P_* , so no leading cancellation is suppressed in this expression.

All coordinates of any two spectral targets lie in $[L, D]$, giving $\omega_{P_*}(t) \leq D - L$. With both channels equal to their collapsed columns, taking every component polynomial to be $(z - L)^d$ or every one to be $(z - D)^d$ gives the same datum P_* . These two targets attain $D - L$, proving (T.9). More generally any constant datum admitting the stochastic factorization of the model has this same spectral diameter, by keeping its factors and weights and choosing the common polynomial at either endpoint. Thus a positive vanishing exponent is not the appropriate invariant at this intersection: the computed constant exact-fibre term persists.

It remains to establish the whole-model transverse assertion. For $d = 1$, $P(T) = A + B/(T - h)$, with A having binary moments $(0, 0, ab)$ and B having zero total and moments $(-ax, -bx, 0)$. The affine normalization calculation gives $\tau(P) \asymp \|B\|_F \asymp x\sqrt{a^2 + b^2}$. The rank-one residue formula gives

$$B_1(P) = \sum_{i=1}^2 2 \|V_b^{-T} e_i\|_2 \|e_i^T U_a^{-1}\|_2 \asymp (ab)^{-1}.$$

Cell floors and clock factors are uniform, so the global observable bound and Theorem K.2 give the upper estimate in (T.10) for all model tangents.

Varying h with a, b, x fixed moves both roots at unit speed and has observation derivative of Fisher norm comparable to $x\sqrt{a^2 + b^2}$. This gives the normalization lower bound. For the other lower bound keep $q = z - h$ and A fixed and add to B a matrix with zero total and binary moments $(0, 0, \varepsilon)$. The determinant of the perturbed numerator, writing $y = z - h$, is

$$\frac{1}{4} \{ ab y^2 + \varepsilon y - ab x^2 \}.$$

Both roots have derivative $-1/(2ab)$ at $\varepsilon = 0$, while the observation derivative has uniformly bounded Fisher norm. This is an admissible model tangent: the pencil has two distinct real roots at the base; its analytic rank-one spectral factors are strictly positive there and remain so for sufficiently small ε . Their positive total, row, and column sums give strict weights and stochastic columns, and the total numerator remains q . The two roots remain interior. The size of this admissible interval may depend on the base point, which is sufficient for the differential assertion. The larger of the two lower bounds controls their sum. This proves (T.10); substituting the power arcs proves (T.11). The complete polynomials have distinct roots for $x > 0$, so the collision group is trivial and the final assertion follows from (G.1). $\square$

For $a, b, x > 0$ in the degree-one normal form, these transverse defect coordinates are themselves recoverable from observations. The three clock ratios give h ; the moment residues give ax and bx , while the constant joint moment gives ab . Thus $x^2 = (ax)(bx)/(ab)$ and similarly for a^2, b^2 . Their positive square roots fix the displayed branch. The exponents in (T.11) therefore describe observable relative orders, not inaccessible choices of latent normalization. At the intersection, (T.9) replaces the diverging differential law. The variational envelope remains valid on other singular faces; the explicit computations here do not identify defect dimensions alone with a universal classification of all of them.

APPENDIX U. SHARP FLAG EXPONENTS AND OBSERVABLE CANCELLATIONS

The common-flag theorem allows noncommuting and defective coefficients, but assumes simultaneous triangularizability. We retain that scope. We now identify a finite cancellation invariant and a condition under which its exponent is attained by positive probability observations. The pole mechanism is classical spectral perturbation theory [27, 24, 30]; the point here is to enforce the scalar normalization constraint on the perturbations and to identify the resulting observation modulus.

Work in the square probability-quotient model of Section N, with $\mathbf{1}^\top M(z) \mathbf{1} = q(z)$, $q(T_j) > 0$, $\tau(P) > 0$, and strictly positive observed cells. A fixed-normalizer slice has numerator perturbations $E(z)$ satisfying $\mathbf{1}^\top E(z) \mathbf{1} = 0$. Its competitors need not share the true flag. Let $\omega_P^{[q]}(t)$ denote the diameter of their spectral targets in the Hellinger ball of radius t with q held fixed. For small t , interpolation bounds their coefficient errors and preserves the invertible leading coefficient; no extra root identification is needed to define this local diameter. Let $\omega_P^{\text{rat}}(t)$ denote the analogous local modulus when the bounded monic normalizer is also allowed to vary in the positive square rational experiment. These two functions are distinguished from the stochastic-model modulus ω_P .

U.1. A finite pole invariant. At a root r of $\det M$, write the triangular form as $T = D_f + N$. For an increasing path $\mathcal{P} = (i_0, \dots, i_h)$, including paths of length zero, put

$$\Phi_{\mathcal{P}}(z) = (-1)^h \frac{\prod_{a < h} N_{i_a i_{a+1}}(z)}{\prod_{a \leq h} f_{i_a}(z)}.$$

For each nonzero edge write $N_{ij}(z) = (z - r)^{e_{ij}} n_{ij}(z)$ and $f_i(z) = (z - r)^{m_i} g_i(z)$, with the residual factors nonzero at r . Let

$$p(\mathcal{P}) = \sum_{a \leq h} m_{i_a} - \sum_{a < h} e_{i_a i_{a+1}}, \quad \psi_{\mathcal{P}} = (-1)^h \frac{\prod_{a < h} n_{i_a i_{a+1}}}{\prod_{a \leq h} g_{i_a}}.$$

An empty numerator product is one. For $s \geq 1$ define the matrix $C_{r,s}$ by

$$(C_{r,s})_{ij} = \sum_{\substack{\mathcal{P}: i \rightsquigarrow j \\ p(\mathcal{P}) \geq s}} [(z - r)^{p(\mathcal{P}) - s}] \psi_{\mathcal{P}}(z). \quad (\text{U.1})$$

The bracket denotes a Taylor coefficient at r . Only finitely many paths and Taylor coefficients, with orders bounded by kd , occur.

Proposition U.1 (Exact cancellation test). *The actual pole order of M^{-1} at r is*

$$\nu_r = \max\{s : C_{r,s} \neq 0\}.$$

Its leading Laurent matrix is $L_r = A_0^{-1} Q C_{r,\nu_r} Q^$ in the notation of (N.1). Both ν_r and L_r are independent of the chosen flag once expressed in the observation coordinates. In particular ν_r is at most the flag-path count at r ; strict inequality is decided by the explicit vanishing conditions (U.1).*

Proof. The finite upper-triangular inverse expansion gives $(T^{-1})_{ij} = \sum_{\mathcal{P}: i \rightsquigarrow j} \Phi_{\mathcal{P}}$. Expanding each summand at r gives precisely (U.1). Multiplication by the constant invertible matrices in $M^{-1} = A_0^{-1} Q T^{-1} Q^*$ cannot change the largest nonzero pole order and gives the stated Laurent matrix. This also proves invariance and the assertion about cancellations. $\square$

For a concrete cancellation, take

$$T(z) = \begin{pmatrix} z^2 & z & 1 \\ 0 & z^2 & z \\ 0 & 0 & z^2 \end{pmatrix}.$$

The direct and two-edge terms in $(T^{-1})_{13}$ are $-z^{-4}$ and z^{-4} and cancel identically. The adjacent entries are $-z^{-3}$, so $\nu_0 = 3$, while the uncorrected three-vertex flag count is 6. This is also realizable as a positive quotient: with strictly positive invertible doubly stochastic L, R , put $M = LTR^T/3$ and $q = z^2 + 2z/3 + 1/3$. Observations at positive clocks are strictly positive. The constant upper-right entry makes the common divisor of the numerator entries one, so $\tau > 0$ at five clocks. Thus the cancellation is not confined to a formal triangular matrix.

Theorem U.2 (Exposure, sharpness, and the observation modulus). *Let $\nu = \max_r \nu_r$. On the fixed-normalizer slice,*

$$\omega_P^{[q]}(t) \leq Ct^{1/\nu}.$$

Suppose some real root r with $\nu_r = \nu$ has an exposed leading Laurent matrix, in the sense

$$L_r \notin \mathbb{C} \mathbf{1} \mathbf{1}^T. \quad (\text{U.2})$$

Then there is a constant rank-one perturbation $E = uv^T$ with $\mathbf{1}^T E \mathbf{1} = 0$ such that $c = v^T L_r u \neq 0$. It preserves the normalizer and the positivity of the observed cells for sufficiently small real perturbation size. Its root displacement at r is

$$|c|^{1/\nu} |\varepsilon|^{1/\nu} + o(|\varepsilon|^{1/\nu}). \quad (\text{U.3})$$

Consequently $\omega_P^{[q]}(t) \asymp t^{1/\nu}$. If, in addition, ν equals the maximal path count ν_Q of the cluster version of Theorem N.1, then

$$\omega_P^{\text{rat}}(t) \asymp t^{1/\nu_Q}. \quad (\text{U.4})$$

The latter assertion allows the normalizer and the competitor flag to vary. It is a theorem for the positive rational extension, not an assertion that its complex-root competitors belong to $\mathcal{K}_d$.

Proof. On the fixed-normalizer slice, interpolation and positive fixed clock denominators give coefficient error $O(t)$ from an observation ball of radius t . Near each isolated root r the true resolvent has norm at most $C|z - r|^{-\nu_r}$. Neumann exclusion on circles of radius Ct^{1/ν_r} , and argument-principle counting along the coefficient homotopy, match the same number of roots. Away from the root circles the resolvent is uniformly bounded. The invertible leading coefficient persists and prevents loss at infinity. The triangle inequality gives the diameter bound.

The rank-one matrices with either $u \perp \mathbf{1}$ or $v \perp \mathbf{1}$ span the hyperplane of matrices with zero entry sum. Its annihilator under $E \mapsto \text{tr}(L_r E)$ is precisely $\mathbb{C} \mathbf{1} \mathbf{1}^T$. Because r and the coefficients are real, L_r is real, and (U.2) yields real u, v with $c \neq 0$. Write $\det M(z) = (z - r)^m b(z)$, where $b(r) \neq 0$ and $m \geq \nu$. The determinant lemma gives, near r ,

$$\det(M + \varepsilon uv^T) = (z - r)^{m-\nu} b(z) \{(z - r)^\nu + \varepsilon(c + O(z - r))\}.$$

Rescaling $z - r = |\varepsilon|^{1/\nu} \zeta$ and applying Rouché's theorem gives ν roots with modulus $|c|^{1/\nu} |\varepsilon|^{1/\nu} + o(|\varepsilon|^{1/\nu})$; the other $m - \nu$ roots remain at r . All base roots in this cluster are equal, proving (U.3). The fixed normalizer is preserved because E has zero entry sum, and strict positivity at the finitely many clocks persists. Furthermore

$$h_w((M + \varepsilon E)/q, M/q) = \frac{1}{2} |\varepsilon| \| (E(T_j)/q(T_j))_j \|_{H_{P,w}} + o(|\varepsilon|).$$

The norm is positive, so this path proves the matching lower bound inside the fixed-normalizer observation ball.

For the final assertion the fixed-normalizer slice supplies the same lower bound in the larger experiment. Since $\tau(P) > 0$ is fixed, the cluster flag theorem gives upper bound $C(t^{1/m_*} + t^{1/\nu_Q})$, where $m_* \leq \nu_Q$ is the largest scalar cluster count. This is $O(t^{1/\nu_Q})$. Apply it from the base to each member of a pair to obtain (U.4). $\square$

When $\nu < \nu_Q$, the improved fixed-normalizer assertion remains valid, but one must not simply substitute ν into the two-scale unknown-normalizer bound. The interpolation gauge can break a cancellation between products of entries. A uniform improvement in that larger experiment requires controlling those gauge-dependent cancellations as well. The exact test above characterizes the pole at the specified datum; Theorem N.1 retains its uniform path certificate for the full gauge perturbation.

U.2. Sharp positive families, including noncommuting coefficients. Let J_k be the nilpotent Jordan shift, and choose strictly positive invertible doubly stochastic matrices L, R , for example $(I + \mathbf{1}\mathbf{1}^\top)/(k+1)$. First set

$$T(z) = (z-r)^d I + \beta J_k, \quad M(z) = LT(z)R^\top/k, \quad q(z) = (z-r)^d + \beta(k-1)/k, \quad \beta > 0.$$

At clocks $T_j > r$, the observations are strictly positive. A constant Jordan entry makes the greatest common divisor of the numerator entries one, so $\tau > 0$ at $2d+1$ clocks. At r the pole order is kd , and the leading Laurent matrix is a nonzero multiple of $R^{-\top}e_1e_k^\top L^{-1}$. It is not proportional to $\mathbf{1}\mathbf{1}^\top$. Thus the exponent $1/(kd)$ is attained by normalizer-preserving positive perturbations, and agrees with the upper bound in the whole positive rational experiment.

For a genuinely noncommuting and defective family, take $d = 2$ and

$$T(z) = z^2 I + \eta z \operatorname{diag}(c_1, \dots, c_k) + \beta J_k, \quad \eta, \beta > 0, \quad c_i > 0 \text{ distinct},$$

with the same L, R and scaling by $1/k$. The scalar normalizer is $z^2 + \eta k^{-1}(\sum_i c_i)z + \beta(k-1)/k$. Positive clocks give strictly positive probability matrices and $\tau > 0$. The coefficient of z does not commute with the constant coefficient, since $[\operatorname{diag}(c_i), J_k] \neq 0$; the constant coefficient in the normalized algebra is similar to the defective matrix βJ_k . Each diagonal polynomial has one root at zero. The longest path there has count k , its top Laurent term cannot cancel, and its Laurent matrix is again a nonzero multiple of $R^{-\top}e_1e_k^\top L^{-1}$. The other roots $-\eta c_i$ are distinct and simple. Theorem U.2 therefore gives $\omega_P^{\text{rat}}(t) \asymp t^{1/k}$, with a linear scalar-normalization term and a matching k -vertex matrix term. This is a sharp common-flag noncommuting law in the observation metric, rather than a claim about arbitrary nontriangularizable coefficient algebras.

APPENDIX V. CONSTRAINED DETERMINANT GEOMETRY AND SCALAR GAUGES

The pole of a resolvent describes unrestricted coefficient perturbations. A probability normalization imposes a linear constraint, and the first nonvanishing perturbation of the determinant need not be linear. We therefore retain every homogeneous part of the determinant. This gives an exact finite invariant, including its leading metric geometry, for the positive rational experiment. No simultaneous triangularization is needed in this section. The common-flag estimates remain the stronger uniform statements when the base datum is allowed to approach a boundary.

Throughout this section, M is a real k by k matrix polynomial of degree d , M_d is invertible, and $q = \mathbf{1}^\top M \mathbf{1}$ is monic of degree d . At the fixed clocks all cells of $P_j = M(T_j)/q(T_j)$ and all $q(T_j)$ are strictly positive. Assume $\ell \geq d+1$ and $\tau(P) > 0$. The normalizer coefficient bounds, when imposed, contain the base in their interior. Competitors in this section are positive rational observations; their determinant roots may be complex. They are not asserted to admit a stochastic factorization as in (2.1).

Let $\mathcal{S}$ be a real linear subspace of

$$\mathcal{S}_{\text{all}} = \{E \in \mathbb{R}[z]^{k \times k} : \deg E \leq d, \mathbf{1}^\top E_d \mathbf{1} = 0\}.$$

The fixed-normalizer space is

$$\mathcal{S}_q = \{E : \deg E \leq d, \mathbf{1}^\top E(z) \mathbf{1} \equiv 0\}.$$

Fix a basis $E_1, \dots, E_n$ of $\mathcal{S}$, write $E_x = \sum_i x_i E_i$, and put $e_x = \mathbf{1}^\top E_x \mathbf{1}$. Thus the actual experiment, including its denominator, is

$$F_{\mathcal{S}}(x)_j = \frac{M(T_j) + E_x(T_j)}{q(T_j) + e_x(T_j)}. \quad (\text{V.1})$$

We take all sufficiently small x in an open coefficient neighbourhood. Let $\omega_P^{\mathcal{S}}(t)$ be the diameter of the determinant-root multisets of the competitors in its Hellinger ball of radius t . The Hellinger convention is $h_w(P, P')^2 = \sum_j w_j \sum_{a,b} (\sqrt{P_{j,ab}} - \sqrt{P'_{j,ab}})^2$, with fixed $w_j > 0$.

V.1. The observation metric in numerator coordinates.

Lemma V.1 (Local rational coefficient chart). *The map (V.1) is an analytic embedding near zero. Its derivative and the induced positive definite norm are*

$$\begin{aligned} (\mathcal{L}E)_j &= \frac{E(T_j) - e(T_j)P_j}{q(T_j)}, \\ \mathcal{I}_{\mathcal{S}}(x) &= \sum_j w_j \sum_{a,b} \frac{(\mathcal{L}E_x)_{j,ab}^2}{P_{j,ab}}. \end{aligned} \quad (\text{V.2})$$

Uniformly on bounded sets of v ,

$$h_w(F_{\mathcal{S}}(tv), P) = \frac{1}{2}t\sqrt{\mathcal{I}_{\mathcal{S}}(v)} + O(t^2).$$

In particular the rescaled parameter balls converge in Hausdorff distance to $\mathcal{K}_{\mathcal{S}} = \{v : \mathcal{I}_{\mathcal{S}}(v) \leq 4\}$.

Proof. The derivative follows by differentiating the quotient. If $\mathcal{L}E = 0$, then $E(T_j) = e(T_j)P_j$ at every clock. Since $\deg e < d$, applying the degree- d residual to this equality gives $\mathcal{N}_P e = 0$. The assumption $\tau(P) > 0$ implies $e = 0$. The values of each entry of E consequently vanish at at least $d+1$ distinct clocks, so $E = 0$. The derivative is injective, and an invertible minor supplies a local analytic inverse on the image. The square-root expansion at the finitely many positive cells gives the formula, with a uniform remainder. Positive definiteness bounds x/t on a small observation ball. The uniform expansion gives the outer limiting inclusion; replacing v by $(1-\epsilon)v$ gives the inner inclusion, and then ϵ tends to zero. This also proves Hausdorff convergence, including the boundary of the ellipsoid. $\square$

V.2. A finite constrained Newton invariant. Expand the determinant exactly, rather than truncating its first variation:

$$\det(M(z) + E_x(z)) = D(z) + \sum_{j=1}^k D_j(z, x), \quad D = \det M, \quad (\text{V.3})$$

where D_j is homogeneous of degree j in x . For a distinct root r of D , let m_r be its multiplicity and write

$$D(r + \zeta) = b_r \zeta^{m_r} + O(\zeta^{m_r+1}), \quad b_r \neq 0, \quad D_j(r + \zeta, x) = \sum_{a=0}^{kd} H_{r,j,a}(x) \zeta^a.$$

Here $H_{r,j,a}$ is a homogeneous polynomial on the full allowed space, not its value on a selected direction. Define

$$\alpha_r(\mathcal{S}) = \min_{\substack{1 \leq j \leq k, 0 \leq a < m_r \\ H_{r,j,a} \neq 0}} \frac{j}{m_r - a}, \quad \alpha(\mathcal{S}) = \min_r \alpha_r(\mathcal{S}). \quad (\text{V.4})$$

An empty minimum is $+\infty$. Only a finite determinant expansion and Taylor coefficients of orders less than $m_r \leq kd$ are required. For algebraic input these are exact algebraic calculations. The statement does not require a tolerance-based test of numerical zero.

Suppose $\alpha = \alpha(\mathcal{S}) < \infty$. For each r put

$$\Pi_{r,v}(\zeta) = b_r \zeta^{m_r} + \sum_{\substack{j+\alpha a = \alpha m_r \\ 1 \leq j \leq k}} H_{r,j,a}(v) \zeta^a. \quad (\text{V.5})$$

If $\alpha_r > \alpha$, this is just $b_r \zeta^{m_r}$. Let $\mathcal{Z}_r(v)$ be its multiset of m_r roots, and define

$$C_{\mathcal{S}} = \max_{v,w \in \mathcal{K}_{\mathcal{S}}} \max_r d_{\infty}(\mathcal{Z}_r(v), \mathcal{Z}_r(w)). \quad (\text{V.6})$$

Different r label different base clusters, not ordered individual roots. In particular this formula already takes the permutation quotient inside each repeated-root block.

Theorem V.2 (Exact constrained observation modulus). *Under the hypotheses above, if $\alpha(\mathcal{S}) < \infty$, then $0 < C_{\mathcal{S}} < \infty$ and*

$$\omega_P^{\mathcal{S}}(t) = C_{\mathcal{S}} t^{\alpha(\mathcal{S})} + o(t^{\alpha(\mathcal{S})}). \quad (\text{V.7})$$

The exponent, the limiting set of root multisets, and its diameter are independent of the basis of $\mathcal{S}$. Constant invertible left and right changes of numerator coordinates give the same result when the normalization functional and allowed space are transported with them. If $\alpha(\mathcal{S}) = \infty$, the spectral target is constant on a coefficient neighbourhood and its local modulus is zero.

Proof. We first give a uniform localization argument. If $\|x\| \leq Ct$, then $|H_{r,j,a}(x)| \leq C_{j,a} t^j$. On $|\zeta| = Rt^{\alpha}$, divide (V.3) by $|\zeta|^{m_r}$. For $a < m_r$, the definition of α gives $j - \alpha(m_r - a) \geq 0$. Thus each such perturbation is bounded by $C_{j,a} R^{-(m_r - a)}$. Terms with $a \geq m_r$ tend to zero uniformly for fixed R , while $D(r + \zeta)/\zeta^{m_r}$ tends to b_r . Choosing R sufficiently large and then t sufficiently small, Rouché's theorem puts exactly m_r roots inside this circle, uniformly over the parameter ball. Small fixed disjoint circles about the distinct base roots and invertibility of the leading coefficient exclude other roots and loss at infinity.

With $x = tv$ and $z = r + t^{\alpha}\zeta$, divide the determinant by $t^{\alpha m_r}$. On bounded ζ -sets, uniformly for bounded v , the resulting polynomial converges to (V.5). Every omitted term has strictly higher order. The leading coefficient of that degree- m_r limiting polynomial is the fixed nonzero b_r . The preceding localization and continuity of polynomial roots in the multiset metric therefore give uniform convergence of all the rescaled clusters to $\mathcal{Z}_r(v)$. One may also obtain uniformity by contradiction: a failing sequence of v has a convergent subsequence, and Rouché counting on small circles about the roots of its limiting polynomial gives convergence on that subsequence.

Lemma V.1 now shows that the sets of rescaled root clusters converge in Hausdorff distance to $\{(\mathcal{Z}_r(v))_r : v \in \mathcal{K}_{\mathcal{S}}\}$. Distinct base clusters cannot be matched to each other once t is small, so their bottleneck distance is the maximum of the within-block distances. Taking diameters proves (V.7) and finiteness of its constant. To prove strict positivity, select r, j, a on a minimizing edge. At that fixed a there is at most one j on the edge. Its nonzero homogeneous polynomial has a real v in the interior of $\mathcal{K}_{\mathcal{S}}$ on which it does not vanish. Then $\Pi_{r,v}$ is not a multiple of ζ^{m_r} and has a nonzero root; at $v = 0$ all roots in this block are zero.

An invertible change of basis in $\mathcal{S}$ composes each homogeneous polynomial with the same invertible linear map. It does not change whether that polynomial vanishes identically, and

it transports the metric ellipsoid. A constant left-right change multiplies every determinant coefficient by one common nonzero factor. This proves the invariance assertions. If every minimum is empty, each perturbed determinant is divisible by $\prod_r (z - r)^{m_r} = D(z)/\det M_d$. Its degree is kd and its leading coefficient stays nonzero. It has exactly the same roots with the same multiplicities throughout the neighbourhood. $\square$

Relation to Laurent exposure. For z outside the base spectrum,

$$D_j(z, x) = D(z) \operatorname{tr} \left(\bigwedge^j (M(z)^{-1} E_x(z)) \right).$$

Consequently, if $\rho_{r,j}$ is the largest pole order at r of this exterior-power trace, as a homogeneous polynomial in x , then

$$\alpha_r(\mathcal{S})^{-1} = \max_{1 \leq j \leq k} \frac{\rho_{r,j}}{j},$$

where a trace with no pole contributes zero. These traces include polynomial-in- z perturbations automatically. The first-order visible Laurent filtration is only the term $j = 1$. It need not determine the answer. Exterior powers here denote the usual induced maps on alternating tensors; their traces are the coefficients of $\det(I + sM^{-1}E_x)$. This identity also gives a direct finite way to compute the constrained invariant without choosing a flag.

V.3. Unknown normalization and gauge covariance. Let ν_r be the pole order of the true inverse M^{-1} at r , and let $\nu = \max_r \nu_r$.

Corollary V.3 (Exact exponent with a free monic normalizer). *On $\mathcal{S}_{\text{all}}$,*

$$\alpha_r(\mathcal{S}_{\text{all}}) = 1/\nu_r, \quad \omega_P^{\text{rat}}(t) = C_{\mathcal{S}_{\text{all}}} t^{1/\nu} + o(t^{1/\nu}). \quad (\text{V.8})$$

This holds also when a flag-path count strictly exceeds ν , and also when the recovered coefficients have no common invariant flag.

Proof. Lemma V.1 bounds the full numerator coefficient error by Ct . Near r the true resolvent is bounded by $C|z - r|^{-\nu_r}$. A Neumann estimate on circles of radius Rt^{1/ν_r} , followed by determinant homotopy counting, gives the corresponding upper displacement bound. For the converse, let $L_r \neq 0$ be the leading Laurent matrix. There are real coordinate vectors u, v such that $c = v^T L_r u \neq 0$, even when r is complex. Constant $E = uv^T$ belongs to $\mathcal{S}_{\text{all}}$; its entry sum is allowed to change the constant coefficient of the monic normalizer. The determinant lemma gives

$$\det(M(z) + \epsilon uv^T) = (z - r)^{m_r - \nu_r} b(z) \{ (z - r)^{\nu_r} + \epsilon(c + O(z - r)) \}, \quad b(r) \neq 0.$$

For real ϵ this yields ν_r roots at displacement $|\epsilon|^{1/\nu_r} + o(|\epsilon|^{1/\nu_r})$. All probability and denominator inequalities remain strict at the finitely many clocks. The observation derivative of this nonzero perturbation is nonzero by the lemma, so its Hellinger distance is comparable to $|\epsilon|$. The upper and lower orders force the claimed exponent; Theorem V.2 supplies the exact leading diameter rather than only its order. $\square$

For precision, this is a *fixed-datum* theorem. Its coefficient inverse constant depends on P , and can diverge as $\tau(P)$ tends to zero. It neither replaces nor weakens the uniform additive Theorem N.1. In particular, it does not insert ν for ν_Q in that theorem while leaving its constants unchanged. The pointwise exponent and the two-scale uniform certificate answer different questions.

Choose any fixed left inverse I_d of degree- d evaluation and set

$$\mathcal{H}_e M = I_d(e(T_j)P_j)_j.$$

For $E \in \mathcal{S}_{\text{all}}$ put $e = \mathbf{1}^T E \mathbf{1}$ and $H = E - \mathcal{H}_e M$.

Proposition V.4 (Covariance of the determinant test under the scalar gauge). *The map*

$$E \longleftrightarrow (e, H), \quad E = \mathcal{H}_e M + H,$$

is a linear isomorphism from $\mathcal{S}_{\text{all}}$ onto $\mathbb{R}[z]_{<d} \times \mathcal{S}_q$. The full homogeneous determinant invariant and the metric limiting set are unchanged by this change of coordinates. Thus cancellations must be computed in the complete polynomial $\det(M + \mathcal{H}_e M + H)$, not separately in selected products along a flag.

Proof. Since every P_j has total sum one and I_d reproduces polynomials of degree at most d , $\mathbf{1}^\top \mathcal{H}_e M \mathbf{1} = I_d(e(T_j)) = e$. It follows that $H \in \mathcal{S}_q$. Conversely, every displayed pair produces E with total e and hence zero degree- d entry sum. The two constructions are inverse and linear. Substitution in the determinant gives a literal polynomial identity. Total homogeneous degree and identically vanishing coefficient polynomials are preserved under an invertible linear substitution. The observation derivative and its ellipsoid transform by the same substitution. Apply Theorem V.2. $\square$

There is also a finite two-scale refinement. Choose coordinates u for e and v for H , and collect the gauge determinant as

$$\det(M + \mathcal{H}_e M + H) = D + \sum_{p+q \geq 1} D_{p,q}(z, u, v), \quad p + q \leq k,$$

with bidegrees (p, q) . Denote the nonzero coefficient of $(z - r)^a$ by $H_{r,p,q,a}(u, v)$ and retain only $a < m_r$. For fixed $A, B > 0$, define

$$\alpha_r(A, B) = \min_{H_{r,p,q,a} \neq 0, a < m_r} \frac{Ap + Bq}{m_r - a}. \quad (\text{V.9})$$

Proposition V.5 (Two-scale Newton law). *In the coefficient boxes $\|e\| \leq s$, $\|H\| \leq t$, the maximal root displacement from the base cluster at r is bounded by*

$$C \max_{H_{r,p,q,a} \neq 0, a < m_r} (s^p t^q)^{1/(m_r - a)}.$$

On $s = \epsilon^A$, $t = \epsilon^B$, the bound is attained up to positive constants; the exponent is (V.9). The empty case has zero displacement. These are coefficient boxes, not independent statistical observation balls for the two errors.

Proof. The coefficients are bounded by $Cs^p t^q$. On a circle whose radius is a sufficiently large multiple of the displayed maximum, division by $(z - r)^{m_r}$ makes the sum of the finitely many perturbations smaller than the base unit. Rouché gives the upper bound and cluster count. For the lower bound take $e = \epsilon^A e_0$ and $H = \epsilon^B H_0$. After rescaling by $\epsilon^{\alpha_r(A,B)}$, the edge polynomial has at least one nonzero nonleading coefficient for a suitable real pair (e_0, H_0) in the interiors of the unit balls. Indeed, different bidegrees are linearly independent as polynomials in the two independent groups of variables. Their sum cannot vanish identically unless each term does. A generic real pair therefore gives a nonzero root of the edge polynomial, proving the matching order. The same compactness argument as in Theorem V.2 is an alternative proof of attainment. If no coefficient is present below m_r , the root is fixed with its full multiplicity. $\square$

The error split (N.6) permits the substitution $s = C\delta/\tau(P)$ and $t = C\delta$ in this finite bound. Unlike the flag-path certificate, its coefficients retain every cancellation at the specified datum. Its constants are datum-dependent. Changing I_d is a linear shear between the two coordinate groups; the total degree invariant is unchanged, whereas the interpretation of separate coefficient boxes must be transported with that shear.

V.4. An invisible first variation with a nonzero modulus.

Proposition V.6 (A quadratic exposure family). *For every $m \geq 3$ there is a strictly positive rational probability experiment with invertible leading coefficient and $\tau > 0$ for which all first-order constrained Laurent coefficients at one root vanish, but*

$$\omega_P^{[q]}(t) \asymp t^{2/m}, \quad \omega_P^{\text{rat}}(t) \asymp t^{1/m}.$$

The full leading constants in both formulas are given by (V.6).

Proof. Choose distinct $0 < a_1 < \dots < a_m$ and clocks $T_j > a_m$, with $\ell \geq 2m + 1$. Set

$$f = z^m, \quad g = \prod_{i=1}^m (z - a_i), \quad M = \frac{1}{4} \begin{pmatrix} f + g & f - g \\ f - g & f + g \end{pmatrix}, \quad q = f.$$

Its leading coefficient is $I/2$. At each clock $0 < g(T_j) < f(T_j)$, so every cell of $M(T_j)/f(T_j)$ is strictly positive. Also f and g are coprime. To check $\tau > 0$ directly, let a polynomial e of degree less than m be in its normalization kernel. Interpolation supplies a polynomial numerator of degree at most m . Multiplying its off-diagonal entry by f shows, at $2m + 1$ clocks and hence identically, that f divides $e(f - g)$. Coprimality gives $f \mid e$, so $e = 0$.

Let $O = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. Then $2O^T M O = \text{diag}(f, g)$. The condition $\mathbf{1}^T E \mathbf{1} \equiv 0$ is exactly the condition that $2O^T E O$ have upper-left entry zero. Write this transformed perturbation as $\begin{pmatrix} 0 & u(z) \\ v(z) & h(z) \end{pmatrix}$, with all three polynomials arbitrary of degree at most m . Apart from the harmless factor $1/4$, its determinant is

$$f(g + h) - uv. \tag{V.10}$$

At zero, the linear part is divisible by z^m , while the quadratic coefficient $-u(0)v(0)$ need not vanish. Equivalently the first-order resolvent trace is h/g , analytic at zero for every allowed polynomial perturbation. Thus $\alpha_0(\mathcal{S}_q) = 2/m$, despite the complete absence of a first-order visible pole. The other roots are simple and a constant h exposes each of them, giving exponent one there. Since $m \geq 3$, the whole spectral exponent is $2/m$. Taking $u = v = \epsilon$, $h = 0$ yields roots with displacement $|g(0)|^{-1/m} |\epsilon|^{2/m} + o(|\epsilon|^{2/m})$ at zero. Positivity persists at the clocks, so these are admissible alternatives.

When the normalizer is free, the transformed upper-left entry may also have an arbitrary constant perturbation a . The determinant becomes $(f + a)(g + h) - uv$. The term $ag(0)$ exposes the full pole of order m ; the coefficient of z^m in q remains one. Corollary V.3 gives exponent $1/m$. This example belongs to the positive rational extension; no real-rooted stochastic factorization of its alternatives is used. $\square$

In the three by three cancellation example of Section U, the true pole order is three although the unweighted flag-path count is six. Its leading Laurent matrix is exposed on $\mathcal{S}_q$. Hence the exact fixed-normalizer and free-normalizer exponents are both $1/3$, not $1/6$. The latter assertion no longer requires equality of the pole and path counts. The proof uses the true numerator and the exact scalar gauge, not a perturbed triangular path bound.

The determinant identity, Newton rescaling and root-counting tools are classical perturbation mechanisms; see [20, 27, 30, 24]. The assertions proved here specify the full probability-normalized perturbation space, include higher-order exposure, pull its limiting set back through the Hellinger metric, and transport it through the unknown scalar normalizer. They classify this local rational experiment, rather than all normalization-kernel or channel-rank singularities of the original stochastic model.

APPENDIX W. RELATION TO THE OTHER OBSERVATION LAWS

In degree one, normalizing raw matrices $c(T)U \operatorname{diag}(a_b(T - \xi_b))V^\top$, with an unknown positive scalar detector $c(T)$, gives (2.1) with $\alpha_b = a_b / \sum_c a_c$. In particular the logarithmic action law is unchanged:

$$\log \frac{p_b(T)}{p_c(T)} = \log \frac{\alpha_b}{\alpha_c} + \log(T - \xi_b) - \log(T - \xi_c).$$

For $k = m_1 = m_2 = 2$, stochastic columns imply

$$\frac{|\det U|}{\sqrt{2}} \leq \sigma_2(U) \leq |\det U|, \quad \frac{|\det V|}{\sqrt{2}} \leq \sigma_2(V) \leq |\det V|.$$

Indeed the largest singular value lies between 1 and $\sqrt{2}$, and the product of the two singular values is the absolute determinant. On an action class with $\xi_2 - \xi_1 \geq \Delta > 0$, Proposition O.4 therefore gives the single inverse power of $|\det U \det V|$. The identity $\alpha_1 = (\xi_2 - h)/(\xi_2 - \xi_1)$ also recovers the relative throughput with a constant depending on this fixed gap.

The shared two-clock spatial experiment has a different nuisance space: the clock factors are common across sites. Spatial calibration can replace a third clock there. Its exact identification and one-clock geometric alternatives, and its local correctly-labelled calibration estimate, are separate from the pointwise projective experiment treated here. The local estimate is not a global bound through collisions of spatial profiles.

The polynomial, rational and compact-surface detector quotients in the historical derivations retain their respective secant and tangent tests, divisor counts, and branched-clock forms of the logarithmic law. Theorem C.1 concerns the specific simultaneous polynomial numerator and its scalar normalization. Its normalization-rank criterion and sharp worst-case $2d + 1$ count do not declare every ambient meromorphic pole-budget bound optimal. In particular, common-factor cancellation in (C.1) is a property of this projective observation, not an identification of all those detector quotients with one another.

Finite-record geometric certificates, component coverings, return-lift constructions, and raw-acquisition laws remain logically separate. Where the recovered actions determine boundary-distance data, metric rigidity is supplied by the stated classical geometric rigidity hypotheses in those derivations. The present results concern the upstream observation inverse. All proofs of the principal results above are contained in this article and require no geometric appendix; the other regimes are retained as an independent archive rather than added as hypotheses to this theorem.

APPENDIX X. OBSERVATION GEOMETRY AND CLASSICAL CONDITIONING

The perturbation of an already specified matrix polynomial and the inversion of a normalized observation are distinct maps. The distinction can be expressed at the level of their differentials, and identifies which ingredients of the preceding proofs are classical.

For a regular matrix polynomial M , a simple finite eigenvalue x , and nonzero left and right null vectors y, v , differentiation gives

$$\dot{x} = -\frac{y^* \dot{M}(x) v}{y^* M'(x) v}.$$

Tisseur's coefficient perturbation framework [29] computes condition numbers by maximizing such a functional over specified coefficient perturbation balls. Anguas–Bueno–Dopico [8] compare nonhomogeneous and homogeneous eigenvalue condition numbers, including the treatment of zero and infinite eigenvalues. Accordingly, one must specify coefficient weights, the eigenvalue metric, and the admissible perturbations before comparing condition numbers. Our leading-invertibility

hypotheses and fixed finite-root disk keep the present target in a finite affine chart; a change to a homogeneous metric is not implicitly made in an observation norm.

At multiple eigenvalues, the Jordan and polynomial spectral background is supplied by [20, 27]. Kressner–Peláez–Moro [24] study structured Hölder conditioning of multiple eigenvalues and matrix pencils. In particular, multiplicity exponents and the effect of restrictions on perturbations belong to an established theory. The Laurent exclusion argument in Proposition M.2 and the argument-principle root counts are instances of this classical machinery. Neither a reciprocal multiplicity exponent nor an expansion of a triangular inverse, by itself, is a novelty claim here.

The additional object is the pullback from the probability observation. In the notation of (K.5), an observation tangent Z produces

$$\dot{M}_Z = Q_1^\top I_d(qZ + r_Z P) Q_2, \quad r_Z = \mathcal{N}_P^\dagger[-(\Omega \otimes I)(q(T_j) \text{vec } Z_j)_j].$$

Substituting this map into the simple-root functional describes the local observation condition. Bounding its two summands independently by a single unstructured coefficient condition can multiply the normalization loss by a second matrix loss. Theorem E.1 instead performs the scalar gauge first on complete-polynomial subspaces. Theorem N.1 performs it on a common invariant flag, even when the coefficient algebra is noncommutative or defective. Only the remaining arbitrary matrix error is then charged the flag-path resolvent exponent. These are observation-level statements about a specific structured pullback, not replacements for coefficient conditioning theory. They do not assert the existence of such a flag for every matrix-polynomial algebra.

Theorem K.2 is the norm of the target differential on the recovered quotient manifold in the specified observation metric. Its Gram-matrix expression is elementary once that manifold and all nearby representatives have been identified. Theorem L.1 uses the classical local asymptotic normality and minimax framework [21, 31]; the finite-permutation Gaussian experiment is the step needed for the unordered target at collisions. The same observable matrix V_H , not the global certificate η_d , governs both its metric and statistical assertions.

Finally, semialgebraic projection, choice and Puiseux expansions in Theorem F.1 and Corollary F.2 are classical [13, 9]. Their application retains all the stochastic and real-rooted inequalities in the admissible arcs. This is essential: the positive sum-of-squares identity in Lemma F.3 improves an anchored real-rooted comparison, whereas the two moving alternatives in Theorem R.3 still give the slower fixed-neighbourhood rate. An unstructured complex coefficient ball, a fixed parameter neighbourhood and a shrinking probability neighbourhood cannot be substituted for one another in these conclusions.

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